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Engineered glycosyltransferases and steviol glycoside glucosylation methods

Abstract

The present invention provides engineered glycosyltransferase (GT) enzymes, polypeptides having GT activity, and polynucleotides encoding these enzymes, as well as vectors and host cells comprising these polynucleotides and polypeptides. The present invention provides engineered sucrose synthase (SuS) enzymes, polypeptides having SuS activity, and polynucleotides encoding these enzymes, as well as vectors and host cells comprising these polynucleotides and polypeptides. The present invention also provides compositions comprising the GT enzymes and methods of using the engineered GT enzymes to make products with β -glucose linkages. The present invention further provides compositions and methods for the production of rebaudiosides (e.g., rebaudioside M, rebaudioside A, rebaudioside I, and rebaudioside D). The present invention also provides compositions comprising the SuS enzymes and methods of using them. Methods for producing GT and SuS enzymes are also provided.

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Background/Summary

(1) The present application is a Divisional of co-pending U.S. patent application Ser. No. 15/886,094, filed Feb. 1, 2018, ABN, which claims priority to U.S. Prov. Pat. Appln. Ser. No. 62/454,417, filed Feb. 3, 2017, and U.S. Prov. Pat. Appln. Ser. No. 62/479,262, filed Mar. 30, 2017, all of which are hereby incorporated by reference in their entireties, for all purposes.

FIELD OF THE INVENTION

(1) The present invention provides engineered glycosyltransferase (GT) enzymes, polypeptides having GT activity, and polynucleotides encoding these enzymes, as well as vectors and host cells comprising these polynucleotides and polypeptides. The present invention provides engineered sucrose synthase (SuS) enzymes, polypeptides having SuS activity, and polynucleotides encoding these enzymes, as well as vectors and host cells comprising these polynucleotides and polypeptides. The present invention also provides compositions comprising the GT enzymes and methods of using the engineered GT enzymes to make products with β -glucose linkages. The present invention further provides compositions and methods for the production of rebaudiosides (e.g., rebaudioside M, rebaudioside A, rebaudioside I, and rebaudioside D). The present invention also provides compositions comprising the SuS enzymes and methods of using them. Methods for producing GT and SuS enzymes are also provided.

REFERENCE TO SEQUENCE LISTING, TABLE OR COMPUTER PROGRAM

(2) The official copy of the Sequence Listing is submitted concurrently with the specification as an ASCII formatted text file via EFS-Web, with a file name of "CX8-162WO3_ST25.txt", a creation date of Jan. 24, 2018, and a size of 31,100 kilobytes. The Sequence Listing filed via EFS-Web is part of the specification and is incorporated in its entirety by reference herein.

BACKGROUND OF THE INVENTION

(3) Glycosyltransferases (GT) are enzymes that post-translationally transfer glycosyl residues from an activated nucleoside sugar to monomeric and polymeric acceptor molecules (e.g., other sugars, proteins, lipids, and other organic substrates). Thus, these enzymes utilize an activated donor sugar substrate that contains a substituted phosphate leaving group. Donor sugar substrates (i.e., the "glycosyl donor") are commonly activated as nucleoside diphosphate sugars. However, other sugars, such as nucleoside monophosphate sugars, lipid phosphates and unsubstituted phosphates are also used (See e.g., Lairson et al., *Ann. Rev. Biochem.*, 77:25.1-25.35 [2008]). GTs are classified as either retaining or inverting enzymes, based on the stereochemistry of the substrates and reaction products. In reactions where the stereochemistry of the donor's anomeric bond is retained (e.g., alpha to alpha), the GT is a retaining enzyme. In reactions where the stereochemistry is inverted (e.g., alpha to beta), the GT is an inverting enzyme. These glycosylated products are involved in various metabolic pathways and processes. Indeed, the biosynthesis of numerous disaccharides, oligosaccharides, and polysaccharides involve the action of various glycosyltransferases. The transfer of a glucosyl moiety can alter the acceptor's bioactivity, solubility, and transport properties within cells. GTs have found use in the targeted synthesis of specific compounds (e.g., glycoconjugates and glycosides), as well as the production of differentially glycosylated drug, biological probes or natural product libraries. In some methods, the large scale use of GTs for glycoconjugate synthesis requires large quantities of glycosyl donors, adding to the cost of such approaches. Nucleotide recycling systems have been developed to allow the resynthesis of glycosyl donors from the released nucleotide. These recycling systems also reduce the amount of nucleotide by-product formed during the reaction, thereby reducing inhibition caused by the GT. Nonetheless, the need remains for improved methods suitable for large-scale production of glycoconjugates by GTs.

SUMMARY OF THE INVENTION

(4) The present invention provides engineered glycosyltransferase (GT) enzymes, polypeptides having GT activity, and polynucleotides encoding these enzymes, as well as vectors and host cells comprising these polynucleotides and polypeptides. The present invention provides engineered sucrose synthase (SuS) enzymes, polypeptides having SuS activity, and polynucleotides encoding these enzymes, as well as vectors and host cells comprising these polynucleotides and polypeptides. The present invention also provides compositions comprising the GT enzymes and methods of using the engineered GT enzymes to make products with β -glucose linkages. The present invention further provides compositions and methods for the production of rebaudiosides (e.g., rebaudioside M, rebaudioside A, rebaudioside I, and rebaudioside D). The present invention also provides compositions comprising the SuS enzymes and methods of using them. Methods for producing GT and SuS enzymes are also provided.

(5) The present invention provides engineered glycosyltransferases comprising polypeptide sequences that have at least 60%, 65%, 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NO:2. In some embodiments, the engineered glycosyltransferase comprises a polypeptide that has 60%, 65%, 70%, 75%, 80%, 85%, 86%,

87%, 88%, 89%, 90%, 91%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NO: 4, 8, 32, 232, 348, 548, 562, 696, 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 3956, 4256, 4550, 4684, 4838, 4876, 5066, 5290, 5372, 5562, 5708, 5976, 6138, 6288, 6468, 6864, 7324, 7388, 7784, and/or 8088. In some additional embodiments, the engineered glycosyltransferase comprises a polypeptide that has at least 90% sequence identity to SEQ ID NO: 4, 8, 32, 232, 348, 548, 562, 696, 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 3956, 4256, 4550, 4684, 4838, 4876, 5066, 5290, 5372, 5562, 5708, 5976, 6138, 6288, 6468, 6864, 7324, 7388, 7784, and/or 8088. In some additional embodiments, the engineered glycosyltransferase comprises a polypeptide that has at least 91% sequence identity to SEQ ID NO: 4, 8, 32, 232, 348, 548, 562, 696, 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 3956, 4256, 4550, 4684, 4838, 4876, 5066, 5290, 5372, 5562, 5708, 5976, 6138, 6288, 6468, 6864, 7324, 7388, 7784, and/or 8088. In some additional embodiments, the engineered glycosyltransferase comprises a polypeptide that has at least 92% sequence identity to SEQ ID NO: 4, 8, 32, 232, 348, 548, 562, 696, 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 3956, 4256, 4550, 4684, 4838, 4876, 5066, 5290, 5372, 5562, 5708, 5976, 6138, 6288, 6468, 6864, 7324, 7388, 7784, and/or 8088. In some additional embodiments, the engineered glycosyltransferase comprises a polypeptide that has at least 93% sequence identity to SEQ ID NO: 4, 8, 32, 232, 348, 548, 562, 696, 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 3956, 4256, 4550, 4684, 4838, 4876, 5066, 5290, 5372, 5562, 5708, 5976, 6138, 6288, 6468, 6864, 7324, 7388, 7784, and/or 8088. In some additional embodiments, the engineered glycosyltransferase comprises a polypeptide that has at least 94% sequence identity to SEQ ID NO: 44, 8, 32, 232, 348, 548, 562, 696, 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 3956, 4256, 4550, 4684, 4838, 4876, 5066, 5290, 5372, 5562, 5708, 5976, 6138, 6288, 6468, 6864, 7324, 7388, 7784, and/or 8088. In some additional embodiments, the engineered glycosyltransferase comprises a polypeptide that has at least 95% sequence identity to SEQ ID NO: 4, 8, 32, 232, 348, 548, 562, 696, 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 3956, 4256, 4550, 4684, 4838, 4876, 5066, 5290, 5372, 5562, 5708, 5976, 6138, 6288, 6468, 6864, 7324, 7388, 7784, and/or 8088. In some additional embodiments, the engineered glycosyltransferase comprises a polypeptide that has at least 96% sequence identity to SEQ ID NO: 4, 8, 32, 232, 348, 548, 562, 696, 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 3956, 4256, 4550, 4684, 4838, 4876, 5066, 5290, 5372, 5562, 5708, 5976, 6138, 6288, 6468, 6864, 7324, 7388, 7784, and/or 8088. In some additional embodiments, the engineered glycosyltransferase comprises a polypeptide that has at least 97% sequence identity to SEQ ID NO: 4, 8, 32, 232, 348, 548, 562, 696, 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 3956, 4256, 4550, 4684, 4838, 4876, 5066, 5290, 5372, 5562, 5708, 5976, 6138, 6288, 6468, 6864, 7324, 7388, 7784, and/or 8088. In some additional embodiments, the engineered glycosyltransferase comprises a polypeptide that has at least 98% sequence identity to SEQ ID NO: 44, 8, 32, 232, 348, 548, 562, 696, 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 3956, 4256, 4550, 4684, 4838, 4876, 5066, 5290, 5372, 5562, 5708, 5976, 6138, 6288, 6468, 6864, 7324, 7388, 7784, and/or 8088. In some additional embodiments, the engineered glycosyltransferase comprises a polypeptide that has at least 99% sequence identity to SEQ ID NO: 44, 8, 32, 232, 348, 548, 562, 696, 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 3956, 4256, 4550, 4684, 4838, 4876, 5066, 5290, 5372, 5562, 5708, 5976, 6138, 6288, 6468, 6864, 7324, 7388, 7784, and/or 8088. In some additional embodiments, the engineered glycosyltransferase comprises a polypeptide selected from SEQ ID NO: 4, 8, 32, 232, 348, 548, 562, 696, 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 3956, 4256, 4550, 4684, 4838, 4876, 5066, 5290, 5372, 5562, 5708, 5976, 6138, 6288, 6468, 6864, 7324, 7388, 7784, and/or 8088. In some additional embodiments, the engineered glycosyltransferase is selected from beta-1,2-glycosyltransferases and beta-1,3-glycosyltransferases. In some further embodiments, the engineered glycosyltransferase preferentially uses a sugar donor other than uracil-diphosphate. In some additional embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a variant engineered glycosyltransferase provided in Table 2.1, 3.1, 5.1, 6.1, 6.3, 8.1, 9.1, 9.2, 9.4, 11.1, 12.1, 14.1, 15.1, 15.2, 15.3, 16.1, 17.1, 43.1, 43.2, 44.2, 45.1, 45.3, 46.1, 46.2, 46.3, 47.1, 47.2, 47.3,

48.1, 48.2, 49.1, 49.2, 49.3, 50.1, 50.2, 50.3, 50.4, 51.1, 51.2, 52.1, 53.1, 53.2, 54.1, 54.2, 54.3, 55.1, 55.2, 55.3, 56.1, 56.2, 56.3, 57.1, 58.1, 58.2, 58.3, 59.1, 59.3, 59.3, 60.1, 60.2, 61.1, 61.2, 62.1, 62.2, 63.1, 63.2, 64.1, 64.2, 65.1, 65.2, 66.1, 66.2, 67.1, 67.2, 67.3, 68.1, 68.2, 69.1, 69.2, 70.1, 70.2, 71.1, 71.2, 71.3, 72.1, 72.2, 72.3, 73.1, 73.2, 74.1, 74.2, 74.3, 75.1, 75.2, 75.3, 77.1, and/or 77.2. The present invention provides engineered glycosyltransferases comprising polypeptide sequences that have at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NOS: 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, 120, 122, 124, 126, 128, 130, 132, 134, 136, 138, 140, 142, 144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 172, 174, 176, 178, 180, 182, 184, 186, 188, 190, 192, 194, 196, 198, 200, 202, 204, 206, 208, 210, 212, 214, 216, 218, 220, 222, 224, 226, 228, 230, 232, 234, 236, 238, 240, 242, 244, 246, 248, 250, 252, 254, 256, 258, 260, 262, 264, 266, 268, 270, 272, 274, 276, 278, 280, 282, 284, 286, 288, 290, 292, 294, 296, 298, 300, 302, 304, 306, 308, 310, 312, 314, 316, 318, 320, 322, 324, 326, 328, 330, 332, 334, 336, 338, 340, 342, 344, 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7938, 7940, 7942, 7944, 7946, 7948, 7950, 7952, 7954, 7956, 7958, 7960, 7962, 7964, 7966, 7968, 7970, 7972, 7974, 7976, 7978, 7980, 7982, 7984, 7986, 7988, 7990, 7992, 7994, 7996, 7998, 8000, 8002, 8004, 8006, 8008, 8010, 8012, 8014, 8016, 8018, 8020, 8022, 8024, 8026, 8028, 8030, 8032, 8034, 8036, 8038, 8040, 8042, 8044, 8046, 8048, 8050, 8052, 8054, 8056, 8058, 8060, 8062, 8064, 8066, 8068, 8070, 8072, 8074, 8076, 8078, 8080, 8082, 8084, 8086, 8088, 8090, 8092, 8094, 8096, 8098, 8100, 8102, 8104, 8106, 8108, 8110, 8112, 8114, 8116, 8118, 8120, 8122, 8124, 8126, 8128, 8130, 8132, 8134, 8136, 8138, 8140, 8142, 8144, 8146, 8148, 8150, 8152, 8154, 8156, 8158, 8160, 8162, 8164, 8166, 8168, 8170, 8172, 8174, 8176, 8178, 8180, 8182, 8184, 8186, 8188, 8190, 8192, 8194, 8196, 8198, 8200, 8202, 8204, 8206, 8208, 8210, 8212, 8214, 8216, 8218, 8220, 8222, 8224, 8226, 8228, 8230, 8232, 8234, 8236, 8238, 8240, 8242, 8244, 8246, 8248, 8250, 8252, 8254, 8256, 8258, 8260, 8262, 8264, 8266, 8268, 8270, 8272, 8274, 8276, 8278, 8280, 8282, 8284, 8286, 8288, 8290, 8292, 8294, 8296, 8298, 8300, 8302, 8304, 8306, 8308, 8310, 8312, 8314, 8316, 8318, 8320, 8322, 8324, 8326, 8328, 8330, 8332, 8334, 8336, 8338, 8340, 8342, 8344, 8346, 8348, 8350, 8352, 8354, 8356, 8358, 8360, 8362, 8364, 8366, 8368, 8370, 8372, 8374, 8376, 8378, 8380, 8382, 8384, 8386, 8388, 8390, 8392, 8394, 8396, 8398, 8400, 8402, 8404, 8406, 8408, 8410, 8412, 8414, 8416, 8418, 8420, 8422, 8424, 8426, 8428, 8430, 8432, 8434, 8436, 8438, 8440, 8442, 8444, 8446, 8448, 8450, 8452, 8454, 8456, 8458, 8460, 8462, 8464, 8466, 8468, 8470, 8472, 8474, 8476, 8478, 8480, 8482, 8484, 8486, 8488, 8490, 8492, 8494, 8496, 8498, 8500, 8502, 8504, 8506, 8508, 8510, 8512, 8514, 8516, 8518, 8520, 8522, 8524, 8526, 8528, 8530, 8532, 8534, 8536, 8538, 8540, 8542, 8544, 8546, 8548, 8550, 8552, 8554, 8556, 8558, 8560, 8562, 8564, 8566, 8568, 8570, 8572, 8574, 8576, 8578, 8580, 8582, 8584, 8586, 8588, 8590, 8592, 8594, 8596, 8598, 8600, 8602, 8604, 8606, 8608, 8610, 8612, 8614, 8616, 8618, 8620, 8622, 8624, 8626, 8628, 8630, 8632, 8634, 8636, 8638, 8640, 8642, 8644, 8646, 8648, 8650, 8652, 8654, 8656, 8658, 8660, 8662, 8664, 8666, 8668, 8670, 8672, 8674, 8676, 8678, 8680, 8682, 8684, 8686, 8688, 8690, 8692, 8694, 8696, 8698, 8700, 8702, 8704, 8706, 8708, 8710, 8712, 8714, 8716, 8718, 8720, 8722, 8724, 8726, 8728, 8730, 8732, 8734, 8736, 8738, 8740, 8742, 8744, 8746, 8748, 8750, 8752, 8754, 8756, 8758, 8760, 8762, 8764, 8766, 8768, 8770, 8772, 8774, 8776, 8778, 8780, 8782, 8784, 8786, 8788, 8790, 8792, 8794, 8796, 8798, 8800, 8802, 8804, 8806, 8808, 8810, 8812, 8814, 8816, 8818, 8820, 8822, 8824, 8826, 8828, 8830, 8832, 8834, 8836, 8838, 8840, 8842, 8844, 8846, 8848, 8850, 8852, 8854, 8856, 8858, 8860, 8862, 8864, 8866, 8868, 8870, 8872, 8874, 8876, 8878, 8880, 8882, 8884, 8886, 8888, 8890, 8892, 8894, 8896, 8898, 8900, 8902, 8904, 8906, 8908, 8910, 8912, 8914, 8916, 8918, 8920, 8922, 8924, 8926, 8928, 8930, 8932, 8934, 8936, 8938, 8940, 8942, 8944, 8946, 8948, 8950, 8952, 8954, 8956, 8958, 8960, 8962, 8964, 8966, 8968, 8970, 8972, 8974, 8976, 8978, 8980, 8982, 8984, 8986, 8988, 8990, 8992, 8994, 8996, 8998, 9000, 9002, 9004, 9006, 9008, 9010, 9012, 9014, 9016, 9018, 9020, 9022, 9024, 9026, 9028, 9030, 9032, 9034, 9036, 9038, 9040, 9042, 9044, 9046, 9048, 9050, 9052, 9054, 9056, 9058, 9060, 9062, 9064, 9066, 9068, 9070, 9072, 9074, 9076, 9078, 9080, 9082, 9084, 9086, 9088, 9090, 9092, 9094, 9096, 9098, 9100, 9102, 9104, 9106, 9108, 9110, 9112, 9114, 9116, 9118, 9120, 9122, 9124, 9126, 9128, 9130, 9132, 9134, 9136, 9138, 9140, 9142, 9144, 9146, 9148, 9150, 9152, 9154, 9156, 9158, 9160, 9162, 9164, 9166, 9168, 9170, 9172, 9174, 9176, 9178, 9180, 9182, 9184, 9186, 9188, 9190, 9192, 9194, 9196, 9198, 9200, 9202, 9204, 9206, 9208, 9210, 9212, 9214, 9216, 9218, 9220, 9222, 9224, 9226, 9228, 9230, 9232, 9234, 9236, 9238, and/or 9240.

(6) In some further embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 10, 10/309, 262, 278/284/311/339/360, 283, 307, 309, 339/361, 344/361, and 361, wherein the positions are numbered with reference to SEQ ID NO:4. In some embodiments, the polypeptide sequence of the engineered Glycosyltransferase comprises at least one mutation or mutation set selected from 10-/309R, 262K, 262L, 278L/284I/311G/339A/360G, 283T, 307V, 309L/N/R/S, 339A/361G, 344I/361G, and 361G, wherein the positions are numbered with reference to SEQ ID NO:4. In some further embodiments, the polypeptide

sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set selected from R10-/V309R, R262K, R262L, Y278L/T284I/R311G/V339A/N360G, S283T, L307V, V309L/N/R/S, V339A/S361G, V344I/S361G, and S361G, wherein the positions are numbered with reference to SEQ ID NO:4. In still some additional embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, and/or 30. In some further embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least identical to any of SEQ ID NOS: 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, and/or 30. In yet some additional embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises SEQ ID NOS: 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, and/or 30.

(7) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 112/172/283/318, 112/261/318, 112/282/283/431, 137/283, 137/283/431, 163/318, 261/283/306/337, 261/283/337, 261/337, 269/318, 282/283, 282/283/431, 283, 283/306/308/360, 283/306/337/426, 283/318/337/360, 283/360, 318, 360, and 431, wherein the positions are numbered with reference to SEQ ID NO:8. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set selected from 112S/172S/283Q/318E, 112S/261S/318E, 112S/282T/283Q/431E, 137K/283Q, 137K/283Q/431E, 163K/318E, 261S/283Q/306V/337F, 261S/283Q/337F, 261S/337S, 269T/318E, 282T/283Q, 282T/283Q/431E, 283Q, 283Q/306V/3085/360G, 283Q/306V/337S/426V, 283Q/318E/337S/360G, 283Q/360G, 318E, 360G, and 431E, wherein the positions are numbered with reference to SEQ ID NO:8. In some further embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set selected from E112S/T172S/T283Q/T318E, E112S/R261S/T318E, E112S/S282T/T283Q/Q431E, N137K/T283Q, N137K/T283Q/Q431E, L163K/T318E, R261S/T283Q/L306V/W337F, R261S/T283Q/W337F, R261S/W337S, Q269T/T318E, S282T/T283Q, S282T/T283Q/Q431E, T283Q, T283Q/L306V/R3085/5360G, T283Q/L306V/W337S/A426V, T283Q/T318E/W337S/S360G, T283Q/S360G, T318E, S360G, and Q431E, wherein the positions are numbered with reference to SEQ ID NO:8. In some additional embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, and/or 70. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, and/or 70. In some additional embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises SEQ ID NOS: 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, and/or 70.

(8) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 4, 6, 22, 64, 74, 84, 87, 97, 106, 110, 112, 137, 139, 154, 159, 169, 179, 191, 195, 198, 199, 207, 233, 259, 261, 262, 306, 347, 356, 396, 417, 421, 427, and 435, wherein the positions are numbered with reference to SEQ ID NO:32. In yet some additional embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set selected from 4P, 6P, 22A/L/H/P, 64P, 74W, 84A/G, 87A/H, 97S, 106D/G/S/T, 1105, 112A/P, 137G, 139P, 154A/L/Q/V, 159M/R, 169T, 179V, 191R, 195G, 198M/S/V, 199A/D/G/K/Q/S, 207L, 233R, 259Q, 261A/H/P/W, 262G, 306V, 347D, 356G, 396R, 417A/R/P, 421V, 427A, and 435Q/R, wherein the positions are numbered with reference to SEQ ID NO:32. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set selected from K4P, E6P, Q22A/L/H/P, F64P, R74W, L84A/G, M87A/H, A97S, L106D/G/S/T, A1105, E112A/P, N137G, R139P, H154A/L/Q/V, Q159M/R, D169T, S179V, S191R, N195G, I198M/S/V, L199A/D/G/K/Q/S, I207L, I233R, H259Q, R261A/H/P/W, T262G, L306V, G347D, S356G, Y396R, E417A/R/P, Y421V, R427A, and V435Q/R, wherein the positions are numbered with reference to SEQ ID NO:32.

(9) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or

more positions selected from 22/84/87/154/198/199/356, 22/84/87/154/198/199/306/356, 22/84/87/154/199/306/356, 22/84/87/154/356, 22/84/87/154/198/199/306/356, 22/84/87/154/199/306/356, 22/84/87/198/199/306/356, 22/84/87/198/199/356, 22/84/87/198/199/306/356, 22/84/87/198/199/356, 22/84/87/199/306/356, 22/84/87/199/356, 22/84/154/198/199/207/306, 22/84/154/198/199/306, 22/84/154/198/199/356, 22/84/154/198/199/356, 22/84/154/198/199/356, 22/84/154/199/356, 22/84/154/199/207, 22/84/154/199/356, 22/84/154/207/306/356, 22/84/154/306/356, 22/84/154/198/199/306/356, 22/84/154/198/199, 22/84/154/198/199/306/356, 22/84/154/199/306/356, 22/84/154/199/356, 22/84/154/199/356, 22/84/198/199/306/356, 22/84/199/356, 22/84/207/356, 22/84/356, 22/84/154/198/199/306/356, 22/87/154/198/199/356, 22/87/154/199/306/356, 22/87/154/322/356, 22/87/154/198/199/356, 22/87/154/199/356, 22/87/154/198/199/207/306/356, 22/87/154/199/207/356, 22/87/154/199/356, 22/87/154/199/306/356, 22/87/198/199/306/356, 22/87/198/199/306/356, 22/87/198/199/207/356, 22/87/198/199/356, 22/87/199/356, 22/87/199/356, 22/154/198/199/207/306/356, 22/154/198/199/356, 22/154/199/356, 22/154/198/199/306/356, 22/154/199/356, 22/154/199/207/306/356, 22/198/199/356, 22/198/199/207/329/356, 22/199/356, 22/207/356, 22/356, and 84/154/198/199, wherein the positions are numbered with reference to SEQ ID NO:32. In some additional embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set selected from 22P/84G/87H/154L/198S/199A/356G, 22P/84G/87H/154L/198S/199S/306V/356G, 22P/84G/87H/154L/199K/306V/356G, 22P/84G/87H/154L/356G, 22P/84G/87H/154V/198S/199S/306V/356G, 22P/84G/87H/154V/199K/306V/356G, 22P/84G/87H/198S/199A/306V/356G, 22P/84G/87H/198S/199K/356G, 22P/84G/87H/198S/199S/306V/356G, 22P/84G/87H/198S/199S/356G, 22P/84G/87H/199A/306V/356G, 22P/84G/87H/199S/356G, 22P/84G/154L/198S/199A/207L/306V, 22P/84G/154L/198S/199A/306V, 22P/84G/154L/198S/199A/356G, 22P/84G/154L/198S/199K/356G, 22P/84G/154L/198S/199S/356G, 22P/84G/154L/199A/356G, 22P/84G/154L/199K/207L, 22P/84G/154L/199S/356G, 22P/84G/154L/207L/306V/356G, 22P/84G/154L/306V/356G, 22P/84G/154V/198S/199A/306V/356G, 22P/84G/154V/198S/199K, 22P/84G/154V/198S/199K/306V/356G, 22P/84G/154V/199A/306V/356G, 22P/84G/154V/199A/356G, 22P/84G/154V/199S/356G, 22P/84G/198S/199K/306V/356G, 22P/84G/199S/356G, 22P/84G/207L/356G, 22P/84G/356G, 22P/84V/154V/198S/199S/306V/356G, 22P/87H/154L/198S/199K/356G, 22P/87H/154L/199A/306V/356G, 22P/87H/154L/322S/356G, 22P/87H/154V/198S/199K/356G, 22P/87H/154V/199S/356G, 22P/87H/154V/198S/199S/207L/306V/356G, 22P/87H/154V/199A/207L/356G, 22P/87H/154V/199K/356G, 22P/87H/154V/199S/306V/356G, 22P/87H/198S/199A/306V/356G, 22P/87H/198S/199K/306V/356G, 22P/87H/198S/199S/207L/356G, 22P/87H/198S/199K/356G, 22P/87H/199A/356G, 22P/87H/199K/356G, 22P/154L/198S/199A/207L/306V/356G, 22P/154L/198S/199A/356G, 22P/154V/198S/199S/306V/356G, 22P/154V/199A/356G, 22P/154V/199K/207L/306V/356G, 22P/198S/199A/356G, 22P/198S/199K/207L/329C/356G, 22P/199A/356G, 22P/207L/356G, 22P/356G, and 84G/154L/198S/199K, wherein the positions are numbered with reference to SEQ ID NO:32. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set selected from Q22P/L84G/M87H/H154L/I198S/L199A/S356G, Q22P/L84G/M87H/H154L/I198S/L199S/L306V/S356G, Q22P/L84G/M87H/H154L/L199K/L306V/S356G, Q22P/L84G/M87H/H154L/S356G, Q22P/L84G/M87H/H154V/I198S/L199S/L306V/S356G, Q22P/L84G/M87H/H154V/L199K/L306V/S356G, Q22P/L84G/M87H/I198S/L199A/L306V/S356G, Q22P/L84G/M87H/I198S/L199K/S356G, Q22P/L84G/M87H/I198S/L199S/L306V/S356G, Q22P/L84G/M87H/I198S/L199S/S356G, Q22P/L84G/M87H/L199A/L306V/S356G, Q22P/L84G/M87H/L199S/S356G, Q22P/L84G/H154L/I198S/L199A/I207L/L306V, Q22P/L84G/H154L/I198S/L199A/L306V, Q22P/L84G/H154L/I198S/L199A/S356G, Q22P/L84G/H154L/I198S/L199K/S356G, Q22P/L84G/H154L/I198S/L199S/S356G, Q22P/L84G/H154L/L199A/S356G, Q22P/L84G/H154L/L199K/I207L, Q22P/L84G/H154L/L199S/S356G, Q22P/L84G/H154L/I207L/L306V/S356G, Q22P/L84G/H154L/L306V/S356G, Q22P/L84G/H154V/I198S/L199A/L306V/S356G,

Q22P/L154V/H154V/L199K/L120P/L154V/I198S/L199K/L306V/S356G,
Q22P/L84G/H154V/L199A/L306V/S356G, Q22P/L84G/H154V/L199A/S356G,
Q22P/L84G/H154V/L199S/S356G, Q22P/L84G/I198S/L199K/L306V/S356G,
Q22P/L84G/L199S/S356G, Q22P/L84G/I207L/S356G, Q22P/L84G/S356G,
Q22P/L84V/H154V/I198S/L199S/L306V/S356G, Q22P/M87H/H154L/I198S/L199K/S356G,
Q22P/M87H/H154L/L199A/L306V/S356G, Q22P/M87H/H154L/P322S/S356G,
Q22P/M87H/H154V/I198S/L199K/S356G, Q22P/M87H/H154V/L199S/S356G,
Q22P/M87H/H154V/I198S/L199S/I207L/L306V/S356G, Q22P/M87H/H154V/L199A/I207L/S356G,
Q22P/M87H/H154V/L199K/S356G, Q22P/M87H/H154V/L199S/L306V/S356G,
Q22P/M87H/I198S/L199A/L306V/S356G, Q22P/M87H/I198S/L199K/L306V/S356G,
Q22P/M87H/I198S/L199S/I207L/S356G, Q22P/M87H/I198S/L199K/S356G,
Q22P/M87H/L199A/S356G, Q22P/M87H/L199K/S356G,
Q22P/H154L/I198S/L199A/I207L/L306V/S356G, Q22P/H154L/I198S/L199A/S356G,
Q22P/H154L/L199A/S356G, Q22P/H154V/I198S/L199S/L306V/S356G, Q22P/H154V/L199A/S356G,
Q22P/H154V/L199K/I207L/L306V/S356G, Q22P/I198S/L199A/S356G,
Q22P/I198S/L199K/I207L/G329C/S356G, Q22P/L199A/S356G, Q22P/I207L/S356G, Q22P/S356G, and
L84G/H154L/I198S/L199K, wherein the positions are numbered with reference to SEQ ID NO:32. In
some further embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a
sequence at least 90% identical to any of SEQ ID NOS: 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100,
102, 104, 106, 108, 110, 112, 114, 116, 118, 120, 122, 124, 126, 128, 130, 132, 134, 136, 138, 140, 142,
144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 172, 174, 176, 178, 180, 182, 184,
186, 188, 190, 192, 194, 196, 198, 200, 202, 204, 206, 208, 210, 212, 214, 216, 218, 220, 222, 224, 226,
228, 230, 232, 234, 236, 238, 240, 242, 244, 246, 248, 250, 252, 254, 256, 258, 260, 262, 264, 266, 268,
270, 272, 274, 276, 278, 280, 282, 284, 286, 288, 290, 292, 294, 296, 298, 300, 302, 304, 306, 308, 310,
312, 314, 316, 318, 320, 322, 324, 326, 328, 330, 332, 334, 336, 338, 340, 342, 344, 346, and 1290. In
some further embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a
sequence at least 95% identical to any of SEQ ID NOS: 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100,
102, 104, 106, 108, 110, 112, 114, 116, 118, 120, 122, 124, 126, 128, 130, 132, 134, 136, 138, 140, 142,
144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 172, 174, 176, 178, 180, 182, 184,
186, 188, 190, 192, 194, 196, 198, 200, 202, 204, 206, 208, 210, 212, 214, 216, 218, 220, 222, 224, 226,
228, 230, 232, 234, 236, 238, 240, 242, 244, 246, 248, 250, 252, 254, 256, 258, 260, 262, 264, 266, 268,
270, 272, 274, 276, 278, 280, 282, 284, 286, 288, 290, 292, 294, 296, 298, 300, 302, 304, 306, 308, 310,
312, 314, 316, 318, 320, 322, 324, 326, 328, 330, 332, 334, 336, 338, 340, 342, 344, 346, and 1290. In
some further embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises
any of SEQ ID NOS: 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114,
116, 118, 120, 122, 124, 126, 128, 130, 132, 134, 136, 138, 140, 142, 144, 146, 148, 150, 152, 154, 156,
158, 160, 162, 164, 166, 168, 170, 172, 174, 176, 178, 180, 182, 184, 186, 188, 190, 192, 194, 196, 198,
200, 202, 204, 206, 208, 210, 212, 214, 216, 218, 220, 222, 224, 226, 228, 230, 232, 234, 236, 238, 240,
242, 244, 246, 248, 250, 252, 254, 256, 258, 260, 262, 264, 266, 268, 270, 272, 274, 276, 278, 280, 282,
284, 286, 288, 290, 292, 294, 296, 298, 300, 302, 304, 306, 308, 310, 312, 314, 316, 318, 320, 322, 324,
326, 328, 330, 332, 334, 336, 338, 340, 342, 344, 346, and 1290.

(10) The present invention further provides engineered glycosyltransferases, wherein the polypeptide
sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or
more positions selected from 64/106/109/112/131/159/233/421/427/431,
64/106/109/112/417/421/427/431, 64/106/109/139/179/417/421/427/431, 64/106/109/233/347/427/431,
64/106/112/347/417/421, 64/106/139/179/233/417/427/431, 64/106/417/421, 64/106/431,
64/106/109/112/131/179/417/427/431, 64/106/109/417/421/427/431/439,
64/106/112/139/159/179/204/396/417, 64/106/112/159/179/417/421, 64/106/204/417/421/427,
64/109/112/139/159/179/417/431, 64/109/112/139/417, 64/109/139/233/417/421,
64/109/159/179/204/233/417/421, 64/109/417/421, 64/109/417/421, 64/139/233/417/427/431/439,
64/139/347/417/421/427/431, 64/417/421/431, 106/109/112/131/159/179/417/421,
106/109/112/131/159/204/347/417/421/427, 106/109/112/204/347/421/439, 106/109/112/261/417/431,
106/109/112/347/427, 106/109/139/427/431, 106/109/417/421/427,

106/112/159/204/233/417/421, 106/112/233/417, 106/112/396/417/421, 106/139/159/233/347/417/421/427/431, 106/233/421/427, 106/417, 106/109/139/347/417/421/427, 106/109, 106/109/139/233/417/421, 106/109/139/417, 106/109/233/427/431, 106/112/159/179/204/417/421, 106/112/159/179/233/417/421/427/439, 106/131/179/233/421/427/431, 106/139/421, 106/347/417/427/431, 109/112/131/159/179/439, 109/112/131/159/417/421, 109/112/139/179/417/427, 109/112/159/179/417/421, 109/112/159/417/427, 109/112/179/204/233/417/421/427, 109/112/179/347/417, 109/112/204/233/417, 109/112/204/427, 109/112/233/417/431, 109/112/417/421/427, 109/112/417/427/431, 109/131/139/179/261/396/421, 109/131/204, 109/139/179/417/421/427, 109/139/179/417/427, 109/179/233/421, 109/204/417/431, 109/417/421, 109/417/427/431, 112/131/179/204/417/421/427, 112/131/179/347/417/421, 112/139/179/204/233/347/427, 112/159/417/421, 112/417/421, 131/179/233/417/427, 139/233/417, 159/347/417/421/431, 179/417/421, 233/417/421/427/431, and 347/417, wherein the positions are numbered with reference to SEQ ID NO:232. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 64P/106S/109R/112P/131P/159R/233R/421V/427L/431D, 64P/106S/109R/112P/417A/421V/427A/431D, 64P/106S/109R/139P/179L/417R/421V/427L/431D, 64P/106S/109R/233R/347D/427A/431D, 64P/106S/112P/347D/417A/421V, 64P/106S/139P/179L/233R/417A/427A/431D, 64P/106S/417R/421V, 64P/106S/431D, 64P/106T/109R/112P/131P/179L/417R/427A/431D, 64P/106T/109R/417A/421V/427L/431D/439P, 64P/106T/112P/139P/159R/179V/204D/396R/417A, 64P/106T/112P/159R/179L/417R/421V, 64P/106T/204D/417A/421V/427A, 64P/109R/112P/139P/159R/179L/417R/431D, 64P/109R/112P/139P/417A, 64P/109R/139P/233R/417R/421V, 64P/109R/159R/179V/204D/233R/417R/421V, 64P/109R/417A/421V, 64P/109R/417R/421V, 64P/139P/233R/417R/427L/431D/439P, 64P/139P/347D/417R/421V/427L/431D, 64P/417R/421V/431D, 106S/109R/112P/131P/159R/179L/417A/421V, 106S/109R/112P/131P/159R/204D/347D/417A/421V/427L, 106S/109R/112P/204D/347D/421V/439P, 106S/109R/112P/261P/417R/431D, 106S/109R/112P/347D/427A, 106S/109R/139P/427A/431D, 106S/109R/417R/421V/427L, 106S/112P/159R/204D/233R/417R/421V/427A/431D, 106S/112P/233R/417R, 106S/112P/396R/417R/421V, 106S/139P/159R/233R/347D/417R/421V/427A/431D, 106S/233R/421V/427A, 106S/417A, 106T/109R/139P/347D/417A/421V/427A, 106T/109R, 106T/109R/139P/233R/417R/421V, 106T/109R/139P/417R, 106T/109R/233R/427A/431D, 106T/112P/159R/179L/204D/417A/421V, 106T/112P/159R/179V/233R/417A/421V/427L/439P, 106T/131P/179L/233R/421V/427L/431D, 106T/139P/421V, 106T/347D/417R/427A/431D, 109R/112P/131P/159R/179V/439P, 109R/112P/131P/159R/417R/421V, 109R/112P/139P/179V/417R/427A, 109R/112P/159R/179V/417R/421V, 109R/112P/159R/417R/427L, 109R/112P/179V/204D/233R/417A/421V/427L, 109R/112P/179V/347D/417R, 109R/112P/204D/233R/417R, 109R/112P/204D/427A, 109R/112P/233R/417A/431D, 109R/112P/417A/421V/427L, 109R/112P/417A/427A/431D, 109R/131P/139P/179V/261P/396R/421V, 109R/131P/204D, 109R/139P/179L/417R/421V/427A, 109R/139P/179L/417R/427L, 109R/179V/233R/421V, 109R/204D/417R/431D, 109R/417R/421V, 109R/417R/427A/431D, 112P/131P/179V/204D/417R/421V/427L, 112P/131P/179L/347D/417R/421V, 112P/139P/179V/204D/233R/347D/427L, 112P/159R/417R/421V, 112P/417R/421V, 131P/179L/233R/417R/427A, 139P/233R/417A, 159R/347D/417A/421V/431D, 179V/417R/421V, 233R/417R/421V/427L/431D, and 347D/417R, wherein the positions are numbered with reference to SEQ ID NO:232. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from F64P/L106S/L109R/E112P/S131P/Q159R/I233R/Y421V/R427L/Q431D, F64P/L106S/L109R/E112P/E417A/Y421V/R427A/Q431D, F64P/L106S/L109R/R139P/S179L/E417R/Y421V/R427L/Q431D, F64P/L106S/L109R/I233R/G347D/R427A/Q431D, F64P/L106S/E112P/G347D/E417A/Y421V, F64P/L106S/R139P/S179L/I233R/E417A/R427A/Q431D, F64P/L106S/E417R/Y421V, F64P/L106S/Q431D, F64P/L106T/L109R/E112P/S131P/S179L/E417R/R427A/Q431D,

F64P/L106T/L109R/E417A/Y421V/R427L/Q431D/K439P,
 F64P/L106T/E112P/R139P/Q159R/S179V/G204D/Y396R/E417A,
 F64P/L106T/E112P/Q159R/S179L/E417R/Y421V, F64P/L106T/G204D/E417A/Y421V/R427A,
 F64P/L109R/E112P/R139P/Q159R/S179L/E417R/Q431D, F64P/L109R/E112P/R139P/E417A,
 F64P/L109R/R139P/I233R/E417R/Y421V, F64P/L109R/Q159R/S179V/G204D/I233R/E417R/Y421V,
 F64P/L109R/E417A/Y421V, F64P/L109R/E417R/Y421V,
 F64P/R139P/I233R/E417R/R427L/Q431D/K439P, F64P/R139P/G347D/E417R/Y421V/R427L/Q431D,
 F64P/E417R/Y421V/Q431D, L106S/L109R/E112P/5131P/Q159R/5179L/E417A/Y421V,
 L106S/L109R/E112P/S131P/Q159R/G204D/G347D/E417A/Y421V/R427L,
 L106S/L109R/E112P/G204D/G347D/Y421V/K439P, L106S/L109R/E112P/R261P/E417R/Q431D,
 L106S/L109R/E112P/G347D/R427A, L106S/L109R/R139P/R427A/Q431D,
 L106S/L109R/E417R/Y421V/R427L,
 L106S/E112P/Q159R/G204D/I233R/E417R/Y421V/R427A/Q431D, L106S/E112P/I233R/E417R,
 L106S/E112P/Y396R/E417R/Y421V,
 L106S/R139P/Q159R/I233R/G347D/E417R/Y421V/R427A/Q431D, L106S/I233R/Y421V/R427A,
 L106S/E417A, L106T/L109R/R139P/G347D/E417A/Y421V/R427A, L106T/L109R,
 L106T/L109R/R139P/I233R/E417R/Y421V, L106T/L109R/R139P/E417R,
 L106T/L109R/I233R/R427A/Q431D, L106T/E112P/Q159R/S179L/G204D/E417A/Y421V,
 L106T/E112P/Q159R/S179V/I233R/E417A/Y421V/R427L/K439P,
 L106T/S131P/S179L/I233R/Y421V/R427L/Q431D, L106T/R139P/Y421V,
 L106T/G347D/E417R/R427A/Q431D, L109R/E112P/S131P/Q159R/S179V/K439P,
 L109R/E112P/S131P/Q159R/E417R/Y421V, L109R/E112P/R139P/S179V/E417R/R427A,
 L109R/E112P/Q159R/S179V/E417R/Y421V, L109R/E112P/Q159R/E417R/R427L,
 L109R/E112P/S179V/G204D/I233R/E417A/Y421V/R427L, L109R/E112P/S179V/G347D/E417R,
 L109R/E112P/G204D/I233R/E417R, L109R/E112P/G204D/R427A,
 L109R/E112P/I233R/E417A/Q431D, L109R/E112P/E417A/Y421V/R427L,
 L109R/E112P/E417A/R427A/Q431D, L109R/S131P/R139P/S179V/R261P/Y396R/Y421V,
 L109R/S131P/G204D, L109R/R139P/S179L/E417R/Y421V/R427A,
 L109R/R139P/S179L/E417R/R427L, L109R/S179V/I233R/Y421V, L109R/G204D/E417R/Q431D,
 L109R/E417R/Y421V, L109R/E417R/R427A/Q431D,
 E112P/S131P/S179V/G204D/E417R/Y421V/R427L, E112P/S131P/S179L/G347D/E417R/Y421V,
 E112P/R139P/S179V/G204D/I233R/G347D/R427L, E112P/Q159R/E417R/Y421V,
 E112P/E417R/Y421V, S131P/S179L/I233R/E417R/R427A, R139P/I233R/E417A,
 Q159R/G347D/E417A/Y421V/Q431D, S179V/E417R/Y421V, I233R/E417R/Y421V/R427L/Q431D,
 and G347D/E417R, wherein the positions are numbered with reference to SEQ ID NO: 232. In some
 further embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a
 sequence at least 90% identical to any of SEQ ID NOS: 348, 350, 352, 354, 356, 358, 360, 362, 364, 366,
 368, 370, 372, 374 376, 378, 380, 382, 384, 386, 388, 390, 392, 394, 396, 398, 400, 402, 404, 406, 408,
 410, 412, 414, 416, 418, 420, 422, 424, 426, 428, 430, 432, 434, 436, 438, 440, 442, 444, 446, 448, 450,
 452, 454, 456, 458, 460, 462, 464, 466, 468, 470, 472, 474, 476, 478, 480, 482, 484, 486, 488, 490, 492,
 494, 496, and/or 498. In some embodiments, the polypeptide sequence of the engineered
 glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 348, 350, 352,
 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, 384, 386, 388, 390, 392, 394,
 396, 398, 400, 402, 404, 406, 408, 410, 412, 414, 416, 418, 420, 422, 424, 426, 428, 430, 432, 434, 436,
 438, 440, 442, 444, 446, 448, 450, 452, 454, 456, 458, 460, 462, 464, 466, 468, 470, 472, 474, 476, 478,
 480, 482, 484, 486, 488, 490, 492, 494, 496, and/or 498. In some additional embodiments, the
 polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 348, 350,
 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374 376, 378, 380, 382, 384, 386, 388, 390, 392,
 394, 396, 398, 400, 402, 404, 406, 408, 410, 412, 414, 416, 418, 420, 422, 424, 426, 428, 430, 432, 434,
 436, 438, 440, 442, 444, 446, 448, 450, 452, 454, 456, 458, 460, 462, 464, 466, 468, 470, 472, 474, 476,
 478, 480, 482, 484, 486, 488, 490, 492, 494, 496, and/or 498.

(11) The present invention further provides engineered glycosyltransferases, wherein the polypeptide
 sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or

more positions selected from 106/112/204/347/396/417, 106/112/204/347/396/417/427, 106/112/204/347/396/417/427/431, 112/204/347/396/417/427, 112/204/347/396/417/427/431, and 204/347/396/417/431, wherein the positions are numbered with reference to SEQ ID NO:348. In some additional embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 106S/112P/204D/347D/396Y/417R, 106S/112P/204D/347D/396Y/417R/427A, 106S/112P/204D/347D/396Y/417R/427A/431D, 112P/204D/347D/396Y/417R/427A, 112P/204D/347D/396Y/417R/427A/431D, and 204D/347D/396Y/417R/431D, wherein the positions are numbered with reference to SEQ ID NO:348. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from L106S/E112P/G204D/G347D/R396Y/E417R, L106S/E112P/G204D/G347D/R396Y/E417R/R427A, L106S/E112P/G204D/G347D/R396Y/E417R/R427A/Q431D, E112P/G204D/G347D/R396Y/E417R/R427A, E112P/G204D/G347D/R396Y/E417R/R427A/Q431D, and G204D/G347D/R396Y/E417R/Q431D, wherein the positions are numbered with reference to SEQ ID NO:348. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 500, 502, 504, 506, 508, and/or 510. In some further embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 500, 502, 504, 506, 508, and/or 510. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises SEQ ID NOS: 500, 502, 504, 506, 508, and/or 510.

(12) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 14/100, 28/44/365/407, 38/118/290/351/375/401/422, 38/178/401, 38/290/351/401/422, 54/413, 74/102/137/161/259/289, 92/118, 98/233, 102/161/250/435, 110/222/250/259/435, 118/156/178/290/375/401/422, 137/161/435, 137/169, 159/169/173/300/424/438, 185/290/401/422, 290/351/401, and 435/438, wherein the positions are numbered with reference to SEQ ID NO:348. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 14V/100F, 28M/44V/365I/407E, 38R/118A/290E/351G/375P/401L/422M, 38R/178V/401L, 38R/290E/351G/401L/422M, 54P/413L, 74W/102K/137G/161L/259S/289S, 92L/118A, 98P/233W, 102K/161L/250A/435E, 110G/222R/250R/259P/435G, 118A/156A/178V/290E/375P/401L/422M, 137G/161L/435R, 137G/169G, 159M/169S/173G/300Q/424E/438A, 185R/290E/401L/422M, 290E/351G/401L, and 435Q/438A, wherein the positions are numbered with reference to SEQ ID NO:348. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 14V/L100F, 128M/I44V/V365I/A407E, K38R/S118A/D290E/A351G/D375P/W401L/I422M, K38R/A178V/W401L, K38R/D290E/A351G/W401L/I422M, T54P/V413L, R74W/R102K/N137G/D161L/H259S/K289S, I92L/S118A, D98P/I233W, R102K/D161L/T250A/V435E, A110G/K222R/T250R/H259P/V435G, S118A/S156A/A178V/D290E/D375P/W401L/I422M, N137G/D161L/V435R, N137G/D169G, Q159M/D169S/R173G/D300Q/Q424E/M438A, K185R/D290E/W401L/I422M, D290E/A351G/W401L, and V435Q/M438A, wherein the positions are numbered with reference to SEQ ID NO:348. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 500, 502, 504, 506, 508, 510, 512, 514, 516, 518, 520, 222, 524, 526, 528, 530, 532, 534, 536, 538, 540, 542, 544, 546, and/or 548. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 500, 502, 504, 506, 508, 510, 512, 514, 516, 518, 520, 222, 524, 526, 528, 530, 532, 534, 536, 538, 540, 542, 544, 546, and/or 548. In some further embodiments, the polypeptide sequences of the engineered glycosyltransferases comprise any of SEQ ID NOS: 500, 502, 504, 506, 508, 510, 512, 514, 516, 518, 520, 222, 524, 526, 528, 530, 532, 534, 536, 538, 540, 542, 544, 546, and/or 548.

(13) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or

more positions selected from 14/28/38/74/100/102/118/161/169/178/233/250/407/422/438,
14/28/38/74/102/156/159/233/250/289/413/422/424/435/438,
14/28/38/100/102/110/159/161/233/259/290/300/351/435, 14/28/38/110/137/161/222/289/401,
14/28/44/74/98/102/137/159/161/185/222/250, 14/28/44/92, 14/28/44/161/169, 14/28/44/375,
14/28/54/161/185/413, 14/28/54/365, 14/28/74/98/100/102/161/173/178/233/250/259/290/407,
14/28/92/100/102/159/161/169/233, 14/28/92/100/102/159/161/233/351/422,
14/28/92/100/102/422/424/435, 14/28/92/100/161/222/233/289/300,
14/28/100/137/156/161/222/259/289/365/401/435/438, 14/28/100/156/161/250,
14/28/102/118/137/161/185/222/250/259/401, 14/28/159/289/290/300, 14/28/159/365/435, 14/28/35,
14/38/74/110/156/161/173/178/222/300, 14/38/100/102/161/173/178/222/250/375/401/413,
44/74/100/102/161/233/365/435/438, 14/74/110/159/161/169/173/250/259/290/375/407/422,
14/74/161/375/401, 14/92/98/100/159/161/259/365/422/424/435, 14/110/156/161/375/401/435/438,
14/159/161/365/435/438, 14/161/222/250/259/289/375/401/413, 14/161/222/250/435/438, 14/161/300,
28, 28/38/92/98/100/102/156/161, 28/44/74/401, 28/44/92/161/222/300/413, 28/44/98/100/102/118,
28/44/118/156/161/222/289/435/M438, 28/44/289/290/351/422, 28/44/435/438, 28/54/92/159/161/290,
28/54/159/290/438, 28/54/250/439, 28/74/156/159/161/178/300/365/435/438, 28/74/156/161/365/407,
28/74/161/290/365, 28/92/100/102/110/161/185/250/300/375/435, 28/92/98/100/110/156/161/401,
28/92/118/159/222/250/259/300/407, 28/156/161/185/435/438, 28/156/161/233/259/300/435,
28/98/100/102/161/185/351/401/435/438, 38/161/300/438,
74/98/100/102/110/118/161/178/250/289/290/300/435/438, 74/98/100/102/118/156/159/161/435/438,
74/156/161/173/178/424/435, 74/375/435, 98/100/118/159/161/300,
98/100/156/159/161/178/259/289/290/351/422, 102/137/159/161/422/424, 118,
137/159/161/185/300/351/365/435, 156/159/161/169, 159/161/222/290/375/407, 159/161/401, and
161/259/289/435/438, wherein the positions are numbered with reference to SEQ ID NO:548. In some
embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one
mutation or mutation set at one or more positions selected from
14V/28M/38R/74/100F/102K/118A/161L/169G/178V/233W/250R/407E/422M/438A,
14V/28M/38R/74W/102K/156A/159M/233W/250A/289S/413L/422M/424E/435R/438A,
14V/28M/38R/100F/102K/110G/159M/161L/233W/259S/290E/300Q/351G/435Q,
14V/28M/38R/110G/137G/161L/222R/289S/401L,
14V/28M/44V/74W/98P/102K/137G/159M/161L/185R/222R/250A, 14V/28M/44V/92L,
14V/28M/44V/161L/169S, 14V/28M/44V/375P, 14V/28M/54P/161L/185R/413L, 14V/28M/54P/365I,
14V/28M/74W/98P/100F/102K/161L/173G/178V/233W/250A/259S/290E/407E,
14V/28M/92L/100F/102K/159M/161L/169S/233W,
14V/28M/92L/100F/102K/159M/161L/233W/351G/422M, 14V/28M/92L/100F/102K/422M/424E/435Q,
14V/28M/92L/100F/161L/222R/233W/289S/300Q,
14V/28M/100F/137G/156A/161L/222R/259S/289S/365I/401L/435R/438A,
14V/28M/100F/156A/161L/250R, 14V/28M/102K/118A/137G/161L/185R/222R/250R/259P/401L,
14V/28M/159M/289S/290E/300Q, 14V/28M/159M/365I/435Q, 14V/28M/351G,
14V/38R/74W/110G/156A/161L/173G/178V/222R/300Q,
14V/38R/100F/102K/161L/173G/178V/222R/250R/375P/401L/413L,
44V/74W/100F/102K/161L/133W/365I/435G/438A,
14V/74W/110G/159M/161L/169G/173G/250A/259P/290E/375P/407E/422M,
14V/74W/161L/375P/401L, 14V/92L/98P/100F/159M/161L/259P/365I/422M/424E/435R,
14V/110G/156A/161L/375P/401L/435EM438A, 14V/159M/161L/365I/435E/438A,
14V/161L/222R/250R/259S/289S/375P/401L/413L, 14V/161L/222R/250R/435Q/438A,
14V/161L/300Q, 28M, 28M/38R/92L/98P/100F/102K/156A/161L, 28M/44V/74W/401L,
28M/44V/92L/161L/222R/300Q/413L, 28M/44V/98P/100F/102K/118A,
28M/44V/118A/156A/161L/222R/289S/435Q/438A, 28M/44V/289S/290E/351G/422M,
28M/44V/435R/438A, 28M/54P/92L/159M/161L/290E, 28M/54P/159M/290E/438A,
28M/54P/250R/439N 28M/74W/156A/159M/161L/178V/300Q/365I/435Q/438A,
28M/74W/156A/161L/365I/407E, 28M/74W/161L/290E/365I,
28M/92L/98P/100F/110G/156A/161L/401L,

28M/92L/118A/159M/222R/250R/259P/300Q/407E,
28M/98P/100F/102K/161L/185R/351G/401L/435E/438A, 28M/156A/161L/185R/435R/438A,
28M/156A/161L/233W/259S/300Q/435R, 38R/161L/300Q/438A,
74W/98P/100F/102K/110G/118A/161L/178V/250R/289S/290E/300Q/435E/438A,
74W/98P/100F/102K/118A/156A/159M/161L/435E/438A, 74W/156A/161L/173G/178V/424E/435E,
74W/375P/435G, 98P/100F/118A/159M/161L/300Q,
98P/100F/156A/159M/161L/178V/259S/289S/290E/351G/422M, 102K/137G/159M/161L/422M/424E,
118A, 137G/159M/161L/185R/300Q/351G/365I/435Q, 156A/159M/161L/169S,
159M/161L/222R/290E/375P/407E, 159M/161L/401L, and 161L/259S/289S/435R/438A, wherein the
positions are numbered with reference to SEQ ID NO:548. In some embodiments, the polypeptide
sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or
more positions selected from
I14V/I28M/K38R/R74W/L100F/R102K/S118A/D161L/D169G/A178V/I233W/T250R/A407E/I422
M/M438A,
I14V/I28M/K38R/R74W/R102K/S156A/Q159M/I233W/T250A/K289S/V413L/I422M/Q424E/V435R/M438A,
I14V/I28M/K38R/L100F/R102K/A110G/Q159M/D161L/I233W/H259S/D290E/D300Q/A351G/V435Q,
I14V/I28M/K38R/A110G/N137G/D161L/K222R/K289S/W401L,
I14V/I28M/I44V/R74W/D98P/R102K/N137G/Q159M/D161L/K185R/K222R/T250A,
I14V/I28M/I44V/I92L, I14V/I28M/I44V/D161L/D169S, I14V/I28M/I44V/D375P,
I14V/I28M/T54P/D161L/K185R/V413L, I14V/I28M/T54P/V365I,
I14V/I28M/R74W/D98P/L100F/R102K/D161L/R173G/A178V/I233W/T250A/H259S/D290E/A407 E,
I14V/I28M/I92L/L100F/R102K/Q159M/D161L/D169S/I233W,
I14V/I28M/I92L/L100F/R102K/Q159M/D161L/I233W/A351G/I422M,
I14V/I28M/I92L/L100F/R102K/I422M/Q424E/V435Q,
I14V/I28M/I92L/L100F/D161L/K222R/I233W/K289S/D300Q,
I14V/I28M/L100F/N137G/S156A/D161L/K222R/H259S/K289S/V365I/W401L/V435R/M438A,
I14V/I28M/L100F/S156A/D161L/T250R,
I14V/I28M/R102K/S118A/N137G/D161L/K185R/K222R/T250R/H259P/W401L,
I14V/I28M/Q159M/K289S/D290E/D300Q, I14V/I28M/Q159M/V365I/V435Q, I14V/I28M/A351G,
I14V/K38R/R74W/A110G/S156A/D161L/R173G/A178V/K222R/D300Q,
I14V/K38R/L100F/R102K/D161L/R173G/A178V/K222R/T250R/D375P/W401L/V413L,
I44V/R74W/L100F/R102K/D161L/I233W/V365I/V435G/M438A,
I14V/R74W/A110G/Q159M/D161L/D169G/R173G/T250A/H259P/D290E/D375P/A407E/I422M,
I14V/R74W/D161L/D375P/W401L,
I14V/I92L/D98P/L100F/Q159M/D161L/H259P/V365I/I422M/Q424E/V435R,
I14V/A110G/S156A/D161L/D375P/W401L/V435E/M438A,
I14V/Q159M/D161L/V365I/V435E/M438A,
I14V/D161L/K222R/T250R/H259S/K289S/D375P/W401L/V413L,
I14V/D161L/K222R/T250R/V435Q/M438A, I14V/D161L/D300Q, I28M,
I28M/K38R/I92L/D98P/L100F/R102K/S156A/D161L, I28M/I44V/R74W/W401L,
I28M/I44V/I92L/D161L/K222R/D300Q/V413L, I28M/I44V/D98P/L100F/R102K/S118A,
I28M/I44V/S118A/S156A/D161L/K222R/K289S/V435Q/M438A,
I28M/I44V/K289S/D290E/A351G/I422M, I28M/I44V/V435R/M438A,
I28M/T54P/I92L/Q159M/D161L/D290E, I28M/T54P/Q159M/D290E/M438A,
I28M/T54P/T250R/K439N, I28M/R74W/S156A/Q159M/D161L/A178V/D300Q/V365I/V435Q/M438A,
I28M/R74W/S156A/D161L/V365I/A407E, I28M/R74W/D161L/D290E/V365I,
I28M/D98P/L100F/R102K/D161L/K185R/A351G/W401L/V435E/M438A,
I28M/I92L/L100F/R102K/A110G/D161L/K185R/T250A/D300Q/D375P/V435Q,
I28M/I92L/S118A/Q159M/K222R/T250R/H259P/D300Q/A407E,
I28M/S156A/D161L/K185R/V435R/M438A, I28M/S156A/D161L/I233W/H259S/D300Q/V435R,
I28M/I92L/D98P/L100F/A110G/S156A/D161L/W401L, K38R/D161L/D300Q/M438A,
R74W/D98P/L100F/R102K/A110G/S118A/D161L/A178V/T250R/K289S/D290E/D300Q/V435E/M

R74W/D98P/L100F/S118A/Q159M/D161L/V435E/M438A,
R74W/S156A/D161L/R173G/A178V/Q424E/V435E, R74W/D375P/V435G,
D98P/L100F/S118A/Q159M/D161L/D300Q,
D98P/L100F/S156A/Q159M/D161L/A178V/H259S/K289S/D290E/A351G/I422M,
R102K/N137G/Q159M/D161L/I422M/Q424E, S118A,
N137G/Q159M/D161L/K185R/D300Q/A351G/V365I/V435Q, S156A/Q159M/D161L/D1695,
Q159M/D161L/K222R/D290E/D375P/A407E, Q159M/D161L/W401L, and
D161L/H259S/K289S/V435R/M438A, wherein the positions are numbered with reference to SEQ ID
NO:548. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase
comprises a sequence at least 90% identical to any of SEQ ID NOS: 550, 552, 554, 556, 558, 560, 562,
564, 566, 568, 570, 572, 574, 576, 578, 580, 582, 584, 586, 588, 590, 592, 594, 596, 598, 600, 602, 604,
606, 608, 610, 612, 614, 616, 618, 620, 622, 624, 626, 628, 630, 632, 634, 636, 638, 640, 642, 644, 646,
648, 650, 652, 654, 656, 658, 660, 662, 664, 666, 668, 670, 672, 674, 676, 678, and 680, wherein the
positions are numbered with reference to SEQ ID NO: 548. In some embodiments, the polypeptide
sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of
SEQ ID NOS: 550, 552, 554, 556, 558, 560, 562, 564, 566, 568, 570, 572, 574, 576, 578, 580, 582, 584,
586, 588, 590, 592, 594, 596, 598, 600, 602, 604, 606, 608, 610, 612, 614, 616, 618, 620, 622, 624, 626,
628, 630, 632, 634, 636, 638, 640, 642, 644, 646, 648, 650, 652, 654, 656, 658, 660, 662, 664, 666, 668,
670, 672, 674, 676, 678, and 680, wherein the positions are numbered with reference to SEQ ID NO: 548.

(14) In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a
sequence any of SEQ ID NOS: 550, 552, 554, 556, 558, 560, 562, 564, 566, 568, 570, 572, 574, 576, 578,
580, 582, 584, 586, 588, 590, 592, 594, 596, 598, 600, 602, 604, 606, 608, 610, 612, 614, 616, 618, 620,
622, 624, 626, 628, 630, 632, 634, 636, 638, 640, 642, 644, 646, 648, 650, 652, 654, 656, 658, 660, 662,
664, 666, 668, 670, 672, 674, 676, 678, and 680, wherein the positions are numbered with reference to
SEQ ID NO: 548.

(15) The present invention also provides engineered glycosyltransferases, wherein the polypeptide
sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or
more positions selected from 19/44/110/191/198/199/208/300/365, 19/75/76/87/92/199/207/208,
19/87/92/191/199/209/413/435, 44/76/197/199/208/351, 44/87/92/137/159/199/209,
44/87/137/159/169/191/199/208, 44/87/137/191/197/209/289/401, 44/87/199/208, 75/76/87/92,
75/76/87/92/290/300, 75/76/87/191, 75/76/87/191/197/199/209/300, 75/87/92/169/207/208/300/413/435,
75/87/110/137/169/191/199/208/209/289/435, 75/87/110/191/197/198/207/208/289/290/300/401/413,
75/87/300, 75/110/197/199/208/290/300/401/413, 76/92/199/209, 87/92/197/198/199/208/300,
87/137/435, 87/169/191/199/207/209/401/413, 87/191/198/199/222/244/289/300/435,
87/92/110/169/199/207/209/290/300, 87/92/159/169/191/198/290/413/435,
87/92/159/191/199/208/209/289/290, 87/92/208/401, 87/435, 92/137/191/199/209,
92/197/199/207/208/401, 137/198/199/207/208/426/435, 137/199/208/209/290/435, 137/365,
159/197/199/207/209, 169/191/197/199/207/208, 169/197/199/207/209/222/300/413/435,
191/207/208/289/290/413/435, and 197/198/199/208/209, wherein the positions are numbered with
reference to SEQ ID NO:562. In some embodiments, the polypeptide sequence of the engineered
glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from
19L/44V/110G/191R/198M/199K/208A/300Q/365I, 19L/75L/76R/87W/92L/199K/207L/208A,
19L/87W/92L/191R/199K/209G/413L/435Q, 44V/76R/197R/199K/208D/351G,
44V/87W/92L/137G/159M/199K/209G, 44V/87W/137G/159M/169S/191R/199K/208D,
44V/87W/137G/191R/197R/209G/2895/401L, 44V/87W/199K/208A, 75L/76R/87W/92L,
75L/76R/87W/92L/290E/300Q, 75L/76R/87W/191R, 75L/76R/87W/191R/197R/199K/209G/300Q,
75L/87W/92L/169S/207L/208A/300Q/413L/435Q,
75L/87W/110G/137G/169S/191R/199K/208A/209G/2895/435Q,
75L/87W/110G/191R/197R/198M/207L/208D/289S/290E/300Q/401L/413L, 75L/87W/300Q,
75L/110G/197R/199K/208A/290E/300Q/401L/413L, 76R/92L/199K/209G,
87L/92L/197R/198M/199K/208D/300Q, 87W/137G/435R,
87W/1695/191R/199K/207L/209G/401L/413L, 87W/191R/198M/199K/222R/244L/2895/300Q/435R,
87W/92L/110G/1695/199K/207L/209G/290E/300Q,

87W/92L/159M/191R/199K/208A/209G/289S/290E/413L/435Q, 87W/92L/208A/401L, 87W/435R, 92L/137G/191R/199K/209G, 92L/197R/199K/207L/208D/401L, 137G/198M/199K/207L/208D/426V/435R, 137G/199K/208A/209G/290E/435R, 137G/365I, 159M/197R/199K/207L/209G, 1695/191R/197R/199K/207L/208D, 1695/197R/199K/207L/209G/222R/300Q/413L/435R, 191R/207L/208A/289S/290E/413L/435Q, and 197R/198M/199K/208D/209G, wherein the positions are numbered with reference to SEQ ID NO:562. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from V19L/I44V/A110G/S191R/I198M/A199K/K208A/D300Q/V365I, V19L/I75L/S76R/M87W/I92L/A199K/I207L/K208A, V19L/M87W/I92L/S191R/A199K/Q209G/V413L/V435Q, I44V/S76R/Q197R/A199K/K208D/A351G, I44V/M87W/I92L/N137G/Q159M/A199K/Q209G, I44V/M87W/N137G/Q159M/G169S/S191R/A199K/K208D, I44V/M87W/N137G/S191R/Q197R/Q209G/K289S/W401L, I44V/M87W/A199K/K208A, I75L/S76R/M87W/I92L, I75L/S76R/M87W/I92L/D290E/D300Q, I75L/S76R/M87W/S191R, I75L/S76R/M87W/S191R/Q197R/A199K/Q209G/D300Q, I75L/M87W/I92L/G169S/I207L/K208A/D300Q/V413L/V435Q, I75L/M87W/A110G/N137G/G169S/S191R/A199K/K208A/Q209G/K289S/V435Q, I75L/M87W/A110G/S191R/Q197R/I198M/I207L/K208D/K289S/D290E/D300Q/W401L/V41 L, I75L/M87W/D300Q, I75L/A110G/Q197R/A199K/K208A/D290E/D300Q/W401L/V413L, S76R/I92L/A199K/Q209G, M87L/I92L/Q197R/I198M/A199K/K208D/D300Q, M87W/N137G/V435R, M87W/G169S/S191R/A199K/I207L/Q209G/W401L/V413L, M87W/S191R/I198M/A199K/K222R/P244L/K289S/D300Q/V435R, M87W/I92L/A110G/G169S/A199K/I207L/Q209G/D290E/D300Q, M87W/I92L/Q159M/G169S/S191R/I198M/D290E/V413L/V435Q, M87W/I92L/Q159M/S191R/A199K/K208A/Q209G/K289S/D290E, M87W/I92L/K208A/W401L, M87W/V435R, I92L/N137G/S191R/A199K/Q209G, I92L/Q197R/A199K/I207L/K208D/W401L, N137G/I198M/A199K/I207L/K208D/A426V/V435R, N137G/A199K/K208A/Q209G/D290E/V435R, N137G/V365I, Q159M/Q197R/A199K/I207L/Q209G, G169S/S191R/Q197R/A199K/I207L/K208D, G169S/Q197R/A199K/I207L/Q209G/K222R/D300Q/V413L/V435R, S191R/I207L/K208A/K289S/D290E/V413L/V435Q, and Q197R/I198M/A199K/K208D/Q209G, wherein the positions are numbered with reference to SEQ ID NO:562. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 682, 684, 686, 688, 690, 692, 694, 696, 698, 700, 702, 704, 706, 708, 710, 712, 714, 716, 718, 720, 722, 724, 726, 728, 730, 732, 734, 736, 738, 740, 742, 744, 746, 748, 750, 752, and/or 754. In some further embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 682, 684, 686, 688, 690, 692, 694, 696, 698, 700, 702, 704, 706, 708, 710, 712, 714, 716, 718, 720, 722, 724, 726, 728, 730, 732, 734, 736, 738, 740, 742, 744, 746, 748, 750, 752, and/or 754. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises SEQ ID NOS: 682, 684, 686, 688, 690, 692, 694, 696, 698, 700, 702, 704, 706, 708, 710, 712, 714, 716, 718, 720, 722, 724, 726, 728, 730, 732, 734, 736, 738, 740, 742, 744, 746, 748, 750, 752, and/or 754.

(16) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 22, 25, 51, 56, 71, 78, 80, 81, 88, 157, 185/208/230/252/255/290/365, 189/206/208/365, 200, 208/365/435, 243, 245, 249, 259, 262/401, 279, 282, 284, 304/322/365/401, 308, 338, 339, 352, 362, 364, 365/401/413/435, 366, and 374, wherein the positions are numbered with reference to SEQ ID NO:696. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 22L, 25L, 25V, 51A, 56L, 71R, 78E, 78F, 78G, 78I, 78K, 78M, 78P, 78Q, 78R, 80L, 81C, 88I, 88K, 88V, 157G, 157Q, 185R/208A/230S/252N/255N/290E/365I, 189L/206K/208A/365I, 200N, 200S, 208A/365I/435Q, 243C, 243L, 243M, 243V, 243Y, 245G, 249E, 249H, 249I, 249M, 249N, 249P, 249S,

292T, 294Y, 259G, 259Y, 259S, 262S/401L, 279G, 284T, 304P/322S/365I/401L, 308F, 308Y, 338C, 339D, 352Q, 362T, 364G, 365I/401L/413L/435Q, 366A, and 374T, wherein the positions are numbered with reference to SEQ ID NO:696. In some additional embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from P22L, I25L, I25V, K51A, N56L, Q71R, L78E, L78F, L78G, L78I, L78K, L78M, L78P, L78Q, L78R, T80L, H81C, R88I, R88K, R88V, L157G, L157Q, K185R/K208A/E230S/5252N/5255N/D290E/V365I, I189L/M206K/K208A/V365I, K200N, K200S, K208A/V365I/V435Q, I243C, I243L, I243M, I243V, I243Y, L245G, L249E, L249H, L249I, L249M, L249N, L249P, L249S, L249T, L249Y, H259G, H259S, H259Y, T262S/W401L, S279G, S282T, S284T, S304P/P322S/V365I/W401L, R308F, R308Y, V338C, P339D, F352Q, L362T, S364G, V365I/W401L/V413L/V435Q, C366A, and S374T, wherein the positions are numbered with reference to SEQ ID NO:696. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 4684, 4686, 4688, 4690, 4692, 4694, 4696, 4698, 4700, 4702, 4704, 4706, 4708, 4710, 4712, 4714, 4716, 4718, 4720, 4722, 4724, 4726, 4728, 4730, 4732, 4734, 4736, 4738, 4740, 4742, 4744, 4746, 4748, 4750, 4752, 4754, 4756, 4758, 4760, 4762, 4764, 4766, 4768, 4770, 4772, 4774, 4776, 4778, 4780, 4782, 4784, 4786, 4788, 4790, 4792, 4794, 4796, 4798, 4800, 4902, 4804, 4806, 4808, 4810, and 4812. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 4684, 4686, 4688, 4690, 4692, 4694, 4696, 4698, 4700, 4702, 4704, 4706, 4708, 4710, 4712, 4714, 4716, 4718, 4720, 4722, 4724, 4726, 4728, 4730, 4732, 4734, 4736, 4738, 4740, 4742, 4744, 4746, 4748, 4750, 4752, 4754, 4756, 4758, 4760, 4762, 4764, 4766, 4768, 4770, 4772, 4774, 4776, 4778, 4780, 4782, 4784, 4786, 4788, 4790, 4792, 4794, 4796, 4798, 4800, 4902, 4804, 4806, 4808, 4810, and 4812. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 4684, 4686, 4688, 4690, 4692, 4694, 4696, 4698, 4700, 4702, 4704, 4706, 4708, 4710, 4712, 4714, 4716, 4718, 4720, 4722, 4724, 4726, 4728, 4730, 4732, 4734, 4736, 4738, 4740, 4742, 4744, 4746, 4748, 4750, 4752, 4754, 4756, 4758, 4760, 4762, 4764, 4766, 4768, 4770, 4772, 4774, 4776, 4778, 4780, 4782, 4784, 4786, 4788, 4790, 4792, 4794, 4796, 4798, 4800, 4902, 4804, 4806, 4808, 4810, and 4812.

(17) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 51/56, 51/56/243/249/282/353/362/366, 51/56/243/249/308/362/364, 51/56/249/353, 51/56/249/362/364, 51/56/249/362/366, 51/243/249/308/353, 51/243/249/348/362/366, 51/249, 51/249/282/284/364, 51/249/282/353/366, 51/249/284/308/362/366, 51/249/353/362/364, 51/353/362, 56, 56/243/249/282/364/366, 56/243/364/366, 56/249, 56/249/284/353, 56/249/353, 56/284/366, 243/249/282/284/362/364/366, 243/249/308/353/366, 243/249/353/362/366, 243/282/353/362/364, 243/282/362/364/366, 243/308/353, 249/353/362/366, 282, 308/366, and 362/366, wherein the positions are numbered with reference to SEQ ID NO: 4684. In some embodiments, the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 51A/56H, 51A/56H/243L/249E/282T/353Y/362M/366S, 51A/56H/243L/249E/308F/362M/364G, 51A/56H/249M/362M/364G, 51A/56H/249M/362M/366S, 51A/56H/249Y/353Y, 51A/243L/249E/308F/353Y, 51A/243L/249E/348S/362M/366V, 51A/249E/353Y/362M/364G, 51A/249M, 51A/249M/282T/284T/364G, 51A/249M/282T/353Y/366S, 51A/249Y/284T/308F/362M/366V, 51A/353Y/362M, 56H, 56H/243L/249E/282T/364G/366V, 56H/243L/364G/366V, 56H/249M/284T/353Y, 56H/249M/353Y, 56H/249Y, 56H/284T/366V, 243L/249E/282T/284T/362M/364G/366S, 243L/249M/308F/353Y/366A, 243L/249Y/353Y/362M/366S, 243L/282T/353Y/362M/364G, 243L/282T/362M/364G/366V, 243L/308F/353Y, 249Y/353Y/362M/366S, 282T, 308F/366A, and 362M/366A, wherein the positions are numbered with reference to SEQ ID NO: 4684. In some embodiments, the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from K51A/N56H, K51A/N56H/I243L/L249E/S282T/W353Y/L362M/C366S, K51A/N56H/I243L/L249E/R308F/L362M/S364G, K51A/N56H/L249M/L362M/S364G, K51A/N56H/L249M/L362M/C366S, K51A/N56H/L249Y/W353Y, K51A/I243L/L249E/R308F/W353Y, K51A/I243L/L249E/A348S/L362M/C366V, K51A/L249E/W353Y/L362M/S364G, K51A/L249M, K51A/L249M/S282T/S284T/S364G, K51A/L249M/S282T/W353Y/C366S,

K51A/L249Y/S284T/R308F/L362M/C366V, K51A/W353Y/L362M, N56H, N56H/I243L/L249E/S282T/S364G/C366V, N56H/I243L/S364G/C366V, N56H/L249M/S284T/W353Y, N56H/L249M/W353Y, N56H/L249Y, N56H/S284T/C366V, I243L/L249E/S282T/S284T/L362M/S364G/C366S, I243L/L249M/R308F/W353Y/C366A, I243L/L249Y/W353Y/L362M/C366S, I243L/S282T/W353Y/L362M/S364G, I243L/S282T/L362M/S364G/C366V, I243L/R308F/W353Y, L249Y/W353Y/L362M/C366S, S282T, R308F/C366A, and L362M/C366A, wherein the positions are numbered with reference to SEQ ID NO: 4684. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 4814, 4816, 4818, 4820, 4822, 4824, 4826, 4828, 4830, 4832, 4834, 4836, 4838, 4840, 4842, 4844, 4846, 4848, 4850, 4852, 4854, 4856, 4858, 4860, 4862, 4864, 4866, 4868, 4870, 4872, and 4874. In some additional embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 4814, 4816, 4818, 4820, 4822, 4824, 4826, 4828, 4830, 4832, 4834, 4836, 4838, 4840, 4842, 4844, 4846, 4848, 4850, 4852, 4854, 4856, 4858, 4860, 4862, 4864, 4866, 4868, 4870, 4872, and 4874. In some further embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 4814, 4816, 4818, 4820, 4822, 4824, 4826, 4828, 4830, 4832, 4834, 4836, 4838, 4840, 4842, 4844, 4846, 4848, 4850, 4852, 4854, 4856, 4858, 4860, 4862, 4864, 4866, 4868, 4870, 4872, and 4874.

(18) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 25, 25/56/353, 25/243/249/259/366, 25/243/249/362, 25/362/366, 51, 51/56/200/243/249/259/338, 51/56/362, 51/71/249/279/284/362/366, 56/243/249, 56/362/366, 70/198/259/313, 85, 88, 88/173, 110, 159, 163, 171, 174, 175, 177, 198, 198/313, 198/313/428, 200/243/249/259, 208, 208/320, 209, 209/234, 222, 226, 234, 234/408, 243/338/362/366, 253, 256, 259, 265, 272, 289, 322, 336, 353/362, 405, 411, 428, and 439, wherein the positions are numbered with reference to SEQ ID NO: 4838. In some embodiments, the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 25L, 25L/56L/353Y, 25V/243M/249I/362M, 25V/243M/249Y/259G/366A, 25V/362M/366S, 51A, 51A/56L/200S/243M/249I/259G/338C, 51A/56L/362M, 51A/71R/249M/279G/284T/362M/366S, 56L/243M/249M, 56L/243M/249Y, 56L/362M/366A, 70S/198D/259E/313S, 85E, 88C/173S, 88I, 110S, 159N, 163K, 171D, 171E, 171P, 171V, 174E, 174S, 175N, 177K, 177P, 177S, 198D, 198D/313S, 198D/313S/428S, 198E, 198S, 198T, 200S/243M/249Y/259G, 208E, 208G, 208I, 208L, 208N, 208T, 208V, 208W/320I, 209A, 209E/234Q, 222P, 226R, 226T, 234A, 234E, 234H/408D, 234T, 243M/338C/362M/366S, 253A, 256A, 259N, 259S, 259T, 265A, 272D, 289H, 289T, 322V, 336E, 353Y/362M, 405Q, 411T, 428K, 439D, and 439Q, wherein the positions are numbered with reference to SEQ ID NO: 4838. In some additional embodiments, the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from I25L, I25L/H56L/W353Y, I25V/L243M/E249I/L362M, I25V/L243M/E249Y/H259G/V366A, I25V/L362M/V366S, K51A, K51A/H56L/K200S/L243M/E249I/H259G/V338C, K51A/H56L/L362M, K51A/Q71R/E249M/S279G/S284T/L362M/V366S, H56L/L243M/E249M, H56L/L243M/E249Y, H56L/L362M/V366A, P70S/I198D/H259E/F313S, A85E, R88C/R173S, R88I, A110S, Q159N, L163K, K171D, K171E, K171P, K171V, L174E, L174S, E175N, Q177K, Q177P, Q177S, I198D, I198D/F313S, I198D/F313S/V428S, I198E, I198S, I198T, K200S/L243M/E249Y/H259G, K208E, K208G, K208I, K208L, K208N, K208T, K208V, K208W/V320I, Q209A, Q209E/R234Q, K222P, E226R, E226T, R234A, R234E, R234H/N408D, R234T, L243M/V338C/L362M/V366S, S253A, L256A, H259N, H259S, H259T, Q265A, P272D, K289H, K289T, P322V, K336E, W353Y/L362M, E405Q, R411T, V428K, K439D, and K439Q, wherein the positions are numbered with reference to SEQ ID NO: 4838. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 4876, 4878, 4880, 4882, 4884, 4886, 4888, 4890, 4892, 4894, 4896, 4898, 4900, 4902, 4904, 4906, 4908, 4910, 4912, 4914, 4916, 4918, 4920, 4922, 4924, 4926, 4928, 4930, 4932, 4934, 4936, 4938, 4940, 4942, 4944, 4946, 4948, 4950, 4952, 4954, 4956, 4958, 4960, 4962, 4964, 4966, 4968, 4970, 4972, 4974, 4976, 4978, 4980, 4982, 4984, 4986, 4988, 4990, 4992, 4994, 4996, 4998, 5000, 5002, 5004, 5006, 5008, 5010, 5012, 5014, and 5016. In some embodiments, the

polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 4876, 4878, 4880, 4882, 4884, 4886, 4888, 4890, 4892, 4894, 4896, 4898, 4900, 4902, 4904, 4906, 4908, 4910, 4912, 4914, 4916, 4918, 4920, 4922, 4924, 4926, 4928, 4930, 4932, 4934, 4936, 4938, 4940, 4942, 4944, 4946, 4948, 4950, 4952, 4954, 4956, 4958, 4960, 4962, 4964, 4966, 4968, 4970, 4972, 4974, 4976, 4978, 4980, 4982, 4984, 4986, 4988, 4990, 4992, 4994, 4996, 4998, 5000, 5002, 5004, 5006, 5008, 5010, 5012, 5014, and 5016. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 4876, 4878, 4880, 4882, 4884, 4886, 4888, 4890, 4892, 4894, 4896, 4898, 4900, 4902, 4904, 4906, 4908, 4910, 4912, 4914, 4916, 4918, 4920, 4922, 4924, 4926, 4928, 4930, 4932, 4934, 4936, 4938, 4940, 4942, 4944, 4946, 4948, 4950, 4952, 4954, 4956, 4958, 4960, 4962, 4964, 4966, 4968, 4970, 4972, 4974, 4976, 4978, 4980, 4982, 4984, 4986, 4988, 4990, 4992, 4994, 4996, 4998, 5000, 5002, 5004, 5006, 5008, 5010, 5012, 5014, and 5016.

(19) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 8, 9, 25, 25/198/209, 25/200/209/338, 25/200/243/249, 25/209/243, 25/209/243/249, 25/209/243/259, 25/209/249/259/366, 25/209/259, 25/209/279/366, 25/209/289/366, 25/249, 25/259/279, 25/259/279/289, 25/279/284, 25/279/284/289, 25/289, 34, 53, 54, 55, 61, 69, 70, 73, 79, 87, 91, 107, 108, 111, 141, 153, 158, 174, 190, 194, 198, 198/200/209/243/249/289, 198/249/338, 200/209/366, 201, 205, 209, 209/249/259/279/338/366, 209/249/259/289, 209/249/279/284/289, 209/249/338, 209/279/289, 209/366, 234/297, 238, 247, 249/259, 252, 253, 254, 256, 259/366, 279, 279/338, 289, 297, 308, 321, 322, 327, 336, 338, 341, 342, 364, 366, 388, 392, 411, 412, 414, 426, 430, 432, 446, and 449, wherein the positions are numbered with reference to SEQ ID NO:4876. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 8D, 8L, 9G, 25H/209E/243M/249Y, 25L, 25L/198D/209E, 25L/200S/209E/338C, 25L/200S/243M/249M, 25L/209E/243M, 25L/209E/243M/249Y, 25L/209E/243M/259G, 25L/209E/249I/259G/366A, 25L/209E/259G, 25L/209E/279G/366A, 25L/209E/289T/366A, 25L/249I, 25L/259G/279G, 25L/259G/279G/289T, 25L/279G/284T, 25L/279G/284T/289T, 25L/289T, 341, 53C, 53L, 53R, 53V, 54P, 54V, 55T, 55W, 61S, 69N, 69Q, 69S, 69T, 70K, 73S, 79G, 79S, 79V, 87L, 87M, 87R, 91Q, 91R, 91T, 107A, 107C, 107T, 108C, 111C, 111G, 141M, 153S, 158Q, 174M, 190R, 194Q, 198D, 198D/200S/209E/243M/249Y/289T, 198D/249M/338C, 200S/209E/366A, 201P, 201T, 205P, 209E, 209E/249I/259G/289T, 209E/249M/338C, 209E/249Y/259G/279G/338C/366A, 209E/249Y/279G/284T/289T, 209E/279G/289T, 209E/366A, 234H/297A, 238K, 247A, 247L, 249M/259G, 252A, 252E, 252Q, 253G, 253P, 254G, 254M, 254P, 256D, 256T, 256W, 259G/366A, 279G, 279G/338C, 289T, 297A, 308C, 308L, 308T, 321D, 322M, 322R, 322T, 327L, 336M, 336Y, 338C, 341V, 342R, 364S, 366A, 388T, 388V, 392H, 392Q, 411E, 411S, 412H, 412T, 414L, 426T, 430R, 432T, 446F, and 449L, wherein the positions are numbered with reference to SEQ ID NO:4876. In some additional embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from TBD, T8L, V9G, I25H/Q209E/L243M/E249Y, I25L, I25L/I198D/Q209E, I25L/K200S/Q209E/V338C, I25L/K200S/L243M/E249M, I25L/Q209E/L243M, I25L/Q209E/L243M/E249Y, I25L/Q209E/L243M/H259G, I25L/Q209E/E249I/H259G/V366A, I25L/Q209E/H259G, I25L/Q209E/S279G/V366A, I25L/Q209E/K289T/V366A, I25L/E249I, I25L/H259G/S279G, I25L/H259G/S279G/K289T, I25L/S279G/S284T, I25L/S279G/S284T/K289T, I25L/K289T, V34I, K53C, K53L, K53R, K53V, T54P, T54V, S55T, S55W, T61S, D69N, D69Q, D69S, D69T, P70K, E73S, P79G, P79S, P79V, W87L, W87M, W87R, I91Q, I91R, I91T, L107A, L107C, L107T, M108C, S111C, S111G, L141M, A153S, P158Q, L174M, K190R, S194Q, I198D, I198D/K200S/Q209E/L243M/E249Y/K289T, I198D/E249M/V338C, K200S/Q209E/V366A, E201P, E201T, K205P, Q209E, Q209E/E249I/H259G/K289T, Q209E/E249M/V338C, Q209E/E249Y/H259G/S279G/V338C/V366A, Q209E/E249Y/S279G/S284T/K289T, Q209E/S279G/K289T, Q209E/V366A, R234H/G297A, A238K, K247A, K247L, E249M/H259G, S252A, S252E, S252Q, S253G, S253P, S254G, S254M, S254P, L256D, L256T, L256W, H259G/V366A, S279G, S279G/V338C, K289T, G297A, R308C, R308L, R308T, E321D, P322M, P322R, P322T, F327L, K336M, K336Y, V338C, Q341V, E342R, G364S, V366A, S388T, S388V, K392H, K392Q, R411E, R411S, R412H, R412T, M414L, A426T, K430R, K432T, S446F, and S449L, wherein the positions are numbered with reference to SEQ ID NO:4876. In some

embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 5018, 5020, 5022, 5024, 5026, 5028, 5030, 5032, 5034, 5036, 5038, 5040, 5042, 5044, 5046, 5048, 5050, 5052, 5054, 5056, 5058, 5060, 5062, 5064, 5066, 5068, 5070, 5072, 5074, 5076, 5078, 5080, 5082, 5084, 5086, 5088, 5090, 5092, 5094, 5096, 5098, 5100, 5102, 5104, 5106, 5108, 5110, 5112, 5114, 5116, 5118, 5120, 5122, 5124, 5126, 5128, 5130, 5132, 5134, 5136, 5138, 5140, 5142, 5144, 5146, 5148, 5150, 5152, 5154, 5156, 5158, 5160, 5162, 5164, 5166, 5168, 5170, 5172, 5174, 5176, 5178, 5180, 5182, 5184, 5186, 5188, 5190, 5192, 5194, 5196, 5198, 5200, 5202, 5204, 5206, 5208, 5210, 5212, 5214, 5216, 5218, 5220, 5222, 5224, 5226, 5228, 5230, 5232, 5234, 5236, 5238, 5240, 5242, 5244, 5246, 5248, 5250, 5252, 5254, 5256, 5258, and 5260. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 5018, 5020, 5022, 5024, 5026, 5028, 5030, 5032, 5034, 5036, 5038, 5040, 5042, 5044, 5046, 5048, 5050, 5052, 5054, 5056, 5058, 5060, 5062, 5064, 5066, 5068, 5070, 5072, 5074, 5076, 5078, 5080, 5082, 5084, 5086, 5088, 5090, 5092, 5094, 5096, 5098, 5100, 5102, 5104, 5106, 5108, 5110, 5112, 5114, 5116, 5118, 5120, 5122, 5124, 5126, 5128, 5130, 5132, 5134, 5136, 5138, 5140, 5142, 5144, 5146, 5148, 5150, 5152, 5154, 5156, 5158, 5160, 5162, 5164, 5166, 5168, 5170, 5172, 5174, 5176, 5178, 5180, 5182, 5184, 5186, 5188, 5190, 5192, 5194, 5196, 5198, 5200, 5202, 5204, 5206, 5208, 5210, 5212, 5214, 5216, 5218, 5220, 5222, 5224, 5226, 5228, 5230, 5232, 5234, 5236, 5238, 5240, 5242, 5244, 5246, 5248, 5250, 5252, 5254, 5256, 5258, and 5260. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 5018, 5020, 5022, 5024, 5026, 5028, 5030, 5032, 5034, 5036, 5038, 5040, 5042, 5044, 5046, 5048, 5050, 5052, 5054, 5056, 5058, 5060, 5062, 5064, 5066, 5068, 5070, 5072, 5074, 5076, 5078, 5080, 5082, 5084, 5086, 5088, 5090, 5092, 5094, 5096, 5098, 5100, 5102, 5104, 5106, 5108, 5110, 5112, 5114, 5116, 5118, 5120, 5122, 5124, 5126, 5128, 5130, 5132, 5134, 5136, 5138, 5140, 5142, 5144, 5146, 5148, 5150, 5152, 5154, 5156, 5158, 5160, 5162, 5164, 5166, 5168, 5170, 5172, 5174, 5176, 5178, 5180, 5182, 5184, 5186, 5188, 5190, 5192, 5194, 5196, 5198, 5200, 5202, 5204, 5206, 5208, 5210, 5212, 5214, 5216, 5218, 5220, 5222, 5224, 5226, 5228, 5230, 5232, 5234, 5236, 5238, 5240, 5242, 5244, 5246, 5248, 5250, 5252, 5254, 5256, 5258, and 5260.

(20) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 88, 88/110, 88/110/159/198/200, 88/110/159/198/234/259/265, 88/110/159/198/259, 88/110/198, 88/110/234, 88/159/198, 88/198, 88/198/200/259, 88/198/259, 88/259, 88/259/265, 110/259/265/411, 159/198/411, 159/259/411, 171/174/175/177/208/320/428/439, 171/175/177/208/320/428, 171/177, 171/177/226/428/439, 171/208/320, 171/208/320/428, 171/208/428, 174/175/177/208/320/428, 174/175/428, 174/320, 198, 198/200, 198/200/234, 198/234, 198/259, 208, 208/320/331/428, 208/320/428, 208/428, 234, 259, 320/428, and 428, wherein the positions are numbered with reference to SEQ ID NO: 5066. In some embodiments, the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 88I, 88I/110D, 88I/110D/159N/198D/234T/259N/265A, 88I/110D/159N/198D/259E, 88I/110D/159N/198E/200S, 88I/110D/198D, 88I/110D/198S, 88I/110D/234T, 88I/159N/198D, 88I/198D/200S/259E, 88I/198D/259E, 88I/198S, 88I/259E/265A, 88I/259N, 110D/259N/265A/411T, 159N/198E/411T, 159N/259E/411T, 171E/177P/226T/428K/439D, 171E/208E/320I, 171E/208E/320I/428K, 171E/208E/428K, 171P/174E/175N/177P/208W/320I/428K/439D, 171P/175N/177P/208L/320I/428K, 171P/177P, 174E/175N/177P/208E/320I/428K, 174E/175N/428K, 174E/320I, 198D, 198D/200S/234E, 198D/234Q, 198E/200S, 198E/259E, 208E, 208E/320I/428K, 208E/428K, 208L/320I/331C/428K, 234E, 259E, 320I/428K, and 428K, wherein the positions are numbered with reference to SEQ ID NO: 5066. In some embodiments, the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from R88I, R88I/A110D, R88I/A110D/Q159N/I198D/R234T/G259N/Q265A, R88I/A110D/Q159N/I198D/G259E, R88I/A110D/Q159N/I198E/K200S, R88I/A110D/I198D, R88I/A110D/I198S, R88I/A110D/R234T, R88I/Q159N/I198D, R88I/I198D/K200S/G259E, R88I/I198D/G259E, R88I/I198S, R88I/G259E/Q265A, R88I/G259N, A110D/G259N/Q265A/R411T, Q159N/I198E/R411T, Q159N/G259E/R411T, K171E/Q177P/E226T/V428K/K439D, K171E/K208E/V320I, K171E/K208E/V320I/V428K, K171E/K208E/V428K, K171P/L174E/E175N/Q177P/K208W/V320I/V428K/K439D,

K171P/E175N/Q177P/K208L/V320I/V428K, K171P/Q177P, L174E/E175N/Q177P/K208E/V320I/V428K, L174E/E175N/V428K, L174E/V320I, I198D, I198D/K200S/R234E, I198D/R234Q, I198E/K200S, I198E/G259E, K208E, K208E/V320I/V428K, K208E/V428K, K208L/V320I/R331C/V428K, R234E, G259E, V320I/V428K, and V428K, wherein the positions are numbered with reference to SEQ ID NO: 5066. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 5262, 5264, 5266, 5268, 5270, 5272, 5274, 5276, 5278, 5280, 5282, 5284, 5286, 5288, 5290, 5292, 5294, 5296, 5298, 5300, 5302, 5304, 5306, 5308, 5310, 5312, 5314, 5316, 5318, 5320, 5322, 5324, 5326, 5328, 5330, 5332, 5334, 5336, 5338, 5340, and 5342. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 5262, 5264, 5266, 5268, 5270, 5272, 5274, 5276, 5278, 5280, 5282, 5284, 5286, 5288, 5290, 5292, 5294, 5296, 5298, 5300, 5302, 5304, 5306, 5308, 5310, 5312, 5314, 5316, 5318, 5320, 5322, 5324, 5326, 5328, 5330, 5332, 5334, 5336, 5338, 5340, and 5342. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 5262, 5264, 5266, 5268, 5270, 5272, 5274, 5276, 5278, 5280, 5282, 5284, 5286, 5288, 5290, 5292, 5294, 5296, 5298, 5300, 5302, 5304, 5306, 5308, 5310, 5312, 5314, 5316, 5318, 5320, 5322, 5324, 5326, 5328, 5330, 5332, 5334, 5336, 5338, 5340, and 5342.

(21) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 2, 7, 25/61/208/252/253, 25/61/428, 25/107/208/320/428, 25/208/247/252/253/364/428, 25/247/249/252/364/428, 38, 53/54/55/153/201, 53/54/308/392, 53/171/308/392, 54, 61/107/208/252/253/254/364/428, 61/208/252/254/428, 64, 68, 73/87/201, 91/201/439/444, 99, 106, 107, 107/208/320/364/428, 107/247/249/252/254/364, 107/247/252, 107/247/428, 109, 159, 169, 171, 172, 177, 179, 190, 190/208/247/252/428, 222, 233, 233/269, 247/249/252/254/320/428, 249/252/253/254, 249/252/254/428, 251, 252/253/254, 253/320, 259, 264, 289, 296, 300, 308, 308/327/439, 317, 318, 320, 320/364/428, 320/428, 347, 404, 408, 417, 424, 427, 428, 428/434, 431, 435, and 438, wherein the positions are numbered with reference to SEQ ID NO: 5290. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 2L, 2N, 7P, 25L/61S/208N/252E/253G, 25L/61S/428K, 25L/107T/208E/320I/428K, 25L/208E/247L/252E/253G/364S/428K, 25L/247L/249I/252E/364S/428K, 38Q, 53R/54V/55T/153S/201P, 53V/54V/308L/392H, 53V/171P/308L/392H, 54A, 61S/107T/208N/252E/253G/254M/364S/428K, 61S/208W/252E/254M/428K, 64S, 68K, 73S/87M/201P, 91T/201P/439D/444H, 99P, 106T, 107A, 107A/208N/320I/364S/428K, 107A/247A/249I/252E/254M/364S, 107A/247A/252E, 107A/247A/428K, 109S, 159L, 169C, 169E, 169L, 169Q, 169V, 171S, 171T, 171V, 172H, 172N, 177P, 179A, 179S, 190R, 190R/208N/247L/252E/428K, 222A, 233C, 233G, 233K, 233L, 233M, 233Q, 233Q/269R, 233V, 247A/249I/252E/254M/320I/428K, 249I/252E/253G/254M, 249I/252E/254M/428K, 251L, 252E/253G/254M, 253G/320I, 259T, 264A, 289S, 296A, 296H, 296Q, 300G, 308L, 308L/327L/439D, 317R, 318T, 320I, 320I/364S/428K, 320I/428K, 347H, 347K, 347P, 347R, 404T, 408R, 417P, 424A, 424W, 427L, 427R, 428E/434N, 428F, 428I, 428K, 428Q, 428R, 428S, 431E, 431R, 435C, 435K, 435M, 435Q, 435R, 435T, and 438Q, wherein the positions are numbered with reference to SEQ ID NO: 5290. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from E2L, E2N, T7P, I25L/T61S/K208N/S252E/S253G, I25L/T61S/V428K, I25L/L107T/K208E/V320I/V428K, I25L/K208E/K247L/5252E/5253G/G364S/V428K, I25L/K247L/Y249I/S252E/G364S/V428K, R38Q, K53R/T54V/S55T/A153S/E201P, K53V/T54V/R308L/K392H, K53V/K171P/R308L/K392H, T54A, T61S/L107T/K208N/S252E/S253G/S254M/G364S/V428K, T61S/K208W/S252E/S254M/V428K, F64S, N68K, E73S/W87M/E201P, I91T/E201P/K439D/Y444H, E99P, S106T, L107A, L107A/K208N/V320I/G364S/V428K, L107A/K247A/Y249I/S252E/S254M/G364S, L107A/K247A/S252E, L107A/K247A/V428K, R109S, N159L, G169C, G169E, G169L, G169Q, G169V, K171S, K171T, K171V, T172H, T172N, Q177P, V179A, V179S, K190R, K190R/K208N/K247L/S252E/V428K, K222A, W233C, W233G, W233K, W233L, W233M, W233Q,

W233Q/Q269R, W233Y, K247A/Y249I/S252E/S254M/V320I/V428K, Y249I/S252E/S253G/S254M, Y249I/S252E/S254M/V428K, A251L, S252E/S253G/S254M, S253G/V320I, E259T, F264A, K289S, R296A, R296H, R296Q, D300G, R308L, R308L/F327L/K439D, S317R, E318T, V320I, V320I/G364S/V428K, V320I/V428K, D347H, D347K, D347P, D347R, G404T, N408R, R417P, Q424A, Q424W, A427L, A427R, V428E/D434N, V428F, V428I, V428K, V428Q, V428R, V428S, D431E, D431R, V435C, V435K, V435M, V435Q, V435R, V435T, and A438Q, wherein the positions are numbered with reference to SEQ ID NO: 5290. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 5344, 5346, 5348, 5350, 5352, 5354, 5356, 5358, 5360, 5362, 5364, 5366, 5368, 5370, 5372, 5374, 5376, 5378, 5380, 5382, 5384, 5386, 5388, 5390, 5392, 5394, 5396, 5398, 5400, 5402, 5404, 5406, 5408, 5410, 5412, 5414, 5416, 5418, 5420, 5422, 5424, 5426, 5428, 5430, 5432, 5434, 5436, 5438, 5440, 5442, 5444, 5446, 5448, 5450, 5452, 5454, 5456, 5458, 5460, 5462, 5464, 5466, 5468, 5470, 5472, 5474, 5476, 5478, 5480, 5482, 5484, 5486, 5488, 5490, 5492, 5494, 5496, 5498, 5500, 5502, 5504, 5506, 5508, 5510, 5512, 5514, 5516, 5518, 5520, 5522, 5524, 5526, 5528, 5530, 5532, 5534, 5536, 5538, 5540, and 5542. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 5344, 5346, 5348, 5350, 5352, 5354, 5356, 5358, 5360, 5362, 5364, 5366, 5368, 5370, 5372, 5374, 5376, 5378, 5380, 5382, 5384, 5386, 5388, 5390, 5392, 5394, 5396, 5398, 5400, 5402, 5404, 5406, 5408, 5410, 5412, 5414, 5416, 5418, 5420, 5422, 5424, 5426, 5428, 5430, 5432, 5434, 5436, 5438, 5440, 5442, 5444, 5446, 5448, 5450, 5452, 5454, 5456, 5458, 5460, 5462, 5464, 5466, 5468, 5470, 5472, 5474, 5476, 5478, 5480, 5482, 5484, 5486, 5488, 5490, 5492, 5494, 5496, 5498, 5500, 5502, 5504, 5506, 5508, 5510, 5512, 5514, 5516, 5518, 5520, 5522, 5524, 5526, 5528, 5530, 5532, 5534, 5536, 5538, 5540, and 5542. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 5344, 5346, 5348, 5350, 5352, 5354, 5356, 5358, 5360, 5362, 5364, 5366, 5368, 5370, 5372, 5374, 5376, 5378, 5380, 5382, 5384, 5386, 5388, 5390, 5392, 5394, 5396, 5398, 5400, 5402, 5404, 5406, 5408, 5410, 5412, 5414, 5416, 5418, 5420, 5422, 5424, 5426, 5428, 5430, 5432, 5434, 5436, 5438, 5440, 5442, 5444, 5446, 5448, 5450, 5452, 5454, 5456, 5458, 5460, 5462, 5464, 5466, 5468, 5470, 5472, 5474, 5476, 5478, 5480, 5482, 5484, 5486, 5488, 5490, 5492, 5494, 5496, 5498, 5500, 5502, 5504, 5506, 5508, 5510, 5512, 5514, 5516, 5518, 5520, 5522, 5524, 5526, 5528, 5530, 5532, 5534, 5536, 5538, 5540, and 5542.

(22) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 2, 49, 73, 87, 87/88, 87/88/94/159/177/208, 87/88/159/198, 87/88/177/198/249, 87/88/208, 87/159/177, 87/159/320, 87/177/249, 87/198, 87/198/199, 87/198/208/320, 87/208, 87/208/320, 87/320, 88, 88/177, 88/177/208, 88/177/320, 88/198, 88/199, 88/199/208, 88/208, 88/208/249/320, 88/320, 113, 134, 135, 158, 159/177/198, 159/177/208, 159/198/208/320, 171, 173, 177/198, 177/208, 195, 214, 222, 253, 256, 257, 268, 272, 289, 300, 302, 330, 348, 374, 392, 399, 408, 411, 412, 437, 439, 445, and 453, wherein the positions are numbered with reference to SEQ ID NO: 5372. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 2G, 2H, 2P, 49A, 73P, 73T, 87M, 87M/88R, 87M/88R/94K/159Q/177P/208E, 87M/88R/159Q/198E, 87M/88R/177P/198E/249I, 87M/88R/208E, 87M/159Q/177P, 87M/159Q/320I, 87M/177P/249I, 87M/198E, 87M/198E/199K, 87M/198E/208E/320I, 87M/208E, 87M/208E/320I, 87M/320I, 88R, 88R/177P, 88R/177P/208E, 88R/177P/320I, 88R/198I, 88R/199K, 88R/199K/208E, 88R/208E, 88R/208E/249I/320I, 88R/320I, 113S, 134G, 134S, 135A, 158D, 158E, 159Q/177P/198E, 159Q/177P/208E, 159Q/198I/208E/320I, 171T, 173N, 173P, 177P/198E, 177P/208E, 195H, 214L, 222R, 253G, 256P, 257A, 257Q, 268A, 268G, 268H, 272K, 289T, 300G, 302R, 330D, 348C, 374K, 374R, 392D, 399D, 408D, 411Q, 412H, 437I, 437T, 437V, 439A, 445T, and 453R, wherein the positions are numbered with reference to SEQ ID NO: 5372. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from E2G, E2H, E2P, F49A, E73P, E73T, W87M, W87M/I88R, W87M/I88R/E94K/N159Q/Q177P/K208E, W87M/I88R/N159Q/D198E, W87M/I88R/Q177P/D198E/Y249I, W87M/I88R/K208E, W87M/N159Q/Q177P, W87M/N159Q/V320I, W87M/Q177P/Y249I, W87M/D198E, W87M/D198E/A199K, W87M/D198E/K208E/V320I,

W87M/Q208E/V320I, W87M/V320I, W87M/V177P, I88R, I88R/Q177P, I88R/Q177P/K208E, I88R/Q177P/V320I, I88R/D198I, I88R/A199K, I88R/A199K/K208E, I88R/K208E, I88R/K208E/Y249I/V320I, I88R/V320I, E113S, D134G, D134S, S135A, P158D, P158E, N159Q/Q177P/D198E, N159Q/Q177P/K208E, N159Q/D198I/K208E/V320I, K171T, R173N, R173P, Q177P/D198E, Q177P/K208E, N195H, S214L, K222R, S253G, L256P, L257A, L257Q, D268A, D268G, D268H, P272K, K289T, D300G, K302R, E330D, A348C, S374K, S374R, K392D, N399D, N408D, R411Q, R412H, L437I, L437T, L437V, K439A, E445T, and Y453R, wherein the positions are numbered with reference to SEQ ID NO: 5372. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 5544, 5546, 5548, 5550, 5552, 5554, 5556, 5558, 5560, 5562, 5564, 5566, 5568, 5570, 5572, 5574, 5576, 5578, 5580, 5582, 5584, 5586, 5588, 5590, 5592, 5594, 5596, 5598, 5600, 5602, 5604, 5606, 5608, 5610, 5612, 5614, 5616, 5618, 5620, 5622, 5624, 5626, 5628, 5630, 5632, 5634, 5636, 5638, 5640, 5642, 5644, 5646, 5648, 5650, 5652, 5654, 5656, 5658, 5660, 5662, 5664, 5666, 5668, 5670, 5672, 5674, 5676, 5678, 5680, 5682, 5684, 5686, 5688, and 5690. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 5544, 5546, 5548, 5550, 5552, 5554, 5556, 5558, 5560, 5562, 5564, 5566, 5568, 5570, 5572, 5574, 5576, 5578, 5580, 5582, 5584, 5586, 5588, 5590, 5592, 5594, 5596, 5598, 5600, 5602, 5604, 5606, 5608, 5610, 5612, 5614, 5616, 5618, 5620, 5622, 5624, 5626, 5628, 5630, 5632, 5634, 5636, 5638, 5640, 5642, 5644, 5646, 5648, 5650, 5652, 5654, 5656, 5658, 5660, 5662, 5664, 5666, 5668, 5670, 5672, 5674, 5676, 5678, 5680, 5682, 5684, 5686, 5688, and 5690. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 5544, 5546, 5548, 5550, 5552, 5554, 5556, 5558, 5560, 5562, 5564, 5566, 5568, 5570, 5572, 5574, 5576, 5578, 5580, 5582, 5584, 5586, 5588, 5590, 5592, 5594, 5596, 5598, 5600, 5602, 5604, 5606, 5608, 5610, 5612, 5614, 5616, 5618, 5620, 5622, 5624, 5626, 5628, 5630, 5632, 5634, 5636, 5638, 5640, 5642, 5644, 5646, 5648, 5650, 5652, 5654, 5656, 5658, 5660, 5662, 5664, 5666, 5668, 5670, 5672, 5674, 5676, 5678, 5680, 5682, 5684, 5686, 5688, and 5690.

(23) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 2/7/107/233/252/253/320, 2/7/107/233/252/253/320/408/428, 2/7/233/252/253/320/427/428, 2/233/252/317/320, 7/107/190/233/252/253/317/408/427, 7/107/233/252/253, 7/107/233/252/317/320, 7/107/233/252/317/408, 7/190/233/252/253/320/427, 7/190/233/252/317/320/427/428, 7/190/233/252/408/427, 7/233/252/253/317/408/427, 7/233/252/253/408, 7/233/252/317/320/427, 7/233/252/317/428, 64/169/201/347/392, 64/172/264/268/347/392/417, 91/94/171/172/201/264/347, 91/201/264/347/392, 94/201/264/347/435, 107/190/233/252, 107/190/233/252/317/320, 107/233/252/253, 107/252/317, 169/171/172/264/392/435, 169/171/201/264/392/435, 169/172/201/264/347, 169/172/201/264/347/392/435, 169/172/201/347/392, 169/172/201/417/435, 171/172/201/264/392/417/435, 171/201/392/417, 190/233/252, 190/233/252/253/317/320/408/428, 190/233/252/317/408, 190/233/252/320/408, 201/264/347/392/417/435, 201/264/347/392/435, 201/264/347/417/435, 233/252/253/317/320/427/428, 233/252/253/408/427, 233/252/317/408, 233/252/317/427, 233/252/320, 233/252/320/408/428, 348/374/435, and 374, wherein the positions are numbered with reference to SEQ ID NO: 5562. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from

2L/7P/233K/252E/253G/320I/427R/428R, 2N/7P/107A/233K/252E/253G/320I, 2N/7P/107A/233K/252E/253G/320I/408R/428R, 2N/233K/252E/317R/320I, 7P/107A/190R/233K/252E/253G/317R/408R/427R, 7P/107A/233K/252E/253G, 7P/107A/233K/252E/317R/320I, 7P/107A/233K/252E/317R/408R, 7P/190R/233K/252E/253G/320I/427L, 7P/190R/233K/252E/317R/320I/427R/428R, 7P/190R/233K/252E/408R/427R, 7P/233K/252E/253G/317R/408R/427L, 7P/233K/252E/253G/408R, 7P/233K/252E/317R/320I/427L, 7P/233K/252E/317R/428R, 64S/169E/201P/347P/392H, 64S/172H/264A/268V/347P/392H/417P, 91T/94E/171T/172H/201P/264A/347G, 91T/201P/264A/347G/392H, 94E/201P/264A/347G/435Q, 107A/190R/233K/252E, 107A/190R/233K/252E/317R/320I, 107A/233K/252E/253G, 107A/252E/317R, 169E/171T/172H/264A/392H/435Q, 169E/171T/201P/264A/392H/435Q, 169E/172H/201P/264A/347G,

169E/172H/201P/392H/435R, 169E/172H/201P/347G/392H, 169E/172H/201P/417P/435R, 171T/172H/201P/264A/392H/417P/435R, 171T/201P/392H/417P, 190R/233K/252E, 190R/233K/252E/253G/317R/320I/408R/428R, 190R/233K/252E/317R/408R, 190R/233K/252E/320I/408R, 201P/264A/347K/392H/417P/435R, 201P/264A/347K/417P/435R, 201P/264A/347P/392H/435Q, 233K/252E/253G/317R/320I/427R/428R, 233K/252E/253G/408R/427L, 233K/252E/317R/408R, 233K/252E/317R/427R, 233K/252E/320I, 233K/252E/320I/408R/428R, 348S/374R/435R, and 374R, wherein the positions are numbered with reference to SEQ ID NO: 5562. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from E2L/T7P/W233K/S252E/S253G/V320I/A427R/K428R, E2N/T7P/L107A/W233K/S252E/S253G/V320I, E2N/T7P/L107A/W233K/S252E/S253G/V320I/N408R/K428R, E2N/W233K/S252E/S317R/V320I, T7P/L107A/K190R/W233K/S252E/S253G/S317R/N408R/A427R, T7P/L107A/W233K/S252E/S253G, T7P/L107A/W233K/S252E/S317R/V320I, T7P/L107A/W233K/S252E/S317R/N408R, T7P/K190R/W233K/S252E/S253G/V320I/A427L, T7P/K190R/W233K/S252E/S317R/V320I/A427R/K428R, T7P/K190R/W233K/S252E/N408R/A427R, T7P/W233K/S252E/S253G/S317R/N408R/A427L, T7P/W233K/S252E/S253G/N408R, T7P/W233K/S252E/S317R/V320I/A427L, T7P/W233K/S252E/S317R/K428R, F64S/G169E/E201P/D347P/K392H, F64S/T172H/F264A/D268V/D347P/K392H/R417P, I91T/K94E/K171T/T172H/E201P/F264A/D347G, I91T/E201P/F264A/D347G/K392H, K94E/E201P/F264A/D347G/V435Q, L107A/K190R/W233K/S252E, L107A/K190R/W233K/S252E/S317R/V320I, L107A/W233K/S252E/S253G, L107A/S252E/S317R, G169E/K171T/T172H/F264A/K392H/V435Q, G169E/K171T/E201P/F264A/K392H/V435Q, G169E/T172H/E201P/F264A/D347G, G169E/T172H/E201P/F264A/D347K/K392H/V435R, G169E/T172H/E201P/D347G/K392H, G169E/T172H/E201P/R417P/V435R, K171T/T172H/E201P/F264A/K392H/R417P/V435R, K171T/E201P/K392H/R417P, K190R/W233K/S252E, K190R/W233K/S252E/S253G/S317R/V320I/N408R/K428R, K190R/W233K/S252E/S317R/N408R, K190R/W233K/S252E/V320I/N408R, E201P/F264A/D347K/K392H/R417P/V435R, E201P/F264A/D347K/R417P/V435R, E201P/F264A/D347P/K392H/V435Q, W233K/S252E/S253G/S317R/V320I/A427R/K428R, W233K/S252E/S253G/N408R/A427L, W233K/S252E/S317R/N408R, W233K/S252E/S317R/A427R, W233K/S252E/V320I, W233K/S252E/V320I/N408R/K428R, A348S/S374R/V435R, and S374R, wherein the positions are numbered with reference to SEQ ID NO: 5562. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 14, 55, 56, 255, 282, 308, 336, 342, 364, 391, 407, and 422, wherein the positions are numbered with reference to SEQ ID NO: 5562. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 141, 55V, 56A, 255L, 282S, 308L, 308Q, 336Q, 342W, 364A, 364S, 391C, 407C, 407V, and 422Q, wherein the positions are numbered with reference to SEQ ID NO: 5562. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from V141, S55V, L56A, S255L, T282S, R308L, R308Q, K336Q, E342W, G364A, G364S, L391C, E407C, E407V, and M422Q, wherein the positions are numbered with reference to SEQ ID NO: 5562. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 5692, 5694, 5696, 5698, 5700, 5702, 5704, 5706, 5708, 5710, 5712, 5714, 5716, 5718, 5720, 5722, 5724, 5726, 5728, 5730, 5732, 5734, 5736, 5738, 5740, 5742, 5744, 5746, 5748, 5750, 5752, 5754, 5756, 5758, 5760, 5762, 5764, 5766, 5768, 5770, 5772, 5774, 5776, 5778, 5780, 5782, 5784, 5786, 5788, 5790, 5792, 5794, 5796, 5798, 5800, 5802, 5804, 5806, 5808, 5810, 5812, and 5814. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 5692, 5694, 5696, 5698, 5700, 5702, 5704, 5706, 5708, 5710, 5712, 5714, 5716, 5718, 5720, 5722, 5724, 5726, 5728, 5730, 5732, 5734, 5736, 5738, 5740, 5742, 5744, 5746, 5748, 5750, 5752, 5754, 5756, 5758, 5760, 5762, 5764, 5766, 5768, 5770, 5772, 5774, 5776, 5778, 5780, 5782, 5784, 5786, 5788, 5790, 5792, 5794, 5796, 5798, 5800, 5802, 5804, 5806, 5808, 5810, 5812, and 5814. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase

comprises any of SEQ ID NOS: 5692, 5694, 5696, 5698, 5700, 5702, 5704, 5706, 5708, 5710, 5712, 5714, 5716, 5718, 5720, 5722, 5724, 5726, 5728, 5730, 5732, 5734, 5736, 5738, 5740, 5742, 5744, 5746, 5748, 5750, 5752, 5754, 5756, 5758, 5760, 5762, 5764, 5766, 5768, 5770, 5772, 5774, 5776, 5778, 5780, 5782, 5784, 5786, 5788, 5790, 5792, 5794, 5796, 5798, 5800, 5802, 5804, 5806, 5808, 5810, 5812, and 5814.

(24) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 2/4/113/158/163/302/364/399/449, 2/4/113/158/163/330/449, 2/4/113/163/449, 2/4/158/163/364/449, 2/4/158/222/257/302/330, 2/4/158/330, 2/4/163/222/302/330/364/449, 2/4/163/257, 2/4/163/257/330/399/449, 2/4/163/330/449, 2/4/163/364, 2/4/163/364/453, 2/4/302, 2/113, 2/113/163, 2/113/163/449, 2/113/330/399, 2/113/449, 2/158/163, 2/158/163/364/399, 2/163/364/453, 2/364/449, 4/113/158/163/330/364/399, 4/113/158/302/330/364, 4/113/163/364/399, 4/158/163/364/399, 4/364/449, 87, 87/95/198, 87/198, 87/266, 87/322, 107/134/135/195/412/417, 107/195/268/322/439, 107/195/272, 107/195/417/439, 107/374/417/439, 113/158/163/364/399, 134, 134/135/195/268/317, 158/163/257/330/364/449, 158/163/302/330/364/399, 158/163/364, 163/257/302/364, 163/302/330/364/449, 163/364, 173, 173/190/233/252/427/437, 173/190/252/257/347/427/437, 173/190/257/374/437, 173/190/257/427/437, 173/233/252/257/427, 173/233/437, 173/252, 173/252/268/437, 173/252/347/411, 173/257/374/437, 173/374/437, 173/427/437, 190/233/252/257/347/411/437, 190/252, 190/252/257/285/427, 190/252/257/411/437, 190/252/257/427, 190/374/427/437, 195/272/320/439, 195/317/320, 198, 198/244, 198/292, 233/252, 233/252/257, 233/252/257/347, 233/252/257/347/411/437, 233/252/257/427/437, 233/252/285, 233/252/285/437, 233/252/411/437, 233/252/437, 233/374, 233/374/437, 252/257, 252/257/347/437, 257/347/411/437, 268/417, 285/347/437, 302/364, 330/364, 347/411/437, 347/427/437, 364/399, 374, 374/411/437, 408/417, 411/437, and 437, wherein the positions are numbered with reference to SEQ ID NO:5708. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 2P/4T/113S/158D/163A/330D/449R, 2P/4T/113S/158D/163R/302R/364S/399D/449R, 2P/4T/113S/163R/449R, 2P/4T/158D/163A/364S/449R, 2P/4T/158D/222R/257Q/302R/330D, 2P/4T/158D/330D, 2P/4T/163A/257Q, 2P/4T/163A/330D/449R, 2P/4T/163A/364S, 2P/4T/163A/364S/453R, 2P/4T/163R/222R/302R/330D/364S/449R, 2P/4T/163R/257Q/330D/399D/449R, 2P/4T/302R, 2P/113S, 2P/113S/163A, 2P/113S/163R/449R, 2P/113S/330D/399D, 2P/113S/449R, 2P/158D/163A/364S/399D, 2P/158D/163R, 2P/163R/364S/453R, 2P/364S/449R, 4T/113S/158D/163A/330D/364S/399D, 4T/113S/158D/302R/330D/364S, 4T/113S/163A/364S/399D, 4T/158D/163A/364S/399D, 4T/364S/449R, 87W, 87W/95L/198E, 87W/198E, 87W/198I, 87W/266L, 87W/322S, 107A/134S/135A/195H/412H/417P, 107A/195H/268A/322L/439P, 107A/195H/272K, 107A/195H/417P/439P, 107A/374T/417P/439P, 113S/158D/163A/364S/399D, 113S/158D/163R/364S/399D, 134S, 134S/135A/195H/268A/317R, 158D/163A/257Q/330D/364S/449R, 158D/163A/364S, 158D/163R/302R/330D/364S/399D, 163A/257Q/302R/364S, 163A/364S, 163R/302R/330D/364S/449R, 173N, 173N/190R/233K/252E/427R/437V, 173N/190R/252E/257A/347G/427R/437I, 173N/190R/257A/374K/437V, 173N/190R/257A/427R/437I, 173N/233K/252E/257A/427R, 173N/233K/437I, 173N/252E, 173N/252E/268H/437V, 173N/252E/347G/411Q, 173N/257A/374K/437V, 173N/374K/437I, 173N/427R/437I, 190R/233K/252E/257A/347G/411Q/437I, 190R/252E, 190R/252E/257A/285Q/427R, 190R/252E/257A/411Q/437I, 190R/252E/257A/427R, 190R/374K/427R/437V, 195H/272K/320I/439P, 195H/317R/320I, 198E, 198E/292P, 198I, 198I/244L, 233K/252E, 233K/252E/257A, 233K/252E/257A/347G, 233K/252E/257A/347G/411Q/437I, 233K/252E/257A/427R/437V, 233K/252E/285Q, 233K/252E/285Q/437V, 233K/252E/411Q/437V, 233K/252E/437V, 233K/374K, 233K/374K/437I, 252E/257A, 252E/257A/347G/437V, 257A/347G/411Q/437V, 268A/417P, 285Q/347G/437V, 302R/364S, 330D/364S, 347G/411Q/437I, 347G/427R/437V, 364S/399D, 374K, 374K/411Q/437I, 374K/411Q/437V, 408D/417P, 411Q/437V, and 437I, wherein the positions are numbered with reference to SEQ ID NO:5708. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from E2P/K4T/E113S/P158D/L163A/E330D/S449R, E2P/K4T/E113S/P158D/L163R/K302R/G364S/N399D/S449R, E2P/K4T/E113S/L163R/S449R,

E2P/K4T/P158D/L163A/G364S/S449R, E2P/K4T/P158D/K222R/L257Q/K302R/E330D, E2P/K4T/P158D/E330D, E2P/K4T/L163A/L257Q, E2P/K4T/L163A/E330D/S449R, E2P/K4T/L163A/G364S, E2P/K4T/L163A/G364S/Y453R, E2P/K4T/L163R/K222R/K302R/E330D/G364S/S449R, E2P/K4T/L163R/L257Q/E330D/N399D/S449R, E2P/K4T/K302R, E2P/E113S, E2P/E113S/L163A, E2P/E113S/L163R/S449R, E2P/E113S/E330D/N399D, E2P/E113S/S449R, E2P/P158D/L163A/G364S/N399D, E2P/P158D/L163R, E2P/L163R/G364S/Y453R, E2P/G364S/S449R, K4T/E113S/P158D/L163A/E330D/G364S/N399D, K4T/E113S/P158D/K302R/E330D/G364S, K4T/E113S/L163A/G364S/N399D, K4T/P158D/L163A/G364S/N399D, K4T/G364S/S449R, M87W, M87W/H95L/D198E, M87W/D198E, M87W/D198I, M87W/W266L, M87W/P322S, L107A/D134S/S135A/N195H/R412H/R417P, L107A/N195H/D268A/P322L/K439P, L107A/N195H/P272K, L107A/N195H/R417P/K439P, L107A/S374T/R417P/K439P, E113S/P158D/L163A/G364S/N399D, E113S/P158D/L163R/G364S/N399D, D134S, D134S/S135A/N195H/D268A/S317R, P158D/L163A/L257Q/E330D/G364S/S449R, P158D/L163A/G364S, P158D/L163R/K302R/E330D/G364S/N399D, L163A/L257Q/K302R/G364S, L163A/G364S, L163R/K302R/E330D/G364S/S449R, R173N, R173N/K190R/W233K/S252E/A427R/L437V, R173N/K190R/S252E/L257A/K347G/A427R/L437I, R173N/K190R/L257A/S374K/L437V, R173N/K190R/L257A/A427R/L437I, R173N/W233K/S252E/L257A/A427R, R173N/W233K/L437I, R173N/S252E, R173N/S252E/D268H/L437V, R173N/S252E/K347G/R411Q, R173N/L257A/S374K/L437V, R173N/S374K/L437I, R173N/A427R/L437I, K190R/W233K/S252E/L257A/K347G/R411Q/L437I, K190R/S252E, K190R/S252E/L257A/E285Q/A427R, K190R/S252E/L257A/R411Q/L437I, K190R/S252E/L257A/A427R, K190R/S374K/A427R/L437V, N195H/P272K/V320I/K439P, N195H/S317R/V320I, D198E, D198E/L292P, D198I, D198I/P244L, W233K/S252E, W233K/S252E/L257A, W233K/S252E/L257A/K347G, W233K/S252E/L257A/K347G/R411Q/L437I, W233K/S252E/L257A/A427R/L437V, W233K/S252E/E285Q, W233K/S252E/E285Q/L437V, W233K/S252E/R411Q/L437V, W233K/S252E/L437V, W233K/S374K, W233K/S374K/L437I, S252E/L257A, S252E/L257A/K347G/L437V, L257A/K347G/R411Q/L437V, D268A/R417P, E285Q/K347G/L437V, K302R/G364S, E330D/G364S, K347G/R411Q/L437I, K347G/A427R/L437V, G364S/N399D, S374K, S374K/R411Q/L437I, S374K/R411Q/L437V, N408D/R417P, R411Q/L437V, and L437I, wherein the positions are numbered with reference to SEQ ID NO:5708. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 5816, 5818, 5820, 5822, 5824, 5826, 5828, 5830, 5832, 5834, 5836, 5838, 5840, 5842, 5844, 5846, 5848, 5850, 5852, 5854, 5856, 5858, 5860, 5862, 5864, 5866, 5868, 5870, 5872, 5874, 5876, 5878, 5880, 5882, 5884, 5886, 5888, 5890, 5892, 5894, 5896, 5898, 5900, 5902, 5904, 5906, 5908, 5910, 5912, 5914, 5916, 5918, 5920, 5922, 5924, 5926, 5928, 5930, 5932, 5934, 5936, 5938, 5940, 5942, 5944, 5946, 5948, 5950, 5952, 5954, 5956, 5958, 5960, 5962, 5964, 5966, 5968, 5970, 5972, 5974, 5976, 5978, 5980, 5982, 5984, 5986, 5988, 5990, 5992, 5994, 5996, 5998, 6000, 6002, 6004, 6006, 6008, 6010, 6012, and 6014. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 5816, 5818, 5820, 5822, 5824, 5826, 5828, 5830, 5832, 5834, 5836, 5838, 5840, 5842, 5844, 5846, 5848, 5850, 5852, 5854, 5856, 5858, 5860, 5862, 5864, 5866, 5868, 5870, 5872, 5874, 5876, 5878, 5880, 5882, 5884, 5886, 5888, 5890, 5892, 5894, 5896, 5898, 5900, 5902, 5904, 5906, 5908, 5910, 5912, 5914, 5916, 5918, 5920, 5922, 5924, 5926, 5928, 5930, 5932, 5934, 5936, 5938, 5940, 5942, 5944, 5946, 5948, 5950, 5952, 5954, 5956, 5958, 5960, 5962, 5964, 5966, 5968, 5970, 5972, 5974, 5976, 5978, 5980, 5982, 5984, 5986, 5988, 5990, 5992, 5994, 5996, 5998, 6000, 6002, 6004, 6006, 6008, 6010, 6012, and 6014. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 5816, 5818, 5820, 5822, 5824, 5826, 5828, 5830, 5832, 5834, 5836, 5838, 5840, 5842, 5844, 5846, 5848, 5850, 5852, 5854, 5856, 5858, 5860, 5862, 5864, 5866, 5868, 5870, 5872, 5874, 5876, 5878, 5880, 5882, 5884, 5886, 5888, 5890, 5892, 5894, 5896, 5898, 5900, 5902, 5904, 5906, 5908, 5910, 5912, 5914, 5916, 5918, 5920, 5922, 5924, 5926, 5928, 5930, 5932, 5934, 5936, 5938, 5940, 5942, 5944, 5946, 5948, 5950, 5952, 5954, 5956, 5958, 5960, 5962, 5964, 5966, 5968, 5970, 5972, 5974, 5976, 5978, 5980, 5982, 5984, 5986, 5988, 5990, 5992, 5994, 5996, 5998, 6000, 6002, 6004, 6006, 6008, 6010, 6012, and 6014.

(25) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 2/4/14/158/282/364/449, 2/4/14/158/282/407, 2/4/14/163/198, 2/4/14/163/282/399, 2/4/14/198/308/407, 2/4/14/282/308/364, 2/4/14/282/308/407, 2/4/14/282/364, 2/4/158/163/364/399/407, 2/4/158/198/282/364/407, 2/4/163/198/282/308/342, 2/4/163/282/342/364, 2/4/163/364/399/449, 2/4/198/282/449, 2/4/198/364/391/449, 2/4/282/342/364/407, 2/14/158/198/407/449, 2/14/163/308/364, 2/14/163/364/407, 2/14/282, 2/14/282/308, 2/14/282/308/364, 2/14/282/399, 2/14/308, 2/14/308/364, 2/158/163/407/449, 2/163/282, 2/163/282/308/364, 2/198/282/308/342, 2/198/282/399, 2/282, 2/282/308/342/407, 2/282/308/391/407, 2/282/399, 2/282/399/407, 2/308, 2/308/364/399, 4/14/158/163/198/282/407, 4/14/158/282/364/391/407, 4/14/163/282, 4/14/163/282/308/342/407/449, 4/14/198/308/364, 4/14/282, 4/14/282/308/364/407/449, 4/14/282/342/399/407, 4/14/364/391, 4/158/282/364/399, 4/163/282/308/407, 4/198/399/407, 4/282/342/364/407, 4/282/364/407, 4/364, 12, 14/158/163/198/364, 14/158/163/282/364, 14/158/198/282/342/364/449, 14/158/364, 14/163/198/282/342/364/449, 14/163/282/308, 14/282, 14/282/308, 14/282/342/364/391, 14/282/364, 14/282/364/391/407/449, 14/282/399/407/449, 14/407, 37, 44, 70, 71/331, 74, 75, 90, 106, 108, 112, 114, 115, 131, 138, 139, 156, 158/282/407, 158/282/407/449, 158/364/399/407/449, 158/364/449, 162, 163/198/282/342, 163/282/308/364/399, 163/282/399/407, 163/282/407/449, 174, 198/282/308/342, 198/282/407, 198/308, 247, 248, 254, 258, 282/308/342/364/449, 282/308/449, 365, 389, 401/402, 416, 427, 429, 432, 433, and 456, wherein the positions are numbered with reference to SEQ ID NO: 5976. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 2P/4T/14I/158D/282S/364A/449R, 2P/4T/14I/158D/282S/407V, 2P/4T/14I/163A/198E, 2P/4T/14I/163A/282S/399D, 2P/4T/14I/198E/308L/407V, 2P/4T/14I/282S/308L/364A, 2P/4T/14I/282S/308L/407V, 2P/4T/14I/282S/364A, 2P/4T/158D/163A/364A/399D/407V, 2P/4T/158D/198E/282S/364A/407V, 2P/4T/163A/198E/282S/308L/342W, 2P/4T/163A/282S/342W/364A, 2P/4T/163A/364A/399D/449R, 2P/4T/198E/282S/449R, 2P/4T/198E/364A/391C/449R, 2P/4T/282S/342W/364A/407V, 2P/14I/158D/198E/407V/449R, 2P/14I/163A/308L/364A, 2P/14I/163A/364A/407V, 2P/14I/282S, 2P/14I/282S/308L, 2P/14I/282S/308L/364A, 2P/14I/282S/399D, 2P/14I/308L, 2P/14I/308L/364A, 2P/158D/163A/407V/449R, 2P/163A/282S, 2P/163A/282S/308L/364A, 2P/198E/282S/308L/342W, 2P/198E/282S/399D, 2P/282S, 2P/282S/308L/342W/407V, 2P/282S/308L/391C/407V, 2P/282S/399D, 2P/282S/399D/407V, 2P/308L, 2P/308L/364A/399D, 4T/14I/158D/163A/198E/282S/407V, 4T/14I/158D/282S/364A/391C/407V, 4T/14I/163A/282S, 4T/14I/163A/282S/308L/342W/407V/449R, 4T/14I/198E/308L/364A, 4T/14I/282S, 4T/14I/282S/308L/364A/407V/449R, 4T/14I/282S/342W/399D/407V, 4T/14I/364A/391C, 4T/158D/282S/364A/399D, 4T/163A/282S/308L/407V, 4T/198E/399D/407V, 4T/282S/342W/364A/407V, 4T/282S/364A/407V, 4T/364A, 12S, 14I/158D/163A/198E/364A, 14I/158D/163A/282S/364A, 14I/158D/198E/282S/342W/364A/449R, 14I/158D/364A, 14I/163A/198E/282S/342W/364A/449R, 14I/163A/282S/308L, 14I/282S, 14I/282S/308L, 14I/282S/342W/364A/391C, 14I/282S/364A, 14I/282S/364A/391C/407V/449R, 14I/282S/399D/407V/449R, 14I/407V, 37R, 44V, 70K, 71L/331K, 74H, 75G, 75M, 90Q, 90S, 90T, 106Y, 108H, 112N, 114P, 115R, 131V, 138V, 139A, 139R, 139S, 156C, 158D/282S/407V, 158D/282S/407V/449R, 158D/364A/399D/407V/449R, 158D/364A/449R, 162A, 163A/198E/282S/342W, 163A/282S/308L/364A/399D, 163A/282S/399D/407V, 163A/282S/407V/449R, 174P, 198E/282S/308L/342W, 198E/282S/407V, 198E/308L, 247C, 247L, 248C, 248L, 248W, 254Q, 258N, 282S/308L/342W/364A/449R, 282S/308L/449R, 365I, 389A, 389E, 389S, 401F/402L, 416L, 416S, 427K, 427R, 429W, 432L, 433L, and 456R, wherein the positions are numbered with reference to SEQ ID NO: 5976. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from E2P/K4T/V14I/P158D/T282S/G364A/S449R, E2P/K4T/V14I/P158D/T282S/E407V, E2P/K4T/V14I/L163A/D198E, E2P/K4T/V14I/L163A/T282S/N399D, E2P/K4T/V14I/D198E/R308L/E407V, E2P/K4T/V14I/T282S/R308L/G364A, E2P/K4T/V14I/T282S/R308L/E407V, E2P/K4T/V14I/T282S/G364A, E2P/K4T/P158D/L163A/G364A/N399D/E407V, E2P/K4T/P158D/D198E/T282S/G364A/E407V,

E2P/K4T/L163A/D198E/T282S/R308L/E342W, E2P/K4T/L163A/T282S/E342W/G364A, E2P/K4T/L163A/G364A/N399D/S449R, E2P/K4T/D198E/T282S/S449R, E2P/K4T/D198E/G364A/L391C/S449R, E2P/K4T/T282S/E342W/G364A/E407V, E2P/V14I/P158D/D198E/E407V/S449R, E2P/V14I/L163A/R308L/G364A, E2P/V14I/L163A/G364A/E407V, E2P/V14I/T282S, E2P/V14I/T282S/R308L, E2P/V14I/T282S/R308L/G364A, E2P/V14I/T282S/N399D, E2P/V14I/R308L, E2P/V14I/R308L/G364A, E2P/P158D/L163A/E407V/S449R, E2P/L163A/T282S, E2P/L163A/T282S/R308L/G364A, E2P/D198E/T282S/R308L/E342W, E2P/D198E/T282S/N399D, E2P/T282S, E2P/T282S/R308L/E342W/E407V, E2P/T282S/R308L/L391C/E407V, E2P/T282S/N399D, E2P/T282S/N399D/E407V, E2P/R308L, E2P/R308L/G364A/N399D, K4T/V14I/P158D/L163A/D198E/T282S/E407V, K4T/V14I/P158D/T282S/G364A/L391C/E407V, K4T/V14I/L163A/T282S, K4T/V14I/L163A/T282S/R308L/E342W/E407V/S449R, K4T/V14I/D198E/R308L/G364A, K4T/V14I/T282S, K4T/V14I/T282S/R308L/G364A/E407V/S449R, K4T/V14I/T282S/E342W/N399D/E407V, K4T/V14I/G364A/L391C, K4T/P158D/T282S/G364A/N399D, K4T/L163A/T282S/R308L/E407V, K4T/D198E/N399D/E407V, K4T/T282S/E342W/G364A/E407V, K4T/T282S/G364A/E407V, K4T/G364A, R12S, V14I/P158D/L163A/D198E/G364A, V14I/P158D/L163A/T282S/G364A, V14I/P158D/D198E/T282S/E342W/G364A/S449R, V14I/P158D/G364A, V14I/L163A/D198E/T282S/E342W/G364A/S449R, V14I/L163A/T282S/R308L, V14I/T282S, V14I/T282S/R308L, V14I/T282S/E342W/G364A/L391C, V14I/T282S/G364A, V14I/T282S/G364A/L391C/E407V/S449R, V14I/T282S/N399D/E407V/S449R, V14I/E407V, S37R, I44V, P70K, Q71L/R331K, W74H, L75G, L75M, P90Q, P90S, P90T, S106Y, M108H, P112N, D114P, E115R, P131V, L138V, P139A, P139R, P139S, S156C, P158D/T282S/E407V, P158D/T282S/E407V/S449R, P158D/G364A/N399D/E407V/S449R, P158D/G364A/S449R, E162A, L163A/D198E/T282S/E342W, L163A/T282S/R308L/G364A/N399D, L163A/T282S/N399D/E407V, L163A/T282S/E407V/S449R, L174P, D198E/T282S/R308L/E342W, D198E/T282S/E407V, D198E/R308L, K247C, K247L, H248C, H248L, H248W, S254Q, D258N, T282S/R308L/E342W/G364A/S449R, T282S/R308L/S449R, V365I, D389A, D389E, D389S, L401F/E402L, D416L, D416S, A427K, A427R, L429W, K432L, A433L, and S456R, wherein the positions are numbered with reference to SEQ ID NO: 5976. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 6016, 6018, 6020, 6022, 6024, 6026, 6028, 6030, 6032, 6034, 6036, 6038, 6040, 6042, 6044, 6046, 6048, 6050, 6052, 6054, 6056, 6058, 6060, 6062, 6064, 6066, 6068, 6070, 6072, 6074, 6076, 6078, 6080, 6082, 6084, 6086, 6088, 6090, 6092, 6094, 6096, 6098, 6100, 6102, 6104, 6106, 6108, 6110, 6112, 6114, 6116, 6118, 6120, 6122, 6124, 6126, 6128, 6130, 6132, 6134, 6136, 6138, 6140, 6142, 6144, 6146, 6148, 6150, 6152, 6154, 6156, 6158, 6160, 6162, 6164, 6166, 6168, 6170, 6172, 6174, 6176, 6178, 6180, 6182, 6184, 6186, 6188, 6190, 6192, 6194, 6196, 6198, 6200, 6202, 6204, 6206, 6208, 6210, 6212, 6214, 6216, 6218, 6220, 6222, 6224, 6226, 6228, 6230, 6232, 6234, 6236, 6238, 6240, 6242, 6244, 6246, 6248, 6250, 6252, 6254, 6256, 6258, and 6260. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 6016, 6018, 6020, 6022, 6024, 6026, 6028, 6030, 6032, 6034, 6036, 6038, 6040, 6042, 6044, 6046, 6048, 6050, 6052, 6054, 6056, 6058, 6060, 6062, 6064, 6066, 6068, 6070, 6072, 6074, 6076, 6078, 6080, 6082, 6084, 6086, 6088, 6090, 6092, 6094, 6096, 6098, 6100, 6102, 6104, 6106, 6108, 6110, 6112, 6114, 6116, 6118, 6120, 6122, 6124, 6126, 6128, 6130, 6132, 6134, 6136, 6138, 6140, 6142, 6144, 6146, 6148, 6150, 6152, 6154, 6156, 6158, 6160, 6162, 6164, 6166, 6168, 6170, 6172, 6174, 6176, 6178, 6180, 6182, 6184, 6186, 6188, 6190, 6192, 6194, 6196, 6198, 6200, 6202, 6204, 6206, 6208, 6210, 6212, 6214, 6216, 6218, 6220, 6222, 6224, 6226, 6228, 6230, 6232, 6234, 6236, 6238, 6240, 6242, 6244, 6246, 6248, 6250, 6252, 6254, 6256, 6258, and 6260.

(26) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 14/113/158, 14/113/158/163, 14/113/158/163/190/257/308/342, 14/113/158/163/437, 14/113/158/308/317/330/449, 14/113/163/190/308/317, 14/113/163/257/308/342/437/449, 14/113/163/257/437, 14/113/163/308/317/320/322/330/449, 14/113/320/437/449, 14/158, 14/158/163/190/257/308/317/330/437, 14/158/163/190/342/437,

14/158/163/257, 14/158/163/257/308/437/449, 14/158/317/320/330/437, 14/163/317/320, 14/190, 14/190/257/317/320/322, 14/257/308/320/322/330, 14/257/308/322/330/437, 14/317, 14/330, 14/449, 19, 29/375, 41, 45, 46, 71, 72, 80, 81, 83, 84, 85, 88, 95, 105, 113/158/163/190/257/437, 113/158/163/190/308/317/322, 113/158/163/190/308/320, 113/158/163/257/308/317/322/437/449, 113/158/190/257/320, 113/158/190/320/322/449, 113/158/320/322/437, 113/257/308, 113/257/317/322/437, 155, 158/163, 158/163/190/257/308/342/449, 158/163/308, 158/163/322/437, 158/308/320/437, 158/320/437/449, 163/308/330/437/449, 168, 190/449, 197, 199, 202, 209, 243, 249, 263, 273, 317/320/322/330, 322/330, 366, 375, and 383, wherein the positions are numbered with reference to SEQ ID NO: 6138. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 14I/113S/158P, 14I/113S/158P/163A, 14I/113S/158P/163A/190R/257Q/308L/342W, 14I/113S/158P/163A/437I, 14I/113S/158P/308L/317R/330D/449R, 14I/113S/163A/190R/308L/317R, 14I/113S/163A/257Q/308L/342W/437I/449R, 14I/113S/163A/257Q/437I, 14I/113S/163A/308L/317R/320I/322L/330D/449R, 14I/113S/320I/437I/449R, 14I/158P, 14I/158P/163A/190R/257Q/308L/317R/330D/437I, 14I/158P/163A/190R/342W/437I, 14I/158P/163A/257Q, 14I/158P/163A/257Q/308L/449R, 14I/158P/257Q/308L/437I/449R, 14I/158P/317R/320I/330D/437I, 14I/163A/317R/320I, 14I/190R, 14I/190R/257Q/317R/320I/322L, 14I/257Q/308L/320I/322L/330D, 14I/257Q/308L/322L/330D/437I, 14I/317R, 14I/330D, 14I/449R, 19Q, 29M/375P, 41A, 45L, 46S, 71V, 72S, 72T, 80P, 81T, 83A, 83K, 83N, 83S, 83T, 84D, 84H, 84N, 85L, 88A, 88C, 88H, 88K, 88T, 95N, 105A, 113S/158P/163A/190R/257Q/437I, 113S/158P/163A/190R/308L/317R/322L, 113S/158P/163A/190R/308L/320I, 113S/158P/163A/257Q/308L/317R/322L/437I/449R, 113S/158P/190R/257Q/320I, 113S/158P/190R/320I/322L/449R, 113S/158P/320I/322L/437I, 113S/257Q/308L, 113S/257Q/317R/322L/437I, 155L, 158P/163A, 158P/163A/190R/257Q/308L/342W/449R, 158P/163A/308L, 158P/163A/322L/437I, 158P/308L/320I/437I, 158P/320I/437I/449R, 163A/308L/330D/437I/449R, 168C, 168T, 190R/449R, 197K, 199E, 199H, 199M, 199Q, 199Y, 202H, 202Q, 202T, 202V, 209T, 243I, 249S, 263T, 273A, 273H, 273R, 317R/320I/322L/330D, 322L/330D, 366C, 366L, 366S, 366T, 366V, 375A, 375P, 375T, 375V, and 383V, wherein the positions are numbered with reference to SEQ ID NO: 6138. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from V14I/E113S/D158P, V14I/E113S/D158P/L163A, V14I/E113S/D158P/L163A/K190R/L257Q/R308L/E342W, V14I/E113S/D158P/L163A/L437I, V14I/E113S/D158P/R308L/S317R/E330D/S449R, V14I/E113S/L163A/K190R/R308L/S317R, V14I/E113S/L163A/L257Q/R308L/E342W/L437I/S449R, V14I/E113S/L163A/L257Q/L437I, V14I/E113S/L163A/R308L/S317R/V320I/P322L/E330D/S449R, V14I/E113S/V320I/L437I/S449R, V14I/D158P, V14I/D158P/L163A/K190R/L257Q/R308L/S317R/E330D/L437I, V14I/D158P/L163A/K190R/E342W/L437I, V14I/D158P/L163A/L257Q, V14I/D158P/L163A/L257Q/R308L/S449R, V14I/D158P/L257Q/R308L/L437I/S449R, V14I/D158P/S317R/V320I/E330D/L437I, V14I/L163A/S317R/V320I, V14I/K190R, V14I/K190R/L257Q/S317R/V320I/P322L, V14I/L257Q/R308L/V320I/P322L/E330D, V14I/L257Q/R308L/P322L/E330D/L437I, V14I/S317R, V14I/E330D, V14I/S449R, V19Q, L29M/D375P, S41A, F45L, H46S, Q71V, D72S, D72T, T80P, H81T, P83A, P83K, P83N, P83S, P83T, G84D, G84H, G84N, A85L, R88A, R88C, R88H, R88K, R88T, H95N, E105A, E113S/D158P/L163A/K190R/L257Q/L437I, E113S/D158P/L163A/K190R/R308L/S317R/P322L, E113S/D158P/L163A/K190R/R308L/V320I, E113S/D158P/L163A/L257Q/R308L/S317R/P322L/L437I/S449R, E113S/D158P/K190R/L257Q/V320I, E113S/D158P/K190R/V320I/P322L/S449R, E113S/D158P/V320I/P322L/L437I, E113S/L257Q/R308L, E113S/L257Q/S317R/P322L/L437I, V155L, D158P/L163A, D158P/L163A/K190R/L257Q/R308L/E342W/S449R, D158P/L163A/R308L, D158P/L163A/P322L/L437I, D158P/R308L/V320I/L437I, D158P/V320I/L437I/S449R, L163A/R308L/E330D/L437I/S449R, P168C, P168T, K190R/S449R, Q197K, A199E, A199H, A199M, A199Q, A199Y, I202H, I202Q, I202T, I202V, E209T, L243I, Y249S, V263T, S273A, S273H, S273R,

317R/V320L/P320L/E330D, A366L, A366L, A366S, A366T, D375A, D375P, D375T, D375V, and N383V, wherein the positions are numbered with reference to SEQ ID NO: 6138.

(27) In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 6262, 6264, 6266, 6268, 6270, 6272, 6274, 6276, 6278, 6280, 6282, 6284, 6286, 6288, 6290, 6292, 6294, 6296, 6298, 6300, 6302, 6304, 6306, 6308, 6310, 6312, 6314, 6316, 6318, 6320, 6322, 6324, 6326, 6328, 6330, 6332, 6334, 6336, 6338, 6340, 6342, 6344, 6346, 6348, 6350, 6352, 6354, 6356, 6358, 6360, 6362, 6364, 6366, 6368, 6370, 6372, 6374, 6376, 6378, 6380, 6382, 6384, 6386, 6388, 6390, 6392, 6394, 6396, 6398, 6400, 6402, 6404, 6406, 6408, 6410, 6412, 6414, 6416, 6418, 6420, 6422, 6424, 6426, 6428, 6430, 6432, 6434, 6436, 6438, 6440, 6442, 6444, 6446, 6448, 6450, 6452, 6454, 6456, 6458, and 6460. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 6262, 6264, 6266, 6268, 6270, 6272, 6274, 6276, 6278, 6280, 6282, 6284, 6286, 6288, 6290, 6292, 6294, 6296, 6298, 6300, 6302, 6304, 6306, 6308, 6310, 6312, 6314, 6316, 6318, 6320, 6322, 6324, 6326, 6328, 6330, 6332, 6334, 6336, 6338, 6340, 6342, 6344, 6346, 6348, 6350, 6352, 6354, 6356, 6358, 6360, 6362, 6364, 6366, 6368, 6370, 6372, 6374, 6376, 6378, 6380, 6382, 6384, 6386, 6388, 6390, 6392, 6394, 6396, 6398, 6400, 6402, 6404, 6406, 6408, 6410, 6412, 6414, 6416, 6418, 6420, 6422, 6424, 6426, 6428, 6430, 6432, 6434, 6436, 6438, 6440, 6442, 6444, 6446, 6448, 6450, 6452, 6454, 6456, 6458, and 6460. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 6262, 6264, 6266, 6268, 6270, 6272, 6274, 6276, 6278, 6280, 6282, 6284, 6286, 6288, 6290, 6292, 6294, 6296, 6298, 6300, 6302, 6304, 6306, 6308, 6310, 6312, 6314, 6316, 6318, 6320, 6322, 6324, 6326, 6328, 6330, 6332, 6334, 6336, 6338, 6340, 6342, 6344, 6346, 6348, 6350, 6352, 6354, 6356, 6358, 6360, 6362, 6364, 6366, 6368, 6370, 6372, 6374, 6376, 6378, 6380, 6382, 6384, 6386, 6388, 6390, 6392, 6394, 6396, 6398, 6400, 6402, 6404, 6406, 6408, 6410, 6412, 6414, 6416, 6418, 6420, 6422, 6424, 6426, 6428, 6430, 6432, 6434, 6436, 6438, 6440, 6442, 6444, 6446, 6448, 6450, 6452, 6454, 6456, 6458, and 6460.

(28) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 37/71/125/174/247/248/427, 37/71/247/331/365/389/401/429, 37/139/329/365/427/433/449, 37/139/427/432/433/449, 37/139/429/432/433, 37/174/401/402/433/449, 37/248/331/389/427/433, 37/248/389/401/429, 37/331/432/433, 71/139/389/427/429/432/433/449, 71/174/329/427/429/432/433, 71/174/365/427/432, 71/248/365/389/401, 71/248/449, 71/254/433/449, 71/427/432/433/449, 71/432/433, 139/156/174/389/401/427/433/449, 139/156/247/365/401/433/449, 139/156/247/389/401/427/433/449, 139/156/248/389, 139/156/248/389/401/416/427/429/433, 139/156/365, 139/174/248/331/389/401/449, 139/174/254, 139/174/365/401/402/427/433/449, 139/247/248/331/401/417/432/449, 139/248/254/449, 139/248/402/416/427/433/449, 139/248/432/433, 139/254/401/416/427/433/449, 139/401/449, 156/248/256, 156/254/331/365/427/432/449, 156/389/401/402/416/432/433, 174/247/248/389/401/432/433, 174/329/432/449, 174/365, 174/389/429/432/433/449, 247/248/401, 247/248/449, 247/331/401/427/432/449, 247/401, 247/427/432, 248/331/427/429/433, 248/365/389/427/429/432/449, 248/401/429/432/433, 248/416, 248/416/449, 248/449, 254/365, 254/427/433, 331/365/429/432/433/449, 365/401/402/429/432/433, 389/401/416/432/449, 389/401/427/432, 401/427, 416/427/433/449, 416/432/433, 416/432/433/449, 427/432, 427/432/449, 432, and 433/449, wherein the positions are numbered with reference to SEQ ID NO: 6288. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 37R/71L/125M/174P/247C/248W/427R, 37R/71L/247C/331K/365I/389E/401F/429W, 37R/139K/329P/365I/427R/433L/449R, 37R/139K/427R/432L/433L/449R, 37R/139K/429W/432L/433L, 37R/174P/401F/402L/433L/449R, 37R/248W/331K/389E/427R/433L, 37R/248W/389A/401F/429W, 37R/331K/432L/433L, 71L/139K/389A/427R/429W/432L/433L/449R, 71L/174P/329P/427R/429W/432L/433L, 71L/174P/365I/427R/432L, 71L/248W/365I/389A/401F, 71L/248W/449R, 71L/254Q/433L/449R, 71L/427R/432L/433L/449R, 71L/432L/433L, 139K/156C/174P/389E/401F/427R/433L/449R, 139K/156C/247C/365I/401F/433L/449R, 139K/156C/247L/389E/401F/427R/433L/449R, 139K/156C/248W/389A/401F/416S/427R/429W/433L, 139K/156C/248W/389E, 139K/156C/365I, 139K/174P/248W/331K/389A/401F/449R, 139K/174P/254Q, 139K/174P/365I/401F/402L/427R/433L/449R, 139K/247C/248W/331K/401F/417Q/432L/449R,

139K/248W/254A/449R, 139K/248W/402L/416L/427R/433L/449R, 139K/248W/432L/433L, 139K/254Q/401F/416S/427R/433L/449R, 139K/401F/449R, 156C/248W/256M, 156C/254Q/331K/365I/427R/432L/449R, 156C/389E/401F/402L/416L/432L/433L, 174P/247C/248W/389A/401F/432L/433L, 174P/329P/432L/449R, 174P/365I, 174P/389E/429W/432L/433L/449R, 247C/248W/401F, 247C/331K/401F/427R/432L/449R, 247C/427R/432L, 247L/248W/449R, 247L/401F, 248W/331K/427R/429W/433L, 248W/365I/389E/427R/429W/432L/449R, 248W/401F/429W/432L/433L, 248W/416L, 248W/416L/449R, 248W/449R, 254Q/365I, 254Q/427R/433L, 331K/365I/429W/432L/433L/449R, 365I/401F/402L/429W/432L/433L, 389A/401F/416S/432L/449R, 389E/401F/427R/432L, 401F/427R, 416L/427R/433L/449R, 416L/432L/433L/449R, 416S/432L/433L, 427R/432L, 427R/432L/449R, 432L, and 433L/449R, wherein the positions are numbered with reference to SEQ ID NO: 6288. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from S37R/Q71L/L125M/L174P/K247C/H248W/A427R, S37R/Q71L/K247C/R331K/V365I/D389E/L401F/L429W, S37R/P139K/G329P/V365I/A427R/A433L/S449R, S37R/P139K/A427R/K432L/A433L/S449R, S37R/P139K/L429W/K432L/A433L, S37R/L174P/L401F/E402L/A433L/S449R, S37R/H248W/R331K/D389E/A427R/A433L, S37R/H248W/D389A/L401F/L429W, S37R/R331K/K432L/A433L, Q71L/P139K/D389A/A427R/L429W/K432L/A433L/S449R, Q71L/L174P/G329P/A427R/L429W/K432L/A433L, Q71L/L174P/V365I/A427R/K432L, Q71L/H248W/V365I/D389A/L401F, Q71L/H248W/S449R, Q71L/S254Q/A433L/S449R, Q71L/A427R/K432L/A433L/S449R, Q71L/K432L/A433L, P139K/S156C/L174P/D389E/L401F/A427R/A433L/S449R, P139K/S156C/K247C/V365I/L401F/A433L/S449R, P139K/S156C/K247L/D389E/L401F/A427R/A433L/S449R, P139K/S156C/H248W/D389A/L401F/D416S/A427R/L429W/A433L, P139K/S156C/H248W/D389E, P139K/S156C/V365I, P139K/L174P/H248W/R331K/D389A/L401F/S449R, P139K/L174P/S254Q, P139K/L174P/V365I/L401F/E402L/A427R/A433L/S449R, P139K/K247C/H248W/R331K/L401F/P417Q/K432L/S449R, P139K/H248W/S254A/S449R, P139K/H248W/E402L/D416L/A427R/A433L/S449R, P139K/H248W/K432L/A433L, P139K/S254Q/L401F/D416S/A427R/A433L/S449R, P139K/L401F/S449R, S156C/H248W/L256M, S156C/S254Q/R331K/V365I/A427R/K432L/S449R, S156C/D389E/L401F/E402L/D416L/K432L/A433L, L174P/K247C/H248W/D389A/L401F/K432L/A433L, L174P/G329P/K432L/S449R, L174P/V365I, L174P/D389E/L429W/K432L/A433L/S449R, K247C/H248W/L401F, K247C/R331K/L401F/A427R/K432L/S449R, K247C/A427R/K432L, K247L/H248W/S449R, K247L/L401F, H248W/R331K/A427R/L429W/A433L, H248W/V365I/D389E/A427R/L429W/K432L/S449R, H248W/L401F/L429W/K432L/A433L, H248W/D416L, H248W/D416L/S449R, H248W/S449R, S254Q/V365I, S254Q/A427R/A433L, R331K/V365I/L429W/K432L/A433L/S449R, V365I/L401F/E402L/L429W/K432L/A433L, D389A/L401F/D416S/K432L/S449R, D389E/L401F/A427R/K432L, L401F/A427R, D416L/A427R/A433L/S449R, D416L/K432L/A433L/S449R, D416S/K432L/A433L, A427R/K432L, A427R/K432L/S449R, K432L, and A433L/S449R, wherein the positions are numbered with reference to SEQ ID NO: 6288. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 5/449, 6, 10, 25, 25/449, 69, 69/449, 87, 87/449, 91, 91/449, 144/449, 153, 153/449, 159, 159/449, 172, 172/449, 212/449, 233, 233/449, 288/449, 303, 317, 347/449, 361, 369, and 421, wherein the positions are numbered with reference to SEQ ID NO: 6288. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 5S/449R, 6P, 10K, 25I/449R, 25M/449R, 25Q/449R, 25S, 69A, 69M/449R, 87A/449R, 87E, 87K, 87Q, 87R, 91L, 91N/449R, 91Q, 91T/449R, 91V, 144Q/449R, 153T/449R, 153V, 159K, 159R/449R, 172S, 172T/449R, 212L/449R, 233A, 233C, 233G, 233L/449R, 233M/449R, 233Q/449R, 233R, 233S, 233V, 288P/449R, 303C, 303V, 317Y, 347P/449R, 361C, 369K, and 421I,

wherein the positions are numbered with reference to SEQ ID NO: 6288. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from T5S/S449R, E6P, R10K, L25I/S449R, L25M/S449R, L25Q/S449R, L25S, D69A, D69M/S449R, M87A/S449R, M87E, M87K, M87Q, M87R, I91L, I91N/S449R, I91Q, I91T/S449R, I91V, M144Q/S449R, A153T/S449R, A153V, Q159K, Q159R/S449R, H172S, H172T/S449R, A212L/S449R, W233A, W233C, W233G, W233L/S449R, W233M/S449R, W233Q/S449R, W233R, W233S, W233V, E288P/S449R, Q303C, Q303V, S317Y, K347P/S449R, T361C, V369K, and V421I, wherein the positions are numbered with reference to SEQ ID NO: 6288. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 6462, 6464, 6466, 6468, 6470, 6472, 6474, 6476, 6478, 6480, 6482, 6484, 6486, 6488, 6490, 6492, 6494, 6496, 6498, 6500, 6502, 6504, 6506, 6508, 6510, 6512, 6514, 6516, 6518, 6520, 6522, 6524, 6526, 6528, 6530, 6532, 6534, 6536, 6538, 6540, 6542, 6544, 6546, 6548, 6550, 6552, 6554, 6556, 6558, 6560, 6562, 6564, 6566, 6568, 6570, 6572, 6574, 6576, 6578, 6580, 6582, 6584, 6586, 6588, 6590, 6592, 6594, 6596, 6598, 6600, 6602, 6604, 6606, 6608, 6610, 6612, 6614, 6616, 6618, 6620, 6622, 6624, 6626, 6628, 6630, 6632, 6634, 6636, 6638, 6640, 6642, 6644, 6646, 6648, 6650, 6652, 6654, 6656, 6658, 6660, 6662, 6664, 6666, 6668, 6670, 6672, 6674, and 6676. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 6462, 6464, 6466, 6468, 6470, 6472, 6474, 6476, 6478, 6480, 6482, 6484, 6486, 6488, 6490, 6492, 6494, 6496, 6498, 6500, 6502, 6504, 6506, 6508, 6510, 6512, 6514, 6516, 6518, 6520, 6522, 6524, 6526, 6528, 6530, 6532, 6534, 6536, 6538, 6540, 6542, 6544, 6546, 6548, 6550, 6552, 6554, 6556, 6558, 6560, 6562, 6564, 6566, 6568, 6570, 6572, 6574, 6576, 6578, 6580, 6582, 6584, 6586, 6588, 6590, 6592, 6594, 6596, 6598, 6600, 6602, 6604, 6606, 6608, 6610, 6612, 6614, 6616, 6618, 6620, 6622, 6624, 6626, 6628, 6630, 6632, 6634, 6636, 6638, 6640, 6642, 6644, 6646, 6648, 6650, 6652, 6654, 6656, 6658, 6660, 6662, 6664, 6666, 6668, 6670, 6672, 6674, and 6676. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 6462, 6464, 6466, 6468, 6470, 6472, 6474, 6476, 6478, 6480, 6482, 6484, 6486, 6488, 6490, 6492, 6494, 6496, 6498, 6500, 6502, 6504, 6506, 6508, 6510, 6512, 6514, 6516, 6518, 6520, 6522, 6524, 6526, 6528, 6530, 6532, 6534, 6536, 6538, 6540, 6542, 6544, 6546, 6548, 6550, 6552, 6554, 6556, 6558, 6560, 6562, 6564, 6566, 6568, 6570, 6572, 6574, 6576, 6578, 6580, 6582, 6584, 6586, 6588, 6590, 6592, 6594, 6596, 6598, 6600, 6602, 6604, 6606, 6608, 6610, 6612, 6614, 6616, 6618, 6620, 6622, 6624, 6626, 6628, 6630, 6632, 6634, 6636, 6638, 6640, 6642, 6644, 6646, 6648, 6650, 6652, 6654, 6656, 6658, 6660, 6662, 6664, 6666, 6668, 6670, 6672, 6674, and 6676.

(29) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 37/41/45/72/81/83/84/88/155/163/168/190/331/375, 37/41/45/72/81/83/84/88/155/263/273/331/375/432, 37/41/45/72/81/83/84/88/155/331/366/375, 37/41/45/72/81/83/84/88/155/331/375, 37/41/45/72/81/83/84/88/248/375, 37/41/45/72/81/83/84/88/331/366/432/433, 37/41/45/72/88/155/168/190/331/366/375/432, 37/41/45/72/88/155/168/190/331/375, 37/41/45/72/88/190/263/273/331/366, 37/41/45/72/155/163/168/190/243/263/273/331/366/432, 37/41/45/72/155/163/168/243/248/273/331/366/432/433, 37/41/45/72/155/163/168/263/366/432/433, 37/41/45/72/155/163/331/366/375, 37/41/45/72/155/190/243/248/273/331/432/433, 37/41/45/72/155/190/248/263/331, 37/41/45/72/155/190/263/331/366, 37/41/45/72/155/190/273/331/366/375, 37/41/45/72/155/190/273/366, 37/41/45/72/155/190/273/366/375/432, 37/41/45/72/155/248/263/273/366, 37/41/45/72/155/263/331/375, 37/41/45/72/155/263/375, 37/41/45/72/155/273/366, 37/41/45/72/155/331/366/375/432/433, 37/41/45/72/190/263/273/331, 37/41/45/72/190/331/366, 37/41/45/72/190/366, 37/41/45/72/263/331/366, 37/41/45/72/263/366/432/433, 37/41/45/72/331/366, 37/41/45/81/83/84/88/155/168/190/243/331/366, 37/41/45/81/83/84/88/155/168/331/375, 37/41/45/81/83/84/88/155/263/273/331/366, 37/41/45/81/83/84/88/163/168/263/273/331/366/375, 37/41/45/155/163/168/263/331/375, 37/41/45/155/168/248/273/331/375, 37/41/45/155/190/331/366/375, 37/41/45/155/366, 37/41/45/155/366/432/433, 37/41/45/155/375, 37/41/45/243/248/273/331, 37/41/45/263/331/375/432, 37/41/45/331/366/432/433, 37/72/81/83/88/155/190, 37/72/197/273/331/375/432,

37/83/263/366/366/375, 37/190/202, 41/45/72/155/263/331/366/375/432/433,
41/45/163/168/243/248/273/366/432, 45/72/84/88/197/375, 45/72/88/366, 45/72/163/202/365/366/375,
45/72/168/243/331/365/366/429/432, 45/84/168/190/199/254/273/365/366,
45/163/168/190/199/366/429/432, 45/163/168/197/263/331/365/366,
72/81/83/84/88/155/163/168/190/366, 72/81/83/84/88/155/190,
72/81/83/84/88/155/190/273/331/366/432, 72/81/83/84/88/155/273/331/375, 72/81/83/84/88/155/366,
72/81/83/84/88/163/168/190/243/263/331/366, 72/81/83/84/88/163/168/263/331/375, 72/81/84/190/248,
72/83/84/88/202/254/273/366/375, 72/83/84/197/202/243/263/365/366, 72/83/88/243/263/331/365/366,
72/155/163/168/190/366/432/433, 72/155/190, 72/155/190/263/331/366, 72/155/190/366,
72/155/273/331/375/432, 72/243/248/263/366/432/433, 72/243/248/273/366/432/433,
72/243/248/366/432/433, 72/248, 81/83/84/88/155/163/168/273/331/375,
81/83/84/88/155/190/263/331/366, 81/83/84/88/155/263/366/375, 81/83/84/88/155/366,
81/83/84/88/190/263/375, 81/83/84/88/263/273/331/366/432, 81/83/169/190/263,
81/83/190/263/365/366, 81/83/202/365/366/402, 81/88/375/402, 83/88/155/273/366/375,
84/155/168/197/199/331/366/375/383/402, 84/168/197/202/263/366, 84/197/366/402,
155/168/190/197/199/366, 155/168/375, 155/263/366/432/433, 190/199/202/331/366, 197/199/202,
197/202/248, 197/248, 199/263/331/365/366, 248/375, 365/366, and 365/375/402, wherein the positions
are numbered with reference to SEQ ID NO: 6468. In some embodiments, the polypeptide sequence of
the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more
positions selected from 37R/41A/45L/72T/81T/83S/84N/88T/155L/163A/168T/190R/331K/375P,
37R/41A/45L/72T/81T/83S/84N/88T/155L/263T/273R/331K/375P/432L,
37R/41A/45L/72T/81T/83S/84N/88T/155L/331K/366C/375P,
37R/41A/45L/72T/81T/83S/84N/88T/155L/331K/375P,
37R/41A/45L/72T/81T/83S/84N/88T/248W/375P,
37R/41A/45L/72T/81T/83S/84N/88T/331K/366V/432L/433A,
37R/41A/45L/72T/88T/155L/168T/190R/331K/366C/375P/432L,
37R/41A/45L/72T/88T/155L/168T/190R/331K/375P,
37R/41A/45L/72T/88T/190R/263T/273R/331K/366C,
37R/41A/45L/72T/155L/163A/168T/190R/243I/263T/273R/331K/366C/432L,
37R/41A/45L/72T/155L/163A/168T/243I/248W/273R/331K/366V/432L/433A,
37R/41A/45L/72T/155L/163A/168T/263T/366C/432L/433A,
37R/41A/45L/72T/155L/163A/331K/366C/375P,
37R/41A/45L/72T/155L/190R/243I/248W/273R/331K/432L/433A,
37R/41A/45L/72T/155L/190R/248W/263T/331K, 37R/41A/45L/72T/155L/190R/263T/331K/366C,
37R/41A/45L/72T/155L/190R/273R/331K/366V/375P, 37R/41A/45L/72T/155L/190R/273R/366C,
37R/41A/45L/72T/155L/190R/273R/366C/375P/432L, 37R/41A/45L/72T/155L/248W/263T/273R/366C,
37R/41A/45L/72T/155L/263T/331K/375P, 37R/41A/45L/72T/155L/263T/375P,
37R/41A/45L/72T/155L/273R/366C, 37R/41A/45L/72T/155L/331K/366V/375P/432L/433A,
37R/41A/45L/72T/190R/263T/273R/331K, 37R/41A/45L/72T/190R/331K/366C,
37R/41A/45L/72T/190R/366C, 37R/41A/45L/72T/263T/331K/366C,
37R/41A/45L/72T/263T/366C/432L/433A, 37R/41A/45L/72T/331K/366C,
37R/41A/45L/81T/83S/84N/88T/155L/168T/190R/243I/331K/366C,
37R/41A/45L/81T/83S/84N/88T/155L/168T/331K/375P,
37R/41A/45L/81T/83S/84N/88T/155L/263T/273R/331K/366C,
37R/41A/45L/81T/83S/84N/88T/163A/168T/263T/273R/331K/366C/375P,
37R/41A/45L/155L/163A/168T/263T/331K/375P, 37R/41A/45L/155L/168T/248W/273R/331K/375P,
37R/41A/45L/155L/190R/331K/366V/375P, 37R/41A/45L/155L/366C,
37R/41A/45L/155L/366C/432L/433A, 37R/41A/45L/155L/375P, 37R/41A/45L/243I/248W/273R/331K,
37R/41A/45L/263T/331K/375P/432L, 37R/41A/45L/331K/366V/432L/433A,
37R/72T/81T/83S/88T/155L/190R, 37R/72T/197K/273R/331K/375P/432L,
37R/83S/263T/365I/366V/375P, 37R/190R/202H,
41A/45L/72T/155L/263T/331K/366V/375P/432L/433A,
41A/45L/163A/168T/243I/248W/273R/366C/432L, 45L/72T/84N/88T/197K/375P, 45L/72T/88T/366C,

45L/72T/163A/202H/366V/375P, 45L/72T/168T/243I/331K/365I/366C/429W/432L,
45L/84N/168T/190R/199Q/254A/273R/365I/366C, 45L/163A/168T/190R/199Q/366C/429W/432L,
45L/163A/168T/197K/263T/331K/365I/366C, 72T/81T/83S/84N/88T/155L/163A/168T/190R/366C,
72T/81T/83S/84N/88T/155L/190R, 72T/81T/83S/84N/88T/155L/190R/273R/331K/366V/432L,
72T/81T/83S/84N/88T/155L/273R/331K/375P, 72T/81T/83S/84N/88T/155L/366C,
72T/81T/83S/84N/88T/163A/168T/190R/243I/263T/331K/366C,
72T/81T/83S/84N/88T/163A/168T/263T/331K/375P, 72T/81T/84N/190R/248W,
72T/83S/84N/88T/202H/254A/273R/366C/375P, 72T/83S/84N/197K/202H/243I/263T/365I/366C,
72T/83S/88T/243I/263T/331K/365I/366C, 72T/155L/163A/168T/190R/366V/432L/433A,
72T/155L/190R, 72T/155L/190R/263T/331K/366C, 72T/155L/190R/366C,
72T/155L/273R/331K/375P/432L, 72T/243I/248W/263T/366V/432L/433A,
72T/243I/248W/273R/366V/432L/433A, 72T/243I/248W/366V/432L/433A, 72T/248W,
81T/83S/84N/88T/155L/163A/168T/273R/331K/375P, 81T/83S/84N/88T/155L/190R/263T/331K/366C,
81T/83S/84N/88T/155L/263T/366V/375P, 81T/83S/84N/88T/155L/366C,
81T/83S/84N/88T/190R/263T/375P, 81T/83S/84N/88T/263T/273R/331K/366C/432L,
81T/83S/169D/190R/263T, 81T/83S/190R/263T/365I/366C, 81T/83S/202H/365I/366C/402L,
81T/88T/375P/402L, 83S/88T/155L/273R/366V/375P,
84N/155L/168T/197K/199Q/331K/366V/375P/383V/402L, 84N/168T/197K/202H/263T/366C,
84N/197K/366C/402L, 155L/168T/190R/197K/199Q/366C, 155L/168T/375P,
155L/263T/366C/432L/433A, 190R/199Q/202H/331K/366C, 197K/199Q/202H, 197K/202H/248W,
197K/248W, 199Q/263T/331K/365I/366C, 248W/375P, 365I/366C, and 365I/375P/402L, wherein the
positions are numbered with reference to SEQ ID NO: 6468. In some embodiments, the polypeptide
sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or
more positions selected from

S37R/541A/F45L/D72T/H81T/P83S/G84N/R88T/V155L/L163A/P168T/K190R/R331K/D375P,
S37R/541A/F45L/D72T/H81T/P83S/G84N/R88T/V155L/V263T/S273R/R331K/D375P/K432L,
S37R/541A/F45L/D72T/H81T/P83S/G84N/R88T/V155L/R331K/A366C/D375P,
S37R/541A/F45L/D72T/H81T/P83S/G84N/R88T/V155L/R331K/D375P,
S37R/S41A/F45L/D72T/H81T/P83S/G84N/R88T/H248W/D375P,
S37R/S41A/F45L/D72T/H81T/P83S/G84N/R88T/R331K/A366V/K432L/L433A,
S37R/541A/F45L/D72T/R88T/V155L/P168T/K190R/R331K/A366C/D375P/K432L,
S37R/541A/F45L/D72T/R88T/V155L/P168T/K190R/R331K/D375P,
S37R/541A/F45L/D72T/R88T/K190R/V263T/S273R/R331K/A366C,
S37R/541A/F45L/D72T/V155L/L163A/P168T/K190R/L243I/V263T/5273R/R331K/A366C/K432L,
S37R/541A/F45L/D72T/V155L/L163A/P168T/L243I/H248W/5273R/R331K/A366V/K432L/L433A,
S37R/S41A/F45L/D72T/V155L/L163A/P168T/V263T/A366C/K432L/L433A,
S37R/541A/F45L/D72T/V155L/L163A/R331K/A366C/D375P,
S37R/541A/F45L/D72T/V155L/K190R/L243I/H248W/5273R/R331K/K432L/L433A,
S37R/S41A/F45L/D72T/V155L/K190R/H248W/V263T/R331K,
S37R/541A/F45L/D72T/V155L/K190R/V263T/R331K/A366C,
S37R/S41A/F45L/D72T/V155L/K190R/S273R/R331K/A366V/D375P,
S37R/S41A/F45L/D72T/V155L/K190R/S273R/A366C,
S37R/S41A/F45L/D72T/V155L/K190R/S273R/A366C/D375P/K432L,
S37R/S41A/F45L/D72T/V155L/H248W/V263T/S273R/A366C,
S37R/541A/F45L/D72T/V155L/V263T/R331K/D375P, S37R/S41A/F45L/D72T/V155L/V263T/D375P,
S37R/S41A/F45L/D72T/V155L/S273R/A366C,
S37R/S41A/F45L/D72T/V155L/R331K/A366V/D375P/K432L/L433A,
S37R/S41A/F45L/D72T/K190R/V263T/S273R/R331K, S37R/S41A/F45L/D72T/K190R/R331K/A366C,
S37R/S41A/F45L/D72T/K190R/A366C, S37R/S41A/F45L/D72T/V263T/R331K/A366C,
S37R/S41A/F45L/D72T/V263T/A366C/K432L/L433A, S37R/S41A/F45L/D72T/R331K/A366C,
S37R/S41A/F45L/H81T/P83S/G84N/R88T/V155L/P168T/K190R/L243I/R331K/A366C,
S37R/S41A/F45L/H81T/P83S/G84N/R88T/V155L/P168T/R331K/D375P,
S37R/S41A/F45L/H81T/P83S/G84N/R88T/V155L/V263T/S273R/R331K/A366C,

S37R/S41A/F45L/H81T/P83S/G84N/R88T/L163A/P168T/V263T/S273R/R331K/A366C/D375P,
S37R/S41A/F45L/V155L/L163A/P168T/V263T/R331K/D375P,
S37R/S41A/F45L/V155L/P168T/H248W/S273R/R331K/D375P,
S37R/S41A/F45L/V155L/K190R/R331K/A366V/D375P, S37R/S41A/F45L/V155L/A366C,
S37R/S41A/F45L/V155L/A366C/K432L/L433A, S37R/S41A/F45L/V155L/D375P,
S37R/S41A/F45L/L243I/H248W/S273R/R331K, S37R/S41A/F45L/V263T/R331K/D375P/K432L,
S37R/S41A/F45L/R331K/A366V/K432L/L433A, S37R/D72T/H81T/P83S/R88T/V155L/K190R,
S37R/D72T/Q197K/S273R/R331K/D375P/K432L, S37R/P83S/V263T/V365I/A366V/D375P,
S37R/K190R/I202H, S41A/F45L/D72T/V155L/V263T/R331K/A366V/D375P/K432L/L433A,
S41A/F45L/L163A/P168T/L243I/H248W/S273R/A366C/K432L,
F45L/D72T/G84N/R88T/Q197K/D375P, F45L/D72T/R88T/A366C,
F45L/D72T/L163A/I202H/V365I/A366V/D375P,
F45L/D72T/P168T/L243I/R331K/V365I/A366C/L429W/K432L,
F45L/G84N/P168T/K190R/A199Q/S254A/S273R/V365I/A366C,
F45L/L163A/P168T/K190R/A199Q/A366C/L429W/K432L,
F45L/L163A/P168T/Q197K/V263T/R331K/V365I/A366C,
D72T/H81T/P83S/G84N/R88T/V155L/L163A/P168T/K190R/A366C,
D72T/H81T/P83S/G84N/R88T/V155L/K190R,
D72T/H81T/P83S/G84N/R88T/V155L/K190R/S273R/R331K/A366V/K432L,
D72T/H81T/P83S/G84N/R88T/V155L/S273R/R331K/D375P,
D72T/H81T/P83S/G84N/R88T/V155L/A366C,
D72T/H81T/P83S/G84N/R88T/L163A/P168T/K190R/L243I/V263T/R331K/A366C,
D72T/H81T/P83S/G84N/R88T/L163A/P168T/V263T/R331K/D375P,
D72T/H81T/G84N/K190R/H248W, D72T/P83S/G84N/R88T/I202H/S254A/S273R/A366C/D375P,
D72T/P83S/G84N/Q197K/I202H/L243I/V263T/V365I/A366C,
D72T/P83S/R88T/L243I/V263T/R331K/V365I/A366C,
D72T/V155L/L163A/P168T/K190R/A366V/K432L/L433A, D72T/V155L/K190R,
D72T/V155L/K190R/V263T/R331K/A366C, D72T/V155L/K190R/A366C,
D72T/V155L/S273R/R331K/D375P/K432L, D72T/L243I/H248W/V263T/A366V/K432L/L433A,
D72T/L243I/H248W/S273R/A366V/K432L/L433A, D72T/L243I/H248W/A366V/K432L/L433A,
D72T/H248W, H81T/P83S/G84N/R88T/V155L/L163A/P168T/S273R/R331K/D375P,
H81T/P83S/G84N/R88T/V155L/K190R/V263T/R331K/A366C,
H81T/P83S/G84N/R88T/V155L/V263T/A366V/D375P, H81T/P83S/G84N/R88T/V155L/A366C,
H81T/P83S/G84N/R88T/K190R/V263T/D375P,
H81T/P83S/G84N/R88T/V263T/S273R/R331K/A366C/K432L, H81T/P83S/E169D/K190R/V263T,
H81T/P83S/K190R/V263T/V365I/A366C, H81T/P83S/I202H/V365I/A366C/E402L,
H81T/R88T/D375P/E402L, P83S/R88T/V155L/S273R/A366V/D375P,
G84N/V155L/P168T/Q197K/A199Q/R331K/A366V/D375P/N383V/E402L,
G84N/P168T/Q197K/I202H/V263T/A366C, G84N/Q197K/A366C/E402L,
V155L/P168T/K190R/Q197K/A199Q/A366C, V155L/P168T/D375P,
V155L/V263T/A366C/K432L/L433A, K190R/A199Q/I202H/R331K/A366C, Q197K/A199Q/I202H,
Q197K/I202H/H248W, Q197K/H248W, A199Q/V263T/R331K/V365I/A366C, H248W/D375P,
V365I/A366C, and V365I/D375P/E402L, wherein the positions are numbered with reference to SEQ ID
NO: 6468. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase
comprises at least one mutation or mutation set at one or more positions selected from 2, 3, 4, 9, 53/437,
61, 64, 72/170, 72/405, 94, 96, 98, 113, 118, 118/120, 120, 129, 134/158, 158, 165, 170, 171, 173, 183,
193, 214, 214/222, 222, 226, 229, 234, 253, 265, 269, 272, 289, 296, 300, 302, 304, 322, 322/407, 330,
390, 395/439, 396, 398, 399, 403, 405, 408, 411, 412, 423, 428, 434, 435, 438, 439, 442, 444, 448, 449,
452, and 454, wherein the positions are numbered with reference to SEQ ID NO: 6468. In some
embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one
mutation or mutation set at one or more positions selected from 2F, 2H, 2K, 2M, 2Q, 2R, 2V, 3F, 3L, 3M,
4N, 4S, 9A, 9G, 9K, 9M, 9S, 53E/437T, 61Q, 64L, 64M, 72E/170A, 72E/405S, 94R, 96C, 96K, 98E,
98S, 98T, 113D, 113G, 113P, 118C, 118T/120V, 118V, 120V, 129P, 134E/158N, 158T, 165L, 170A,

170G, 170H, 170P, 171A, 171L, 171P, 171Q, 173I, 173K, 173L, 173P, 173S, 183I, 183L, 183P, 193F, 214K, 214R, 214R/222H, 222A, 222N, 222Q, 222R, 226S, 229M, 229Q, 234N, 234S, 234T, 253D, 253E, 253N, 253P, 253T, 253V, 265H, 269L, 269M, 269N, 269R, 272S, 289D, 289G, 289N, 289R, 296Q, 300A, 300E, 302G, 304K, 304P, 322A, 322G, 322K, 322P, 322P/407I, 322S, 322T, 322V/407I, 330S, 390I, 390R, 395I/439V, 396T, 396V, 398S, 399P, 399Q, 399S, 403V, 405A, 405D, 405P, 405S, 405T, 408D, 408K, 408S, 411H, 411K, 411T, 412K, 423T, 428E, 428G, 428I, 428L, 428N, 428Q, 428R, 428S, 428T, 428V, 428Y, 434E, 434G, 435A, 435D, 435E, 435G, 435I, 435K, 435L, 435N, 435S, 435V, 435Y, 438E, 438R, 438S, 439A, 439E, 439G, 439H, 439M, 439R, 439W, 442F, 442T, 444A, 448K, 448Q, 449G, 449L, 449S, 452T, and 454V, wherein the positions are numbered with reference to SEQ ID NO: 6468. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from P2F, P2H, P2K, P2M, P2Q, P2R, P2V, N3F, N3L, N3M, T4N, T4S, V9A, V9G, V9K, V9M, V9S, K53E/I437T, S61Q, F64L, F64M, D72E/D170A, D72E/E405S, K94R, G96C, G96K, D98E, D98S, D98T, S113D, S113G, S113P, A118C, A118T/L120V, A118V, L120V, A129P, D134E/P158N, P158T, Y165L, D170A, D170G, D170H, D170P, D170V, K171A, K171L, K171P, K171Q, R173I, R173K, R173L, R173S, M183I, M183L, M183P, Y193F, S214K, S214R, S214R/K222H, K222A, K222N, K222Q, K222R, E226S, L229M, L229Q, R234N, R234S, R234T, S253D, S253E, S253N, S253P, S253T, S253V, Q265H, Q269L, Q269M, Q269N, Q269R, P272S, K289D, K289G, K289N, K289R, R296Q, D300A, D300E, K302G, S304K, S304P, L322A, L322G, L322K, L322P, L322P/V407I, L322S, L322T, L322V/V407I, E330S, V390I, V390R, V395I/P439V, Y396T, Y396V, E398S, N399P, N399Q, N399S, R403V, E405A, E405D, E405P, E405S, E405T, N408D, N408K, N408S, R411H, R411K, R411T, R412K, R423T, K428E, K428G, K428I, K428L, K428N, K428Q, K428R, K428S, K428T, K428V, K428Y, D434E, D434G, R435A, R435D, R435E, R435G, R435I, R435K, R435L, R435N, R435S, R435V, R435Y, A438E, A438R, A438S, P439A, P439E, P439G, P439H, P439M, P439R, P439W, S442F, S442T, Y444A, E448K, E448Q, R449G, R449L, R449S, S452T, and I454V, wherein the positions are numbered with reference to SEQ ID NO: 6468. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 6678, 6680, 6682, 6684, 6686, 6688, 6690, 6692, 6694, 6696, 6698, 6700, 6702, 6704, 6706, 6708, 6710, 6712, 6714, 6716, 6718, 6720, 6722, 6724, 6726, 6728, 6730, 6732, 6734, 6736, 6738, 6740, 6742, 6744, 6746, 6748, 6750, 6752, 6754, 6756, 6758, 6760, 6762, 6764, 6766, 6768, 6770, 6772, 6774, 6776, 6778, 6780, 6782, 6784, 6786, 6788, 6790, 6792, 6794, 6796, 6798, 6800, 6802, 6804, 6806, 6808, 6810, 6812, 6814, 6816, 6818, 6820, 6822, 6824, 6826, 6828, 6830, 6832, 6834, 6836, 6838, 6840, 6842, 6844, 6846, 6848, 6850, 6852, 6854, 6856, 6858, 6860, 6862, 6864, 6866, 6868, 6870, 6872, 6874, 6876, 6878, 6880, 6882, 6884, 6886, 6888, 6890, 6892, 6894, 6896, 6898, 6900, 6902, 6904, 6906, 6908, 6910, 6912, 6914, 6916, 6918, 6920, 6922, 6924, 6926, 6928, 6930, 6932, 6934, 6936, 6938, 6940, 6942, 6944, 6946, 6948, 6950, 6952, 6954, 6956, 6958, 6960, 6962, 6964, 6966, 6968, 6970, 6972, 6974, 6976, 6978, 6980, 6982, 6984, 6986, 6988, 6990, 6992, 6994, 6996, 6998, 7000, 7002, 7004, 7006, 7008, 7010, 7012, 7014, 7016, 7018, 7020, 7022, 7024, 7026, 7028, 7030, 7032, 7034, 7036, 7038, 7040, 7042, 7044, 7046, 7048, 7050, 7052, 7054, 7056, 7058, 7060, 7062, 7064, 7066, 7068, 7070, 7072, 7074, 7076, 7078, 7080, 7082, 7084, 7086, 7088, 7090, 7092, 7094, 7096, 7098, 7100, 7102, 7104, 7106, 7108, 7110, 7112, 7114, 7116, 7118, 7120, 7122, 7124, 7126, 7128, 7130, 7132, 7134, 7136, 7138, 7140, 7142, 7144, 7146, 7148, 7150, 7152, 7154, 7156, 7158, 7160, 7162, 7164, 7166, 7168, 7170, 7172, 7174, 7176, 7178, 7180, 7182, 7184, 7186, 7188, 7190, 7192, 7194, 7196, 7198, 7200, 7202, 7204, 7206, 7208, 7210, 7212, and 7214. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 6678, 6680, 6682, 6684, 6686, 6688, 6690, 6692, 6694, 6696, 6698, 6700, 6702, 6704, 6706, 6708, 6710, 6712, 6714, 6716, 6718, 6720, 6722, 6724, 6726, 6728, 6730, 6732, 6734, 6736, 6738, 6740, 6742, 6744, 6746, 6748, 6750, 6752, 6754, 6756, 6758, 6760, 6762, 6764, 6766, 6768, 6770, 6772, 6774, 6776, 6778, 6780, 6782, 6784, 6786, 6788, 6790, 6792, 6794, 6796, 6798, 6800, 6802, 6804, 6806, 6808, 6810, 6812, 6814, 6816, 6818, 6820, 6822, 6824, 6826, 6828, 6830, 6832, 6834, 6836, 6838, 6840, 6842, 6844, 6846, 6848, 6850, 6852, 6854, 6856, 6858, 6860, 6862, 6864, 6866, 6868, 6870, 6872, 6874, 6876, 6878, 6880, 6882, 6884, 6886, 6888, 6890, 6892, 6894, 6896, 6898, 6900, 6902, 6904, 6906, 6908, 6910, 6912, 6914, 6916, 6918, 6920, 6922, 6924, 6926, 6928, 6930, 6932, 6934, 6936, 6938, 6940, 6942, 6944, 6946, 6948, 6950, 6952, 6954, 6956, 6958, 6960, 6962, 6964, 6966, 6968, 6970, 6972, 6974, 6976, 6978, 6980, 6982, 6984, 6986,

6998, 6999, 6999, 6999, 6996, 6996, 6998, 7000, 7002, 7004, 7006, 7008, 7010, 7012, 7014, 7016, 7018, 7020, 7022, 7024, 7026, 7028, 7030, 7032, 7034, 7036, 7038, 7040, 7042, 7044, 7046, 7048, 7050, 7052, 7054, 7056, 7058, 7060, 7062, 7064, 7066, 7068, 7070, 7072, 7074, 7076, 7078, 7080, 7082, 7084, 7086, 7088, 7090, 7092, 7094, 7096, 7098, 7100, 7102, 7104, 7106, 7108, 7110, 7112, 7114, 7116, 7118, 7120, 7122, 7124, 7126, 7128, 7130, 7132, 7134, 7136, 7138, 7140, 7142, 7144, 7146, 7148, 7150, 7152, 7154, 7156, 7158, 7160, 7162, 7164, 7166, 7168, 7170, 7172, 7174, 7176, 7178, 7180, 7182, 7184, 7186, 7188, 7190, 7192, 7194, 7196, 7198, 7200, 7202, 7204, 7206, 7208, 7210, 7212, and 7214. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 6678, 6680, 6682, 6684, 6686, 6688, 6690, 6692, 6694, 6696, 6698, 6700, 6702, 6704, 6706, 6708, 6710, 6712, 6714, 6716, 6718, 6720, 6722, 6724, 6726, 6728, 6730, 6732, 6734, 6736, 6738, 6740, 6742, 6744, 6746, 6748, 6750, 6752, 6754, 6756, 6758, 6760, 6762, 6764, 6766, 6768, 6770, 6772, 6774, 6776, 6778, 6780, 6782, 6784, 6786, 6788, 6790, 6792, 6794, 6796, 6798, 6800, 6802, 6804, 6806, 6808, 6810, 6812, 6814, 6816, 6818, 6820, 6822, 6824, 6826, 6828, 6830, 6832, 6834, 6836, 6838, 6840, 6842, 6844, 6846, 6848, 6850, 6852, 6854, 6856, 6858, 6860, 6862, 6864, 6866, 6868, 6870, 6872, 6874, 6876, 6878, 6880, 6882, 6884, 6886, 6888, 6890, 6892, 6894, 6896, 6898, 6900, 6902, 6904, 6906, 6908, 6910, 6912, 6914, 6916, 6918, 6920, 6922, 6924, 6926, 6928, 6930, 6932, 6934, 6936, 6938, 6940, 6942, 6944, 6946, 6948, 6950, 6952, 6954, 6956, 6958, 6960, 6962, 6964, 6966, 6968, 6970, 6972, 6974, 6976, 6978, 6980, 6982, 6984, 6986, 6988, 6990, 6992, 6994, 6996, 6998, 7000, 7002, 7004, 7006, 7008, 7010, 7012, 7014, 7016, 7018, 7020, 7022, 7024, 7026, 7028, 7030, 7032, 7034, 7036, 7038, 7040, 7042, 7044, 7046, 7048, 7050, 7052, 7054, 7056, 7058, 7060, 7062, 7064, 7066, 7068, 7070, 7072, 7074, 7076, 7078, 7080, 7082, 7084, 7086, 7088, 7090, 7092, 7094, 7096, 7098, 7100, 7102, 7104, 7106, 7108, 7110, 7112, 7114, 7116, 7118, 7120, 7122, 7124, 7126, 7128, 7130, 7132, 7134, 7136, 7138, 7140, 7142, 7144, 7146, 7148, 7150, 7152, 7154, 7156, 7158, 7160, 7162, 7164, 7166, 7168, 7170, 7172, 7174, 7176, 7178, 7180, 7182, 7184, 7186, 7188, 7190, 7192, 7194, 7196, 7198, 7200, 7202, 7204, 7206, 7208, 7210, 7212, and 7214.

(30) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 5/6/25/317, 5/6/69/288/303, 5/6/91/288/317, 5/25/91/212/303/317, 5/25/91/263, 5/25/288, 5/69/91/212/288, 5/69/91/212/303, 5/91/288/303, 5/91/303, 5/91/317/421, 5/288, 6/91/212/288/303/369/421, 6/212/288/303, 6/288, 25/91/212/288, 25/91/263/288/303, 25/91/303/317/369, 25/91/317/369, 25/263/317, 87/144, 87/144/159/361, 87/144/159/361/433, 87/433, 88/144, 88/144/159, 91/263/317/369, 91/288/303/317/369, 91/288/317/369/421, 91/303, 91/317, 91/317/369, 109/144/153/155/159/433, 144, 212/288, 263/288/303/317, and 288, wherein the positions are numbered with reference to SEQ ID NO: 6864. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 5S/6P/25I/317Y, 5S/6P/69A/288P/303V, 5S/6P/91Q/288P/317Y, 5S/25Q/91Q/212L/303V/317Y, 5S/25Q/91Q/263T, 5S/25Q/288P, 5S/69A/91Q/212L/288P, 5S/69A/91Q/212L/303V, 5S/91Q/288P/303V, 5S/91Q/303V, 5S/91Q/317Y/421I, 5S/288P, 6P/91Q/212L/288P/303V/369K/421I, 6P/212L/288P/303V, 6P/288P, 25Q/91Q/212L/288P, 25Q/91Q/263T/288P/303V, 25Q/91Q/317Y/369K, 25Q/91T/303V/317Y/369K, 25Q/263T/317Y, 87K/144Q, 87K/144Q/159K/361C, 87K/144Q/159K/361C/433A, 87K/433A, 88T/144Q, 88T/144Q/159K, 91Q/263T/317Y/369K, 91Q/288P/303V/317Y/369K, 91Q/288P/317Y/369K/421I, 91Q/303V, 91Q/317Y, 91Q/317Y/369K, 109W/144Q/153V/155L/159K/433A, 144Q, 212L/288P, 263T/288P/303V/317Y, and 288P, wherein the positions are numbered with reference to SEQ ID NO: 6864. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from T5S/E6P/L25I/S317Y, T5S/E6P/D69A/E288P/Q303V, T5S/E6P/I91Q/E288P/S317Y, T5S/L25Q/I91Q/A212L/Q303V/5317Y, T5S/L25Q/I91Q/V263T, T5S/L25Q/E288P, T5S/D69A/I91Q/A212L/E288P, T5S/D69A/I91Q/A212L/Q303V, T5S/I91Q/E288P/Q303V, T5S/I91Q/Q303V, T5S/I91Q/S317Y/V421I, T5S/E288P, E6P/I91Q/A212L/E288P/Q303V/V369K/V421I, E6P/A212L/E288P/Q303V, E6P/E288P, L25Q/I91Q/A212L/E288P, L25Q/I91Q/V263T/E288P/Q303V, L25Q/I91Q/5317Y/V369K, L25Q/I91T/Q303V/S317Y/V369K, L25Q/V263T/S317Y, M87K/M144Q, M87K/M144Q/Q159K/T361C, M87K/M144Q/Q159K/T361C/L433A, M87K/L433A, R88T/M144Q, R88T/M144Q/Q159K, I91Q/V263T/S317Y/V369K, I91Q/E288P/Q303V/S317Y/V369K, I91Q/E288P/5317Y/V369K/V421I,

I91Q/Q303V, I91Q/317Y, I91Q/V317Y/V369K, R109W/M144Q/A153V/V155L/Q159K/L433A, M144Q, A212L/E288P, V263T/E288P/Q303V/S317Y, and E288P, wherein the positions are numbered with reference to SEQ ID NO: 6864. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 7362, 7364, 7366, 7368, 7370, 7372, 7374, 7376, 7378, 7380, 7382, 7384, 7386, 7388, 7390, 7392, 7394, 7396, 7398, 7400, 7402, 7404, 7406, 7408, 7410, 7412, 7414, 7416, 7418, 7420, 7422, 7424, 7426, 7428, 7430, 7432, 7434, and 7436, wherein the positions are numbered with reference to SEQ ID NO: 6864. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 7362, 7364, 7366, 7368, 7370, 7372, 7374, 7376, 7378, 7380, 7382, 7384, 7386, 7388, 7390, 7392, 7394, 7396, 7398, 7400, 7402, 7404, 7406, 7408, 7410, 7412, 7414, 7416, 7418, 7420, 7422, 7424, 7426, 7428, 7430, 7432, 7434, and 7436. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 7362, 7364, 7366, 7368, 7370, 7372, 7374, 7376, 7378, 7380, 7382, 7384, 7386, 7388, 7390, 7392, 7394, 7396, 7398, 7400, 7402, 7404, 7406, 7408, 7410, 7412, 7414, 7416, 7418, 7420, 7422, 7424, 7426, 7428, 7430, 7432, 7434, and 7436.

(31) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 2/3/433/435/442, 2/81, 9/37/113/396, 9/87, 9/144/331, 37/113/144/396, 37/233, 53/144/233/269/331/428, 53/233/269/390/395/396, 53/234, 61/120/156/163/165/197/303/308, 61/120/159/300/308/407, 61/120/300/303/407, 61/156/159/163/165/243/248/253/300/303/308, 61/156/163/243/248/300/303/308/407, 61/156/163/300/365/411, 61/156/197/253, 61/163/300/303/308/405/407/411, 61/165/248/253/407/411, 61/214, 61/214/300/308, 61/243/300/308/407/411, 61/300/303/308, 61/300/303/405, 61/365/405, 69, 69/81, 69/134, 69/263, 69/263/434/438/439, 69/439, 81, 81/134, 81/222, 81/222/263/322/435/442, 81/433/435/438/442, 85/156/159/243/248/253/308/405/407/411, 87/144, 87/144/396, 113/233/234, 120/156/159/169/197/214/303/308/365/405/407, 120/156/159/248/300/308, 120/156/248/303/308/411, 120/159, 120/159/165/197, 120/159/197/308/407/411, 120/159/197/365/411, 120/197/253/300/308, 120/253/300/303/308/407, 120/308/407/411, 134/222/263, 144/234/269, 156/165/248/300/303/308, 156/197/248/300/411, 156/214/308/411, 156/248/253/308, 159/163/165/197/214/243/300/303/308/407, 163/197/253/300/303/308/365, 171/263, 183/233/234/331/428, 197/253/308/407, 197/300/303/308/365, 197/300/308/411, 222, 222/263/435/442, 233/396/399, 263, 269, 269/428/437, 300/303/308, 300/308/405/411, 308, and 322, wherein the positions are numbered with reference to SEQ ID NO: 7388. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 2K/3M/433A/435E/442F, 2K/81T, 9G/37S/113D/396T, 9G/87K, 9G/144Q/331R, 37S/113D/144Q/396T, 37S/233S, 53E/144Q/233Q/269R/331R/428I, 53E/233Q/269R/390I/395I/396T, 53E/234N, 61Q/120V/156S/163A/165L/197K/303V/308L, 61Q/120V/159K/300A/308L/407I, 61Q/120V/300A/303V/407I, 61Q/156S/159K/163A/165L/243I/248W/253T/300A/303V/308L, 61Q/156S/163A/243I/248W/300A/303V/308L/407I, 61Q/156S/163A/300A/365I/411T, 61Q/156S/197K/253E, 61Q/163A/300A/303V/308L/405P/407I/411T, 61Q/165L/248W/253T/407I/411T, 61Q/214R, 61Q/214R/300A/308L, 61Q/243I/300A/308L/407I/411T, 61Q/300A/303V/308L, 61Q/300A/303V/405P, 61Q/365I/405P, 69A, 69A/81T, 69A/134E, 69A/263T, 69A/263T/434E/438S/439H, 69A/439H, 81T, 81T/134E, 81T/222A, 81T/222A/263T/322S/435E/442F, 81T/433A/435I/438R/442F, 85V/156S/159K/243I/248W/253T/308L/405P/407I/411T, 87K/144Q, 87K/144Q/396T, 113D/233Q/234N, 120V/156S/159K/169D/197K/214R/303V/308L/365I/405P/407I, 120V/156S/159K/248W/300A/308L, 120V/156S/248W/303V/308L/411T, 120V/159K, 120V/159K/165L/197K, 120V/159K/197K/308L/407I/411T, 120V/159K/197K/365I/411T, 120V/197K/253T/300A/308L, 120V/253T/300A/303V/308L/407I, 120V/308L/407I/411T, 134E/222A/263T, 144Q/234N/269R, 156S/165L/248W/300A/303V/308L, 156S/197K/248W/300A/411T, 156S/214R/308L/411T, 156S/248W/253T/308L, 159K/163A/165L/197K/214R/243I/300A/303V/308L/407I, 163A/197K/253E/300A/303V/308L/365I, 171P/263T, 183L/233Q/234N/331R/428I, 197K/253T/308L/407I, 197K/300A/303V/308L/365I, 197K/300A/308L/411T, 222A, 222A/263T/435I/442F, 233Q/396T/399Q, 263T, 269R, 269R/428I/437L,

300A/303V/308L/308L/405P/411T, 308L, and 322S, wherein the positions are numbered with reference to SEQ ID NO: 7388. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from P2K/N3M/L433A/R435E/S442F, P2K/H81T, V9G/R37S/S113D/Y396T, V9G/M87K, V9G/M144Q/K331R, R37S/S113D/M144Q/Y396T, R37S/W233S, K53E/M144Q/W233Q/Q269R/K331R/K4281, K53E/W233Q/Q269R/V390I/V395I/Y396T, K53E/R234N, S61Q/L120V/C156S/L163A/Y165L/Q197K/Q303V/R308L, S61Q/L120V/Q159K/D300A/R308L/V407I, S61Q/L120V/D300A/Q303V/V407I, S61Q/C156S/Q159K/L163A/Y165L/L243I/H248W/5253T/D300A/Q303V/R308L, S61Q/C156S/L163A/L243I/H248W/D300A/Q303V/R308L/V407I, S61Q/C156S/L163A/D300A/V365I/R411T, S61Q/C156S/Q197K/S253E, S61Q/L163A/D300A/Q303V/R308L/E405P/V407I/R411T, S61Q/Y165L/H248W/S253T/V407I/R411T, S61Q/S214R, S61Q/S214R/D300A/R308L, S61Q/L243I/D300A/R308L/V407I/R411T, S61Q/D300A/Q303V/R308L, S61Q/D300A/Q303V/E405P, S61Q/V365I/E405P, D69A, D69A/H81T, D69A/D134E, D69A/V263T, D69A/V263T/D434E/A438S/P439H, D69A/P439H, H81T, H81T/D134E, H81T/K222A, H81T/K222A/V263T/L322S/R435E/S442F, H81T/L433A/R435I/A438R/S442F, A85V/C156S/Q159K/L243I/H248W/S253T/R308L/E405P/V407I/R411T, M87K/M144Q, M87K/M144Q/Y396T, S113D/W233Q/R234N, L120V/C156S/Q159K/E169D/Q197K/S214R/Q303V/R308L/V365I/E405P/V407I, L120V/C156S/Q159K/H248W/D300A/R308L, L120V/C156S/H248W/Q303V/R308L/R411T, L120V/Q159K, L120V/Q159K/Y165L/Q197K, L120V/Q159K/Q197K/R308L/V407I/R411T, L120V/Q159K/Q197K/V365I/R411T, L120V/Q197K/S253T/D300A/R308L, L120V/5253T/D300A/Q303V/R308L/V407I, L120V/R308L/V407I/R411T, D134E/K222A/V263T, M144Q/R234N/Q269R, C156S/Y165L/H248W/D300A/Q303V/R308L, C156S/Q197K/H248W/D300A/R411T, C156S/S214R/R308L/R411T, C156S/H248W/S253T/R308L, Q159K/L163A/Y165L/Q197K/S214R/L243I/D300A/Q303V/R308L/V407I, L163A/Q197K/5253E/D300A/Q303V/R308L/V365I, K171P/V263T, M183L/W233Q/R234N/K331R/K4281, Q197K/S253T/R308L/V407I, Q197K/D300A/Q303V/R308L/V365I, Q197K/D300A/R308L/R411T, K222A, K222A/V263T/R435I/S442F, W233Q/Y396T/N399Q, V263T, Q269R, Q269R/K428I/I437L, D300A/Q303V/R308L, D300A/R308L/E405P/R411T, R308L, and L322S, wherein the positions are numbered with reference to SEQ ID NO: 7388. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 41, 56, 61, 72, 76, 87, 88, 107, 139, 156, 338, and 407, wherein the positions are numbered with reference to SEQ ID NO: 7388. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 41E, 56D, 61D, 61E, 72P, 76S, 87E, 88L, 88M, 107L, 107V, 139N, 156S, 338V, and 407T, wherein the positions are numbered with reference to SEQ ID NO: 7388. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from A41E, L56D, S61D, S61E, T72P, R76S, M87E, R88L, R88M, A107L, A107V, K139N, C156S, C338V, and V407T, wherein the positions are numbered with reference to SEQ ID NO: 7388. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 7946, 7948, 7950, 7952, 7954, 7956, 7958, 7960, 7962, 7964, 7966, 7968, 7970, 7972, 7974, 7976, 7978, 7980, 7982, 7984, 7986, 7988, 7990, 7992, 7994, 7996, 7998, 8000, 8002, 8004, 8006, 8008, 8010, 8012, 8014, 8016, 8018, 8020, 8022, 8024, 8026, 8028, 8030, 8032, 8034, 8036, 8038, 8040, 8042, 8044, 8046, 8048, 8050, 8052, 8054, 8056, 8058, 8060, 8062, 8064, 8066, 8068, 8070, 8072, 8074, 8076, 8078, 8080, 8082, 8084, 8086, 8088, 8090, 8340, 8342, 8344, 8346, 8348, 8350, 8352, 8354, 8356, 8358, 8360, 8362, 8364, 8366, and 8368. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 7946, 7948, 7950, 7952, 7954, 7956, 7958, 7960, 7962, 7964, 7966, 7968, 7970, 7972, 7974, 7976, 7978, 7980, 7982, 7984, 7986, 7988, 7990, 7992, 7994, 7996, 7998, 8000, 8002, 8004, 8006, 8008, 8010, 8012, 8014, 8016, 8018, 8020, 8022, 8024, 8026, 8028, 8030, 8032, 8034, 8036, 8038, 8040, 8042, 8044, 8046, 8048, 8050, 8052, 8054, 8056, 8058, 8060, 8062, 8064, 8066,

8068, 8070, 8072, 8076, 8080, 8082, 8084, 8086, 8088, 8090, 8340, 8342, 8344, 8346, 8348, 8350, 8352, 8354, 8356, 8358, 8360, 8362, 8364, 8366, and 8368. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 7946, 7948, 7950, 7952, 7954, 7956, 7958, 7960, 7962, 7964, 7966, 7968, 7970, 7972, 7974, 7976, 7978, 7980, 7982, 7984, 7986, 7988, 7990, 7992, 7994, 7996, 7998, 8000, 8002, 8004, 8006, 8008, 8010, 8012, 8014, 8016, 8018, 8020, 8022, 8024, 8026, 8028, 8030, 8032, 8034, 8036, 8038, 8040, 8042, 8044, 8046, 8048, 8050, 8052, 8054, 8056, 8058, 8060, 8062, 8064, 8066, 8068, 8070, 8072, 8074, 8076, 8078, 8080, 8082, 8084, 8086, 8088, 8090, 8340, 8342, 8344, 8346, 8348, 8350, 8352, 8354, 8356, 8358, 8360, 8362, 8364, 8366, and 8368.

(32) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 37/72/76/81, 37/72/76/107/156/331, 37/72/195/331, 53/81/195/197, 69/72/76/107, 72/76/107/156, 72/76/107/195/197, 72/269, 81/107/195, 87, 87/91, 87/91/94, 87/91/94/120, 87/91/94/233, 87/91/94/233/259, 87/91/94/263/389, 87/91/120/233, 87/91/120/233/411/431/435/437, 87/91/144/259, 87/91/163, 87/91/163/233, 87/91/163/233/263, 87/91/163/389, 87/91/233, 87/91/233/389, 87/91/259/263/389/428/431/435/437, 87/91/263/389, 87/91/322, 87/91/389, 87/94, 87/94/144/263/428/435, 87/94/263, 87/163/233, 87/233, 87/233/259, 87/233/322/389/411, 87/233/389, 87/259, 87/263, 87/428/431/435, 87/435, 91, 91/94, 91/94/120/233, 91/94/120/233/389/431/438, 91/94/233/259, 91/120, 91/120/233, 91/233, 91/233/259/389, 91/233/389, 94/233/411, 144/389, 163/233, 195/197, 197, 233, 233/259/263, 233/259/389, 233/438, 259, 263, 263/389, 322, 389, and 428/431/435, wherein the positions are numbered with reference to SEQ ID NO: 8088. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 37S/72P/76S/81T, 37S/72P/76S/107L/156S/331R, 37S/72P/195Q/331R, 53E/81T/195Q/197K, 69A/72P/76S/107L, 72P/76S/107L/156S, 72P/76S/107L/195Q/197K, 72P/269R, 81T/107L/195Q, 87A, 87A/91L, 87A/91L/94C/233Q, 87A/91L/94C/233Q/259T, 87A/91L/120L/233Q, 87A/91L/120L/233Q/411T/431M/435E/437L, 87A/91L/163A, 87A/91L/233Q, 87A/91L/233Q/389L, 87A/91L/259T/263T/389L/428I/431M/435I/437L, 87A/91L/322S, 87A/94C, 87A/94C/263T, 87A/233Q, 87A/233Q/259T, 87A/233Q/322S/389L/411T, 87A/233Q/389L, 87A/259T, 87A/263T, 87A/428I/431M/435E, 87A/435E, 87K, 87K/91L, 87K/91L/94C, 87K/91L/94C/120L, 87K/91L/94C/233Q, 87K/91L/94C/263T/389L, 87K/91L/120L/233Q, 87K/91L/144Q/259T, 87K/91L/163A/233Q, 87K/91L/163A/233Q/263T, 87K/91L/163A/389L, 87K/91L/233Q, 87K/91L/263T/389L, 87K/91L/389L, 87K/94C/144Q/263T/428I/435E, 87K/163A/233Q, 91L, 91L/94C, 91L/94C/120L/233Q, 91L/94C/120L/233Q/389L/431M/438S, 91L/94C/233Q/259T, 91L/120L, 91L/120L/233Q, 91L/233Q, 91L/233Q/259T/389L, 91L/233Q/389L, 94C/233Q/411T, 144Q/389L, 163A/233Q, 195Q/197K, 197K, 233Q, 233Q/259T/263T, 233Q/259T/389L, 233Q/438S, 259T, 263T, 263T/389L, 322S, 389L, and 428I/431M/435E, wherein the positions are numbered with reference to SEQ ID NO: 8088. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from R37S/T72P/R76S/H81T, R37S/T72P/R76S/A107L/C156S/K331R, R37S/T72P/H195Q/K331R, K53E/H81T/H195Q/Q197K, D69A/T72P/R76S/A107L, T72P/R76S/A107L/C156S, T72P/R76S/A107L/H195Q/Q197K, T72P/Q269R, H81T/A107L/H195Q, M87A, M87A/Q91L, M87A/Q91L/K94C/W233Q, M87A/Q91L/K94C/W233Q/E259T, M87A/Q91L/V120L/W233Q, M87A/Q91L/V120L/W233Q/R411T/D431M/R435E/I437L, M87A/Q91L/L163A, M87A/Q91L/W233Q, M87A/Q91L/W233Q/E389L, M87A/Q91L/E259T/V263T/E389L/K428I/D431M/R435I/I437L, M87A/Q91L/L322S, M87A/K94C, M87A/K94C/V263T, M87A/W233Q, M87A/W233Q/E259T, M87A/W233Q/L322S/E389L/R411T, M87A/W233Q/E389L, M87A/E259T, M87A/V263T, M87A/K428I/D431M/R435E, M87A/R435E, M87K, M87K/Q91L, M87K/Q91L/K94C, M87K/Q91L/K94C/V120L, M87K/Q91L/K94C/W233Q, M87K/Q91L/K94C/V263T/E389L, M87K/Q91L/V120L/W233Q, M87K/Q91L/M144Q/E259T, M87K/Q91L/L163A/W233Q, M87K/Q91L/L163A/W233Q/V263T, M87K/Q91L/L163A/E389L, M87K/Q91L/W233Q, M87K/Q91L/V263T/E389L, M87K/Q91L/E389L, M87K/K94C/M144Q/V263T/K428I/R435E, M87K/L163A/W233Q, Q91L, Q91L/K94C, Q91L/K94C/V120L/W233Q, Q91L/K94C/V120L/W233Q/E389L/D431M/A438S, Q91L/K94C/W233Q/E259T, Q91L/V120L,

Q91L/V120L/W233Q, Q91L/W233Q/E259T/E389L, Q91L/W233Q/E389L, K94C/W233Q/R411T, M144Q/E389L, L163A/W233Q, H195Q/Q197K, Q197K, W233Q, W233Q/E259T/V263T, W233Q/E259T/E389L, W233Q/A438S, E259T, V263T, V263T/E389L, L322S, E389L, and K428I/D431M/R435E, wherein the positions are numbered with reference to SEQ ID NO: 8088. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 55, 111, 252, 255, 324, 328, 413, and 451, wherein the positions are numbered with reference to SEQ ID NO: 8088. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 55G, 111T, 252P, 255T, 324D, 324G, 328T, 413L, and 451Q, wherein the positions are numbered with reference to SEQ ID NO: 8088. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from S55G, S111T, S252P, S255T, P324D, P324G, L328T, V413L, and V451Q, wherein the positions are numbered with reference to SEQ ID NO: 8088. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 8482, 8484, 8486, 8488, 8490, 8492, 8494, 8496, 8498, 8500, 8502, 8504, 8506, 8508, 8510, 8512, 8514, 8516, 8518, 8520, 8522, 8524, 8526, 8528, 8530, 8532, 8534, 8536, 8538, 8540, 8542, 8544, 8546, 8548, 8550, 8552, 8554, 8556, 8558, 8560, 8562, 8564, 8566, 8568, 8570, 8572, 8574, 8576, 8578, 8580, 8582, 8584, 8586, 8588, 8590, 8592, 8594, 8596, 8598, 8600, 8602, 8604, 8606, 8608, 8610, 8612, 8614, 8616, 8618, 8620, 8622, 9224, 9226, 9228, 9230, 9232, 9234, 9236, 9238, and 9240. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 8482, 8484, 8486, 8488, 8490, 8492, 8494, 8496, 8498, 8500, 8502, 8504, 8506, 8508, 8510, 8512, 8514, 8516, 8518, 8520, 8522, 8524, 8526, 8528, 8530, 8532, 8534, 8536, 8538, 8540, 8542, 8544, 8546, 8548, 8550, 8552, 8554, 8556, 8558, 8560, 8562, 8564, 8566, 8568, 8570, 8572, 8574, 8576, 8578, 8580, 8582, 8584, 8586, 8588, 8590, 8592, 8594, 8596, 8598, 8600, 8602, 8604, 8606, 8608, 8610, 8612, 8614, 8616, 8618, 8620, 8622, 9224, 9226, 9228, 9230, 9232, 9234, 9236, 9238, and 9240. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 8482, 8484, 8486, 8488, 8490, 8492, 8494, 8496, 8498, 8500, 8502, 8504, 8506, 8508, 8510, 8512, 8514, 8516, 8518, 8520, 8522, 8524, 8526, 8528, 8530, 8532, 8534, 8536, 8538, 8540, 8542, 8544, 8546, 8548, 8550, 8552, 8554, 8556, 8558, 8560, 8562, 8564, 8566, 8568, 8570, 8572, 8574, 8576, 8578, 8580, 8582, 8584, 8586, 8588, 8590, 8592, 8594, 8596, 8598, 8600, 8602, 8604, 8606, 8608, 8610, 8612, 8614, 8616, 8618, 8620, 8622, 9224, 9226, 9228, 9230, 9232, 9234, 9236, 9238, and 9240.

(33) In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, and/or 108. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, and/or 108. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises SEQ ID NOS: 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, and/or 108.

(34) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 69/173/175/243/246/354/365/383/399, 69/173/243/383/399, 56/191/354/383/399, 70/225/246/409/413, 70/115/225/409, 70/225/413, 70/225/247, 74/310/396/424, 74/396, and 173/175/191/365/383/399, wherein the positions are numbered with reference to SEQ ID NO: 758. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 69H/173N/175S/243A/246K/354I/365I/383V/399A, 69H/173N/243A/383V/399A, 56T/191D/354I/383V/399A, 70L/225G/246P/409K/413V, 70L/1155/225G/409K, 70L/225G/413V, 70L/225G/247G, 74T/310D/396E/424S, 74T/396E, and 173H/175S/191D/365I/383V/399A, wherein the positions are numbered with reference to SEQ ID NO: 758. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from R69H/Y173N/P175S/V243A/M246K/L354I/M365I/M383V/T399A, R69H/Y173N/V243A/M383V/T399A, I56T/N191D/L354I/M383V/T399A,

F70L/N225G/M246P/E409K/I413V, F70L/Q115S/N225G/E409K, F70L/N225G/I413V, F70L/N225G/E247G, H74T/K310D/G396E/N424S, H74T/G396E, and Y173H/P175S/N191D/M365I/M383V/T399A, wherein the positions are numbered with reference to SEQ ID NO:758. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 770, 772, 774, 776, 778, 780, 782, 784, 786, and/or 788. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 770, 772, 774, 776, 778, 780, 782, 784, 786, and/or 788. In some additional embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 770, 772, 774, 776, 778, 780, 782, 784, 786, and/or 788.

(35) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 24/28, 24/28/262, 24/28/262/264, 24/28/262/264/423, 24/264/294, 28, 28/262/264, 28/423, 69/173/175/243/246/354/365/383/399, 69/173/243/383/399, 56/191/354/383/399, 70/115/225/409, 70/225/246/409/413, 70/225/247, 70/225/413, 74/310/396/424, 74/396, 159, 173/175/191/365/383/399, 199, 262/264, and 264/29, wherein the positions are numbered with reference to SEQ ID NO:770. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 24E/28N, 24E/28N/262Y, 24E/28N/262Y/264S, 24E/28N/262Y/264S/423E, 24E/264S/294V, 28N, 28N/262Y/264S, 28N/423E, 69H/173N/175S/243A/246K/354I/365I/383V/399A, 69H/173N/243A/383V/399A, 56T/191D/354I/383V/399A, 70L/1155/225G/409K, 70L/225G/246P/409K/413V, 70L/225G/247G, 70L/225G/413V, 74T/310D/396E/424S, 74T/396E, 159R, 173H/175S/191D/365I/383V/399A, 199H, 262Y/264S, and 264S/291V, wherein the positions are numbered with reference to SEQ ID NO:770. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from Y24E/S28N, Y24E/S28N/F262Y, Y24E/S28N/F262Y/C264S, Y24E/S28N/F262Y/C264S/K423E, Y24E/C264S/A294V, S28N, S28N/F262Y/C264S, S28N/K423E, R69H/Y173N/P175S/V243A/M246K/L354I/M365I/M383V/T399A, R69H/Y173N/V243A/M383V/T399A, I56T/N191D/L354I/M383V/T399A, F70L/Q115S/N225G/E409K, F70L/N225G/M246P/E409K/I413V, F70L/N225G/E247G, F70L/N225G/I413V, H74T/K310D/G396E/N424S, H74T/G396E, F156R, Y173H/P175S/N191D/M365I/M383V/T399A, G199H, F262Y/C264S, and C264S/I291V, wherein the positions are numbered with reference to SEQ ID NO:770. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 790, 792, 794, 796, 798, 800, 802, 804, 806, 808, 1292, and/or 1294. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 790, 792, 794, 796, 798, 800, 802, 804, 806, 808, 1292, and/or 1294. In some additional embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 790, 792, 794, 796, 798, 800, 802, 804, 806, 808, 1292, and/or 1294.

(36) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 24, 28, 32, 264, 269, 325, 341, 351, and 366, wherein the positions are numbered with reference to SEQ ID NO:770. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 24L/V, 28G/K/L, 32C/R/S, 264A/G, 269S/W, 325G/H, 341V, 351L, and 366L/Q/T, wherein the positions are numbered with reference to SEQ ID NO:770. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from Y24L/V, S28G/K/L, N32C/R/S, C264A/G, Y269S/W, K325G/H, F341V, M351L, and H366L/Q/T, wherein the positions are numbered with reference to SEQ ID NO:770. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 810, 812, 814, 816, 818, 820, 822, 824, 826, 828, 830, 832, 834, 836, 838, 840, 842, 844, and/or 846. In some embodiments, the polypeptide sequence of the

engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 810, 812, 814, 816, 818, 820, 822, 824, 826, 828, 830, 832, 834, 836, 838, 840, 842, 844, and/or 846. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 810, 812, 814, 816, 818, 820, 822, 824, 826, 828, 830, 832, 834, 836, 838, 840, 842, 844, and/or 846.

(37) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 56/69/70/175/191/246, 56/69/70/175/191/246/354, 56/69/70/175/354, 56/69/70/191/246/354/365, 56/69/70/246/354, 56/69/175/191/246/354, 56/69/175/246/354, 56/69/175/246/354/365, 56/69/246/354, 56/69/246/365, 56/70/175/191/246, 56/70/175/191/246/354, 56/70/175/191/354/365, 56/70/175/246/354, 56/70/175/246/365, 56/70/191/246/354/365, 56/70/191/354, 56/175/246, 56/175/354, 56/175/354/365, 56/191/246/354, 56/246, 56/246/354/365, 56/354, 69, 69/70/175/191/246/354/365, 69/70/191/246/354/365, 69/70/246, 69/70/354/365, 69/175/191/246/354, 69/175/354, 69/246/354/365, 69/354, 70, 70/175/191/246/354/365, 70/175/191/354/365, 70/191/246/354/365, 70/191/246/365, 126, 126/220, 126/403, 175, 175/191, 175/191/246/354/365, 175/191/354, 175/191/354/365, 175/246/354, 175/246/354/365, 175/354, 191/246/354, 246/354, 354, and 354/365, wherein the positions are numbered with reference to SEQ ID NO:770. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 56T/69Q/70L/175S/191D/246K, 56T/69Q/70L/175S/191D/246K/354I, 56T/69Q/70L/175S/354I, 56T/69Q/70L/191D/246K/354I/365I, 56T/69Q/70L/246K/354I, 56T/69Q/175S/191D/246K/354I, 56T/69Q/175S/246K/354I, 56T/69Q/175S/246K/354I/365I, 56T/69Q/246K/354I, 56T/69Q/246K/M65I, 56T/70L/175S/191D/246K, 56T/70L/175S/191D/246K/354I, 56T/70L/175S/191D/354I/365I, 56T/70L/175S/246K/354I, 56T/70L/175S/246K/365I, 56T/70L/191D/246K/354I/365I, 56T/70L/191D/354I, 56T/175S/246K, 56T/175S/354I, 56T/175S/354I/365I, 56T/191D/246K/354I, 56T/246K, 56T/246K/354I/365I, 56T/354I, 69Q, 69Q/70L/175S/191D/246K/354I/365I, 69Q/70L/191D/246K/354I/365I, 69Q/70L/246K, 69Q/70L/354I/365I, 69Q/175S/191D/246K/354I, 69Q/175S/354I, 69Q/246K/354I/365I, 69Q/354I, 70L, 70L/175S/191D/246K/354I/365I, 70L/175S/191D/354I/365I, 70L/191D/246K/354I/365I, 70L/191D/246K/365I, 126F, 126F/220L, 126F/403R, 175S, 175S/191D, 175S/191D/246K/354I/365I, 175S/191D/354I, 175S/191D/354I/365I, 175S/246K/354I, 175S/246K/354I/365I, 175S/354I, 191D/246K/354I, 246K/354I, 354I, and 354I/365I, wherein the positions are numbered with reference to SEQ ID NO:770. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from I56T/H69Q/F70L/P175S/N191D/M246K, I56T/H69Q/F70L/P175S/N191D/M246K/L354I, I56T/H69Q/F70L/P175S/L354I, I56T/H69Q/F70L/N191D/M246K/L354I/M365I, I56T/H69Q/F70L/M246K/L354I, I56T/H69Q/P175S/N191D/M246K/L354I, I56T/H69Q/P175S/M246K/L354I, I56T/H69Q/P175S/M246K/L354I/M365I, I56T/H69Q/M246K/L354I, I56T/H69Q/M246K/M365I, I56T/F70L/P175S/N191D/M246K, I56T/F70L/P175S/N191D/M246K/L354I, I56T/F70L/P175S/N191D/L354I/M365I, I56T/F70L/P175S/M246K/L354I, I56T/F70L/P175S/M246K/M365I, I56T/F70L/N191D/M246K/L354I/M365I, I56T/F70L/N191D/L354I, I56T/P175S/M246K, I56T/P175S/L354I, I56T/P175S/L354I/M365I, I56T/N191D/M246K/L354I, I56T/M246K, I56T/M246K/L354I/M365I, I56T/L354I, H69Q, H69Q/F70L/P175S/N191D/M246K/L354I/M365I, H69Q/F70L/N191D/M246K/L354I/M365I, H69Q/F70L/M246K, H69Q/F70L/L354I/M365I, H69Q/P175S/N191D/M246K/L354I, H69Q/P175S/L354I, H69Q/M246K/L354I/M365I, H69Q/L354I, F70L, F70L/P175S/N191D/M246K/L354I/M365I, F70L/P175S/N191D/L354I/M365I, F70L/N191D/M246K/L354I/M365I, F70L/N191D/M246K/M365I, I126F, I126F/C220L, I126F/K403R, P175S, P175S/N191D, P175S/N191D/M246K/L354I/M365I, P175S/N191D/L354I, P175S/N191D/L354I/M365I, P175S/M246K/L354I, P175S/M246K/L354I/M365I, P175S/L354I, N191D/M246K/L354I, M246K/L354I, L354I, and L354I/M365I, wherein the positions are numbered with reference to SEQ ID NO:770. In some additional embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 848, 850, 852, 854, 856, 858, 860, 862, 864, 866, 868, 870, 872, 874, 876, 878, 880, 882, 884, 886, 888, 890,

892, 896, 898, 900, 902, 904, 906, 908, 910, 912, 914, 916, 918, 920, 922, 924, 926, 928, 930, 932, 934, 936, 938, 940, 942, 944, 946, 948, 950, and/or 952. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 848, 850, 852, 854, 856, 858, 860, 862, 864, 866, 868, 870, 872, 874, 876, 878, 880, 882, 884, 886, 888, 890, 892, 894, 896, 898, 900, 902, 904, 906, 908, 910, 912, 914, 916, 918, 920, 922, 924, 926, 928, 930, 932, 934, 936, 938, 940, 942, 944, 946, 948, 950, and/or 952. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 848, 850, 852, 854, 856, 858, 860, 862, 864, 866, 868, 870, 872, 874, 876, 878, 880, 882, 884, 886, 888, 890, 892, 894, 896, 898, 900, 902, 904, 906, 908, 910, 912, 914, 916, 918, 920, 922, 924, 926, 928, 930, 932, 934, 936, 938, 940, 942, 944, 946, 948, 950, and/or 952.

(38) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequences of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 24/32/264/269/330/402/403, 24/32/264/269/382/403/406, 24/32/264/330, 24/32/269, 24/32/269/382/385/389/402/406, 24/32/269/403, 24/32/330, 24/264/269/389/402/406, 70/126/175/191/246/325/354/366, 70/126/175/325/330/351/354/366, 70/126/191/246/325/351/354/366/423, 70/126/191/246/325/351/354/423, 70/126/191/246/325/351/366/423, 70/126/191/246/354/366, 70/126/246/330/366, 70/126/246/354/366, 126/211/220/253/316/342, 126/211/220/275/279/323, 126/211/260/423, 126/220/260, 126/220/316, 56/220/260/423, 220/260, and 220/260/423, wherein the positions are numbered with reference to SEQ ID NO:792. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 24L/32S/264A/269S/330Q/402V/403R, 24L/32S/264A/269S/382G/403R/406M, 24L/32S/264A/330Q, 24L/32S/269S, 24L/32S/269S/382G/385V/389E/402I/406M, 24L/32S/269S/403R, 24L/32S/330Q, 24L/264A/269S/389E/402V/406M, 70L/126F/175S/191D/246K/325H/354I/366Q, 70L/126F/175S/325H/330Q/351L/354I/366Q, 70L/126F/191D/246K/325H/351L/354I/366Q/423K, 70L/126F/191D/246K/325H/351L/354I/423K, 70L/126F/191D/246K/325H/351L/366Q/423K, 70L/126F/191D/246K/354I/366Q, 70L/126F/246K/330Q/366Q, 70L/126F/246K/354I/366Q, 126F/211E/220L/253D/316V/342L, 126F/211E/220L/275Q/279L/323V, 126F/211E/260V/423K, 126F/220L/260V, 126F/220L/316V, 56V/220L/260V/423K, 220L/260V, and 220L/260V/423K, wherein the positions are numbered with reference to SEQ ID NO:792. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from E24L/N32S/S264A/Y269S/P330Q/L402V/K403R, E24L/N32S/S264A/Y269S/A382G/K403R/I406M, E24L/N32S/S264A/P330Q, E24L/N32S/Y269S, E24L/N32S/Y269S/A382G/I385V/D389E/L402I/I406M, E24L/N32S/Y269S/K403R, E24L/N32S/P330Q, E24L/S264A/Y269S/D389E/L402V/I406M, F70L/I126F/P175S/N191D/M246K/K325H/L354I/H366Q, F70L/I126F/P175S/K325H/P330Q/M351L/L354I/H366Q, F70L/I126F/N191D/M246K/K325H/M351L/L354I/H366Q/E423K, F70L/I126F/N191D/M246K/K325H/M351L/L354I/E423K, F70L/I126F/N191D/M246K/K325H/M351L/H366Q/E423K, F70L/I126F/N191D/M246K/L354I/H366Q, F70L/I126F/M246K/P330Q/H366Q, F70L/I126F/M246K/L354I/H366Q, I126F/T211E/C220L/G253D/I316V/I342L, I126F/T211E/C220L/D275Q/V279L/L323V, I126F/T211E/T260V/E423K, I126F/C220L/T260V, I126F/C220L/I316V, I56V/C220L/T260V/E423K, C220L/T260V, and C220L/T260V/E423K, wherein the positions are numbered with reference to SEQ ID NO:792. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 954, 956, 958, 960, 962, 964, 966, 968, 970, 972, 974, 976, 978, 980, 982, 984, 986, 988, 990, 992, 994, 996, 998, and/or 1000. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 954, 956, 958, 960, 962, 964, 966, 968, 970, 972, 974, 976, 978, 980, 982, 984, 986, 988, 990, 992, 994, 996, 998, and/or 1000. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 954, 956, 958, 960, 962, 964, 966, 968, 970, 972, 974, 976, 978, 980, 982, 984, 986, 988, 990, 992, 994, 996, 998, and/or 1000.

(39) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 24/32/97/162/202, 24/32/126/198/201/367, 24/32/126/198/202/220/226/260/269S, 24/32/146/226, 24/32/198/201/220/226, 24/32/198/201/226, 24/97/260/367, 32/97/198/202/226/260, 32/202/367, 32/226/367, 56/175/197/211/330/382/385, 56/175/197/253/385/389, 56/175/264/382/385/389/402/406, 56/197/264/279/330/382/389/402/403/406, 56/211/253/316/323, 56/211/264/316/389, 56/211/279/323/330/402, 56/264/385/389, 97/202/367, 162/220/226/367, 175/197/211/264/330, 175/211/264/279/316/323, 175/211/279/323/330/402/403/406, 175/211/323/382/402/403/406, 175/211/403/406, 175/264/316/323/330/403/406, 175/264/316/389/402, 175/264/323/330, 197/211/316/342/406, 197/211/316/382/389/402/403, 197/211/402, 197/279/323, 198/201/367, 198/202/220/269/367, 201/202/367, 211/382/406, 211/385/389, 211/402/403, and 389, wherein the positions are numbered with reference to SEQ ID NO:954. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 24L/32S/97G/162R/202G, 24L/32S/126A/198P/201G/367W, 24L/32S/126A/198P/202G/220L/226V/260V/269S, 24L/32S/146A/226V, 24L/32S/198P/201G/220L/226V, 24L/32S/198P/201G/226V, 24L/97G/260V/367W, 32S/97G/198P/202G/226V/260V, 32S/202G/367W, 32S/226V/367W, 56V/175S/197P/211E/330Q/382G/385V, 56V/175S/197P/253D/385V/389E, 56V/175S/264A/382G/385V/389E/402V/406M, 56V/197P/264A/279L/330Q/382G/389E/402V/403R/406M, 56V/211E/253D/316V/323V, 56V/211E/264A/316V/389E, 56V/211E/279L/323V/330Q/402I, 56V/264A/385V/389E, 97G/202G/367W, 162R/220L/226V/367W, 175S/197P/211E/264A/330Q, 175S/211E/264A/279L/316V/323V, 175S/211E/279L/323V/330Q/402I/403R/406M, 175S/211E/323V/382G/402I/403R/406M, 175S/211E/403R/406M, 175S/264A/316V/323V/330Q/403R/406M, 175S/264A/316V/389E/402V, 175S/264A/323V/330Q, 197P/211E/316V/342L/406M, 197P/211E/316V/382G/389E/402I/403R, 197P/211E/402I, 197P/279L/323V, 198P/201G/367W, 198P/202G/220L/269S/367W, 201G/202G/367W, 211E/382G/406M, 211E/385V/389E, 211E/402I/403R, and D389E, wherein the positions are numbered with reference to SEQ ID NO:954. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from E24L/N32S/H97G/N162R/Q202G, E24L/N32S/F126A/E198P/M201G/N367W, E24L/N32S/F126A/E198P/Q202G/C220L/W226V/T260V/Y269S, E24L/N32S/L146A/W226V, E24L/N32S/E198P/M201G/C220L/W226V, E24L/N32S/E198P/M201G/W226V, E24L/H97G/T260V/N367W, N32S/H97G/E198P/Q202G/W226V/T260V, N32S/Q202G/N367W, N32S/W226V/N367W, I56V/P175S/A197P/T211E/P330Q/A382G/I385V, I56V/P175S/A197P/G253D/I385V/D389E, I56V/P175S/S264A/A382G/I385V/D389E/L402V/I406M, I56V/A197P/S264A/V279L/P330Q/A382G/D389E/L402V/K403R/I406M, I56V/T211E/G253D/I316V/L323V, I56V/T211E/S264A/I316V/D389E, I56V/T211E/V279L/L323V/P330Q/L402I, I56V/S264A/I385V/D389E, H97G/Q202G/N367W, N162R/C220L/W226V/N367W, P175S/A197P/T211E/S264A/P330Q, P175S/T211E/S264A/V279L/I316V/L323V, P175S/T211E/V279L/L323V/P330Q/L402I/K403R/I406M, P175S/T211E/L323V/A382G/L402I/K403R/I406M, P175S/T211E/K403R/I406M, P175S/S264A/I316V/L323V/P330Q/K403R/I406M, P175S/S264A/I316V/D389E/L402V, P175S/S264A/L323V/P330Q, A197P/T211E/I316V/I342L/I406M, A197P/T211E/I316V/A382G/D389E/L402I/K403R, A197P/T211E/L402I, A197P/V279L/L323V, E198P/M201G/N367W, E198P/Q202G/C220L/Y269S/N367W, M201G/Q202G/N367W, T211E/A382G/I406M, T211E/I385V/D389E, T211E/L402I/K403R, and D389E, wherein the positions are numbered with reference to SEQ ID NO: 954. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS:1002, 1004, 1006, 1008, 1010, 1012, 1014, 1016, 1018, 1020, 1022, 1024, 1026, 1028, 1030, 1032, 1034, 1036, 1038, 1040, 1042, 1044, 1046, 1048, 1050, 1052, 1054, 1056, 1058, 1060, 1062, 1064, 1066, 1068, 1070, 1072, 1074, 1076, and/or 1078. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 1002, 1004,

1006, 1008, 1010, 1012, 1014, 1016, 1018, 1020, 1022, 1024, 1026, 1028, 1030, 1032, 1034, 1036, 1038, 1040, 1042, 1044, 1046, 1048, 1050, 1052, 1054, 1056, 1058, 1060, 1062, 1064, 1066, 1068, 1070, 1072, 1074, 1076, and/or 1078. In some additional embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 1002, 1004, 1006, 1008, 1010, 1012, 1014, 1016, 1018, 1020, 1022, 1024, 1026, 1028, 1030, 1032, 1034, 1036, 1038, 1040, 1042, 1044, 1046, 1048, 1050, 1052, 1054, 1056, 1058, 1060, 1062, 1064, 1066, 1068, 1070, 1072, 1074, 1076, and/or 1078.

(40) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 24/32/175/211/260/330/403/406, 24/32/175/226/253/275/316, 24/32/201/226/260/264/330/402/406, 24/162/175/198/211/226/275/316/323/389, 24/162/175/201/275/316, 24/162/198/201/211/226/323/351, 24/162/198/201/226/351, 24/162/201/253/264/351/402/406, 24/175/198/211/226/253/316, 24/175/201/275/316/351, 24/175/201/316, 24/175/211/220/260/275/330/389, 24/175/211/253/316, 24/175/211/316/330, 24/175/226/323/351, 24/198/201, 24/198/201/211, 24/198/201/211/220/260, 24/198/201/220/275/389/402/406, 24/198/201/226/330/389, 24/198/201/351, 24/201/211/253/323/351/366/389/402/403, 24/201/226/253/402/403/406, 24/226/330/351/403, 175/198/201/211/226/260/264/323/402/406, and 175/198/226/260/351/402/403/406, wherein the positions are numbered with reference to SEQ ID NO: 1054. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 24L/32S/175S/211E/260V/330Q/403R/406M, 24L/32S/175S/226V/253D/275Q/316V, 24L/32S/201G/226V/260V/264A/330Q/402I/406M, 24L/162R/175S/198P/211E/226V/275Q/316V/323V/389E, 24L/162R/175S/201G/275Q/316V, 24L/162R/198P/201G/211E/226V/323V/351M, 24L/162R/198P/201G/226V/351M, 24L/162R/201G/253D/264A/351M/402I/406M, 24L/175S/198P/211E/226V/253D/316V, 24L/175S/201G/275Q/316V/351M, 24L/175S/201G/316V, 24L/175S/211E/220L/260V/275Q/330Q/389E, 24L/175S/211E/253D/316V, 24L/175S/211E/316V/330Q, 24L/175S/226V/323V/351M, 24L/198P/201G, 24L/198P/201G/211E, 24L/198P/201G/211E/220L/260V, 24L/198P/201G/220L/275Q/389E/402I/406M, 24L/198P/201G/226V/330Q/389E, 24L/198P/201G/351M, 24L/201G/211E/253D/323V/351M/366H/389E/402I/403R, 24L/201G/226V/253D/402I/403R/406M, 24L/226V/330Q/351M/403R, 175S/198P/201G/211E/226V/260V/264A/323V/402I/406M, and 175S/198P/226V/260V/351M/402I/403R/406M, wherein the positions are numbered with reference to SEQ ID NO: 1054. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from E24L/N32S/P175S/T211E/T260V/P330Q/K403R/I406M, E24L/N32S/P175S/W226V/G253D/D275Q/I316V, E24L/N32S/M201G/W226V/T260V/S264A/P330Q/L402I/I406M, E24L/N162R/P175S/E198P/T211E/W226V/D275Q/I316V/L323V/D389E, E24L/N162R/P175S/M201G/D275Q/I316V, E24L/N162R/E198P/M201G/T211E/W226V/L323V/L351M, E24L/N162R/E198P/M201G/W226V/L351M, E24L/N162R/M201G/G253D/S264A/L351M/L402I/I406M, E24L/P175S/E198P/T211E/W226V/G253D/I316V, E24L/P175S/M201G/D275Q/I316V/L351M, E24L/P175S/M201G/I316V, E24L/P175S/T211E/C220L/T260V/D275Q/P330Q/D389E, E24L/P175S/T211E/G253D/I316V, E24L/P175S/T211E/I316V/P330Q, E24L/P175S/W226V/L323V/L351M, E24L/E198P/M201G, E24L/E198P/M201G/T211E, E24L/E198P/M201G/T211E/C220L/T260V, E24L/E198P/M201G/C220L/D275Q/D389E/L402I/I406M, E24L/E198P/M201G/W226V/P330Q/D389E, E24L/E198P/M201G/L351M, E24L/M201G/T211E/G253D/L323V/L351M/Q366H/D389E/L402I/K403R, E24L/M201G/W226V/G253D/L402I/K403R/I406M, E24L/W226V/P330Q/L351M/K403R, P175S/E198P/M201G/T211E/W226V/T260V/S264A/L323V/L402I/I406M, and P175S/E198P/W226V/T260V/L351M/L402I/K403R/I406M, wherein the positions are numbered with reference to SEQ ID NO: 1054. In some embodiments, the polypeptide sequence of the engineered

glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 2596, 2598, 2600, 2602, 2604, 2606, 2608, 2610, 2612, 2614, 2616, 2618, 2620, 2622, 2624, 2626, 2628, 2630, 2632, 2634, 2636, 2638, 2640, 2642, 2644, and 2646. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 2596, 2598, 2600, 2602, 2604, 2606, 2608, 2610, 2612, 2614, 2616, 2618, 2620, 2622, 2624, 2626, 2628, 2630, 2632, 2634, 2636, 2638, 2640, 2642, 2644, and 2646. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 2596, 2598, 2600, 2602, 2604, 2606, 2608, 2610, 2612, 2614, 2616, 2618, 2620, 2622, 2624, 2626, 2628, 2630, 2632, 2634, 2636, 2638, 2640, 2642, 2644, and 2646.

(41) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 21/127/129/161, 21/127/129/161/162, 21/127/129/162/199/200, 127/129/161/162/199, 127/129/161/199/200, 127/129/162, 156, 156/161, 156/161/162, 156/162/199, and 156/199/200, wherein the positions are numbered with reference to SEQ ID NO: 1002. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 21Y/127H/129A/161S, 21Y/127H/129A/161S/162G, 21Y/127H/129A/162T/199H/200A, 127H/129A/162T, 127Q/129A/161S/162G/199H, 127Q/129A/161S/199H/200A, 156R, 156R/161S, 156R/161S/162G, 156R/161S/162T, 156R/162G/199H, and 156R/199H/200A, wherein the positions are numbered with reference to SEQ ID NO: 1002. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from W21Y/L127H/P129A/K161S, W21Y/L127H/P129A/K161S/N162G, W21Y/L127H/P129A/N162T/G199H/N200A, L127H/P129A/N162T, L127Q/P129A/K161S/N162G/G199H, L127Q/P129A/K161S/G199H/N200A, F156R, F156R/K161S, F156R/K161S/N162G, F156R/K161S/N162T, F156R/N162G/G199H, and F156R/G199H/N200A, wherein the positions are numbered with reference to SEQ ID NO: 1002. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 2648, 2650, 2652, 2654, 2656, 2658, 2660, 2662, 2664, 2666, 2668, 2670, 2672, 2674, 2676, 2678, 2680, 2682, and 2684. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 2648, 2650, 2652, 2654, 2656, 2658, 2660, 2662, 2664, 2666, 2668, 2670, 2672, 2674, 2676, 2678, 2680, 2682, and 2684. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 2648, 2650, 2652, 2654, 2656, 2658, 2660, 2662, 2664, 2666, 2668, 2670, 2672, 2674, 2676, 2678, 2680, 2682, and 2684.

(42) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 2/7/12/15/175/260/318, 7/12/400/435, 7/318/451, 12/15/57/71/175/260/400/402, 12/15/57/220/254/260/318/402, 12/15/57/318/402/435, 12/15/318/400/402/406, 12/57/175/451, 12/175/260/264/318/400, 12/175/400/402/406, 12/318/402/404/406/451, 12/318/402/404/451, 15/175/318/400/402, 57/175/220/260/264/402, 57/175/404, 57/220/260/400/402/406, 57/260/400/402/404, 57/400/402, 57/402, 152/192/195, 160/186/195, 195, 260, and 400/402, wherein the positions are numbered with reference to SEQ ID NO: 2600. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 2-/7E/12S/15K/175S/260V/318D, 7E/12S/400Q/435V, 7E/318D/451N, 12S/15K/57K/71I/175S/260V/400Q/402I, 12S/15K/57K/220L/254K/260V/318D/402I, 12S/15K/57K/318D/402I/435V, 12S/15K/318D/400Q/402I/406M, 12S/57K/175S/451N, 12S/175S/260V/264A/318D/400Q, 12S/175S/400Q/402I/406M, 12S/318D/402I/404S/406M/451N, 12S/318D/402I/404S/451N, 15K/175S/318D/400Q/402I, 57K/175S/220L/260V/264A/402I, 57K/175S/404G, 57K/220L/260V/400Q/402I/406M, 57K/260V/400Q/402I/404S, 57K/400Q/402I, 57K/402I, 152V/192D/195P, 160V/186M/195P, 195P, 260V, and 400Q/402I, wherein the positions are numbered with reference to SEQ ID NO: 2600. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions

selected from H2-/H7E/A12S/R15K/P175S/T260V/E318D, H7E/A12S/E400Q/A435V, H7E/E318D/K451N, A12S/R15K/I57K/V71I/P175S/T260V/E400Q/L402I, A12S/R15K/I57K/C220L/T254K/T260V/E318D/L402I, A12S/R15K/I57K/E318D/L402I/A435V, A12S/R15K/E318D/E400Q/L402I/I406M, A12S/I57K/P175S/K451N, A12S/P175S/T260V/S264A/E318D/E400Q, A12S/P175S/E400Q/L402I/I406M, A12S/E318D/L402I/D404S/I406M/K451N, A12S/E318D/L402I/D404S/K451N, R15K/P175S/E318D/E400Q/L402I, I57K/P175S/C220L/T260V/S264A/L402I, I57K/P175S/D404G, I57K/C220L/T260V/E400Q/L402I/I406M, I57K/T260V/E400Q/L402I/D404S, I57K/E400Q/L402I, I57K/L402I, L152V/E192D/R195P, L160V/F186M/R195P, R195P, T260V, and E400Q/L402I, wherein the positions are numbered with reference to SEQ ID NO: 2600. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 32, 135, 148, 152, 186, 237, 239, 240, 323, 325, 326, 327, 330, 331, and 356, wherein the positions are numbered with reference to SEQ ID NO: 2600. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 32R, 135A, 148A, 152V, 186V, 237T, 239E, 239F, 239Y, 240A, 240P, 323L, 325G, 325R, 326M, 327V, 330A, 331C, 331H, 331S, and 356G, wherein the positions are numbered with reference to SEQ ID NO: 2600. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from N32R, V135A, S148A, L152V, F186V, D237T, I239E, I239F, I239Y, T240A, T240P, V323L, H325G, H325R, F326M, A327V, P330A, R331C, R331H, R331S, and F356G, wherein the positions are numbered with reference to SEQ ID NO: 2600. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 2686, 2688, 2690, 2692, 2694, 2696, 2698, 2700, 2702, 2704, 2706, 2708, 2710, 2712, 2714, 2716, 2718, 2720, 2722, 2724, 2726, 2728, 2730, 2732, 2734, 2736, 2738, 2740, 2742, 2744, 2746, 2748, 2750, 2752, 2754, 2756, 2758, 2760, 2762, 2764, 2766, 2768, 2770, 2772, and 2774. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 2686, 2688, 2690, 2692, 2694, 2696, 2698, 2700, 2702, 2704, 2706, 2708, 2710, 2712, 2714, 2716, 2718, 2720, 2722, 2724, 2726, 2728, 2730, 2732, 2734, 2736, 2738, 2740, 2742, 2744, 2746, 2748, 2750, 2752, 2754, 2756, 2758, 2760, 2762, 2764, 2766, 2768, 2770, 2772, and 2774. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 2686, 2688, 2690, 2692, 2694, 2696, 2698, 2700, 2702, 2704, 2706, 2708, 2710, 2712, 2714, 2716, 2718, 2720, 2722, 2724, 2726, 2728, 2730, 2732, 2734, 2736, 2738, 2740, 2742, 2744, 2746, 2748, 2750, 2752, 2754, 2756, 2758, 2760, 2762, 2764, 2766, 2768, 2770, 2772, and 2774.

(43) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 14/23/274/326/329, 14/31/184/274/322/326/329/330, 14/56/184/185/194/238/239/274/329/399, 14/56/184/194/234/315/326/329/399/401, 14/56/194/238/315/325/326/329/330/399, 14/56/252/274/315/326/329/330/401, 14/184/185/194/234/325/326/329, 14/184/185/194/388/399/401, 14/184/194/355/399, 14/185/194/238/399, 14/185/236/238/239/274/322/326/329/355/399/401, 14/194, 14/322/326/330, 14/326/330, 14/355, 14/355/399, 23/31/147/184/185/238/252/325/329/330/388/401, 31/56/315/329/330, 65, 65/114/132/238, 65/238/240, 147/236/238/243/315/329/330/399/401, 185/194/236/239/325/326, 223/412, and 238, wherein the positions are numbered with reference to SEQ ID NO: 2718. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 14R/23Q/274Q/326V/329A, 14R/31R/184A/274Q/322L/326V/329A/330H, 14R/56K/184A/185M/194P/238M/239A/274Q/329A/399Q, 14R/56K/184A/194P/234Y/315V/326V/329Q/399Q/401I, 14R/56K/194P/238M/315V/325M/326V/329A/330H/399Q, 14R/56K/252D/274Q/315V/326V/329A/330H/401I, 14R/184A/185M/194P/234Y/325M/326V/329A, 14R/184A/185M/194P/388E/399Q/401I, 14R/184A/194P/355G/399Q, 14R/185M/194P/238M/399Q, 14R/185M/236T/238M/239A/274Q/322L/326V/329Q/355G/399Q/401I, 14R/194P,

14R/322L/326V/330H, 14R/355G, 14R/355G/399Q, 23Q/31R/147A/184A/185M/238M/252D/325M/329A/330H/388E/401I, 31R/56K/315V/329A/330H, 65D, 65D/114E/132R/238M, 65D/238T/240S, 147A/236T/238M/243G/315V/329A/330H/399Q/401I, 185M/194P/236T/239A/325M/326V, 223T/412S, and 238M, wherein the positions are numbered with reference to SEQ ID NO: 2718. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from K14R/L23Q/D274Q/A326V/P329A, K14R/N31R/M184A/D274Q/V322L/A326V/P329A/R330H, K14R/I56K/M184A/F185M/R194P/I238M/T239A/D274Q/P329A/E399Q, K14R/I56K/M184A/R194P/F234Y/I315V/A326V/P329Q/E399Q/L401I, K14R/I56K/R194P/I238M/I315V/F325M/A326V/P329A/R330H/E399Q, K14R/I56K/G252D/D274Q/I315V/A326V/P329A/R330H/L401I, K14R/M184A/F185M/R194P/F234Y/F325M/A326V/P329A, K14R/M184A/F185M/R194P/D388E/E399Q/L401I, K14R/M184A/R194P/F355G/E399Q, K14R/F185M/R194P/I238M/E399Q, K14R/F185M/D236T/I238M/T239A/D274Q/V322L/A326V/P329Q/F355G/E399Q/L401I, K14R/R194P, K14R/V322L/A326V/R330H, K14R/A326V/R330H, K14R/F355G, K14R/F355G/E399Q, L23Q/N31R/S147A/M184A/F185M/I238M/G252D/F325M/P329A/R330H/D388E/L401I, N31R/I56K/I315V/P329A/R330H, E65D, E65D/Q114E/H132R/I238M, E65D/I238T/N240S, S147A/D236T/I238M/D243G/I315V/P329A/R330H/E399Q/L401I, F185M/R194P/D236T/T239A/F325M/A326V, S223T/I412S, and I238M, wherein the positions are numbered with reference to SEQ ID NO: 2718. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 11, 45, 55, 56, 58, 65, 104, 113, 114, 132, 135, 138, 165, 238, 256, 273, 286, 309, 391, 422, 430, and 449, wherein the positions are numbered with reference to SEQ ID NO: 2718. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 11G, 11Q, 45F, 45V, 55L, 56T, 58R, 65N, 65S, 104L, 113V, 114R, 132Q, 132S, 135L, 138G, 138K, 165P, 238G, 256P, 273R, 286R, 309E, 309H, 391R, 422R, 430L, 430V, and 449F, wherein the positions are numbered with reference to SEQ ID NO: 2718. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from S11G, S11Q, L45F, L45V, I55L, I56T, K58R, E65N, E65S, M104L, L113V, Q114R, H132Q, H132S, N135L, N138G, N138K, E165P, I238G, E256P, E273R, N286R, K309E, K309H, N391R, K422R, E430L, E430V, and Y449F, wherein the positions are numbered with reference to SEQ ID NO: 2718. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 2776, 2778, 2780, 2782, 2784, 2786, 2788, 2790, 2792, 2794, 2796, 2798, 2800, 2802, 2804, 2806, 2808, 2810, 2812, 2814, 2816, 2818, 2820, 2822, 2824, 2826, 2828, 2830, 2832, 2834, 2836, 2838, 2840, 2842, 2844, 2846, 2848, 2850, 2852, 2854, 2856, 2858, 2860, 2862, 2864, 2866, 2868, 2870, 2872, 2874, 2876, 2878, 2880, and 2882. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 2776, 2778, 2780, 2782, 2784, 2786, 2788, 2790, 2792, 2794, 2796, 2798, 2800, 2802, 2804, 2806, 2808, 2810, 2812, 2814, 2816, 2818, 2820, 2822, 2824, 2826, 2828, 2830, 2832, 2834, 2836, 2838, 2840, 2842, 2844, 2846, 2848, 2850, 2852, 2854, 2856, 2858, 2860, 2862, 2864, 2866, 2868, 2870, 2872, 2874, 2876, 2878, 2880, and 2882. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 2776, 2778, 2780, 2782, 2784, 2786, 2788, 2790, 2792, 2794, 2796, 2798, 2800, 2802, 2804, 2806, 2808, 2810, 2812, 2814, 2816, 2818, 2820, 2822, 2824, 2826, 2828, 2830, 2832, 2834, 2836, 2838, 2840, 2842, 2844, 2846, 2848, 2850, 2852, 2854, 2856, 2858, 2860, 2862, 2864, 2866, 2868, 2870, 2872, 2874, 2876, 2878, 2880, and 2882.

(44) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 23/31/185/324/401, 23/31/185/355, 31/134/185/252/274/324/388, 31/134/238/252/322/324/388, 31/134/252/324, 31/184/185/238/239/322, 31/236, 31/322, 31/388, 134/184/185/234/236/239/274/324/388, 184/185/322, 234/236/238/322/324/355, 236, 238/324/329/355, 322/324, and 324/329/355/401, numbered with reference to SEQ ID NO: 2814. In some embodiments,

the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 23Q/31R/185M/324G/401I, 23Q/31R/185M/355G, 31R/134A/185M/252D/274Q/324G/388E, 31R/134A/238E/252D/322L/324G/388E, 31R/134A/252D/324G, 31R/184A/185M/238Y/239A/322L, 31R/236T, 31R/322L, 31R/388E, 134A/184A/185M/234Y/236T/239A/274Q/324G/388E, 184A/185M/322L, 234Y/236T/238E/322L/324G/355G, 236T, 238Y/324G/329Q/355G, 322L/324G, and 324G/329Q/355G/401I, numbered with reference to SEQ ID NO: 2814. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from L23Q/N31R/F185M/H324G/L401I, L23Q/N31R/F185M/F355G, N31R/V134A/F185M/G252D/D274Q/H324G/D388E, N31R/V134A/M238E/G252D/V322L/H324G/D388E, N31R/V134A/G252D/H324G, N31R/M184A/F185M/M238Y/T239A/V322L, N31R/D236T, N31R/V322L, N31R/D388E, V134A/M184A/F185M/F234Y/D236T/T239A/D274Q/H324G/D388E, M184A/F185M/V322L, F234Y/D236T/M238E/V322L/H324G/F355G, D236T, M238Y/H324G/A329Q/F355G, V322L/H324G, and H324G/A329Q/F355G/L401I, numbered with reference to SEQ ID NO: 2814. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 122, 164, 176, 177, 316, 325, 400, 425, 426, 427, 440, and 446, numbered with reference to SEQ ID NO: 2814. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 122L, 164H, 164M, 164R, 176K, 176L, 176N, 176R, 177A, 316R, 325L, 400V, 425R, 426A, 426R, 427R, 440R, and 446R, numbered with reference to SEQ ID NO: 2814. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from I122L, V164H, V164M, V164R, V176K, V176L, V176N, V176R, E177A, G316R, M325L, T400V, K425R, S426A, S426R, I427R, I440R, and S446R, numbered with reference to SEQ ID NO: 2814. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 2884, 2886, 2888, 2890, 2892, 2894, 2896, 2898, 2900, 2902, 2904, 2906, 2908, 2910, 2912, 2914, 2916, 2918, 2920, 2922, 2924, 2926, 2928, 2830, 2932, 2934, 2936, 2938, 2940, 2942, 2944, 2946, 2948, and 2950, numbered with reference to SEQ ID NO: 2814. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 2884, 2886, 2888, 2890, 2892, 2894, 2896, 2898, 2900, 2902, 2904, 2906, 2908, 2910, 2912, 2914, 2916, 2918, 2920, 2922, 2924, 2926, 2928, 2830, 2932, 2934, 2936, 2938, 2940, 2942, 2944, 2946, 2948, and 2950. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 2884, 2886, 2888, 2890, 2892, 2894, 2896, 2898, 2900, 2902, 2904, 2906, 2908, 2910, 2912, 2914, 2916, 2918, 2920, 2922, 2924, 2926, 2928, 2830, 2932, 2934, 2936, 2938, 2940, 2942, 2944, 2946, 2948, and 2950.

(45) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 11, 11/45/58/132/138/286, 11/58, 11/58/65/104/107/138/286, 11/58/65/107/135/223/286/391/430, 11/58/65/135/138, 11/58/104/107/114/138/223/391, 11/58/114/286/309/391, 11/58/132/165/286/391/422/430, 11/58/138/309, 11/65/104/107/138/165/286/391/430, 11/65/107/135/165/391/430, 11/65/132/135/138/223/391, 11/65/132/135/252, 11/65/132/391/430, 11/65/135/138, 11/104/132/138/309/391, 11/104/132/138/391, 11/107/114/223/309/430, 11/107/138, 11/114/135/138/223, 11/114/223/252/286/391, 11/132/135/138/223/286, 11/132/138/223/286/391, 11/138/165/223/309, 58/65/138, 58/65/138/165, 58/65/165/309/430, 58/104/114/165/391, 107/114/132/138, 107/430, 135/138/165/309/430, and 223/309, wherein the positions are numbered with reference to SEQ ID NO: 2884. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 11G/45V/58R/132Q/138G/286R, 11G/58R/104L/107G/114R/138K/223T/391R, 11G/58R/114R/286R/309H/391R, 11G/58R/132Q/165P/286R/391R/4225/430L, 11G/65N/132Q/391R/430L, 11G/104L/132Q/138G/309H/391R, 11G/107G/114R/223T/309H/430L, 11Q, 11Q/58R, 11Q/58R/65N/104L/107G/138G/286R, 11Q/58R/65N/107G/135L/223T/286R/391R/430L,

58R/236T, 107G/122L/326T/425R/46A/446R, 107G/164H/236T/400V/446R, 107G/176R/322L/325L/440R/446R, 107G/176R/400V/425R/427R/440R, 107G/236T, 107G/440R, 122L/164H/176R/324G/400V, 122L/164H/400V/440R, 122L/164H/400V/440R/446R, 122L/176R/236T/400V, 122L/176R/324G/400V/440R, 122L/400V, 122L/425R/426R/446R, 122L/440R/446R, 139V, 139V/252D, 164H, 164H/271G/425R/426R, 164H/322L/324G/325L/400V/425R/440R/446R, 164H/400V, 176R/400V, 189R, and 324G/400V/425R/426R/440R/446R, wherein the positions are numbered with reference to SEQ ID NO: 3016. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from G8S/K448A, K58R/P107G/I122L/V176R/D236T/H324G/M325L/T400V/S426R/I427R, K58R/P107G/I122L/D236T/H324G/T400V/K425R/S446R, K58R/P107G/I122L/V322L/T400V/K425R/I427R/I440R/S446R, K58R/P107G/V164H/T400V, K58R/P107G/D236T/T400V, K58R/P107G/T400V, K58R/I122L/V164H/D236T/S446R, K58R/I122L/V176R/D236T/T400V/S446R, K58R/I122L/V176R/V322L/H324G/S426A/I427R, K58R/I122L/V322L/M325L, K58R/I122L/M325L, K58R/I122L/I440R, K58R/V164H/V176R, K58R/V164H/H324G/K425R/I427R, K58R/V176R/D236T, K58R/D236T, P107G/I122L/D236T/K425R/S426A/S446R, P107G/V164H/D236T/T400V/S446R, P107G/V176R/V322L/M325L/I440R/S446R, P107G/V176R/T400V/K425R/I427R/I440R, P107G/D236T, P107G/I440R, I122L/V164H/V176R/H324G/T400V, I122L/V164H/T400V/I440R, I122L/V164H/T400V/I440R/S446R, I122L/V176R/D236T/T400V, I122L/V176R/H324G/T400V/I440R, I122L/T400V, I122L/K425R/S426R/S446R, I122L/I440R/S446R, I139V, I139V/G252D, V164H, V164H/S271G/K425R/S426R, V164H/V322L/H324G/M325L/T400V/K425R/I440R/S446R, V164H/T400V, V176R/T400V, P189R, and H324G/T400V/K425R/S426R/I440R/S446R, wherein the positions are numbered with reference to SEQ ID NO: 3016. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 9, 65, 106, 115, 116, 172, 178, 200, 210, 213, 240, 242, 245, 255, 324/423, 385, 408, 409, 411, 412, 415, 416, 423, and 447, wherein the positions are numbered with reference to SEQ ID NO: 3016. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 9C, 9M, 65S, 106A, 115D, 115R, 116V, 172R, 172S, 178K, 200A, 200T, 200V, 210L, 210V, 213G, 240C, 240E, 240L, 240P, 240V, 242I, 242Y, 245M, 255P, 324R/423R, 385R, 408P, 409L, 411Q, 412R, 415A, 415H, 415K, 415R, 416R, 423A, 423R, 447L, and 447R, wherein the positions are numbered with reference to SEQ ID NO: 3016. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from S9C, S9M, N65S, K106A, N115D, N115R, L116V, N172R, N172S, Q178K, G200A, G200T, G200V, E210L, E210V, A213G, N240C, N240E, N240L, N240P, N240V, A242I, A242Y, K245M, D255P, H324R/N423R, V385R, E408P, T409L, E411Q, 1412R, G415A, G415H, G415K, G415R, K416R, N423A, N423R, N447L, and N447R, wherein the positions are numbered with reference to SEQ ID NO: 3016. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 3018, 3020, 3022, 3024, 3026, 3028, 3030, 3032, 3034, 3036, 3038, 3040, 3042, 3044, 3046, 3048, 3050, 3052, 3054, 3056, 3058, 3060, 3062, 3064, 3066, 3068, 3070, 3072, 3074, 3076, 3078, 3080, 3082, 3084, 3086, 3088, 3090, 3092, 3094, 3096, 3098, 3100, 3102, 3104, 3106, 3108, 3110, 3112, 3114, 3116, 3118, 3120, 3122, 3124, 3126, 3128, 3130, 3132, 3134, 3136, 3138, 3140, 3142, 3144, 3146, 3148, 3150, 3152, 3154, 3156, 3158, 3160, 3162, 3164, 3166, 3168, 3170, 3172, 3174, 3176, 3178, and 3180. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 3018, 3020, 3022, 3024, 3026, 3028, 3030, 3032, 3034, 3036, 3038, 3040, 3042, 3044, 3046, 3048, 3050, 3052, 3054, 3056, 3058, 3060, 3062, 3064, 3066, 3068, 3070, 3072, 3074, 3076, 3078, 3080, 3082, 3084, 3086, 3088, 3090, 3092, 3094, 3096, 3098, 3100, 3102, 3104, 3106, 3108, 3110, 3112, 3114, 3116, 3118, 3120, 3122, 3124, 3126, 3128, 3130, 3132, 3134, 3136, 3138, 3140, 3142, 3144, 3146, 3148, 3150, 3152, 3154, 3156, 3158, 3160, 3162, 3164, 3166, 3168, 3170, 3172, 3174, 3176, 3178, and 3180. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 3018, 3020, 3022, 3024, 3026, 3028, 3030, 3032, 3034, 3036, 3038, 3040, 3042, 3044, 3046, 3048,

3050, 3052, 3054, 3056, 3058, 3060, 3062, 3064, 3066, 3068, 3070, 3072, 3074, 3076, 3078, 3080, 3082, 3084, 3086, 3088, 3090, 3092, 3094, 3096, 3098, 3100, 3102, 3104, 3106, 3108, 3110, 3112, 3114, 3116, 3118, 3120, 3122, 3124, 3126, 3128, 3130, 3132, 3134, 3136, 3138, 3140, 3142, 3144, 3146, 3148, 3150, 3152, 3154, 3156, 3158, 3160, 3162, 3164, 3166, 3168, 3170, 3172, 3174, 3176, 3178, and 3180.

(47) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from, 8, 8/107/139/164/415/416/440/448,

8/107/139/255/322/325/415/416/440/448, 8/107/164/255, 8/107/164/423/440, 8/139, 8/139/164/200/236/240/322/440/448, 8/139/164/236/416/423/440, 8/139/189/240/325/416, 8/139/200/236/240/255/423, 8/139/255/415/416, 8/164, 8/164/189/200/322/324/325/416/448, 8/164/240/423, 8/164/252/255/448, 8/164/448, 8/236/240/252/448, 8/240/423, 8/252/255/322/325/448, 12/164/440, 107/139, 107/139/200/240/322/324/325/448, 107/236/240/325/440, 107/240/252/423/448, 107/423, 139, 139/164/236/240, 139/255/325/415/440, 164/189, 164/189/236/240, 164/189/240/252/415/423, 164/200/236, 164/200/236/240/324/416/440, 164/200/236/255/322/324/423/440, 164/236, 164/236/240/440, 164/236/423, 164/322/325, 164/322/325/416/423, 164/416/448, 200/236/322/325/416, 200/322/325/415/448, 236/415/416, 240, 240/252/255/322/415/416, 240/252/255/415/448, 255/423, 325, and 415/416/448, wherein the positions are numbered with reference to SEQ ID NO: 3082. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 2-, 8S, 8S/107G/139V/164H/415A/416R/440R/448A, 8S/107G/139V/255P/322L/325L/415A/416R/440R/448A, 8S/107G/164H/255P, 8S/107G/164H/423R/440R, 8S/139V, 85/139V/164H/200A/236T/240E/322L/440R/448A, 85/139V/164H/236T/416R/423R/440R, 8S/139V/189R/240E/325L/416R, 85/139V/200A/236T/240E/255P/423R, 8S/139V/255P/415A/416R, 8S/164H, 8S/164H/189R/200A/322L/324G/325L/416R/448A, 8S/164H/240E/423R, 8S/164H/252D/255P/448A, 8S/164H/448A, 8S/236T/240E/252D/448A, 8S/240E/423R, 8S/252D/255P/322L/325L/448A, 12S/164H/440R, 107G/139V, 107G/139V/200A/240E/322L/324G/325L/448A, 107G/236T/240E/325L/440R, 107G/240E/252D/423R/448A, 107G/423R, 139V, 139V/164H/236T/240E, 139V/255P/325L/415A/440R, 164H/189R, 164H/189R/236T/240E, 164H/189R/240E/252D/415A/423R, 164H/200A/236T, 164H/200A/236T/240E/324G/416R/440R, 164H/200A/236T/255P/322L/324G/423R/440R, 164H/236T, 164H/236T/240E/440R, 164H/236T/423R, 164H/322L/325L, 164H/322L/325L/416R/423R, 164H/416R/448A, 200A/236T/322L/325L/416R, 200A/322L/325L/415A/448A, 236T/415A/416R, 240E, 240E/252D/255P/322L/415A/416R, 240E/252D/255P/415A/448A, 255P/423R, 325L, and 415A/416R/448A, wherein the positions are numbered with reference to SEQ ID NO: 3082. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from H2-, G8S, G8S/P107G/I139V/V164H/G415A/K416R/I440R/K448A, G8S/P107G/I139V/D255P/V322L/M325L/G415A/K416R/I440R/K448A, G8S/P107G/V164H/D255P, G8S/P107G/V164H/N423R/I440R, G8S/I139V, G8S/I139V/V164H/G200A/D236T/N240E/V322L/I440R/K448A, G8S/I139V/V164H/D236T/K416R/N423R/I440R, G8S/I139V/P189R/N240E/M325L/K416R, G8S/I139V/G200A/D236T/N240E/D255P/N423R, G8S/I139V/D255P/G415A/K416R, G8S/V164H, G8S/V164H/P189R/G200A/V322L/H324G/M325L/K416R/K448A, G8S/V164H/N240E/N423R, G8S/V164H/G252D/D255P/K448A, G8S/V164H/K448A, G8S/D236T/N240E/G252D/K448A, G8S/N240E/N423R, G8S/G252D/D255P/V322L/M325L/K448A, T12S/V164H/I440R, P107G/I139V, P107G/I139V/G200A/N240E/V322L/H324G/M325L/K448A, P107G/D236T/N240E/M325L/I440R, P107G/N240E/G252D/N423R/K448A, P107G/N423R, I139V, I139V/V164H/D236T/N240E, I139V/D255P/M325L/G415A/I440R, V164H/P189R, V164H/P189R/D236T/N240E, V164H/P189R/N240E/G252D/G415A/N423R, V164H/G200A/D236T, V164H/G200A/D236T/N240E/H324G/K416R/I440R, V164H/G200A/D236T/D255P/V322L/H324G/N423R/I440R, V164H/D236T, V164H/D236T/N240E/I440R, V164H/D236T/N423R, V164H/V322L/M325L, V164H/V322L/M325L/K416R/N423R, V164H/K416R/K448A, G200A/D236T/V322L/M325L/K416R,

G200A/V322L/G415A/K448A, N240E, N240E/G252D/D255P/V322L/G415A/K416R, N240E/G252D/D255P/G415A/K448A, D255P/N423R, M325L, and G415A/K416R/K448A, wherein the positions are numbered with reference to SEQ ID NO: 3082. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 2, 3, 8, 34, 72, 73, 75, 113, 114, 186, 189, 221, 235, 237, 239, 256, 286, 299, 305, 309, 312, 313, 323, 355, 389, 406, 422, 438, and 446, wherein the positions are numbered with reference to SEQ ID NO: 3082. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 2N, 2S, 2T, 3I, 8S, 34R, 72Y, 73A, 73P, 73T, 73V, 75H, 113I, 114V, 186G, 186I, 189S, 221K, 235M, 237L, 237M, 237V, 239A, 256I, 256L, 256S, 256T, 286L, 286S, 299A, 299L, 299R, 299V, 305G, 309R, 312S, 312T, 312V, 313D, 323P, 355A, 389F, 389G, 406F, 406G, 406N, 406Q, 422C, 422R, 422S, 438T, 446H, and 446P, wherein the positions are numbered with reference to SEQ ID NO: 3082. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from H2N, H2S, H2T, H3I, G8S, K34R, L72Y, H73A, H73P, H73T, H73V, P75H, L113I, Q114V, E186G, E186I, P189S, E221K, Q235M, P237L, P237M, P237V, T239A, E256I, E256L, E256S, E256T, N286L, N286S, E299A, E299L, E299R, E299V, D305G, K309R, L312S, L312T, L312V, E313D, D323P, F355A, D389F, D389G, T406F, T406G, T406N, T406Q, K422C, K422R, K422S, E438T, R446H, and R446P, wherein the positions are numbered with reference to SEQ ID NO: 3082. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 20/126/128/155/161, 20/126/128/160/161, 20/126/128/160/161/198, 20/126/128/161, 20/126/128/161/198/199, 20/126/155/160/198, 20/126/155/161, 20/126/160, 20/126/160/161, 20/126/161, 20/128/155/160/161/198, 20/128/155/161/199, 20/128/155/199, 20/128/160/161/198, 20/155/160, 20/155/160/161, 20/155/161, 20/155/161/199, 20/160/161/198, 121/126/128/161/369, 126/128/155/160/161, 126/128/155/160/161/199, 126/128/155/161/199, 126/128/160/161/369, 126/128/160/198/369, 126/128/160/199/369, 126/128/160/369, 126/128/161/199/369, 126/128/161/369, 126/128/369, 126/160/161/199/369, 126/160/198/369, 126/196/198/369, 126/198/369, 126/199/369, 126/369, 128/155/160/161, 128/155/160/161/199, 128/155/160/198, 128/155/199, 128/160/161/369, 128/161/199/369, 128/198/199/369, 128/199/369, 128/369, 155/161/198/199, 155/199, 160/161/369, 161/198/369, 161/369, and 199/369, wherein the positions are numbered with reference to SEQ ID NO: 3082. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 20Y/126H/128A/155R/161T, 20Y/126H/128A/161T, 20Y/126H/155R/160S/198P, 20Y/126H/160S, 20Y/126H/160S/161T, 20Y/126Q/128A/160S/161T, 20Y/126Q/128A/160S/161T/198H, 20Y/126Q/128A/161T/198H/199A, 20Y/126Q/155R/161G, 20Y/126Q/161G, 20Y/128A/155R/160S/161T/198H, 20Y/128A/155R/161T/199A, 20Y/128A/155R/199A, 20Y/128A/160S/161T/198P, 20Y/155R/160S, 20Y/155R/160S/161G, 20Y/155R/161T, 20Y/155R/161T/199A, 20Y/160S/161T/198H, 121F/126H/128A/161T/369N, 126H/128A/155R/160S/161T, 126H/128A/160S/198H/369N, 126H/128A/161G/199A/369N, 126H/128A/161G/369N, 126H/128A/369N, 126H/160S/161G/199A/369N, 126H/160S/198P/369N, 126H/196V/198P/369N, 126Q/128A/155R/160S/161T/199A, 126Q/128A/155R/161G/199A, 126Q/128A/160S/161T/369N, 126Q/128A/160S/199A/369N, 126Q/128A/160S/369N, 126Q/198P/369N, 126Q/199A/369N, 126Q/369N, 128A/155R/160S/161T, 128A/155R/160S/161T/199A, 128A/155R/160S/198P, 128A/155R/199A, 128A/160S/161G/369N, 128A/161G/199A/369N, 128A/198H/199A/369N, 128A/199A/369N, 128A/369N, 155R/161G/198H/199A, 155R/199A, 160S/161G/369N, 160S/161T/369N, 161T/198P/369N, 161T/369N, and 199A/369N, wherein the positions are numbered with reference to SEQ ID NO: 3082. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from W20Y/L126H/P128A/F155R/R161T, W20Y/L126H/P128A/R161T, W20Y/L126H/F155R/K160S/G198P, W20Y/L126H/K160S, W20Y/L126H/K160S/R161T, W20Y/L126Q/P128A/K160S/R161T, W20Y/L126Q/P128A/K160S/R161T/G198H, W20Y/L126Q/P128A/R161T/G198H/N199A, W20Y/L126Q/F155R/R161G, W20Y/L126Q/R161G,

W20Y/P128A/F155R/N199A, W20Y/P128A/F155R/R161T/N199A, W20Y/P128A/F155R/N199A, W20Y/P128A/K160S/R161T/G198P, W20Y/F155R/K160S, W20Y/F155R/K160S/R161G, W20Y/F155R/R161T, W20Y/F155R/R161T/N199A, W20Y/K160S/R161T/G198H, V121F/L126H/P128A/R161T/P369N, L126H/P128A/F155R/K160S/R161T, L126H/P128A/K160S/G198H/P369N, L126H/P128A/R161G/N199A/P369N, L126H/P128A/R161G/P369N, L126H/P128A/P369N, L126H/K160S/R161G/N199A/P369N, L126H/K160S/G198P/P369N, L126H/A196V/G198P/P369N, L126Q/P128A/F155R/K160S/R161T/N199A, L126Q/P128A/F155R/R161G/N199A, L126Q/P128A/K160S/R161T/P369N, L126Q/P128A/K160S/N199A/P369N, L126Q/P128A/K160S/P369N, L126Q/G198P/P369N, L126Q/N199A/P369N, L126Q/P369N, P128A/F155R/K160S/R161T, P128A/F155R/K160S/R161T/N199A, P128A/F155R/K160S/G198P, P128A/F155R/N199A, P128A/K160S/R161G/P369N, P128A/R161G/N199A/P369N, P128A/G198H/N199A/P369N, P128A/N199A/P369N, P128A/P369N, F155R/R161G/G198H/N199A, F155R/N199A, K160S/R161G/P369N, K160S/R161T/P369N, R161T/G198P/P369N, R161T/P369N, and N199A/P369N, wherein the positions are numbered with reference to SEQ ID NO: 3082. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 3182, 3184, 3186, 3188, 3190, 3192, 3194, 3196, 3198, 3200, 3202, 3204, 3206, 3208, 3210, 3212, 3214, 3216, 3218, 3220, 3222, 3224, 3226, 3228, 3230, 3232, 3234, 3236, 3238, 3240, 3242, 3244, 3246, 3248, 3250, 3252, 3254, 3256, 3258, 3260, 3262, 3264, 3266, 3268, 3270, 3272, 3274, 3276, 3278, 3280, 3282, 3284, 3286, 3288, 3290, 3292, 3294, 3296, 3298, 3300, 3302, 3304, 3306, 3308, 3310, 3312, 3314, 3316, 3318, 3320, 3322, 3324, 3326, 3328, 3330, 3332, 3334, 3336, 3338, 3340, 3342, 3344, 3346, 3348, 3350, 3352, 3354, 3356, 3358, 3360, 3362, 3364, 3366, 3368, 3370, 3372, 3374, 3376, 3378, 3380, 3382, 3384, 3386, 3388, 3390, 3392, 3394, 3396, 3398, 3400, 3402, 3404, 3406, 3408, 3410, 3412, 3414, 3416, 3418, 3420, 3422, 3424, 3426, 3428, 3430, 3432, 3434, 3436, 3438, 3440, 3442, 3444, 3446, 3448, 3450, 3452, 3454, 3456, 3458, 3460, 3462, 3464, 3466, 3468, 3470, 3472, 3474, 3476, 3478, 3480, 3482, 3484, 3486, 3488, and 3490. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 3182, 3184, 3186, 3188, 3190, 3192, 3194, 3196, 3198, 3200, 3202, 3204, 3206, 3208, 3210, 3212, 3214, 3216, 3218, 3220, 3222, 3224, 3226, 3228, 3230, 3232, 3234, 3236, 3238, 3240, 3242, 3244, 3246, 3248, 3250, 3252, 3254, 3256, 3258, 3260, 3262, 3264, 3266, 3268, 3270, 3272, 3274, 3276, 3278, 3280, 3282, 3284, 3286, 3288, 3290, 3292, 3294, 3296, 3298, 3300, 3302, 3304, 3306, 3308, 3310, 3312, 3314, 3316, 3318, 3320, 3322, 3324, 3326, 3328, 3330, 3332, 3334, 3336, 3338, 3340, 3342, 3344, 3346, 3348, 3350, 3352, 3354, 3356, 3358, 3360, 3362, 3364, 3366, 3368, 3370, 3372, 3374, 3376, 3378, 3380, 3382, 3384, 3386, 3388, 3390, 3392, 3394, 3396, 3398, 3400, 3402, 3404, 3406, 3408, 3410, 3412, 3414, 3416, 3418, 3420, 3422, 3424, 3426, 3428, 3430, 3432, 3434, 3436, 3438, 3440, 3442, 3444, 3446, 3448, 3450, 3452, 3454, 3456, 3458, 3460, 3462, 3464, 3466, 3468, 3470, 3472, 3474, 3476, 3478, 3480, 3482, 3484, 3486, 3488, and 3490. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 3182, 3184, 3186, 3188, 3190, 3192, 3194, 3196, 3198, 3200, 3202, 3204, 3206, 3208, 3210, 3212, 3214, 3216, 3218, 3220, 3222, 3224, 3226, 3228, 3230, 3232, 3234, 3236, 3238, 3240, 3242, 3244, 3246, 3248, 3250, 3252, 3254, 3256, 3258, 3260, 3262, 3264, 3266, 3268, 3270, 3272, 3274, 3276, 3278, 3280, 3282, 3284, 3286, 3288, 3290, 3292, 3294, 3296, 3298, 3300, 3302, 3304, 3306, 3308, 3310, 3312, 3314, 3316, 3318, 3320, 3322, 3324, 3326, 3328, 3330, 3332, 3334, 3336, 3338, 3340, 3342, 3344, 3346, 3348, 3350, 3352, 3354, 3356, 3358, 3360, 3362, 3364, 3366, 3368, 3370, 3372, 3374, 3376, 3378, 3380, 3382, 3384, 3386, 3388, 3390, 3392, 3394, 3396, 3398, 3400, 3402, 3404, 3406, 3408, 3410, 3412, 3414, 3416, 3418, 3420, 3422, 3424, 3426, 3428, 3430, 3432, 3434, 3436, 3438, 3440, 3442, 3444, 3446, 3448, 3450, 3452, 3454, 3456, 3458, 3460, 3462, 3464, 3466, 3468, 3470, 3472, 3474, 3476, 3478, 3480, 3482, 3484, 3486, 3488, and 3490.

(48) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 106/164/172/200/409, 106/164/172/210/240/415, 106/164/172/242/409/416, 106/164/200/210/242, 106/164/200/210/415, 106/164/200/240/408/409/416, 106/164/210/240/408/409/416, 106/164/210/408/409, 106/164/240/415, 106/164/409/415/423, 106/172/200/210/240/242/408/415/416/423, 106/172/200/210/242, 106/172/240,

106/172/240/242/409/415, 106/172/242, 106/172/240/416/423, 106/172/408/409, 106/172/409/423, 106/200/210/409, 106/210/240, 106/210/240/408/415, 106/240/242, 106/240/242/409, 106/242/408/409, 164/172/200/210/242/409, 164/172/240/242/415, 164/172/242/408, 164/200/408/415/416, 164/240/242/408, 164/240/242/415/423, 164/423, 172/200/210/240/408, 172/200/240, 172/200/408/416, 172/210/415, 172/240, 172/240/242/415, 172/240/409, 200/210/240/242/409, 200/240/242/407, 200/242/415/416/423, 200/408/409, 210/240, 240, 240/242, 240/242/408/416, 240/242/415/423, 240/408, 240/415, and 409/415, wherein the positions are numbered with reference to SEQ ID NO: 3244. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 106A/164H/172R/200A/409L, 106A/164H/172R/210V/240E/415R, 106A/164H/172R/242I/409L/416R, 106A/164H/200A/210V/242I, 106A/164H/200A/210V/415A, 106A/164H/200A/240E/408P/409L/416R, 106A/164H/210V/240E/408P/409L/416R, 106A/164H/210V/408P/409L, 106A/164H/240E/415A, 106A/164H/409L/415A/423R, 106A/172R/200A/210V/240E/242I/408P/415A/416R/423R, 106A/172R/200A/210V/242I, 106A/172R/240E, 106A/172R/240E/242I/409L/415R, 106A/172R/242I, 106A/172R/242I/416R/423R, 106A/172R/408P/409L, 106A/172R/409L/423R, 106A/200A/210V/409L, 106A/210V/240E, 106A/210V/240E/408P/415R, 106A/240E/242I, 106A/240E/242I/409L, 106A/242I/408P/409L, 164H/172R/200A/210V/242I/409L, 164H/172R/240E/242I/415R, 164H/172R/242I/408P, 164H/200A/408P/415A/416R, 164H/240E/242I/408P, 164H/240E/242I/415R/423R, 164H/423R, 172R/200A/210V/240E/408P, 172R/200A/240E, 172R/200A/408P/416R, 172R/210V/415A, 172R/240E, 172R/240E/242I/415R, 172R/240E/409L, 200A/210V/240E/242I/409L, 200A/240E/242I/407S, 200A/242I/415A/416R/423R, 200A/408P/409L, 210V/240E, 240E, 240E/242I, 240E/242I/408P/416R, 240E/242I/415A/423R, 240E/408P, 240E/415R, and 409L/415R, wherein the positions are numbered with reference to SEQ ID NO: 3244. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from K106A/V164H/N172R/G200A/T409L, K106A/V164H/N172R/E210V/N240E/G415R, K106A/V164H/N172R/A242I/T409L/K416R, K106A/V164H/G200A/E210V/A242I, K106A/V164H/G200A/E210V/G415A, K106A/V164H/G200A/N240E/E408P/T409L/K416R, K106A/V164H/E210V/N240E/E408P/T409L/K416R, K106A/V164H/E210V/E408P/T409L, K106A/V164H/N240E/G415A, K106A/V164H/T409L/G415A/N423R, K106A/N172R/G200A/E210V/N240E/A242I/E408P/G415A/K416R/N423R, K106A/N172R/G200A/E210V/A242I, K106A/N172R/N240E, K106A/N172R/N240E/A242I/T409L/G415R, K106A/N172R/A242I, K106A/N172R/A242I/K416R/N423R, K106A/N172R/E408P/T409L, K106A/N172R/T409L/N423R, K106A/G200A/E210V/T409L, K106A/E210V/N240E, K106A/E210V/N240E/E408P/G415R, K106A/N240E/A242I, K106A/N240E/A242I/T409L, K106A/A242I/E408P/T409L, V164H/N172R/G200A/E210V/A242I/T409L, V164H/N172R/N240E/A242I/G415R, V164H/N172R/A242I/E408P, V164H/G200A/E408P/G415A/K416R, V164H/N240E/A242I/E408P, V164H/N240E/A242I/G415R/N423R, V164H/N423R, N172R/G200A/E210V/N240E/E408P, N172R/G200A/N240E, N172R/G200A/E408P/K416R, N172R/E210V/G415A, N172R/N240E, N172R/N240E/A242I/G415R, N172R/N240E/T409L, G200A/E210V/N240E/A242I/T409L, G200A/N240E/A242I/G407S, G200A/A242I/G415A/K416R/N423R, G200A/E408P/T409L, E210V/N240E, N240E, N240E/A242I, N240E/A242I/E408P/K416R, N240E/A242I/G415A/N423R, N240E/E408P, N240E/G415R, and T409L/G415R, wherein the positions are numbered with reference to SEQ ID NO: 3244. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 14, 35, 42, 46, 49, 105, 134, 143, 179, 181, 232, 278, 290, 336, 373, 381, 401, and 441, wherein the positions are numbered with reference to SEQ ID NO: 3244. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 14K, 35D, 42F, 42I, 42V, 46T, 46V, 49A, 49M, 49P, 49Q, 49S, 105A, 134A, 134C, 134S, 143P, 179A, 179D, 179T, 181L, 232T, 278I, 278L, 290L, 336A, 373R, 381G, 401V, 441I, and 441R, wherein the positions are numbered with reference to SEQ ID NO: 3244. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one

mutation or mutation set at one or more positions selected from R14K, Q35D, L42F, L42I, L42V, C46T, C46V, L49A, L49M, L49P, L49Q, L49S, S105A, V134A, V134C, V134S, K143P, V179A, V179D, V179T, M181L, P232T, V278I, V278L, I290L, S336A, K373R, A381G, L401V, Q441I, and Q441R, wherein the positions are numbered with reference to SEQ ID NO: 3244. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 3492, 3494, 3496, 3498, 3500, 3502, 3504, 3506, 3508, 3510, 3512, 3514, 3516, 3518, 3520, 3522, 3524, 3526, 3528, 3530, 3532, 3534, 3536, 3538, 3540, 3542, 3544, 3546, 3548, 3550, 3552, 3554, 3556, 3558, 3560, 3562, 3564, 3566, 3568, 3570, 3572, 3574, 3576, 3578, 3580, 3582, 3584, 3586, 3588, 3590, 3592, 3594, 3596, 3598, 3600, 3602, 3604, 3606, 3608, 3610, 3612, 3614, 3616, 3618, 3620, 3622, 3624, 3626, 3628, 3630, 3632, 3634, 3636, 3638, 3640, 3642, 3644, 3646, 3648, 3650, and 3652. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 3492, 3494, 3496, 3498, 3500, 3502, 3504, 3506, 3508, 3510, 3512, 3514, 3516, 3518, 3520, 3522, 3524, 3526, 3528, 3530, 3532, 3534, 3536, 3538, 3540, 3542, 3544, 3546, 3548, 3550, 3552, 3554, 3556, 3558, 3560, 3562, 3564, 3566, 3568, 3570, 3572, 3574, 3576, 3578, 3580, 3582, 3584, 3586, 3588, 3590, 3592, 3594, 3596, 3598, 3600, 3602, 3604, 3606, 3608, 3610, 3612, 3614, 3616, 3618, 3620, 3622, 3624, 3626, 3628, 3630, 3632, 3634, 3636, 3638, 3640, 3642, 3644, 3646, 3648, 3650, and 3652. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 3492, 3494, 3496, 3498, 3500, 3502, 3504, 3506, 3508, 3510, 3512, 3514, 3516, 3518, 3520, 3522, 3524, 3526, 3528, 3530, 3532, 3534, 3536, 3538, 3540, 3542, 3544, 3546, 3548, 3550, 3552, 3554, 3556, 3558, 3560, 3562, 3564, 3566, 3568, 3570, 3572, 3574, 3576, 3578, 3580, 3582, 3584, 3586, 3588, 3590, 3592, 3594, 3596, 3598, 3600, 3602, 3604, 3606, 3608, 3610, 3612, 3614, 3616, 3618, 3620, 3622, 3624, 3626, 3628, 3630, 3632, 3634, 3636, 3638, 3640, 3642, 3644, 3646, 3648, 3650, and 3652.

(49) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 21, 91, 125, 127, 130/187, 143, 143/150, 145, 152, 156, 186, 187, 195, 197, 200, 201, 202, 264, 268, 364, 365, and 415, wherein the positions are numbered with reference to SEQ ID NO: 3346. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 21M, 21P, 91M, 125C, 125M, 125V, 127G, 130T/187Q, 143H, 143P/150S, 145W, 152L, 156P, 156Q, 186N, 186V, 187S, 195F, 195G, 195R, 195S, 195T, 195Y, 197D, 197L, 197Q, 197W, 200E, 200P, 200R, 200T, 201A, 201R, 201S, 201W, 202A, 202W, 264S, 264T, 268F, 268Q, 268W, 364S, 364W, 365L, and 415D, wherein the positions are numbered with reference to SEQ ID NO: 3346. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from L21M, L21P, P91M, F125C, F125M, F125V, Q127G, A130T/K187Q, K143H, K143P/A150S, L145W, F152L, F156P, F156Q, E186N, E186V, K187S, L195F, L195G, L195R, L195S, L195T, L195Y, P197D, P197L, P197Q, P197W, G200E, G200P, G200R, G200T, G201A, G201R, G201S, G201W, I202A, I202W, F264S, F264T, Y268F, Y268Q, Y268W, M364S, M364W, Q365L, and G415D, wherein the positions are numbered with reference to SEQ ID NO: 3346. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 3760, 3762, 3764, 3766, 3768, 3770, 3772, 3774, 3776, 3778, 3780, 3782, 3784, 3786, 3788, 3790, 3792, 3794, 3796, 3798, 3800, 3802, 3804, 3806, 3808, 3810, 3812, 3814, 3816, 3818, 3820, 3822, 3824, 3826, 3828, 3830, 3832, 3834, 3836, 3838, 3840, 3842, 3844, 3846, 3848, and 3850. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 3760, 3762, 3764, 3766, 3768, 3770, 3772, 3774, 3776, 3778, 3780, 3782, 3784, 3786, 3788, 3790, 3792, 3794, 3796, 3798, 3800, 3802, 3804, 3806, 3808, 3810, 3812, 3814, 3816, 3818, 3820, 3822, 3824, 3826, 3828, 3830, 3832, 3834, 3836, 3838, 3840, 3842, 3844, 3846, 3848, and 3850. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase any of SEQ ID NOS: 3760, 3762, 3764, 3766, 3768, 3770, 3772, 3774, 3776, 3778, 3780, 3782, 3784, 3786, 3788, 3790, 3792, 3794, 3796, 3798, 3800, 3802, 3804, 3806, 3808, 3810, 3812, 3814, 3816, 3818, 3820, 3822, 3824, 3826, 3828, 3830, 3832, 3834, 3836, 3838, 3840, 3842, 3844, 3846, 3848, and 3850.

(50) The present invention also provides engineered glycosyltransferases, wherein the polypeptide

sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 2/5/113/186/237/406, 2/73/186/237, 2/189/286, 2/286/355/416/422, 5/34/73/113/406, 5/73/186/406, 5/113/237/406, 5/186/237/245/256/406, 5/186/237/406, 5/256/406, 34/113/186/237/406, 72/73/172/235/240/242/438, 72/172/239/240/242/408, 72/172/240/242, 73/172, 73/172/235/239/240, 73/172/235/239/242, 73/172/239/240, 73/172/239/240/242, 73/172/240/242/408, 73/172/240/408, 73/186/237/406, 73/235/240/323, 73/235/240/408, 73/237, 73/239/242, 113/186/406, 172, 172/188/323, 172/235, 172/235/239/240/242, 172/235/239/408, 172/235/240, 172/235/240/242/438, 172/239/240/242/323/408, 172/240, 172/240/299/323, 186, 186/237, 186/237/286, 186/406, 189/333/355/421, 235/239/240, 235/240/242, 237, 239/240, 239/240/242/256/323, 239/240/242/323/408, 239/240/408, 239/242/408, 240/242/256/438, and 242/408, wherein the positions are numbered with reference to SEQ ID NO: 3502. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 275N/113I/186G/237V/406Q, 2-/73A/186G/237V, 2S/189S/286Q, 2S/286Q/355A/416R/422R, 5N/34R/73A/113I/406N, 5N/73A/186G/406N, 5N/113I/237V/406N, 5N/186G/237V/245T/256I/406Q, 5N/186G/237V/406Q, 5N/256L/406Q, 34R/113I/186G/237L/406N, 72Y/73P/172R/235M/240E/242I/438T, 72Y/172R/239A/240E/242I/408P, 72Y/172R/240E/242I, 73A/186G/237L/406Q, 73A/237V, 73P/172R, 73P/172R/235M/239A/240E, 73P/172R/235M/239A/242I, 73P/172R/239A/240E, 73P/172R/239A/240E/242I, 73P/172R/240E/242I/408P, 73P/172R/240E/408P, 73P/235M/240E/323P, 73P/235M/240E/408P, 73P/239A/242I, 113I/186G/406Q, 172R, 172R/188G/323P, 172R/235M, 172R/235M/239A/240E/242I, 172R/235M/239A/408P, 172R/235M/240E, 172R/235M/240E/242I/438T, 172R/239A/240E/242I/323P/408P, 172R/240E, 172R/240E/299A/323P, 186G, 186G/237L, 186G/237V/286S, 186G/406N, 186G/406Q, 189S/333H/355A/421Q, 235M/239A/240E, 235M/240E/242I, 237L, 239A/240E, 239A/240E/242I/256S/323P, 239A/240E/242I/323P/408P, 239A/240E/408P, 239A/242I/408P, 240E/242I/256S/438T, and 242I/408P, wherein the positions are numbered with reference to SEQ ID NO: 3502. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from H2-/H5N/L113I/E186G/P237V/T406Q, H2-/H73A/E186G/P237V, H2S/P189S/N286Q, H2S/N286Q/F355A/K416R/K422R, H5N/K34R/H73A/L113I/T406N, H5N/H73A/E186G/T406N, H5N/L113I/P237V/T406N, H5N/E186G/P237V/K245T/E256I/T406Q, H5N/E186G/P237V/T406Q, H5N/E256L/T406Q, K34R/L113I/E186G/P237L/T406N, L72Y/H73P/N172R/Q235M/N240E/A242I/E438T, L72Y/N172R/T239A/N240E/A242I/E408P, L72Y/N172R/N240E/A242I, H73A/E186G/P237L/T406Q, H73A/P237V, H73P/N172R, H73P/N172R/Q235M/T239A/N240E, H73P/N172R/Q235M/T239A/A242I, H73P/N172R/T239A/N240E, H73P/N172R/T239A/N240E/A242I, H73P/N172R/N240E/A242I/E408P, H73P/N172R/N240E/E408P, H73P/Q235M/N240E/D323P, H73P/Q235M/N240E/E408P, H73P/T239A/A242I, L113I/E186G/T406Q, N172R, N172R/E188G/D323P, N172R/Q235M, N172R/Q235M/T239A/N240E/A242I, N172R/Q235M/T239A/E408P, N172R/Q235M/N240E, N172R/Q235M/N240E/A242I/E438T, N172R/T239A/N240E/A242I/D323P/E408P, N172R/N240E, N172R/N240E/E299A/D323P, E186G, E186G/P237L, E186G/P237V/N286S, E186G/T406N, E186G/T406Q, P189S/N333H/F355A/S421Q, Q235M/T239A/N240E, Q235M/N240E/A242I, P237L, T239A/N240E, T239A/N240E/A242I/E256S/D323P, T239A/N240E/A242I/D323P/E408P, T239A/N240E/E408P, T239A/A242I/E408P, N240E/A242I/E256S/E438T, and A242I/E408P, wherein the positions are numbered with reference to SEQ ID NO: 3502. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 96, 127, 132, 144, 153, 155, 156, 186, 187, 196, 199, and 200, wherein the positions are numbered with reference to SEQ ID NO: 3502. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 96A, 96P, 127I, 127L, 127V, 132H, 132K, 132T, I44V, 153A, 153G, 155M, 156W, 186A, 186G, 186R, 187A, 187R, 187T, 196S, 199A, 199S, 199Y, and 200S, wherein the positions are numbered with reference to SEQ ID NO: 3502. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from G96A, G96P, Q127I, Q127L, Q127V, Q132H, Q132K, Q132T, I144V, S153A, S153G, F155M, F156W, E186A, E186G, E186R, K187A, K187R, K187T, A196S, N199A, N199S,

N199Y, and A200S, wherein the positions are numbered with reference to SEQ ID NO: 3502.

(51) In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 3654, 3656, 3658, 3660, 3662, 3664, 3666, 3668, 3670, 3672, 3674, 3676, 3678, 3680, 3682, 3684, 3686, 3688, 3690, 3692, 3694, 3696, 3698, 3700, 3702, 3704, 3706, 3708, 3710, 3712, 3714, 3716, 3718, 3720, 3722, 3724, 3726, 3728, 3730, 3732, 3734, 3736, 3738, 3740, 3742, 3744, 3746, 3748, 3750, 3752, 3754, 3756, 3758, 3852, 3854, 3856, 3858, 3860, 3862, 3864, 3866, 3868, 3870, 3872, 3874, 3876, 3878, 3880, 3882, 3884, 3886, 3888, 3890, 3892, 3894, 3896, and 3898. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 3654, 3656, 3658, 3660, 3662, 3664, 3666, 3668, 3670, 3672, 3674, 3676, 3678, 3680, 3682, 3684, 3686, 3688, 3690, 3692, 3694, 3696, 3698, 3700, 3702, 3704, 3706, 3708, 3710, 3712, 3714, 3716, 3718, 3720, 3722, 3724, 3726, 3728, 3730, 3732, 3734, 3736, 3738, 3740, 3742, 3744, 3746, 3748, 3750, 3752, 3754, 3756, 3758, 3852, 3854, 3856, 3858, 3860, 3862, 3864, 3866, 3868, 3870, 3872, 3874, 3876, 3878, 3880, 3882, 3884, 3886, 3888, 3890, 3892, 3894, 3896, and 3898. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 3654, 3656, 3658, 3660, 3662, 3664, 3666, 3668, 3670, 3672, 3674, 3676, 3678, 3680, 3682, 3684, 3686, 3688, 3690, 3692, 3694, 3696, 3698, 3700, 3702, 3704, 3706, 3708, 3710, 3712, 3714, 3716, 3718, 3720, 3722, 3724, 3726, 3728, 3730, 3732, 3734, 3736, 3738, 3740, 3742, 3744, 3746, 3748, 3750, 3752, 3754, 3756, 3758, 3852, 3854, 3856, 3858, 3860, 3862, 3864, 3866, 3868, 3870, 3872, 3874, 3876, 3878, 3880, 3882, 3884, 3886, 3888, 3890, 3892, 3894, 3896, and 3898.

(52) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 2/5/35/105/143/237/373/416/422, 2/5/35/143/232/237/416/422/441, 2/5/35/232/278/373/416, 2/5/105/143/232/373/416/422, 2/5/278, 2/105/143/232/237/278/373/441, 2/143/232/373/441, 2/143/373/441, 5/35/232/373/416/422, 5/105/232/237/373/416/441, 5/105/237/278/422/441, 5/105/237/416/422/441, 5/143/189/232/237/441, 5/143/232/237/278, 5/143/232/237/416/422, 5/143/232/373/422/441, 5/143/237/278/373/416/422, 5/143/373, 5/189/237/278/373/416/422/441, 5/232/416/422, 5/237/373/422, 5/373/416, 26/42, 26/42/46/49, 26/42/46/49/134, 26/42/46/49/134/186, 26/42/46/134, 26/42/49/134, 26/42/49/134/186/355, 26/42/49/134/401, 26/42/134/401, 26/49/134, 26/134, 26/134/186/355/401, 26/134/355, 26/134/401, 26/355/401, 35, 35/105/189/232/373/416, 42/46/49/97/134/401, 42/46/49/134, 42/46/49/134/186, 42/46/49/134/186/355, 42/46/49/134/235/355/401, 42/46/49/134/401, 42/46/49/186, 42/49/134/186/355, 42/49/134/186/401, 42/49/134/355/401, 42/134/186, 42/186/355, 46/49/134, 49/134/355, 105/143/232/237/373/416/422/441, 105/143/278/373, 105/189/237, 105/237/278/373/416/422, 105/237/373/422/441, 105/373, 105/373/441, 134/401, 143, 143/189, 143/189/232/422/441, 143/189/237/373/416/422, 143/232, 143/237, 143/237/422/441, 186/355, 189/232/373/416/422/441, 189/237/278/416/441, 232/237, 237/373, 237/373/441, and 237/422, wherein the positions are numbered with reference to SEQ ID NO: 3696. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 2-/5N/35D/105A/143P/237V/373R/416R/422R, 2-/5N/35D/143P/232T/237L/416R/422R/441R, 2-/5N/35D/232T/278L/373R/416R, 2-/5N/105A/143P/232T/373R/416R/422R, 2-/5N/278L, 2-/105A/143P/232T/237L/278L/373R/441R, 2-/143P/232T/373R/441R, 2-/143P/373R/441R, 5N/35D/232T/373R/416R/422R, 5N/105A/232T/237L/373R/416R/441R, 5N/105A/237L/278I/422R/441R, 5N/105A/237V/416R/422R/441R, 5N/143P/189S/232T/237V/441R, 5N/143P/232T/237L/278L, 5N/143P/232T/237L/416R/422R, 5N/143P/232T/373R/422R/441R, 5N/143P/237L/278L/373R/416R/422R, 5N/143P/373R, 5N/189S/237V/278I/373R/416R/422R/441R, 5N/232T/416R/422R, 5N/237L/373R/422R, 5N/373R/416R, 26V/42I/49Q/134C/186G/355A, 26V/42I/49Q/134C/401V, 26V/42V, 26V/42V/46V/49A, 26V/42V/46V/49A/134A, 26V/42V/46V/49A/134C/186G, 26V/42V/46V/134A, 26V/42V/49S/134C, 26V/42V/134C/401V, 26V/49Q/134A, 26V/134A, 26V/134A/186G/355A/401V, 26V/134C/355A, 26V/134T/401V, 26V/355A/401V, 35D, 35D/105A/189S/232T/373R/416R, 42I/46V/49A/134A, 42I/46V/49S/134C/186G, 42I/46V/49S/186G, 42V/46V/49A/134A/401V, 42V/46V/49A/134T/186G/355A, 42V/46V/49P/134C/235R/355A/401V, 42V/46V/49S/97P/134C/401V, 42V/49A/134C/186G/355A,

42V/49S/134A/186G/401V, 42V/49S/134C/355A/401V, 42V/134C/186G, 42V/186G/355A, 46V/49Q/134T, 49S/134C/355A, 105A/143P/232T/237V/373R/416R/422R/441R, 105A/143P/278L/373R, 105A/189S/237V, 105A/237L/278L/373R/416R/422R, 105A/237L/373R/422R/441R, 105A/373R, 105A/373R/441R, 134C/401V, 143P, 143P/189S, 143P/189S/232T/422R/441R, 143P/189S/237L/373R/416R/422R, 143P/232T, 143P/237L/422R/441R, 143P/237V, 186G/355A, 189S/232T/373R/416R/422R/441R, 189S/237L/278I/416R/441R, 232T/237L, 237L/422R, 237V/373R, and 237V/373R/441R, wherein the positions are numbered with reference to SEQ ID NO: 3696. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from H2-/H5N/Q35D/S105A/K143P/P237V/K373R/K416R/K422R, H2-/H5N/Q35D/K143P/P232T/P237L/K416R/K422R/Q441R, H2-/H5N/Q35D/P232T/V278L/K373R/K416R, H2-/H5N/S105A/K143P/P232T/K373R/K416R/K422R, H2-/H5N/V278L, H2-/S105A/K143P/P232T/P237L/V278L/K373R/Q441R, H2-/K143P/P232T/K373R/Q441R, H2-/K143P/K373R/Q441R, H5N/Q35D/P232T/K373R/K416R/K422R, H5N/S105A/P232T/P237L/K373R/K416R/Q441R, H5N/S105A/P237L/V278I/K422R/Q441R, H5N/S105A/P237V/K416R/K422R/Q441R, H5N/K143P/P189S/P232T/P237V/Q441R, H5N/K143P/P232T/P237L/V278L, H5N/K143P/P232T/P237L/K416R/K422R, H5N/K143P/P232T/K373R/K422R/Q441R, H5N/K143P/P237L/V278L/K373R/K416R/K422R, H5N/K143P/K373R, H5N/P189S/P237V/V278I/K373R/K416R/K422R/Q441R, H5N/P232T/K416R/K422R, H5N/P237L/K373R/K422R, H5N/K373R/K416R, 126V/L42I/L49Q/V134C/E186G/F355A, 126V/L42I/L49Q/V134C/L401V, 126V/L42V, 126V/L42V/C46V/L49A, 126V/L42V/C46V/L49A/V134A, 126V/L42V/C46V/L49A/V134C/E186G, 126V/L42V/C46V/V134A, 126V/L42V/L49S/V134C, 126V/L42V/V134C/L401V, 126V/L49Q/V134A, 126V/V134A, 126V/V134A/E186G/F355A/L401V, 126V/V134C/F355A, 126V/V134T/L401V, 126V/F355A/L401V, Q35D, Q35D/S105A/P189S/P232T/K373R/K416R, L42I/C46V/L49A/V134A, L42I/C46V/L49S/V134C/E186G, L42I/C46V/L49S/E186G, L42V/C46V/L49A/V134A/L401V, L42V/C46V/L49A/V134T/E186G/F355A, L42V/C46V/L49P/V134C/Q235R/F355A/L401V, L42V/C46V/L49S/T97P/V134C/L401V, L42V/L49A/V134C/E186G/F355A, L42V/L49S/V134A/E186G/L401V, L42V/L49S/V134C/F355A/L401V, L42V/V134C/E186G, L42V/E186G/F355A, C46V/L49Q/V134T, L49S/V134C/F355A, S105A/K143P/P232T/P237V/K373R/K416R/K422R/Q441R, S105A/K143P/V278L/K373R, S105A/P189S/P237V, S105A/P237L/V278L/K373R/K416R/K422R, S105A/P237L/K373R/K422R/Q441R, S105A/K373R, S105A/K373R/Q441R, V134C/L401V, K143P, K143P/P189S, K143P/P189S/P232T/K422R/Q441R, K143P/P189S/P237L/K373R/K416R/K422R, K143P/P232T, K143P/P237L/K422R/Q441R, K143P/P237V, E186G/F355A, P189S/P232T/K373R/K416R/K422R/Q441R, P189S/P237L/V278I/K416R/Q441R, P232T/P237L, P237L/K422R, P237V/K373R, and P237V/K373R/Q441R, wherein the positions are numbered with reference to SEQ ID NO: 3696. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 3, 8, 50, 61, 62, 101, 137, 158, 161, 164, 176, 193, 223, 223/243, 235, 237, 239, 240, 243, 244, 248, 249, 301, 323, 330, 352, 364, 426, and 427, wherein the positions are numbered with reference to SEQ ID NO: 3696. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 3N, 8R, 8V, 50P, 50V, 61N, 61Y, 62A, 101L, 137R, 158W, 161L, 164A, 164E, 164F, 164L, 176P, 176Q, 176T, 193F, 223A, 223L/243G, 235R, 237A, 237L, 239F, 239G, 239M, 239P, 239Q, 239Y, 240V, 243G, 243H, 243R, 243S, 243Y, 244N, 248T, 249G, 249H, 301A, 323T, 330C, 352A, 364Q, 426L, 427S, and 427T, wherein the positions are numbered with reference to SEQ ID NO: 3696. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from H3N, S8R, S8V, 150P, I50V, E61N, E61Y, K62A, A101L, Q137R, F158W, R161L, H164A, H164E, H164F, H164L, R176P, R176Q, R176T, D193F, T223A, T223L/D243G, Q235R, P237A, P237L, T239F, T239G, T239M, T239P, T239Q, T239Y, E240V, D243G, D243H, D243R, D243S, D243Y, D244N, I248T, D249G, D249H, Q301A, D323T, H330C, S352A, M364Q, A426L, R427S, and

R427T, wherein the positions are numbered with reference to SEQ ID NO: 3696. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 3900, 3902, 3904, 3906, 3908, 3910, 3912, 3914, 3916, 3918, 3920, 3922, 3924, 3926, 3928, 3930, 3932, 3934, 3936, 3938, 3940, 3942, 3944, 3946, 3948, 3950, 3952, 3954, 3956, 3958, 3960, 3962, 3964, 3966, 3968, 3970, 3972, 3974, 3976, 3978, 3980, 3982, 3984, 3986, 3988, 3990, 3992, 3994, 3996, 3998, 4000, 4002, 4004, 4006, 4008, 4010, 4012, 4014, 4016, 4018, 4020, 4022, 4024, 4026, 4028, 4030, 4032, 4034, 4036, 4038, 4040, 4042, 4044, 4046, 4048, 4050, 4052, 4054, 4056, 4058, 4060, 4062, 4064, 4066, 4068, 4070, 4072, 4074, 4076, 4078, 4080, 4082, 4084, 4086, 4088, 4090, 4092, 4094, 4096, 4098, 4100, 4102, 4104, 4106, 4108, 4110, 4112, 4114, 4116, 4118, 4120, 4122, 4124, 4126, 4128, 4130, 4132, 4134, 4136, 4138, 4140, 4142, 4144, and 4146. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 3900, 3902, 3904, 3906, 3908, 3910, 3912, 3914, 3916, 3918, 3920, 3922, 3924, 3926, 3928, 3930, 3932, 3934, 3936, 3938, 3940, 3942, 3944, 3946, 3948, 3950, 3952, 3954, 3956, 3958, 3960, 3962, 3964, 3966, 3968, 3970, 3972, 3974, 3976, 3978, 3980, 3982, 3984, 3986, 3988, 3990, 3992, 3994, 3996, 3998, 4000, 4002, 4004, 4006, 4008, 4010, 4012, 4014, 4016, 4018, 4020, 4022, 4024, 4026, 4028, 4030, 4032, 4034, 4036, 4038, 4040, 4042, 4044, 4046, 4048, 4050, 4052, 4054, 4056, 4058, 4060, 4062, 4064, 4066, 4068, 4070, 4072, 4074, 4076, 4078, 4080, 4082, 4084, 4086, 4088, 4090, 4092, 4094, 4096, 4098, 4100, 4102, 4104, 4106, 4108, 4110, 4112, 4114, 4116, 4118, 4120, 4122, 4124, 4126, 4128, 4130, 4132, 4134, 4136, 4138, 4140, 4142, 4144, and 4146. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 3900, 3902, 3904, 3906, 3908, 3910, 3912, 3914, 3916, 3918, 3920, 3922, 3924, 3926, 3928, 3930, 3932, 3934, 3936, 3938, 3940, 3942, 3944, 3946, 3948, 3950, 3952, 3954, 3956, 3958, 3960, 3962, 3964, 3966, 3968, 3970, 3972, 3974, 3976, 3978, 3980, 3982, 3984, 3986, 3988, 3990, 3992, 3994, 3996, 3998, 4000, 4002, 4004, 4006, 4008, 4010, 4012, 4014, 4016, 4018, 4020, 4022, 4024, 4026, 4028, 4030, 4032, 4034, 4036, 4038, 4040, 4042, 4044, 4046, 4048, 4050, 4052, 4054, 4056, 4058, 4060, 4062, 4064, 4066, 4068, 4070, 4072, 4074, 4076, 4078, 4080, 4082, 4084, 4086, 4088, 4090, 4092, 4094, 4096, 4098, 4100, 4102, 4104, 4106, 4108, 4110, 4112, 4114, 4116, 4118, 4120, 4122, 4124, 4126, 4128, 4130, 4132, 4134, 4136, 4138, 4140, 4142, 4144, and 4146.

(53) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 5/73/113/186/187/373/423, 5/73/143/144/179/186/187/422, 5/73/144/179/186/373/423, 5/73/144/187/373/422, 5/143/144/179/181/186/187/373, 5/144/179/181/373/422, 5/144/373/422, 49/96/127/132/156/196, 49/96/127/153/278, 49/96/132/153/155/156/237/278/406, 49/96/132/153/199/200, 49/96/132/155, 49/96/132/155/237/278, 49/96/155/199/200/406, 49/153/155, 73/113/143/179/181/422, 73/113/179/181/186/187, 73/143/144/179/186/187/373/423, 73/179/181/186/187/373, 73/179/181/186/373/422, 73/181/186/187, 96/127/132/153/278, 96/127/153/155, 96/132/153/155, 96/132/153/155/156, 96/132/153/155/156/200, 96/132/153/156, 96/132/196/199, 96/132/278, 96/153/155/156, 96/153/155/199/200/237, 96/153/406, 113/143/179/186/187, 113/144/186/423, 113/144/373, 113/181/186/373/422, 113/373/422, 127/132, 127/132/153/156, 127/132/155/156/406, 127/153/155/199/200/237, 127/406, 132/153/155, 132/153/237/406, 132/155, 132/237, 143/144/179/181/186/187/422, 143/179/181/186/187/422/423, 144/179/181/186/187/373, 144/179/186/187/373, 153/155/156/237, 153/155/196/199/237, 153/199/406, 153/237, 155/199, 179/181/186, 179/181/186/187/423, 179/186/187, 179/187/373/422, 181/186/187/422/423, and 373/423, wherein the positions are numbered with reference to SEQ ID NO: 3956. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 5N/73A/113I/186G/187R/373R/423R, 5N/73A/143P/I44V/179D/186G/187R/422R, 5N/73A/I44V/179D/186A/373R/423R, 5N/73A/I44V/187R/373R/422R, 5N/143P/I44V/179T/181L/186G/187R/373R, 5N/I44V/179T/181L/373R/422R, 5N/I44V/373R/422R, 49S/96A/127I/132K/156W/196S, 49S/96A/132H/153A/199S/200S, 49S/96P/127I/153A/278L, 49S/96P/132H/153A/155M/156W/237L/278L/406Q, 49S/96P/132H/155M, 49S/96P/132K/155M/237L/278L, 49S/96P/155M/199S/200S/406Q, 49S/153A/155M, 73A/113I/143P/179T/181L/422R, 73A/113I/179T/181L/186A/187R,

73A/143P/179D/186G/187R/373R, 73A/179D/181L/186A/373R/422R, 73A/179D/181L/186G/187R/373R, 73A/181L/186A/187T, 96A/132H/153A/156W, 96A/132K/153A/155M/156W, 96A/132K/153A/155M/156W/200S, 96A/153A/155M/156W, 96P/127I/132H/153A/278L, 96P/127I/153A/155M, 96P/132H/153A/155M/156W, 96P/132H/196S/199S, 96P/132K/153A/155M, 96P/132K/278L, 96P/153A/155M/199S/200S/237L, 96P/153A/406Q, 113I/143P/179D/186G/187R, 113I/144V/186G/423R, 113I/144V/373R, 113I/181L/186A/373R/422R, 113I/373R/422R, 127I/132H, 127I/132H/153A/156W, 127I/132H/155M/156W/406Q, 127I/153A/155M/199S/200S/237L, 127I/406Q, 132H/153A/237L/406Q, 132H/155M, 132K/153A/155M, 132K/237L, 143P/I44V/179T/181L/186G/187R/422R, 143P/179T/181L/186G/187R/422R/423R, I44V/179T/181L/186A/187R/373R, I44V/179T/186A/187T/373R, 153A/155M/156W/237L, 153A/155M/196S/199S/237L, 153A/199S/406Q, 153A/237L, 155M/199S, 179D/181L/186G/187R/423R, 179D/187T/373R/422R, 179T/181L/186A, 179T/186A/187T, 181L/186G/187R/422R/423R, and 373R/423R, wherein the positions are numbered with reference to SEQ ID NO: 3956. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from H5N/P73A/L113I/E186G/K187R/K373R/N423R, H5N/P73A/K143P/I144V/V179D/E186G/K187R/K422R, H5N/P73A/I144V/V179D/E186A/K373R/N423R, H5N/P73A/I144V/K187R/K373R/K422R, H5N/K143P/I144V/V179T/M181L/E186G/K187R/K373R, H5N/I144V/V179T/M181L/K373R/K422R, H5N/I144V/K373R/K422R, A49S/G96A/Q127I/Q132K/F156W/A196S, A49S/G96A/Q132H/S153A/N199S/A200S, A49S/G96P/Q127I/S153A/V278L, A49S/G96P/Q132H/S153A/F155M/F156W/P237L/V278L/T406Q, A49S/G96P/Q132H/F155M, A49S/G96P/Q132K/F155M/P237L/V278L, A49S/G96P/F155M/N199S/A200S/T406Q, A49S/S153A/F155M, P73A/L113I/K143P/V179T/M181L/K422R, P73A/L113I/V179T/M181L/E186A/K187R, P73A/K143P/I144V/V179D/E186G/K187T/K373R/N423R, P73A/V179D/M181L/E186A/K373R/K422R, P73A/V179D/M181L/E186G/K187R/K373R, P73A/M181L/E186A/K187T, G96A/Q132H/S153A/F156W, G96A/Q132K/S153A/F155M/F156W, G96A/Q132K/S153A/F155M/F156W/A200S, G96A/S153A/F155M/F156W, G96P/Q127I/Q132H/S153A/V278L, G96P/Q127I/S153A/F155M, G96P/Q132H/S153A/F155M/F156W, G96P/Q132H/A196S/N199S, G96P/Q132K/S153A/F155M, G96P/Q132K/V278L, G96P/S153A/F155M/N199S/A200S/P237L, G96P/S153A/T406Q, L113I/K143P/V179D/E186G/K187R, L113I/I144V/E186G/N423R, L113I/I144V/K373R, L113I/M181L/E186A/K373R/K422R, L113I/K373R/K422R, Q127I/Q132H, Q127I/Q132H/S153A/F156W, Q127I/Q132H/F155M/F156W/T406Q, Q127I/S153A/F155M/N199S/A200S/P237L, Q127I/T406Q, Q132H/S153A/P237L/T406Q, Q132H/F155M, Q132K/S153A/F155M, Q132K/P237L, K143P/I144V/V179T/M181L/E186G/K187R/K422R, K143P/V179T/M181L/E186G/K187R/K422R/N423R, I144V/V179T/M181L/E186A/K187R/K373R, I144V/V179T/E186A/K187T/K373R, S153A/F155M/F156W/P237L, S153A/F155M/A196S/N199S/P237L, S153A/N199S/T406Q, S153A/P237L, F155M/N199S, V179D/M181L/E186G/K187R/N423R, V179D/K187T/K373R/K422R, V179T/M181L/E186A, V179T/E186A/K187T, M181L/E186G/K187R/K422R/N423R, and K373R/N423R, wherein the positions are numbered with reference to SEQ ID NO: 3956. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 11, 12, 41, 44, 44/187, 45, 55, 56, 57, 65, 66, 70, 72, 73, 74/238, 82, 83, 85, 103, 111, 113, 114, 117, 132, 135, 138, 140, 159, 160, 162, 167, 182, 214, 220, 222, 223, 226, 236, 238, 256, 286, 299, 309, 387, 388, 389, 391, 393, 406, 408, 412, 418, 422, 429, 430, 449, and 450, wherein the positions are numbered with reference to SEQ ID NO: 3956. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 11D, 11L, 121, 12L, 41H, 44H, 44S/187T, 45Y, 55A, 55G, 55L, 56R, 57G, 57H, 57P, 65C, 65G, 65P, 65Q, 65R, 66L, 70W, 72V, 73H, 73S, 74M/238V, 82S, 83R, 85W, 103H, 103Q, 111F, 113S, 114G, 114I, 117N, 117S, 132G, 135E, 138K, 138S, 138T, 140C, 140G, 159N, 160D, 160Q, 160T, 162L, 167L, 182I, 182L, 182Q, 182T, 182V, 182Y, 214H, 214T, 220G, 220S, 222F, 223D, 226Q, 236L, 236T, 236V, 238A, 238G, 238I, 238R, 238S, 238T, 256R, 286G, 299V, 309P, 309R, 309S, 309T,

387P, 387R, 388A, 388S, 389A, 389L, 389P, 391A, 391L, 391S, 393P, 393T, 406G, 406S, 408L, 408T, 412G, 412R, 418A, 418H, 418M, 422L, 429D, 430P, 449G, 449H, 449L, 449R, 449S, 450N, and 450V, wherein the positions are numbered with reference to SEQ ID NO: 3956. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from Q11D, Q11L, T121, T12L, F41H, Y44H, Y44S/K187T, L45Y, I55A, I55G, I55L, K56R, K57G, K57H, K57P, N65C, N65G, N65P, N65Q, N65R, S66L, V70W, L72V, P73H, P73S, L74M/M238V, P82S, H83R, H85W, K103H, K103Q, K111F, L113S, Q114G, Q114I, K117N, K117S, Q132G, L135E, G138K, G138S, G138T, P140C, P140G, L159N, K160D, K160Q, K160T, P162L, P167L, R182I, R182L, R182Q, R182T, R182V, R182Y, K214H, K214T, T220G, T220S, L222F, T223D, K226Q, D236L, D236T, D236V, M238A, M238G, M238I, M238R, M238S, M238T, E256R, N286G, E299V, K309P, K309R, K309S, K309T, D387P, D387Q, D387R, E388A, E388S, D389A, D389G, D389L, D389V, R391A, R391L, R391S, R391V, H393P, H393T, T406G, T406S, P408L, P408T, I412G, I412R, R418A, R418H, R418M, K422L, E429D, E430P, Y449G, Y449H, Y449L, Y449R, Y449S, K450N, and K450V, wherein the positions are numbered with reference to SEQ ID NO: 3956. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 4148, 4150, 4152, 4154, 4156, 4158, 4160, 4162, 4164, 4166, 4168, 4170, 4172, 4174, 4176, 4178, 4180, 4182, 4184, 4186, 4188, 4190, 4192, 4194, 4196, 4198, 4200, 4202, 4204, 4206, 4208, 4210, 4212, 4214, 4216, 4218, 4220, 4222, 4224, 4226, 4228, 4230, 4232, 4234, 4236, 4238, 4240, 4242, 4244, 4246, 4248, 4250, 4252, 4254, 4256, 4258, 4260, 4262, 4264, 4266, 4268, 4270, 4272, 4274, 4276, 4278, 4280, 4282, 4284, 4286, 4288, 4290, 4292, 4294, 4296, 4298, 4300, 4302, 4304, 4306, 4308, 4310, 4312, 4314, 4316, 4318, 4320, 4322, 4324, 4326, 4328, 4330, 4332, 4334, 4336, 4338, 4340, 4342, 4344, 4346, 4348, 4350, 4352, 4354, 4356, 4358, 4360, 4362, 4364, 4366, 4368, 4370, 4372, 4374, 4376, 4378, 4380, 4382, 4384, 4386, 4388, 4390, 4392, 4394, 4396, 4398, 4400, 4402, 4404, 4406, 4408, 4410, 4412, 4414, 4416, 4418, 4420, 4422, 4424, 4426, 4428, 4430, 4432, 4434, 4436, 4438, 4440, 4442, 4444, 4446, 4448, 4450, 4452, 4454, 4456, 4458, 4460, 4462, 4464, 4466, 4468, 4470, 4472, 4474, 4476, 4478, 4480, 4482, 4484, 4486, 4488, 4490, 4492, 4494, and 4496. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 4148, 4150, 4152, 4154, 4156, 4158, 4160, 4162, 4164, 4166, 4168, 4170, 4172, 4174, 4176, 4178, 4180, 4182, 4184, 4186, 4188, 4190, 4192, 4194, 4196, 4198, 4200, 4202, 4204, 4206, 4208, 4210, 4212, 4214, 4216, 4218, 4220, 4222, 4224, 4226, 4228, 4230, 4232, 4234, 4236, 4238, 4240, 4242, 4244, 4246, 4248, 4250, 4252, 4254, 4256, 4258, 4260, 4262, 4264, 4266, 4268, 4270, 4272, 4274, 4276, 4278, 4280, 4282, 4284, 4286, 4288, 4290, 4292, 4294, 4296, 4298, 4300, 4302, 4304, 4306, 4308, 4310, 4312, 4314, 4316, 4318, 4320, 4322, 4324, 4326, 4328, 4330, 4332, 4334, 4336, 4338, 4340, 4342, 4344, 4346, 4348, 4350, 4352, 4354, 4356, 4358, 4360, 4362, 4364, 4366, 4368, 4370, 4372, 4374, 4376, 4378, 4380, 4382, 4384, 4386, 4388, 4390, 4392, 4394, 4396, 4398, 4400, 4402, 4404, 4406, 4408, 4410, 4412, 4414, 4416, 4418, 4420, 4422, 4424, 4426, 4428, 4430, 4432, 4434, 4436, 4438, 4440, 4442, 4444, 4446, 4448, 4450, 4452, 4454, 4456, 4458, 4460, 4462, 4464, 4466, 4468, 4470, 4472, 4474, 4476, 4478, 4480, 4482, 4484, 4486, 4488, 4490, 4492, 4494, and 4496. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 4148, 4150, 4152, 4154, 4156, 4158, 4160, 4162, 4164, 4166, 4168, 4170, 4172, 4174, 4176, 4178, 4180, 4182, 4184, 4186, 4188, 4190, 4192, 4194, 4196, 4198, 4200, 4202, 4204, 4206, 4208, 4210, 4212, 4214, 4216, 4218, 4220, 4222, 4224, 4226, 4228, 4230, 4232, 4234, 4236, 4238, 4240, 4242, 4244, 4246, 4248, 4250, 4252, 4254, 4256, 4258, 4260, 4262, 4264, 4266, 4268, 4270, 4272, 4274, 4276, 4278, 4280, 4282, 4284, 4286, 4288, 4290, 4292, 4294, 4296, 4298, 4300, 4302, 4304, 4306, 4308, 4310, 4312, 4314, 4316, 4318, 4320, 4322, 4324, 4326, 4328, 4330, 4332, 4334, 4336, 4338, 4340, 4342, 4344, 4346, 4348, 4350, 4352, 4354, 4356, 4358, 4360, 4362, 4364, 4366, 4368, 4370, 4372, 4374, 4376, 4378, 4380, 4382, 4384, 4386, 4388, 4390, 4392, 4394, 4396, 4398, 4400, 4402, 4404, 4406, 4408, 4410, 4412, 4414, 4416, 4418, 4420, 4422, 4424, 4426, 4428, 4430, 4432, 4434, 4436, 4438, 4440, 4442, 4444, 4446, 4448, 4450, 4452, 4454, 4456, 4458, 4460, 4462, 4464, 4466, 4468, 4470, 4472, 4474, 4476, 4478, 4480, 4482, 4484, 4486, 4488, 4490, 4492, 4494, and 4496.

(54) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 8/375, 50/137/189/375, 62/153/155/156/159, 62/153/155/156/159/427,

62/153/155/199/406, 137/164/375, 153, 153/155/156, 153/155/156/159/199/238/406, 153/155/237/238/239/406, 153/155/238/239, 153/155/427, 153/156/159, 153/156/199/237/427, 153/156/427, 153/159/237/238/352, 153/176/181/427, 155/156, 155/156/176/181/199, 155/156/176/238/427, 155/181/199/238/406, 156/199, 156/199/427, 164/375/433, 223/375, 375, and 427, wherein the positions are numbered with reference to SEQ ID NO: 4256. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 8V/375L, 50V/137R/189G/375L, 62A/153A/155M/156W/159V, 62A/153A/155M/156W/159V/427T, 62A/153A/155M/199S/406Q, 137R/164E/375L, 153A, 153A/155M/156W, 153A/155M/156W/159V/199S/238G/406Q, 153A/155M/237L/238T/239F/406Q, 153A/155M/238T/239F, 153A/155M/427T, 153A/156W/159V, 153A/156W/199S/237L/427T, 153A/156W/427T, 153A/159V/237L/238G/352A, 153A/176Q/181L/427S, 155M/156W, 155M/156W/176Q/181L/199S, 155M/156W/176Q/238T/427S, 155M/181L/199S/238G/406Q, 156W/199S, 156W/199S/427T, 164E/375L/433D, 223A/375L, 375L, and 427S, wherein the positions are numbered with reference to SEQ ID NO: 4256. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from S8V/I375L, I50V/Q137R/P189G/I375L, K62A/S153A/F155M/F156W/L159V, K62A/S153A/F155M/F156W/L159V/R427T, K62A/S153A/F155M/N199S/T406Q, Q137R/H164E/I375L, S153A, S153A/F155M/F156W, S153A/F155M/F156W/L159V/N199S/M238G/T406Q, S153A/F155M/P237L/M238T/T239F/T406Q, S153A/F155M/M238T/T239F, S153A/F155M/R427T, S153A/F156W/L159V, S153A/F156W/N199S/P237L/R427T, S153A/F156W/R427T, S153A/L159V/P237L/M238G/S352A, S153A/R176Q/M181L/R427S, F155M/F156W, F155M/F156W/R176Q/M181L/N199S, F155M/F156W/R176Q/M238T/R427S, F155M/M181L/N199S/M238G/T406Q, F156W/N199S, F156W/N199S/R427T, H164E/I375L/N433D, T223A/I375L, I375L, and R427S, wherein the positions are numbered with reference to SEQ ID NO: 4256. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 7, 9, 10, 12, 53, 65, 68, 99, 106, 110, 115, 116, 131, 132, 136, 170, 178, 190, 192, 194, 200, 220, 238, 242, 245, 257, 272, 280, 302, 304, 335, 385, 395, 399, 402, 408, 412, 416, 423, 445, 447, and 449, wherein the positions are numbered with reference to SEQ ID NO: 4256. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 7H, 9G, 10D, 12S, 53N, 65F, 68I, 99R, 106K, 106R, 106S, 110G, 115A, 115T, 116I, 131C, 131S, 132G, 132R, 132T, 136D, 136G, 136R, 136S, 170E, 178L, 190T, 192A, 192L, 192P, 194F, 200S, 220M, 220Q, 220S, 238I, 238L, 242L, 245P, 257H, 272H, 280W, 302P, 302S, 304I, 335K, 335R, 385A, 385C, 385P, 385S, 395Q, 399K, 402R, 408E, 412R, 416R, 423L, 445H, 445K, 445R, 447L, 449K, 449L, 449Q, and 449R, wherein the positions are numbered with reference to SEQ ID NO: 4256. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from G7H, S9G, G10D, T12S, E53N, N65F, H68I, H99R, A106K, A106R, A106S, S110G, N115A, N115T, L116I, E131C, E131S, Q132G, Q132R, Q132T, E136D, E136G, E136R, E136S, A170E, Q178L, D190T, E192A, E192L, E192P, P194F, A200S, T220M, T220Q, T220S, M238I, M238L, I242L, K245P, N257H, R272H, F280W, N302P, N302S, E304I, P335K, P335R, V385A, V385C, V385P, V385S, G395Q, Q399K, K402R, P408E, I412R, K416R, R423L, N445H, N445K, N445R, N447L, Y449K, Y449L, Y449Q, and Y449R, wherein the positions are numbered with reference to SEQ ID NO: 4256. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 4498, 4500, 4502, 4504, 4506, 4508, 4510, 4512, 4514, 4516, 4518, 4520, 4522, 4524, 4526, 4528, 4530, 4532, 4534, 4536, 4538, 4540, 4542, 4544, 4546, 4548, 4550, 4552, 4554, 4556, 4558, 4560, 4562, 4564, 4566, 4568, 4570, 4572, 4574, 4576, 4578, 4580, 4582, 4584, 4586, 4588, 4590, 4592, 4594, 4596, 4598, 4600, 4602, 4604, 4606, 4608, 4610, 4612, 4614, 4616, 4618, 4620, 4622, 4624, 4626, 4628, 4630, 4632, 4634, 4636, 4638, 4640, 4642, 4644, 4646, 4648, 4650, 4652, 4654, 4656, 4658, 4660, 4662, 4664, 4666, 4668, 4670, 4672, 4674, 4676, 4678, 4680, and 4682. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 4498, 4500, 4502, 4504, 4506, 4508, 4510, 4512, 4514, 4516, 4518, 4520, 4522, 4524, 4526, 4528, 4530, 4532, 4534, 4536, 4538, 4540, 4542, 4544, 4546, 4548,

4552, 4554, 4556, 4558, 4560, 4562, 4564, 4566, 4568, 4570, 4572, 4574, 4576, 4578, 4580, 4582, 4584, 4586, 4588, 4590, 4592, 4594, 4596, 4598, 4600, 4602, 4604, 4606, 4608, 4610, 4612, 4614, 4616, 4618, 4620, 4622, 4624, 4626, 4628, 4630, 4632, 4634, 4636, 4638, 4640, 4642, 4644, 4646, 4648, 4650, 4652, 4654, 4656, 4658, 4660, 4662, 4664, 4666, 4668, 4670, 4672, 4674, 4676, 4678, 4680, and 4682. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 4498, 4500, 4502, 4504, 4506, 4508, 4510, 4512, 4514, 4516, 4518, 4520, 4522, 4524, 4526, 4528, 4530, 4532, 4534, 4536, 4538, 4540, 4542, 4544, 4546, 4548, 4550, 4552, 4554, 4556, 4558, 4560, 4562, 4564, 4566, 4568, 4570, 4572, 4574, 4576, 4578, 4580, 4582, 4584, 4586, 4588, 4590, 4592, 4594, 4596, 4598, 4600, 4602, 4604, 4606, 4608, 4610, 4612, 4614, 4616, 4618, 4620, 4622, 4624, 4626, 4628, 4630, 4632, 4634, 4636, 4638, 4640, 4642, 4644, 4646, 4648, 4650, 4652, 4654, 4656, 4658, 4660, 4662, 4664, 4666, 4668, 4670, 4672, 4674, 4676, 4678, 4680, and 4682.

(55) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 11/41/44/127/132/406, 11/41/44/127/278/406, 11/41/45/73/127/278/406/408, 11/41/45/127/278/406, 11/41/132/278, 11/44/45/127, 11/44/73/127/132/135, 11/44/73/127/132/135/406/408, 11/44/127, 11/44/132/278/406, 11/45/73/127/135, 11/45/127/132/135/138/182, 11/45/406, 11/73/127/132/135/182/278, 11/73/127/132/278/406/408, 11/73/127/132/406/408, 11/73/132, 11/73/132/135/408, 11/127/132/135, 11/127/132/135/182, 11/127/132/135/406, 11/127/132/138/182, 11/127/132/182, 11/132/135/406, 11/182/406, 41/44/45/73/127/132/278, 41/45/127/132/135/278/406, 41/73/132/135/406, 44/45/73/132/135/406, 44/45/127/132/135/138/182/406/408, 44/45/127/132/278, 44/73/127/135, 44/73/127/135/182/278/406/408, 44/73/132/135/406/408, 45/73/127/132/135/406/408, 45/73/132, 45/73/132/135/182, 45/73/278/406/408, 45/127/132, 45/127/132/135/182, 45/127/132/135/182/406/408, 45/127/135, 45/132/408, 45/406, 56/309/449, 57, 65/114/422/427, 65/114/427, 65/143, 65/143/235/427, 73/127, 73/127/132, 127, 127/132/135/182, 127/132/135/182/406, 132/135, 132/135/182, 132/135/406, 132/406, 164, 164/220/309/449, 164/220/449, 164/449, 182, 220/449, 309, 309/449, 406, 418/427, 427, 427/429, and 449, wherein the positions are numbered with reference to SEQ ID NO: 4550. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 11L/41H/44H/127I/132K/406G, 11L/41H/44H/127I/278L/406S, 11L/41H/45Y/73H/127I/278L/406G/408A, 11L/41H/45Y/127I/278L/406G, 11L/41H/132K/278L, 11L/44H/45Y/127I, 11L/44H/73H/127I/132K/135E, 11L/44H/73S/127I/132K/135E/406G/408A, 11L/44H/127I, 11L/44H/132K/278L/406G, 11L/45Y/73H/127I/135E, 11L/45Y/127I/132K/135E/138T/182T, 11L/45Y/406G, 11L/73H/127I/132K/278L/406G/408A, 11L/73H/127I/132K/406G/408A, 11L/73H/132K, 11L/73S/127I/132K/135E/182L/278L, 11L/73S/132K/135E/408A, 11L/127I/132K/135E, 11L/127I/132K/135E/182L, 11L/127I/132K/135E/406S, 11L/127I/132K/138T/182L, 11L/127I/132K/182T, 11L/132K/135E/406G, 11L/182L/406S, 41H/44H/45Y/73H/127I/132K/278L, 41H/45Y/127I/132K/135E/278L/406S, 41H/73H/132K/135E/406G, 44H/45Y/73H/132K/135E/406G, 44H/45Y/127I/132K/135E/138T/182L/406S/408A, 44H/45Y/127I/132K/278L, 44H/73H/127I/135E, 44H/73S/127I/135E/182L/278L/406G/408A, 44H/73S/132K/135E/406G/408A, 45Y/73H/127I/132K/135E/406G/408A, 45Y/73H/132K, 45Y/73H/132K/135E/182L, 45Y/73S/278L/406G/408A, 45Y/127I/132K, 45Y/127I/132K/135E/182L, 45Y/127I/132K/135E/182T/406G/408A, 45Y/127I/135E, 45Y/132K/408A, 45Y/406G, 56R/309P/449L, 57G, 65G/114I/422L/427S, 65G/114I/427S, 65G/143K, 65G/143K/235R/427S, 73H/127I, 73H/127I/132K, 127I, 127I/132K/135E/182I/406G, 127I/132K/135E/182L, 132K/135E, 132K/135E/182I, 132K/135E/406G, 132K/406S, 164E, 164E/220G/309S/449H, 164E/220G/449L, 164E/449H, 182L, 220G/449H, 309P/449L, 309R, 309T/449H, 406G, 418A/427S, 427S, 427S/429D, and 449L, wherein the positions are numbered with reference to SEQ ID NO: 4550. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from Q11L/F41H/Y44H/Q127I/Q132K/T406G, Q11L/F41H/Y44H/Q127I/V278L/T406S, Q11L/F41H/L45Y/A73H/Q127I/V278L/T406G/P408A, Q11L/F41H/L45Y/Q127I/V278L/T406G, Q11L/F41H/Q132K/V278L, Q11L/Y44H/L45Y/Q127I, Q11L/Y44H/A73H/Q127I/Q132K/L135E,

Q11L/Y43H/Q132K/L135E/T406G/P408A, Q11L/Y44H/Q127I,
Q11L/Y44H/Q132K/V278L/T406G, Q11L/L45Y/A73H/Q127I/L135E,
Q11L/L45Y/Q127I/Q132K/L135E/G138T/R182T, Q11L/L45Y/T406G,
Q11L/A73H/Q127I/Q132K/V278L/T406G/P408A, Q11L/A73H/Q127I/Q132K/T406G/P408A,
Q11L/A73H/Q132K, Q11L/A73S/Q127I/Q132K/L135E/R182L/V278L,
Q11L/A73S/Q132K/L135E/P408A, Q11L/Q127I/Q132K/L135E, Q11L/Q127I/Q132K/L135E/R182L,
Q11L/Q127I/Q132K/L135E/T406S, Q11L/Q127I/Q132K/G138T/R182L, Q11L/Q127I/Q132K/R182T,
Q11L/Q132K/L135E/T406G, Q11L/R182L/T406S, F41H/Y44H/L45Y/A73H/Q127I/Q132K/V278L,
F41H/L45Y/Q127I/Q132K/L135E/V278L/T406S, F41H/A73H/Q132K/L135E/T406G,
Y44H/L45Y/A73H/Q132K/L135E/T406G,
Y44H/L45Y/Q127I/Q132K/L135E/G138T/R182L/T406S/P408A, Y44H/L45Y/Q127I/Q132K/V278L,
Y44H/A73H/Q127I/L135E, Y44H/A73S/Q127I/L135E/R182L/V278L/T406G/P408A,
Y44H/A73S/Q132K/L135E/T406G/P408A, L45Y/A73H/Q127I/Q132K/L135E/T406G/P408A,
L45Y/A73H/Q132K, L45Y/A73H/Q132K/L135E/R182L, L45Y/A73S/V278L/T406G/P408A,
L45Y/Q127I/Q132K, L45Y/Q127I/Q132K/L135E/R182L,
L45Y/Q127I/Q132K/L135E/R182T/T406G/P408A, L45Y/Q127I/L135E, L45Y/Q132K/P408A,
L45Y/T406G, K56R/K309P/Y449L, K57G, N65G/Q114I/K422L/R427S, N65G/Q114I/R427S,
N65G/P143K, N65G/P143K/Q235R/R427S, A73H/Q127I, A73H/Q127I/Q132K, Q127I,
Q127I/Q132K/L135E/R182I/T406G, Q127I/Q132K/L135E/R182L, Q132K/L135E,
Q132K/L135E/R182I, Q132K/L135E/T406G, Q132K/T406S, H164E, H164E/T220G/K309S/Y449H,
H164E/T220G/Y449L, H164E/Y449H, R182L, T220G/Y449H, K309P/Y449L, K309R, K309T/Y449H,
T406G, R418A/R427S, R427S, R427S/E429D, and Y449L, wherein the positions are numbered with
reference to SEQ ID NO: 4550. In some embodiments, the polypeptide sequence of the engineered
glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 7216, 7218,
7220, 7222, 7224, 7226, 7228, 7230, 7232, 7234, 7236, 7238, 7240, 7242, 7244, 7246, 7248, 7250, 7252,
7254, 7256, 7258, 7260, 7262, 7264, 7266, 7268, 7270, 7272, 7274, 7276, 7278, 7280, 7282, 7284, 7286,
7288, 7290, 7292, 7294, 7296, 7298, 7300, 7302, 7304, 7306, 7308, 7310, 7312, 7314, 7316, 7318, 7320,
7322, 7324, 7326, 7328, 7330, 7332, 7334, 7336, 7338, 7340, 7342, 7344, 7346, 7348, 7350, 7352, 7354,
7356, 7358, and 7360. In some embodiments, the polypeptide sequence of the engineered
glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 7216, 7218,
7220, 7222, 7224, 7226, 7228, 7230, 7232, 7234, 7236, 7238, 7240, 7242, 7244, 7246, 7248, 7250, 7252,
7254, 7256, 7258, 7260, 7262, 7264, 7266, 7268, 7270, 7272, 7274, 7276, 7278, 7280, 7282, 7284, 7286,
7288, 7290, 7292, 7294, 7296, 7298, 7300, 7302, 7304, 7306, 7308, 7310, 7312, 7314, 7316, 7318, 7320,
7322, 7324, 7326, 7328, 7330, 7332, 7334, 7336, 7338, 7340, 7342, 7344, 7346, 7348, 7350, 7352, 7354,
7356, 7358, and 7360. In some embodiments, the polypeptide sequence of the engineered
glycosyltransferase comprises any of SEQ ID NOS: 7216, 7218, 7220, 7222, 7224, 7226, 7228, 7230,
7232, 7234, 7236, 7238, 7240, 7242, 7244, 7246, 7248, 7250, 7252, 7254, 7256, 7258, 7260, 7262, 7264,
7266, 7268, 7270, 7272, 7274, 7276, 7278, 7280, 7282, 7284, 7286, 7288, 7290, 7292, 7294, 7296, 7298,
7300, 7302, 7304, 7306, 7308, 7310, 7312, 7314, 7316, 7318, 7320, 7322, 7324, 7326, 7328, 7330, 7332,
7334, 7336, 7338, 7340, 7342, 7344, 7346, 7348, 7350, 7352, 7354, 7356, 7358, and 7360.

(56) The present invention also provides engineered glycosyltransferases, wherein the polypeptide
sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or
more positions selected from 7/10/73/127/132/385/423, 10, 10/53/73/99, 10/53/272/423/427, 10/53/423,
10/73, 10/73/127, 10/127/132/385, 10/132/302/385, 53/73/200/423, 53/127/132/385, 53/127/136/385,
53/132/136/302/423/427, 53/132/302/385/423/427, 65/106/445/447/449, 65/143/220,
65/220/309/445/447, 65/220/445/449, 65/399/406/447/449, 65/402/406/445/449, 65/445/447/449,
65/447/449, 73, 73/127/427, 73/132/136/385/427, 73/385, 73/385/427, 106/220/399/402/406,
115/116/278, 115/170/190/238/412, 115/190/194, 115/245/278, 116/170/190, 116/190/406/408, 116/238,
116/238/245, 116/416, 127, 132, 143/220/445/447, 143/309/402, 143/309/445/447/449, 170,
170/190/192/194, 170/192, 170/192/194/278, 170/192/194/335, 170/194, 170/194/335/416, 170/238,
170/335, 170/335/416, 190, 190/192/194, 190/194, 190/194/238/245, 190/194/335/416, 190/194/412,
190/245/412, 192/194, 192/194/242/406/408, 194, 200, 200/385, 220/399, 220/445, 220/445/447,
257/385, 272/302, 272/385, 278, 302/385, 309/399/449, 309/445/447/449, 385, 385/427, 399/406,

399/406/449, 402/445/449, 406/445/447/449, 406/445/449, 423/427, 445/447/449, 445/449, 447, and 449, wherein the positions are numbered with reference to SEQ ID NO: 7324. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 7V/10D/73H/127I/132G/385P/423L, 10D, 10D/53N/73H/99R, 10D/53N/272H/423L/427S, 10D/53N/423L, 10D/73H, 10D/73H/127I, 10D/127I/132R/385P, 10D/132R/302P/385P, 53N/73H/200S/423L, 53N/127I/132R/385S, 53N/127I/136R/385L, 53N/132G/136R/302P/423L/427S, 53N/132G/302P/385S/423L/427S, 65G/106R/445K/447L/449L, 65G/143K/220Q, 65G/220Q/309P/445R/447L, 65G/220Q/445R/449K, 65G/220Q/445R/449R, 65G/399K/406G/447L/449R, 65G/402R/406G/445R/449R, 65G/445K/447L/449R, 65G/447L/449L, 73H, 73H/127I/427S, 73H/132R/136R/385S/427S, 73H/385S, 73H/385S/427S, 106R/220Q/399K/402R/406G, 115A/170E/190T/238L/412R, 115A/190T/194F, 115A/245P/278L, 115T/116I/278L, 116I/170E/190T, 116I/190T/406G/408A, 116I/238L, 116I/238L/245P, 116I/416R, 127I, 132G, 143K/220Q/445K/447L, 143K/309P/402R, 143K/309P/445K/447L/449L, 170E, 170E/190T/192L/194F, 170E/192L, 170E/192L/194F/278L, 170E/192L/194F/335K, 170E/194F, 170E/194F/335R/416R, 170E/238L, 170E/335K, 170E/335K/416R, 190T, 190T/192L/194F, 190T/194F, 190T/194F/238L/245P, 190T/194F/335R/416R, 190T/194F/412R, 190T/245P/412R, 192L/194F, 192L/194F/242L/406G/408A, 194F, 200S, 200S/385S, 220M/445K, 220Q/399K, 220Q/445K/447L, 257H/385S, 272H/302P, 272H/385P, 278L, 302P/385P, 302P/385S, 309P/399K/449L, 309P/445K/447L/449L, 385S, 385S/427S, 399K/406G, 399K/406G/449L, 402R/445R/449L, 406G/445K/449R, 406G/445R/447L/449L, 406G/445R/449K, 423L/427S, 445K/449L, 445R/447L/449L, 445R/449L, 447L, and 449L, wherein the positions are numbered with reference to SEQ ID NO: 7324. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from G7V/G10D/A73H/Q127I/K132G/V385P/R423L, G10D, G10D/E53N/A73H/H99R, G10D/E53N/R272H/R423L/R427S, G10D/E53N/R423L, G10D/A73H, G10D/A73H/Q127I, G10D/Q127I/K132R/V385P, G10D/K132R/N302P/V385P, E53N/A73H/A200S/R423L, E53N/Q127I/K132R/V385S, E53N/Q127I/E136R/V385L, E53N/K132G/E136R/N302P/R423L/R427S, E53N/K132G/N302P/V385S/R423L/R427S, N65G/A106R/N445K/N447L/Y449L, N65G/P143K/T220Q, N65G/T220Q/K309P/N445R/N447L, N65G/T220Q/N445R/Y449K, N65G/T220Q/N445R/Y449R, N65G/Q399K/S406G/N447L/Y449R, N65G/K402R/S406G/N445R/Y449R, N65G/N445K/N447L/Y449R, N65G/N447L/Y449L, A73H, A73H/Q127I/R427S, A73H/K132R/E136R/V385S/R427S, A73H/V385S, A73H/V385S/R427S, A106R/T220Q/Q399K/K402R/S406G, N115A/A170E/D190T/M238L/I412R, N115A/D190T/P194F, N115A/K245P/V278L, N115T/L116I/V278L, L116I/A170E/D190T, L116I/D190T/S406G/P408A, L116I/M238L, L116I/M238L/K245P, L116I/K416R, Q127I, K132G, P143K/T220Q/N445K/N447L, P143K/K309P/K402R, P143K/K309P/N445K/N447L/Y449L, A170E, A170E/D190T/E192L/P194F, A170E/E192L, A170E/E192L/P194F/V278L, A170E/E192L/P194F/P335K, A170E/P194F, A170E/P194F/P335R/K416R, A170E/M238L, A170E/P335K, A170E/P335K/K416R, D190T, D190T/E192L/P194F, D190T/P194F, D190T/P194F/M238L/K245P, D190T/P194F/P335R/K416R, D190T/P194F/I412R, D190T/K245P/I412R, E192L/P194F, E192L/P194F/I242L/S406G/P408A, P194F, A200S, A200S/V385S, T220M/N445K, T220Q/Q399K, T220Q/N445K/N447L, N257H/V385S, R272H/N302P, R272H/V385P, V278L, N302P/V385P, N302P/V385S, K309P/Q399K/Y449L, K309P/N445K/N447L/Y449L, V385S, V385S/R427S, Q399K/S406G, Q399K/S406G/Y449L, K402R/N445R/Y449L, S406G/N445K/Y449R, S406G/N445R/N447L/Y449L, S406G/N445R/Y449K, R423L/R427S, N445K/Y449L, N445R/N447L/Y449L, N445R/Y449L, N447L, and Y449L, wherein the positions are numbered with reference to SEQ ID NO: 7324. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 2, 4, 11/44, 11/71, 11/287, 11/441, 37, 40, 42, 42/138, 42/141, 43, 46, 47, 48/398, 49, 49/406/408, 51, 64, 71, 76, 97, 100, 108, 108/172, 109, 112, 117/157/301, 118, 118/406/408, 119, 119/172, 130, 133, 134, 141, 157, 169, 172/420, 172/437, 179, 181, 259, 274, 275, 287, 288, 333/398, 333/406/408, 338, 356, 357, 376, 381, 385, 394/420, 396, 397/406/408, 398, 401, 406/408, 410, 417, 420, 434, 437, and 441, wherein the positions are numbered with reference to SEQ ID NO: 7324. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least

one mutation or mutation set at one or more positions selected from 2T, 4F, 11L/44H, 11L/71P, 11L/71T, 11L/287S, 11L/441S, 37G, 40Q, 42A, 42A/141P, 42D/138C, 42G, 43A, 43L, 43M, 43T, 46C, 46L, 46Q, 46T, 47N, 48S/398V, 49G, 49K, 49N/406G/408A, 51M, 51Q, 51V, 64A, 64F, 64G, 64L, 64Q, 64R, 64V, 71T, 76H, 76L, 97L, 97V, 100R, 108G/172H, 108T, 109L, 109V, 109Y, 112N, 112Q, 112R, 112T, 112V, 117R/157T/301R, 118A, 118A/406G/408A, 118I, 118L, 118S, 118V, 119A, 119G, 119S, 119T/172H, 130S, 133H, 133L, 133R, 133S, 134S, 141G, 141Q, 141S, 157C, 157E, 157T, 169E, 172H/420L, 172H/437D, 179I, 179R, 181H, 181T, 259T, 274Q, 274T, 275L, 287A, 287K, 287L, 287S, 288P, 333S/398T, 333S/406G/408A, 338S, 356S, 356T, 357T, 376G, 376M, 376N, 376R, 376S, 381C, 385A, 394H/420W, 396A, 396S, 397L/406G/408A, 398H, 398K, 398L, 398M, 398R, 401T, 406G/408A, 410R, 417C, 417S, 420F, 420L, 434C, 437G, 437V, and 441L, wherein the positions are numbered with reference to SEQ ID NO: 7324. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from H2T, H4F, Q11L/Y44H, Q11L/E71P, Q11L/E71T, Q11L/V287S, Q11L/Q441S, A37G, G40Q, V42A, V42A/A141P, V42D/G138C, V42G, I43A, I43L, I43M, I43T, V46C, V46L, V46Q, V46T, S47N, T48S/A398V, A49G, A49K, A49N/S406G/P408A, N51M, N51Q, N51V, S64A, S64F, S64G, S64L, S64Q, S64R, S64V, E71T, E76H, E76L, T97L, T97V, K100R, N108G/R172H, N108T, F109L, F109V, F109Y, I112N, I112Q, I112R, I112T, I112V, K117R/N157T/Q301R, P118A, P118A/S406G/P408A, P118I, P118L, P118S, P118V, D119A, D119G, D119S, D119T/R172H, A130S, V133H, V133L, V133R, V133S, A134S, A141G, A141Q, A141S, N157C, N157E, N157T, P169E, R172H/I420L, R172H/E437D, D179I, D179R, M181H, M181T, V259T, D274Q, D274T, M275L, V287A, V287K, V287L, V287S, N288P, N333S/A398T, N333S/S406G/P408A, G338S, G356S, G356T, V357T, V376G, V376M, V376N, V376R, V376S, A381C, V385A, R394H/I420W, E396A, E396S, I397L/S406G/P408A, A398H, A398K, A398L, A398M, A398R, L401T, S406G/P408A, G410R, V417C, V417S, I420F, I420L, A434C, E437G, E437V, and Q441L, wherein the positions are numbered with reference to SEQ ID NO: 7324. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 7766, 7768, 7770, 7772, 7774, 7776, 7778, 7780, 7782, 7784, 7786, 7788, 7790, 7792, 7794, 7796, 7798, 7800, 7812, 7814, 7816, 7818, 7820, 7822, 7824, 7826, 7828, 7830, 7832, 7834, 7836, 7838, 7840, 7842, 7844, 7846, 7848, 7850, 7852, 7854, 7856, 7858, 7860, 7862, 7864, 7866, 7868, 7870, 7872, 7874, 7876, 7878, 7880, 7882, 7884, 7886, 7888, 7890, 7892, 7894, 7896, 7898, 7900, 7902, 7904, 7906, 7908, 7910, 7912, 7914, 7916, 7918, 7920, 7922, 7924, 7926, 7928, 7930, 7932, 7934, 7936, 7938, 7940, 7942, 7944, 8092, 8094, 8096, 8098, 8100, 8102, 8104, 8106, 8108, 8110, 8112, 8114, 8116, 8118, 8120, 8122, 8124, 8126, 8128, 8130, 8132, 8134, 8136, 8138, 8140, 8142, 8144, 8146, 8148, 8150, 8152, 8154, 8156, 8158, 8160, 8162, 8164, 8166, 8168, 8170, 8172, 8174, 8176, 8178, 8180, 8182, 8184, 8186, 8188, 8190, 8192, 8194, 8196, 8198, 8200, 8202, 8204, 8206, 8208, 8210, 8212, 8214, 8216, 8218, 8220, 8222, 8224, 8226, 8228, 8230, 8232, 8234, 8236, 8238, 8240, 8242, 8244, 8246, 8248, 8250, 8252, 8254, 8256, 8258, 8260, 8262, 8264, 8266, 8268, 8270, 8272, 8274, 8276, 8278, 8280, 8282, 8284, 8286, 8288, 8290, 8292, 8294, 8296, 8298, 8300, 8302, 8304, 8306, 8308, 8310, 8312, 8314, 8316, 8318, 8320, 8322, 8324, 8326, 8328, 8330, 8332, 8334, 8336, and 8338. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 7766, 7768, 7770, 7772, 7774, 7776, 7778, 7780, 7782, 7784, 7786, 7788, 7790, 7792, 7794, 7796, 7798, 7800, 7812, 7814, 7816, 7818, 7820, 7822, 7824, 7826, 7828, 7830, 7832, 7834, 7836, 7838, 7840, 7842, 7844, 7846, 7848, 7850, 7852, 7854, 7856, 7858, 7860, 7862, 7864, 7866, 7868, 7870, 7872, 7874, 7876, 7878, 7880, 7882, 7884, 7886, 7888, 7890, 7892, 7894, 7896, 7898, 7900, 7902, 7904, 7906, 7908, 7910, 7912, 7914, 7916, 7918, 7920, 7922, 7924, 7926, 7928, 7930, 7932, 7934, 7936, 7938, 7940, 7942, 7944, 8092, 8094, 8096, 8098, 8100, 8102, 8104, 8106, 8108, 8110, 8112, 8114, 8116, 8118, 8120, 8122, 8124, 8126, 8128, 8130, 8132, 8134, 8136, 8138, 8140, 8142, 8144, 8146, 8148, 8150, 8152, 8154, 8156, 8158, 8160, 8162, 8164, 8166, 8168, 8170, 8172, 8174, 8176, 8178, 8180, 8182, 8184, 8186, 8188, 8190, 8192, 8194, 8196, 8198, 8200, 8202, 8204, 8206, 8208, 8210, 8212, 8214, 8216, 8218, 8220, 8222, 8224, 8226, 8228, 8230, 8232, 8234, 8236, 8238, 8240, 8242, 8244, 8246, 8248, 8250, 8252, 8254, 8256, 8258, 8260, 8262, 8264, 8266, 8268, 8270, 8272, 8274, 8276, 8278, 8280, 8282, 8284, 8286, 8288, 8290, 8292, 8294, 8296, 8298, 8300, 8302, 8304, 8306, 8308, 8310, 8312, 8314, 8316, 8318, 8320, 8322, 8324, 8326, 8328, 8330, 8332, 8334, 8336, and 8338. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 7766, 7768, 7770, 7772,

7774, 7776, 7778, 7780, 7782, 7784, 7786, 7788, 7790, 7792, 7794, 7796, 7798, 7800, 7812, 7814, 7816, 7818, 7820, 7822, 7824, 7826, 7828, 7830, 7832, 7834, 7836, 7838, 7840, 7842, 7844, 7846, 7848, 7850, 7852, 7854, 7856, 7858, 7860, 7862, 7864, 7866, 7868, 7870, 7872, 7874, 7876, 7878, 7880, 7882, 7884, 7886, 7888, 7890, 7892, 7894, 7896, 7898, 7900, 7902, 7904, 7906, 7908, 7910, 7912, 7914, 7916, 7918, 7920, 7922, 7924, 7926, 7928, 7930, 7932, 7934, 7936, 7938, 7940, 7942, 7944, 8092, 8094, 8096, 8098, 8100, 8102, 8104, 8106, 8108, 8110, 8112, 8114, 8116, 8118, 8120, 8122, 8124, 8126, 8128, 8130, 8132, 8134, 8136, 8138, 8140, 8142, 8144, 8146, 8148, 8150, 8152, 8154, 8156, 8158, 8160, 8162, 8164, 8166, 8168, 8170, 8172, 8174, 8176, 8178, 8180, 8182, 8184, 8186, 8188, 8190, 8192, 8194, 8196, 8198, 8200, 8202, 8204, 8206, 8208, 8210, 8212, 8214, 8216, 8218, 8220, 8222, 8224, 8226, 8228, 8230, 8232, 8234, 8236, 8238, 8240, 8242, 8244, 8246, 8248, 8250, 8252, 8254, 8256, 8258, 8260, 8262, 8264, 8266, 8268, 8270, 8272, 8274, 8276, 8278, 8280, 8282, 8284, 8286, 8288, 8290, 8292, 8294, 8296, 8298, 8300, 8302, 8304, 8306, 8308, 8310, 8312, 8314, 8316, 8318, 8320, 8322, 8324, 8326, 8328, 8330, 8332, 8334, 8336, and 8338.

(57) The present invention also provides engineered glycosyltransferases, wherein the polypeptide sequence of the engineered glycosyltransferases comprise at least one mutation or mutation set at one or more positions selected from 11, 11/64/109, 11/445, 42/43, 42/44/71/73/116, 43/73/141, 46/47/51, 46/51, 47/49, 47/51, 64, 64/65/109, 64/65/112, 64/112, 64/134, 64/445, 65/112, 65/112/445, 71, 71/73, 71/73/141, 71/141, 71/302, 73, 73/116/141, 73/141, 73/302, 109, 109/112, 109/115/118, 109/134, 109/406, 112, 112/445, 116/287, 127, 127/169/172, 127/169/287, 127/169/376/398/399, 127/169/398/399, 127/287, 127/376, 141, 141/302, 169/172/287, 169/172/288, 169/172/288/398/399/420/423/427, 169/172/398/399, 169/287, 169/398, 169/398/399, 287, 287/288, 287/376, 287/398, 287/399, 287/420/423/427, 288, 288/376/398, 288/398, 288/399, 302, 376, 376/398, 376/399, 398, 398/399, 398/399/420, 398/427, 399, and 420, wherein the positions are numbered with reference to SEQ ID NO: 7784. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 11L, 11L/64R/109V, 11L/445K, 42A/44H/71T/73H/116I, 42G/43M, 43T/73H/141P, 46L/47N/51Q, 46T/51Q, 47N/49G, 47N/51V, 64R, 64R/65G/109V, 64R/65G/112N, 64R/65G/112T, 64R/112N, 64R/134S, 64R/445K, 65G/112N, 65G/112N/445K, 71T, 71T/73H, 71T/73H/141G, 71T/73H/141S, 71T/141P, 71T/302P, 73H, 73H/116I/141P, 73H/141G, 73H/141P, 73H/141S, 73H/302P, 109L/112T, 109V, 109V/112T, 109V/115A/118V, 109V/134S, 109V/406G, 112N/445K, 112T, 116I/287S, 127I, 127I/169E/172H, 127I/169E/287S, 127I/169E/376M/398M/399K, 127I/169E/398L/399K, 127I/287S, 127I/376M, 141G, 141G/302P, 141P, 141S, 141S/302P, 169E/172H/287S, 169E/172H/288P, 169E/172H/288P/398L/399K/420F/423L/427S, 169E/172H/398M/399K, 169E/287S, 169E/398M/399K, 169E/398T, 287L/288P, 287L/399K, 287M, 287S, 287S/376M, 287S/398T, 287S/399K, 287S/420F/423L/427S, 288P, 288P/376S/398L, 288P/398K, 288P/399K, 302P, 376M, 376M/398L, 376M/398M, 376M/399K, 398L, 398L/399K, 398L/427S, 398M, 398M/399K, 398M/399K/420F, 398R, 398R/399K, 398T/399K, 399K, and 420F, wherein the positions are numbered with reference to SEQ ID NO: 7784. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from Q11L, Q11L/S64R/F109V, Q11L/N445K, V42A/Y44H/E71T/A73H/L116I, V42G/I43M, I43T/A73H/A141P, V46L/S47N/N51Q, V46T/N51Q, S47N/A49G, S47N/N51V, S64R, S64R/N65G/F109V, S64R/N65G/I112N, S64R/N65G/I112T, S64R/I112N, S64R/A134S, S64R/N445K, N65G/I112N, N65G/I112N/N445K, E71T, E71T/A73H, E71T/A73H/A141G, E71T/A73H/A141S, E71T/A141P, E71T/N302P, A73H, A73H/L116I/A141P, A73H/A141G, A73H/A141P, A73H/A141S, A73H/N302P, F109L/I112T, F109V, F109V/I112T, F109V/N115A/P118V, F109V/A134S, F109V/S406G, I112N/N445K, I112T, L116I/V287S, Q127I, Q127I/P169E/R172H, Q127I/P169E/V287S, Q127I/P169E/V376M/A398M/Q399K, Q127I/P169E/A398L/Q399K, Q127I/V287S, Q127I/V376M, A141G, A141G/N302P, A141P, A141S, A141S/N302P, P169E/R172H/V287S, P169E/R172H/N288P, P169E/R172H/N288P/A398L/Q399K/I420F/R423L/R427S, P169E/R172H/A398M/Q399K, P169E/V287S, P169E/A398M/Q399K, P169E/A398T, V287L/N288P, V287L/Q399K, V287M, V287S, V287S/V376M, V287S/A398T, V287S/Q399K, V287S/I420F/R423L/R427S, N288P, N288P/V376S/A398L, N288P/A398K, N288P/Q399K, N302P, V376M, V376M/A398L, V376M/A398M, V376M/Q399K, A398L, A398L/Q399K, A398L/R427S, A398M, A398M/Q399K,

A398M/Q399K/I420F, A398R, A398R/Q399K, A398T/Q399K, Q399K, and I420F, wherein the positions are numbered with reference to SEQ ID NO: 7784. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 10, 10/144, 10/199, 13, 14, 15, 15/394, 16, 22, 36, 89, 93, 96, 116, 116/123, 116/143, 116/350, 123, 125, 127, 143, 144, 149, 156, 186, 187, 197, 198, 199, 201, 202, 203, 268, 287, 324, 331, and 350, wherein the positions are numbered with reference to SEQ ID NO: 7784. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from 10D, 10D/144L, 10D/199G, 13Q, 14Q, 15A, 15L/394H, 16A, 16G, 16T, 22G, 36M, 89A, 93A, 96M, 116I, 116I/123N, 116I/123S, 116I/143R, 116I/350L, 123A, 123G, 123L, 123S, 123V, 125L, 127A, 127G, 127V, 143G, 144G, 144Q, 144S, 149S, 156V, 186N, 187G, 187Y, 197H, 198Q, 199G, 199P, 199R, 199S, 199Y, 201A, 201K, 201L, 201N, 202A, 202Y, 203T, 268W, 287A, 324R, 331C, 331V, and 350L, wherein the positions are numbered with reference to SEQ ID NO: 7784. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises at least one mutation or mutation set at one or more positions selected from G10D, G10D/V144L, G10D/N199G, L13Q, R14Q, V15A, V15L/R394H, L16A, L16G, L16T, A22G, L36M, G89A, H93A, G96M, L116I, L116I/Y123N, L116I/Y123S, L116I/P143R, L116I/M350L, Y123A, Y123G, Y123L, Y123S, Y123V, F125L, Q127A, Q127G, Q127V, P143G, V144G, V144Q, V144S, A149S, F156V, G186N, T187G, T187Y, P197H, G198Q, N199G, N199P, N199R, N199S, N199Y, G201A, G201K, G201L, G201N, I202A, I202Y, M203T, Y268W, V287A, H324R, I331C, I331V, and M350L, wherein the positions are numbered with reference to SEQ ID NO: 7784. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 90% identical to any of SEQ ID NOS: 8624, 8626, 8628, 8630, 8632, 8634, 8636, 8638, 8640, 8642, 8644, 8646, 8648, 8650, 8652, 8654, 8656, 8658, 8660, 8662, 8664, 8666, 8668, 8670, 8672, 8674, 8676, 8678, 8680, 8682, 8684, 8686, 8688, 8690, 8692, 8694, 8696, 8698, 8700, 8702, 8704, 8706, 8708, 8710, 8712, 8714, 8716, 8718, 8720, 8722, 8724, 8726, 8728, 8730, 8732, 8734, 8736, 8738, 8740, 8742, 8744, 8746, 8748, 8750, 8752, 8754, 8756, 8758, 8760, 8762, 8764, 8766, 8768, 8770, 8772, 8774, 8776, 8778, 8780, 8782, 8784, 8786, 8788, 8790, 8792, 8794, 8796, 9108, 9110, 9112, 9114, 9116, 9118, 9120, 9122, 9124, 9126, 9128, 9130, 9132, 9134, 9136, 9138, 9140, 9142, 9144, 9146, 9148, 9150, 9152, 9154, 9156, 9158, 9160, 9162, 9164, 9166, 9168, 9170, 9172, 9174, 9176, 9178, 9180, 9182, 9184, 9186, 9188, 9190, 9192, 9194, 9196, 9198, 9200, 9202, 9204, 9206, 9208, 9210, 9212, 9214, 9216, 9218, 9220, and 9222. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises a sequence at least 95% identical to any of SEQ ID NOS: 8624, 8626, 8628, 8630, 8632, 8634, 8636, 8638, 8640, 8642, 8644, 8646, 8648, 8650, 8652, 8654, 8656, 8658, 8660, 8662, 8664, 8666, 8668, 8670, 8672, 8674, 8676, 8678, 8680, 8682, 8684, 8686, 8688, 8690, 8692, 8694, 8696, 8698, 8700, 8702, 8704, 8706, 8708, 8710, 8712, 8714, 8716, 8718, 8720, 8722, 8724, 8726, 8728, 8730, 8732, 8734, 8736, 8738, 8740, 8742, 8744, 8746, 8748, 8750, 8752, 8754, 8756, 8758, 8760, 8762, 8764, 8766, 8768, 8770, 8772, 8774, 8776, 8778, 8780, 8782, 8784, 8786, 8788, 8790, 8792, 8794, 8796, 9108, 9110, 9112, 9114, 9116, 9118, 9120, 9122, 9124, 9126, 9128, 9130, 9132, 9134, 9136, 9138, 9140, 9142, 9144, 9146, 9148, 9150, 9152, 9154, 9156, 9158, 9160, 9162, 9164, 9166, 9168, 9170, 9172, 9174, 9176, 9178, 9180, 9182, 9184, 9186, 9188, 9190, 9192, 9194, 9196, 9198, 9200, 9202, 9204, 9206, 9208, 9210, 9212, 9214, 9216, 9218, 9220, and 9222. In some embodiments, the polypeptide sequence of the engineered glycosyltransferase comprises any of SEQ ID NOS: 8624, 8626, 8628, 8630, 8632, 8634, 8636, 8638, 8640, 8642, 8644, 8646, 8648, 8650, 8652, 8654, 8656, 8658, 8660, 8662, 8664, 8666, 8668, 8670, 8672, 8674, 8676, 8678, 8680, 8682, 8684, 8686, 8688, 8690, 8692, 8694, 8696, 8698, 8700, 8702, 8704, 8706, 8708, 8710, 8712, 8714, 8716, 8718, 8720, 8722, 8724, 8726, 8728, 8730, 8732, 8734, 8736, 8738, 8740, 8742, 8744, 8746, 8748, 8750, 8752, 8754, 8756, 8758, 8760, 8762, 8764, 8766, 8768, 8770, 8772, 8774, 8776, 8778, 8780, 8782, 8784, 8786, 8788, 8790, 8792, 8794, 8796, 9108, 9110, 9112, 9114, 9116, 9118, 9120, 9122, 9124, 9126, 9128, 9130, 9132, 9134, 9136, 9138, 9140, 9142, 9144, 9146, 9148, 9150, 9152, 9154, 9156, 9158, 9160, 9162, 9164, 9166, 9168, 9170, 9172, 9174, 9176, 9178, 9180, 9182, 9184, 9186, 9188, 9190, 9192, 9194, 9196, 9198, 9200, 9202, 9204, 9206, 9208, 9210, 9212, 9214, 9216, 9218, 9220, and 9222.

(58) The present invention also provides engineered sucrose synthase comprising a polypeptide sequence that is at least 60%, 65%, 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NO:72. In some embodiments the

present invention provides engineered sucrose synthase comprising a polypeptide sequence that is at least 60%, 65%, 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NO: 74, 1080, 1158, 1222, 1392, 1456, 1582, 1764, 1804, 1840, 2064, 2432, 2510, 7506, and/or 8420. In some embodiments the present invention provides engineered sucrose synthase comprising a polypeptide sequence that is at least 90% or more sequence identity to SEQ ID NO: 74, 1080, 1158, 1222, 1392, 1456, 1582, 1764, 1804, 1840, 2064, 2432, 2510, 7506, and/or 8420. In some embodiments the present invention provides engineered sucrose synthase comprising a polypeptide sequence that is at least 91% or more sequence identity to SEQ ID NO: 74, 1080, 1158, 1222, 1392, 1456, 1582, 1764, 1804, 1840, 2064, 2432, 2510, 7506, and/or 8420. In some embodiments the present invention provides engineered sucrose synthase comprising a polypeptide sequence that is at least 92% or more sequence identity to SEQ ID NO: 74, 1080, 1158, 1222, 1392, 1456, 1582, 1764, 1804, 1840, 2064, 2432, 2510, 7506, and/or 8420. In some embodiments the present invention provides engineered sucrose synthase comprising a polypeptide sequence that is at least 93% or more sequence identity to SEQ ID NO: 74, 1080, 1158, 1222, 1392, 1456, 1582, 1764, 1804, 1840, 2064, 2432, 2510, 7506, and/or 8420. In some embodiments the present invention provides engineered sucrose synthase comprising a polypeptide sequence that is at least 94% or more sequence identity to SEQ ID NO: 74, 1080, 1158, 1222, 1392, 1456, 1582, 1764, 1804, 1840, 2064, 2432, 2510, 7506, and/or 8420. In some embodiments the present invention provides engineered sucrose synthase comprising a polypeptide sequence that is at least 95% or more sequence identity to SEQ ID NO: 74, 1080, 1158, 1222, 1392, 1456, 1582, 1764, 1804, 1840, 2064, 2432, 2510, 7506, and/or 8420. In some embodiments the present invention provides engineered sucrose synthase comprising a polypeptide sequence that is at least 96% or more sequence identity to SEQ ID NO: 74, 1080, 1158, 1222, 1392, 1456, 1582, 1764, 1804, 1840, 2064, 2432, 2510, 7506, and/or 8420. In some embodiments the present invention provides engineered sucrose synthase comprising a polypeptide sequence that is at least 97% or more sequence identity to SEQ ID NO: 74, 1080, 1158, 1222, 1392, 1456, 1582, 1764, 1804, 1840, 2064, 2432, 2510, 7506, and/or 8420. In some embodiments the present invention provides engineered sucrose synthase comprising a polypeptide sequence that is at least 98% or more sequence identity to SEQ ID NO: 74, 1080, 1158, 1222, 1392, 1456, 1582, 1764, 1804, 1840, 2064, 2432, 2510, 7506, and/or 8420. In some embodiments the present invention provides engineered sucrose synthase comprising a polypeptide sequence that is at least 99% or more sequence identity to SEQ ID NO: 74, 1080, 1158, 1222, 1392, 1456, 1582, 1764, 1804, 1840, 2064, 2432, 2510, 7506, and/or 8420. In some embodiments the present invention provides engineered sucrose synthase comprising a polypeptide sequence of SEQ ID NO: 74, 1080, 1158, 1222, 1392, 1456, 1582, 1764, 1804, 1840, 2064, 2432, 2510, 7506, and/or 8420. In additional embodiments, the present invention provides sucrose synthases provided in Table 18.1, 19.1, 19.2, 20.1, 20.2, 20.3, 31.2, 31.3, 32.1, 32.2, 33.1, 33.2, 34.1, 34.2, 35.1, 35.2, 36.1, 36.2, 37.1, 37.2, 37.3, 38.1, 38.2, 38.3, 39.1, 39.2, 39.3, 40.1, 40.2, 40.3, 41.1, 41.2, 42.1, and/or 42.2. The present invention also provides engineered sucrose synthase comprising a polypeptide sequence that has at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NO: 1080, 1082, 1084, 1086, 1088, 1090, 1092, 1094, 1096, 1098, 1100, 1102, 1104, 1106, 1108, 1110, 1112, 1114, 1116, 1118, 1120, 1122, 1124, 1126, 1128, 1130, 1132, 1134, 1136, 1138, 1140, 1142, 1144, 1146, 1148, 1150, 1152, 1154, 1156, 1158, 1160, 1162, 1164, 1166, 1168, 1170, 1172, 1174, 1176, 1178, 1180, 1182, 1184, 1186, 1188, 1190, 1192, 1194, 1196, 1198, 1200, 1202, 1204, 1206, 1208, 1210, 1212, 1214, 1216, 1218, 1220, 1222, 1224, 1226, 1228, 1230, 1232, 1234, 1236, 1238, 1240, 1242, 1244, 1246, 1248, 1250, 1252, 1254, 1256, 1258, 1260, 1262, 1264, 1266, 1268, 1270, 1272, 1274, 1276, 1278, 1280, 1282, 1284, 1286, 1288, 1296, 1298, 1300, 1302, 1304, 1306, 1308, 1310, 1312, 1314, 1316, 1318, 1320, 1322, 1324, 1326, 1328, 1330, 1332, 1334, 1336, 1338, 1340, 1342, 1344, 1346, 1348, 1350, 1352, 1354, 1356, 1358, 1360, 1362, 1364, 1366, 1368, 1370, 1372, 1374, 1376, 1378, 1380, 1382, 1384, 1386, 1388, 1390, 1392, 1394, 1396, 1398, 1400, 1402, 1404, 1406, 1408, 1410, 1412, 1414, 1416, 1418, 1420, 1422, 1424, 1426, 1428, 1430, 1432, 1434, 1436, 1438, 1440, 1442, 1444, 1446, 1448, 1450, 1452, 1454, 1456, 1458, 1460, 1462, 1464, 1466, 1468, 1470, 1472, 1474, 1476, 1478, 1480, 1482, 1484, 1486, 1488, 1490, 1492, 1494, 1496, 1498, 1500, 1502, 1504, 1506, 1508, 1510, 1512, 1514, 1516, 1518, 1520, 1522, 1524, 1526, 1528, 1530, 1532, 1534, 1536, 1538, 1540, 1542, 1544, 1546, 1548, 1550, 1552, 1554, 1556, 1558, 1560, 1562, 1564, 1566, 1568, 1570, 1572, 1574, 1576, 1578, 1580, 1582, 1584, 1586, 1588, 1590, 1592, 1594, 1596, 1598, 1600, 1602, 1604, 1606, 1608, 1610, 1612,

1614, 1616, 1618, 1620, 1622, 1624, 1626, 1628, 1630, 1632, 1634, 1636, 1638, 1640, 1642, 1644, 1646, 1648, 1650, 1652, 1654, 1656, 1658, 1660, 1662, 1664, 1666, 1668, 1670, 1672, 1674, 1676, 1678, 1680, 1682, 1684, 1686, 1688, 1690, 1692, 1694, 1696, 1698, 1700, 1702, 1704, 1706, 1708, 1710, 1712, 1714, 1716, 1718, 1720, 1722, 1724, 1726, 1728, 1730, 1732, 1734, 1736, 1738, 1740, 1742, 1744, 1746, 1748, 1750, 1752, 1754, 1756, 1758, 1760, 1762, 1764, 1766, 1768, 1770, 1772, 1774, 1776, 1778, 1780, 1782, 1784, 1786, 1788, 1790, 1792, 1794, 1796, 1798, 1800, 1802, 1804, 1806, 1808, 1810, 1812, 1814, 1816, 1818, 1820, 1822, 1824, 1826, 1828, 1830, 1832, 1834, 1836, 1838, 1840, 1842, 1844, 1846, 1848, 1850, 1852, 1854, 1856, 1858, 1860, 1862, 1864, 1866, 1868, 1870, 1872, 1874, 1876, 1878, 1880, 1882, 1884, 1886, 1888, 1890, 1892, 1894, 1896, 1898, 1900, 1902, 1904, 1906, 1908, 1910, 1912, 1914, 1916, 1918, 1920, 1922, 1924, 1926, 1928, 1930, 1932, 1934, 1936, 1938, 1940, 1942, 1944, 1946, 1948, 1950, 1952, 1954, 1956, 1958, 1960, 1962, 1964, 1966, 1968, 1970, 1972, 1974, 1976, 1978, 1980, 1982, 1984, 1986, 1988, 1990, 1992, 1994, 1996, 1998, 2000, 2002, 2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022, 2024, 2026, 2028, 2030, 2032, 2034, 2036, 2038, 2040, 2042, 2044, 2046, 2048, 2050, 2052, 2054, 2056, 2058, 2060, 2062, 2064, 2066, 2068, 2070, 2072, 2074, 2076, 2078, 2080, 2082, 2084, 2086, 2088, 2090, 2092, 2094, 2096, 2098, 2100, 2102, 2104, 2106, 2108, 2110, 2112, 2114, 2116, 2118, 2120, 2122, 2124, 2126, 2128, 2130, 2132, 2134, 2136, 2138, 2140, 2142, 2144, 2146, 2148, 2150, 2152, 2154, 2156, 2158, 2160, 2162, 2164, 2166, 2168, 2170, 2172, 2174, 2176, 2178, 2180, 2182, 2184, 2186, 2188, 2190, 2192, 2194, 2196, 2198, 2200, 2202, 2204, 2206, 2208, 2210, 2212, 2214, 2216, 2218, 2220, 2222, 2224, 2226, 2228, 2230, 2232, 2234, 2236, 2238, 2240, 2242, 2244, 2246, 2248, 2250, 2252, 2254, 2256, 2258, 2260, 2262, 2264, 2266, 2268, 2270, 2272, 2274, 2276, 2278, 2280, 2282, 2284, 2286, 2288, 2290, 2292, 2294, 2296, 2298, 2300, 2302, 2304, 2306, 2308, 2310, 2312, 2314, 2316, 2318, 2320, 2322, 2324, 2326, 2328, 2330, 2332, 2334, 2336, 2338, 2340, 2342, 2344, 2346, 2348, 2350, 2352, 2354, 2356, 2358, 2360, 2362, 2364, 2366, 2368, 2370, 2372, 2374, 2376, 2378, 2380, 2382, 2384, 2386, 2388, 2390, 2392, 2394, 2396, 2398, 2400, 2402, 2404, 2406, 2408, 2410, 2412, 2414, 2416, 2418, 2420, 2422, 2424, 2426, 2428, 2430, 2432, 2434, 2436, 2438, 2440, 2442, 2444, 2446, 2448, 2450, 2452, 2454, 2456, 2458, 2460, 2462, 2464, 2466, 2468, 2470, 2472, 2474, 2476, 2478, 2480, 2482, 2484, 2486, 2488, 2490, 2492, 2494, 2496, 2498, 2500, 2502, 2504, 2506, 2508, 2510, 2512, 2514, 2516, 2518, 2520, 2522, 2524, 2526, 2528, 2530, 2532, 2534, 2536, 2538, 2540, 2542, 2544, 2546, 2548, 2550, 2552, 2554, 2556, 2558, 2560, 2562, 2564, 2566, 2568, 2570, 2572, 2574, 2576, 2578, 2580, 2582, 2584, 2586, 2588, 2590, 2592, 2594, 7438, 7440, 7442, 7444, 7446, 7448, 7450, 7452, 7454, 7456, 7458, 7460, 7462, 7464, 7466, 7468, 7470, 7472, 7474, 7476, 7478, 7480, 7482, 7484, 7486, 7488, 7490, 7492, 7494, 7496, 7498, 7500, 7502, 7504, 7506, 7508, 7510, 7512, 7514, 7516, 7518, 7520, 7522, 7524, 7526, 7528, 7530, 7532, 7534, 7536, 7538, 7540, 7542, 7544, 7546, 7548, 7550, 7552, 7554, 7556, 7558, 7560, 7562, 7564, 7566, 7568, 7570, 7572, 7574, 7576, 7578, 7580, 7582, 7584, 7586, 7588, 7590, 7592, 7594, 7596, 7598, 7600, 7602, 7604, 7606, 7608, 7610, 7612, 7614, 7616, 7618, 7620, 7622, 7624, 7626, 7628, 7630, 7632, 7634, 7636, 7638, 7640, 7642, 7644, 7646, 7648, 7650, 7652, 7654, 7656, 7658, 7660, 7662, 7664, 7666, 7668, 7670, 7672, 7674, 7676, 7678, 7680, 7682, 7684, 7686, 7688, 7690, 7692, 7694, 7696, 7698, 7700, 7702, 7704, 7706, 7708, 7710, 7712, 7714, 7716, 7718, 7720, 7722, 7724, 7726, 7728, 7730, 7732, 7734, 7736, 7738, 7740, 7742, 7744, 7746, 7748, 7750, 7752, 7754, 7756, 7758, 7760, 7762, 7764, 8370, 8372, 8374, 8376, 8378, 8380, 8382, 8384, 8386, 8388, 8390, 8392, 8394, 8396, 8398, 8400, 8402, 8404, 8406, 8408, 8410, 8412, 8414, 8416, 8418, 8420, 8422, 8424, 8426, 8428, 8430, 8432, 8434, 8436, 8438, 8440, 8442, 8444, 8446, 8448, 8450, 8452, 8454, 8456, 8458, 8460, 8462, 8464, 8466, 8468, 8470, 8472, 8474, 8476, 8478, 8480, 8798, 8800, 8802, 8804, 8806, 8808, 8810, 8812, 8814, 8816, 8818, 8820, 8822, 8824, 8826, 8828, 8830, 8832, 8834, 8836, 8838, 8840, 8842, 8844, 8846, 8848, 8850, 8852, 8854, 8856, 8858, 8860, 8862, 8864, 8866, 8868, 8870, 8872, 8874, 8876, 8878, 8880, 8882, 8884, 8886, 8888, 8890, 8892, 8894, 8896, 8898, 8900, 8902, 8904, 8906, 8908, 8910, 8912, 8914, 8916, 8918, 8920, 8922, 8924, 8926, 8928, 8930, 8932, 8934, 8936, 8938, 8940, 8942, 8944, 8946, 8948, 8950, 8952, 8954, 8956, 8958, 8960, 8962, 8964, 8966, 8968, 8970, 8972, 8974, 8976, 8978, 8980, 8982, 8984, 8986, 8988, 8990, 8992, 8994, 8996, 8998, 9000, 9002, 9004, 9006, 9008, 9010, 9012, 9014, 9016, 9018, 9020, 9022, 9024, 9026, 9028, 9030, 9032, 9034, 9036, 9038, 9040, 9042, 9044, 9046, 9048, 9050, 9052, 9054, 9056, 9058, 9060, 9062, 9064, 9066, 9068, 9070, 9072, 9074, 9076, 9078, 9080, 9082, 9084, 9086, 9088, 9090, 9092, 9094, 9096, 9098, 9100, 9102, 9104, and/or 9106.

(59) The present invention also provides engineered sucrose synthases, wherein the polypeptide sequence

of the engineered sucrose synthases comprise a mutation set at one or more positions selected from 4/9/349/532, 4/13/113/343/532, 4/13/113/532, 4/33/47/52/343/532, 4/47/52/532, 4/113/532, 4/13/113, 4/13/532, 4/33/113, 4/343, 7, 8, 44, 95, 117/440, 136, 221, 343/532, 440, 444, 478, 532, 583, 611, 615, 615/789, 695, 722, and 788, wherein the positions are numbered with reference to SEQ ID NO:74. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 4E/9T/349H/532S, 4E/13R/113Q/343H/532S, 4E/13R/113Q/532S, 4E/33Q/47H/52D/343H/532S, 4E/47H/52D/532S, 4E/113Q/532S, 4E/13R/113Q, 4E/13R/532S, 4E/33Q/113Q, 4E/343H, 7T, 8M, 44K, 95S, 117D/440T, 136S, 221A/H, 343S/532S, 440P/T, 444K/T, 478T/V, 532R/T, 583Q, 611V, 615C/E/T/V, 615L/789D, 695L, 722Y, and 788E, wherein the positions are numbered with reference to SEQ ID NO:74. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from A4E/L9T/Q349H/F532S, A4E/P13R/I113Q/V343H/F532S, A4E/P13R/I113Q/F532S, A4E/Y33Q/L47H/A52D/V343H/F532S, A4E/L47H/A52D/F532S, A4E/I113Q/F532S, A4E/P13R/I113Q, A4E/P13R/F532S, A4E/Y33Q/I113Q, A4E/V343H, Q7T, Q8M, R44K, Q95S, G117D/R440T, R136S, R221A/H, V343S/F532S, R440P/T, Q444K/T, R478T/V, F532R/T, R583Q, R611V, R615C/E/T/V, R615L/A789D, V695L, R722Y, and H788E, wherein the positions are numbered with reference to SEQ ID NO:74. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 90% identical to any of SEQ ID NOS: 1080, 1082, 1084, 1086, 1088, 1090, 1092, 1094, 1096, 1098, 1100, 1102, 1104, 1106, 1108, 1110, 1112, 1114, 1116, 1118, 1120, 1122, 1124, 1126, 1128, 1130, 1132, 1134, 1136, 1138, 1140, 1142, 1144, 1146, 1148, 1150, and/or 1152. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 95% identical to any of SEQ ID NOS: 1080, 1082, 1084, 1086, 1088, 1090, 1092, 1094, 1096, 1098, 1100, 1102, 1104, 1106, 1108, 1110, 1112, 1114, 1116, 1118, 1120, 1122, 1124, 1126, 1128, 1130, 1132, 1134, 1136, 1138, 1140, 1142, 1144, 1146, 1148, 1150, and/or 1152. In some further embodiments, the polypeptide sequence of the engineered sucrose synthase comprises any of SEQ ID NOS: 1080, 1082, 1084, 1086, 1088, 1090, 1092, 1094, 1096, 1098, 1100, 1102, 1104, 1106, 1108, 1110, 1112, 1114, 1116, 1118, 1120, 1122, 1124, 1126, 1128, 1130, 1132, 1134, 1136, 1138, 1140, 1142, 1144, 1146, 1148, 1150, and/or 1152.

(60) The present invention also provides engineered sucrose synthases, wherein the polypeptide sequence of the engineered sucrose synthases comprise at least one mutation or mutation set at one or more positions selected from 8/221, 47/221, 68/129/248, 68/129/248/595/600/756, 68/146/248/387/506/550, 68/189/272/316/477/719/756, 75/105/154/215/264/345, 75/105/345/410/769, 75/105/530, 75/345/530, 85/170/225/266/534, 87/125/230/267/375/464/708, 93/129/506/550/595/719/756, 93/477/635, 95/136/788, 95/201/478/583/724/788, 95/385/478/583/788, 95/440/478/724/788/792, 95/444/478/603/792, 95/444/478/724/788, 95/478/724, 98/250, 113/225/266/415, 126/314/499/549/589/755, 136/440/444/478/603, 136/440/444/478/583/788, 136/444/478/583/788/792, 225/372/534, 266, 306/358/703/776, 358/636/737, 440/444/583/724/788, 440/478, and 466, wherein the positions are numbered with reference to SEQ ID NO:1080. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 8M/221H, 47L/221H, 68A/129E/248A, 68A/129E/248A/595V/600I/756V, 68A/146N/248A/387I/506P/550H, 68A/189R/272L/316I/477K/719C/756V, 75V/105E/154H/215F/264V/345T, 75V/105E/345T/410S/769R, 75V/105E/530L, 75V/345T/530L, 85V/170L/225E/266N/534H, 87E/125E/230D/267V/375Y/464F/708A, 93V/129E/506P/550H/595V/719C/756V, 93V/477K/635S, 95S/136S/788E, 95S/201E/478V/583Q/724H/788E, 95S/385L/478V/583Q/788E, 95S/440T/478V/724H/788E/792S, 95S/444T/478V/603S/792S, 95S/444T/478V/724H/788E, 95S/478T/724H, 98V/250D, 113L/225E/266N/415K, 126L/314L/499H/549E/589E/755G, 136S/440P/444T/478T/603S, 136S/440T/444T/478V/583Q/788E, 136S/444T/478V/583Q/788E/792S, 225E/372V/534H, 266N, 306L/358E/703Y/776E, 358E/636Q/737I, 440P/444T/583Q/724H/788E, 440T/478V, and 466I, wherein the positions are numbered with reference to SEQ ID NO:1080. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from Q8M/R221H, H47L/R221H, V68A/R129E/S248A, V68A/R129E/S248A/I595V/V600I/I756V, V68A/D146N/S248A/V387I/S506P/R550H,

V68A/R189R/G128R/I271K/V105E/R154H/I215F/I264V/A345T, M75V/A105E/A345T/T410S/Q769R, M75V/A105E/P530L, M75V/A345T/P530L, R85V/I170L/A225E/R266N/E534H, I87E/T125E/N230D/I267V/W375Y/I464F/T708A, R93V/R129E/S506P/R550H/I595V/A719C/I756V, R93V/D477K/A635S, Q95S/R136S/H788E, Q95S/Q201E/R478V/R583Q/K724H/H788E, Q95S/R385L/R478V/R583Q/H788E, Q95S/R440T/R478V/K724H/H788E/M792S, Q95S/Q444T/R478V/G603S/M792S, Q95S/Q444T/R478V/K724H/H788E, Q95S/R478T/K724H, L98V/S250D, I113L/A225E/R266N/R415K, V126L/V314L/N499H/D549E/G589E/R755G, R136S/R440P/Q444T/R478T/G603S, R136S/R440T/Q444T/R478V/R583Q/H788E, R136S/Q444T/R478V/R583Q/H788E/M792S, A225E/I372V/E534H, R266N, R306L/P358E/N703Y/Q776E, P358E/L636Q/V737I, R440P/Q444T/R583Q/K724H/H788E, R440T/R478V, and V466I, wherein the positions are numbered with reference to SEQ ID NO: 1080. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 90% identical to any of SEQ ID NOS: 1192, 1194, 1196, 1198, 1200, 1202, 1204, 1206, 1208, 1210, 1212, 1214, 1216, 1218, and/or 1220. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 95% identical to any of SEQ ID NOS: 1192, 1194, 1196, 1198, 1200, 1202, 1204, 1206, 1208, 1210, 1212, 1214, 1216, 1218, and/or 1220. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises any of SEQ ID NOS: 1192, 1194, 1196, 1198, 1200, 1202, 1204, 1206, 1208, 1210, 1212, 1214, 1216, 1218, and/or 1220.

(61) The present invention also provides engineered sucrose synthases, wherein the polypeptide sequence of the engineered sucrose synthases comprise at least one mutation or mutation set at one or more positions selected from 8/68/95/98/358/478/595/724/792, 8/68/98/221/248/250/440/477/534/595/724, 8/68/788, 8/93/95/98/136/221/595/600/788, 8/93/95/113/250/440/595/600/724/788, 8/95/98/440/478/534/600/788, 8/136/248/478/788, 47/75/85/105/125/129/170/635, 47/75/85/105/375/756/776, 47/75/85/264/267/372/415/635, 47/75/85/87/129/375/776, 47/75/85/87/170/372/756, 47/85/105/129/201/230/267/583, 47/85/125/372/583/635/756, 47/85/170/756, 47/85/87/105/125/635, 47/85/87/154/756, 47/125/129/375/756/776, 47/129/170/635, 47/154/372/375/583/635/708/756, 68/93/95/358/440/444/478/534/595/603, 68/93/95/444/788, 68/93/98/136/248/250/358/440/534/724, 75/85/87/105/264/267/583/708, 75/85/129/154/264/375, 85/125/215/375/415/635/776, 85/87/105/215/267/756, 85/87/129/375/756/776, 87/125/129/170/230/756, 87/154/306/375/756, 93/95/98/534/792, 95/440/444/724/788, 129/215/372/756, and 170/264/267, wherein the positions are numbered with reference to SEQ ID NO:1158. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 8M/68A/95S/98V/358E/478T/595V/724H/792S, 8M/68A/98V/221H/248A/250D/440P/477K/534H/595V/724H, 8M/68A/788E, 8M/93V/95S/98V/136S/221H/595V/600I/788E, 8M/93V/95S/113I/250D/440T/595V/600I/724H/788E, 8M/95S/98V/440P/478V/534H/600I/788E, 8M/136S/248A/478V/788E, 47L/75V/85V/105E/125E/129E/170L/635S, 47L/75V/85V/105E/375Y/756V/776E, 47L/75V/85V/264V/267V/372V/415R/635S, H47L/M75V/R85V/I87E/R129E/W375Y/Q776E, 47L/75V/85V/87E/170L/372V/756V, 47L/85V/105E/129E/201E/230D/267V/583Q, 47L/85V/125E/372V/583Q/635S/756V, 47L/85V/170L/756V, 47L/85V/87E/105E/125E/635S, 47L/85V/87E/154H/756V, 47L/125E/129E/375Y/756V/776E, 47L/129E/170L/635S, 47L/154H/372V/375Y/583Q/635S/708A/756V, 68A/93V/95S/358E/440T/444T/478V/534H/595V/603S, 68A/93V/95S/444T/788E, 68A/93V/98V/136S/248A/250D/358E/440P/534H/724H, 75V/85V/87E/105E/264V/267V/583Q/708A, 75V/85V/129E/154H/264V/375Y, 85V/125E/215F/375Y/415R/635S/776E, 85V/87E/105E/215F/267V/756V, 85V/87E/129E/375Y/756V/776E, 87E/125E/129E/170L/230D/756V, 87E/154H/306L/375Y/756V, 93V/95S/98V/534H/792S, 95S/440P/444T/724H/788E, 129E/215F/372V/756V, 170L/264V/267V, 8M/68A/95S/98V/358E/478T/595V/724H/792S, 8M/68A/98V/221H/248A/250D/440P/477K/534H/595V/724H, 8M/68A/788E, 8M/93V/95S/98V/136S/221H/595V/600I/788E, 8M/93V/95S/113I/250D/440T/595V/600I/724H/788E, 8M/95S/98V/440P/478V/534H/600I/788E, 8M/136S/248A/478V/788E, 47L/75V/85V/105E/125E/129E/170L/635S, 47L/75V/85V/105E/375Y/756V/776E,

47L/75V/264V/372V/415R/635S, 47L/75V/85V/87E/129E/375Y/776E, 47L/75V/85V/87E/170L/372V/756V, 47L/85V/105E/129E/201E/230D/267V/583Q, 47L/85V/125E/372V/583Q/635S/756V, 47L/85V/170L/756V, 47L/85V/87E/105E/125E/635S, 47L/85V/87E/154H/756V, 47L/125E/129E/375Y/756V/776E, 47L/129E/170L/635S, 47L/154H/372V/375Y/583Q/635S/708A/756V, 68A/93V/95S/358E/440T/444T/478V/534H/595V/603S, 68A/93V/95S/444T/788E, 68A/93V/98V/136S/248A/250D/358E/440P/534H/724H, 75V/85V/87E/105E/264V/267V/583Q/708A, 75V/85V/129E/154H/264V/375Y, 85V/125E/215F/375Y/415R/635S/776E, 85V/87E/105E/215F/267V/756V, 85V/87E/129E/375Y/756V/776E, 87E/125E/129E/170L/230D/756V, 87E/154H/306L/375Y/756V, 93V/95S/98V/534H/792S, 95S/440P/444T/724H/788E, 129E/215F/372V/756V, and 170L/264V/267V, wherein the positions are numbered with reference to SEQ ID NO:1158. In some further embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from Q8M/V68A/Q95S/L98V/P358E/R478T/I595V/K724H/M792S, Q8M/V68A/L98V/R221H/S248A/S250D/R440P/D477K/E534H/I595V/K724H, Q8M/V68A/H788E, Q8M/R93V/Q95S/L98V/R136S/R221H/I595V/V600I/H788E, Q8M/R93V/Q95S/L113I/S250D/R440T/I595V/V600I/K724H/H788E, Q8M/Q95S/L98V/R440P/R478V/E534H/V600I/H788E, Q8M/R136S/S248A/R478V/H788E, H47L/M75V/R85V/A105E/T125E/R129E/I170L/A635S, H47L/M75V/R85V/A105E/W375Y/I756V/Q776E, H47L/M75V/R85V/I264V/I267V/I372V/K415R/A635S, H47L/M75V/R85V/I87E/R129E/W375Y/Q776E, H47L/M75V/R85V/I87E/I170L/I372V/I756V, H47L/R85V/A105E/R129E/Q201E/N230D/I267V/R583Q, H47L/R85V/T125E/I372V/R583Q/A635S/I756V, H47L/R85V/I170L/I756V, H47L/R85V/I87E/A105E/T125E/A635S, H47L/R85V/I87E/R154H/I756V, H47L/T125E/R129E/W375Y/I756V/Q776E, H47L/R129E/I170L/A635S, H47L/R154H/I372V/W375Y/R583Q/A635S/T708A/I756V, V68A/R93V/Q95S/P358E/R440T/Q444T/R478V/E534H/I595V/G603S, V68A/R93V/Q95S/Q444T/H788E, V68A/R93V/L98V/R136S/S248A/S250D/P358E/R440P/E534H/K724H, M75V/R85V/I87E/A105E/I264V/I267V/R583Q/T708A, M75V/R85V/R129E/R154H/I264V/W375Y, R85V/T125E/I215F/W375Y/K415R/A635S/Q776E, R85V/I87E/A105E/I215F/I267V/I756V, R85V/I87E/R129E/W375Y/I756V/Q776E, I87E/T125E/R129E/I170L/N230D/I756V, I87E/R154H/R306L/W375Y/I756V, R93V/Q95S/L98V/E534H/M792S, Q95S/R440P/Q444T/K724H/H788E, R129E/I215F/I372V/I756V, and I170L/I264V/I267V, wherein the positions are numbered with reference to SEQ ID NO: 1158. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 90% identical to any of SEQ ID NOS: 1222, 1224, 1226, 1228, 1230, 1232, 1234, 1236, 1238, 1240, 1242, 1244, 1246, 1248, 1250, 1252, 1254, 1256, 1258, 1260, 1262, 1264, 1266, 1268, 1270, 1272, 1274, 1276, 1278, 1280, 1282, 1284, 1286, and/or 1288. In some further embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 95% identical to any of SEQ ID NOS: 1222, 1224, 1226, 1228, 1230, 1232, 1234, 1236, 1238, 1240, 1242, 1244, 1246, 1248, 1250, 1252, 1254, 1256, 1258, 1260, 1262, 1264, 1266, 1268, 1270, 1272, 1274, 1276, 1278, 1280, 1282, 1284, 1286, and/or 1288. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises any of SEQ ID NOS: 1222, 1224, 1226, 1228, 1230, 1232, 1234, 1236, 1238, 1240, 1242, 1244, 1246, 1248, 1250, 1252, 1254, 1256, 1258, 1260, 1262, 1264, 1266, 1268, 1270, 1272, 1274, 1276, 1278, 1280, 1282, 1284, 1286, and/or 1288.

(62) The present invention also provides engineered sucrose synthases, wherein the polypeptide sequence of the engineered sucrose synthases comprise at least one mutation or mutation set at one or more positions selected from 47/68/93/98/358/440, 47/68/154/372/375, 47/93/98/136/154/772/776, 47/93/98/154/372/375/776, 47/93/98/358/583/635, 47/93/129/136/154/250/372/534/635/724, 47/93/129/136/375/534/583, 47/93/358/372/375/440/724, 47/93/358/372/375/776, 47/98/129/358/372/375/438/534, 47/98/129/375/534/635/724/776, 47/98/372/375, 47/125/154, 47/129/136/372/375/534, 47/129/248/250/372/375/534/724, 47/136/583/776, 47/358/440/724,

47/358/635/776, 47/358/635/776, 68/93/98/129/358/375/724, 68/93/154/358/372/440/776, 68/129/440, 68/129/583/724, 68/136/724, 68/154/358/375, 68/154/534/635, 68/375/440/534/724/776, 93/98/125/154/248, 93/98/125/154/250/440, 93/98/129/154/248, 93/98/154/250/358/375/534, 93/98/154/635/776, 93/98/534, 93/125/154/440/534, 93/129/250/358/372/375/583, 93/154/248/724, 93/358/534/635, 98/129/375/583, 98/534/583/635, 129/136/154/248/250/372/375/534, 129/136/375, 129/154/248/250/358/375/534/635, 129/358/372/635, 154/250/358/375/583, 154/358/375/534/776, 154/534/635/724, 372/375/776, 375/635, and 534/583, wherein the positions are numbered with reference to SEQ ID NO: 1222. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 47L/68A/93V/98V/358E/440P, 47L/68A/154H/372V/375Y, 47L/93V/98V/136S/154H/772G/776E, 47L/93V/98V/154H/372V/375Y/776E, 47L/93V/98V/358E/583Q/635S, 47L/93V/129E/136S/154H/250D/372V/534H/635S/724H, 47L/93V/129E/136S/375Y/534H/583Q, 47L/93V/358E/372V/375Y/440P/724H, 47L/93V/358E/372V/375Y/776E, 47L/98V/129E/358E/372V/375Y/438Q/534H, 47L/98V/129E/375Y/534H/635S/724H/776E, 47L/98V/372V/375Y, 47L/125E/154H, 47L/129E/136S/372V/375Y/534H, 47L/129E/248A/250D/372V/375Y/534H/724H, 47L/136S/583Q/776E, 47L/358E/440P/724H, 47L/358E/635S/776E, 47L/372V/635S/776E, 68A/93V/98V/129E/358E/375Y/724H, 68A/93V/154H/358E/372V/440P/776E, 68A/129E/440P, 68A/129E/583Q/724H, 68A/136S/724H, 68A/154H/358E/375Y, 68A/154H/534H/635S, 68A/375Y/440P/534H/724H/776E, 93V/98V/125E/154H/248A, 93V/98V/125E/154H/250D/440P, 93V/98V/129E/154H/248A, 93V/98V/154H/250D/358E/375Y/534H, 93V/98V/154H/635S/776E, 93V/98V/534H, 93V/125E/154H/440P/534H, 93V/129E/250D/358E/372V/375Y/583Q, 93V/154H/248A/724H, 93V/358E/534H/635S, 98V/129E/375Y/583Q, 98V/534H/583Q/635S, 129E/136S/154H/248A/250D/372V/375Y/534H, 129E/136S/375Y, 129E/154H/248A/250D/358E/375Y/534H/635S, 129E/358E/372V/635S, 154H/250D/358E/375Y/583Q, 154H/358E/375Y/534H/776E, 154H/534H/635S/724H, 372V/375Y/776E, 375Y/635S, and 534H/583Q, wherein the positions are numbered with reference to SEQ ID NO: 1222. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from H47L/V68A/R93V/L98V/P358E/R440P, H47L/V68A/R154H/I372V/W375Y, H47L/R93V/L98V/R136S/R154H/E772G/Q776E, H47L/R93V/L98V/R154H/I372V/W375Y/Q776E, H47L/R93V/L98V/P358E/R583Q/A635S, H47L/R93V/R129E/R136S/R154H/S250D/I372V/E534H/A635S/K724H, H47L/R93V/R129E/R136S/W375Y/E534H/R583Q, H47L/R93V/P358E/I372V/W375Y/R440P/K724H, H47L/R93V/P358E/I372V/W375Y/Q776E, H47L/L98V/R129E/P358E/I372V/W375Y/H438Q/E534H, H47L/L98V/R129E/W375Y/E534H/A635S/K724H/Q776E, H47L/L98V/I372V/W375Y, H47L/T125E/R154H, H47L/R129E/R136S/I372V/W375Y/E534H, H47L/R129E/S248A/S250D/I372V/W375Y/E534H/K724H, H47L/R136S/R583Q/Q776E, H47L/P358E/R440P/K724H, H47L/P358E/A635S/Q776E, H47L/I372V/A635S/Q776E, V68A/R93V/L98V/R129E/P358E/W375Y/K724H, V68A/R93V/R154H/P358E/I372V/R440P/Q776E, V68A/R129E/R440P, V68A/R129E/R583Q/K724H, V68A/R136S/K724H, V68A/R154H/P358E/W375Y, V68A/R154H/E534H/A635S, V68A/W375Y/R440P/E534H/K724H/Q776E, R93V/L98V/T125E/R154H/S248A, R93V/L98V/T125E/R154H/S250D/R440P, R93V/L98V/R129E/R154H/S248A, R93V/L98V/R154H/S250D/P358E/W375Y/E534H, R93V/L98V/R154H/A635S/Q776E, R93V/L98V/E534H, R93V/T125E/R154H/R440P/E534H, R93V/R129E/S250D/P358E/I372V/W375Y/R583Q, R93V/R154H/S248A/K724H, R93V/P358E/E534H/A635S, L98V/R129E/W375Y/R583Q, L98V/E534H/R583Q/A635S, R129E/R136S/R154H/S248A/S250D/I372V/W375Y/E534H, R129E/R136S/W375Y, R129E/R154H/S248A/S250D/P358E/W375Y/E534H/A635S, R129E/P358E/I372V/A635S, R154H/S250D/P358E/W375Y/R583Q, R154H/P358E/W375Y/E534H/Q776E, R154H/E534H/A635S/K724H, I372V/W375Y/Q776E, W375Y/A635S, and E534H/R583Q, wherein the positions are numbered with reference to SEQ ID NO: 1222. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 90% identical to any of SEQ

ID NOS: 1296, 1298, 1300, 1302, 1304, 1306, 1308, 1310, 1312, 1314, 1316, 1320, 1322, 1324, 1326, 1328, 1330, 1332, 1334, 1336, 1338, 1340, 1342, 1344, 1346, 1348, 1350, 1352, 1354, 1356, 1358, 1360, 1362, 1364, 1366, 1368, 1370, 1372, 1374, 1376, 1378, 1380, 1382, 1384, 1386, 1388, 1390, and 1392. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 95% identical to any of SEQ ID NOS: 1296, 1298, 1300, 1302, 1304, 1306, 1308, 1310, 1312, 1314, 1316, 1318, 1320, 1322, 1324, 1326, 1328, 1330, 1332, 1334, 1336, 1338, 1340, 1342, 1344, 1346, 1348, 1350, 1352, 1354, 1356, 1358, 1360, 1362, 1364, 1366, 1368, 1370, 1372, 1374, 1376, 1378, 1380, 1382, 1384, 1386, 1388, 1390, and 1392. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises any of SEQ ID NOS: 1296, 1298, 1300, 1302, 1304, 1306, 1308, 1310, 1312, 1314, 1316, 1318, 1320, 1322, 1324, 1326, 1328, 1330, 1332, 1334, 1336, 1338, 1340, 1342, 1344, 1346, 1348, 1350, 1352, 1354, 1356, 1358, 1360, 1362, 1364, 1366, 1368, 1370, 1372, 1374, 1376, 1378, 1380, 1382, 1384, 1386, 1388, 1390, and 1392.

(63) The present invention also provides engineered sucrose synthases, wherein the polypeptide sequence of the engineered sucrose synthases comprise at least one mutation or mutation set at one or more positions selected from 17/357/364/434/519/684, 17/357/434/519/684, 17/434/684, 17/684, 54/97/118/307/694/727/738, 68/98/129/136, 68/98/129/136/154, 68/98/129/136/154/534, 68/98/129/154/534, 68/98/129/154/635, 68/98/136/154/534/635, 68/98/136/154/635, 68/98/154, 68/98/154/534, 68/98/154/534/635, 68/129/136, 68/129/136/154, 68/129/136/154/464/635, 68/129/136/534/635, 68/129/154, 68/129/154/765, 68/136/154/534/635, 68/136/534/635, 68/136/635, 68/154, 68/154/534/635, 68/154/635, 97/118/442/694/727/738, 98/129/136/154, 98/129/136/154/635, 98/129/136/534, 98/129/136/635, 98/129/154, 98/129/154/534/635, 98/129/534/635, 98/136/154/635, 98/136/534/635, 98/136/635, 98/154, 98/154/534, 122, 129/136, 129/136/154/635, 129/136/534, 129/136/635, 129/154/534, 129/154/635, 129/635, 132/136/154/534/635, 136/154/635, 136/534/635, 136/603, 136/635, 154, 154/635, 157, 160, 161, 167, 253, 285, 381, 519, 550, 563, 564, 635, and 785, wherein the positions are numbered with reference to SEQ ID NO: 1392. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 17D/357K/364R/434H/519T/684H, 17D/357K/434H/519T/684H, 17D/434H/684H, 17D/684H, 54D/97V/118N/307E/694N/727E/738E, 68A/98V/129E/136S, 68A/98V/129E/136S/154H, 68A/98V/129E/136S/154H/534H, 68A/98V/129E/154H/534H, 68A/98V/129E/154H/635S, 68A/98V/136S/154H/534H/635S, 68A/98V/136S/154H/635S, 68A/98V/154H, 68A/98V/154H/534H, 68A/98V/154H/534H/635S, 68A/129E/136S, 68A/129E/136S/154H, 68A/129E/136S/154H/464F/635S, 68A/129E/136S/534H/635S, 68A/129E/154H, 68A/129E/154H/765H, 68A/136S/154H/534H/635S, 68A/136S/534H/635S, 68A/136S/635S, 68A/154H, 68A/154H/534H/635S, 68A/154H/635S, 97V/118N/442N/694N/727E/738E, 98V/129E/136S/154H, 98V/129E/136S/154H/635S, 98V/129E/136S/534H, 98V/129E/136S/635S, 98V/129E/154H, 98V/129E/154H/534H/635S, 98V/129E/534H/635S, 98V/136S/154H/635S, 98V/136S/534H/635S, 98V/136S/635S, 98V/154H, 98V/154H/534H, 122D, 122E, 129E/136S, 129E/136S/154H/635S, 129E/136S/534H, 129E/136S/635S, 129E/154H/534H, 129E/154H/635S, 129E/635S, 132C/136S/154H/534H/635S, 136S/154H/635S, 136S/534H/635S, 136S/603D, 136S/635S, 154H, 154H/635S, 157A, 157F, 160A, 160E, 160M, 160N, 160S, 160W, 161Q, 167E, 253G, 253T, 253V, 285A, 381S, 519A, 519G, 519L, 519S, 519T, 550I, 550M, 550Q, 550S, 563V, 564A, 635D, 635E, 635R, and 785D, wherein the positions are numbered with reference to SEQ ID NO: 1392. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from Y17D/Y357K/P364R/Y434H/F519T/F684H, Y17D/Y357K/Y434H/F519T/F684H, Y17D/Y434H/F684H, Y17D/F684H, G54D/A97V/A118N/N307E/G694N/L727E/A738E, V68A/L98V/R129E/R136S, V68A/L98V/R129E/R136S/R154H, V68A/L98V/R129E/R136S/R154H/E534H, V68A/L98V/R129E/R154H/E534H, V68A/L98V/R129E/R154H/A635S, V68A/L98V/R136S/R154H/E534H/A635S, V68A/L98V/R136S/R154H/A635S, V68A/L98V/R154H, V68A/L98V/R154H/E534H, V68A/L98V/R154H/E534H/A635S, V68A/R129E/R136S, V68A/R129E/R136S/R154H, V68A/R129E/R136S/R154H/I464F/A635S, V68A/R129E/R136S/E534H/A635S, V68A/R129E/R154H, V68A/R129E/R154H/D765H, V68A/R136S/R154H/E534H/A635S, V68A/R136S/E534H/A635S, V68A/R136S/A635S, V68A/R154H,

V68A/R154H/E534S, V68A/R154H/A635S, A97V/A118N/H442N/G694N/L727E/A738E, L98V/R129E/R136S/R154H, L98V/R129E/R136S/R154H/A635S, L98V/R129E/R136S/E534H, L98V/R129E/R136S/A635S, L98V/R129E/R154H, L98V/R129E/R154H/E534H/A635S, L98V/R129E/E534H/A635S, L98V/R136S/R154H/A635S, L98V/R136S/E534H/A635S, L98V/R136S/A635S, L98V/R154H, L98V/R154H/E534H, A122D, A122E, R129E/R136S, R129E/R136S/R154H/A635S, R129E/R136S/E534H, R129E/R136S/A635S, R129E/R154H/E534H, R129E/R154H/A635S, R129E/A635S, R132C/R136S/R154H/E534H/A635S, R136S/R154H/A635S, R136S/E534H/A635S, R136S/G603D, R136S/A635S, R154H, R154H/A635S, G157A, G157F, F160A, F160E, F160M, F160N, F160S, F160W, S161Q, R167E, A253G, A253T, A253V, P285A, Q381S, F519A, F519G, F519L, F519S, F519T, R550I, R550M, R550Q, R550S, L563V, S564A, A635D, A635E, A635R, and P785D, wherein the positions are numbered with reference to SEQ ID NO: 1392. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 90% identical to any of SEQ ID NOS: 1394, 1396, 1398, 1400, 1402, 1404, 1406, 1408, 1410, 1412, 1414, 1416, 1418, 1420, 1422, 1424, 1426, 1428, 1430, 1432, 1434, 1436, 1438, 1440, 1442, 1444, 1446, 1448, 1450, 1452, 1454, 1456, 1458, 1460, 1462, 1464, 1466, 1468, 1470, 1472, 1474, 1476, 1478, 1480, 1482, 1484, 1486, 1488, 1490, 1492, 1494, 1496, 1498, 1500, 1502, 1504, 1506, 1508, 1510, 1512, 1514, 1516, 1518, 1520, 1522, 1524, 1526, 1528, 1530, 1532, 1534, 1536, 1538, 1540, 1542, 1544, 1546, 1548, 1550, 1552, 1554, 1556, 1558, 1560, 1562, 1564, and 1566. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 95% identical to any of SEQ ID NOS: 1394, 1396, 1398, 1400, 1402, 1404, 1406, 1408, 1410, 1412, 1414, 1416, 1418, 1420, 1422, 1424, 1426, 1428, 1430, 1432, 1434, 1436, 1438, 1440, 1442, 1444, 1446, 1448, 1450, 1452, 1454, 1456, 1458, 1460, 1462, 1464, 1466, 1468, 1470, 1472, 1474, 1476, 1478, 1480, 1482, 1484, 1486, 1488, 1490, 1492, 1494, 1496, 1498, 1500, 1502, 1504, 1506, 1508, 1510, 1512, 1514, 1516, 1518, 1520, 1522, 1524, 1526, 1528, 1530, 1532, 1534, 1536, 1538, 1540, 1542, 1544, 1546, 1548, 1550, 1552, 1554, 1556, 1558, 1560, 1562, 1564, and 1566. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises any of SEQ ID NOS: 1394, 1396, 1398, 1400, 1402, 1404, 1406, 1408, 1410, 1412, 1414, 1416, 1418, 1420, 1422, 1424, 1426, 1428, 1430, 1432, 1434, 1436, 1438, 1440, 1442, 1444, 1446, 1448, 1450, 1452, 1454, 1456, 1458, 1460, 1462, 1464, 1466, 1468, 1470, 1472, 1474, 1476, 1478, 1480, 1482, 1484, 1486, 1488, 1490, 1492, 1494, 1496, 1498, 1500, 1502, 1504, 1506, 1508, 1510, 1512, 1514, 1516, 1518, 1520, 1522, 1524, 1526, 1528, 1530, 1532, 1534, 1536, 1538, 1540, 1542, 1544, 1546, 1548, 1550, 1552, 1554, 1556, 1558, 1560, 1562, 1564, and 1566.

(64) The present invention also provides engineered sucrose synthases, wherein the polypeptide sequence of the engineered sucrose synthases comprise at least one mutation or mutation set at one or more positions selected from 17/54/97/136/329/550/684/738, 17/54/97/329/524/684, 17/54/161/519/727/738, 17/54/524/550/727, 17/161/434/524/766, 17/434/524/684, 17/434/738, 17/442/524/550/684/721, 17/727, 17/738, 54/97/161/434/442, 54/97/434/524/550/684/727, 54/136/442/550, 54/434/524/738, 97/136/519/550/727/738, and 329/550/684/727/738, wherein the positions are numbered with reference to SEQ ID NO: 1456. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 17D/54D/97V/136P/329Q/550I/684H/738E, 17D/54D/97V/329Q/524T/684H, 17D/54D/161T/519T/727E/738E, 17D/54D/524T/550I/727E, 17D/161T/434H/524T/766H, 17D/434H/524T/684H, 17D/434H/738E, 17D/442N/524T/550I/684H/721K, 17D/727E, 17D/738E, 54D/97V/161T/434H/442N, 54D/97V/434H/524T/550I/684H/727E, 54D/136P/442N/550I, 54D/434H/524T/738E, 97V/136P/519T/550I/727E/738E, and 329Q/550I/684H/727E/738E, wherein the positions are numbered with reference to SEQ ID NO: 1456. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from Y17D/G54D/A97V/R136P/E329Q/R550I/F684H/A738E, Y17D/G54D/A97V/E329Q/A524T/F684H, Y17D/G54D/S161T/F519T/L727E/A738E, Y17D/G54D/A524T/R550I/L727E, Y17D/S161T/Y434H/A524T/R766H, Y17D/Y434H/A524T/F684H, Y17D/Y434H/A738E, Y17D/H442N/A524T/R550I/F684H/E721K, Y17D/L727E, Y17D/A738E, G54D/A97V/S161T/Y434H/H442N, G54D/A97V/Y434H/A524T/R550I/F684H/L727E, G54D/R136P/H442N/R550I, G54D/Y434H/A524T/A738E, A97V/R136P/F519T/R550I/L727E/A738E, and E329Q/R550I/F684H/L727E/A738E, wherein the positions are numbered with reference to SEQ ID

NO: In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 90% identical to any of SEQ ID NOS: 1568, 1570, 1572, 1574, 1576, 1578, 1580, 1582, 1584, 1586, 1588, 1590, 1592, 1594, 1596, and 1598. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 95% identical to any of SEQ ID NOS: 1568, 1570, 1572, 1574, 1576, 1578, 1580, 1582, 1584, 1586, 1588, 1590, 1592, 1594, 1596, and 1598. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises any of SEQ ID NOS: 1568, 1570, 1572, 1574, 1576, 1578, 1580, 1582, 1584, 1586, 1588, 1590, 1592, 1594, 1596, and 1598.

(65) The present invention also provides engineered sucrose synthases, wherein the polypeptide sequence of the engineered sucrose synthases comprise at least one mutation or mutation set at one or more positions selected from 14/137/356/745, 14/570, 14/570/745, 26/117/365, 26/164/165/213/586, 71/158/222/356, 71/222/236, 71/319/356/606, 117/158/213/332/608, 117/164/707, 117/213/365/517, 117/311/332, 117/608, 122, 122/160/161/167/550, 122/160/161/282/381/550, 122/160/161/282/381/550/636, 122/160/161/282/550, 122/160/161/550/636, 122/160/167/282/381/550/636, 122/160/282/381, 122/160/282/381/550, 122/160/282/550, 122/160/381/550, 122/160/381/550/636, 122/160/550, 122/160/550/636, 122/161/550, 122/167, 122/167/550, 122/282/381/550, 122/282/550, 122/282/550/636, 122/381/706, 122/550, 137/319/570, 157/253/519, 160/161, 160/161/282/381/550, 160/161/282/550, 160/161/550/636/735, 160/167/282/381/636, 160/282, 160/282/381/550, 160/282/550/636, 160/381/550/636/681, 161/282/550/636, 161/381/550, 165/311, 167/282/636, 167/550, 213/365/517/707, 236, 253/519, 253/519/563, 253/519/635, 253/563/635, 270/322/517, 270/367/452/517/613/700/750, 270/452/517/700/750, 270/570, 282/381/550, 282/550, 356/570, 381/550, 517, 517/562/750, 517/640, 519/563, 550, 550/636, and 562, wherein the positions are numbered with reference to SEQ ID NO: 1582. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 14D/137K/356H/745L, 14D/570H, 14D/570H/745L, 26E/117E/365E, 26E/164E/165E/213E/586E, 71Q/158T/222L/356H, 71Q/222L/236S, 71Q/319K/356H/606S, 117E/158E/213E/332E/608E, 117E/164E/707E, 117E/213E/365E/517E, 117E/311E/332E, 117E/608E, 122D, 122D/160M/282M/550S, 122D/160W/161Q/167E/550M, 122D/160W/161Q/282M/550Q, 122D/160W/282M/381S/550M, 122D/160W/550M, 122D/160W/550Q/636Q, 122D/167E, 122D/282M/381S/550M, 122D/282M/550Q, 122D/550M, 122E/160I/167E/282M/381H/550Q/636Q, 122E/160M/161Q/282M/381S/550M/636Q, 122E/160M/550Q/636Q, 122E/160W/161Q/167E/550S, 122E/160W/161Q/282M/381S/550S, 122E/160W/161Q/550M/636Q, 122E/160W/282M/381S, 122E/160W/381S/550M/636Q, 122E/160W/381S/550Q, 122E/161Q/550Q, 122E/167E/550M, 122E/282M/550M/636Q, 122E/282M/550Q, 122E/282M/550S, 122E/381S/706K, 122E/550Q, 137K/319K/570H, 157A/253T/519L, 160M/161Q, 160M/161Q/282M/381S/550M, 160M/161Q/282M/550Q, 160M/282M, 160M/282M/381S/550M, 160M/282M/550M/636Q, 160W/161Q/282M/550M, 160W/161Q/282M/550Q, 160W/161Q/550Q/636Q/735V, 160W/167E/282M/381S/636Q, 160W/282M/381S/550M, 160W/381S/550Q/636Q/681V, 161Q/282M/550Q/636Q, 161Q/381S/550Q, 165E/311E, 167E/282M/636Q, 167E/550Q, 213E/365E/517E/707E, 236S, 253G/519L, 253T/519L, 253T/519L/563V, 253T/519L/635D, 253T/519L/635E, 253T/563V/635R, 253V/519G, 253V/519L, 270L/322V/517A, 270L/367V/452Y/517A/613Q/700F/750M, 270L/452Y/517A/700F/750M, 270L/570H, 282M/381S/550S, 282M/550Q, 356H/570H, 381S/550Q, 381S/550S, 517A, 517A/562I/750M, 517A/640N, 519L/563V, 550M/636Q, 550Q, and 562I, wherein the positions are numbered with reference to SEQ ID NO: 1582. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from R14D/R137K/R356H/R745L, R14D/R570H, R14D/R570H/R745L, S26E/G117E/H365E, S26E/A164E/A165E/A213E/R586E, R71Q/R158T/R222L/R356H, R71Q/R222L/R236S, R71Q/R319K/R356H/R606S, G117E/R158E/A213E/G332E/A608E, G117E/A164E/A707E, G117E/A213E/H365E/P517E, G117E/Q311E/G332E, G117E/A608E, A122D, A122D/F160M/L282M/R550S, A122D/F160W/T161Q/R167E/R550M, A122D/F160W/T161Q/L282M/R550Q, A122D/F160W/L282M/Q381S/R550M, A122D/F160W/R550M, A122D/F160W/R550Q/L636Q, A122D/R167E, A122D/L282M/Q381S/R550M, A122D/L282M/R550Q,

A122D/R550M, A122E/F160I/R167E/L282M/Q381H/R550Q/L636Q, A122E/F160M/T161Q/L282M/Q381S/R550M/L636Q, A122E/F160M/R550Q/L636Q, A122E/F160W/T161Q/R167E/R550S, A122E/F160W/T161Q/L282M/Q381S/R550S, A122E/F160W/T161Q/R550M/L636Q, A122E/F160W/L282M/Q381S, A122E/F160W/Q381S/R550M/L636Q, A122E/F160W/Q381S/R550Q, A122E/T161Q/R550Q, A122E/R167E/R550M, A122E/L282M/R550M/L636Q, A122E/L282M/R550Q, A122E/L282M/R550S, A122E/Q381S/E706K, A122E/R550Q, R137K/R319K/R570H, G157A/A253T/T519L, F160M/T161Q, F160M/T161Q/L282M/Q381S/R550M, F160M/T161Q/L282M/R550Q, F160M/L282M, F160M/L282M/Q381S/R550M, F160M/L282M/R550M/L636Q, F160W/T161Q/L282M/R550M, F160W/T161Q/L282M/R550Q, F160W/T161Q/R550Q/L636Q/A735V, F160W/R167E/L282M/Q381S/L636Q, F160W/L282M/Q381S/R550M, F160W/Q381S/R550Q/L636Q/A681V, T161Q/L282M/R550Q/L636Q, T161Q/Q381S/R550Q, A165E/Q311E, R167E/L282M/L636Q, R167E/R550Q, A213E/H365E/P517E/A707E, R236S, A253G/T519L, A253T/T519L, A253T/T519L/L563V, A253T/T519L/S635D, A253T/T519L/S635E, A253T/L563V/S635R, A253V/T519G, A253V/T519L, V270L/I322V/P517A, V270L/I367V/F452Y/P517A/E613Q/I700F/L750M, V270L/F452Y/P517A/I700F/L750M, V270L/R570H, L282M/Q381S/R550S, L282M/R550Q, R356H/R570H, Q381S/R550Q, Q381S/R550S, P517A, P517A/L562I/L750M, P517A/T640N, T519L/L563V, R550M/L636Q, R550Q, and L562I, wherein the positions are numbered with reference to SEQ ID NO: 1582. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 90% identical to any of SEQ ID NOS: 1600, 1602, 1604, 1606, 1608, 1610, 1612, 1614, 1616, 1618, 1620, 1622, 1624, 1626, 1628, 1630, 1632, 1634, 1636, 1638, 1640, 1642, 1644, 1646, 1648, 1650, 1652, 1654, 1656, 1658, 1660, 1662, 1664, 1666, 1668, 1670, 1672, 1674, 1676, 1678, 1680, 1682, 1684, 1686, 1688, 1690, 1692, 1694, 1696, 1698, 1700, 1702, 1704, 1706, 1708, 1710, 1712, 1714, 1716, 1718, 1720, 1722, 1724, 1726, 1728, 1730, 1732, 1734, 1736, 1738, 1740, 1742, 1744, 1746, 1748, 1750, 1752, 1754, 1756, 1758, 1760, 1762, 1764, 1766, 1768, 1770, and 1772. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 95% identical to any of SEQ ID NOS: 1600, 1602, 1604, 1606, 1608, 1610, 1612, 1614, 1616, 1618, 1620, 1622, 1624, 1626, 1628, 1630, 1632, 1634, 1636, 1638, 1640, 1642, 1644, 1646, 1648, 1650, 1652, 1654, 1656, 1658, 1660, 1662, 1664, 1666, 1668, 1670, 1672, 1674, 1676, 1678, 1680, 1682, 1684, 1686, 1688, 1690, 1692, 1694, 1696, 1698, 1700, 1702, 1704, 1706, 1708, 1710, 1712, 1714, 1716, 1718, 1720, 1722, 1724, 1726, 1728, 1730, 1732, 1734, 1736, 1738, 1740, 1742, 1744, 1746, 1748, 1750, 1752, 1754, 1756, 1758, 1760, 1762, 1764, 1766, 1768, 1770, and 1772. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises any of SEQ ID NOS: 1600, 1602, 1604, 1606, 1608, 1610, 1612, 1614, 1616, 1618, 1620, 1622, 1624, 1626, 1628, 1630, 1632, 1634, 1636, 1638, 1640, 1642, 1644, 1646, 1648, 1650, 1652, 1654, 1656, 1658, 1660, 1662, 1664, 1666, 1668, 1670, 1672, 1674, 1676, 1678, 1680, 1682, 1684, 1686, 1688, 1690, 1692, 1694, 1696, 1698, 1700, 1702, 1704, 1706, 1708, 1710, 1712, 1714, 1716, 1718, 1720, 1722, 1724, 1726, 1728, 1730, 1732, 1734, 1736, 1738, 1740, 1742, 1744, 1746, 1748, 1750, 1752, 1754, 1756, 1758, 1760, 1762, 1764, 1766, 1768, 1770, and 1772.

(66) The present invention also provides engineered sucrose synthases, wherein the polypeptide sequence of the engineered sucrose synthases comprise at least one mutation or mutation set at one or more positions selected from 63/536, 117/122/270/540/681, 181/536/548, 181/536/548/705, 181/548/705, 270/681, 347/532, 347/536/548/705, 407/570/681, 407/681, 536, 536/548, 536/548/699, 536/705, 548, 548/580, 548/705, 580, 681, 699, and 705, wherein the positions are numbered with reference to SEQ ID NO: 1764. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 63I/536L, 117E/122D/270L/540M/681A, 181N/536L/548P, 181N/536L/548P/705M, 181N/548P/705P, 270L/681A, 347R/532Y, 347R/536L/548P/705P, 407I/570H/681A, 407T/681A, 536L, 536L/548P, 536L/548P/699F, 536L/705M, 548P, 548P/580M, 548P/705P, 580M, 681A, 699F, 705M, and 705P, wherein the positions are numbered with reference to SEQ ID NO: 1764. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from A63I/E536L, G117E/A122D/V270L/L540M/V681A, G181N/E536L/A548P, G181N/E536L/A548P/H705M, G181N/A548P/H705P, V270L/V681A, N347R/S532Y,

N347R/L5436L/A548P/H705P, L407I/V570H/V681A, L407T/V681A, E536L/A548P, E536L/A548P/H699F, E536L/H705M, A548P, A548P/L580M, A548P/H705P, L580M, V681A, H699F, H705M, and H705P, wherein the positions are numbered with reference to SEQ ID NO: 1764. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 90% identical to any of SEQ ID NOS: 1774, 1776, 1778, 1780, 1782, 1784, 1786, 1788, 1790, 1792, 1794, 1796, 1798, 1800, 1802, 1804, 1806, 1808, 1810, 1812, 1814, 1816, 1818, and 1820. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 95% identical to any of SEQ ID NOS: 1774, 1776, 1778, 1780, 1782, 1784, 1786, 1788, 1790, 1792, 1794, 1796, 1798, 1800, 1802, 1804, 1806, 1808, 1810, 1812, 1814, 1816, 1818, and 1820. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises any of SEQ ID NOS: 1774, 1776, 1778, 1780, 1782, 1784, 1786, 1788, 1790, 1792, 1794, 1796, 1798, 1800, 1802, 1804, 1806, 1808, 1810, 1812, 1814, 1816, 1818, and 1820.

(67) The present invention also provides engineered sucrose synthases, wherein the polypeptide sequence of the engineered sucrose synthases comprise at least one mutation or mutation set at one or more positions selected from 13, 17, 18, 30, 37, 52, 57, 60, 71, 85, 87, 90, 98, 99, 118, 129, 164, 180, 183, 347/434/517/562/640/681, 347/434/532/562/640/681, 347/434/550/562/681, 347/434/681, 347/517/532/681, 347/532/550/640/681/699, 347/536/562/681, 347/550/580/681, 347/550/681, 347/681, 365, 388, 389, 415, 433, 434/517/532/681, 517/681, 531, 532/681, 535, 536/580/681, 539, 562/681, 589, 606, 608, 707, 711, 727, 738, 748, 765, 769, and 789, wherein the positions are numbered with reference to SEQ ID NO: 1804. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 13H, 13K, 13N, 17R, 18G, 30H, 37G, 37R, 52G, 52P, 52R, 52W, 57R, 57W, 60L, 71G, 71Q, 85H, 87H, 87L, 90H, 98L, 99I, 118G, 118N, 129A, 129G, 129T, 164S, 164T, 180P, 183P, 347R/434H/517A/562I/640N/681A, 347R/434H/532Y/562I/640N/681A, 347R/434H/550I/562I/681A, 347R/434H/681A, 347R/517A/532Y/681A, 347R/532Y/550I/640N/681A/699F, 347R/536L/562I/681A, 347R/550I/580M/681A, 347R/550I/681A, 347R/681A, 365W, 388K, 388R, 389G, 415H, 433K, 433P, 434H/517A/532Y/681A, 517A/681A, 531A, 531R, 531T, 532Y/681A, 535A, 535H, 535S, 536L/580M/681A, 539A, 539R, 562I/681A, 589S, 606A, 606H, 606I, 606L, 606M, 606Q, 606V, 608P, 707G, 711K, 727K, 738S, 748T, 765A, 765S, 769K, 769R, 789N, and 789R, wherein the positions are numbered with reference to SEQ ID NO: 1804. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from P13H, P13K, P13N, D17R, A18G, S30H, Q37G, Q37R, D52G, D52P, D52R, D52W, P57R, P57W, D60L, R71G, R71Q, V85H, E87H, E87L, R90H, V98L, E99I, A118G, A118N, E129A, E129G, E129T, A164S, A164T, D180P, N183P, N347R/Y434H/P517A/L562I/T640N/V681A, N347R/Y434H/S532Y/L562I/T640N/V681A, N347R/Y434H/Q550I/L562I/V681A, N347R/Y434H/V681A, N347R/P517A/S532Y/V681A, N347R/5532Y/Q550I/T640N/V681A/H699F, N347R/E536L/L562I/V681A, N347R/Q550I/L580M/V681A, N347R/Q550I/V681A, N347R/V681A, H365W, L388K, L388R, A389G, K415H, L433K, L433P, Y434H/P517A/S532Y/V681A, P517A/V681A, S531A, S531R, S531T, S532Y/V681A, P535A, P535H, P535S, E536L/L580M/V681A, S539A, S539R, L562I/V681A, G589S, R606A, R606H, R606I, R606L, R606M, R606Q, R606V, A608P, A707G, R711K, E727K, E738S, E748T, D765A, D765S, Q769K, Q769R, A789N, and A789R, wherein the positions are numbered with reference to SEQ ID NO: 1804. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 90% identical to any of SEQ ID NOS: 1822, 1824, 1826, 1828, 1830, 1832, 1834, 1836, 1838, 1840, 1842, 1844, 1846, 1848, 1850, 1852, 1854, 1856, 1858, 1860, 1862, 1864, 1866, 1868, 1870, 1872, 1874, 1876, 1878, 1880, 1882, 1884, 1886, 1888, 1890, 1892, 1894, 1896, 1898, 1900, 1902, 1904, 1906, 1908, 1910, 1912, 1914, 1916, 1918, 1920, 1922, 1924, 1926, 1928, 1930, 1932, 1934, 1936, 1938, 1940, 1942, 1944, 1946, 1948, 1950, 1952, 1954, 1956, 1958, 1960, 1962, 1964, 1966, 1968, 1970, 1972, 1974, 1976, 1978, 1980, 1982, and 1984. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 95% identical to any of SEQ ID NOS: 1822, 1824, 1826, 1828, 1830, 1832, 1834, 1836, 1838, 1840, 1842, 1844, 1846, 1848, 1850, 1852, 1854, 1856, 1858, 1860, 1862, 1864, 1866, 1868, 1870, 1872, 1874, 1876, 1878, 1880, 1882, 1884, 1886, 1888, 1890, 1892, 1894, 1896, 1898, 1900, 1902, 1904, 1906, 1908, 1910, 1912, 1914, 1916, 1918, 1920, 1922, 1924, 1926, 1928, 1930, 1932, 1934, 1936, 1938, 1940, 1942,

1944, 1946, 1954, 1956, 1958, 1960, 1962, 1964, 1966, 1968, 1970, 1972, 1974, 1976, 1978, 1980, 1982, and 1984. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises any of SEQ ID NOS: 1822, 1824, 1826, 1828, 1830, 1832, 1834, 1836, 1838, 1840, 1842, 1844, 1846, 1848, 1850, 1852, 1854, 1856, 1858, 1860, 1862, 1864, 1866, 1868, 1870, 1872, 1874, 1876, 1878, 1880, 1882, 1884, 1886, 1888, 1890, 1892, 1894, 1896, 1898, 1900, 1902, 1904, 1906, 1908, 1910, 1912, 1914, 1916, 1918, 1920, 1922, 1924, 1926, 1928, 1930, 1932, 1934, 1936, 1938, 1940, 1942, 1944, 1946, 1948, 1950, 1952, 1954, 1956, 1958, 1960, 1962, 1964, 1966, 1968, 1970, 1972, 1974, 1976, 1978, 1980, 1982, and 1984.

(68) The present invention also provides engineered sucrose synthases, wherein the polypeptide sequence of the engineered sucrose synthases comprise at least one mutation or mutation set at one or more positions selected from 17/52/87/118/129/388/589/738/765, 17/52/87/118/129/589/738, 17/52/87/118/129/589/738/765, 17/52/87/118/129/589/765, 17/52/87/129/388/589, 17/52/87/129/388/589/738, 17/52/87/129/738, 17/52/87/388/589/765, 17/52/87/589/738/765, 17/52/118/129/265/589/765, 17/52/118/129/388/589/738/765, 17/52/118/129/589/738/765, 17/52/118/129/738/765, 17/52/118/388/589/738, 17/52/118/388/589/738/765, 17/52/118/388/738/765, 17/52/129/388/589/738, 17/52/129/388/589/738/765, 17/52/129/589, 17/52/129/589/738, 17/52/129/589/765, 17/52/129/653/738/765, 17/52/129/738, 17/52/129/738/765, 17/52/388/589/738, 17/52/388/589/738/765, 17/52/589/738/765, 17/52/589/765, 17/87/118/388/738, 17/87/129/388/738, 17/118/129/388/738/765, 17/129/589, 17/129/589/738, 17/129/589/738/765, 17/129/738/765, 17/388/589/738, 17/589/738, 17/589/765, 17/738/765, 52/84/129/388/738/765, 52/87/118/129/388/765, 52/87/118/388/589/738, 52/87/118/589/738/765, 52/87/129/388/738/765, 52/87/129/765, 52/87/589/738, 52/87/738, 52/118/129/589/738, 52/118/129/765, 52/118/388/589/738, 52/118/388/738/765, 52/129/589/738/765, 52/129/589/765, 52/129/738, 52/388/738/765, 52/589/738, 52/589/738/765, 52/738/765, 84/129/589/738/765, 87/118/129/765, 87/129/388/589, and 589/738/765, wherein the positions are numbered with reference to SEQ ID NO: 1840. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 17R/52G/87H/118N/129G/589S/738S, 17R/52G/87H/118N/129T/589S/765S, 17R/52G/87H/129T/388K/589S/738S, 17R/52G/87H/129T/738S, 17R/52G/87H/388K/589S/765S, 17R/52G/118N/129T/388K/589S/738S/765S, 17R/52G/118N/388K/738S/765S, 17R/52G/129G/589S/738S, 17R/52G/129T/388K/589S/738S, 17R/52G/129T/589S, 17R/52G/129T/653H/738S/765S, 17R/52G/589S/765S, 17R/52P/87H/118N/129T/388K/589S/738S/765S, 17R/52P/87H/118N/129T/589S/738S/765S, 17R/52P/87H/129G/388K/589S, 17R/52P/87H/129T/388K/589S/738S, 17R/52P/87H/589S/738S/765S, 17R/52P/118N/129T/265T/589S/765S, 17R/52P/118N/129T/589S/738S/765S, 17R/52P/118N/129T/738S/765S, 17R/52P/118N/388K/589S/738S, 17R/52P/118N/388K/589S/738S/765S, 17R/52P/129G/388K/589S/738S/765S, 17R/52P/129G/738S/765S, 17R/52P/129T/589S/765S, 17R/52P/129T/738S, 17R/52P/388K/589S/738S, 17R/52P/388K/589S/738S/765S, 17R/52P/589S/738S/765S, 17R/87H/118N/388K/738S, 17R/87H/129T/388K/738S, 17R/118N/129T/388K/738S/765S, 17R/129T/589S, 17R/129T/589S/738S, 17R/129T/589S/738S/765S, 17R/129T/738S/765S, 17R/388K/589S/738S, 17R/589S/738S, 17R/589S/765S, 17R/738S/765S, 52G/84A/129T/388K/738S/765S, 52G/87H/118N/589S/738S/765S, 52G/87H/129T/388K/738S/765S, 52G/87H/738S, 52G/118N/388K/589S/738S, 52G/118N/388K/738S/765S, 52G/129G/589S/765S, 52G/388K/738S/765S, 52G/589S/738S/765S, 52G/738S/765S, 52P/87H/118N/129T/388K/765S, 52P/87H/118N/388K/589S/738S, 52P/87H/129G/765S, 52P/87H/589S/738S, 52P/118N/129T/589S/738S, 52P/118N/129T/765S, 52P/129T/589S/738S/765S, 52P/129T/738S, 52P/589S/738S, 52P/589S/738S/765S, 84A/129T/589S/738S/765S, 87H/118N/129T/765S, 87H/129T/388K/589S, and 589S/738S/765S, wherein the positions are numbered with reference to SEQ ID NO: 1840. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from D17R/D52G/E87H/A118N/E129G/G589S/E738S, D17R/D52G/E87H/A118N/E129T/G589S/D765S, D17R/D52G/E87H/E129T/L388K/G589S/E738S, D17R/D52G/E87H/E129T/E738S, D17R/D52G/E87H/L388K/G589S/D765S, D17R/D52G/A118N/E129T/L388K/G589S/E738S/D765S, D17R/D52G/A118N/L388K/E738S/D765S,

D17R/D52G/E129G/E589S/E738S, D17R/D52G/E129T/L388K/G589S/E738S,
D17R/D52G/E129T/G589S, D17R/D52G/E129T/R653H/E738S/D765S, D17R/D52G/G589S/D765S,
D17R/D52P/E87H/A118N/E129T/L388K/G589S/E738S/D765S,
D17R/D52P/E87H/A118N/E129T/G589S/E738S/D765S, D17R/D52P/E87H/E129G/L388K/G589S,
D17R/D52P/E87H/E129T/L388K/G589S/E738S, D17R/D52P/E87H/G589S/E738S/D765S,
D17R/D52P/A118N/E129T/S265T/G589S/D765S, D17R/D52P/A118N/E129T/G589S/E738S/D765S,
D17R/D52P/A118N/E129T/E738S/D765S, D17R/D52P/A118N/L388K/G589S/E738S,
D17R/D52P/A118N/L388K/G589S/E738S/D765S, D17R/D52P/E129G/L388K/G589S/E738S/D765S,
D17R/D52P/E129G/E738S/D765S, D17R/D52P/E129T/G589S/D765S, D17R/D52P/E129T/E738S,
D17R/D52P/L388K/G589S/E738S, D17R/D52P/L388K/G589S/E738S/D765S,
D17R/D52P/G589S/E738S/D765S, D17R/E87H/A118N/L388K/E738S,
D17R/E87H/E129T/L388K/E738S, D17R/A118N/E129T/L388K/E738S/D765S, D17R/E129T/G589S,
D17R/E129T/G589S/E738S, D17R/E129T/G589S/E738S/D765S, D17R/E129T/E738S/D765S,
D17R/L388K/G589S/E738S, D17R/G589S/E738S, D17R/G589S/D765S, D17R/E738S/D765S,
D52G/G84A/E129T/L388K/E738S/D765S, D52G/E87H/A118N/G589S/E738S/D765S,
D52G/E87H/E129T/L388K/E738S/D765S, D52G/E87H/E738S, D52G/A118N/L388K/G589S/E738S,
D52G/A118N/L388K/E738S/D765S, D52G/E129G/G589S/D765S, D52G/L388K/E738S/D765S,
D52G/G589S/E738S/D765S, D52G/E738S/D765S, D52P/E87H/A118N/E129T/L388K/D765S,
D52P/E87H/A118N/L388K/G589S/E738S, D52P/E87H/E129G/D765S, D52P/E87H/G589S/E738S,
D52P/A118N/E129T/G589S/E738S, D52P/A118N/E129T/D765S, D52P/E129T/G589S/E738S/D765S,
D52P/E129T/E738S, D52P/G589S/E738S, D52P/G589S/E738S/D765S,
G84A/E129T/G589S/E738S/D765S, E87H/A118N/E129T/D765S, E87H/E129T/L388K/G589S, and
G589S/E738S/D765S, wherein the positions are numbered with reference to SEQ ID NO: 1840. In some
embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one
mutation or mutation set at one or more positions selected from 14, 15, 18/362, 20, 24, 26, 33, 33/154, 46,
50, 54, 58, 59, 59/72, 79, 81, 92, 93, 97/154, 104, 105, 130, 134, 154, 165, 175, 185, 212, 213, 218, 241,
256, 263, 316, 319, 349, 360, 362, 364, 390, 393, 434, 480, 498, 530, 534, 534/739, 542, 603, and 652,
wherein the positions are numbered with reference to SEQ ID NO: 1840. In some embodiments, the
polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set
at one or more positions selected from 14K, 15A, 15I, 15P, 18V/362A, 20M, 24S, 26A, 26E, 26I, 26T,
33H/154C, 33L, 33P, 33S, 46G, 46I, 46R, 46T, 46V, 50R, 54M, 58M, 59A, 59C, 59N/72N, 59R, 59S,
59V, 59W, 79H, 79Y, 81G, 81I, 81L, 92G, 93T, 97V/154S, 104T, 105S, 130Y, 134A, 134P, 154A, 154E,
154R, 154S, 165I, 165L, 165T, 175G, 175T, 185L, 212Y, 213V, 218A, 218N, 218Q, 218S, 218T, 218V,
241T, 256G, 263S, 263Y, 316H, 316T, 319S, 349D, 349R, 349T, 360D, 360E, 360R, 362E, 364S, 390M,
393H, 434G, 434R, 480P, 480V, 498L, 498Q, 530F, 534G, 534K, 534L, 534R, 534T, 534W, 534W/739K,
542W, 603A, 603E, 603H, 603Q, 603S, 652K, 652L, 652R, 652S, and 652T, wherein the positions are
numbered with reference to SEQ ID NO: 1840. In some embodiments, the polypeptide sequence of the
engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions
selected from R14K, S15A, S15I, S15P, A18V/I362A, L20M, V24S, S26A, S26E, S26I, S26T,
Q33H/H154C, Q33L, Q33P, Q33S, Q46G, Q46I, Q46R, Q46T, Q46V, G50R, D54M, L58M, E59A,
E59C, E59N/D72N, E59R, E59S, E59V, E59W, W79H, W79Y, P81G, P81I, P81L, H92G, V93T,
A97V/H154S, D104T, E105S, D130Y, V134A, V134P, H154A, H154E, H154R, H154S, A165I, A165L,
A165T, S175G, S175T, M185L, W212Y, A213V, D218A, D218N, D218Q, D218S, D218T, D218V,
L241T, S256G, M263S, M263Y, V316H, V316T, R319S, Q349D, Q349R, Q349T, G360D, G360E,
G360R, I362E, P364S, E390M, S393H, Y434G, Y434R, I480P, I480V, E498L, E498Q, P530F, E534G,
E534K, E534L, E534R, E534T, E534W, E534W/E739K, F542W, G603A, G603E, G603H, G603Q,
G603S, G652K, G652L, G652R, G652S, and G652T, wherein the positions are numbered with reference
to SEQ ID NO: 1840. In some embodiments, the polypeptide sequence of the engineered sucrose
synthase comprises a sequence at least 90% identical to any of SEQ ID NOS: 1986, 1988, 1990, 1992,
1994, 1996, 1998, 2000, 2002, 2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022, 2024, 2026,
2028, 2030, 2032, 2034, 2036, 2038, 2040, 2042, 2044, 2046, 2048, 2050, 2052, 2054, 2056, 2058, 2060,
2062, 2064, 2066, 2068, 2070, 2072, 2074, 2076, 2078, 2080, 2082, 2084, 2086, 2088, 2090, 2092, 2094,
2096, 2098, 2100, 2102, 2104, 2106, 2108, 2110, 2112, 2114, 2116, 2118, 2120, 2122, 2124, 2126, 2128,

2130, 2132, 2134, 2136, 2138, 2140, 2142, 2144, 2146, 2148, 2150, 2152, 2154, 2156, 2158, 2160, 2162, 2164, 2166, 2168, 2170, 2172, 2174, 2176, 2178, 2180, 2182, 2184, 2186, 2188, 2190, 2192, 2194, 2196, 2198, 2200, 2202, 2204, 2206, 2208, 2210, 2212, 2214, 2216, 2218, 2220, 2222, 2224, 2226, 2228, 2230, 2232, 2234, 2236, 2238, 2240, 2242, 2244, 2246, 2248, 2250, 2252, 2254, 2256, 2258, 2260, 2262, 2264, 2266, 2268, 2270, 2272, 2274, 2276, 2278, 2280, 2282, 2284, 2286, 2288, 2290, 2292, 2294, 2296, 2298, 2300, 2302, 2304, 2306, 2308, 2310, 2312, 2314, 2316, and 2318. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 95% identical to any of SEQ ID NOS: 1986, 1988, 1990, 1992, 1994, 1996, 1998, 2000, 2002, 2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022, 2024, 2026, 2028, 2030, 2032, 2034, 2036, 2038, 2040, 2042, 2044, 2046, 2048, 2050, 2052, 2054, 2056, 2058, 2060, 2062, 2064, 2066, 2068, 2070, 2072, 2074, 2076, 2078, 2080, 2082, 2084, 2086, 2088, 2090, 2092, 2094, 2096, 2098, 2100, 2102, 2104, 2106, 2108, 2110, 2112, 2114, 2116, 2118, 2120, 2122, 2124, 2126, 2128, 2130, 2132, 2134, 2136, 2138, 2140, 2142, 2144, 2146, 2148, 2150, 2152, 2154, 2156, 2158, 2160, 2162, 2164, 2166, 2168, 2170, 2172, 2174, 2176, 2178, 2180, 2182, 2184, 2186, 2188, 2190, 2192, 2194, 2196, 2198, 2200, 2202, 2204, 2206, 2208, 2210, 2212, 2214, 2216, 2218, 2220, 2222, 2224, 2226, 2228, 2230, 2232, 2234, 2236, 2238, 2240, 2242, 2244, 2246, 2248, 2250, 2252, 2254, 2256, 2258, 2260, 2262, 2264, 2266, 2268, 2270, 2272, 2274, 2276, 2278, 2280, 2282, 2284, 2286, 2288, 2290, 2292, 2294, 2296, 2298, 2300, 2302, 2304, 2306, 2308, 2310, 2312, 2314, 2316, and 2318.

In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises any of SEQ ID NOS: 1986, 1988, 1990, 1992, 1994, 1996, 1998, 2000, 2002, 2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022, 2024, 2026, 2028, 2030, 2032, 2034, 2036, 2038, 2040, 2042, 2044, 2046, 2048, 2050, 2052, 2054, 2056, 2058, 2060, 2062, 2064, 2066, 2068, 2070, 2072, 2074, 2076, 2078, 2080, 2082, 2084, 2086, 2088, 2090, 2092, 2094, 2096, 2098, 2100, 2102, 2104, 2106, 2108, 2110, 2112, 2114, 2116, 2118, 2120, 2122, 2124, 2126, 2128, 2130, 2132, 2134, 2136, 2138, 2140, 2142, 2144, 2146, 2148, 2150, 2152, 2154, 2156, 2158, 2160, 2162, 2164, 2166, 2168, 2170, 2172, 2174, 2176, 2178, 2180, 2182, 2184, 2186, 2188, 2190, 2192, 2194, 2196, 2198, 2200, 2202, 2204, 2206, 2208, 2210, 2212, 2214, 2216, 2218, 2220, 2222, 2224, 2226, 2228, 2230, 2232, 2234, 2236, 2238, 2240, 2242, 2244, 2246, 2248, 2250, 2252, 2254, 2256, 2258, 2260, 2262, 2264, 2266, 2268, 2270, 2272, 2274, 2276, 2278, 2280, 2282, 2284, 2286, 2288, 2290, 2292, 2294, 2296, 2298, 2300, 2302, 2304, 2306, 2308, 2310, 2312, 2314, 2316, and 2318.

(69) The present invention also provides engineered sucrose synthases, wherein the polypeptide sequence of the engineered sucrose synthases comprise at least one mutation or mutation set at one or more positions selected from 57/71/87/347/434/562/606, 57/71/129/180/434/536/562, 57/71/129/434/531/536/562, 57/71/129/531/532/536/539/606, 57/71/562/606/711/789, 57/71/789, 57/87/180/531/532/562/606/612/711, 57/87/347/562, 57/90/129/562, 57/90/129/562/711, 57/96/129/180/531/532/550/562, 57/129/347/531/532/539/562/711/747, 57/129/347/536/550/562/711/789, 57/129/347/550/711, 57/129/531/539/562/789, 57/129/536/606/789, 57/129/606, 57/180/562, 57/180/562/606/612, 57/347/434/531/532/539/789, 57/434/550/562/606/612/789, 57/531/532/536/562, 57/562/606/711, 57/562/711, 71/129/180/347/531/539/550, 71/129/180/434/532/536/539/711/789, 71/129/531, 71/129/606, 71/347/532/550/562/711, 71/347/536/562/612/789, 71/536/539/562, 87/189/532/536/562/711/789, 87/347/531/606/789, 87/347/536/539/550, 90/129/539/550/606, 129/180/434/562/711/789, 129/180/606/711/789, 129/347/562, 129/536/539/562, 129/539/562/789, 129/550, 129/550/562, 129/562/606/711, 180/532, 180/550/606, 347/531/550/711, 347/536/539/550/711/789, 347/536/562/606/612, 347/550/562/606, 434/531/539/550/562/711, 434/550, 531/532/536/539/562/711/789, 531/532/536/550/562/606/789, 531/532/562/606/711, 532/539/550, and 562/711, wherein the positions are numbered with reference to SEQ ID NO: 2064. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 57W/71Q/87H/347R/434H/562I/606H, 57W/71Q/129G/531R/532Y/536L/539R/606H, 57W/71Q/129T/180P/434H/536L/562I, 57W/71Q/129T/434H/531R/536L/562I, 57W/71Q/562I/606M/711K/789N, 57W/71Q/789N, 57W/87H/180P/531R/532Y/562I/606M/612A/711K, 57W/87H/347R/562I, 57W/90H/129G/562I, 57W/90H/129T/562I/711K, 57W/96Q/129T/180P/531R/532Y/550I/562I, 57W/129G/531R/539R/562I/789N, 57W/129G/536L/606M/789N,

57W/129T/347R/531R/532Y/539R/562I/711K/747V, 57W/129T/347R/536L/550I/562I/711K/789N, 57W/129T/347R/550I/711K, 57W/129T/606M, 57W/180P/562I, 57W/180P/562I/606H/612A, 57W/347R/434H/531R/532Y/539R/789N, 57W/434H/550I/562I/606H/612A/789N, 57W/531R/532Y/536L/562I, 57W/562I/606M/711K, 57W/562I/711K, 71Q/129T/180P/347R/531R/539R/550I, 71Q/129T/180P/434H/532Y/536L/539R/711K/789N, 71Q/129T/531R, 71Q/129T/606M, 71Q/347R/532Y/550I/562I/711K, 71Q/347R/536L/562I/612A/789N, 71Q/536L/539R/562I, 87H/189D/532Y/536L/562I/711K/789N, 87H/347R/531R/606M/789N, 87H/347R/536L/539R/550I, 90H/129T/539R/550I/606H, 129G/550I, 129G/550I/562I, 129G/562I/606M/711K, 129T/180P/434H/562I/711K/789N, 129T/180P/606M/711K/789N, 129T/347R/562I, 129T/536L/539R/562I, 129T/539R/562I/789N, 129T/550I/562I, 180P/532Y, 180P/550I/606M, 347R/531R/550I/711K, 347R/536L/539R/550I/711K/789N, 347R/536L/562I/606M/612A, 347R/550I/562I/606H, 434H/531R/539R/550I/562I/711K, 434H/550I, 531R/532Y/536L/539R/562I/711K/789N, 531R/532Y/536L/550I/562I/606M/789N, 531R/532Y/562I/606H/711K, 532Y/539R/550I, and 562I/711K, wherein the positions are numbered with reference to SEQ ID NO: 2064. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from P57W/R71Q/E87H/N347R/Y434H/L562I/R606H, P57W/R71Q/E129G/5531R/S532Y/E536L/S539R/R606H, P57W/R71Q/E129T/D180P/Y434H/E536L/L562I, P57W/R71Q/E129T/Y434H/S531R/E536L/L562I, P57W/R71Q/L562I/R606M/R711K/A789N, P57W/R71Q/A789N, P57W/E87H/D180P/5531R/S532Y/L562I/R606M/E612A/R711K, P57W/E87H/N347R/L562I, P57W/R90H/E129G/L562I, P57W/R90H/E129T/L562I/R711K, P57W/L96Q/E129T/D180P/5531R/5532Y/Q550I/L562I, P57W/E129G/S531R/S539R/L562I/A789N, P57W/E129G/E536L/R606M/A789N, P57W/E129T/N347R/5531R/5532Y/5539R/L562I/R711K/A747V, P57W/E129T/N347R/E536L/Q550I/L562I/R711K/A789N, P57W/E129T/N347R/Q550I/R711K, P57W/E129T/R606M, P57W/D180P/L562I, P57W/D180P/L562I/R606H/E612A, P57W/N347R/Y434H/S531R/S532Y/S539R/A789N, P57W/Y434H/Q550I/L562I/R606H/E612A/A789N, P57W/S531R/S532Y/E536L/L562I, P57W/L562I/R606M/R711K, P57W/L562I/R711K, R71Q/E129T/D180P/N347R/S531R/S539R/Q550I, R71Q/E129T/D180P/Y434H/S532Y/E536L/S539R/R711K/A789N, R71Q/E129T/S531R, R71Q/E129T/R606M, R71Q/N347R/S532Y/Q550I/L562I/R711K, R71Q/N347R/E536L/L562I/E612A/A789N, R71Q/E536L/S539R/L562I, E87H/G189D/S532Y/E536L/L562I/R711K/A789N, E87H/N347R/S531R/R606M/A789N, E87H/N347R/E536L/S539R/Q550I, R90H/E129T/S539R/Q550I/R606H, E129G/Q550I, E129G/Q550I/L562I, E129G/L562I/R606M/R711K, E129T/D180P/Y434H/L562I/R711K/A789N, E129T/D180P/R606M/R711K/A789N, E129T/N347R/L562I, E129T/E536L/S539R/L562I, E129T/S539R/L562I/A789N, E129T/Q550I/L562I, D180P/S532Y, D180P/Q550I/R606M, N347R/S531R/Q550I/R711K, N347R/E536L/S539R/Q550I/R711K/A789N, N347R/E536L/L562I/R606M/E612A, N347R/Q550I/L562I/R606H, Y434H/S531R/S539R/Q550I/L562I/R711K, Y434H/Q550I, 5531R/S532Y/E536L/S539R/L562I/R711K/A789N, S531R/S532Y/E536L/Q550I/L562I/R606M/A789N, S531R/S532Y/L562I/R606H/R711K, S532Y/S539R/Q550I, and L562I/R711K, wherein the positions are numbered with reference to SEQ ID NO: 2064. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 21, 25/112, 41, 89, 91, 112, 186, 200, 226, 259, 318, 330, 485, 487, 641, 674, 684, 688, 763, and 764, wherein the positions are numbered with reference to SEQ ID NO: 2064.

(70) In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 21Q, 25T/112W, 41K, 89L, 89M, 91C, 91G, 112Q, 112R, 186V, 200A, 226V, 259G, 318A, 330A, 485A, 485S, 487I, 487K, 487R, 487T, 487V, 641L, 674A, 684G, 684H, 684M, 684T, 688A, 688F, 688G, 688H, 688Q, 763L, and 764R, wherein the positions are numbered with reference to SEQ ID NO: 2064. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from R21Q, A25T/G112W, A41K, V89L, V89M, I91C, I91G, G112Q,

G1126R, L126V, L226V, D259G, P318A, S330A, G485A, Q487I, Q487K, Q487R, Q487T, Q487V, V641L, S674A, F684G, F684H, F684M, F684T, L688A, L688F, L688G, L688H, L688Q, V763L, and L764R, wherein the positions are numbered with reference to SEQ ID NO: 2064. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 90% identical to any of SEQ ID NOS: 2320, 2322, 2324, 2326, 2328, 2330, 2332, 2334, 2336, 2338, 2340, 2342, 2344, 2346, 2348, 2350, 2352, 2354, 2356, 2358, 2360, 2362, 2364, 2366, 2368, 2370, 2372, 2374, 2376, 2378, 2380, 2382, 2384, 2386, 2388, 2390, 2392, 2394, 2396, 2398, 2400, 2402, 2404, 2406, 2408, 2410, 2412, 2414, 2416, 2418, 2420, 2422, 2424, 2426, 2428, 2430, 2432, 2434, 2436, 2438, 2440, 2442, 2444, 2446, 2448, 2450, 2452, 2454, 2456, 2458, 2460, 2462, 2464, 2466, 2468, 2470, 2472, 2474, 2476, 2478, 2480, 2482, 2484, 2486, 2488, 2490, 2492, 2494, 2496, 2498, 2500, and 2502. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 95% identical to any of SEQ ID NOS: 2320, 2322, 2324, 2326, 2328, 2330, 2332, 2334, 2336, 2338, 2340, 2342, 2344, 2346, 2348, 2350, 2352, 2354, 2356, 2358, 2360, 2362, 2364, 2366, 2368, 2370, 2372, 2374, 2376, 2378, 2380, 2382, 2384, 2386, 2388, 2390, 2392, 2394, 2396, 2398, 2400, 2402, 2404, 2406, 2408, 2410, 2412, 2414, 2416, 2418, 2420, 2422, 2424, 2426, 2428, 2430, 2432, 2434, 2436, 2438, 2440, 2442, 2444, 2446, 2448, 2450, 2452, 2454, 2456, 2458, 2460, 2462, 2464, 2466, 2468, 2470, 2472, 2474, 2476, 2478, 2480, 2482, 2484, 2486, 2488, 2490, 2492, 2494, 2496, 2498, 2500, and 2502. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises any of SEQ ID NOS: 2320, 2322, 2324, 2326, 2328, 2330, 2332, 2334, 2336, 2338, 2340, 2342, 2344, 2346, 2348, 2350, 2352, 2354, 2356, 2358, 2360, 2362, 2364, 2366, 2368, 2370, 2372, 2374, 2376, 2378, 2380, 2382, 2384, 2386, 2388, 2390, 2392, 2394, 2396, 2398, 2400, 2402, 2404, 2406, 2408, 2410, 2412, 2414, 2416, 2418, 2420, 2422, 2424, 2426, 2428, 2430, 2432, 2434, 2436, 2438, 2440, 2442, 2444, 2446, 2448, 2450, 2452, 2454, 2456, 2458, 2460, 2462, 2464, 2466, 2468, 2470, 2472, 2474, 2476, 2478, 2480, 2482, 2484, 2486, 2488, 2490, 2492, 2494, 2496, 2498, 2500, and 2502.

(71) The present invention also provides engineered sucrose synthases, wherein the polypeptide sequence of the engineered sucrose synthases comprise at least one mutation or mutation set at one or more positions selected from 33/47/59/81/175/530/534/550/606, 33/58/59/81/130/480/530/534/550/652, 33/58/59/480/530/534/550, 33/58/154/480/534/550/603/606, 33/59/480/530/534/550/606, 33/79/81/175/530/534, 33/79/81/175/530/534/603, 33/79/154/480/530/534/550, 33/81/130/480/530/534/550, 33/81/175/530/534/542/550/652, 33/130/530/534/550, 33/154/480/530/534/603/606, 33/154/534, 33/530/534/550, 58/59/79/175/480/534/550/652, 59/154/530/534/550, 79/81/480/530/534/550/603/606/652, 81/480/530/534/550, and 130/480/530/534/550/603/606, wherein the positions are numbered with reference to SEQ ID NO: 2432.

In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from

33H/47P/59A/81L/175G/530F/534W/550I/606M,
33H/58M/59A/81G/130Y/480V/530F/534W/550I/652K, 33H/58M/59A/480P/530F/534W/550I,
33H/58M/154A/480P/534W/550I/603A/606M, 33H/59A/480P/530F/534W/550I/606M,
33H/79H/81G/175G/530F/534W, 33H/79H/81L/175G/530F/534W/603Q,
33H/79H/154A/480P/530F/534W/550I, 33H/81G/130Y/480V/530F/534W/550I,
33H/81I/175G/530F/534W/542W/550I/652R, 33H/130Y/530F/534W/550I,
33H/154A/480P/530F/534W/603Q/606M, 33H/154A/534W, 33H/530F/534W/550I,
58M/59A/79H/175G/480P/534W/550I/652R, 59A/154A/530F/534W/550I,
79H/81I/480P/530F/534W/550I/603E/606M/652R, 81G/480V/530F/534W/550I, and
130Y/480V/530F/534W/550I/603Q/606M, wherein the positions are numbered with reference to SEQ ID

NO: 2432. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from

Q33H/L47P/E59A/P81L/S175G/P530F/E534W/Q550I/R606M,
Q33H/L58M/E59A/P81G/D130Y/I480V/P530F/E534W/Q550I/G652K,
Q33H/L58M/E59A/I480P/P530F/E534W/Q550I,
Q33H/L58M/H154A/I480P/E534W/Q550I/G603A/R606M,
Q33H/E59A/I480P/P530F/E534W/Q550I/R606M, Q33H/W79H/P81G/S175G/P530F/E534W,
Q33H/W79H/P81L/S175G/P530F/E534W/G603Q, Q33H/W79H/H154A/I480P/P530F/E534W/Q550I,

Q33H/P81I/S175G/P530F/E534W/Q550I, Q33H/P81I/S175G/P530F/E534W/F542W/Q550I/G652R, Q33H/D130Y/P530F/E534W/Q550I, Q33H/H154A/I480P/P530F/E534W/G603Q/R606M, Q33H/H154A/E534W, Q33H/P530F/E534W/Q550I, L58M/E59A/W79H/S175G/I480P/E534W/Q550I/G652R, E59A/H154A/P530F/E534W/Q550I, W79H/P81I/I480P/P530F/E534W/Q550I/G603E/R606M/G652R, P81G/I480V/P530F/E534W/Q550I, and D130Y/I480V/P530F/E534W/Q550I/G603Q/R606M, wherein the positions are numbered with reference to SEQ ID NO: 2432. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 25, 42, 70, 75, 77, 106, 199, 265, 267, 380, 410, 561, 642, and 758, wherein the positions are numbered with reference to SEQ ID NO: 2432. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 25E, 25G, 25L, 42H, 42S, 42T, 70H, 70N, 70R, 70S, 70V, 75T, 75W, 77L, 77W, 106W, 199A, 265A, 265Q, 267I, 380T, 410S, 561I, 561V, 642V, 758Q, and 758R, wherein the positions are numbered with reference to SEQ ID NO: 2432. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from A25E, A25G, A25L, D42H, D42S, D42T, F70H, F70N, F70R, F70S, F70V, M75T, M75W, F77L, F77W, Y106W, T199A, S265A, S265Q, V267I, A380T, T410S, L561I, L561V, A642V, G758Q, and G758R, wherein the positions are numbered with reference to SEQ ID NO: 2432. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 90% identical to any of SEQ ID NOS: 2504, 2506, 2508, 2510, 2512, 2514, 2516, 2518, 2520, 2522, 2524, 2526, 2528, 2530, 2532, 2534, 2536, 2538, 2540, 2542, 2544, 2546, 2548, 2550, 2552, 2554, 2556, 2558, 2560, 2562, 2564, 2566, 2568, 2570, 2572, 2574, 2576, 2578, 2580, 2582, 2584, 2586, 2588, 2590, 2592, and 2594. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 95% identical to any of SEQ ID NOS: 2504, 2506, 2508, 2510, 2512, 2514, 2516, 2518, 2520, 2522, 2524, 2526, 2528, 2530, 2532, 2534, 2536, 2538, 2540, 2542, 2544, 2546, 2548, 2550, 2552, 2554, 2556, 2558, 2560, 2562, 2564, 2566, 2568, 2570, 2572, 2574, 2576, 2578, 2580, 2582, 2584, 2586, 2588, 2590, 2592, and 2594. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises any of SEQ ID NOS: 2504, 2506, 2508, 2510, 2512, 2514, 2516, 2518, 2520, 2522, 2524, 2526, 2528, 2530, 2532, 2534, 2536, 2538, 2540, 2542, 2544, 2546, 2548, 2550, 2552, 2554, 2556, 2558, 2560, 2562, 2564, 2566, 2568, 2570, 2572, 2574, 2576, 2578, 2580, 2582, 2584, 2586, 2588, 2590, 2592, and 2594.

(72) The present invention also provides engineered sucrose synthases, wherein the polypeptide sequence of the engineered sucrose synthases comprise at least one mutation or mutation set at one or more positions selected from 41, 41/71, 41/71/112, 41/71/112/259/485/487/684/688, 41/71/112/259/485/688, 41/71/259/485/532, 41/71/485, 41/71/485/532/684, 41/71/487, 41/71/487/532/684, 41/71/532, 41/71/532/684, 41/71/684, 41/84/259/485/487, 41/91/112/485, 41/91/112/485/487/532/684, 41/91/112/485/532/684, 41/91/485, 41/112, 41/112/259/485/487, 41/112/259/487/532/684, 41/112/485/684, 41/112/487/684, 41/112/532, 41/112/684/688, 41/259/485, 41/259/485/487, 41/259/485/487/532/684, 41/259/485/487/684/688, 41/259/532, 41/485, 41/485/487, 41/485/487/684/688, 41/485/532, 41/485/532/688, 41/485/684/688, 41/487/684, 41/532, 41/684, 41/684/688, 44/112/684/688, 71/112/259/485/487/684, 71/112/485/688, 71/485/684/688, 71/532, 71/684/688, 112, 112/259, 112/259/532/684/688, 112/259/684/688, 112/485/684, 112/485/684/688, 226/487/684/688, 259/485/487/684, 259/485/532, 259/487/684/688, 259/532, 485, 485/487, 485/487/532, 485/487/532/684, 485/487/684, 485/487/684/688, 485/532, 485/684, 485/684/688, 532, 532/684/688, 684, and 684/688, wherein the positions are numbered with reference to SEQ ID NO: 2510. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 41K, 41K/71Q, 41K/71Q/112Q, 41K/71Q/112Q/259G/485A/688Q, 41K/71Q/112W/259G/485S/487R/684H/688Q, 41K/71Q/259G/485A/532Y, 41K/71Q/485A, 41K/71Q/485A/532Y/684H, 41K/71Q/487R, 41K/71Q/487R/532Y/684H, 41K/71Q/532Y, 41K/71Q/532Y/684H, 41K/71Q/684H, 41K/84A/259G/485A/487I, 41K/91C/112Q/485S, 41K/91C/112Q/485S/487K/532Y/684H, 41K/91C/112Q/485S/532Y/684H, 41K/91C/485S, 41K/112Q/259G/485S/487R, 41K/112Q/259G/487R/532Y/684H, 41K/112Q/485S/684H, 41K/112Q/487I/684H, 41K/112Q/532Y,

41K/112Q/684H/688A, 259G/485A/487I, 41K/259G/485A/487R/532Y/684H, 41K/259G/485A/487R/684H/688A, 41K/259G/485S, 41K/259G/532Y, 41K/485A, 41K/485A/487K, 41K/485A/487R, 41K/485A/532Y, 41K/485A/532Y/688G, 41K/485S/487I/684M/688G, 41K/485S/684M/688A, 41K/487I/684M, 41K/487R/684M, 41K/532Y, 41K/684H, 41K/684M/688A, 44C/112W/684H/688A, 71Q/112Q/259G/485A/487R/684M, 71Q/112W/485S/688A, 71Q/485S/684M/688Q, 71Q/532Y, 71Q/684M/688A, 112Q, 112Q/259G/532Y/684M/688G, 112Q/259G/684M/688Q, 112Q/485S/684H, 112Q/485S/684H/688G, 112Q/485S/684M/688G, 112W, 112W/259G, 226V/487I/684M/688A, 259G/485S/487K/684H, 259G/485S/532Y, 259G/487I/684M/688G, 259G/532Y, 485A, 485A/487R, 485A/532Y, 485A/684H/688G, 485A/684M, 485S, 485S/487I/684H/688Q, 485S/487I/684M, 485S/487K/532Y, 485S/487R/532Y/684H, 485S/684M/688G, 532Y, 532Y/684H/688Q, 684H, and 684M/688Q, wherein the positions are numbered with reference to SEQ ID NO: 2510. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from A41K, A41K/R71Q, A41K/R71Q/G112Q, A41K/R71Q/G112Q/D259G/G485A/L688Q, A41K/R71Q/G112W/D259G/G485S/Q487R/F684H/L688Q, A41K/R71Q/D259G/G485A/S532Y, A41K/R71Q/G485A, A41K/R71Q/G485A/S532Y/F684H, A41K/R71Q/Q487R, A41K/R71Q/Q487R/S532Y/F684H, A41K/R71Q/S532Y, A41K/R71Q/S532Y/F684H, A41K/R71Q/F684H, A41K/G84A/D259G/G485A/Q487I, A41K/I91C/G112Q/G485S, A41K/I91C/G112Q/G485S/Q487K/S532Y/F684H, A41K/I91C/G112Q/G485S/S532Y/F684H, A41K/I91C/G485S, A41K/G112Q/D259G/G485S/Q487R, A41K/G112Q/D259G/Q487R/S532Y/F684H, A41K/G112Q/G485S/F684H, A41K/G112Q/Q487I/F684H, A41K/G112Q/S532Y, A41K/G112Q/F684H/L688A, A41K/G112W, A41K/D259G/G485A/Q487I, A41K/D259G/G485A/Q487R/S532Y/F684H, A41K/D259G/G485A/Q487R/F684H/L688A, A41K/D259G/G485S, A41K/D259G/S532Y, A41K/G485A, A41K/G485A/Q487K, A41K/G485A/Q487R, A41K/G485A/S532Y, A41K/G485A/S532Y/L688G, A41K/G485S/Q487I/F684M/L688G, A41K/G485S/F684M/L688A, A41K/Q487I/F684M, A41K/Q487R/F684M, A41K/S532Y, A41K/F684H, A41K/F684M/L688A, R44C/G112W/F684H/L688A, R71Q/G112Q/D259G/G485A/Q487R/F684M, R71Q/G112W/G485S/L688A, R71Q/G485S/F684M/L688Q, R71Q/S532Y, R71Q/F684M/L688A, G112Q, G112Q/D259G/S532Y/F684M/L688G, G112Q/D259G/F684M/L688Q, G112Q/G485S/F684H, G112Q/G485S/F684H/L688G, G112Q/G485S/F684M/L688G, G112W, G112W/D259G, P226V/Q487I/F684M/L688A, D259G/G485S/Q487K/F684H, D259G/G485S/S532Y, D259G/Q487I/F684M/L688G, D259G/S532Y, G485A, G485A/Q487R, G485A/S532Y, G485A/F684H/L688G, G485A/F684M, G485S, G485S/Q487I/F684H/L688Q, G485S/Q487I/F684M, G485S/Q487K/S532Y, G485S/Q487R/S532Y/F684H, G485S/F684M/L688G, S532Y, S532Y/F684H/L688Q, F684H, and F684M/L688Q, wherein the positions are numbered with reference to SEQ ID NO: 2510. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 7/12, 12, 27, 29, 44, 45, 47, 48, 51, 55, 72, 95, 100, 116, 136, 139, 176, 178, 198, 201, 205, 205/485, 207, 208, 280, 303, 317, 343, 358, 361, 440, 478, 611, 615, 630, 675, 724, 756, and 788, wherein the positions are numbered with reference to SEQ ID NO: 2510. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 7K/12L, 12C, 12N, 12Q, 12S, 12V, 27R, 29A, 29G, 29L, 29P, 44L, 44V, 45A, 45C, 45G, 45L, 45S, 45V, 47C, 47D, 47I, 47L, 47N, 47T, 47V, 48A, 48V, 51P, 51S, 55D, 55S, 72E, 95D, 95L, 95N, 95T, 95V, 100K, 100P, 100Q, 116A, 116F, 136A, 136F, 136K, 136N, 136P, 136Q, 139K, 176M, 176R, 176T, 176V, 178H, 198A, 198R, 201S, 205R/485S, 205S, 207K, 208K, 280G, 303V, 317I, 343A, 343N, 358P, 358S, 361A, 361L, 361T, 440R, 478H, 478I, 478Q, 611A, 615C, 615K, 615L, 615M, 630M, 675C, 724G, 724K, 724S, 756C, and 788K, wherein the positions are numbered with reference to SEQ ID NO: 2510. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from Q7K/D12L, D12C, D12N, D12Q, D12S, D12V, Q27R, D29A, D29G, D29L, D29P, R44L, R44V, E45A, E45C, E45G, E45L, E45S, E45V, P47C, P47D, P47I, P47L, P47N, P47T, P47V, P48A, P48V, Y51P, Y51S, I55D, I55S, D72E, Q95D, Q95L, Q95N, Q95T, Q95V, E100K, E100P, E100Q, L116A, L116F, R136A, R136F, R136K, R136N,

R136P, R136Q, L176M, L176T, L176V, R178H, Q198A, Q198R, Q201S, T205R/G485S, T205S, P207K, R208K, K280G, R303V, E317I, H343A, H343N, E358P, E358S, R361A, R361L, R361T, P440R, R478H, R478I, R478Q, R611A, R615C, R615K, R615L, R615M, L630M, 5675C, H724G, H724K, H724S, V756C, and H788K, wherein the positions are numbered with reference to SEQ ID NO: 2510. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 90% identical to any of SEQ ID NOS: 7438, 7440, 7442, 7444, 7446, 7448, 7450, 7452, 7454, 7456, 7458, 7460, 7462, 7464, 7466, 7468, 7470, 7472, 7474, 7476, 7478, 7480, 7482, 7484, 7486, 7488, 7490, 7492, 7494, 7496, 7498, 7500, 7502, 7504, 7506, 7508, 7510, 7512, 7514, 7516, 7518, 7520, 7522, 7524, 7526, 7528, 7530, 7532, 7534, 7536, 7538, 7540, 7542, 7544, 7546, 7548, 7550, 7552, 7554, 7556, 7558, 7560, 7562, 7564, 7566, 7568, 7570, 7572, 7574, 7576, 7578, 7580, 7582, 7584, 7586, 7588, 7590, 7592, 7594, 7596, 7598, 7600, 7602, 7604, 7606, 7608, 7610, 7612, 7614, 7616, 7618, 7620, 7622, 7624, 7626, 7628, 7630, 7632, 7634, 7636, 7638, 7640, 7642, 7644, 7646, 7648, 7650, 7652, 7654, 7656, 7658, 7660, 7662, 7664, 7666, 7668, 7670, 7672, 7674, 7676, 7678, 7680, 7682, 7684, 7686, 7688, 7690, 7692, 7694, 7696, 7698, 7700, 7702, 7704, 7706, 7708, 7710, 7712, 7714, 7716, 7718, 7720, 7722, 7724, 7726, 7728, 7730, 7732, 7734, 7736, 7738, 7740, 7742, 7744, 7746, 7748, 7750, 7752, 7754, 7756, 7758, 7760, 7762, and 7764. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 95% identical to any of SEQ ID NOS: 7438, 7440, 7442, 7444, 7446, 7448, 7450, 7452, 7454, 7456, 7458, 7460, 7462, 7464, 7466, 7468, 7470, 7472, 7474, 7476, 7478, 7480, 7482, 7484, 7486, 7488, 7490, 7492, 7494, 7496, 7498, 7500, 7502, 7504, 7506, 7508, 7510, 7512, 7514, 7516, 7518, 7520, 7522, 7524, 7526, 7528, 7530, 7532, 7534, 7536, 7538, 7540, 7542, 7544, 7546, 7548, 7550, 7552, 7554, 7556, 7558, 7560, 7562, 7564, 7566, 7568, 7570, 7572, 7574, 7576, 7578, 7580, 7582, 7584, 7586, 7588, 7590, 7592, 7594, 7596, 7598, 7600, 7602, 7604, 7606, 7608, 7610, 7612, 7614, 7616, 7618, 7620, 7622, 7624, 7626, 7628, 7630, 7632, 7634, 7636, 7638, 7640, 7642, 7644, 7646, 7648, 7650, 7652, 7654, 7656, 7658, 7660, 7662, 7664, 7666, 7668, 7670, 7672, 7674, 7676, 7678, 7680, 7682, 7684, 7686, 7688, 7690, 7692, 7694, 7696, 7698, 7700, 7702, 7704, 7706, 7708, 7710, 7712, 7714, 7716, 7718, 7720, 7722, 7724, 7726, 7728, 7730, 7732, 7734, 7736, 7738, 7740, 7742, 7744, 7746, 7748, 7750, 7752, 7754, 7756, 7758, 7760, 7762, and 7764. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises any of SEQ ID NOS: 7438, 7440, 7442, 7444, 7446, 7448, 7450, 7452, 7454, 7456, 7458, 7460, 7462, 7464, 7466, 7468, 7470, 7472, 7474, 7476, 7478, 7480, 7482, 7484, 7486, 7488, 7490, 7492, 7494, 7496, 7498, 7500, 7502, 7504, 7506, 7508, 7510, 7512, 7514, 7516, 7518, 7520, 7522, 7524, 7526, 7528, 7530, 7532, 7534, 7536, 7538, 7540, 7542, 7544, 7546, 7548, 7550, 7552, 7554, 7556, 7558, 7560, 7562, 7564, 7566, 7568, 7570, 7572, 7574, 7576, 7578, 7580, 7582, 7584, 7586, 7588, 7590, 7592, 7594, 7596, 7598, 7600, 7602, 7604, 7606, 7608, 7610, 7612, 7614, 7616, 7618, 7620, 7622, 7624, 7626, 7628, 7630, 7632, 7634, 7636, 7638, 7640, 7642, 7644, 7646, 7648, 7650, 7652, 7654, 7656, 7658, 7660, 7662, 7664, 7666, 7668, 7670, 7672, 7674, 7676, 7678, 7680, 7682, 7684, 7686, 7688, 7690, 7692, 7694, 7696, 7698, 7700, 7702, 7704, 7706, 7708, 7710, 7712, 7714, 7716, 7718, 7720, 7722, 7724, 7726, 7728, 7730, 7732, 7734, 7736, 7738, 7740, 7742, 7744, 7746, 7748, 7750, 7752, 7754, 7756, 7758, 7760, 7762, and 7764.

(73) The present invention also provides engineered sucrose synthases, wherein the polypeptide sequence of the engineered sucrose synthases comprise at least one mutation or mutation set at one or more positions selected from 8/25/55/95/208/358/440/517/788, 12/45/47, 12/45/47/48/51/136/142/630, 12/45/47/51/136/139/630/758, 12/45/47/136/139/142/675/758, 12/45/51/136/139/630/675/756/758, 12/45/51/630/756, 12/48/51/136/139/758, 12/136/139/142/756/758, 12/136/142, 12/630/756, 25/29/208/440, 25/100/154/208/440/517/705/788, 25/517, 29/208/361/517/788, 42/198/199/480/532/539/561, 42/198/532/561, 42/198/532/561/724, 42/199/480/532/561, 42/259/480/561, 42/259/480/652, 42/480/561, 42/480/561/724, 42/561, 47/51/136/756/758, 55, 55/410/440/603/788, 55/517, 55/517/788, 70, 70/642, 77/176/487/615/642, 95/603, 106/199/539/561/652, 116, 136/139/142, 154/361/440/517/603/788, 176, 198/199/480/561, 198/199/480/561/724, 198/199/561/724, 198/480/561, 198/480/724, 199/532/539/561/652/724, 259/480, 267/611/642, 280/440/517, 380, 480/561/652, 480/561/652/724/764, 480/724, 517, 532/539/561, 532/561/724, 603, and 642, wherein the positions are numbered with reference to SEQ ID NO: 7506. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 8R/25E/55D/95T/208K/358S/440V/517P/788K,

12N/45A/47I/51P/136P/139K/630M/758R, 12N/45A/51P/136P/139K/630M/675C/756C/758Q, 12N/48A/51P/136Q/139K/758Q, 12N/136Q/142N, 12S/45A/51P/630M/756C, 12S/630M/756C, 12V/45A/47I/48A/51S/136Q/142N/630M, 12V/45A/47I/136Q/139K/142N/675C/758Q, 12V/45A/47L, 12V/136Q/139K/142N/756C/758Q, 25E/29P/208K/440R, 25E/100K/154A/208K/440R/517P/705M/788K, 25E/517P, 29P/208K/361T/517P/788K, 42T/198R/199A/480V/532Y/539R/561V, 42T/198R/532Y/561I/724K, 42T/198R/532Y/561V, 42T/199A/480V/532Y/561V, 42T/259G/480V/561V, 42T/259G/480V/652K, 42T/480P/561I, 42T/480V/561I/724K, 42T/561I, 47I/51S/136Q/756C/758Q, 55D, 55D/410S/440R/603E/788K, 55D/517P, 55D/517P/788K, 70N, 70N/642V, 77L/176M/487K/615M/642V, 95T/603E, 106W/199A/539R/561I/652R, 116F, 136P/139K/142N, 154A/361T/440R/517P/603E/788K, 176T, 198R/199A/480P/561I, 198R/199A/480P/561V/724K, 198R/199A/561V/724G, 198R/480P/561I, 198R/480V/724K, 199A/532Y/539R/561I/652R/724S, 259G/480V, 267I/611Q/642V, 280G/440R/517P, 380T, 480V/561V/652R, 480V/561V/652R/724K/764R, 480V/724K, 517P, 532Y/539R/561V, 532Y/561I/724K, 603E, and 642V, wherein the positions are numbered with reference to SEQ ID NO: 7506. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from Q8R/A25E/155D/Q95T/R208K/E358S/P440V/A517P/H788K, D12N/E45A/P47I/Y51P/R136P/R139K/L630M/G758R, D12N/E45A/Y51P/R136P/R139K/L630M/S675C/V756C/G758Q, D12N/P48A/Y51P/R136Q/R139K/G758Q, D12N/R136Q/S142N, D12S/E45A/Y51P/L630M/V756C, D12S/L630M/V756C, D12V/E45A/P47I/P48A/Y51S/R136Q/S142N/L630M, D12V/E45A/P47I/R136Q/R139K/S142N/S675C/G758Q, D12V/E45A/P47L, D12V/R136Q/R139K/S142N/V756C/G758Q, A25E/D29P/R208K/P440R, A25E/E100K/H154A/R208K/P440R/A517P/P705M/H788K, A25E/A517P, D29P/R208K/R361T/A517P/H788K, D42T/Q198R/T199A/I480V/S532Y/S539R/L561V, D42T/Q198R/S532Y/L561I/H724K, D42T/Q198R/S532Y/L561V, D42T/T199A/I480V/S532Y/L561V, D42T/D259G/I480V/L561V, D42T/D259G/I480V/G652K, D42T/I480P/L561I, D42T/I480V/L561I/H724K, D42T/L561I, P47I/Y51S/R136Q/V756C/G758Q, I55D, 155D/T410S/P440R/G603E/H788K, 155D/A517P, 155D/A517P/H788K, F70N, F70N/A642V, F77L/L176M/Q487K/R615M/A642V, Q95T/G603E, Y106W/T199A/S539R/L561I/G652R, L116F, R136P/R139K/S142N, H154A/R361T/P440R/A517P/G603E/H788K, L176T, Q198R/T199A/I480P/L561I, Q198R/T199A/I480P/L561V/H724K, Q198R/T199A/L561V/H724G, Q198R/I480P/L561I, Q198R/I480V/H724K, T199A/5532Y/5539R/L561I/G652R/H724S, D259G/I480V, V267I/R611Q/A642V, K280G/P440R/A517P, A380T, I480V/L561V/G652R, I480V/L561V/G652R/H724K/L764R, I480V/H724K, A517P, S532Y/S539R/L561V, S532Y/L561I/H724K, G603E, and A642V, wherein the positions are numbered with reference to SEQ ID NO: 7506. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 90% identical to any of SEQ ID NOS: 8370, 8372, 8374, 8376, 8378, 8380, 8382, 8384, 8386, 8388, 8390, 8392, 8394, 8396, 8398, 8400, 8402, 8404, 8406, 8408, 8410, 8412, 8414, 8416, 8418, 8420, 8422, 8424, 8426, 8428, 8430, 8432, 8434, 8436, 8438, 8440, 8442, 8444, 8446, 8448, 8450, 8452, 8454, 8456, 8458, 8460, 8462, 8464, 8466, 8468, 8470, 8472, 8474, 8476, 8478, and 8480. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 95% identical to any of SEQ ID NOS: 8370, 8372, 8374, 8376, 8378, 8380, 8382, 8384, 8386, 8388, 8390, 8392, 8394, 8396, 8398, 8400, 8402, 8404, 8406, 8408, 8410, 8412, 8414, 8416, 8418, 8420, 8422, 8424, 8426, 8428, 8430, 8432, 8434, 8436, 8438, 8440, 8442, 8444, 8446, 8448, 8450, 8452, 8454, 8456, 8458, 8460, 8462, 8464, 8466, 8468, 8470, 8472, 8474, 8476, 8478, and 8480. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises any of SEQ ID NOS: 8370, 8372, 8374, 8376, 8378, 8380, 8382, 8384, 8386, 8388, 8390, 8392, 8394, 8396, 8398, 8400, 8402, 8404, 8406, 8408, 8410, 8412, 8414, 8416, 8418, 8420, 8422, 8424, 8426, 8428, 8430, 8432, 8434, 8436, 8438, 8440, 8442, 8444, 8446, 8448, 8450, 8452, 8454, 8456, 8458, 8460, 8462, 8464, 8466, 8468, 8470, 8472, 8474, 8476, 8478, and 8480.

(74) The present invention also provides engineered sucrose synthases, wherein the polypeptide sequence of the engineered sucrose synthases comprise at least one mutation or mutation set at one or more

positions selected from 12/45/95/136/139/199/517/630/756, 12/45/95/136/756, 12/45/136/139/199/517/603, 12/45/136/139/208/603/630/756, 12/45/136/139/517/603/756, 12/45/136/139/517/630/642/756, 12/45/136/139/517/756, 12/45/136/139/603/756, 12/45/136/139/642/756, 12/45/136/176/517/603/630/642, 12/45/136/208/517/630/756, 12/45/136/517/603/642/756/789, 12/45/136/517/630/642/756, 12/45/136/603/756, 12/45/136/630/642, 12/45/139/176/208/517/603/630, 12/45/139/199/208/603, 12/45/139/517/756, 12/45/139/756, 12/45/176/603/630/642/756, 12/45/199/208/517/603/630/756, 12/45/208/517/603/642/756, 12/95/136/139/517/603/756, 12/95/139/517/630/756, 12/95/139/517/642, 12/95/139/630/642, 12/95/199/517/642, 12/95/517/630/756, 12/95/630/756, 12/136, 12/136/139/176/517/603/630, 12/136/139/176/517/603/756, 12/136/139/176/630/756, 12/136/139/176/642, 12/136/139/176/756, 12/136/139/199/208/517/603/756, 12/136/139/199/208/517/630/642/756, 12/136/139/208/517, 12/136/139/517, 12/136/139/517/603/630/642/756, 12/136/139/517/603/630/756, 12/136/139/517/603/756, 12/136/139/517/630, 12/136/139/517/630/642, 12/136/139/517/630/642/756, 12/136/139/517/756, 12/136/139/603/630, 12/136/139/603/630/642, 12/136/139/603/630/756, 12/136/139/603/642/756, 12/136/139/630, 12/136/139/630/642/756, 12/136/139/630/756, 12/136/139/642/756, 12/136/176/208/517/603/630/756, 12/136/176/517/642, 12/136/176/603/756, 12/136/199/208/517/603/642/756, 12/136/199/208/630/642, 12/136/199/517/756, 12/136/208, 12/136/208/603/642, 12/136/517/603/630/642, 12/136/517/603/630/756, 12/136/517/630, 12/136/517/630/642, 12/136/517/642, 12/136/517/642/756, 12/136/517/756, 12/136/603/630/642/756, 12/136/603/630/756, 12/136/603/642, 12/136/603/642/756, 12/136/603/756, 12/136/630, 12/136/630/756, 12/139/176/517/603/630/756, 12/139/176/630/756, 12/139/199/208/642, 12/139/199/517/630/756, 12/139/208/517, 12/139/208/642, 12/139/517/603/642/756, 12/139/517/630/642/756, 12/139/517/642, 12/139/603/642/756, 12/139/603/756, 12/139/630/642, 12/139/630/642/756, 12/139/630/756, 12/176/517/603/630, 12/176/517/630/642/756, 12/176/517/756, 12/176/603/630/756, 12/176/603/756, 12/199/208/517/642, 12/199/630/642/756, 12/199/642/756, 12/199/756, 12/208/517/603/623/630/642, 12/208/603/630/756, 12/208/630/756, 12/517/603/630/642/756, 12/517/603/630/756, 12/517/603/756, 12/517/630/642/756, 12/517/642/756, 12/603/630/756, 12/603/642/756, 12/603/756, 12/630/642/756, 12/630/756, 25/176/198/532/539, 116/142/198/434/440, 136/139/176/199/208/517/630/642, 136/139/176/517/630/642/756, 136/139/199/517/603/756, 136/139/208/517/630/756, 136/139/208/603/630/756, 136/139/517/603/630/642/756, 136/139/517/603/642/756, 136/139/517/603/756, 136/139/517/756, 136/139/603, 136/139/630/642/756, 136/517/756, 136/603/756, 136/630/642, 136/630/756, 136/642, 136/642/756, 136/756, 139/199/208/517/630/756, 139/199/517/642, 139/208/517/630/642/756, 139/517/603/756, 139/517/630/756, 139/642/756, 154/532/652/788, 199/517/603/630/756, 208/517/630/642/756, 517/603/630/642/756, 517/630/756, 603/630/756, and 603/756, wherein the positions are numbered with reference to SEQ ID NO: 8420. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from 12N/45A/95T/136Q/139K/199A/517P/630M/756C, 12N/45A/136Q/139K/199A/517P/603E, 12N/45A/136Q/139K/517P/603E/756C, 12N/45A/136Q/139K/517P/630M/642V/756C, 12N/45A/136Q/139K/603E/756C, 12N/45A/136Q/139K/642V/756C, 12N/45A/136Q/208K/517P/630M/756C, 12N/45A/136Q/517P/630M/642V/756C, 12N/45A/136Q/603E/756C, 12N/45A/136Q/630M/642V, 12N/45A/139K/199A/208K/603E, 12N/45A/139K/517P/756C, 12N/45A/199A/208K/517P/603E/630M/756C, 12N/95T/136Q/139K/517P/603Q/756C, 12N/95T/139K/517P/642V, 12N/95T/139K/630M/642V, 12N/95T/199A/517P/642V, 12N/95T/517P/630M/756C, 12N/136Q, 12N/136Q/139K/176T/517P/603Q/756C, 12N/136Q/139K/176T/630M/756C, 12N/136Q/139K/176T/642V, 12N/136Q/139K/199A/208K/517P/630M/642V/756C, 12N/136Q/139K/517P, 12N/136Q/139K/517P/630M/642V, 12N/136Q/139K/517P/630M/642V/756C, 12N/136Q/139K/517P/756C, 12N/136Q/139K/603E/630M, 12N/136Q/139K/603Q/642V/756C, 12N/136Q/139K/630M, 12N/136Q/139K/642V/756C, 12N/136Q/176T/603Q/756C, 12N/136Q/199A/208K/517P/603E/642V/756C, 12N/136Q/199A/517P/756C, 12N/136Q/208K, 12N/136Q/517P/630M, 12N/136Q/603E/642V, 12N/136Q/603E/642V/756C,

12N/136Q/603Q/630M/756C, 12N/136Q/603Q/642V/756C, 12N/136Q/603Q/756C,
12N/139K/199A/208K/642V, 12N/139K/199A/517P/630M/756C, 12N/139K/208K/517P,
12N/139K/517P/603Q/642V/756C, 12N/139K/603E/642V/756C, 12N/139K/603Q/756C,
12N/139K/630M/642V, 12N/139K/630M/642V/756C, 12N/139K/630M/756C,
12N/176T/517P/603E/630M, 12N/176T/517P/630M/642V/756C, 12N/176T/517P/756C,
12N/199A/208K/517P/642V, 12N/199A/630M/642V/756C, 12N/199A/642V/756C,
12N/208K/517P/603E/623N/630M/642V, 12N/208K/630M/756C, 12N/517P/603E/756C,
12N/517P/603Q/630M/756C, 12N/517P/630M/642V/756C, 12N/603E/642V/756C,
12N/603Q/630M/756C, 12N/603Q/642V/756C, 12N/603Q/756C, 12N/630M/642V/756C,
12S/45A/95T/136Q/756C, 12S/45A/136Q/139K/208K/603E/630M/756C,
12S/45A/136Q/139K/517P/756C, 12S/45A/136Q/176T/517P/603E/630M/642V,
12S/45A/136Q/517P/603Q/642V/756C/789V, 12S/45A/139K/176T/208K/517P/603E/630M,
12S/45A/139K/517P/756C, 12S/45A/139K/756C, 12S/45A/176T/603Q/630M/642V/756C,
12S/45A/208K/517P/603E/642V/756C, 12S/95T/139K/517P/630M/756C, 12S/95T/630M/756C,
12S/136Q/139K/176T/517P/603Q/630M, 12S/136Q/139K/176T/756C,
12S/136Q/139K/199A/208K/517P/603E/756C, 12S/136Q/139K/208K/517P,
12S/136Q/139K/517P/603E/630M/642V/756C, 12S/136Q/139K/517P/603E/630M/756C,
12S/136Q/139K/517P/603E/756C, 12S/136Q/139K/517P/603Q/630M/642V/756C,
12S/136Q/139K/517P/603Q/756C, 12S/136Q/139K/517P/630M, 12S/136Q/139K/517P/630M/642V,
12S/136Q/139K/603Q/630M/642V, 12S/136Q/139K/603Q/630M/756C,
12S/136Q/139K/630M/642V/756C, 12S/136Q/139K/630M/756C,
12S/136Q/176T/208K/517P/603E/630M/756C, 12S/136Q/176T/517P/642V,
12S/136Q/199A/208K/630M/642V, 12S/136Q/208K/603Q/642V, 12S/136Q/517P/603Q/630M/642V,
12S/136Q/517P/603Q/630M/756C, 12S/136Q/517P/630M/642V, 12S/136Q/517P/642V,
12S/136Q/517P/642V/756C, 12S/136Q/517P/756C, 12S/136Q/603E/756C,
12S/136Q/603Q/630M/642V/756C, 12S/136Q/630M, 12S/136Q/630M/756C,
12S/139K/176T/517P/603Q/630M/756C, 12S/139K/176T/630M/756C, 12S/139K/208K/642V,
12S/139K/517P/630M/642V/756C, 12S/139K/517P/642V, 12S/139K/603Q/756C, 12S/176T/603E/756C,
12S/176T/603Q/630M/756C, 12S/199A/756C, 12S/208K/603Q/630M/756C,
12S/517P/603Q/630M/642V/756C, 12S/517P/630M/642V/756C, 12S/517P/642V/756C,
12S/603Q/630M/756C, 12S/630M/756C, 25E/176M/198R/532Y/539R, 116F/142N/198R/434H/440R,
136Q/139K/176T/199A/208K/517P/630M/642V, 136Q/139K/176T/517P/630M/642V/756C,
136Q/139K/199A/517P/603E/756C, 136Q/139K/208K/517P/630M/756C,
136Q/139K/208K/603Q/630M/756C, 136Q/139K/517P/603E/642V/756C,
136Q/139K/517P/603Q/630M/642V/756C, 136Q/139K/517P/603Q/756C, 136Q/139K/517P/756C,
136Q/139K/603Q, 136Q/139K/630M/642V/756C, 136Q/517P/756C, 136Q/603Q/756C,
136Q/630M/642V, 136Q/630M/756C, 136Q/642V, 136Q/642V/756C, 136Q/756C,
139K/199A/208K/517P/630M/756C, 139K/199A/517P/642V, 139K/208K/517P/630M/642V/756C,
139K/517P/603Q/756C, 139K/517P/630M/756C, 139K/642V/756C, 154A/532Y/652R/788K,
199A/517P/603Q/630M/756C, 208K/517P/630M/642V/756C, 517P/603E/630M/642V/756C,
517P/630M/756C, 603E/630M/756C, and 603Q/756C, wherein the positions are numbered with
reference to SEQ ID NO: 8420. In some embodiments, the polypeptide sequence of the engineered
sucrose synthase comprises at least one mutation or mutation set at one or more positions selected from
D12N/E45A/Q95T/R136Q/R139K/T199A/A517P/L630M/V756C,
D12N/E45A/R136Q/R139K/T199A/A517P/G603E, D12N/E45A/R136Q/R139K/A517P/G603E/V756C,
D12N/E45A/R136Q/R139K/A517P/L630M/A642V/V756C,
D12N/E45A/R136Q/R139K/G603E/V756C, D12N/E45A/R136Q/R139K/A642V/V756C,
D12N/E45A/R136Q/R208K/A517P/L630M/V756C,
D12N/E45A/R136Q/A517P/L630M/A642V/V756C, D12N/E45A/R136Q/G603E/V756C,
D12N/E45A/R136Q/L630M/A642V, D12N/E45A/R139K/T199A/R208K/G603E,
D12N/E45A/R139K/A517P/V756C, D12N/E45A/T199A/R208K/A517P/G603E/L630M/V756C,
D12N/Q95T/R136Q/R139K/A517P/G603Q/V756C, D12N/Q95T/R139K/A517P/A642V,
D12N/Q95T/R139K/L630M/A642V, D12N/Q95T/T199A/A517P/A642V,

D12N/Q95T/A517P/L630M/V756C, D12N/R136Q,
D12N/R136Q/R139K/L176T/A517P/G603Q/V756C, D12N/R136Q/R139K/L176T/L630M/V756C,
D12N/R136Q/R139K/L176T/A642V,
D12N/R136Q/R139K/T199A/R208K/A517P/L630M/A642V/V756C, D12N/R136Q/R139K/A517P,
D12N/R136Q/R139K/A517P/L630M/A642V, D12N/R136Q/R139K/A517P/L630M/A642V/V756C,
D12N/R136Q/R139K/A517P/V756C, D12N/R136Q/R139K/G603E/L630M,
D12N/R136Q/R139K/G603Q/A642V/V756C, D12N/R136Q/R139K/L630M,
D12N/R136Q/R139K/A642V/V756C, D12N/R136Q/L176T/G603Q/V756C,
D12N/R136Q/T199A/R208K/A517P/G603E/A642V/V756C, D12N/R136Q/T199A/A517P/V756C,
D12N/R136Q/R208K, D12N/R136Q/A517P/L630M, D12N/R136Q/G603E/A642V,
D12N/R136Q/G603E/A642V/V756C, D12N/R136Q/G603Q/L630M/V756C,
D12N/R136Q/G603Q/A642V/V756C, D12N/R136Q/G603Q/V756C,
D12N/R139K/T199A/R208K/A642V, D12N/R139K/T199A/A517P/L630M/V756C,
D12N/R139K/R208K/A517P, D12N/R139K/A517P/G603Q/A642V/V756C,
D12N/R139K/G603E/A642V/V756C, D12N/R139K/G603Q/V756C, D12N/R139K/L630M/A642V,
D12N/R139K/L630M/A642V/V756C, D12N/R139K/L630M/V756C,
D12N/L176T/A517P/G603E/L630M, D12N/L176T/A517P/L630M/A642V/V756C,
D12N/L176T/A517P/V756C, D12N/T199A/R208K/A517P/A642V,
D12N/T199A/L630M/A642V/V756C, D12N/T199A/A642V/V756C,
D12N/R208K/A517P/G603E/H623N/L630M/A642V, D12N/R208K/L630M/V756C,
D12N/A517P/G603E/V756C, D12N/A517P/G603Q/L630M/V756C,
D12N/A517P/L630M/A642V/V756C, D12N/G603E/A642V/V756C, D12N/G603Q/L630M/V756C,
D12N/G603Q/A642V/V756C, D12N/G603Q/V756C, D12N/L630M/A642V/V756C,
D12S/E45A/Q95T/R136Q/V756C, D12S/E45A/R136Q/R139K/R208K/G603E/L630M/V756C,
D12S/E45A/R136Q/R139K/A517P/V756C, D12S/E45A/R136Q/L176T/A517P/G603E/L630M/A642V,
D12S/E45A/R136Q/A517P/G603Q/A642V/V756C/A789V,
D12S/E45A/R139K/L176T/R208K/A517P/G603E/L630M, D12S/E45A/R139K/A517P/V756C,
D12S/E45A/R139K/V756C, D12S/E45A/L176T/G603Q/L630M/A642V/V756C,
D12S/E45A/R208K/A517P/G603E/A642V/V756C, D12S/Q95T/R139K/A517P/L630M/V756C,
D12S/Q95T/L630M/V756C, D12S/R136Q/R139K/L176T/A517P/G603Q/L630M,
D12S/R136Q/R139K/L176T/V756C, D12S/R136Q/R139K/T199A/R208K/A517P/G603E/V756C,
D12S/R136Q/R139K/R208K/A517P, D12S/R136Q/R139K/A517P/G603E/L630M/A642V/V756C,
D12S/R136Q/R139K/A517P/G603E/L630M/V756C, D12S/R136Q/R139K/A517P/G603E/V756C,
D12S/R136Q/R139K/A517P/G603Q/L630M/A642V/V756C,
D12S/R136Q/R139K/A517P/G603Q/V756C, D12S/R136Q/R139K/A517P/L630M,
D12S/R136Q/R139K/A517P/L630M/A642V, D12S/R136Q/R139K/G603Q/L630M/A642V,
D12S/R136Q/R139K/G603Q/L630M/V756C, D12S/R136Q/R139K/L630M/A642V/V756C,
D12S/R136Q/R139K/L630M/V756C, D12S/R136Q/L176T/R208K/A517P/G603E/L630M/V756C,
D12S/R136Q/L176T/A517P/A642V, D12S/R136Q/T199A/R208K/L630M/A642V,
D12S/R136Q/R208K/G603Q/A642V, D12S/R136Q/A517P/G603Q/L630M/A642V,
D12S/R136Q/A517P/G603Q/L630M/V756C, D12S/R136Q/A517P/L630M/A642V,
D12S/R136Q/A517P/A642V, D12S/R136Q/A517P/A642V/V756C, D12S/R136Q/A517P/V756C,
D12S/R136Q/G603E/V756C, D12S/R136Q/G603Q/L630M/A642V/V756C, D12S/R136Q/L630M,
D12S/R136Q/L630M/V756C, D12S/R139K/L176T/A517P/G603Q/L630M/V756C,
D12S/R139K/L176T/L630M/V756C, D12S/R139K/R208K/A642V,
D12S/R139K/A517P/L630M/A642V/V756C, D12S/R139K/A517P/A642V,
D12S/R139K/G603Q/V756C, D12S/L176T/G603E/V756C, D12S/L176T/G603Q/L630M/V756C,
D12S/T199A/V756C, D12S/R208K/G603Q/L630M/V756C,
D12S/A517P/G603Q/L630M/A642V/V756C, D12S/A517P/L630M/A642V/V756C,
D12S/A517P/A642V/V756C, D12S/G603Q/L630M/V756C, D12S/L630M/V756C,
A25E/L176M/Q198R/S532Y/S539R, L116F/S142N/Q198R/Y434H/P440R,
R136Q/R139K/L176T/T199A/R208K/A517P/L630M/A642V,
R136Q/R139K/L176T/A517P/L630M/A642V/V756C, R136Q/R139K/T199A/A517P/G603E/V756C,

R136Q/R139K/R208K/A517P/L630M/V756C, R136Q/R139K/R208K/G603Q/L630M/V756C, R136Q/R139K/A517P/G603E/A642V/V756C, R136Q/R139K/A517P/G603Q/L630M/A642V/V756C, R136Q/R139K/A517P/G603Q/V756C, R136Q/R139K/A517P/V756C, R136Q/R139K/G603Q, R136Q/R139K/L630M/A642V/V756C, R136Q/A517P/V756C, R136Q/G603Q/V756C, R136Q/L630M/A642V, R136Q/L630M/V756C, R136Q/A642V, R136Q/A642V/V756C, R136Q/V756C, R139K/T199A/R208K/A517P/L630M/V756C, R139K/T199A/A517P/A642V, R139K/R208K/A517P/L630M/A642V/V756C, R139K/A517P/G603Q/V756C, R139K/A517P/L630M/V756C, R139K/A642V/V756C, H154A/S532Y/G652R/H788K, T199A/A517P/G603Q/L630M/V756C, R208K/A517P/L630M/A642V/V756C, A517P/G603E/L630M/A642V/V756C, A517P/L630M/V756C, G603E/L630M/V756C, and G603Q/V756C, wherein the positions are numbered with reference to SEQ ID NO: 8420. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence that is at least 90% identical to any of SEQ ID NOS: 8798, 8800, 8802, 8804, 8806, 8808, 8810, 8812, 8814, 8816, 8818, 8820, 8822, 8824, 8826, 8828, 8830, 8832, 8834, 8836, 8838, 8840, 8842, 8844, 8846, 8848, 8850, 8852, 8854, 8856, 8858, 8860, 8862, 8864, 8866, 8868, 8870, 8872, 8874, 8876, 8878, 8880, 8882, 8884, 8886, 8888, 8890, 8892, 8894, 8896, 8898, 8900, 8902, 8904, 8906, 8908, 8910, 8912, 8914, 8916, 8918, 8920, 8922, 8924, 8926, 8928, 8930, 8932, 8934, 8936, 8938, 8940, 8942, 8944, 8946, 8948, 8950, 8952, 8954, 8956, 8958, 8960, 8962, 8964, 8966, 8968, 8970, 8972, 8974, 8976, 8978, 8980, 8982, 8984, 8986, 8988, 8990, 8992, 8994, 8996, 8998, 9000, 9002, 9004, 9006, 9008, 9010, 9012, 9014, 9016, 9018, 9020, 9022, 9024, 9026, 9028, 9030, 9032, 9034, 9036, 9038, 9040, 9042, 9044, 9046, 9048, 9050, 9052, 9054, 9056, 9058, 9060, 9062, 9064, 9066, 9068, 9070, 9072, 9074, 9076, 9078, 9080, 9082, 9084, 9086, 9088, 9090, 9092, 9094, 9096, 9098, 9100, 9102, 9104, and 9106. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises a sequence at least 95% identical to any of SEQ ID NOS: 8798, 8800, 8802, 8804, 8806, 8808, 8810, 8812, 8814, 8816, 8818, 8820, 8822, 8824, 8826, 8828, 8830, 8832, 8834, 8836, 8838, 8840, 8842, 8844, 8846, 8848, 8850, 8852, 8854, 8856, 8858, 8860, 8862, 8864, 8866, 8868, 8870, 8872, 8874, 8876, 8878, 8880, 8882, 8884, 8886, 8888, 8890, 8892, 8894, 8896, 8898, 8900, 8902, 8904, 8906, 8908, 8910, 8912, 8914, 8916, 8918, 8920, 8922, 8924, 8926, 8928, 8930, 8932, 8934, 8936, 8938, 8940, 8942, 8944, 8946, 8948, 8950, 8952, 8954, 8956, 8958, 8960, 8962, 8964, 8966, 8968, 8970, 8972, 8974, 8976, 8978, 8980, 8982, 8984, 8986, 8988, 8990, 8992, 8994, 8996, 8998, 9000, 9002, 9004, 9006, 9008, 9010, 9012, 9014, 9016, 9018, 9020, 9022, 9024, 9026, 9028, 9030, 9032, 9034, 9036, 9038, 9040, 9042, 9044, 9046, 9048, 9050, 9052, 9054, 9056, 9058, 9060, 9062, 9064, 9066, 9068, 9070, 9072, 9074, 9076, 9078, 9080, 9082, 9084, 9086, 9088, 9090, 9092, 9094, 9096, 9098, 9100, 9102, 9104, and 9106. In some embodiments, the polypeptide sequence of the engineered sucrose synthase comprises any of SEQ ID NOS: 8798, 8800, 8802, 8804, 8806, 8808, 8810, 8812, 8814, 8816, 8818, 8820, 8822, 8824, 8826, 8828, 8830, 8832, 8834, 8836, 8838, 8840, 8842, 8844, 8846, 8848, 8850, 8852, 8854, 8856, 8858, 8860, 8862, 8864, 8866, 8868, 8870, 8872, 8874, 8876, 8878, 8880, 8882, 8884, 8886, 8888, 8890, 8892, 8894, 8896, 8898, 8900, 8902, 8904, 8906, 8908, 8910, 8912, 8914, 8916, 8918, 8920, 8922, 8924, 8926, 8928, 8930, 8932, 8934, 8936, 8938, 8940, 8942, 8944, 8946, 8948, 8950, 8952, 8954, 8956, 8958, 8960, 8962, 8964, 8966, 8968, 8970, 8972, 8974, 8976, 8978, 8980, 8982, 8984, 8986, 8988, 8990, 8992, 8994, 8996, 8998, 9000, 9002, 9004, 9006, 9008, 9010, 9012, 9014, 9016, 9018, 9020, 9022, 9024, 9026, 9028, 9030, 9032, 9034, 9036, 9038, 9040, 9042, 9044, 9046, 9048, 9050, 9052, 9054, 9056, 9058, 9060, 9062, 9064, 9066, 9068, 9070, 9072, 9074, 9076, 9078, 9080, 9082, 9084, 9086, 9088, 9090, 9092, 9094, 9096, 9098, 9100, 9102, 9104, and 9106.

(75) The present invention also provides engineered glycosyltransferases that are NDP-glycosyltransferases, wherein the engineered NDP-glycosyltransferase are NDP-glycosyltransferases selected from ADP-glucose-dependent glycosyltransferases (AGTs), CDP-glucose-dependent glycosyltransferases (CGTs), GDP-glucose-dependent glycosyltransferase (GGTs), TDP-glucose-dependent glycosyltransferases (TGTs), and IDP-glucose-dependent glycosyltransferase (IGTs). In some embodiments, the engineered NDP-glycosyltransferase is an ADP-glucose-dependent glycosyltransferase. In some embodiments, the engineered NDP-glycotransferase is not an UDP-glucose-dependent glycosyltransferase.

(76) The present invention also provides engineered polynucleotides encoding at least one engineered glycosyltransferase polypeptide provided herein. In some embodiments, the engineered polynucleotide

encoding at least one engineered glycosyltransferase provided herein comprises at least 60%, 65%, 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NO: 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 75, 77, 79, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, 119, 121, 123, 125, 127, 129, 131, 133, 135, 137, 139, 141, 143, 145, 147, 149, 151, 153, 155, 157, 159, 161, 163, 165, 167, 169, 171, 173, 175, 177, 179, 181, 183, 185, 187, 189, 191, 193, 195, 197, 199, 201, 203, 205, 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 239, 241, 243, 245, 247, 249, 251, 253, 255, 257, 259, 261, 263, 265, 267, 269, 271, 273, 275, 277, 279, 281, 283, 285, 287, 289, 291, 293, 295, 297, 299, 301, 303, 305, 307, 309, 311, 313, 315, 317, 319, 321, 323, 325, 327, 329, 331, 333, 335, 337, 339, 341, 343, 345, 347, 349, 351, 353, 355, 357, 359, 361, 363, 365, 367, 369, 371, 373, 375, 377, 379, 381, 383, 385, 387, 389, 391, 393, 395, 397, 399, 401, 403, 405, 407, 409, 411, 413, 415, 417, 419, 421, 423, 425, 427, 429, 431, 433, 435, 437, 439, 441, 443, 445, 447, 449, 451, 453, 455, 457, 459, 461, 463, 465, 467, 469, 471, 473, 475, 477, 479, 481, 483, 485, 487, 489, 491, 493, 495, 497, 499, 501, 503, 505, 507, 509, 511, 513, 515, 517, 519, 521, 523, 525, 527, 529, 531, 533, 535, 537, 539, 541, 543, 545, 547, 549, 551, 553, 555, 557, 559, 561, 563, 565, 567, 569, 571, 573, 575, 577, 579, 581, 583, 585, 587, 589, 591, 593, 595, 597, 599, 601, 603, 605, 607, 609, 611, 613, 615, 617, 619, 621, 623, 625, 627, 629, 631, 633, 635, 637, 639, 641, 643, 645, 647, 649, 651, 653, 655, 657, 659, 661, 663, 665, 667, 669, 671, 673, 675, 677, 679, 681, 683, 685, 687, 689, 691, 693, 695, 697, 699, 701, 703, 705, 707, 709, 711, 713, 715, 717, 719, 721, 723, 725, 727, 729, 731, 733, 735, 737, 739, 741, 743, 745, 747, 749, 751, 753, 769, 771, 773, 775, 777, 779, 781, 783, 785, 787, 789, 791, 793, 795, 797, 799, 801, 803, 805, 807, 809, 811, 813, 815, 817, 819, 821, 823, 825, 827, 829, 831, 833, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 859, 861, 863, 865, 867, 869, 871, 873, 875, 877, 879, 881, 883, 885, 887, 889, 891, 893, 895, 897, 899, 901, 903, 905, 907, 909, 911, 913, 915, 917, 919, 921, 923, 925, 927, 929, 931, 933, 935, 937, 939, 941, 943, 945, 947, 949, 951, 953, 955, 957, 959, 961, 963, 965, 967, 969, 971, 973, 975, 977, 979, 981, 983, 985, 987, 989, 991, 993, 995, 997, 999, 1001, 1003, 1005, 1007, 1009, 1011, 1013, 1015, 1017, 1019, 1021, 1023, 1025, 1027, 1029, 1031, 1033, 1035, 1037, 1039, 1041, 1043, 1045, 1047, 1049, 1051, 1053, 1055, 1057, 1059, 1061, 1063, 1065, 1067, 1069, 1071, 1073, 1075, 1077, 1289, 1291, 1293, 2595, 2597, 2599, 2601, 2603, 2605, 2607, 2609, 2611, 2613, 2615, 2617, 2619, 2621, 2623, 2625, 2627, 2629, 2631, 2633, 2635, 2637, 2639, 2641, 2643, 2645, 2647, 2649, 2651, 2653, 2655, 2657, 2659, 2661, 2663, 2665, 2667, 2669, 2671, 2673, 2675, 2677, 2679, 2681, 2683, 2685, 2687, 2689, 2691, 2693, 2695, 2697, 2699, 2701, 2703, 2705, 2707, 2709, 2711, 2713, 2715, 2717, 2719, 2721, 2723, 2725, 2727, 2729, 2731, 2733, 2735, 2737, 2739, 2741, 2743, 2745, 2747, 2749, 2751, 2753, 2755, 2757, 2759, 2761, 2763, 2765, 2767, 2769, 2771, 2773, 2775, 2777, 2779, 2781, 2783, 2785, 2787, 2789, 2791, 2793, 2795, 2797, 2799, 2801, 2803, 2805, 2807, 2809, 2811, 2813, 2815, 2817, 2819, 2821, 2823, 2825, 2827, 2829, 2831, 2833, 2835, 2837, 2839, 2841, 2843, 2845, 2847, 2849, 2851, 2853, 2855, 2857, 2859, 2861, 2863, 2865, 2867, 2869, 2871, 2873, 2875, 2877, 2879, 2881, 2883, 2885, 2887, 2889, 2891, 2893, 2895, 2897, 2899, 2901, 2903, 2905, 2907, 2909, 2911, 2913, 2915, 2917, 2919, 2921, 2923, 2925, 2927, 2829, 2931, 2933, 2935, 2937, 2939, 2941, 2943, 2945, 2947, 2949, 2951, 2953, 2955, 2957, 2959, 2961, 2963, 2965, 2967, 2969, 2971, 2973, 2975, 2977, 2979, 2981, 2983, 2985, 2987, 2989, 2991, 2993, 2995, 2997, 2999, 3001, 3003, 3005, 3007, 3009, 3011, 3013, 3015, 3017, 3019, 3021, 3023, 3025, 3027, 3029, 3031, 3033, 3035, 3037, 3039, 3041, 3043, 3045, 3047, 3049, 3051, 3053, 3055, 3057, 3059, 3061, 3063, 3065, 3067, 3069, 3071, 3073, 3075, 3077, 3079, 3081, 3083, 3085, 3087, 3089, 3091, 3093, 3095, 3097, 3099, 3101, 3103, 3105, 3107, 3109, 3111, 3113, 3115, 3117, 3119, 3121, 3123, 3125, 3127, 3129, 3131, 3133, 3135, 3137, 3139, 3141, 3143, 3145, 3147, 3149, 3151, 3153, 3155, 3157, 3159, 3161, 3163, 3165, 3167, 3169, 3171, 3173, 3175, 3177, 3179, 3181, 3183, 3185, 3187, 3189, 3191, 3193, 3195, 3197, 3199, 3201, 3203, 3205, 3207, 3209, 3211, 3213, 3215, 3217, 3219, 3221, 3223, 3225, 3227, 3229, 3231, 3233, 3235, 3237, 3239, 3241, 3243, 3245, 3247, 3249, 3251, 3253, 3255, 3257, 3259, 3261, 3263, 3265, 3267, 3269, 3271, 3273, 3275, 3277, 3279, 3281, 3283, 3285, 3287, 3289, 3291, 3293, 3295, 3297, 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(77) The present invention also provides engineered polynucleotides encoding at least one engineered sucrose synthase polypeptide provided herein. In some embodiments, the engineered polynucleotide encoding at least one engineered sucrose synthase provided herein comprises at least 60%, 65%, 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NO: 1079, 1081, 1083, 1085, 1087, 1089, 1091, 1093, 1095, 1097, 1099, 1101, 1103, 1105, 1107, 1109, 1111, 1113, 1115, 1117, 1119, 1121, 1123, 1125, 1127, 1129, 1131, 1133, 1135, 1137, 1139, 1141, 1143, 1145, 1147, 1149, 1151, 1153, 1155, 1157, 1159, 1161, 1163, 1165, 1167, 1169, 1171, 1173, 1175, 1177, 1179, 1181, 1183, 1185, 1187, 1189, 1191, 1193, 1195, 1197, 1199, 1201, 1203, 1205, 1207, 1209, 1211, 1213, 1215, 1217, 1219, 1221, 1223, 1225, 1227, 1229, 1231, 1233, 1235, 1237, 1239, 1241, 1243, 1245, 1247, 1249, 1251, 1253, 1255, 1257, 1259, 1261, 1263, 1265, 1267, 1269, 1271, 1273, 1275, 1277, 1279, 1281, 1283, 1285, 1287, 1295, 1297, 1299, 1301, 1303, 1305, 1307,

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(78) The present invention also provides vectors comprising at least one engineered polynucleotide provided herein. In some embodiments, the vectors further comprise at least one control sequence.

(79) The present invention also provides host cells comprising at least one engineered polypeptide provided herein. In some embodiments, the host cells comprise at least one vector provided herein. In some embodiments, the host cell is selected from eukaryotic and prokaryotic organisms. In some additional embodiments, the host cell is *E. coli*.

(80) The present invention also provides methods for producing at least one engineered glycosyltransferase variant provided herein, comprising culturing the host cell provided herein, under conditions such that the engineered glycosyltransferase variant is produced by the host cell. In some embodiments, the methods further comprise the step of recovering the engineered glycosyltransferase variant.

(81) The present invention also provides compositions comprising at least one engineered glycosyltransferase variant and/or sucrose synthase variant provided herein. In some embodiments, the compositions comprise at least one engineered glycosyltransferase variant provided herein.

(82) The present invention also provides methods for producing at least one engineered sucrose synthase variant provided herein, comprising culturing the host cell provided herein, under conditions such that the engineered sucrose synthase variant is produced by the host cell. In some embodiments, the methods further comprise the step of recovering the engineered sucrose synthase variant. In some embodiments, the compositions comprise at least one engineered sucrose synthase variant provided herein.

(83) The present invention also provides methods for glycosylation of a substrate comprising: providing at least one substrate, at least one engineered glycosyl transferase selected from the even numbered sequences provided herein; contacting the substrate with the glycosyltransferase under conditions such that the substrate is glycosylated to produce at least one glycosylated product. In some embodiments, the substrate comprises at least one steviol glycoside. In some embodiments, the glycosylated product comprises at least one mono-glycosylated and/or polyglycosylated product. It is not intended that the present invention be limited to any limitations regarding the extent of glycosylation of the product (e.g., diglycosylated, triglycosylated, and products with higher glycosylation levels find use in the present invention).

(84) The present invention provides methods for producing rebaudioside M, comprising providing a rebaudioside D and/or rebaudioside I substrate, NDP-glucose, and a least one engineered glycosyltransferase provided herein, combining the rebaudioside D and rebaudioside I substrate, NDP-glucose, and the glycosyltransferase under conditions such that rebaudioside M is produced. In some additional embodiments, the present invention provides methods for producing rebaudioside M, comprising providing a rebaudioside D substrate, NDP-glucose, and a least one engineered glycosyltransferase provided herein, combining the rebaudioside D substrate, NDP-glucose, and glycosyltransferase under conditions such that rebaudioside M is produced. In some further embodiments, the present invention provides methods for producing rebaudioside M, comprising providing a rebaudioside I substrate, NDP-glucose, and a least one engineered glycosyltransferase provided herein, combining the rebaudioside I substrate, NDP-glucose, and glycosyltransferase under conditions such that rebaudioside M is produced. In some of the present embodiments, the NDP-glucose is selected from ADP-glucose, CDP-glucose, TDP-glucose, GDP-glucose, and/or IDT-glucose. In some additional embodiments, the NDP-glucose is not UDP-glucose.

(85) The present invention provides methods for producing rebaudioside A and/or rebaudioside I, comprising providing a stevioside substrate, NDP-glucose, and at least one engineered glycosyltransferase provided herein, combining the stevioside substrate, NDP-glucose, and glycosyltransferase under conditions such that rebaudioside A and/or rebaudioside I is produced. In some

embodiments, the present invention provides methods for producing rebaudioside A, comprising providing a stevioside substrate, NDP-glucose, and at least one engineered glycosyltransferase provided herein, combining the stevioside substrate, NDP-glucose, and glycosyltransferase under conditions such that rebaudioside A is produced. In some additional embodiments, the present invention provides methods for producing rebaudioside I, comprising providing a stevioside substrate, NDP-glucose, and at least one engineered glycosyltransferase provided herein, combining the stevioside substrate, NDP-glucose, and glycosyltransferase under conditions such that rebaudioside I is produced. In some of the present embodiments, the NDP-glucose is selected from ADP-glucose, CDP-glucose, TDP-glucose, GDP-glucose, and/or IDT-glucose. In some additional embodiments, the NDP-glucose is not UDP-glucose.

(86) The present invention also provides methods for producing rebaudioside D, comprising providing a stevioside substrate, NDP-glucose, and at least one engineered glycosyltransferase provided herein, combining the stevioside substrate, NDP-glucose, and glycosyltransferase under conditions such that rebaudioside D is produced. In some of the present embodiments, the NDP-glucose is selected from ADP-glucose, CDP-glucose, TDP-glucose, GDP-glucose, and/or IDT-glucose. In some additional embodiments, the NDP-glucose is not UDP-glucose.

(87) The present invention also provides methods for producing rebaudioside M comprising providing a rebaudioside D and/or rebaudioside I substrate, ADP-glucose, and at least one engineered glycosyltransferase provided herein, combining the rebaudioside D and/or rebaudioside I substrate, ADP-glucose, and glycosyltransferase under conditions such that rebaudioside M is produced. In some embodiments, the present invention provides methods for producing rebaudioside M comprising providing a rebaudioside D substrate, ADP-glucose, and at least one engineered glycosyltransferase of provided herein, combining the rebaudioside D substrate, ADP-glucose, and glycosyltransferase under conditions such that rebaudioside M is produced. In some additional embodiments, the present invention provides methods for producing rebaudioside M comprising providing a rebaudioside I substrate, ADP-glucose, and at least one engineered glycosyltransferase provided herein, combining the rebaudioside I substrate, ADP-glucose, and glycosyltransferase under conditions such that rebaudioside M is produced. In some of the present embodiments, the engineered glycosyltransferase comprises an ADP-glycosyltransferase.

(88) The present invention also provides methods for producing rebaudioside A and/or rebaudioside I, comprising providing a stevioside substrate, ADP-glucose, and at least one engineered glycosyltransferase provided herein, combining the stevioside substrate, ADP-glucose, and glycosyltransferase under conditions such that rebaudioside A and/or rebaudioside I is produced. In some embodiments, the present invention provides methods for producing rebaudioside A, comprising providing a stevioside substrate, ADP-glucose, and at least one engineered glycosyltransferase provided herein, combining the stevioside substrate, ADP-glucose, and glycosyltransferase under conditions such that rebaudioside A is produced. In some additional embodiments, the present invention provides methods for producing rebaudioside I, comprising providing a stevioside substrate, ADP-glucose, and at least one engineered glycosyltransferase provided herein, combining the stevioside substrate, ADP-glucose, and glycosyltransferase under conditions such that rebaudioside I is produced. In some of the present embodiments, the engineered glycosyltransferase comprises an ADP-glycosyltransferase.

(89) The present invention also provides methods for producing rebaudioside D, comprising providing a stevioside substrate, ADP-glucose, and at least one engineered glycosyltransferase provided herein, combining the stevioside substrate, ADP-glucose, and glycosyltransferase under conditions such that rebaudioside D is produced. In some of the present embodiments, the engineered glycosyltransferase comprises an ADP-glycosyltransferase.

(90) The present invention also provides methods for producing rebaudioside M comprising providing a rebaudioside D substrate, NDP, sucrose, a sucrose synthase, and at least one engineered glycosyltransferase provided herein, combining the rebaudioside D substrate, NDP, sucrose, sucrose synthase, and glycosyltransferase under conditions such that rebaudioside M is produced. In some of the present embodiments, the engineered glycosyltransferase comprises an ADP-glycosyltransferase. In some additional embodiments, the NDP is selected from ADP, CDP, TDP, GDP, and/or IDT. In some additional embodiments, the NDP is not UDP.

(91) The present invention also provides methods for producing rebaudioside A and/or rebaudioside I

comprising providing a stevioside substrate, NDP, sucrose, a sucrose synthase, and at least one engineered glycosyltransferase provided herein, combining the stevioside substrate, NDP, sucrose, sucrose synthase, and glycosyltransferase under conditions such that rebaudioside A and/or rebaudioside I is produced. In some embodiments, the present invention provides methods for producing rebaudioside A, comprising providing a stevioside substrate, NDP, sucrose, a sucrose synthase, and at least one engineered glycosyltransferase provided herein, combining the stevioside substrate, NDP, sucrose, sucrose synthase, and glycosyltransferase under conditions such that rebaudioside A is produced. In some additional embodiments, the present invention provides methods for producing rebaudioside I, comprising providing a stevioside substrate, NDP, sucrose, a sucrose synthase, and at least one engineered ADP-glycosyltransferase provided herein, combining the stevioside substrate, NDP, sucrose, sucrose synthase and an ADP-glycosyltransferase under conditions such that rebaudioside I is produced. In some of the present embodiments, the engineered glycosyltransferase comprises an ADP-glycosyltransferase. In some additional embodiments, the NDP is selected from ADP, CDP, TDP, GDP, and/or IDT. In some additional embodiments, the NDP is not UDP.

(92) The present invention also provides methods for producing rebaudioside D, comprising providing a stevioside substrate, NDP, sucrose, a sucrose synthase, and at least one engineered glycosyltransferase provided herein, combining the stevioside substrate, NDP, sucrose, sucrose synthase, and glycosyltransferase under conditions such that rebaudioside D is produced. In some of the present embodiments, the engineered glycosyltransferase comprises an ADP-glycosyltransferase. In some additional embodiments, the NDP is selected from ADP, CDP, TDP, GDP, and/or IDT. In some additional embodiments, the NDP is not UDP.

(93) The present invention also provides methods for producing rebaudioside M, comprising providing a stevioside substrate comprising at least one stevioside and/or a mixture of steviosides and rebA, NDP, sucrose, a sucrose synthase, and at least one engineered glycosyltransferase provided herein, combining the stevioside substrate, NDP, sucrose, sucrose synthase, and glycosyltransferase under conditions such that rebaudioside M is produced. In some of the present embodiments, the engineered glycosyltransferase comprises an ADP-glycosyltransferase. In some additional embodiments, the NDP is selected from ADP, CDP, TDP, GDP, and/or IDT. In some additional embodiments, the NDP is not UDP.

(94) The present invention also provides methods for producing rebaudioside M, comprising providing a stevioside substrate, NDP, sucrose, at least one sucrose synthase, and at least one engineered glycosyltransferase provided herein, combining the stevioside substrate, NDP, and glycosyltransferase under conditions such that rebaudioside A is first produced, rebaudioside D and/or rebaudioside I is then produced, and rebaudioside M finally produced. In some of the present embodiments, the engineered glycosyltransferase comprises an ADP-glycosyltransferase. In some additional embodiments, the NDP is selected from ADP, CDP, TDP, GDP, and/or IDT. In some additional embodiments, the NDP is not UDP.

(95) In some embodiments of the methods of the present invention for the production of rebaudioside(s), the methods are conducted as one-pot reactions. In some additional embodiments, the methods are conducted sequentially. In some additional embodiments, the methods further comprise repeating the steps of the methods. In some further embodiments, sucrose is recycled during repeated steps. In some additional embodiments, at least one engineered glycosyltransferase and/or other reaction components are recycled. In some additional embodiments, the stevioside substrate is extracted from *Stevia rebaudiana*, while in some alternative embodiments, the stevioside substrate is synthetically produced, and still some further embodiments, the stevioside substrate is a mixture of steviosides that are either naturally and/or synthetically produced. In some further embodiments of the methods, at least one engineered glycosyltransferase is immobilized. In some additional embodiments of the methods, sucrose synthase is immobilized. In still some additional embodiments, at least one glycosyltransferase and/or the sucrose synthase is immobilized. In some further embodiments of the methods, a reaction product that includes fructose is produced. In some embodiments, the fructose is removed from the reaction product. In yet some additional embodiments, the methods further comprises a washing step. In some embodiments, the washing step comprises exposing the rebaudioside M, rebaudioside A, rebaudioside I, and/or rebaudioside D produced from the method to a solvent. In some embodiments, the solvent is water. In still further embodiments, the methods further comprise at least one column chromatography step. In some embodiments, at least one column chromatography step is conducted on the rebaudioside M, rebaudioside

A, rebaudiside I, and/or rebaudiside D produced by the method. In some additional embodiments of the methods, the at least one engineered glycosyltransferase is a beta-1,2 glycosyltransferase provided herein. In some further embodiments of the methods, at least one engineered glycosyltransferase is a beta-1,3 glycosyltransferase provided herein. In yet some additional embodiments of the methods, at least one engineered glycosyltransferase is a beta-1,2 glycosyltransferase provided herein and at least one additional engineered glycosyltransferase is a beta-1,3 glycosyltransferase provided herein. In some further embodiments of the methods, at least one engineered sucrose synthase provided herein finds use.

(96) In the methods provided herein, at least one engineered glycosyltransferase comprising a polypeptide sequence that has at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NO: 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, 120, 122, 124, 126, 128, 130, 132, 134, 136, 138, 140, 142, 144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 172, 174, 176, 178, 180, 182, 184, 186, 188, 190, 192, 194, 196, 198, 200, 202, 204, 206, 208, 210, 212, 214, 216, 218, 220, 222, 224, 226, 228, 230, 232, 234, 236, 238, 240, 242, 244, 246, 248, 250, 252, 254, 256, 258, 260, 262, 264, 266, 268, 270, 272, 274, 276, 278, 280, 282, 284, 286, 288, 290, 292, 294, 296, 298, 300, 302, 304, 306, 308, 310, 312, 314, 316, 318, 320, 322, 324, 326, 328, 330, 332, 334, 336, 338, 340, 342, 344, 346, 348, 350, 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, 384, 386, 388, 390, 392, 394, 396, 398, 400, 402, 404, 406, 408, 410, 412, 414, 416, 418, 420, 422, 424, 426, 428, 430, 432, 434, 436, 438, 440, 442, 444, 446, 448, 450, 452, 454, 456, 458, 460, 462, 464, 466, 468, 470, 472, 474, 476, 478, 480, 482, 484, 486, 488, 490, 492, 494, 496, 498, 500, 502, 504, 506, 508, 510, 512, 514, 516, 518, 520, 522, 524, 526, 528, 530, 532, 534, 536, 538, 540, 542, 544, 546, 548, 550, 552, 554, 556, 558, 560, 562, 564, 566, 568, 570, 572, 574, 576, 578, 580, 582, 584, 586, 588, 590, 592, 594, 596, 598, 600, 602, 604, 606, 608, 610, 612, 614, 616, 618, 620, 622, 624, 626, 628, 630, 632, 634, 636, 638, 640, 642, 644, 646, 648, 650, 652, 654, 656, 658, 660, 662, 664, 666, 668, 670, 672, 674, 676, 678, 680, 682, 684, 686, 688, 690, 692, 694, 696, 698, 700, 702, 704, 706, 708, 710, 712, 714, 716, 718, 720, 722, 724, 726, 728, 730, 732, 734, 736, 738, 740, 742, 744, 746, 748, 750, 752, 754, 770, 772, 774, 776, 778, 780, 782, 784, 786, 788, 790, 792, 794, 796, 798, 800, 802, 804, 806, 808, 810, 812, 814, 816, 818, 820, 822, 824, 826, 828, 830, 832, 834, 836, 838, 840, 842, 844, 846, 848, 850, 852, 854, 856, 858, 860, 862, 864, 866, 868, 870, 872, 874, 876, 878, 880, 882, 884, 886, 888, 890, 892, 894, 896, 898, 900, 902, 904, 906, 908, 910, 912, 914, 916, 918, 920, 922, 924, 926, 928, 930, 932, 934, 936, 938, 940, 942, 944, 946, 948, 950, 952, 954, 956, 958, 960, 962, 964, 966, 968, 970, 972, 974, 976, 978, 980, 982, 984, 986, 988, 990, 992, 994, 996, 998, 1000, 1002, 1004, 1006, 1008, 1010, 1012, 1014, 1016, 1018, 1020, 1022, 1024, 1026, 1028, 1030, 1032, 1034, 1036, 1038, 1040, 1042, 1044, 1046, 1048, 1050, 1052, 1054, 1056, 1058, 1060, 1062, 1064, 1066, 1068, 1070, 1072, 1074, 1076, 1078, 1290, 1292, 1294, 2596, 2598, 2600, 2602, 2604, 2606, 2608, 2610, 2612, 2614, 2616, 2618, 2620, 2622, 2624, 2626, 2628, 2630, 2632, 2634, 2636, 2638, 2640, 2642, 2644, 2646, 2648, 2650, 2652, 2654, 2656, 2658, 2660, 2662, 2664, 2666, 2668, 2670, 2672, 2674, 2676, 2678, 2680, 2682, 2684, 2686, 2688, 2690, 2692, 2694, 2696, 2698, 2700, 2702, 2704, 2706, 2708, 2710, 2712, 2714, 2716, 2718, 2720, 2722, 2724, 2726, 2728, 2730, 2732, 2734, 2736, 2738, 2740, 2742, 2744, 2746, 2748, 2750, 2752, 2754, 2756, 2758, 2760, 2762, 2764, 2766, 2768, 2770, 2772, 2774, 2776, 2778, 2780, 2782, 2784, 2786, 2788, 2790, 2792, 2794, 2796, 2798, 2800, 2802, 2804, 2806, 2808, 2810, 2812, 2814, 2816, 2818, 2820, 2822, 2824, 2826, 2828, 2830, 2832, 2834, 2836, 2838, 2840, 2842, 2844, 2846, 2848, 2850, 2852, 2854, 2856, 2858, 2860, 2862, 2864, 2866, 2868, 2870, 2872, 2874, 2876, 2878, 2880, 2882, 2884, 2886, 2888, 2890, 2892, 2894, 2896, 2898, 2900, 2902, 2904, 2906, 2908, 2910, 2912, 2914, 2916, 2918, 2920, 2922, 2924, 2926, 2928, 2830, 2932, 2934, 2936, 2938, 2940, 2942, 2944, 2946, 2948, 2950, 2952, 2954, 2956, 2958, 2960, 2962, 2964, 2966, 2968, 2970, 2972, 2974, 2976, 2978, 2980, 2982, 2984, 2986, 2988, 2990, 2992, 2994, 2996, 2998, 3000, 3002, 3004, 3006, 3008, 3010, 3012, 3014, 3016, 3018, 3020, 3022, 3024, 3026, 3028, 3030, 3032, 3034, 3036, 3038, 3040, 3042, 3044, 3046, 3048, 3050, 3052, 3054, 3056, 3058, 3060, 3062, 3064, 3066, 3068, 3070, 3072, 3074, 3076, 3078, 3080, 3082, 3084, 3086, 3088, 3090, 3092, 3094, 3096, 3098, 3100, 3102, 3104, 3106, 3108, 3110, 3112, 3114, 3116, 3118, 3120, 3122, 3124, 3126, 3128, 3130, 3132, 3134, 3136, 3138, 3140, 3142, 3144, 3146, 3148, 3150, 3152, 3154, 3156, 3158, 3160, 3162, 3164, 3166, 3168, 3170, 3172, 3174, 3176, 3178, 3180, 3182, 3184, 3186, 3188, 3190, 3192, 3194, 3196, 3198, 3200, 3202, 3204, 3206, 3208, 3210, 3212, 3214, 3216, 3218, 3220, 3222, 3224,

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0028, 5030, 5032, 5034, 5036, 5038, 5040, 5042, 5044, 5046, 5048, 5050, 5052, 5054, 5056, 5058, 5060, 5062, 5064, 5066, 5068, 5070, 5072, 5074, 5076, 5078, 5080, 5082, 5084, 5086, 5088, 5090, 5092, 5094, 5096, 5098, 5100, 5102, 5104, 5106, 5108, 5110, 5112, 5114, 5116, 5118, 5120, 5122, 5124, 5126, 5128, 5130, 5132, 5134, 5136, 5138, 5140, 5142, 5144, 5146, 5148, 5150, 5152, 5154, 5156, 5158, 5160, 5162, 5164, 5166, 5168, 5170, 5172, 5174, 5176, 5178, 5180, 5182, 5184, 5186, 5188, 5190, 5192, 5194, 5196, 5198, 5200, 5202, 5204, 5206, 5208, 5210, 5212, 5214, 5216, 5218, 5220, 5222, 5224, 5226, 5228, 5230, 5232, 5234, 5236, 5238, 5240, 5242, 5244, 5246, 5248, 5250, 5252, 5254, 5256, 5258, 5260, 5262, 5264, 5266, 5268, 5270, 5272, 5274, 5276, 5278, 5280, 5282, 5284, 5286, 5288, 5290, 5292, 5294, 5296, 5298, 5300, 5302, 5304, 5306, 5308, 5310, 5312, 5314, 5316, 5318, 5320, 5322, 5324, 5326, 5328, 5330, 5332, 5334, 5336, 5338, 5340, 5342, 5344, 5346, 5348, 5350, 5352, 5354, 5356, 5358, 5360, 5362, 5364, 5366, 5368, 5370, 5372, 5374, 5376, 5378, 5380, 5382, 5384, 5386, 5388, 5390, 5392, 5394, 5396, 5398, 5400, 5402, 5404, 5406, 5408, 5410, 5412, 5414, 5416, 5418, 5420, 5422, 5424, 5426, 5428, 5430, 5432, 5434, 5436, 5438, 5440, 5442, 5444, 5446, 5448, 5450, 5452, 5454, 5456, 5458, 5460, 5462, 5464, 5466, 5468, 5470, 5472, 5474, 5476, 5478, 5480, 5482, 5484, 5486, 5488, 5490, 5492, 5494, 5496, 5498, 5500, 5502, 5504, 5506, 5508, 5510, 5512, 5514, 5516, 5518, 5520, 5522, 5524, 5526, 5528, 5530, 5532, 5534, 5536, 5538, 5540, 5542, 5544, 5546, 5548, 5550, 5552, 5554, 5556, 5558, 5560, 5562, 5564, 5566, 5568, 5570, 5572, 5574, 5576, 5578, 5580, 5582, 5584, 5586, 5588, 5590, 5592, 5594, 5596, 5598, 5600, 5602, 5604, 5606, 5608, 5610, 5612, 5614, 5616, 5618, 5620, 5622, 5624, 5626, 5628, 5630, 5632, 5634, 5636, 5638, 5640, 5642, 5644, 5646, 5648, 5650, 5652, 5654, 5656, 5658, 5660, 5662, 5664, 5666, 5668, 5670, 5672, 5674, 5676, 5678, 5680, 5682, 5684, 5686, 5688, 5690, 5692, 5694, 5696, 5698, 5700, 5702, 5704, 5706, 5708, 5710, 5712, 5714, 5716, 5718, 5720, 5722, 5724, 5726, 5728, 5730, 5732, 5734, 5736, 5738, 5740, 5742, 5744, 5746, 5748, 5750, 5752, 5754, 5756, 5758, 5760, 5762, 5764, 5766, 5768, 5770, 5772, 5774, 5776, 5778, 5780, 5782, 5784, 5786, 5788, 5790, 5792, 5794, 5796, 5798, 5800, 5802, 5804, 5806, 5808, 5810, 5812, 5814, 5816, 5818, 5820, 5822, 5824, 5826, 5828, 5830, 5832, 5834, 5836, 5838, 5840, 5842, 5844, 5846, 5848, 5850, 5852, 5854, 5856, 5858, 5860, 5862, 5864, 5866, 5868, 5870, 5872, 5874, 5876, 5878, 5880, 5882, 5884, 5886, 5888, 5890, 5892, 5894, 5896, 5898, 5900, 5902, 5904, 5906, 5908, 5910, 5912, 5914, 5916, 5918, 5920, 5922, 5924, 5926, 5928, 5930, 5932, 5934, 5936, 5938, 5940, 5942, 5944, 5946, 5948, 5950, 5952, 5954, 5956, 5958, 5960, 5962, 5964, 5966, 5968, 5970, 5972, 5974, 5976, 5978, 5980, 5982, 5984, 5986, 5988, 5990, 5992, 5994, 5996, 5998, 6000, 6002, 6004, 6006, 6008, 6010, 6012, 6014, 6016, 6018, 6020, 6022, 6024, 6026, 6028, 6030, 6032, 6034, 6036, 6038, 6040, 6042, 6044, 6046, 6048, 6050, 6052, 6054, 6056, 6058, 6060, 6062, 6064, 6066, 6068, 6070, 6072, 6074, 6076, 6078, 6080, 6082, 6084, 6086, 6088, 6090, 6092, 6094, 6096, 6098, 6100, 6102, 6104, 6106, 6108, 6110, 6112, 6114, 6116, 6118, 6120, 6122, 6124, 6126, 6128, 6130, 6132, 6134, 6136, 6138, 6140, 6142, 6144, 6146, 6148, 6150, 6152, 6154, 6156, 6158, 6160, 6162, 6164, 6166, 6168, 6170, 6172, 6174, 6176, 6178, 6180, 6182, 6184, 6186, 6188, 6190, 6192, 6194, 6196, 6198, 6200, 6202, 6204, 6206, 6208, 6210, 6212, 6214, 6216, 6218, 6220, 6222, 6224, 6226, 6228, 6230, 6232, 6234, 6236, 6238, 6240, 6242, 6244, 6246, 6248, 6250, 6252, 6254, 6256, 6258, 6260, 6262, 6264, 6266, 6268, 6270, 6272, 6274, 6276, 6278, 6280, 6282, 6284, 6286, 6288, 6290, 6292, 6294, 6296, 6298, 6300, 6302, 6304, 6306, 6308, 6310, 6312, 6314, 6316, 6318, 6320, 6322, 6324, 6326, 6328, 6330, 6332, 6334, 6336, 6338, 6340, 6342, 6344, 6346, 6348, 6350, 6352, 6354, 6356, 6358, 6360, 6362, 6364, 6366, 6368, 6370, 6372, 6374, 6376, 6378, 6380, 6382, 6384, 6386, 6388, 6390, 63

6832, 6834, 6836, 6838, 6840, 6842, 6844, 6846, 6848, 6850, 6852, 6854, 6856, 6858, 6860, 6862, 6864, 6866, 6868, 6870, 6872, 6874, 6876, 6878, 6880, 6882, 6884, 6886, 6888, 6890, 6892, 6894, 6896, 6898, 6900, 6902, 6904, 6906, 6908, 6910, 6912, 6914, 6916, 6918, 6920, 6922, 6924, 6926, 6928, 6930, 6932, 6934, 6936, 6938, 6940, 6942, 6944, 6946, 6948, 6950, 6952, 6954, 6956, 6958, 6960, 6962, 6964, 6966, 6968, 6970, 6972, 6974, 6976, 6978, 6980, 6982, 6984, 6986, 6988, 6990, 6992, 6994, 6996, 6998, 7000, 7002, 7004, 7006, 7008, 7010, 7012, 7014, 7016, 7018, 7020, 7022, 7024, 7026, 7028, 7030, 7032, 7034, 7036, 7038, 7040, 7042, 7044, 7046, 7048, 7050, 7052, 7054, 7056, 7058, 7060, 7062, 7064, 7066, 7068, 7070, 7072, 7074, 7076, 7078, 7080, 7082, 7084, 7086, 7088, 7090, 7092, 7094, 7096, 7098, 7100, 7102, 7104, 7106, 7108, 7110, 7112, 7114, 7116, 7118, 7120, 7122, 7124, 7126, 7128, 7130, 7132, 7134, 7136, 7138, 7140, 7142, 7144, 7146, 7148, 7150, 7152, 7154, 7156, 7158, 7160, 7162, 7164, 7166, 7168, 7170, 7172, 7174, 7176, 7178, 7180, 7182, 7184, 7186, 7188, 7190, 7192, 7194, 7196, 7198, 7200, 7202, 7204, 7206, 7208, 7210, 7212, 7214, 7216, 7218, 7220, 7222, 7224, 7226, 7228, 7230, 7232, 7234, 7236, 7238, 7240, 7242, 7244, 7246, 7248, 7250, 7252, 7254, 7256, 7258, 7260, 7262, 7264, 7266, 7268, 7270, 7272, 7274, 7276, 7278, 7280, 7282, 7284, 7286, 7288, 7290, 7292, 7294, 7296, 7298, 7300, 7302, 7304, 7306, 7308, 7310, 7312, 7314, 7316, 7318, 7320, 7322, 7324, 7326, 7328, 7330, 7332, 7334, 7336, 7338, 7340, 7342, 7344, 7346, 7348, 7350, 7352, 7354, 7356, 7358, 7360, 7362, 7364, 7366, 7368, 7370, 7372, 7374, 7376, 7378, 7380, 7382, 7384, 7386, 7388, 7390, 7392, 7394, 7396, 7398, 7400, 7402, 7404, 7406, 7408, 7410, 7412, 7414, 7416, 7418, 7420, 7422, 7424, 7426, 7428, 7430, 7432, 7434, 7436, 7766, 7768, 7770, 7772, 7774, 7776, 7778, 7780, 7782, 7784, 7786, 7788, 7790, 7792, 7794, 7796, 7798, 7800, 7812, 7814, 7816, 7818, 7820, 7822, 7824, 7826, 7828, 7830, 7832, 7834, 7836, 7838, 7840, 7842, 7844, 7846, 7848, 7850, 7852, 7854, 7856, 7858, 7860, 7862, 7864, 7866, 7868, 7870, 7872, 7874, 7876, 7878, 7880, 7882, 7884, 7886, 7888, 7890, 7892, 7894, 7896, 7898, 7900, 7902, 7904, 7906, 7908, 7910, 7912, 7914, 7916, 7918, 7920, 7922, 7924, 7926, 7928, 7930, 7932, 7934, 7936, 7938, 7940, 7942, 7944, 7946, 7948, 7950, 7952, 7954, 7956, 7958, 7960, 7962, 7964, 7966, 7968, 7970, 7972, 7974, 7976, 7978, 7980, 7982, 7984, 7986, 7988, 7990, 7992, 7994, 7996, 7998, 8000, 8002, 8004, 8006, 8008, 8010, 8012, 8014, 8016, 8018, 8020, 8022, 8024, 8026, 8028, 8030, 8032, 8034, 8036, 8038, 8040, 8042, 8044, 8046, 8048, 8050, 8052, 8054, 8056, 8058, 8060, 8062, 8064, 8066, 8068, 8070, 8072, 8074, 8076, 8078, 8080, 8082, 8084, 8086, 8088, 8090, 8092, 8094, 8096, 8098, 8100, 8102, 8104, 8106, 8108, 8110, 8112, 8114, 8116, 8118, 8120, 8122, 8124, 8126, 8128, 8130, 8132, 8134, 8136, 8138, 8140, 8142, 8144, 8146, 8148, 8150, 8152, 8154, 8156, 8158, 8160, 8162, 8164, 8166, 8168, 8170, 8172, 8174, 8176, 8178, 8180, 8182, 8184, 8186, 8188, 8190, 8192, 8194, 8196, 8198, 8200, 8202, 8204, 8206, 8208, 8210, 8212, 8214, 8216, 8218, 8220, 8222, 8224, 8226, 8228, 8230, 8232, 8234, 8236, 8238, 8240, 8242, 8244, 8246, 8248, 8250, 8252, 8254, 8256, 8258, 8260, 8262, 8264, 8266, 8268, 8270, 8272, 8274, 8276, 8278, 8280, 8282, 8284, 8286, 8288, 8290, 8292, 8294, 8296, 8298, 8300, 8302, 8304, 8306, 8308, 8310, 8312, 8314, 8316, 8318, 8320, 8322, 8324, 8326, 8328, 8330, 8332, 8334, 8336, 8338, 8340, 8342, 8344, 8346, 8348, 8350, 8352, 8354, 8356, 8358, 8360, 8362, 8364, 8366, 8368, 8482, 8484, 8486, 8488, 8490, 8492, 8494, 8496, 8498, 8500, 8502, 8504, 8506, 8508, 8510, 8512, 8514, 8516, 8518, 8520, 8522, 8524, 8526, 8528, 8530, 8532, 8534, 8536, 8538, 8540, 8542, 8544, 8546, 8548, 8550, 8552, 8554, 8556, 8558, 8560, 8562, 8564, 8566, 8568, 8570, 8572, 8574, 8576, 8578, 8580, 8582, 8584, 8586, 8588, 8590, 8592, 8594, 8596, 8598, 8600, 8602, 8604, 8606, 8608, 8610, 8612, 8614, 8616, 8618, 8620, 8622, 8624, 8626, 8628, 8630, 8632, 8634, 8636, 8638, 8640, 8642, 8644, 8646, 8648, 8650, 8652, 8654, 8656, 8658, 8660, 8662, 8664, 8666, 8668, 8670, 8672, 8674, 8676, 8678, 8680, 8682, 8684, 8686, 8688, 8690, 8692, 8694, 8696, 8698, 8700, 8702, 8704, 8706, 8708, 8710, 8712, 8714, 8716, 8718, 8720, 8722, 8724, 8726, 8728, 8730, 8732, 8734, 8736, 8738, 8740, 8742, 8744, 8746, 8748, 8750, 8752, 8754, 8756, 8758, 8760, 8762, 8764, 8766, 8768, 8770, 8772, 8774, 8776, 8778, 8780, 8782, 8784, 8786, 8788, 8790, 8792, 8794, 8796, 9108, 9110, 9112, 9114, 9116, 9118, 9120, 9122, 9124, 9126, 9128, 9130, 9132, 9134, 9136, 9138, 9140, 9142, 9144, 9146, 9148, 9150, 9152, 9154, 9156, 9158, 9160, 9162, 9164, 9166, 9168, 9170, 9172, 9174, 9176, 9178, 9180, 9182, 9184, 9186, 9188, 9190, 9192, 9194, 9196, 9198, 9200, 9202, 9204, 9206, 9208, 9210, 9212, 9214, 9216, 9218, 9220, 9222, 9224, 9226, 9228, 9230, 9232, 9234, 9236, 9238, and/or 9240 finds use.

(97) In the methods provided herein, at least one engineered glycosyltransferase comprising polypeptide sequence that has at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NO: 1080, 1082, 1084, 1086, 1088, 1090, 1092, 1094, 1096, 1098, 1100, 1102, 1104, 1106, 1108, 1110, 1112, 1114, 1116, 1118, 1120, 1122, 1124, 1126, 1128, 1130, 1132, 1134, 1136, 1138,

1140, 1142, 1144, 1146, 1148, 1150, 1152, 1154, 1156, 1158, 1160, 1162, 1164, 1166, 1168, 1170, 1172, 1174, 1176, 1178, 1180, 1182, 1184, 1186, 1188, 1190, 1192, 1194, 1196, 1198, 1200, 1202, 1204, 1206, 1208, 1210, 1212, 1214, 1216, 1218, 1220, 1222, 1224, 1226, 1228, 1230, 1232, 1234, 1236, 1238, 1240, 1242, 1244, 1246, 1248, 1250, 1252, 1254, 1256, 1258, 1260, 1262, 1264, 1266, 1268, 1270, 1272, 1274, 1276, 1278, 1280, 1282, 1284, 1286, 1288, 1296, 1298, 1300, 1302, 1304, 1306, 1308, 1310, 1312, 1314, 1316, 1318, 1320, 1322, 1324, 1326, 1328, 1330, 1332, 1334, 1336, 1338, 1340, 1342, 1344, 1346, 1348, 1350, 1352, 1354, 1356, 1358, 1360, 1362, 1364, 1366, 1368, 1370, 1372, 1374, 1376, 1378, 1380, 1382, 1384, 1386, 1388, 1390, 1392, 1394, 1396, 1398, 1400, 1402, 1404, 1406, 1408, 1410, 1412, 1414, 1416, 1418, 1420, 1422, 1424, 1426, 1428, 1430, 1432, 1434, 1436, 1438, 1440, 1442, 1444, 1446, 1448, 1450, 1452, 1454, 1456, 1458, 1460, 1462, 1464, 1466, 1468, 1470, 1472, 1474, 1476, 1478, 1480, 1482, 1484, 1486, 1488, 1490, 1492, 1494, 1496, 1498, 1500, 1502, 1504, 1506, 1508, 1510, 1512, 1514, 1516, 1518, 1520, 1522, 1524, 1526, 1528, 1530, 1532, 1534, 1536, 1538, 1540, 1542, 1544, 1546, 1548, 1550, 1552, 1554, 1556, 1558, 1560, 1562, 1564, 1566, 1568, 1570, 1572, 1574, 1576, 1578, 1580, 1582, 1584, 1586, 1588, 1590, 1592, 1594, 1596, 1598, 1600, 1602, 1604, 1606, 1608, 1610, 1612, 1614, 1616, 1618, 1620, 1622, 1624, 1626, 1628, 1630, 1632, 1634, 1636, 1638, 1640, 1642, 1644, 1646, 1648, 1650, 1652, 1654, 1656, 1658, 1660, 1662, 1664, 1666, 1668, 1670, 1672, 1674, 1676, 1678, 1680, 1682, 1684, 1686, 1688, 1690, 1692, 1694, 1696, 1698, 1700, 1702, 1704, 1706, 1708, 1710, 1712, 1714, 1716, 1718, 1720, 1722, 1724, 1726, 1728, 1730, 1732, 1734, 1736, 1738, 1740, 1742, 1744, 1746, 1748, 1750, 1752, 1754, 1756, 1758, 1760, 1762, 1764, 1766, 1768, 1770, 1772, 1774, 1776, 1778, 1780, 1782, 1784, 1786, 1788, 1790, 1792, 1794, 1796, 1798, 1800, 1802, 1804, 1806, 1808, 1810, 1812, 1814, 1816, 1818, 1820, 1822, 1824, 1826, 1828, 1830, 1832, 1834, 1836, 1838, 1840, 1842, 1844, 1846, 1848, 1850, 1852, 1854, 1856, 1858, 1860, 1862, 1864, 1866, 1868, 1870, 1872, 1874, 1876, 1878, 1880, 1882, 1884, 1886, 1888, 1890, 1892, 1894, 1896, 1898, 1900, 1902, 1904, 1906, 1908, 1910, 1912, 1914, 1916, 1918, 1920, 1922, 1924, 1926, 1928, 1930, 1932, 1934, 1936, 1938, 1940, 1942, 1944, 1946, 1948, 1950, 1952, 1954, 1956, 1958, 1960, 1962, 1964, 1966, 1968, 1970, 1972, 1974, 1976, 1978, 1980, 1982, 1984, 1986, 1988, 1990, 1992, 1994, 1996, 1998, 2000, 2002, 2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022, 2024, 2026, 2028, 2030, 2032, 2034, 2036, 2038, 2040, 2042, 2044, 2046, 2048, 2050, 2052, 2054, 2056, 2058, 2060, 2062, 2064, 2066, 2068, 2070, 2072, 2074, 2076, 2078, 2080, 2082, 2084, 2086, 2088, 2090, 2092, 2094, 2096, 2098, 2100, 2102, 2104, 2106, 2108, 2110, 2112, 2114, 2116, 2118, 2120, 2122, 2124, 2126, 2128, 2130, 2132, 2134, 2136, 2138, 2140, 2142, 2144, 2146, 2148, 2150, 2152, 2154, 2156, 2158, 2160, 2162, 2164, 2166, 2168, 2170, 2172, 2174, 2176, 2178, 2180, 2182, 2184, 2186, 2188, 2190, 2192, 2194, 2196, 2198, 2200, 2202, 2204, 2206, 2208, 2210, 2212, 2214, 2216, 2218, 2220, 2222, 2224, 2226, 2228, 2230, 2232, 2234, 2236, 2238, 2240, 2242, 2244, 2246, 2248, 2250, 2252, 2254, 2256, 2258, 2260, 2262, 2264, 2266, 2268, 2270, 2272, 2274, 2276, 2278, 2280, 2282, 2284, 2286, 2288, 2290, 2292, 2294, 2296, 2298, 2300, 2302, 2304, 2306, 2308, 2310, 2312, 2314, 2316, 2318, 2320, 2322, 2324, 2326, 2328, 2330, 2332, 2334, 2336, 2338, 2340, 2342, 2344, 2346, 2348, 2350, 2352, 2354, 2356, 2358, 2360, 2362, 2364, 2366, 2368, 2370, 2372, 2374, 2376, 2378, 2380, 2382, 2384, 2386, 2388, 2390, 2392, 2394, 2396, 2398, 2400, 2402, 2404, 2406, 2408, 2410, 2412, 2414, 2416, 2418, 2420, 2422, 2424, 2426, 2428, 2430, 2432, 2434, 2436, 2438, 2440, 2442, 2444, 2446, 2448, 2450, 2452, 2454, 2456, 2458, 2460, 2462, 2464, 2466, 2468, 2470, 2472, 2474, 2476, 2478, 2480, 2482, 2484, 2486, 2488, 2490, 2492, 2494, 2496, 2498, 2500, 2502, 2504, 2506, 2508, 25

8396, 8398, 8400, 8402, 8404, 8406, 8408, 8410, 8412, 8414, 8416, 8418, 8420, 8422, 8424, 8426, 8428, 8430, 8432, 8434, 8436, 8438, 8440, 8442, 8444, 8446, 8448, 8450, 8452, 8454, 8456, 8458, 8460, 8462, 8464, 8466, 8468, 8470, 8472, 8474, 8476, 8478, 8480, 8482, 8484, 8486, 8488, 8490, 8492, 8494, 8496, 8498, 8500, 8502, 8504, 8506, 8508, 8510, 8512, 8514, 8516, 8518, 8520, 8522, 8524, 8526, 8528, 8530, 8532, 8534, 8536, 8538, 8540, 8542, 8544, 8546, 8548, 8550, 8552, 8554, 8556, 8558, 8560, 8562, 8564, 8566, 8568, 8570, 8572, 8574, 8576, 8578, 8580, 8582, 8584, 8586, 8588, 8590, 8592, 8594, 8596, 8598, 8600, 8602, 8604, 8606, 8608, 8610, 8612, 8614, 8616, 8618, 8620, 8622, 8624, 8626, 8628, 8630, 8632, 8634, 8636, 8638, 8640, 8642, 8644, 8646, 8648, 8650, 8652, 8654, 8656, 8658, 8660, 8662, 8664, 8666, 8668, 8670, 8672, 8674, 8676, 8678, 8680, 8682, 8684, 8686, 8688, 8690, 8692, 8694, 8696, 8698, 8700, 8702, 8704, 8706, 8708, 8710, 8712, 8714, 8716, 8718, 8720, 8722, 8724, 8726, 8728, 8730, 8732, 8734, 8736, 8738, 8740, 8742, 8744, 8746, 8748, 8750, 8752, 8754, 8756, 8758, 8760, 8762, 8764, 8766, 8768, 8770, 8772, 8774, 8776, 8778, 8780, 8782, 8784, 8786, 8788, 8790, 8792, 8794, 8796, 8798, 8800, 8802, 8804, 8806, 8808, 8810, 8812, 8814, 8816, 8818, 8820, 8822, 8824, 8826, 8828, 8830, 8832, 8834, 8836, 8838, 8840, 8842, 8844, 8846, 8848, 8850, 8852, 8854, 8856, 8858, 8860, 8862, 8864, 8866, 8868, 8870, 8872, 8874, 8876, 8878, 8880, 8882, 8884, 8886, 8888, 8890, 8892, 8894, 8896, 8898, 8900, 8902, 8904, 8906, 8908, 8910, 8912, 8914, 8916, 8918, 8920, 8922, 8924, 8926, 8928, 8930, 8932, 8934, 8936, 8938, 8940, 8942, 8944, 8946, 8948, 8950, 8952, 8954, 8956, 8958, 8960, 8962, 8964, 8966, 8968, 8970, 8972, 8974, 8976, 8978, 8980, 8982, 8984, 8986, 8988, 8990, 8992, 8994, 8996, 8998, 9000, 9002, 9004, 9006, 9008, 9010, 9012, 9014, 9016, 9018, 9020, 9022, 9024, 9026, 9028, 9030, 9032, 9034, 9036, 9038, 9040, 9042, 9044, 9046, 9048, 9050, 9052, 9054, 9056, 9058, 9060, 9062, 9064, 9066, 9068, 9070, 9072, 9074, 9076, 9078, 9080, 9082, 9084, 9086, 9088, 9090, 9092, 9094, 9096, 9098, 9100, 9102, 9104, and/or 9106 finds use.

(98) The present invention also provides rebaudiosides and compositions comprising the rebaudiosides produced according to the methods provided herein. In some embodiments, the rebaudioside is rebaudioside M, while in some alternative embodiments, the rebaudioside is rebaudioside A, and in still further embodiments, the rebaudioside is rebaudioside I, and in some additional embodiments, the rebaudioside is rebaudioside D. In some further embodiments, the present invention provides a mixture of rebaudiosides produced according to the methods provided herein, including mixtures of rebaudioside M, rebaudioside A, rebaudioside I, and/or rebaudioside D, in any combination of concentrations of rebaudiosides and other components of interest. In some embodiments, the present invention provides compositions comprising mixtures of rebaudioside M, rebaudioside A, rebaudioside I, and/or rebaudioside D, in any combination of concentrations of rebaudiosides and other components of interest. Indeed, it is not intended that the present invention be limited to any particular combination or mixture of rebaudioside(s) produced according to the methods of the present invention.

(99) The present invention provides methods for producing rebaudioside M, comprising providing a rebaudioside D and/or rebaudioside I substrate, NDP-glucose, and a least one engineered NDP-glycosyltransferase provided herein, combining the rebaudioside D and rebaudioside I substrate, NDP-glucose and an NDP-glycosyltransferase under conditions such that rebaudioside M is produced.

(100) The present invention also provides methods for producing rebaudioside M, comprising providing a rebaudioside D substrate, NDP-glucose, and a least one engineered NDP-glycosyltransferase provided herein, combining the rebaudioside D substrate, NDP-glucose and an NDP-glycosyltransferase under conditions such that rebaudioside M is produced.

(101) The present invention also provides methods for producing rebaudioside M, comprising providing a rebaudioside I substrate, NDP-glucose, and a least one engineered NDP-glycosyltransferase provided herein, combining the rebaudioside I substrate, NDP-glucose and an NDP-glycosyltransferase under conditions such that rebaudioside M is produced.

(102) The present invention also provides methods for producing rebaudioside A and/or rebaudioside I, comprising providing a stevioside substrate, NDP-glucose, and at least one engineered NDP-glycosyltransferase provided herein, combining the stevioside substrate, NDP-glucose and an NDP-glycosyltransferase under conditions such that rebaudioside A and/or rebaudioside I is produced.

(103) The present invention also provides methods for producing rebaudioside A, comprising providing a stevioside substrate, NDP-glucose, and at least one engineered NDP-glycosyltransferase provided herein, combining the stevioside substrate, NDP-glucose and an NDP-glycosyltransferase under conditions such that rebaudioside A is produced.

(104) The present invention also provides methods for producing rebaudioside I, comprising providing a stevioside substrate, NDP-glucose, and at least one engineered NDP-glycosyltransferase provided herein, combining the stevioside substrate, NDP-glucose and an NDP-glycosyltransferase under conditions such that rebaudioside I is produced.

(105) The present invention also provides methods for producing rebaudioside D, comprising providing a stevioside substrate, NDP-glucose, and at least one engineered NDP-glycosyltransferase provided herein, combining the stevioside substrate, NDP-glucose and an NDP-glycosyltransferase under conditions such that rebaudioside D is produced.

(106) The present invention also provides methods for producing rebaudioside M comprising providing a

rebaudioside D and/or rebaudioside I substrate, ADP-glucose, and at least one engineered ADP-glycosyltransferase provided herein, combining the rebaudioside D and/or rebaudioside I substrate, ADP-glucose and an ADP-glycosyltransferase under conditions such that rebaudioside M is produced.

(107) The present invention also provides methods for producing rebaudioside M comprising providing a rebaudioside D substrate, ADP-glucose, and at least one engineered ADP-glycosyltransferase provided herein, combining the rebaudioside D substrate, ADP-glucose and an ADP-glycosyltransferase under conditions such that rebaudioside M is produced.

(108) The present invention also provides methods for producing rebaudioside M comprising providing a rebaudioside I substrate, ADP-glucose, and at least one engineered ADP-glycosyltransferase provided herein, combining the rebaudioside I substrate, ADP-glucose and an ADP-glycosyltransferase under conditions such that rebaudioside M is produced.

(109) The present invention also provides methods for producing rebaudioside A and/or rebaudioside I, comprising providing a stevioside substrate, ADP-glucose, and at least one engineered ADP-glycosyltransferase provided herein, combining the stevioside substrate, ADP-glucose and an ADP-glycosyltransferase under conditions such that rebaudioside A and/or rebaudioside I is produced.

(110) The present invention also provides methods for producing rebaudioside A, comprising providing a stevioside substrate, ADP-glucose, and at least one engineered ADP-glycosyltransferase provided herein, combining the stevioside substrate, ADP-glucose and an ADP-glycosyltransferase under conditions such that rebaudioside A is produced.

(111) The present invention also provides methods for producing rebaudioside I, comprising providing a stevioside substrate, ADP-glucose, and at least one engineered ADP-glycosyltransferase provided herein, combining the stevioside substrate, ADP-glucose and an ADP-glycosyltransferase under conditions such that rebaudioside I is produced.

(112) The present invention also provides methods for producing rebaudioside D, comprising providing a stevioside substrate, ADP-glucose, and at least one engineered ADP-glycosyltransferase provided herein, and/or 118, combining the stevioside substrate, ADP-glucose and an ADP-glycosyltransferase under conditions such that rebaudioside D is produced.

(113) The present invention also provides methods for producing rebaudioside M comprising providing a rebaudioside D substrate, NDP, sucrose, a sucrose synthase, and at least one engineered ADP-glycosyltransferase provided herein, combining the rebaudioside D substrate, NDP, sucrose, sucrose synthase and an ADP-glycosyltransferase under conditions such that rebaudioside M is produced. In some embodiments, the sucrose synthase is an engineered sucrose synthase provided herein. In some embodiments, a sugar other than sucrose finds use, in combination with the appropriate synthase.

(114) The present invention also provides methods for producing rebaudioside A and/or rebaudioside I comprising providing a stevioside substrate, NDP, sucrose, a sucrose synthase, and at least one engineered ADP-glycosyltransferase provided herein, combining the stevioside substrate, NDP, sucrose, sucrose synthase and an ADP-glycosyltransferase under conditions such that rebaudioside A and/or rebaudioside I is produced. In some embodiments, the sucrose synthase is an engineered sucrose synthase provided herein. In some embodiments, a sugar other than sucrose finds use, in combination with the appropriate synthase.

(115) The present invention also provides methods for producing rebaudioside A, comprising providing a stevioside substrate, NDP, sucrose, a sucrose synthase, and at least one engineered ADP-glycosyltransferase provided herein, combining the stevioside substrate, NDP, sucrose, sucrose synthase and an ADP-glycosyltransferase under conditions such that rebaudioside A is produced. In some embodiments, the sucrose synthase is an engineered sucrose synthase provided herein. In some embodiments, a sugar other than sucrose finds use, in combination with the appropriate synthase.

(116) The present invention also provides methods for producing rebaudioside I, comprising providing a stevioside substrate, NDP, sucrose, a sucrose synthase, and at least one engineered ADP-glycosyltransferase provided herein, combining the stevioside substrate, NDP, sucrose, sucrose synthase and an ADP-glycosyltransferase under conditions such that rebaudioside A and/or rebaudioside I is produced. In some embodiments, the sucrose synthase is an engineered sucrose synthase provided herein. In some embodiments, a sugar other than sucrose finds use, in combination with the appropriate synthase.

(117) The present invention also provides methods for producing rebaudioside D, comprising providing a

stevioside substrate, NDP, sucrose, a sucrose synthase, and at least one engineered ADP-glycosyltransferase provided herein, combining the stevioside substrate, NDP, sucrose, sucrose synthase and an ADP-glycosyltransferase under conditions such that rebaudioside D is produced. In some embodiments, sucrose synthase is an engineered sucrose synthase provided herein. In some embodiments, a sugar other than sucrose finds use, in combination with the appropriate synthase.

(118) The present invention also provides methods for producing rebaudioside M, comprising providing a stevioside substrate comprising at least one stevioside and/or a mixture of steviosides and rebA, NDP, sucrose, a sucrose synthase, and at least one engineered ADP-glycosyltransferase provided herein, combining the stevioside substrate, NDP, sucrose, sucrose synthase and an ADP-glycosyltransferase under conditions such that rebaudioside D is produced. In some embodiments, the sucrose synthase is an engineered sucrose synthase provided herein. In some embodiments, a sugar other than sucrose finds use, in combination with the appropriate synthase.

(119) The present invention also provides methods of producing rebaudioside M, comprising providing a stevioside substrate, NDP, sucrose, at least one sucrose synthase, and at least one engineered ADP-glycosyltransferase provided herein, combining the stevioside substrate, NDP, and an ADP-glycosyltransferase under conditions such that rebaudioside A is first produced, rebaudioside D and/or rebaudioside I is then produced, and rebaudioside M finally produced. In some embodiments, the methods further comprise sucrose and sucrose synthase. In some embodiments, the sucrose synthase is an engineered sucrose synthase provided herein.

(120) In some embodiments, the methods provided by the present invention are conducted as one-pot reaction, while in some alternative embodiments, the methods are conducted in multiple reaction vessels. In some embodiments, the methods are conducted in single and/or multiple reaction vessels in a sequential manner. In some embodiments, the method steps are repeated (i.e., there are multiple iterations of some or all of the steps of the methods). In some embodiments, the sucrose is recycled during repeated steps. In some additional embodiments, engineered glycosyltransferase and/or other reaction components (e.g., co-factors) are recycled. In some embodiments of the methods, the stevioside substrate is extracted from *Stevia rebaudiana*, while in some alternative methods, the stevioside substrate is synthetically produced. In some further embodiments, the glycosyltransferase is immobilized. In some additional embodiments, the sucrose synthase is immobilized. In some further embodiments, the glycosyltransferase and/or the sucrose synthase is immobilized. In some embodiments of the methods, fructose is produced. In some additional embodiments, fructose is removed from the reaction products.

(121) The present invention also provides compositions comprising at least one engineered glycosyltransferase variant provided herein. The present invention also provides compositions comprising at least one non-naturally occurring glycosyltransferase variant as provided herein.

(122) The present invention also provides methods for glycosylation of a substrate to produce a beta-glycosylated product, comprising the steps of: providing at least one glycosyl group donor, a least one glycosyl group acceptor, and at least one glycosyltransferase enzyme; contacting the glycosyl group donor and glycosyl group acceptor with the glycosyltransferase enzyme under conditions such that the glycosyl group acceptor is glycosylated to produce at least one product having beta-glucose linkages. In some embodiments of the methods, the glycosyl group donor is a nucleotide diphosphate sugar, for example adenine diphosphoglucose (ADP-glucose). In some further embodiments of the methods, the glycosyl group acceptor is selected from glycosyl, alkoxy, carboxy, aminocarbonyl, heteroalkyl, heteroalkenyl, heteroalkynyl, carboxyalkyl, aminoalkyl, haloalkyl, alkylthioalkyl, heterocycloalkyl, heteroaryl, and heteroarylalkyl groups. In some yet additional embodiments of the methods, the product having beta-glucose linkages is a steviol glycoside. In some further embodiments of the methods, the glycosyl group acceptor is rebaudioside D, the glycosyl group donor is ADP-glucose, and the product having beta-glucose linkages is rebaudioside M. In some further embodiments of the methods, the glycosyl group acceptor is stevioside, the glycosyl group donor is ADP-glucose, and the product having beta-glucose linkages is rebaudioside A or rebaudioside I.

(123) The present invention also provides methods for production of nucleoside diphosphoglucose, comprising the steps of: providing a nucleoside diphosphate-dependent synthase, a nucleoside diphosphate, and a disaccharide, trisaccharide, or oligosaccharide substrate of the synthase; contacting the synthase, nucleoside diphosphate, and saccharide under conditions such that the saccharide is cleaved to

produce a lower molecular weight saccharide and nucleoside diphosphoglucose. In some embodiments of the method, this method is combined with the previously described method. In some additional embodiments of the method, the nucleoside diphosphate is ADP, the nucleoside diphosphoglucose is ADP-glucose, and the synthase substrate is sucrose.

Description

DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 provides an enzymatic reaction scheme in which a glycosyltransferase catalyzes the transfer of a glucosyl group from a nucleoside diphosphoglucose (NDP-glucose), for example ADP-glucose, to an acceptor, for example R—OH, where R is any glycosyl, alkoxy, carboxy, aminocarbonyl, heteroalkyl, heteroalkenyl, heteroalkynyl, carboxyalkyl, aminoalkyl, haloalkyl, alkylthioalkyl, heterocycloalkyl, heteroaryl, or heteroarylalkyl group. In a further embodiment, R—OH is a stevioside or rebaudioside D, and the product is rebaudioside A, rebaudioside I, or rebaudioside M. A nucleoside diphosphate dependent synthase catalyzes the transfer of a glucosyl group from a glucose donor (e.g., sucrose), to a nucleoside diphosphate, regenerating NDP-glucose and releasing a byproduct (e.g., fructose).

(2) FIG. 2 provides the structure of rebaudioside M with the carbons numbered.

(3) FIG. 3 provides the structure of rebaudioside I with the carbons numbered.

DESCRIPTION OF THE INVENTION

(4) The present invention provides engineered glycosyltransferase (GT) enzymes, polypeptides having GT activity, and polynucleotides encoding these enzymes, as well as vectors and host cells comprising these polynucleotides and polypeptides. The present invention provides engineered sucrose synthase (SuS) enzymes, polypeptides having SuS activity, and polynucleotides encoding these enzymes, as well as vectors and host cells comprising these polynucleotides and polypeptides. The present invention also provides compositions comprising the GT enzymes and methods of using the engineered GT enzymes to make products with β -glucose linkages. The present invention further provides compositions and methods for the production of rebaudiosides (e.g., rebaudioside M, rebaudioside A, rebaudioside I, and rebaudioside D). The present invention also provides compositions comprising the SuS enzymes and methods of using them. Methods for producing GT and SuS enzymes are also provided.

(5) Unless defined otherwise, all technical and scientific terms used herein generally have the same meaning as commonly understood by one of ordinary skill in the art to which this invention pertains. Generally, the nomenclature used herein and the laboratory procedures of cell culture, molecular genetics, microbiology, organic chemistry, analytical chemistry and nucleic acid chemistry described below are those well-known and commonly employed in the art. Such techniques are well-known and described in numerous texts and reference works well known to those of skill in the art. Standard techniques, or modifications thereof, are used for chemical syntheses and chemical analyses. All patents, patent applications, articles and publications mentioned herein, both supra and infra, are hereby expressly incorporated herein by reference.

(6) Although any suitable methods and materials similar or equivalent to those described herein find use in the practice of the present invention, some methods and materials are described herein. It is to be understood that this invention is not limited to the particular methodology, protocols, and reagents described, as these may vary, depending upon the context they are used by those of skill in the art. Accordingly, the terms defined immediately below are more fully described by reference to the invention as a whole.

(7) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the present invention. The section headings used herein are for organizational purposes only and not to be construed as limiting the subject matter described. Numeric ranges are inclusive of the numbers defining the range. Thus, every numerical range disclosed herein is intended to encompass every narrower numerical range that falls within such broader numerical range, as if such narrower numerical ranges were all expressly written herein. It is also intended that every maximum (or minimum) numerical limitation disclosed herein includes every lower (or higher) numerical limitation, as if such lower (or higher) numerical limitations were expressly written herein.

Abbreviations

(8) The abbreviations used for the genetically encoded amino acids are conventional and are as follows:

(9) TABLE-US-00001 Three-Letter One-Letter Amino Acid Abbreviation
Alanine Ala A
Arginine Arg R
Asparagine Asn N
Aspartate Asp D
Cysteine Cys C
Glutamate Glu E
Glutamine Gln Q
Glycine Gly G
Histidine His H
Isoleucine Ile I
Leucine Leu L
Lysine Lys K
Methionine Met M
Phenylalanine Phe F
Proline Pro P
Serine Ser S
Threonine Thr T
Tryptophan Trp W
Tyrosine Tyr Y
Valine Val V

(10) When the three-letter abbreviations are used, unless specifically preceded by an “L” or a “D” or clear from the context in which the abbreviation is used, the amino acid may be in either the L- or D-configuration about α -carbon (C.sub. α). For example, whereas “Ala” designates alanine without specifying the configuration about the α -carbon, “D-Ala” and “L-Ala” designate D-alanine and L-alanine, respectively. When the one-letter abbreviations are used, upper case letters designate amino acids in the L-configuration about the α -carbon and lower case letters designate amino acids in the D-configuration about the α -carbon. For example, “A” designates L-alanine and “a” designates D-alanine. When polypeptide sequences are presented as a string of one-letter or three-letter abbreviations (or mixtures thereof), the sequences are presented in the amino (N) to carboxy (C) direction in accordance with common convention.

(11) The abbreviations used for the genetically encoding nucleosides are conventional and are as follows: adenosine (A); guanosine (G); cytidine (C); thymidine (T); and uridine (U). Unless specifically delineated, the abbreviated nucleosides may be either ribonucleosides or 2'-deoxyribonucleosides. The nucleosides may be specified as being either ribonucleosides or 2'-deoxyribonucleosides on an individual basis or on an aggregate basis. When nucleic acid sequences are presented as a string of one-letter abbreviations, the sequences are presented in the 5' to 3' direction in accordance with common convention, and the phosphates are not indicated.

Definitions

(12) In reference to the present invention, the technical and scientific terms used in the descriptions herein will have the meanings commonly understood by one of ordinary skill in the art, unless specifically defined otherwise. Accordingly, the following terms are intended to have the following meanings.

(13) As used herein, the singular forms “a”, “an” and “the” include plural referents unless the context clearly indicates otherwise. Thus, for example, reference to “a polypeptide” includes more than one polypeptide.

(14) Similarly, “comprise,” “comprises,” “comprising” “include,” “includes,” and “including” are interchangeable and not intended to be limiting. Thus, as used herein, the term “comprising” and its cognates are used in their inclusive sense (i.e., equivalent to the term “including” and its corresponding cognates).

(15) It is to be further understood that where descriptions of various embodiments use the term “comprising,” those skilled in the art would understand that in some specific instances, an embodiment can be alternatively described using language “consisting essentially of” or “consisting of.”

(16) As used herein, the term “about” means an acceptable error for a particular value. In some instances “about” means within 0.05%, 0.5%, 1.0%, or 2.0%, of a given value range. In some instances, “about” means within 1, 2, 3, or 4 standard deviations of a given value.

(17) As used herein, “EC” number refers to the Enzyme Nomenclature of the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology (NC-IUBMB). The IUBMB biochemical classification is a numerical classification system for enzymes based on the chemical reactions they catalyze.

(18) As used herein, “ATCC” refers to the American Type Culture Collection whose biorepository collection includes genes and strains.

(19) As used herein, “NCBI” refers to National Center for Biological Information and the sequence databases provided therein.

(20) “Protein,” “polypeptide,” and “peptide” are used interchangeably herein to denote a polymer of at least two amino acids covalently linked by an amide bond, regardless of length or post-translational modification (e.g., glycosylation or phosphorylation). Included within this definition are D- and L-amino acids, and mixtures of D- and L-amino acids, as well as polymers comprising D- and L-amino acids, and

mixtures of D- and L-amino acids.

(21) “Amino acids” are referred to herein by either their commonly known three-letter symbols or by the one-letter symbols recommended by IUPAC-IUB Biochemical Nomenclature Commission. Nucleotides, likewise, may be referred to by their commonly accepted single letter codes.

(22) As used herein, “polynucleotide” and “nucleic acid” refer to two or more nucleotides that are covalently linked together. The polynucleotide may be wholly comprised of ribonucleotides (i.e., RNA), wholly comprised of 2' deoxyribonucleotides (i.e., DNA), or comprised of mixtures of ribo- and 2' deoxyribonucleotides. While the nucleosides will typically be linked together via standard phosphodiester linkages, the polynucleotides may include one or more non-standard linkages. The polynucleotide may be single-stranded or double-stranded, or may include both single-stranded regions and double-stranded regions. Moreover, while a polynucleotide will typically be composed of the naturally occurring encoding nucleobases (i.e., adenine, guanine, uracil, thymine and cytosine), it may include one or more modified and/or synthetic nucleobases, such as, for example, inosine, xanthine, hypoxanthine, etc. In some embodiments, such modified or synthetic nucleobases are nucleobases encoding amino acid sequences.

(23) As used herein, “nucleoside” refers to glycosylamines comprising a nucleobase (i.e., a nitrogenous base), and a 5-carbon sugar (e.g., ribose or deoxyribose). Non-limiting examples of nucleosides include cytidine, uridine, adenosine, guanosine, thymidine, and inosine. In contrast, the term “nucleotide” refers to the glycosylamines comprising a nucleobase, a 5-carbon sugar, and one or more phosphate groups. In some embodiments, nucleosides can be phosphorylated by kinases to produce nucleotides.

(24) As used herein, “nucleoside diphosphate” refers to glycosylamines comprising a nucleobase (i.e., a nitrogenous base), a 5-carbon sugar (e.g., ribose or deoxyribose), and a diphosphate (i.e., pyrophosphate) moiety. In some embodiments herein, “nucleoside diphosphate” is abbreviated as “NDP.” Non-limiting examples of nucleoside diphosphates include cytidine diphosphate (CDP), uridine diphosphate (UDP), adenosine diphosphate (ADP), guanosine diphosphate (GDP), thymidine diphosphate (TDP), and inosine diphosphate. The terms “nucleoside” and “nucleotide” may be used interchangeably in some contexts.

(25) As used herein, “coding sequence” refers to that portion of a nucleic acid (e.g., a gene) that encodes an amino acid sequence of a protein.

(26) As used herein, the terms “biocatalysis,” “biocatalytic,” “biotransformation,” and “biosynthesis” refer to the use of enzymes to perform chemical reactions on organic compounds.

(27) As used herein, “glycosyltransferase” (GT) refers to a polypeptide having an enzymatic capability of transferring glycosyl residues from an activated sugar donor to monomeric and polymeric acceptor molecules. In some embodiments, the glycosyltransferases are referred to as “glycosyltransferase variants” or “glycosyltransferase combinatorial variants.” In some embodiments, “glycosyltransferase” refers to an UDP-glucuronosyltransferase enzyme of the classification EC 2.4.1.17, which catalyzes the transfer of glucose from UDP- α -D-glucuronate (also known as UDP-glucose) to an acceptor, releasing UDP and forming acceptor β -D-glucuronoside. The Carbohydrate-Active Enzymes database (CAZy) provides a continuously updated list of the glycosyltransferase families. In some embodiments, the glycosyltransferases include, but are not limited to, enzymes classified in the GT1 family. In some preferred embodiments, the glycosyltransferase variants of the present invention preferentially utilize ADP-glucose. In some additional embodiments, the glycosyltransferase enzymes of the present invention do not utilize UDP-glucose. In some further embodiments, the glycosyltransferase variants of the present invention utilize ADP-glucose, CDP-glucose, TDP-glucose, GDP-glucose, and/or IDP-glucose, but not UDP-glucose. Thus, in some preferred embodiments, the present invention provides ADP-glucose-dependent glycosyltransferases (ADP-glycosyltransferases; AGTs), CDP-glucose-dependent glycosyltransferases (CDP-glycosyltransferases; CGTs), GDP-glucose-dependent glycosyltransferases (GDP-glycosyltransferases; GGTs), TDP-glucose-dependent glycosyltransferases (TDP-glycosyltransferases; TGTs), and IDP-glucose-dependent glycosyltransferase (IDP-glycosyltransferases; IGTs).

(28) As used herein, “NDP-glycosyltransferase” (NDP-GT) refers to a polypeptide having an enzymatic capability of transferring glycosyl residues from an activated sugar donor that is an NDP to monomeric and polymeric acceptor molecules. In some embodiments, NDP-glycosyltransferases are generally referred to as “glycosyltransferases.” Indeed, the term “glycosyltransferase” as used herein encompasses NDP-glycosyltransferases, including, but not limited to ADP-glucose-dependent glycosyltransferases

(ADP-glycosyltransferases; AGTs), CDP-glucose-dependent glycosyltransferases (CDP-glycosyltransferases; CGTs), GDP-glucose-dependent glycosyltransferase (GDP-glycosyltransferases; GGTs), TDP-glucose-dependent glycosyltransferases (TDP-glycosyltransferases; TGTs), and IDP-glucose-dependent glycosyltransferase (IDP-glycosyltransferases; IGTs). In some embodiments, the glycosyltransferase enzymes of the present invention utilize ADP-glucose, CDP-glucose, TDP-glucose, GDP-glucose, and/or IDT-glucose, but not UDP-glucose. In some additional embodiments the enzymes are referred to as “variants” or “combinatorial variants” (e.g., ADP-glycosyltransferase variants).

(29) As used herein, “transglycosylation” refers to a reaction in which a glycosyl residue is transferred from a disaccharide, trisaccharide, or oligosaccharide donor to an aglycosylated or glycosylated acceptor molecule.

(30) As used herein, “transglucosylation” refers to a transglycosylation reaction in which the glycosyl residue that is transferred is a glucose and the disaccharide, trisaccharide, or oligosaccharide donor contains glucose.

(31) As used herein, “glycosylation” refers to the formation of a glycosidic linkage between a glycosyl residue and an acceptor molecule.

(32) As used herein, “glucosylation” refers to the formation of a glycosidic linkage between a glucose residue and an acceptor molecule.

(33) As used herein, “glycosyl” refers to an organic group that is a univalent free radical or substituent structure obtained by removing the hemiacetal hydroxyl group from the cyclic form of a monosaccharide, lower oligosaccharide or oligosaccharide derivative. Glycosyl groups react with inorganic acids (e.g., phosphoric acid) to form esters (e.g., glucose 1-phosphate).

(34) As used herein, “glycoside” refers to a molecule in which a carbohydrate (e.g., sugar) is bound to another functional group by a glycosidic bond. Glycosides can be hydrolyzed to produce a sugar and a non-sugar (i.e., aglycone) component.

(35) As used herein, the term “steviol glycoside” refers to a glycoside of steviol, including but not limited to, naturally occurring steviol glycosides (e.g., stevioside, steviolmonoside, steviolbioside, rubusoside, dulcoside B, dulcoside A, rebaudioside B, rebaudioside G, rebaudioside C, rebaudioside F, rebaudioside A, rebaudioside I, rebaudioside E, rebaudioside H, rebaudioside L, rebaudioside K, rebaudioside J, rebaudioside M (also referred to as rebaudioside X), rebaudioside D, rebaudioside N, rebaudioside O), and synthetic steviol glycosides (e.g., enzymatically glucosylated steviol glycosides), and combinations thereof. The chemical structures of steviol and its glycosides are below (See, WO 2013/176738).

(36) As used herein, “stevioside substrate” refers to any suitable material comprising at least one steviol glycoside.

(37) TABLE-US-00002 Chemical Structure of Steviol and Its Glycosides  Embedded image Glycoside R.sub.1 R.sub.2 Steviol H H Steviolmonoside H Glcβ1- Steviol Glcβ1- H monoglucosyl ester Rubusoside Glcβ1- Glcβ1- Steviolbioside H Glcβ (1-2) Glcβ1- Dulcoside A Glcβ1- Rhaα(1-2) Glcβ1- Stevioside Glcβ1- Glcβ (1-2) Glcβ1- Rebaudioside B H Glcβ (1-2)[Glcβ (1-3)] Glcβ1- Rebaudioside C Glcβ1- Rhaα(1-2)[Glcβ (1-3)] Glcβ1- Rebaudioside A Glcβ1- Glcβ (1-2)[Glcβ (1-3)] Glcβ1- Rebaudioside D Glcβ (1-2) Glcβ1- Glcβ (1-2)[Glcβ (1-3)] Glcβ1- Rebaudioside I Glcβ (1-3) Glcβ1- Glcβ (1-2)[Glcβ (1-3)] Glcβ1- Rebaudioside M Glcβ (1-2)[Glcβ Glcβ (1-2)[Glcβ (1-3)] Glcβ1- (1-3)] Glcβ1- (Glc = glucose, Rha = rhamnose)

(38) As used herein, “sucrose synthase” refers to a glycosyltransferase enzyme (EC 2.4.1.1.13) that reversibly catalyzes the chemical reaction NDP-glucose+D-fructose to NDP and sucrose. In some embodiments, the present invention provides variants of *Acidithiobacillus caldus* sucrose synthase (“AcSuS”). In some embodiments, these enzymes are referred to as “sucrose synthase variants,” “SuS,” “SUS,” “SuS variants,” “SUS variants,” “sucrose synthase combinatorial variants,” or “SuS combinatorial variants,” or “SUS combinatorial variants.” In some embodiments, these variants preferentially utilize NDPs other than uridine (i.e., ADP-glucose, CDP-glucose, TDP-glucose, GDP-glucose, and/or IDP-glucose are utilized, rather than UDP-glucose). In some embodiments, these variants do not utilize UDP-glucose.

(39) As used herein, the term “one-pot reaction” refers to the production of rebaudioside of interest in one reaction vessel. In some embodiments, the term is used in reference to the production of rebM from a starting material, including but not limited to as rebA and/or steviosides with the intermediate production

of other rebaudiosides (e.g., rebD and/or rebI). In some embodiments, the conversion of stevioside to RebA, RebA to RebD and/or RebI and RebD and/or RebI to Reb M, are conducted as a multiple enzyme cascade in one reaction vessel.

(40) As used herein, “wild-type” and “naturally-occurring” refer to the form found in nature. For example a wild-type polypeptide or polynucleotide sequence is a sequence present in an organism that can be isolated from a source in nature and which has not been intentionally modified by human manipulation.

(41) As used herein, “recombinant,” “engineered,” and “non-naturally occurring” when used with reference to a cell, nucleic acid, or polypeptide, refers to a material, or a material corresponding to the natural or native form of the material, that has been modified in a manner that would not otherwise exist in nature. In some embodiments, the cell, nucleic acid or polypeptide is identical a naturally occurring cell, nucleic acid or polypeptide, but is produced or derived from synthetic materials and/or by manipulation using recombinant techniques. Non-limiting examples include, among others, recombinant cells expressing genes that are not found within the native (non-recombinant) form of the cell or express native genes that are otherwise expressed at a different level.

(42) The term “percent (%) sequence identity” is used herein to refer to comparisons among polynucleotides or polypeptides, and are determined by comparing two optimally aligned sequences over a comparison window, wherein the portion of the polynucleotide or polypeptide sequence in the comparison window may comprise additions or deletions (i.e., gaps) as compared to the reference sequence for optimal alignment of the two sequences. The percentage may be calculated by determining the number of positions at which the identical nucleic acid base or amino acid residue occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison and multiplying the result by 100 to yield the percentage of sequence identity. Alternatively, the percentage may be calculated by determining the number of positions at which either the identical nucleic acid base or amino acid residue occurs in both sequences or a nucleic acid base or amino acid residue is aligned with a gap to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison and multiplying the result by 100 to yield the percentage of sequence identity. Those of skill in the art appreciate that there are many established algorithms available to align two sequences. Optimal alignment of sequences for comparison can be conducted by any suitable method, including, but not limited to the local homology algorithm of Smith and Waterman (Smith and Waterman, *Adv. Appl. Math.*, 2:482 [1981]), by the homology alignment algorithm of Needleman and Wunsch (Needleman and Wunsch, *J. Mol. Biol.*, 48:443 [1970]), by the search for similarity method of Pearson and Lipman (Pearson and Lipman, *Proc. Natl. Acad. Sci. USA* 85:2444 [1988]), by computerized implementations of these algorithms (e.g., GAP, BESTFIT, FASTA, and TFASTA in the GCG Wisconsin Software Package), or by visual inspection, as known in the art. Examples of algorithms that are suitable for determining percent sequence identity and sequence similarity include, but are not limited to the BLAST and BLAST 2.0 algorithms, which are described by Altschul et al. (See Altschul et al., *J. Mol. Biol.*, 215: 403-410 [1990]; and Altschul et al., *Nucl. Acids Res.*, 3389-3402 [1977], respectively). Software for performing BLAST analyses is publicly available through the National Center for Biotechnology Information website. This algorithm involves first identifying high scoring sequence pairs (HSPs) by identifying short words of length W in the query sequence, which either match or satisfy some positive-valued threshold score T when aligned with a word of the same length in a database sequence. T is referred to as, the neighborhood word score threshold (See, Altschul et al., *supra*). These initial neighborhood word hits act as seeds for initiating searches to find longer HSPs containing them. The word hits are then extended in both directions along each sequence for as far as the cumulative alignment score can be increased. Cumulative scores are calculated using, for nucleotide sequences, the parameters M (reward score for a pair of matching residues; always >0) and N (penalty score for mismatching residues; always <0). For amino acid sequences, a scoring matrix is used to calculate the cumulative score. Extension of the word hits in each direction are halted when: the cumulative alignment score falls off by the quantity X from its maximum achieved value; the cumulative score goes to zero or below, due to the accumulation of one or more negative-scoring residue alignments; or the end of either sequence is reached. The BLAST algorithm parameters W, T, and X determine the sensitivity and speed of the alignment. The BLASTN program (for nucleotide sequences) uses as defaults a wordlength (W) of 11, an

expectation (E) of 10, M=5, N=-4, and a comparison of both strands. For amino acid sequences, the BLASTP program uses as defaults a wordlength (W) of 3, an expectation (E) of 10, and the BLOSUM62 scoring matrix (See, Henikoff and Henikoff, Proc. Natl. Acad. Sci. USA 89:10915 [1989]). Exemplary determination of sequence alignment and % sequence identity can employ the BESTFIT or GAP programs in the GCG Wisconsin Software package (Accelrys, Madison Wis.), using default parameters provided.

(43) As used herein, “reference sequence” refers to a defined sequence used as a basis for a sequence and/or activity comparison. A reference sequence may be a subset of a larger sequence, for example, a segment of a full-length gene or polypeptide sequence. Generally, a reference sequence is at least 20 nucleotide or amino acid residues in length, at least 25 residues in length, at least 50 residues in length, at least 100 residues in length or the full length of the nucleic acid or polypeptide. Since two polynucleotides or polypeptides may each (1) comprise a sequence (i.e., a portion of the complete sequence) that is similar between the two sequences, and (2) may further comprise a sequence that is divergent between the two sequences, sequence comparisons between two (or more) polynucleotides or polypeptides are typically performed by comparing sequences of the two polynucleotides or polypeptides over a “comparison window” to identify and compare local regions of sequence similarity. In some embodiments, a “reference sequence” can be based on a primary amino acid sequence, where the reference sequence is a sequence that can have one or more changes in the primary sequence.

(44) As used herein, “comparison window” refers to a conceptual segment of at least about 20 contiguous nucleotide positions or amino acid residues wherein a sequence may be compared to a reference sequence of at least 20 contiguous nucleotides or amino acids and wherein the portion of the sequence in the comparison window may comprise additions or deletions (i.e., gaps) of 20 percent or less as compared to the reference sequence (which does not comprise additions or deletions) for optimal alignment of the two sequences. The comparison window can be longer than 20 contiguous residues, and includes, optionally 30, 40, 50, 100, or longer windows.

(45) As used herein, “corresponding to,” “reference to,” and “relative to” when used in the context of the numbering of a given amino acid or polynucleotide sequence refer to the numbering of the residues of a specified reference sequence when the given amino acid or polynucleotide sequence is compared to the reference sequence. In other words, the residue number or residue position of a given polymer is designated with respect to the reference sequence rather than by the actual numerical position of the residue within the given amino acid or polynucleotide sequence. For example, a given amino acid sequence, such as that of an engineered glycosyltransferase, can be aligned to a reference sequence by introducing gaps to optimize residue matches between the two sequences. In these cases, although the gaps are present, the numbering of the residue in the given amino acid or polynucleotide sequence is made with respect to the reference sequence to which it has been aligned.

(46) As used herein, “substantial identity” refers to a polynucleotide or polypeptide sequence that has at least 80 percent sequence identity, at least 85 percent identity, at least between 89 to 95 percent sequence identity, or more usually, at least 99 percent sequence identity as compared to a reference sequence over a comparison window of at least 20 residue positions, frequently over a window of at least 30-50 residues, wherein the percentage of sequence identity is calculated by comparing the reference sequence to a sequence that includes deletions or additions which total 20 percent or less of the reference sequence over the window of comparison. In some specific embodiments applied to polypeptides, the term “substantial identity” means that two polypeptide sequences, when optimally aligned, such as by the programs GAP or BESTFIT using default gap weights, share at least 80 percent sequence identity, preferably at least 89 percent sequence identity, at least 95 percent sequence identity or more (e.g., 99 percent sequence identity). In some embodiments, residue positions that are not identical in sequences being compared differ by conservative amino acid substitutions.

(47) As used herein, “amino acid difference” and “residue difference” refer to a difference in the amino acid residue at a position of a polypeptide sequence relative to the amino acid residue at a corresponding position in a reference sequence. In some cases, the reference sequence has a histidine tag, but the numbering is maintained relative to the equivalent reference sequence without the histidine tag. The positions of amino acid differences generally are referred to herein as “X_n,” where n refers to the corresponding position in the reference sequence upon which the residue difference is based. For

example, a “residue difference at position X93 as compared to SEQ ID NO:4” refers to a difference of the amino acid residue at the polypeptide position corresponding to position 93 of SEQ ID NO:4. Thus, if the reference polypeptide of SEQ ID NO:4 has a serine at position 93, then a “residue difference at position X93 as compared to SEQ ID NO:4” an amino acid substitution of any residue other than serine at the position of the polypeptide corresponding to position 93 of SEQ ID NO:4. In most instances herein, the specific amino acid residue difference at a position is indicated as “XnY” where “Xn” specified the corresponding position as described above, and “Y” is the single letter identifier of the amino acid found in the engineered polypeptide (i.e., the different residue than in the reference polypeptide). In some instances (e.g., in the Tables presented in the Examples), the present invention also provides specific amino acid differences denoted by the conventional notation “AnB”, where A is the single letter identifier of the residue in the reference sequence, “n” is the number of the residue position in the reference sequence, and B is the single letter identifier of the residue substitution in the sequence of the engineered polypeptide. In some instances, a polypeptide of the present invention can include one or more amino acid residue differences relative to a reference sequence, which is indicated by a list of the specified positions where residue differences are present relative to the reference sequence. In some embodiments, where more than one amino acid can be used in a specific residue position of a polypeptide, the various amino acid residues that can be used are separated by a “/” (e.g., X307H/X307P or X307H/P). The slash may also be used to indicate multiple substitutions within a given variant (i.e., there is more than one substitution present in a given sequence, such as in a combinatorial variant). In some embodiments, the present invention includes engineered polypeptide sequences comprising one or more amino acid differences comprising conservative or non-conservative amino acid substitutions. In some additional embodiments, the present invention provides engineered polypeptide sequences comprising both conservative and non-conservative amino acid substitutions.

(48) As used herein, “conservative amino acid substitution” refers to a substitution of a residue with a different residue having a similar side chain, and thus typically involves substitution of the amino acid in the polypeptide with amino acids within the same or similar defined class of amino acids. By way of example and not limitation, in some embodiments, an amino acid with an aliphatic side chain is substituted with another aliphatic amino acid (e.g., alanine, valine, leucine, and isoleucine); an amino acid with a hydroxyl side chain is substituted with another amino acid with a hydroxyl side chain (e.g., serine and threonine); an amino acids having aromatic side chains is substituted with another amino acid having an aromatic side chain (e.g., phenylalanine, tyrosine, tryptophan, and histidine); an amino acid with a basic side chain is substituted with another amino acid with a basis side chain (e.g., lysine and arginine); an amino acid with an acidic side chain is substituted with another amino acid with an acidic side chain (e.g., aspartic acid or glutamic acid); and/or a hydrophobic or hydrophilic amino acid is replaced with another hydrophobic or hydrophilic amino acid, respectively.

(49) As used herein, “non-conservative substitution” refers to substitution of an amino acid in the polypeptide with an amino acid with significantly differing side chain properties. Non-conservative substitutions may use amino acids between, rather than within, the defined groups and affects (a) the structure of the peptide backbone in the area of the substitution (e.g., proline for glycine) (b) the charge or hydrophobicity, or (c) the bulk of the side chain. By way of example and not limitation, an exemplary non-conservative substitution can be an acidic amino acid substituted with a basic or aliphatic amino acid; an aromatic amino acid substituted with a small amino acid; and a hydrophilic amino acid substituted with a hydrophobic amino acid.

(50) As used herein, “deletion” refers to modification to the polypeptide by removal of one or more amino acids from the reference polypeptide. Deletions can comprise removal of 1 or more amino acids, 2 or more amino acids, 5 or more amino acids, 10 or more amino acids, 15 or more amino acids, or 20 or more amino acids, up to 10% of the total number of amino acids, or up to 20% of the total number of amino acids making up the reference enzyme while retaining enzymatic activity and/or retaining the improved properties of an engineered glycosyltransferase enzyme. Deletions can be directed to the internal portions and/or terminal portions of the polypeptide. In various embodiments, the deletion can comprise a continuous segment or can be discontinuous. Deletions are typically indicated by “–” in amino acid sequences.

(51) As used herein, “insertion” refers to modification to the polypeptide by addition of one or more

amino acids can be in the reference polypeptide. Insertions can be in the internal portions of the polypeptide, or to the carboxy or amino terminus. Insertions as used herein include fusion proteins as is known in the art. The insertion can be a contiguous segment of amino acids or separated by one or more of the amino acids in the naturally occurring polypeptide.

(52) A “functional fragment” and “biologically active fragment” are used interchangeably herein to refer to a polypeptide that has an amino-terminal and/or carboxy-terminal deletion(s) and/or internal deletions, but where the remaining amino acid sequence is identical to the corresponding positions in the sequence to which it is being compared (e.g., a full-length engineered glycosyltransferase of the present invention) and that retains substantially all of the activity of the full-length polypeptide.

(53) As used herein, “isolated polypeptide” refers to a polypeptide which is substantially separated from other contaminants that naturally accompany it (e.g., protein, lipids, and polynucleotides). The term embraces polypeptides which have been removed or purified from their naturally-occurring environment or expression system (e.g., within a host cell or via in vitro synthesis). The recombinant glycosyltransferase polypeptides may be present within a cell, present in the cellular medium, or prepared in various forms, such as lysates or isolated preparations. As such, in some embodiments, the recombinant glycosyltransferase polypeptides can be an isolated polypeptide.

(54) As used herein, “substantially pure polypeptide” or “purified protein” refers to a composition in which the polypeptide species is the predominant species present (i.e., on a molar or weight basis it is more abundant than any other individual macromolecular species in the composition), and is generally a substantially purified composition when the object species comprises at least about 50 percent of the macromolecular species present by mole or % weight. However, in some embodiments, the composition comprising glycosyltransferase comprises glycosyltransferase that is less than 50% pure (e.g., about 10%, about 20%, about 30%, about 40%, or about 50%) Generally, a substantially pure glycosyltransferase composition comprises about 60% or more, about 70% or more, about 80% or more, about 90% or more, about 95% or more, and about 98% or more of all macromolecular species by mole or % weight present in the composition. In some embodiments, the object species is purified to essential homogeneity (i.e., contaminant species cannot be detected in the composition by conventional detection methods) wherein the composition consists essentially of a single macromolecular species. Solvent species, small molecules (<500 Daltons), and elemental ion species are not considered macromolecular species. In some embodiments, the isolated recombinant glycosyltransferase polypeptides are substantially pure polypeptide compositions.

(55) As used herein, “improved enzyme property” refers to at least one improved property of an enzyme. In some embodiments, the present invention provides engineered glycosyltransferase polypeptides that exhibit an improvement in any enzyme property as compared to a reference glycosyltransferase polypeptide and/or a wild-type glycosyltransferase polypeptide, and/or another engineered glycosyltransferase polypeptide. Thus, the level of “improvement” can be determined and compared between various glycosyltransferase polypeptides, including wild-type, as well as engineered glycosyltransferases. Improved properties include, but are not limited, to such properties as increased protein expression, increased thermoactivity, increased thermostability, increased pH activity, increased stability, increased enzymatic activity, increased substrate specificity or affinity, increased specific activity, increased resistance to substrate or end-product inhibition, increased chemical stability, improved chemoselectivity, improved solvent stability, increased tolerance to acidic pH, increased tolerance to proteolytic activity (i.e., reduced sensitivity to proteolysis), reduced aggregation, increased solubility, and altered temperature profile. In additional embodiments, the term is used in reference to the at least one improved property of sucrose synthase enzymes. In some embodiments, the present invention provides engineered sucrose synthase polypeptides that exhibit an improvement in any enzyme property as compared to a reference sucrose synthase polypeptide and/or a wild-type sucrose synthase polypeptide, and/or another engineered sucrose synthase polypeptide. Thus, the level of “improvement” can be determined and compared between various sucrose synthase polypeptides, including wild-type, as well as engineered sucrose synthases.

(56) As used herein, “increased enzymatic activity” and “enhanced catalytic activity” refer to an improved property of the engineered polypeptides, which can be represented by an increase in specific activity (e.g., product produced/time/weight protein) or an increase in percent conversion of the substrate

to the product (e.g., percent conversion of starting amount of substrate to product in a specified time period using a specified amount of enzyme) as compared to the reference enzyme. In some embodiments, the terms refer to an improved property of engineered glycosyltransferase polypeptides provided herein, which can be represented by an increase in specific activity (e.g., product produced/time/weight protein) or an increase in percent conversion of the substrate to the product (e.g., percent conversion of starting amount of substrate to product in a specified time period using a specified amount of glycosyltransferase) as compared to the reference glycosyltransferase enzyme. In some embodiments, the terms are used in reference to improved sucrose synthase enzymes provided herein. Exemplary methods to determine enzyme activity of the engineered glycosyltransferases and sucrose synthases of the present invention are provided in the Examples. Any property relating to enzyme activity may be affected, including the classical enzyme properties of $K_{sub.m}$, $V_{sub.max}$ or $k_{sub.cat}$, changes of which can lead to increased enzymatic activity. For example, improvements in enzyme activity can be from about 1.1 fold the enzymatic activity of the corresponding wild-type enzyme, to as much as 2-fold, 5-fold, 10-fold, 20-fold, 25-fold, 50-fold, 75-fold, 100-fold, 150-fold, 200-fold or more enzymatic activity than the naturally occurring glycosyltransferase or another engineered glycosyltransferase from which the glycosyltransferase polypeptides were derived.

(57) As used herein, “conversion” refers to the enzymatic conversion (or biotransformation) of a substrate(s) to the corresponding product(s). “Percent conversion” refers to the percent of the substrate that is converted to the product within a period of time under specified conditions. Thus, the “enzymatic activity” or “activity” of a glycosyltransferase polypeptide can be expressed as “percent conversion” of the substrate to the product in a specific period of time.

(58) Enzymes with “generalist properties” (or “generalist enzymes”) refer to enzymes that exhibit improved activity for a wide range of substrates, as compared to a parental sequence. Generalist enzymes do not necessarily demonstrate improved activity for every possible substrate. In some embodiments, the present invention provides glycosyltransferase variants with generalist properties, in that they demonstrate similar or improved activity relative to the parental gene for a wide range of sterically and electronically diverse substrates. In addition, the generalist enzymes provided herein were engineered to be improved across a wide range of diverse molecules to increase the production of metabolites/products.

(59) The term “stringent hybridization conditions” is used herein to refer to conditions under which nucleic acid hybrids are stable. As known to those of skill in the art, the stability of hybrids is reflected in the melting temperature ($T_{sub.m}$) of the hybrids. In general, the stability of a hybrid is a function of ion strength, temperature, G/C content, and the presence of chaotropic agents. The T values for polynucleotides can be calculated using known methods for predicting melting temperatures (See e.g., Baldino et al., *Meth. Enzymol.*, 168:761-777 [1989]; Bolton et al., *Proc. Natl. Acad. Sci. USA* 48:1390 [1962]; Bresslauer et al., *Proc. Natl. Acad. Sci. USA* 83:8893-8897 [1986]; Freier et al., *Proc. Natl. Acad. Sci. USA* 83:9373-9377 [1986]; Kierzek et al., *Biochem.*, 25:7840-7846 [1986]; Rychlik et al., *Nucl. Acids Res.*, 18:6409-6412 [1990] (erratum, *Nucl. Acids Res.*, 19:698 [1991]); Sambrook et al., *supra*; Suggs et al., 1981, in *Developmental Biology Using Purified Genes*, Brown et al. [eds.], pp. 683-693, Academic Press, Cambridge, Mass. [1981]; and Wetmur, *Crit. Rev. Biochem. Mol. Biol.* 26:227-259 [1991]). In some embodiments, the polynucleotide encodes the polypeptide disclosed herein and hybridizes under defined conditions, such as moderately stringent or highly stringent conditions, to the complement of a sequence encoding an engineered glycosyltransferase enzyme of the present invention.

(60) As used herein, “hybridization stringency” relates to hybridization conditions, such as washing conditions, in the hybridization of nucleic acids. Generally, hybridization reactions are performed under conditions of lower stringency, followed by washes of varying but higher stringency. The term “moderately stringent hybridization” refers to conditions that permit target-DNA to bind a complementary nucleic acid that has about 60% identity, preferably about 75% identity, about 85% identity to the target DNA, with greater than about 90% identity to target-polynucleotide. Exemplary moderately stringent conditions are conditions equivalent to hybridization in 50% formamide, 5×Denhart's solution, 5×SSPE, 0.2% SDS at 42° C., followed by washing in 0.2×SSPE, 0.2% SDS, at 42° C. “High stringency hybridization” refers generally to conditions that are about 10° C. or less from the thermal melting temperature $T_{sub.m}$ as determined under the solution condition for a defined polynucleotide sequence. In some embodiments, a high stringency condition refers to conditions that permit hybridization of only

those nucleic acid sequences that form stable hybrids in 0.018M NaCl at 65° C. (i.e., if a hybrid is not stable in 0.018M NaCl at 65° C., it will not be stable under high stringency conditions, as contemplated herein). High stringency conditions can be provided, for example, by hybridization in conditions equivalent to 50% formamide, 5×Denhart's solution, 5×SSPE, 0.2% SDS at 42° C., followed by washing in 0.1×SSPE, and 0.1% SDS at 65° C. Another high stringency condition is hybridizing in conditions equivalent to hybridizing in 5×SSC containing 0.1% (w/v) SDS at 65° C. and washing in 0.1×SSC containing 0.1% SDS at 65° C. Other high stringency hybridization conditions, as well as moderately stringent conditions, are described in the references cited above.

(61) As used herein, “codon optimized” refers to changes in the codons of the polynucleotide encoding a protein to those preferentially used in a particular organism such that the encoded protein is efficiently expressed in the organism of interest. Although the genetic code is degenerate in that most amino acids are represented by several codons, called “synonyms” or “synonymous” codons, it is well known that codon usage by particular organisms is nonrandom and biased towards particular codon triplets. This codon usage bias may be higher in reference to a given gene, genes of common function or ancestral origin, highly expressed proteins versus low copy number proteins, and the aggregate protein coding regions of an organism's genome. In some embodiments, the polynucleotides encoding the glycosyltransferase enzymes may be codon optimized for optimal production in the host organism selected for expression.

(62) As used herein, “preferred,” “optimal,” and “high codon usage bias” codons when used alone or in combination refer(s) interchangeably to codons that are used at higher frequency in the protein coding regions than other codons that code for the same amino acid. The preferred codons may be determined in relation to codon usage in a single gene, a set of genes of common function or origin, highly expressed genes, the codon frequency in the aggregate protein coding regions of the whole organism, codon frequency in the aggregate protein coding regions of related organisms, or combinations thereof. Codons whose frequency increases with the level of gene expression are typically optimal codons for expression. A variety of methods are known for determining the codon frequency (e.g., codon usage, relative synonymous codon usage) and codon preference in specific organisms, including multivariate analysis, for example, using cluster analysis or correspondence analysis, and the effective number of codons used in a gene (See e.g., GCG CodonPreference, Genetics Computer Group Wisconsin Package; CodonW, Peden, University of Nottingham; McInerney, *Bioinform.*, 14:372-73 [1998]; Stenico et al., *Nucl. Acids Res.*, 22:2437-46 [1994]; and Wright, *Gene* 87:23-29 [1990]). Codon usage tables are available for many different organisms (See e.g., Wada et al., *Nucl. Acids Res.*, 20:2111-2118 [1992]; Nakamura et al., *Nucl. Acids Res.*, 28:292 [2000]; Duret, et al., *supra*; Henaut and Danchin, in *Escherichia coli and Salmonella*, Neidhardt, et al. (eds.), ASM Press, Washington D.C., p. 2047-2066 [1996]). The data source for obtaining codon usage may rely on any available nucleotide sequence capable of coding for a protein. These data sets include nucleic acid sequences actually known to encode expressed proteins (e.g., complete protein coding sequences-CDS), expressed sequence tags (ESTs), or predicted coding regions of genomic sequences (See e.g., Mount, *Bioinformatics: Sequence and Genome Analysis*, Chapter 8, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. [2001]; Uberbacher, *Meth. Enzymol.*, 266:259-281 [1996]; and Tiwari et al., *Comput. Appl. Biosci.*, 13:263-270 [1997]).

(63) As used herein, “control sequence” includes all components, which are necessary or advantageous for the expression of a polynucleotide and/or polypeptide of the present invention. Each control sequence may be native or foreign to the nucleic acid sequence encoding the polypeptide. Such control sequences include, but are not limited to, a leader, polyadenylation sequence, propeptide sequence, promoter sequence, signal peptide sequence, initiation sequence and transcription terminator. At a minimum, the control sequences include a promoter, and transcriptional and translational stop signals. The control sequences may be provided with linkers for the purpose of introducing specific restriction sites facilitating ligation of the control sequences with the coding region of the nucleic acid sequence encoding a polypeptide.

(64) “Operably linked” is defined herein as a configuration in which a control sequence is appropriately placed (i.e., in a functional relationship) at a position relative to a polynucleotide of interest such that the control sequence directs or regulates the expression of the polynucleotide and/or polypeptide of interest.

(65) “Promoter sequence” refers to a nucleic acid sequence that is recognized by a host cell for expression

of a polynucleotide of interest, such as a coding sequence. The promoter sequence contains transcriptional control sequences, which mediate the expression of a polynucleotide of interest. The promoter may be any nucleic acid sequence which shows transcriptional activity in the host cell of choice including mutant, truncated, and hybrid promoters, and may be obtained from genes encoding extracellular or intracellular polypeptides either homologous or heterologous to the host cell.

(66) The phrase “suitable reaction conditions” refers to those conditions in the enzymatic conversion reaction solution (e.g., ranges of enzyme loading, substrate loading, temperature, pH, buffers, co-solvents, etc.) under which a glycosyltransferase polypeptide of the present invention is capable of converting a substrate to the desired product compound. Some exemplary “suitable reaction conditions” are provided herein.

(67) As used herein, “loading,” such as in “compound loading” or “enzyme loading” refers to the concentration or amount of a component in a reaction mixture at the start of the reaction.

(68) As used herein, “substrate” in the context of an enzymatic conversion reaction process refers to the compound or molecule acted on by the engineered enzymes provided herein (e.g., engineered glycosyltransferase polypeptides).

(69) As used herein, the terms “biomass,” “biomass substrate,” “cellulosic biomass,” “cellulosic feedstock,” and “cellulosic substrate” refer to any materials that contain cellulose. Biomass can be derived from plants, animals, or microorganisms, and may include, but is not limited to agricultural, industrial, and forestry residues, industrial and municipal wastes, and terrestrial and aquatic crops grown for energy purposes. Examples of cellulosic substrates include, but are not limited to, wood, wood pulp, paper pulp, corn fiber, corn grain, corn cobs, crop residues such as corn husks, corn stover, grasses, wheat, wheat straw, barley, barley straw, hay, rice, rice straw, switchgrass, waste paper, paper and pulp processing waste, woody or herbaceous plants, fruit or vegetable pulp, corn cobs, distillers grain, grasses, rice hulls, cotton, hemp, flax, sisal, sugar cane bagasse, sorghum, soy, switchgrass, components obtained from milling of grains, trees, branches, roots, leaves, wood chips, sawdust, shrubs and bushes, vegetables, fruits, and flowers and any suitable mixtures thereof. In some embodiments, the cellulosic biomass comprises, but is not limited to cultivated crops (e.g., grasses, including C4 grasses, such as switch grass, cord grass, rye grass, miscanthus, reed canary grass, or any combination thereof), sugar processing residues, for example, but not limited to, bagasse (e.g., sugar cane bagasse, beet pulp [e.g., sugar beet], or a combination thereof), agricultural residues (e.g., soybean stover, corn stover, corn fiber, rice straw, sugar cane straw, rice, rice hulls, barley straw, corn cobs, wheat straw, canola straw, oat straw, oat hulls, corn fiber, hemp, flax, sisal, cotton, or any combination thereof), fruit pulp, vegetable pulp, distillers' grains, forestry biomass (e.g., wood, wood pulp, paper pulp, recycled wood pulp fiber, sawdust, hardwood, such as aspen wood, softwood, or a combination thereof). Furthermore, in some embodiments, the cellulosic biomass comprises cellulosic waste material and/or forestry waste materials, including but not limited to, paper and pulp processing waste, newsprint, cardboard and the like. In some embodiments, the cellulosic biomass comprises one species of fiber, while in some alternative embodiments, the cellulosic biomass comprises a mixture of fibers that originate from different cellulosic biomasses. In some embodiments, the biomass may also comprise transgenic plants that express ligninase and/or cellulase enzymes (See e.g., US 2008/0104724, incorporated by reference herein).

(70) As used herein, the term “slurry” refers to an aqueous solution in which are dispersed one or more solid components, such as a cellulosic substrate.

(71) As used herein, “increasing” yield of a product (e.g., a steviol glycoside) from a reaction occurs when a particular component present during the reaction (e.g., a GH enzyme) causes more product to be produced, compared with a reaction conducted under the same conditions with the same substrate and other substituents, but in the absence of the component of interest.

(72) As used herein, “hydrolyzing” cellulose or other polysaccharide occurs when at least some of the glycosidic bonds between two monosaccharides present in the substrate are hydrolyzed, thereby detaching from each other the two monomers that were previously bonded.

(73) A reaction is said to be “substantially free” of a particular enzyme if the amount of that enzyme compared with other enzymes that participate in catalyzing the reaction is less than about 2%, about 1%, or about 0.1% (wt/wt).

(74) As used herein, “fractionating” a liquid (e.g., a culture broth) means applying a separation process

(e.g., salt precipitation, column chromatography, size exclusion, and filtration) or a combination of such processes to provide a solution in which a desired protein (e.g., rebaudiosides) comprises a greater percentage of total protein in the solution than in the initial liquid product.

(75) As used herein, “starting composition” refers to any composition that comprises at least one substrate. In some embodiments, the starting composition comprises any cellulosic substrate.

(76) In some alternative embodiments, the term “starting composition” refers to any composition comprising at least one steviol glycoside, wherein one or more of the steviol glycosides act as substrate(s) for a biotransformation. In some embodiments, the starting composition is provided as an aqueous solution. In some embodiments, the starting composition comprises at least one steviol glycoside selected from stevioside, steviolmonoside, steviolbioside, rubusoside, dulcoside B, dulcoside A, rebaudioside B, rebaudioside G, rebaudioside C, rebaudioside F, rebaudioside A, rebaudioside I, rebaudioside E, rebaudioside H, rebaudioside L, rebaudioside K, rebaudioside J, rebaudioside M (also referred to as rebaudioside X), rebaudioside D, rebaudioside N, rebaudioside O, and synthetic steviol glycosides (e.g., enzymatically glucosylated steviol glycosides). In some embodiments, the starting composition comprises two or more steviol glycosides. In some embodiments, the starting composition comprises an extract obtained from purification of *Stevia rebaudiana* plant material (e.g., leaves). In some alternative embodiments, the starting composition comprises commercially available *stevia* extract(s). Additional starting compositions comprise by-products of processes used to isolate and purify steviol glycosides. In some embodiments, the starting composition comprises purified or partially purified steviol glycoside substrate(s). In some embodiments, the starting composition comprises greater than about 99% of a particular steviol glycoside by weight.

(77) In some embodiments, the starting composition comprises at least one glycoside and a cellulosic component as the substrate to produce at least one steviol glycoside (e.g., rebaudioside A, D, etc.).

(78) As used herein, “product” in the context of an enzymatic conversion process refers to the compound or molecule resulting from the action of an enzymatic polypeptide on a substrate. As used herein, in some embodiments, the term refers to the compound or molecule resulting from the action of the glycosyltransferase polypeptide on a substrate. In some embodiments, the product provided by the present invention is a steviol glycoside. In some embodiments, the product comprises at least one steviol glycoside selected from stevioside, steviolmonoside, steviolbioside, rubusoside, dulcoside B, dulcoside A, rebaudioside B, rebaudioside G, rebaudioside C, rebaudioside F, rebaudioside A, rebaudioside I, rebaudioside E, rebaudioside H, rebaudioside L, rebaudioside K, rebaudioside J, rebaudioside M (also referred to as rebaudioside X), rebaudioside D, rebaudioside N, rebaudioside O, and synthetic steviol glycosides (e.g., enzymatically glucosylated steviol glycosides).

(79) As used herein the term “culturing” refers to the growing of a population of microbial cells under any suitable conditions (e.g., using a liquid, gel or solid medium).

(80) Recombinant polypeptides can be produced using any suitable methods known in the art. Genes encoding the wild-type polypeptide of interest can be cloned in vectors, such as plasmids, and expressed in desired hosts, such as *E. coli*, etc. Variants of recombinant polypeptides can be generated by various methods known in the art. Indeed, there is a wide variety of different mutagenesis techniques well known to those skilled in the art. In addition, mutagenesis kits are also available from many commercial molecular biology suppliers. Methods are available to make specific substitutions at defined amino acids (site-directed), specific or random mutations in a localized region of the gene (regio-specific), or random mutagenesis over the entire gene (e.g., saturation mutagenesis). Numerous suitable methods are known to those in the art to generate enzyme variants, including but not limited to site-directed mutagenesis of single-stranded DNA or double-stranded DNA using PCR, cassette mutagenesis, gene synthesis, error-prone PCR, shuffling, and chemical saturation mutagenesis, or any other suitable method known in the art. Mutagenesis and directed evolution methods can be readily applied to enzyme-encoding polynucleotides to generate variant libraries that can be expressed, screened, and assayed. Any suitable mutagenesis and directed evolution methods find use in the present invention and are well known in the art (See e.g., U.S. Pat. Nos. 5,605,793, 5,811,238, 5,830,721, 5,834,252, 5,837,458, 5,928,905, 6,096,548, 6,117,679, 6,132,970, 6,165,793, 6,180,406, 6,251,674, 6,265,201, 6,277,638, 6,287,861, 6,287,862, 6,291,242, 6,297,053, 6,303,344, 6,309,883, 6,319,713, 6,319,714, 6,323,030, 6,326,204, 6,335,160, 6,335,198, 6,344,356, 6,352,859, 6,355,484, 6,358,740, 6,358,742, 6,365,377, 6,365,408, 6,368,861,

6,372,497, 6,337,186, 6,376,246, 6,379,964, 6,387,702, 6,391,552, 6,391,640, 6,395,547, 6,406,855, 6,406,910, 6,413,745, 6,413,774, 6,420,175, 6,423,542, 6,426,224, 6,436,675, 6,444,468, 6,455,253, 6,479,652, 6,482,647, 6,483,011, 6,484,105, 6,489,146, 6,500,617, 6,500,639, 6,506,602, 6,506,603, 6,518,065, 6,519,065, 6,521,453, 6,528,311, 6,537,746, 6,573,098, 6,576,467, 6,579,678, 6,586,182, 6,602,986, 6,605,430, 6,613,514, 6,653,072, 6,686,515, 6,703,240, 6,716,631, 6,825,001, 6,902,922, 6,917,882, 6,946,296, 6,961,664, 6,995,017, 7,024,312, 7,058,515, 7,105,297, 7,148,054, 7,220,566, 7,288,375, 7,384,387, 7,421,347, 7,430,477, 7,462,469, 7,534,564, 7,620,500, 7,620,502, 7,629,170, 7,702,464, 7,747,391, 7,747,393, 7,751,986, 7,776,598, 7,783,428, 7,795,030, 7,853,410, 7,868,138, 7,783,428, 7,873,477, 7,873,499, 7,904,249, 7,957,912, 7,981,614, 8,014,961, 8,029,988, 8,048,674, 8,058,001, 8,076,138, 8,108,150, 8,170,806, 8,224,580, 8,377,681, 8,383,346, 8,457,903, 8,504,498, 8,589,085, 8,762,066, 8,768,871, 9,593,326, and all related US, as well as PCT and non-US counterparts; Ling et al., *Anal. Biochem.*, 254(2):157-78 [1997]; Dale et al., *Meth. Mol. Biol.*, 57:369-74 [1996]; Smith, *Ann. Rev. Genet.*, 19:423-462 [1985]; Botstein et al., *Science*, 229:1193-1201 [1985]; Carter, *Biochem. J.*, 237:1-7 [1986]; Kramer et al., *Cell*, 38:879-887 [1984]; Wells et al., *Gene*, 34:315-323 [1985]; Minshull et al., *Curr. Op. Chem. Biol.*, 3:284-290 [1999]; Christians et al., *Nat. Biotechnol.*, 17:259-264 [1999]; Cramer et al., *Nature*, 391:288-291 [1998]; Cramer et al., *Nat. Biotechnol.*, 15:436-438 [1997]; Zhang et al., *Proc. Nat. Acad. Sci. U.S.A.*, 94:4504-4509 [1997]; Cramer et al., *Nat. Biotechnol.*, 14:315-319 [1996]; Stemmer, *Nature*, 370:389-391 [1994]; Stemmer, *Proc. Nat. Acad. Sci. USA*, 91:10747-10751 [1994]; WO 95/22625; WO 97/0078; WO 97/35966; WO 98/27230; WO 00/42651; WO 01/75767; and WO 2009/152336, all of which are incorporated herein by reference).

(81) In some embodiments, the enzyme clones obtained following mutagenesis treatment are screened by subjecting the enzyme preparations to a defined temperature (or other assay conditions) and measuring the amount of enzyme activity remaining after heat treatments or other suitable assay conditions. Clones containing a polynucleotide encoding a polypeptide are then isolated from the gene, sequenced to identify the nucleotide sequence changes (if any), and used to express the enzyme in a host cell. Measuring enzyme activity from the expression libraries can be performed using any suitable method known in the art (e.g., standard biochemistry techniques, such as HPLC analysis).

(82) After the variants are produced, they can be screened for any desired property (e.g., high or increased activity, or low or reduced activity, increased thermal activity, increased thermal stability, and/or acidic pH stability, etc.). In some embodiments, “recombinant glycosyltransferase polypeptides” (also referred to herein as “engineered glycosyltransferase polypeptides,” “variant glycosyltransferase enzymes,” “glycosyltransferase variants,” and “glycosyltransferase combinatorial variants”) find use. In some embodiments, “recombinant sucrose synthase polypeptides” (also referred to as “engineered sucrose synthase polypeptides,” “variant sucrose synthase enzymes,” “sucrose synthase variants,” and “sucrose synthase combinatorial variants”) find use.

(83) As used herein, a “vector” is a DNA construct for introducing a DNA sequence into a cell. In some embodiments, the vector is an expression vector that is operably linked to a suitable control sequence capable of effecting the expression in a suitable host of the polypeptide encoded in the DNA sequence. In some embodiments, an “expression vector” has a promoter sequence operably linked to the DNA sequence (e.g., transgene) to drive expression in a host cell, and in some embodiments, also comprises a transcription terminator sequence.

(84) As used herein, the term “expression” includes any step involved in the production of the polypeptide including, but not limited to, transcription, post-transcriptional modification, translation, and post-translational modification. In some embodiments, the term also encompasses secretion of the polypeptide from a cell.

(85) As used herein, the term “produces” refers to the production of proteins and/or other compounds by cells. It is intended that the term encompass any step involved in the production of polypeptides including, but not limited to, transcription, post-transcriptional modification, translation, and post-translational modification. In some embodiments, the term also encompasses secretion of the polypeptide from a cell.

(86) As used herein, an amino acid or nucleotide sequence (e.g., a promoter sequence, signal peptide, terminator sequence, etc.) is “heterologous” to another sequence with which it is operably linked if the two sequences are not associated in nature. For example a “heterologous polynucleotide” is any

polynucleotide that is introduced into a host cell by laboratory techniques, and includes polynucleotides that are removed from a host cell, subjected to laboratory manipulation, and then reintroduced into a host cell.

(87) As used herein, the terms “host cell” and “host strain” refer to suitable hosts for expression vectors comprising DNA provided herein (e.g., the polynucleotides encoding the glycosyltransferase variants). In some embodiments, the host cells are prokaryotic or eukaryotic cells that have been transformed or transfected with vectors constructed using recombinant DNA techniques as known in the art.

(88) The term “analogue” means a polypeptide having more than 70% sequence identity but less than 100% sequence identity (e.g., more than 75%, 78%, 80%, 83%, 85%, 88%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% sequence identity) with a reference polypeptide. In some embodiments, analogues means polypeptides that contain one or more non-naturally occurring amino acid residues including, but not limited, to homoarginine, ornithine and norvaline, as well as naturally occurring amino acids. In some embodiments, analogues also include one or more D-amino acid residues and non-peptide linkages between two or more amino acid residues.

(89) The term “effective amount” means an amount sufficient to produce the desired result. One of general skill in the art may determine what the effective amount by using routine experimentation.

(90) The terms “isolated” and “purified” are used to refer to a molecule (e.g., an isolated nucleic acid, polypeptide, etc.) or other component that is removed from at least one other component with which it is naturally associated. The term “purified” does not require absolute purity, rather it is intended as a relative definition.

(91) As used herein, “stereoselectivity” refers to the preferential formation in a chemical or enzymatic reaction of one stereoisomer over another. Stereoselectivity can be partial, where the formation of one stereoisomer is favored over the other, or it may be complete where only one stereoisomer is formed. When the stereoisomers are enantiomers, the stereoselectivity is referred to as enantioselectivity, the fraction (typically reported as a percentage) of one enantiomer in the sum of both. It is commonly alternatively reported in the art (typically as a percentage) as the enantiomeric excess (“e.e.”) calculated therefrom according to the formula $[\text{major enantiomer} - \text{minor enantiomer}] / [\text{major enantiomer} + \text{minor enantiomer}]$. Where the stereoisomers are diastereoisomers, the stereoselectivity is referred to as diastereoselectivity, the fraction (typically reported as a percentage) of one diastereomer in a mixture of two diastereomers, commonly alternatively reported as the diastereomeric excess (“d.e.”). Enantiomeric excess and diastereomeric excess are types of stereomeric excess.

(92) As used herein, “regioselectivity” and “regioselective reaction” refer to a reaction in which one direction of bond making or breaking occurs preferentially over all other possible directions. Reactions can completely (100%) regioselective if the discrimination is complete, substantially regioselective (at least 75%), or partially regioselective (x %, wherein the percentage is set dependent upon the reaction of interest), if the product of reaction at one site predominates over the product of reaction at other sites.

(93) As used herein, “thermostable” refers to a glycosyltransferase polypeptide that maintains similar activity (more than 60% to 80% for example) after exposure to elevated temperatures (e.g., 40-80° C.) for a period of time (e.g., 0.5-24 h) compared to the wild-type enzyme exposed to the same elevated temperature.

(94) As used herein, “solvent stable” refers to a glycosyltransferase polypeptide that maintains similar activity (more than e.g., 60% to 80%) after exposure to varying concentrations (e.g., 5-99%) of solvent (ethanol, isopropyl alcohol, dimethylsulfoxide [DMSO], tetrahydrofuran, 2-methyltetrahydrofuran, acetone, toluene, butyl acetate, methyl tert-butyl ether, etc.) for a period of time (e.g., 0.5-24 h) compared to the wild-type enzyme exposed to the same concentration of the same solvent.

(95) As used herein, “thermo- and solvent stable” refers to a glycosyltransferase polypeptide that is both thermostable and solvent stable.

(96) As used herein, “reductant” refers to a compound or agent capable of converting Fe³⁺ to Fe²⁺. An exemplary reductant is ascorbic acid, which is generally in the form of L-ascorbic acid.

(97) As used herein, “optional” and “optionally” mean that the subsequently described event or circumstance may or may not occur, and that the description includes instances where the event or circumstance occurs and instances in which it does not. One of ordinary skill in the art would understand that with respect to any molecule described as containing one or more optional substituents, only

sterically practical and/or synthetically feasible compounds are meant to be included. "Optionally substituted" refers to all subsequent modifiers in a term or series of chemical groups.

(98) Glycosylation

(99) Glycosylation can alter many properties of natural and synthetic products including stability, pharmacodynamics, solubility, and membrane transport. The present invention provides compositions, methods and enzymes suitable for generating new glycosylated compounds from various aglycone and glycosylated substrates. In some embodiments, the present invention provides means to efficiently generate known glycosylated compounds from easily obtained precursors. In some cases, glycosylation is achieved through chemical synthesis methods. However, these methods typically require undesirable chemicals and processes and can result in mixed products (e.g., with linkages in incorrect positions and/or with undesired anomeric configurations). Furthermore, carbohydrate chemistry requires multiple protection and deprotection steps.

(100) In contrast, glycosylating enzymes can be active under mild conditions and can confer high positional selectivity and stereospecificity in a single step. Many naturally-occurring glycosylated metabolites are generated in vivo using glycosyltransferases that transfer sugar moieties from various sugar nucleosides. Many molecules, including many secondary metabolites with antimicrobial, antitumor, natural sweetness properties, etc., comprise non-ribosomal peptide, polyketide, or terpenoid backbones modified with β -glycosidic linkages. Many of the diterpene glycosides extracted from the plant, *Stevia rebaudiana* Bertoni, contain β -linked glucose molecules. Naturally, these molecules are glycosylated in vivo using UDP-glucose dependent glycosyl transferase enzymes. The present invention provides a method (See, FIG. 1), in which a new engineered glycosyltransferase is used to transfer the glucose moiety from a nucleoside diphosphoglucose to a substrate (e.g., rebaudioside D or stevioside), to produce one or more β -glucose linked products (e.g., rebaudioside M, rebaudioside A, or rebaudioside I). However, when used in vitro, the UDP-glucose can be prohibitively expensive and/or unavailable. In the some additional embodiments, a synthase (e.g., sucrose synthase or trehalose synthase) acts in the reverse direction to form a nucleoside diphosphoglucose compound from a nucleoside diphosphate and a glucose donor (e.g., sucrose, trehalose, or starch).

(101) Thus, glycosylation finds use in the production of natural sweeteners, such as those derived from the sweet herb, *Stevia rebaudiana* Bertoni. As indicated above, this plant produces a number of diterpene glycosides which feature high intensity sweetness and sensory properties superior to those of many other high potency sweeteners. The above-mentioned sweet glycosides, have a common aglycone (i.e., steviol), and differ by the number and type of carbohydrate residues at the C13 and C19 positions. Steviol glycosides differ from each other not only in their molecular structure, but also by their taste properties. Usually, stevioside is reported to be 89-143 times sweeter than sucrose, while rebaudioside A is reported to be between 85 and 242 times sweeter than sucrose (See e.g., Kasai et al., Nippon Kagaku Kaishi, 1981:726-735 [1981]). Of these common compounds, rebaudioside A has the least astringent, the least bitter, and the least persistent aftertaste. Thus, it has the most favorable sensory attributes of the major steviol glycosides and has been commercialized. However, rebaudioside A only constitutes a smaller fraction (about 20%) of total glycosides isolated from *Stevia rebaudiana* Bertoni, with stevioside (about 70%) and minor steviol glycosides making up the rest (See e.g., FAO, Chemical and Technical Assessment, 63.sup.rd JECFA, Steviol Glycosides [2004]). The naturally occurring but even less abundant compound rebaudioside M, also known as rebaudioside X, is 200-350 times sweeter than sucrose and has a reduced aftertaste relative to rebaudioside A (See e.g., Prakash et al., Food, 3:162-175 [2014]). Thus, there is interest in the commercialization of rebaudioside M, for example as a natural sweetener, but currently no viable commercial route to synthesize this compound.

(102) Engineered Glycosyltransferase Polypeptides

(103) The present invention provides glycosyltransferase polypeptides, polynucleotides encoding the polypeptides, methods of preparing the polypeptides, and methods for using the polypeptides. Where the description relates to polypeptides, it is to be understood that it also describes the polynucleotides encoding the polypeptides. In some embodiments, the present invention provides engineered, non-naturally occurring GT enzymes with improved properties as compared to wild-type GT enzymes. Any suitable reaction conditions find use in the present invention. In some embodiments, methods are used to analyze the improved properties of the engineered polypeptides to carry out the transferase reaction. In

some embodiments, the reaction conditions are modified with regard to concentrations or amounts of polypeptide, substrate, co-substrate, buffer, co-solvent, pH, conditions including temperature and reaction time, and/or conditions with the polypeptide immobilized on a solid support, as further described below and in the Examples.

(104) In some embodiments, the engineered GT polypeptides described herein have improved properties as compared to wild-type GT enzymes such as in the conversion of steviol glycosides to further glycosylated steviol glycosides (e.g., stevioside to rebaudioside A or rebaudioside D to rebaudioside M) and in the use of adenine diphosphoglucose or other nucleoside diphosphates. In some embodiments, the engineered GT enzymes comprise amino acid sequences having one or more residue differences as compared to SEQ ID NO: 4, 8, 32, 232, 348, 548, 562, 696, 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 4256, 4684, 4838, 4876, 5066, 5290, 5372, 5562, 5708, 5976, 6138, 6288, 6468, 684, 7388, and/or 8088. In some embodiments, the engineered GT enzymes are beta-1,2 glycosyltransferase variants having one or more residue differences as compared to SEQ ID NO: 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 3956, 4256, 4550, 7324, and/or 7784. In some embodiments, the engineered GT enzymes are beta-1,3 glycosyltransferase variants having one or more residue differences as compared to SEQ ID NO: 4, 8, 32, 232, 348, 548, 562, 696, 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 3956, 4256, 4550, 4684, 4838, 4876, 5066, 5290, 5372, 5562, 5708, 5976, 6138, 6288, 6468, 6864, 7324, 7388, and/or 8088. In some further embodiments, engineered GT polypeptides are circular permuted proteins in which a peptide linker is incorporated at the genetic level between the N- and C-termini and new amino acid positions are selected as the location of the new N- and C-termini. In some embodiments, the circular permuted GT enzymes comprise amino acid sequences having one or more residue differences as compared to SEQ ID NO: 4, 8, 32, 232, 348, 548, 562, 696, 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 3956, 4256, 4550, 4684, 4838, 4876, 5066, 5290, 5372, 5562, 5708, 5976, 6138, 6288, 6468, 6864, 7324, 7388, 7784, and/or 8088. In some embodiments, the circular permuted GT enzymes comprise amino acid sequences having one or more residue differences as compared to SEQ ID NO: 32. In some embodiments, the GT enzymes comprise at least one sequence set forth herein. In some embodiments, the present invention provides engineered glycosyltransferases comprising polypeptide sequences that have at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NOS: 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, 120, 122, 124, 126, 128, 130, 132, 134, 136, 138, 140, 142, 144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 172, 174, 176, 178, 180, 182, 184, 186, 188, 190, 192, 194, 196, 198, 200, 202, 204, 206, 208, 210, 212, 214, 216, 218, 220, 222, 224, 226, 228, 230, 232, 234, 236, 238, 240, 242, 244, 246, 248, 250, 252, 254, 256, 258, 260, 262, 264, 266, 268, 270, 272, 274, 276, 278, 280, 282, 284, 286, 288, 290, 292, 294, 296, 298, 300, 302, 304, 306, 308, 310, 312, 314, 316, 318, 320, 322, 324, 326, 328, 330, 332, 334, 336, 338, 340, 342, 344, 346, 348, 350, 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, 384, 386, 388, 390, 392, 394, 396, 398, 400, 402, 404, 406, 408, 410, 412, 414, 416, 418, 420, 422, 424, 426, 428, 430, 432, 434, 436, 438, 440, 442, 444, 446, 448, 450, 452, 454, 456, 458, 460, 462, 464, 466, 468, 470, 472, 474, 476, 478, 480, 482, 484, 486, 488, 490, 492, 494, 496, 498, 500, 502, 504, 506, 508, 510, 512, 514, 516, 518, 520, 522, 524, 526, 528, 530, 532, 534, 536, 538, 540, 542, 544, 546, 548, 550, 552, 554, 556, 558, 560, 562, 564, 566, 568, 570, 572, 574, 576, 578, 580, 582, 584, 586, 588, 590, 592, 594, 596, 598, 600, 602, 604, 606, 608, 610, 612, 614, 616, 618, 620, 622, 624, 626, 628, 630, 632, 634, 636, 638, 640, 642, 644, 646, 648, 650, 652, 654, 656, 658, 660, 662, 664, 666, 668, 670, 672, 674, 676, 678, 680, 682, 684, 686, 688, 690, 692, 694, 696, 698, 700, 702, 704, 706, 708, 710, 712, 714, 716, 718, 720, 722, 724, 726, 728, 730, 732, 734, 736, 738, 740, 742, 744, 746, 748, 750, 752, 754, 770, 772, 774, 776, 778, 780, 782, 784, 786, 788, 790, 792, 794, 796, 798, 800, 802, 804, 806, 808, 810, 812, 814, 816, 818, 820, 822, 824, 826, 828, 830, 832, 834, 836, 838, 840, 842, 844, 846, 848, 850, 852, 854, 856, 858, 860, 862, 864, 866, 868, 870, 872, 874, 876, 878, 880, 882, 884, 886, 888, 890, 892, 894, 896, 898, 900, 902, 904, 906, 908, 910, 912, 914, 916, 918, 920, 922, 924, 926, 928, 930, 932, 934, 936, 938, 940, 942, 944, 946, 948, 950, 952, 954, 956, 958, 960, 962, 964, 966, 968, 970, 972, 974, 976, 978, 980, 982, 984, 986, 988, 990, 992, 994, 996, 998, 1000, 1002, 1004, 1006, 1008, 1010, 1012, 1014, 1016,

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In some additional embodiments, the present invention further provides engineered sucrose synthases comprising polypeptide sequences that have at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NOS: 1080, 1082, 1084, 1086, 1088, 1090, 1092, 1094, 1096, 1098, 1100, 1102, 1104, 1106, 1108, 1110, 1112, 1114, 1116, 1118, 1120, 1122, 1124, 1126, 1128, 1130, 1132, 1134, 1136, 1138, 1140, 1142, 1144, 1146, 1148, 1150, 1152, 1154, 1156, 1158, 1160, 1162, 1164, 1166, 1168, 1170, 1172, 1174, 1176, 1178, 1180, 1182, 1184, 1186, 1188, 1190, 1192, 1194, 1196, 1198, 1200, 1202, 1204, 1206, 1208, 1210, 1212, 1214, 1216, 1218, 1220, 1222, 1224, 1226, 1228, 1230, 1232, 1234, 1236, 1238, 1240, 1242, 1244, 1246, 1248, 1250, 1252, 1254, 1256, 1258, 1260, 1262, 1264, 1266, 1268, 1270, 1272, 1274, 1276, 1278, 1280, 1282, 1284, 1286, 1288, 1296, 1298, 1300, 1302, 1304, 1306, 1308, 1310, 1312, 1314, 1316, 1318, 1320, 1322, 1324, 1326, 1328, 1330, 1332, 1334, 1336, 1338, 1340, 1342, 1344, 1346, 1348, 1350, 1352, 1354, 1356, 1358, 1360, 1362, 1364, 1366, 1368, 1370, 1372, 1374, 1376, 1378, 1380, 1382, 1384, 1386, 1388, 1390, 1392, 1394, 1396, 1398, 1400, 1402, 1404, 1406, 1408, 1410, 1412, 1414, 1416, 1418, 1420, 1422, 1424, 1426, 1428, 1430, 1432, 1434, 1436, 1438, 1440, 1442, 1444, 1446, 1448, 1450, 1452, 1454, 1456, 1458, 1460, 1462, 1464, 1466, 1468, 1470, 1472, 1474, 1476, 1478, 1480, 1482, 1484, 1486, 1488, 1490, 1492, 1494, 1496, 1498, 1500, 1502, 1504, 1506, 1508, 1510, 1512, 1514, 1516, 1518, 1520, 1522, 1524, 1526, 1528, 1530, 1532, 1534, 1536, 1538, 1540, 1542, 1544, 1546, 1548, 1550, 1552, 1554, 1556, 1558, 1560, 1562, 1564, 1566, 1568, 1570, 1572, 1574, 1576, 1578, 1580, 1582, 1584, 1586, 1588, 1590, 1592, 1594, 1596, 1598, 1600, 1602, 1604, 1606, 1608, 1610, 1612, 1614, 1616, 1618, 1620, 1622, 1624, 1626, 1628, 1630, 1632, 1634, 1636, 1638, 1640, 1642, 1644, 1646, 1648, 1650, 1652, 1654, 1656, 1658, 1660, 1662, 1664, 1666, 1668, 1670, 1672, 1674, 1676, 1678, 1680, 1682, 1684, 1686, 1688, 1690, 1692, 1694, 1696, 1698, 1700, 1702, 1704, 1706, 1708, 1710, 1712, 1714, 1716, 1718, 1720, 1722, 1724, 1726, 1728, 1730, 1732, 1734, 1736, 1738, 1740, 1742, 1744, 1746, 1748, 1750, 1752, 1754, 1756, 1758, 1760, 1762, 1764, 1766, 1768, 1770, 1772, 1774, 1776, 1778, 1780, 1782, 1784, 1786, 1788, 1790, 1792, 1794, 1796, 1798, 1800, 1802, 1804, 1806, 1808, 1810, 1812, 1814, 1816, 1818, 1820, 1822, 1824, 1826, 1828, 1830, 1832, 1834, 1836, 1838, 1840, 1842, 1844, 1846, 1848, 1850, 1852, 1854, 1856, 1858, 1860, 1862, 1864, 1866, 1868, 1870, 1872, 1874, 1876, 1878, 1880, 1882, 1884, 1886, 1888, 1890, 1892, 1894, 1896, 1898, 1900, 1902, 1904, 1906, 1908, 1910, 1912, 1914, 1916, 1918, 1920, 1922, 1924, 1926, 1928, 1930, 1932, 1934, 1936, 1938, 1940, 1942, 1944, 1946, 1948, 1950, 1952, 1954, 1956, 1958, 1960, 1962, 1964, 1966, 1968, 1970, 1972, 1974, 1976, 1978, 1980, 1982, 1984, 1986, 1988, 1990, 1992, 1994, 1996, 1998, 2000, 2002, 2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022, 2024, 2026, 2028, 2030, 2032, 2034, 2036, 2038, 2040, 2042, 2044, 2046, 2048, 2050, 2052, 2054, 2056, 2058, 2060, 2062, 2064, 2066, 2068, 2070, 2072, 2074, 2076, 2078, 2080, 2082, 2084, 2086, 2088, 2090, 2092, 2094, 2096, 2098, 2100, 2102, 2104, 2106, 2108, 2110, 2112, 2114, 2116, 2118, 2120, 2122, 2124, 2126, 2128, 2130, 2132, 2134, 2136, 2138, 2140, 2142, 2144, 2146, 2148, 2150, 2152, 2154, 2156, 2158, 2160, 2162, 2164, 2166, 2168, 2170, 2172, 2174, 2176, 2178, 2180, 2182, 2184, 2186, 2188, 2190, 2192, 2194, 2196, 2198, 2200, 2202, 2204, 2206, 2208, 2210, 2212, 2214, 2216, 2218, 2220, 2222, 2224, 2226, 2228, 2230, 2232, 2234, 2236, 2238, 2240, 2242, 2244, 2246, 2248, 2250, 2252,

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(105) In some embodiments, additional reaction components or additional techniques are utilized to supplement the reaction conditions. In some embodiments, these include taking measures to stabilize or prevent inactivation of the enzyme, reduce product inhibition, shift reaction equilibrium to glucosylated product formation.

(106) In some further embodiments, any of the above described process for the conversion of substrate compound to product compound can further comprise one or more steps selected from: extraction; isolation; purification; and crystallization, filtration, or lyophilization of product compound. Methods, techniques, and protocols for extracting, isolating, purifying, and/or crystallizing the glucosylated product from biocatalytic reaction mixtures produced by the above disclosed processes are known to the ordinary artisan and/or accessed through routine experimentation. Additionally, illustrative methods are provided in the Examples below.

(107) Engineered Sucrose Synthase Polypeptides

(108) The present invention provides engineered sucrose synthase (SuS) polypeptides, polynucleotides encoding the polypeptides, methods of preparing the polypeptides, and methods for using the polypeptides. Where the description relates to polypeptides, it is to be understood that it also describes the polynucleotides encoding the polypeptides. In some embodiments, the present invention provides engineered, non-naturally occurring SuS enzymes with improved properties as compared to wild-type SuS enzymes. Any suitable reaction conditions find use in the present invention. In some embodiments, methods are used to analyze the improved properties of the engineered polypeptides to carry out the synthase reaction. In some embodiments, the reaction conditions are modified with regard to

concentrations or amounts of engineered SuS, substrate(s), solvent(s), pH, conditions including temperature and reaction time, and/or conditions with the engineered SuS polypeptide immobilized on a solid support, as further described below and in the Examples.

(109) In some embodiments, the engineered SuS polypeptides described herein have improved properties as compared to wild-type SuS enzymes such as in the reactions described herein. In some embodiments, the engineered SuS enzymes comprise amino acid sequences having one or more residue differences as compared to SEQ ID NO: 74, 1080, 1158, 1222, 1392, 1456, 1582, 1764, 1804, 1840, 2064, 2432, 2510, 7506, and/or 8420.

(110) In some embodiments, the present invention provides engineered SuS enzymes, wherein the polypeptide of the SuS enzymes comprise at least 60%, 65%, 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NO:1080, 1082, 1084, 1086, 1088, 1090, 1092, 1094, 1096, 1098, 1100, 1102, 1104, 1106, 1108, 1110, 1112, 1114, 1116, 1118, 1120, 1122, 1124, 1126, 1128, 1130, 1132, 1134, 1136, 1138, 1140, 1142, 1144, 1146, 1148, 1150, 1152, 1154, 1156, 1158, 1160, 1162, 1164, 1166, 1168, 1170, 1172, 1174, 1176, 1178, 1180, 1182, 1184, 1186, 1188, 1190, 1192, 1194, 1196, 1198, 1200, 1202, 1204, 1206, 1208, 1210, 1212, 1214, 1216, 1218, 1220, 1222, 1224, 1226, 1228, 1230, 1232, 1234, 1236, 1238, 1240, 1242, 1244, 1246, 1248, 1250, 1252, 1254, 1256, 1258, 1260, 1262, 1264, 1266, 1268, 1270, 1272, 1274, 1276, 1278, 1280, 1282, 1284, 1286, 1288, 1296, 1298, 1300, 1302, 1304, 1306, 1308, 1310, 1312, 1314, 1316, 1318, 1320, 1322, 1324, 1326, 1328, 1330, 1332, 1334, 1336, 1338, 1340, 1342, 1344, 1346, 1348, 1350, 1352, 1354, 1356, 1358, 1360, 1362, 1364, 1366, 1368, 1370, 1372, 1374, 1376, 1378, 1380, 1382, 1384, 1386, 1388, 1390, 1392, 1394, 1396, 1398, 1400, 1402, 1404, 1406, 1408, 1410, 1412, 1414, 1416, 1418, 1420, 1422, 1424, 1426, 1428, 1430, 1432, 1434, 1436, 1438, 1440, 1442, 1444, 1446, 1448, 1450, 1452, 1454, 1456, 1458, 1460, 1462, 1464, 1466, 1468, 1470, 1472, 1474, 1476, 1478, 1480, 1482, 1484, 1486, 1488, 1490, 1492, 1494, 1496, 1498, 1500, 1502, 1504, 1506, 1508, 1510, 1512, 1514, 1516, 1518, 1520, 1522, 1524, 1526, 1528, 1530, 1532, 1534, 1536, 1538, 1540, 1542, 1544, 1546, 1548, 1550, 1552, 1554, 1556, 1558, 1560, 1562, 1564, 1566, 1568, 1570, 1572, 1574, 1576, 1578, 1580, 1582, 1584, 1586, 1588, 1590, 1592, 1594, 1596, 1598, 1600, 1602, 1604, 1606, 1608, 1610, 1612, 1614, 1616, 1618, 1620, 1622, 1624, 1626, 1628, 1630, 1632, 1634, 1636, 1638, 1640, 1642, 1644, 1646, 1648, 1650, 1652, 1654, 1656, 1658, 1660, 1662, 1664, 1666, 1668, 1670, 1672, 1674, 1676, 1678, 1680, 1682, 1684, 1686, 1688, 1690, 1692, 1694, 1696, 1698, 1700, 1702, 1704, 1706, 1708, 1710, 1712, 1714, 1716, 1718, 1720, 1722, 1724, 1726, 1728, 1730, 1732, 1734, 1736, 1738, 1740, 1742, 1744, 1746, 1748, 1750, 1752, 1754, 1756, 1758, 1760, 1762, 1764, 1766, 1768, 1770, 1772, 1774, 1776, 1778, 1780, 1782, 1784, 1786, 1788, 1790, 1792, 1794, 1796, 1798, 1800, 1802, 1804, 1806, 1808, 1810, 1812, 1814, 1816, 1818, 1820, 1822, 1824, 1826, 1828, 1830, 1832, 1834, 1836, 1838, 1840, 1842, 1844, 1846, 1848, 1850, 1852, 1854, 1856, 1858, 1860, 1862, 1864, 1866, 1868, 1870, 1872, 1874, 1876, 1878, 1880, 1882, 1884, 1886, 1888, 1890, 1892, 1894, 1896, 1898, 1900, 1902, 1904, 1906, 1908, 1910, 1912, 1914, 1916, 1918, 1920, 1922, 1924, 1926, 1928, 1930, 1932, 1934, 1936, 1938, 1940, 1942, 1944, 1946, 1948, 1950, 1952, 1954, 1956, 1958, 1960, 1962, 1964, 1966, 1968, 1970, 1972, 1974, 1976, 1978, 1980, 1982, 1984, 1986, 1988, 1990, 1992, 1994, 1996, 1998, 2000, 2002, 2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022, 2024, 2026, 2028, 2030, 2032, 2034, 2036, 2038, 2040, 2042, 2044, 2046, 2048, 2050, 2052, 2054, 2056, 2058, 2060, 2062, 2064, 2066, 2068, 2070, 2072, 2074, 2076, 2078, 2080, 2082, 2084, 2086, 2088, 2090, 2092, 2094, 2096, 2098, 2100, 2102, 2104, 2106, 2108, 2110, 2112, 2114, 2116, 2118, 2120, 2122, 2124, 2126, 2128, 2130, 2132, 2134, 2136, 2138, 2140, 2142, 2144, 2146, 2148, 2150, 2152, 2154, 2156, 2158, 2160, 2162, 2164, 2166, 2168, 2170, 2172, 2174, 2176, 2178, 2180, 2182, 2184, 2186, 2188, 2190, 2192, 2194, 2196, 2198, 2200, 2202, 2204, 2206, 2208, 2210, 2212, 2214, 2216, 2218, 2220, 2222, 2224, 2226, 2228, 2230, 2232, 2234, 2236, 2238, 2240, 2242, 2244, 2246, 2248, 2250, 2252, 2254, 2256, 2258, 2260, 2262, 2264, 2266, 2268, 2270, 2272, 2274, 2276, 2278, 2280, 2282, 2284, 2286, 2288, 2290, 2292, 2294, 2296, 2298, 2300, 2302, 2304, 2306, 2308, 2310, 2312, 2314, 2316, 2318, 2320, 2322, 2324, 2326, 2328, 2330, 2332, 2334, 2336, 2338, 2340, 2342, 2344, 2346, 2348, 2350, 2352, 2354, 2356, 2358, 2360, 2362, 2364, 2366, 2368, 2370, 2372, 2374, 2376, 2378, 2380, 2382, 2384, 2386, 2388, 2390, 2392, 2394, 2396, 2398, 2400, 2402, 2404, 2406, 2408, 2410, 2412, 2414, 2416, 2418, 2420, 2422, 2424, 2426, 2428, 2430, 2432, 2434, 2436, 2438, 2440, 2442, 2444, 2446, 2448, 2450, 2452, 2454, 2456, 2458, 2460, 2462, 2464, 2466, 2468, 2470, 2472, 2474, 2476, 2478, 2480, 2482, 2484, 2486, 2488, 2490, 2492, 2494, 2496, 2498, 2500, 2502, 2504, 2506, 2508, 2510,

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(111) In some embodiments, additional reaction components or additional techniques are utilized to supplement the reaction conditions. In some embodiments, these include taking measures to stabilize or prevent inactivation of the enzyme, reduce product inhibition, shift reaction equilibrium to glucosylated product formation.

(112) In some further embodiments, any of the above described process for the conversion of substrate compound to product compound can further comprise one or more steps selected from: extraction, isolation, purification, crystallization, filtration, and/or lyophilization of product compound(s). Methods, techniques, and protocols for extracting, isolating, purifying, and/or crystallizing the product(s) (e.g., rebaudiosides) from biocatalytic reaction mixtures produced by the processes provided herein are known to the ordinary artisan and/or accessed through routine experimentation. Additionally, illustrative methods are provided in the Examples below.

(113) Polynucleotides Encoding Engineered Polypeptides, Expression Vectors and Host Cells

(114) The present invention provides polynucleotides encoding the engineered enzyme polypeptides described herein. In some embodiments, the polynucleotides are operatively linked to one or more heterologous regulatory sequences that control gene expression to create a recombinant polynucleotide capable of expressing the polypeptide. In some embodiments, expression constructs containing at least one heterologous polynucleotide encoding the engineered enzyme polypeptide(s) is introduced into appropriate host cells to express the corresponding enzyme polypeptide(s).

(115) As will be apparent to the skilled artisan, availability of a protein sequence and the knowledge of the codons corresponding to the various amino acids provide a description of all the polynucleotides capable of encoding the subject polypeptides. The degeneracy of the genetic code, where the same amino acids are encoded by alternative or synonymous codons, allows an extremely large number of nucleic acids to be made, all of which encode an engineered enzyme (e.g., GT or SuS) polypeptide. Thus, the present invention provides methods and compositions for the production of each and every possible variation of enzyme polynucleotides that could be made that encode the enzyme polypeptides described herein by selecting combinations based on the possible codon choices, and all such variations are to be considered specifically disclosed for any polypeptide described herein, including the amino acid sequences presented in the Examples (e.g., in the various Tables).

(116) In some embodiments, the codons are preferably optimized for utilization by the chosen host cell for protein production. For example, preferred codons used in bacteria are typically used for expression in bacteria. Consequently, codon optimized polynucleotides encoding the engineered enzyme polypeptides contain preferred codons at about 40%, 50%, 60%, 70%, 80%, 90%, or greater than 90% of the codon positions in the full length coding region.

(117) In some embodiments, the enzyme polynucleotide encodes an engineered polypeptide having enzyme activity with the properties disclosed herein, wherein the polypeptide comprises an amino acid sequence having at least 60%, 65%, 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more identity to a reference sequence selected from the SEQ ID NOS provided herein, or the amino acid sequence of any variant (e.g., those provided in the Examples), and one or more residue differences as compared to the reference polynucleotide(s), or the amino acid sequence of any variant as disclosed in the Examples (for example 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more amino acid residue positions). In some embodiments, the reference polypeptide sequence is selected from SEQ ID NOS: 4, 8, 32, 232, 348, 548, 562, 696, 758, 770, 792, 954, 1002, 1054, 2600, 2718, 2814, 2884, 3016, 3082, 3244, 3346, 3502, 3696, 3956, 4256, 4550, 4684, 4838, 4876, 5066, 5290, 5372, 5562, 5708, 5976, 6138, 6288, 6468, 6864, 7324, 7388, 7784, and/or 8088. In some alternative embodiments, the reference polypeptide sequence is selected from SEQ ID NOS: 74, 1080, 1158, 1222, 1392, 1456, 1582, 1764, 1804, 1840, 2064, 2432, 2510, 7506, and/or 8420.

(118) In some embodiments, the reference polynucleotide sequence is selected from SEQ ID NOS: 3, 7, 31, 231, 347, 547, 561, 695, 757, 769, 791, 953, 1001, 1053, 2599, 2717, 2813, 2883, 3015, 3081, 3243, 3345, 3501, 3695, 3955, 4255, 4549, 4683, 4837, 4875, 5065, 5289, 5371, 5561, 5707, 5975, 6137, 6287, 6467, 6863, 7323, 7387, 7783, and/or 8087. In some embodiments, the reference polynucleotide sequence is selected from SEQ ID NOS: 757, 769, 791, 953, 1001, 1053, 2599, 2717, 2813, 2883, 3015, 3081, 3243, 3345, 3501, 3695, 3955, 4255, 4549, 7323, and/or 7783, while in some alternative embodiments, the reference polynucleotide sequence is selected from SEQ ID NOS: 3, 7, 31, 231, 347, 547, 561, 695, 4683, 4837, 4875, 5065, 5289, 5371, 5561, 5707, 5975, 6137, 6287, 6467, 6863, 7387, and/or 8087. In still some additional embodiments, the reference polynucleotide sequence is selected from SEQ ID NOS: 73, 1079, 1157, 1221, 1391, 1455, 1581, 1763, 1803, 1839, 2063, 2431, 2509, 7505, and/or 8419.

(119) In some embodiments, the engineered polynucleotide encoding at least one engineered glycosyltransferase provided herein comprises at least 60%, 65%, 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NO: 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 75, 77, 79, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, 119, 121, 123, 125, 127, 129, 131, 133, 135, 137, 139, 141, 143, 145, 147, 149, 151, 153, 155, 157, 159, 161, 163, 165, 167, 169, 171, 173, 175, 177, 179, 181, 183, 185, 187, 189, 191, 193, 195, 197, 199, 201, 203, 205, 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 239, 241, 243, 245, 247, 249, 251, 253, 255, 257, 259, 261, 263, 265, 267, 269, 271, 273, 275, 277, 279, 281, 283, 285, 287, 289, 291, 293, 295, 297, 299, 301, 303, 305, 307, 309, 311, 313, 315, 317, 319, 321, 323, 325, 327, 329, 331, 333, 335, 337, 339, 341, 343, 345, 347, 349, 351, 353, 355, 357, 359, 361, 363, 365, 367, 369, 371, 373, 375, 377, 379, 381, 383, 385, 387, 389, 391, 393, 395, 397, 399, 401, 403, 405, 407, 409, 411, 413, 415, 417, 419, 421, 423, 425, 427, 429, 431, 433, 435, 437, 439, 441, 443, 445, 447, 449, 451, 453, 455, 457, 459, 461, 463, 465, 467, 469, 471, 473, 475, 477, 479, 481, 483, 485, 487, 489, 491, 493, 495, 497, 499, 501, 503, 505, 507, 509, 511, 513, 515, 517, 519, 521, 523, 525, 527, 529, 531, 533, 535, 537, 539, 541, 543, 545, 547, 549, 551, 553, 555, 557, 559, 561, 563, 565, 567, 569, 571, 573, 575, 577, 579, 581, 583, 585, 587, 589, 591, 593, 595, 597, 599, 601, 603, 605, 607, 609, 611, 613, 615, 617, 619, 621, 623, 625, 627, 629, 631, 633, 635, 637, 639, 641, 643, 645, 647, 649, 651, 653, 655, 657, 659, 661, 663, 665, 667, 669, 671, 673, 675, 677, 679, 681, 683, 685, 687, 689, 691, 693, 695, 697, 699, 701, 703, 705, 707, 709, 711, 713, 715, 717, 719, 721, 723, 725, 727, 729, 731, 733, 735, 737, 739, 741, 743, 745, 747, 749, 751, 753, 769, 771, 773, 775, 777, 779, 781, 783, 785, 787, 789, 791, 793, 795, 797, 799, 801, 803, 805, 807, 809, 811, 813, 815, 817, 819, 821, 823, 825, 827, 829, 831, 833, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 859, 861, 863, 865, 867, 869, 871, 873, 875, 877, 879, 881, 883, 885, 887, 889, 891, 893, 895, 897, 899, 901, 903, 905, 907, 909, 911, 913, 915, 917, 919, 921, 923, 925, 927, 929, 931, 933, 935, 937, 939, 941, 943, 945, 947, 949, 951, 953, 955, 957, 959, 961, 963, 965, 967,

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8231, 8233, 8235, 8237, 8239, 8241, 8243, 8245, 8247, 8249, 8251, 8253, 8255, 8257, 8259, 8261, 8263, 8265, 8267, 8269, 8271, 8273, 8275, 8277, 8279, 8281, 8283, 8285, 8287, 8289, 8291, 8293, 8295, 8297, 8299, 8301, 8303, 8305, 8307, 8309, 8311, 8313, 8315, 8317, 8319, 8321, 8323, 8325, 8327, 8329, 8331, 8333, 8335, 8337, 8339, 8341, 8343, 8345, 8347, 8349, 8351, 8353, 8355, 8357, 8359, 8361, 8363, 8365, 8367, 8481, 8483, 8485, 8487, 8489, 8491, 8493, 8495, 8497, 8489, 8501, 8503, 8505, 8507, 8509, 8511, 8513, 8515, 8517, 8519, 8521, 8523, 8525, 8527, 8529, 8531, 8533, 8535, 8537, 8539, 8541, 8543, 8545, 8547, 8549, 8551, 8553, 8555, 8557, 8559, 8561, 8563, 8565, 8567, 8569, 8571, 8573, 8575, 8577, 8579, 8581, 8583, 8585, 8587, 8589, 8591, 8593, 8595, 8597, 8599, 8601, 8603, 8605, 8607, 8609, 8611, 8613, 8615, 8617, 8619, 8621, 8623, 8625, 8627, 8629, 8631, 8633, 8635, 8637, 8639, 8641, 8643, 8645, 8647, 8649, 8651, 8653, 8655, 8657, 8659, 8661, 8663, 8665, 8667, 8669, 8671, 8673, 8675, 8677, 8679, 8681, 8683, 8685, 8687, 8689, 8691, 8693, 8695, 8697, 8699, 8701, 8703, 8705, 8707, 8709, 8711, 8713, 8715, 8717, 8719, 8721, 8723, 8725, 8727, 8729, 8731, 8733, 8735, 8737, 8739, 8741, 8743, 8745, 8747, 8749, 8751, 8753, 8755, 8757, 8759, 8761, 8763, 8765, 8767, 8769, 8771, 8773, 8775, 8777, 8779, 8781, 8783, 8785, 8787, 8789, 8791, 8793, 8795, 9107, 9109, 9111, 9113, 9115, 9117, 9119, 9121, 9123, 9125, 9127, 9129, 9131, 9133, 9135, 9137, 9139, 9141, 9143, 9145, 9147, 9149, 9151, 9153, 9155, 9157, 9159, 9161, 9163, 9165, 9167, 9169, 9171, 9173, 9175, 9177, 9179, 9181, 9183, 9185, 9187, 9189, 9191, 9193, 9195, 9197, 9199, 9201, 9203, 9205, 9207, 9209, 9211, 9213, 9215, 9217, 9219, 9221, 9223, 9225, 9227, 9229, 9231, 9233, 9235, 9237, and/or 9239.

(120) In some additional embodiments, the engineered polynucleotide encoding at least one engineered sucrose synthase provided herein comprises at least 60%, 65%, 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NO: 1079, 1081, 1083, 1085, 1087, 1089, 1091, 1093, 1095, 1097, 1099, 1101, 1103, 1105, 1107, 1109, 1111, 1113, 1115, 1117, 1119, 1121, 1123, 1125, 1127, 1129, 1131, 1133, 1135, 1137, 1139, 1141, 1143, 1145, 1147, 1149, 1151, 1153, 1155, 1157, 1159, 1161, 1163, 1165, 1167, 1169, 1171, 1173, 1175, 1177, 1179, 1181, 1183, 1185, 1187, 1189, 1191, 1193, 1195, 1197, 1199, 1201, 1203, 1205, 1207, 1209, 1211, 1213, 1215, 1217, 1219, 1221, 1223, 1225, 1227, 1229, 1231, 1233, 1235, 1237, 1239, 1241, 1243, 1245, 1247, 1249, 1251, 1253, 1255, 1257, 1259, 1261, 1263, 1265, 1267, 1269, 1271, 1273, 1275, 1277, 1279, 1281, 1283, 1285, 1287, 1295, 1297, 1299, 1301, 1303, 1305, 1307, 1309, 1311, 1313, 1315, 1317, 1319, 1321, 1323, 1325, 1327, 1329, 1331, 1333, 1335, 1337, 1339, 1341, 1343, 1345, 1347, 1349, 1351, 1353, 1355, 1357, 1359, 1361, 1363, 1365, 1367, 1369, 1371, 1373, 1375, 1377, 1379, 1381, 1383, 1385, 1387, 1389, 1391, 1393, 1395, 1397, 1399, 1401, 1403, 1405, 1407, 1409, 1411, 1413, 1415, 1417, 1419, 1421, 1423, 1425, 1427, 1429, 1431, 1433, 1435, 1437, 1439, 1441, 1443, 1445, 1447, 1449, 1451, 1453, 1455, 1457, 1459, 1461, 1463, 1465, 1467, 1469, 1471, 1473, 1475, 1477, 1479, 1481, 1483, 1485, 1487, 1489, 1491, 1493, 1495, 1497, 1499, 1501, 1503, 1505, 1507, 1509, 1511, 1513, 1515, 1517, 1519, 1521, 1523, 1525, 1527, 1529, 1531, 1533, 1535, 1537, 1539, 1541, 1543, 1545, 1547, 1549, 1551, 1553, 1555, 1557, 1559, 1561, 1563, 1565, 1567, 1569, 1571, 1573, 1575, 1577, 1579, 1581, 1583, 1585, 1587, 1589, 1591, 1593, 1595, 1597, 1599, 1601, 1603, 1605, 1607, 1609, 1611, 1613, 1615, 1617, 1619, 1621, 1623, 1625, 1627, 1629, 1631, 1633, 1635, 1637, 1639, 1641, 1643, 1645, 1647, 1649, 1651, 1653, 1655, 1657, 1659, 1661, 1663, 1665, 1667, 1669, 1671, 1673, 1675, 1677, 1679, 1681, 1683, 1685, 1687, 1689, 1691, 1693, 1695, 1697, 1699, 1701, 1703, 1705, 1707, 1709, 1711, 1713, 1715, 1717, 1719, 1721, 1723, 1725, 1727, 1729, 1731, 1733, 1735, 1737, 1739, 1741, 1743, 1745, 1747, 1749, 1751, 1753, 1755, 1757, 1759, 1761, 1763, 1765, 1767, 1769, 1771, 1773, 1775, 1777, 1779, 1781, 1783, 1785, 1787, 1789, 1791, 1793, 1795, 1797, 1799, 1801, 1803, 1805, 1807, 1809, 1811, 1813, 1815, 1817, 1819, 1821, 1823, 1825, 1827, 1829, 1831, 1833, 1835, 1837, 1839, 1841, 1843, 1845, 1847, 1849, 1851, 1853, 1855, 1857, 1859, 1861, 1863, 1865, 1867, 1869, 1871, 1873, 1875, 1877, 1879, 1881, 1883, 1885, 1887, 1889, 1891, 1893, 1895, 1897, 1899, 1901, 1903, 1905, 1907, 1909, 1911, 1913, 1915, 1917, 1919, 1921, 1923, 1925, 1927, 1929, 1931, 1933, 1935, 1937, 1939, 1941, 1943, 1945, 1947, 1949, 1951, 1953, 1955, 1957, 1959, 1961, 1963, 1965, 1967, 1969, 1971, 1973, 1975, 1977, 1979, 1981, 1983, 1985, 1987, 1989, 1991, 1993, 1995, 1997, 1999, 2001, 2003, 2005, 2007, 2009, 2011, 2013, 2015, 2017, 2019, 2021, 2023, 2025, 2027, 2029, 2031, 2033, 2035, 2037, 2039, 2041, 2043, 2045, 2047, 2049, 2051, 2053, 2055, 2057, 2059, 2061, 2063, 2065, 2067, 2069, 2071, 2073, 2075, 2077, 2079, 2081, 2083, 2085, 2087, 2089, 2091, 2093, 2095, 2097, 2099, 2101, 2103, 2105, 2107, 2109, 2111, 2113, 2115, 2117, 2119, 2121, 2123, 2125, 2127, 2129, 2131, 2133, 2135, 2137, 2139, 2141, 2143, 2145, 2147, 2149, 2151, 2153, 2155, 2157, 2159, 2161, 2163, 2165, 2167, 2169, 2171,

2173, 2175, 2177, 2179, 2181, 2183, 2185, 2187, 2189, 2191, 2193, 2195, 2197, 2199, 2201, 2203, 2205, 2207, 2209, 2211, 2213, 2215, 2217, 2219, 2221, 2223, 2225, 2227, 2229, 2231, 2233, 2235, 2237, 2239, 2241, 2243, 2245, 2247, 2249, 2251, 2253, 2255, 2257, 2259, 2261, 2263, 2265, 2267, 2269, 2271, 2273, 2275, 2277, 2279, 2281, 2283, 2285, 2287, 2289, 2291, 2293, 2295, 2297, 2299, 2301, 2303, 2305, 2307, 2309, 2311, 2313, 2315, 2317, 2319, 2321, 2323, 2325, 2327, 2329, 2331, 2333, 2335, 2337, 2339, 2341, 2343, 2345, 2347, 2349, 2351, 2353, 2355, 2357, 2359, 2361, 2363, 2365, 2367, 2369, 2371, 2373, 2375, 2377, 2379, 2381, 2383, 2385, 2387, 2389, 2391, 2393, 2395, 2397, 2399, 2401, 2403, 2405, 2407, 2409, 2411, 2413, 2415, 2417, 2419, 2421, 2423, 2425, 2427, 2429, 2431, 2433, 2435, 2437, 2439, 2441, 2443, 2445, 2447, 2449, 2451, 2453, 2455, 2457, 2459, 2461, 2463, 2465, 2467, 2469, 2471, 2473, 2475, 2477, 2479, 2481, 2483, 2485, 2487, 2489, 2491, 2493, 2495, 2497, 2499, 2501, 2503, 2505, 2507, 2509, 2511, 2513, 2515, 2517, 2519, 2521, 2523, 2525, 2527, 2529, 2531, 2533, 2535, 2537, 2539, 2541, 2543, 2545, 2547, 2549, 2551, 2553, 2555, 2557, 2559, 2561, 2563, 2565, 2567, 2569, 2571, 2573, 2575, 2577, 2579, 2581, 2583, 2585, 2587, 2589, 2591, 2593, 7437, 7439, 7441, 7443, 7445, 7447, 7449, 7451, 7453, 7455, 7457, 7459, 7461, 7463, 7465, 7467, 7469, 7471, 7473, 7475, 7477, 7479, 7481, 7483, 7485, 7487, 7489, 7491, 7493, 7495, 7497, 7499, 7501, 7503, 7505, 7507, 7509, 7511, 7513, 7515, 7517, 7519, 7521, 7523, 7525, 7527, 7529, 7531, 7533, 7535, 7537, 7539, 7541, 7543, 7545, 7547, 7549, 7551, 7553, 7555, 7557, 7559, 7561, 7563, 7565, 7567, 7569, 7571, 7573, 7575, 7577, 7579, 7581, 7583, 7585, 7587, 7589, 7591, 7593, 7595, 7597, 7599, 7601, 7603, 7605, 7607, 7609, 7611, 7613, 7615, 7617, 7619, 7621, 7623, 7625, 7627, 7629, 7631, 7633, 7635, 7637, 7639, 7641, 7643, 7645, 7647, 7649, 7651, 7653, 7655, 7657, 7659, 7661, 7663, 7665, 7667, 7669, 7671, 7673, 7675, 7677, 7679, 7681, 7683, 7685, 7687, 7689, 7691, 7693, 7695, 7697, 7699, 7701, 7703, 7705, 7707, 7709, 7711, 7713, 7715, 7717, 7719, 7721, 7723, 7725, 7727, 7729, 7731, 7733, 7735, 7737, 7739, 7741, 7743, 7745, 7747, 7749, 7751, 7753, 7755, 7757, 7759, 7761, 7763, 8369, 8371, 8373, 8375, 8377, 8379, 8381, 8383, 8385, 8387, 8389, 8391, 8393, 8395, 8397, 8399, 8401, 8403, 8405, 8407, 8409, 8411, 8413, 8415, 8417, 8419, 8421, 8423, 8425, 8427, 8429, 8431, 8433, 8435, 8437, 8439, 8441, 8443, 8445, 8447, 8449, 8451, 8453, 8455, 8457, 8459, 8461, 8463, 8465, 8467, 8469, 8471, 8473, 8475, 8477, 8479, 8797, 8799, 8801, 8803, 8805, 8807, 8809, 8811, 8813, 8815, 8817, 8819, 8821, 8823, 8825, 8827, 8829, 8831, 8833, 8835, 8837, 8839, 8841, 8843, 8845, 8847, 8849, 8851, 8853, 8855, 8857, 8859, 8861, 8863, 8865, 8867, 8869, 8871, 8873, 8875, 8877, 8879, 8881, 8883, 8885, 8887, 8889, 8891, 8893, 8895, 8897, 8899, 8901, 8903, 8905, 8907, 8909, 8911, 8913, 8915, 8917, 8919, 8921, 8923, 8925, 8927, 8929, 8931, 8933, 8935, 8937, 8939, 8941, 8943, 8945, 8947, 8949, 8951, 8953, 8955, 8957, 8959, 8961, 8963, 8965, 8967, 8969, 8971, 8973, 8975, 8977, 8979, 8981, 8983, 8985, 8987, 8989, 8991, 8993, 8995, 8997, 8999, 9001, 9003, 9005, 9007, 9009, 9011, 9013, 9015, 9017, 9019, 9021, 9023, 9025, 9027, 9029, 9031, 9033, 9035, 9037, 9039, 9041, 9043, 9045, 9047, 9049, 9051, 9053, 9055, 9057, 9059, 9061, 9063, 9065, 9067, 9069, 9071, 9073, 9075, 9077, 9079, 9081, 9083, 9085, 9087, 9089, 9091, 9093, 9095, 9097, 9099, 9101, 9103, and/or 9105.

(121) In some embodiments, the polynucleotides are capable of hybridizing under highly stringent conditions to a reference polynucleotide sequence selected from any polynucleotide sequence provided herein, or a complement thereof, or a polynucleotide sequence encoding any of the variant enzyme polypeptides provided herein. In some embodiments, the polynucleotide capable of hybridizing under highly stringent conditions encodes a enzyme polypeptide comprising an amino acid sequence that has one or more residue differences as compared to a reference sequence.

(122) In some embodiments, an isolated polynucleotide encoding any of the engineered enzyme polypeptides herein is manipulated in a variety of ways to facilitate expression of the enzyme polypeptide. In some embodiments, the polynucleotides encoding the enzyme polypeptides comprise expression vectors where one or more control sequences is present to regulate the expression of the enzyme polynucleotides and/or polypeptides. Manipulation of the isolated polynucleotide prior to its insertion into a vector may be desirable or necessary depending on the expression vector utilized. Techniques for modifying polynucleotides and nucleic acid sequences utilizing recombinant DNA methods are well known in the art. In some embodiments, the control sequences include among others, promoters, leader sequences, polyadenylation sequences, propeptide sequences, signal peptide sequences, and transcription terminators. In some embodiments, suitable promoters are selected based on the host cells selection. For bacterial host cells, suitable promoters for directing transcription of the nucleic acid constructs of the present disclosure, include, but are not limited to promoters obtained from the *E. coli* lac

operon, *Streptomyces coelicolor* agarase gene (dagA), *Bacillus subtilis* levansucrase gene (sacB), *Bacillus licheniformis* alpha-amylase gene (amyL), *Bacillus stearothermophilus* maltogenic amylase gene (amyM), *Bacillus amyloliquefaciens* alpha-amylase gene (amyQ), *Bacillus licheniformis* penicillinase gene (penP), *Bacillus subtilis* xylA and xylB genes, and prokaryotic beta-lactamase gene (See e.g., Villa-Kamaroff et al., Proc. Natl Acad. Sci. USA 75: 3727-3731 [1978]), as well as the tac promoter (See e.g., DeBoer et al., Proc. Natl Acad. Sci. USA 80: 21-25 [1983]). Exemplary promoters for filamentous fungal host cells, include, but are not limited to promoters obtained from the genes for *Aspergillus oryzae* TAKA amylase, *Rhizomucor miehei* aspartic proteinase, *Aspergillus niger* neutral alpha-amylase, *Aspergillus niger* acid stable alpha-amylase, *Aspergillus niger* or *Aspergillus awamori* glucoamylase (glaA), *Rhizomucor miehei* lipase, *Aspergillus oryzae* alkaline protease, *Aspergillus oryzae* triose phosphate isomerase, *Aspergillus nidulans* acetamidase, and *Fusarium oxysporum* trypsin-like protease (See e.g., WO 96/00787), as well as the NA2-tpi promoter (a hybrid of the promoters from the genes for *Aspergillus niger* neutral alpha-amylase and *Aspergillus oryzae* triose phosphate isomerase), and mutant, truncated, and hybrid promoters thereof. Exemplary yeast cell promoters can be from the genes can be from the genes for *Saccharomyces cerevisiae* enolase (ENO-1), *Saccharomyces cerevisiae* galactokinase (GAL1), *Saccharomyces cerevisiae* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (ADH2/GAP), and *Saccharomyces cerevisiae* 3-phosphoglycerate kinase. Other useful promoters for yeast host cells are known in the art (See e.g., Romanos et al., Yeast 8:423-488 [1992]).

(123) In some embodiments, the control sequence is also a suitable transcription terminator sequence (i.e., a sequence recognized by a host cell to terminate transcription). In some embodiments, the terminator sequence is operably linked to the 3' terminus of the nucleic acid sequence encoding the enzyme polypeptide. Any suitable terminator which is functional in the host cell of choice finds use in the present invention. Exemplary transcription terminators for filamentous fungal host cells can be obtained from the genes for *Aspergillus oryzae* TAKA amylase, *Aspergillus niger* glucoamylase, *Aspergillus nidulans* anthranilate synthase, *Aspergillus niger* alpha-glucosidase, and *Fusarium oxysporum* trypsin-like protease. Exemplary terminators for yeast host cells can be obtained from the genes for *Saccharomyces cerevisiae* enolase, *Saccharomyces cerevisiae* cytochrome C (CYC1), and *Saccharomyces cerevisiae* glyceraldehyde-3-phosphate dehydrogenase. Other useful terminators for yeast host cells are known in the art (See e.g., Romanos et al., supra).

(124) In some embodiments, the control sequence is also a suitable leader sequence (i.e., a non-translated region of an mRNA that is important for translation by the host cell). In some embodiments, the leader sequence is operably linked to the 5' terminus of the nucleic acid sequence encoding the enzyme polypeptide. Any suitable leader sequence that is functional in the host cell of choice find use in the present invention. Exemplary leaders for filamentous fungal host cells are obtained from the genes for *Aspergillus oryzae* TAKA amylase, and *Aspergillus nidulans* triose phosphate isomerase. Suitable leaders for yeast host cells are obtained from the genes for *Saccharomyces cerevisiae* enolase (ENO-1), *Saccharomyces cerevisiae* 3-phosphoglycerate kinase, *Saccharomyces cerevisiae* alpha-factor, and *Saccharomyces cerevisiae* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (ADH2/GAP).

(125) In some embodiments, the control sequence is also a polyadenylation sequence (i.e., a sequence operably linked to the 3' terminus of the nucleic acid sequence and which, when transcribed, is recognized by the host cell as a signal to add polyadenosine residues to transcribed mRNA). Any suitable polyadenylation sequence which is functional in the host cell of choice finds use in the present invention. Exemplary polyadenylation sequences for filamentous fungal host cells include, but are not limited to the genes for *Aspergillus oryzae* TAKA amylase, *Aspergillus niger* glucoamylase, *Aspergillus nidulans* anthranilate synthase, *Fusarium oxysporum* trypsin-like protease, and *Aspergillus niger* alpha-glucosidase. Useful polyadenylation sequences for yeast host cells are known (See e.g., Guo and Sherman, Mol. Cell. Bio., 15:5983-5990 [1995]).

(126) In some embodiments, the control sequence is also a signal peptide (i.e., a coding region that codes for an amino acid sequence linked to the amino terminus of a polypeptide and directs the encoded polypeptide into the cell's secretory pathway). In some embodiments, the 5' end of the coding sequence of the nucleic acid sequence inherently contains a signal peptide coding region naturally linked in translation reading frame with the segment of the coding region that encodes the secreted polypeptide. Alternatively,

in some embodiments, the 5' end of the coding sequence contains a signal peptide coding region that is foreign to the coding sequence. Any suitable signal peptide coding region which directs the expressed polypeptide into the secretory pathway of a host cell of choice finds use for expression of the engineered polypeptide(s). Effective signal peptide coding regions for bacterial host cells are the signal peptide coding regions include, but are not limited to those obtained from the genes for *Bacillus* NC1B 11837 maltogenic amylase, *Bacillus stearothermophilus* alpha-amylase, *Bacillus licheniformis* subtilisin, *Bacillus licheniformis* beta-lactamase, *Bacillus stearothermophilus* neutral proteases (nprT, nprS, nprM), and *Bacillus subtilis* prsA. Further signal peptides are known in the art (See e.g., Simonen and Palva, Microbiol. Rev., 57:109-137 [1993]). In some embodiments, effective signal peptide coding regions for filamentous fungal host cells include, but are not limited to the signal peptide coding regions obtained from the genes for *Aspergillus oryzae* TAKA amylase, *Aspergillus niger* neutral amylase, *Aspergillus niger* glucoamylase, *Rhizomucor miehei* aspartic proteinase, *Humicola insolens* cellulase, and *Humicola lanuginosa* lipase. Useful signal peptides for yeast host cells include, but are not limited to those from the genes for *Saccharomyces cerevisiae* alpha-factor and *Saccharomyces cerevisiae* invertase.

(127) In some embodiments, the control sequence is also a propeptide coding region that codes for an amino acid sequence positioned at the amino terminus of a polypeptide. The resultant polypeptide is referred to as a "proenzyme," "propolypeptide," or "zymogen." A propolypeptide can be converted to a mature active polypeptide by catalytic or autocatalytic cleavage of the propeptide from the propolypeptide. The propeptide coding region may be obtained from any suitable source, including, but not limited to the genes for *Bacillus subtilis* alkaline protease (aprE), *Bacillus subtilis* neutral protease (nprT), *Saccharomyces cerevisiae* alpha-factor, *Rhizomucor miehei* aspartic proteinase, and *Myceliophthora thermophila* lactase (See e.g., WO 95/33836). Where both signal peptide and propeptide regions are present at the amino terminus of a polypeptide, the propeptide region is positioned next to the amino terminus of a polypeptide and the signal peptide region is positioned next to the amino terminus of the propeptide region.

(128) In some embodiments, regulatory sequences are also utilized. These sequences facilitate the regulation of the expression of the polypeptide relative to the growth of the host cell. Examples of regulatory systems are those that cause the expression of the gene to be turned on or off in response to a chemical or physical stimulus, including the presence of a regulatory compound. In prokaryotic host cells, suitable regulatory sequences include, but are not limited to the lac, tac, and trp operator systems. In yeast host cells, suitable regulatory systems include, but are not limited to the ADH2 system or GAL1 system. In filamentous fungi, suitable regulatory sequences include, but are not limited to the TAKA alpha-amylase promoter, *Aspergillus niger* glucoamylase promoter, and *Aspergillus oryzae* glucoamylase promoter.

(129) In another aspect, the present invention is directed to a recombinant expression vector comprising a polynucleotide encoding an engineered enzyme polypeptide, and one or more expression regulating regions such as a promoter and a terminator, a replication origin, etc., depending on the type of hosts into which they are to be introduced. In some embodiments, the various nucleic acid and control sequences described herein are joined together to produce recombinant expression vectors which include one or more convenient restriction sites to allow for insertion or substitution of the nucleic acid sequence encoding the enzyme polypeptide at such sites. Alternatively, in some embodiments, the nucleic acid sequence of the present invention is expressed by inserting the nucleic acid sequence or a nucleic acid construct comprising the sequence into an appropriate vector for expression. In some embodiments involving the creation of the expression vector, the coding sequence is located in the vector so that the coding sequence is operably linked with the appropriate control sequences for expression.

(130) The recombinant expression vector may be any suitable vector (e.g., a plasmid or virus), that can be conveniently subjected to recombinant DNA procedures and bring about the expression of the enzyme polynucleotide sequence. The choice of the vector typically depends on the compatibility of the vector with the host cell into which the vector is to be introduced. The vectors may be linear or closed circular plasmids.

(131) In some embodiments, the expression vector is an autonomously replicating vector (i.e., a vector that exists as an extra-chromosomal entity, the replication of which is independent of chromosomal replication, such as a plasmid, an extra-chromosomal element, a minichromosome, or an artificial

(132) The vector may contain any means for assuring self-replication. In some alternative embodiments, the vector is one in which, when introduced into the host cell, it is integrated into the genome and replicated together with the chromosome(s) into which it has been integrated. Furthermore, in some embodiments, a single vector or plasmid, or two or more vectors or plasmids which together contain the total DNA to be introduced into the genome of the host cell, and/or a transposon is utilized. (132) In some embodiments, the expression vector contains one or more selectable markers, which permit easy selection of transformed cells. A "selectable marker" is a gene, the product of which provides for biocide or viral resistance, resistance to heavy metals, prototrophy to auxotrophs, and the like. Examples of bacterial selectable markers include, but are not limited to the *dal* genes from *Bacillus subtilis* or *Bacillus licheniformis*, or markers, which confer antibiotic resistance such as ampicillin, kanamycin, chloramphenicol or tetracycline resistance. Suitable markers for yeast host cells include, but are not limited to ADE2, HIS3, LEU2, LYS2, MET3, TRP1, and URA3. Selectable markers for use in filamentous fungal host cells include, but are not limited to, *amdS* (acetamidase; e.g., from *A. nidulans* or *A. oryzae*), *argB* (ornithine carbamoyltransferases), *bar* (phosphinothricin acetyltransferase; e.g., from *S. hygrosopicus*), *hph* (hygromycin phosphotransferase), *niaD* (nitrate reductase), *pyrG* (orotidine-5'-phosphate decarboxylase; e.g., from *A. nidulans* or *A. oryzae*), *sC* (sulfate adenylyltransferase), and *trpC* (anthranilate synthase), as well as equivalents thereof. In another aspect, the present invention provides a host cell comprising at least one polynucleotide encoding at least one engineered enzyme polypeptide of the present invention, the polynucleotide(s) being operatively linked to one or more control sequences for expression of the engineered enzyme enzyme(s) in the host cell. Host cells suitable for use in expressing the polypeptides encoded by the expression vectors of the present invention are well known in the art and include but are not limited to, bacterial cells, such as *E. coli*, *Vibrio fluvialis*, *Streptomyces* and *Salmonella typhimurium* cells; fungal cells, such as yeast cells (e.g., *Saccharomyces cerevisiae* or *Pichia pastoris* (ATCC Accession No. 201178)); insect cells such as *Drosophila* S2 and *Spodoptera* Sf9 cells; animal cells such as CHO, COS, BHK, 293, and Bowes melanoma cells; and plant cells. Exemplary host cells also include various *Escherichia coli* strains (e.g., W3110 (Δ fhuA) and BL21).

(133) Accordingly, in another aspect, the present invention provides methods of producing the engineered enzyme polypeptides, where the methods comprise culturing a host cell capable of expressing a polynucleotide encoding the engineered enzyme polypeptide under conditions suitable for expression of the polypeptide. In some embodiments, the methods further comprise the steps of isolating and/or purifying the enzyme polypeptides, as described herein.

(134) Appropriate culture media and growth conditions for host cells are well known in the art. It is contemplated that any suitable method for introducing polynucleotides for expression of the enzyme polypeptides into cells will find use in the present invention. Suitable techniques include, but are not limited to electroporation, biolistic particle bombardment, liposome mediated transfection, calcium chloride transfection, and protoplast fusion.

(135) Various features and embodiments of the present invention are illustrated in the following representative examples, which are intended to be illustrative, and not limiting.

EXPERIMENTAL

(136) The following Examples, including experiments and results achieved, are provided for illustrative purposes only and are not to be construed as limiting the present invention. Indeed, there are various suitable sources for many of the reagents and equipment described below. It is not intended that the present invention be limited to any particular source for any reagent or equipment item.

(137) In the experimental disclosure below, the following abbreviations apply: M (molar); mM (millimolar), μ M and μ M (micromolar); nM (nanomolar); mol (moles); gm and g (gram); mg (milligrams); μ g and μ g (micrograms); L and l (liter); ml and mL (milliliter); cm (centimeters); mm (millimeters); μ m and μ m (micrometers); sec. (seconds); min(s) (minute(s)); h(s) and hr(s) (hour(s)); U (units); MW (molecular weight); rpm (rotations per minute); psi and PSI (pounds per square inch); ° C. (degrees Centigrade); RT and rt (room temperature); CV (coefficient of variability); CAM and cam (chloramphenicol); PMBS (polymyxin B sulfate); IPTG (isopropyl β -D-1-thiogalactopyranoside); LB (Luria broth); TB (terrific broth); SFP (shake flask powder); CDS (coding sequence); DNA (deoxyribonucleic acid); RNA (ribonucleic acid); nt (nucleotide; polynucleotide); aa (amino acid; polypeptide); *E. coli* W3110 (commonly used laboratory *E. coli* strain, available from the *Coli* Genetic

Stock Center [CGSC], New Haven, Conn.); AcSus (*Acidithiobacillus caldus* sucrose synthase); SUS, SuS, and SuSy (sucrose synthase, also known as sucrose synthetase); NDP (nucleoside diphosphate); adenosine diphosphate (ADP); cytidine diphosphate (CDP); guanosine diphosphate (GDP); thymidine diphosphate (TDP); uridine diphosphate (UDP); inosine diphosphate (IDP); GT (glycosyltransferase); UGT (UDP-glucose-dependent glycosyltransferase); NGT (NDP-nucleoside diphosphate-dependent glycosyltransferase); AGT (ADP-glucose-dependent glycosyltransferase); CGT (CDP-glucose-dependent glycosyltransferase); GGT (GDP-glucose-dependent glycosyltransferase); TGT (TDP-glucose-dependent glycosyltransferase); IGT (IDP-glucose-dependent glycosyltransferase); UGT (UDP-glucose-dependent glycosyltransferase); reb (rebaudioside); rebA (rebaudioside A); rebD (rebaudioside D); rebI (rebaudioside I); rebM (rebaudioside M); "Reb A 60" is a ~1:2 mixture of stevioside and rebaudioside A respectively; HTP (high throughput); HPLC (high pressure liquid chromatography); HPLC-UV (HPLC-Ultraviolet Visible Detector); ¹H NMR (proton nuclear magnetic resonance spectroscopy); HSQC NMR (heteronuclear single quantum coherence spectroscopy NMR); COSY NMR (homonuclear correlation spectroscopy NMR); Acorn (Acorn NMR, Livermore, Calif.); FIOPC (fold improvements over positive control); Sigma and Sigma-Aldrich (Sigma-Aldrich, St. Louis, Mo.; Difco (Difco Laboratories, BD Diagnostic Systems, Detroit, Mich.); Microfluidics (Microfluidics, Westwood, Mass.); ChromaDex (ChromaDex, Inc., Irvine, Calif.); Life Technologies (Life Technologies, a part of Fisher Scientific, Waltham, Mass.); Amresco (Amresco, LLC, Solon, Ohio); Carbosynth (Carbosynth, Ltd., Berkshire, UK); Varian (Varian Medical Systems, Palo Alto, Calif.); Agilent (Agilent Technologies, Inc., Santa Clara, Calif.); and Thermotron (Thermotron, Inc., Holland, Mich.).

Example 1

Synthesis, Optimization, and Assaying of UGT Enzymes with Glucosylation Activity

(138) In this Example, methods used in the synthesis, optimization and assaying of UGT enzymes with glucosylation activity are described.

(139) Gene Synthesis and Optimization:

(140) The polynucleotide sequence (SEQ ID NO: 1) encoding the wild-type *Stevia rebaudiana* polypeptide (SEQ ID NO: 2) reported to glucosylate steviolbioside to rebaudioside B and glucosylate stevioside to rebaudioside A (See e.g., Richman et al., Plant J., 41:56-67 [2005]), was codon-optimized and synthesized as the gene of SEQ ID NO: 3. This synthetic gene (SEQ ID NO: 3) was cloned into a pCK110900 vector system (See e.g., US Pat. Appln. Publn. No. 2006/0195947, which is hereby incorporated by reference herein) and subsequently expressed in *E. coli* W3110 (Δ fhuA). The *E. coli* strain W3110 expressed the UGT enzymes under the control of the lac promoter.

(141) Production of Shake Flask Powders (SFP):

(142) A shake-flask procedure was used to generate the glycosyltransferase polypeptide shake flask powders (SFP) for characterization assays used in the biocatalytic processes described herein. Shake flask powder (SFP) preparation of enzymes provides a more purified preparation (e.g., up to >30% of total protein) of the enzyme as compared to the cell lysate used in HTP assays and also allows for the use of more concentrated enzyme solutions. A single colony of *E. coli* containing a plasmid encoding an engineered polypeptide of interest was inoculated into 5 mL Luria Bertani broth containing 30 μ g/ml chloramphenicol and 1% glucose. Cells were grown overnight (at least 16 hours) in an incubator at 30° C. with shaking at 250 rpm. The culture was diluted into 250 mL Terrific Broth (12 g/L bacto-tryptone, 24 g/L yeast extract, 4 mL/L glycerol, 65 mM potassium phosphate, pH 7.0, 1 mM MgSO₄) containing 30 μ g/ml CAM, in a 1 L flask to an optical density of 600 nm (OD₆₀₀) of 0.2 and allowed to grow at 30° C.

(143) Expression of the glycosyltransferase gene was induced by addition of IPTG to a final concentration of 1 mM when the OD₆₀₀ of the culture was 0.6 to 0.8. Incubation was then continued overnight (at least 16 hours). Cells were harvested by centrifugation (5000 rpm, 15 min, 4° C.) and the supernatant discarded. The cell pellet was resuspended in two volumes of 25 mM triethanolamine buffer, pH 7.5, and passed through a MICROFLUIDIZER® high pressure homogenizer (Microfluidics), with standard *E. coli* lysis settings and maintained at 4° C. Cell debris was removed by centrifugation (10,000 rpm, 45 minutes, 4° C.). The cleared lysate supernatant was collected and frozen at -80° C. and then either His-affinity purified and dialyzed to produce purified protein or lyophilized to produce a dry shake-flask powder of crude protein.

(144) Assay of SFP for Stevioside Glucosylation:

(145) SFP was reconstituted to provide 20 g/L powder. Then, 50 μ L of these stocks were diluted in 200 μ L total reaction volume of 50 mM Tris-HCl buffer, pH 7.5, with 3 mM MgSO₄ and 1 mM stevioside (ChromaDex, >94% purity), with 2 mM uridine diphosphoglucose. The reaction was performed at 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 16-18h.

(146) HPLC-MS/MS Analysis:

(147) The reaction described above was quenched with 0.5 volume/volume acetonitrile with 0.2% formic acid and precipitated by centrifugation. Glycosylated stevioside products were detected in the supernatant by LC-MS/MS with the following instrument and parameters:

(148) TABLE-US-00003 TABLE 1.1 HPLC-MS/MS Analysis of Steviol Glycosides Instrument Agilent HPLC 1200 series, Sciex 4000 QTrap Column Poroshell 120 EC C18 50 $\times$ 3.0 mm, 2.7 μ m with Poroshell 120 EC C18 5 $\times$ 3.0, 2.7 μ m guard column (Agilent Technologies) Mobile phase Gradient (A: 0.1% formic acid in water, B: 0.1% formic acid in methanol) Time (m) % B 0 60 0.50 60 1.00 70 4.33 70 5.00 95 5.33 9 5.34 60 6.00 60 Flow rate 0.8 mL/m Run time 6 m Peak retention times Rebaudioside A: 2.35 m Column temperature 40° C. Injection volume 10 μ L MS detection MRM 990/828 (for steviol tetraglycosides, e.g., rebaudioside A), 1152/828 (for steviol pentaglycosides, e.g., rebaudioside D), 1314/828 (steviol hexaglycosides, e.g., rebaudioside M), 828/666 (for steviol triglycosides, e.g., stevioside), 666/504 (steviol diglycosides, e.g., rubusoside) MS conditions MODE: MRM; CUR: 30; IS: 4750; CAD: high; TEM: 550° C.; GS1: 50; GS2: 50; DP: 150; EP: 10; CXP: 14; DT: 50 ms for each transition. For the first three transitions CE: 85; for the last two transitions CE: 60.

(149) Activity was detected for SEQ ID NO:4. High conversion (i.e., >95%), of stevioside to rebaudioside A was observed in the LC-MS/MS analysis of the assay samples described above.

Example 2

GT Variants of SEQ ID NO: 4

(150) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 4 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 3 (i.e., SEQ ID NO:4) was carried out by constructing libraries of variant genes in which positions associated with certain structural features of the enzyme were subjected to mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a first round (“Round 1”) of 12 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(151) HTP Growth, Expression, and Lysate Preparation

(152) Cells were picked into 96-well plates and grown overnight in LB media containing 1% glucose and 30 μ g/mL CAM, 30° C., 200 rpm, 85% humidity. Then, 20 μ L of overnight growth were transferred to a deep-well plate containing 380 μ L TB growth media containing 30 μ g/mL CAM, induced with 1 mM IPTG, and incubated for 18 h at 30° C., 200 rpm, 85% humidity. Cell cultures were centrifuged at 4000 rpm, 4° C. for 10 m, and the media discarded. Cell pellets thus obtained were frozen at -80° C., and lysed in 250 μ L lysis buffer (0.5 g/L lysozyme and 0.5 g/L PMBS in 20 mM Tris-HCl buffer, pH 7.5) with low-speed shaking for 2 h on titre-plate shaker at room temperature. The plates were then centrifuged at 4000 rpm and 4° C. for 20 mM and the cleared lysate supernatants were used in the HTP assay reactions described below.

(153) HTP Assay for Glucose Transfer from ADP-Glucose to Stevioside:

(154) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 3 variants with lysate loading of 50 μ L lysate in 200 μ L reactions and with substrate loading of 1 mM stevioside (ChromaDex, >94% purity), from a 20 mM stock solution in 50% ethanol and co-substrate loading of 0.5 mM ADP-glucose (Sigma, >93% purity). The following reaction conditions were used: 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl₂, 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 18 h. The reactions were quenched with 100 μ L/well acetonitrile with 0.2% formic acid, centrifuged 10 m at 4° C., and the supernatants were analyzed by HPLC-MS/MS as described in Example 1, Table 1.1.

(155) Formation of rebaudioside A from stevioside in the presence of wild-type UGT76G1 (SEQ ID NO:4) with ADP-glucose was indistinguishable from a no enzyme control. In contrast to the wild-type enzyme of SEQ ID NO:4, glucosyltransferase variant polypeptides were identified that produced rebaudioside A from stevioside with ADP-glucose. The engineered polypeptides are listed in Table 2.1.

Although the parent and variant constructs contain an N-terminal histidine tag for affinity purification, the mutations were numbered relative to the untagged reference sequence for clarity. Shake-flask scale cultures were grown for protein purification as described in Example 1. Variants with the following amino acid mutations shown in Table 2.1, relative to SEQ ID NO:4 were analyzed.

(156) TABLE-US-00004 TABLE 2.1 Variants and RebA Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 4) RebA.sup.a 5/6 V309R ++ 7/8 R10-/V309R ++ 9/10 S361G ++ 11/12 V309S ++ 13/14 L307V + 15/16 S283T + 17/18 V309L + 19/20

Y278L/T284I/R311G/V339A/N360G + 21/22 V309N + 23/24 R262L + 25/26 V339A/S361G + 27/28 V344I/S361G + 29/30 R262K + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 4, and defined as follows: “+” = production at least that of the reference polypeptide, but less than 1.3-fold; and “++” = at least 1.3-fold, but less than 1.6-fold increased production, as compared to the reference polypeptide.

Purified Protein Characterization Assay and Analysis for Glucosyl Transfer from ADP-Glucose to Stevioside or Rebaudioside D:

(157) First, 250 mL shake flask cultures were grown, induced, lysed, histidine-affinity purified, dialyzed, and diluted 1:1 with glycerol as described in Example 1, to produce purified protein. Then, 50 μ L of these proteins were diluted in 200 μ L total reaction volume of 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl.sub.2, with 1 mM stevioside (ChromaDex, >94% purity) or rebaudioside D (Sigma, >93% purity) and 0.5 mM ADP-glucose. The reaction was performed at 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 16 h. The reaction described above was quenched with 0.5 volume/volume acetonitrile with 0.2% formic acid and precipitated by centrifugation. Glycosylated products were detected in the supernatant by LC-MS/MS following 1:10 dilution in water with the instrument and parameters described in Example 1, Table 1.1.

(158) TABLE-US-00005 TABLE 2.2 Purified Round 1 Variants and RebA and RebM Levels SEQ ID NO: Amino Acid Differences Increased Increased (nt/aa) (Relative to SEQ ID NO: 4) RebA.sup.a RebM.sup.a 7/8 R10-/V309R +++ + 9/10 S361G ++ - 11/12 V309S ++ + 13/14 L307V ++ - 15/16 S283T +++ - 29/30 R262K - - .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 4, and defined as follows: “-” = production less than 1.5-fold; “+” = production at least 1.5-fold, but less than 2.5-fold; “++” = at least 2.5-fold, but less than 3.5-fold; and “+++” = at least 3.5-fold increased production, as compared to the reference polypeptide.

(159) Variants with mutations R10- and V309R, S361G, V309S, L307V, and S283T (SEQ ID NOS: 8, 10, 12, 14, and 16) produced rebaudioside A from stevioside, at levels above the above negative control and SEQ ID NO: 2 levels with ADP-glucose. The variant with mutations R10- and V309R (SEQ ID NO: 8) and the variant with mutation V3095 (SEQ ID NO: 12) produced rebaudioside M from rebaudioside D above negative control and SEQ ID NO: 2 levels with ADP-glucose. Thus, these engineered ADP-glycosyltransferase enzymes provide new biocatalytic reagents for use in new methods for the β -glucosylation of stevioside to rebaudioside A, and rebaudioside D to rebaudioside M. The variant with mutations R10- and V309R (SEQ ID NO: 8) had the highest activity on both stevioside and rebaudioside D with ADP-glucose as a co-substrate. Thus, the encoding polynucleotide (SEQ ID NO: 7) was selected for further directed evolution.

Example 3

ADP-Glycosyltransferase Variants of SEQ ID NO: 8

(160) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 8, for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 7 (i.e., SEQ ID NO:8) was carried out by constructing libraries of variant genes in which mutations associated with improved activity in Round 1 were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described in Example 2 to provide a second round (“Round 2”) of 20 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. The engineered polypeptides are listed in Table 3.1. Although the parent and variant constructs contain an N-terminal histidine tag for affinity purification, the mutations were numbered relative to the untagged reference sequence for clarity. Shake-flask scale cultures were grown for protein purification as described in Example 1 for variants with the amino acid mutations shown in Table 3.1, relative to SEQ ID NO:8, as

indicated below, were analyzed.

(161) TABLE-US-00006 TABLE 3.1 Variants and RebA Levels SEQ ID NO: Amino Acid Differences (nt/aa) (Relative to SEQ ID NO: 8) Increased RebA.sup.a 31/32 T283Q/T318E/W337S/S360G +++ 33/34 E112S/T172S/T283Q/T318E ++ 35/36 T283Q ++ 37/38 T283Q/S360G ++ 39/40 T283Q/L306V/R308S/S360G + 41/42 S282T/T283Q/Q431E + 43/44 N137K/T283Q + 45/46 T318E + 47/48 E112S/R261S/T318E + 49/50 N137K/T283Q/Q431E + 51/52 S282T/T283Q + 53/54 L163K/T318E + 55/56 T283Q/L306V/W337S/A426V + 57/58 R261S/T283Q/W337F + 59/60 E112S/S282T/T283Q/Q431E + 61/62 R261S/W337S + 63/64 Q431E + 65/66 S360G + 67/68 Q269T/T318E + 69/70 R261S/T283Q/L306V/W337F + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 7, and defined as follows: “+” = production at least that of the reference polypeptide, but less than 1.5-fold; “++” = at least 1.5-fold, but less than 2-fold; and “+++” = at least 2-fold increased production, as compared to the reference polypeptide.

Purified Protein Characterization Assay and Analysis for Glucosyl Transfer from ADP-Glucose to Stevioside or Rebaudioside D

(162) Proteins were purified, assayed, and analyzed as described in Example 2.

(163) TABLE-US-00007 TABLE 3.2 Purified Round 2 Variants and RebA and RebM Levels SEQ ID NO: Amino Acid Differences Increased Increased (nt/aa) (Relative to SEQ ID NO: 8) RebA.sup.a RebM.sup.a 31/32 T283Q/T318E/W337S/S360G +++ +++ 33/34 E112S/T172S/T283Q/T318E ++ 35/36 T283Q + 37/38 T283Q/S360G ++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 7, and defined as follows: “+” = production at least that of the reference polypeptide, but less than 2-fold; “++” = at least 2-fold, but less than 3-fold; and “+++” = at least 3-fold increased production, as compared to the reference polypeptide.

(164) All variants in Table 3.2 (SEQ ID NOS: 32, 34, 36, and 38) produced rebaudioside A from stevioside and rebaudioside M from rebaudioside D with ADP-glucose, at greater quantities than SEQ ID NO: 8. The variant with mutations T283Q, T318E, W337S, and S360G (SEQ ID NO: 32) had the highest activity on both stevioside and rebaudioside D with ADP-glucose as a co-substrate. Thus, the encoding polynucleotide (SEQ ID NO: 31) was selected for further directed evolution.

(165) Purified Protein Characterization Assay and Analysis for Glucosyl Transfer from NDP-Glucose to Stevioside

(166) To profile the nucleotide diphosphate specificity of SEQ ID NO: 4 relative to SEQ ID NO: 32, the following experiment was performed. First, 50 μ L purified protein was diluted in 200 μ L total reaction volume of 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl.sub.2, with 1 mM stevioside (ChromaDex, >94% purity) and 0.5 mM ADP-glucose, UDP-glucose, or TDP-glucose. The reaction was performed at 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 16 h. The reaction described above was quenched with 0.5 volume/volume acetonitrile with 0.2% formic acid and precipitated by centrifugation. Glycosylated products were detected in the supernatant by LC-MS/MS following 1:10 dilution in water with the instrument and parameters described in Example 1, Table 1.1. SEQ ID NO: 32 produced 13-fold more rebaudioside A with ADP-glucose than did SEQ ID NO: 4, 90% as much with UDP-glucose, and 22% as much with TDP-glucose. The glycosyltransferase encoded by SEQ ID NO: 31 (i.e., SEQ ID NO:32) has substantially modified NDP-glucose specificity, as compared to the glycosyltransferase encoded by SEQ ID NO: 3 (i.e., SEQ ID NO:4).

(167) Determination of Specific Activity of GT Encoded by SEQ ID NO: 31 on Rebaudioside D and NDP-Glucose

(168) To profile the nucleotide diphosphate specificity of SEQ ID NO: 32, the following experiment was performed. First, 5 μ L purified protein was diluted in 100 μ L total reaction volume of 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl.sub.2, with 1 mM rebaudioside D (ChromaDex, >93% purity) and 2 mM ADP-glucose, UDP-glucose, TDP-glucose, or GDP-glucose. The reaction was performed at 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1-18 h. The reaction described above was quenched with 0.5 volume/volume acetonitrile with 0.2% formic acid and precipitated by centrifugation. Glycosylated products were detected in the supernatant by LC-MS/MS, following 1:10 dilution in water with the instrument and parameters described in Example 1, Table 1.1. The glycosyltransferase encoded by SEQ ID NO: 31 (i.e., SEQ ID NO:32), produced 50% as much rebaudioside M with GDP-glucose

relative to ADP-glucose, 70% as much with UDP-glucose, and an amount below the detection limit with TDP-glucose. The specific activity (μmol RebM formed per mg purified protein per min) of GT SEQ ID NO: 32 with rebaudioside D and ADP-glucose was 1.4-fold higher than the specific activity with UDP-glucose. Therefore, the glycosyltransferase encoded by SEQ ID NO: 31 (i.e., SEQ ID NO:32) is a new adenine diphosphoglucose-dependent glycosyltransferase, or “AGT.”

Example 4

Transformation of Stevioside to Rebaudioside A with Engineered AGT SEQ ID NO: 32 and In Situ Formation of ADP-Glucose

(169) In this Example, experiments to assess the in situ formation of ADP-glucose for glucosylation of steviol glycosides (See, FIG. 1) are described.

(170) Gene Synthesis and Optimization

(171) The polynucleotide sequences encoding the wild-type *Acidithiobacillus caldus* sucrose synthase and *Thermosynechococcus elongatus* sucrose synthase polypeptides (SEQ ID NOS: 72 and 74, respectively) reported to preferentially utilize ADP-glucose to donate a glucose to fructose to form sucrose in a reversible conversion (See e.g., Diricks et al., Appl. Microbiol. Biotechnol., 99:8465-74 [2015] and Figueroa et al., FEBS Lett., 587: 165-9 [2013]), were codon-optimized and synthesized as the gene of SEQ ID NOS: 71 and 73. These synthetic genes (SEQ ID NOS: 71 and 73) were individually cloned into a pCK110900 vector system (See e.g., US Pat. Appln. Publn. No. 2006/0195947, which is hereby incorporated by reference herein), and subsequently expressed in *E. coli* W3110 (ΔfhuA). The *E. coli* strain W3110 expressed the enzymes under the control of the lac promoter. The strains were grown at shake-flask scale and lysed for His-affinity protein purification, which was performed as described in Example 1.

(172) Coupled Assay with Purified AGT and Sucrose Synthase

(173) To examine the potential for NDP recycling by coupling a sucrose synthase (SuS) with the AGT encoded by SEQ ID NO: 31 (i.e., SEQ ID NO:32), the following experiment was performed. First, 20 μL purified sucrose synthase polypeptide (SEQ ID NO: 72 or 74) and 30 μL purified AGT polypeptide (SEQ ID NO: 32) were diluted in 200 μL total reaction volume of 50 mM Tris-HCl buffer, pH 7.5, 5 mM MgCl_2 , with 1 mM stevioside (ChromaDex, >94% purity), 200 mM sucrose, and 5 mM adenosine diphosphate (ADP), cytidine diphosphate (CDP), guanosine diphosphate (GDP), thymidine diphosphate (TDP), or uridine diphosphate (UDP) (Sigma, all >93% purity). The reaction was performed at 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 16 h. The reaction described above was quenched with 0.5 volume/volume acetonitrile with 0.2% formic acid and precipitated by centrifugation.

(174) Glycosylated products were detected in the supernatant by LC-MS/MS following 1:10 dilution in water with the instrument and parameters described in Example 1, Table 1.1.

(175) For SEQ ID NO: 32 and SuS (SEQ ID NO: 74), comparable amounts of rebaudioside A were formed with ADP, UDP, and GDP, and almost no rebaudioside A was formed with CDP and TDP. For SEQ ID NO: 32 and SuS (SEQ ID NO: 72), the level of rebaudioside A formed with ADP was comparable to that formed with SuS (SEQ ID NO: 74) with ADP. The amount of rebaudioside A formed with UDP was less than 20% of that formed with ADP, and almost no rebaudioside A was formed with GT (SEQ ID NO: 32) and SuS (SEQ ID NO: 72) and CDP, GDP, or TDP. These results demonstrate that both SEQ ID NOS: 72 and 74 are capable of generating ADP-glucose in situ from sucrose and find use with AGT (SEQ ID NO: 32) for glucosylation of steviol glycosides with sucrose and ADP as co-substrates instead of the more expensive substrate ADP-glucose. In addition, SEQ ID NO: 72 can also be used with GDP and UDP. In some embodiments involving ADP-selective coupling systems, SEQ ID NO: 74 finds use.

Example 5

ADP-Glycosyltransferase Circular Permuted Variants of SEQ ID NO: 32

(176) In this Example, experiments for design, construction, and evaluation of GT polypeptides derived from SEQ ID NO: 32 for glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 31 was carried out by constructing libraries of variant genes in which the N- and C-termini were linked in the encoding sequence and positions associated with certain structural features of the enzyme were selected as the new N-terminus of the protein. This library of circular permuted variants was then plated, grown, and screened using the high-throughput (HTP) assay

described below to provide a “Round 3.2” of 17 engineered GT circular permuted variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. None of the variants had activity higher than the unpermuted SEQ ID NO: 32, but the 17 engineered polypeptides listed in Table 5.1 had activity higher than a no enzyme negative control.

(177) HTP Assay for Glucose Transfer from ADP-Glucose to Stevioside

(178) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 31 variants with lysate loading of 25 μ L lysate in 200 μ L reactions and with substrate loading of 1 mM stevioside (ChromaDex, >94% purity), from a 20 mM stock solution in 50% ethanol and co-substrate loading of 0.5 mM ADP-glucose (Sigma, >93% purity). The following reaction conditions were used: 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl₂, 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. The reactions were quenched with 100 μ L/well acetonitrile with 0.2% formic acid, centrifuged 10 m at 4° C., and the supernatants were analyzed by HPLC-MS/MS as described in Example 1, Table 1.1.

(179) The engineered polypeptides analyzed are listed in Table 5.1. Although the parent and variant constructs contain an N-terminal histidine tag for affinity purification, the amino acids were numbered relative to the untagged reference sequence for clarity. Shake-flask scale cultures were grown for protein purification as described in Example 1, for variants with the following first amino acids relative to SEQ ID NO 32: 71, 170, 259, and 401. These variants represented circular permutants from the most distinct regions of the protein.

(180) TABLE-US-00008 TABLE 5.1 Circular Permuted Round 3.2 Variants First Amino Acid Position
SEQ ID NO: (nt/aa) (Relative to SEQ ID NO: 32) 75/76 70 77/78 71 79/80 72 81/82 73 83/84 74
85/86 75 87/88 169 89/90 170 91/92 171 93/94 174 95/96 194 97/98 198 99/100 259 101/102 260
103/104 370 105/106 401 107/108 403

Determination of Specific Activity of Purified Circular Permuted GTs on Rebaudioside D and ADP-Glucose

(181) First, 10 μ L purified protein was diluted in 100 μ L total reaction volume of 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl₂, with 1 mM rebaudioside D (ChromaDex, >93% purity) and 2 mM ADP-glucose. The reaction was performed at 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1-4 h. The reaction described above was quenched with 0.5 volume/volume acetonitrile with 0.2% formic acid and precipitated by centrifugation.

(182) Glycosylated products were detected in the supernatant by LC-MS/MS following 1:10 dilution in water with the instrument and parameters described in Example 1, Table 1.1. Of the four circular permutants, SEQ ID NO: 106 had the highest specific activity (μ mol RebM formed per mg purified protein per min), followed by SEQ ID NO: 100, and SEQ ID NO: 90. SEQ ID NO: 78 had barely detectable activity. Thus, the glucosyltransferase encoded by SEQ ID NO: 105 (i.e., SEQ ID NO:106) was identified in these experiments as the best candidate circular permuted AGT for further directed evolution.

Example 6

ADP-Glycosyltransferase Variants of SEQ ID NO: 32

(183) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 32 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO:31 (i.e., SEQ ID NO:32) was carried out by constructing libraries of variant genes in which positions associated with certain structural features of the enzyme were subjected to mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a third round (“Round 3.1”) of 60 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(184) HTP Assay for Glucose Transfer from ADP-Glucose to Stevioside or Rebaudioside D

(185) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 32 variants with lysate loading of 25 μ L lysate in 200 μ L reactions and with substrate loading of 0.5 mM rebaudioside D or 1 mM stevioside (ChromaDex, >94% purity), from a 20 mM stock solution in 50% ethanol and co-substrate loading of 0.5 mM ADP-glucose (Sigma, >93% purity). The following reaction conditions were used: 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl₂, 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. The reactions were quenched with 100 μ L/well acetonitrile

with 0.2% formic acid, centrifuged 10 m at 4° C., and the supernatants were analyzed by HPLC-MS/MS as described in Example 1, Table 1.1.

(186) The engineered polypeptides are listed in Table 6.1. Although the parent and variant constructs contain an N-terminal histidine tag for affinity purification, the mutations were numbered relative to the untagged reference sequence for clarity. Shake-flask scale cultures were grown for protein purification as described in Example 1 for variants with the amino acid mutations shown in Table 6.1, relative to SEQ ID NO: 32.

(187) TABLE-US-00009 TABLE 6.1 Round 3.1 Variants and RebA/RebM Levels Amino Acid Differences SEQ ID NO: (Relative to SEQ ID Increased Increased (nt/aa) NO: 32) RebA.sup.a
RebM.sup.a 109/110 L84A + 111/112 L199G + 113/114 L84G +++ 115/116 M87A + 117/118 L199Q + 119/120 L199D + 121/122 I207L + 123/124 M87H ++ 125/126 I198M + 127/128 L199S +++ 129/130 H154Q + 131/132 S191R + 133/134 I198S + 135/136 L199K +++ 137/138 H154L +++ 139/140 L199A +++ 141/142 H154V ++ 143/144 I198V + 145/146 H154A + 147/148 Q22A ++ 149/150 Q22L + 151/152 Q22P +++ 153/154 S356G +++ 155/156 Q22H + 157/158 L306V + 159/160 T262G + 161/162 D169T + 163/164 G347D + 165/166 S179V + 167/168 Q159M + 169/170 I233R + 171/172 Y396R ++ 173/174 E6P + 175/176 R139P ++ 177/178 Y421V + 179/180 R261W + 181/182 L106S ++ 183/184 N137G ++ 185/186 A97S ++ 187/188 R74W + 189/190 A110S + 191/192 R261P +++ 193/194 Q159R + 195/196 N195G + 197/198 E417R + 199/200 L106T + 201/202 R427A + 203/204 F64P + 205/206 H259Q + 207/208 R261H + 209/210 E417P + 211/212 K4P + 213/214 V435Q + 215/216 L106G + 217/218 L106D + 219/220 R261A + 221/222 E417A + 223/224 E112P + 225/226 V435R + 227/228 E112A +
.sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 32, and defined as follows: “–” = production less than the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 2-fold; “++” = at least 2-fold, but less than 3-fold; and “+++” = at least 3-fold increased production, as compared to the reference polypeptide.

Purified Protein Characterization Assay and Analysis for Glucosyl Transfer from ADP-Glucose to Stevioside or Rebaudioside D

(188) First, 1 µL purified protein was diluted in 100 µL total reaction volume of 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl.sub.2, with 1 mM stevioside (Chromadex, >94% purity) or rebaudioside D (ChromaDex, >93% purity) and 2 mM ADP-glucose. The reaction was performed at 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1-30 h. The reaction described above was quenched with 0.5 volume/volume acetonitrile with 0.2% formic acid and precipitated by centrifugation. Glycosylated products were detected in the supernatant by LC-MS/MS following 1:10 dilution in water with the instrument and parameters described in Example 1, Table 1.1. Specific activities described in Table 6.2 for stevioside and rebaudioside D were determined as µmol product formed per min per mg purified protein from the linear portion of the reaction progress curve.

(189) TABLE-US-00010 TABLE 6.2 Round 3.1 Variant Specific Activity for Stevioside and RebD SEQ ID NO: Amino Acid Differences µmol RebA/ µmol RebM/ (nt/aa) (Relative to SEQ ID NO: 32) min-mg.sup.a min-mg.sup.a 113/114 L84G + ++ 123/124 M87H – – 127/128 L199S – ++ 151/152 Q22P + +++ 153/154 S356G + +++ 157/158 L306V – ++ .sup.aLevels of increased specific activity were determined relative to the reference polypeptide of SEQ ID NO: 32, and defined as follows: “–” = activity less than 1.5-fold; “+” = activity at least 1.5-fold, but less than 3-fold; “++” = at least 3-fold, but less than 6-fold; and “+++” = at least 6-fold increased activity, as compared to the reference polypeptide.

(190) Directed evolution of the GT encoded by SEQ ID NO:31 (i.e., SEQ ID NO:32) was continued by constructing libraries of variant genes in which mutations associated with improved activity above were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described above with rebaudioside D to provide Round 3.3 with 59 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. The engineered polypeptides are listed in Table 6.3. Shake-flask scale cultures were grown for protein purification as described in Example 1 for variants with the amino acid mutations as shown in Table 6.3, relative to SEQ ID NO: 32.

(191) TABLE-US-00011 TABLE 6.3 Round 3.3 Variants and RebM Levels SEQ ID NO: Amino Acid Differences (nt/aa) (Relative to SEQ ID NO: 32) Increased RebM.sup.a 229/230 Q22P/L84G/H154L/I198S/L199A/S356G +++++ 231/232 Q22P/L84G/H154L/L199A/S356G +++++

233/234 Q22P/L84G/H154L/I198S/L199S/S356G +++++ 235/236
 Q22P/L84V/H154V/I198S/L199S/L306V/S356G +++++ 237/238
 Q22P/L84G/H154V/I198S/L199A/L306V/S356G +++++ 239/240
 Q22P/L84G/H154L/I198S/L199K/S356G +++++ 241/242 Q22P/L84G/H154L/L199S/S356G +++++
 243/244 Q22P/L84G/M87H/H154L/I198S/L199S/L306V/S356G +++++ 245/246
 Q22P/L84G/H154V/L199A/L306V/S356G +++++ 247/248
 Q22P/L84G/M87H/H154L/I198S/L199A/S356G +++++ 249/250
 Q22P/L84G/H154V/I198S/L199K/L306V/S356G +++++ 251/252
 Q22P/L84G/M87H/H154L/L199K/L306V/S356G +++++ 253/254 Q22P/L84G/H154V/L199S/S356G
 +++++ 255/256 Q22P/M87H/H154L/L199A/L306V/S356G +++ 257/258
 Q22P/L84G/M87H/H154V/I198S/L199S/L306V/S356G +++ 259/260
 Q22P/L84G/I198S/L199K/L306V/S356G +++ 261/262 Q22P/L84G/H154V/L199A/S356G +++ 263/264
 Q22P/H154L/I198S/L199A/I207L/L306V/S356G +++ 265/266 Q22P/H154L/L199A/S356G +++
 267/268 Q22P/H154L/I198S/L199A/S356G +++ 269/270
 Q22P/L84G/M87H/H154V/L199K/L306V/S356G +++ 271/272
 Q22P/M87H/H154V/I198S/L199S/I207L/L306V/S356G +++ 273/274
 Q22P/L84G/H154L/I207L/L306V/S356G +++ 275/276 Q22P/L84G/H154L/L306V/S356G +++ 277/278
 Q22P/M87H/H154V/L199S/S356G ++ 279/280 Q22P/M87H/L199A/S356G ++ 281/282
 Q22P/H154V/I198S/L199S/L306V/S356G ++ 283/284 Q22P/L84G/M87H/L199A/L306V/S356G ++
 285/286 Q22P/M87H/H154V/L199A/I207L/S356G ++ 287/288 Q22P/L84G/L199S/S356G ++ 289/290
 Q22P/L84G/H154L/I198S/L199A/I207L/L306V ++ 291/292 Q22P/M87H/H154L/I198S/L199K/S356G
 ++ 293/294 Q22P/L84G/M87H/I198S/L199K/S356G ++ 295/296 Q22P/H154V/L199A/S356G ++
 297/298 Q22P/M87H/H154V/L199K/S356G ++ 299/300 Q22P/H154V/L199K/I207L/L306V/S356G ++
 301/302 Q22P/L84G/I207L/S356G ++ 303/304 Q22P/M87H/H154V/L199S/L306V/S356G ++ 305/306
 Q22P/M87H/L199K/S356G ++ 307/308 Q22P/M87H/I198S/L199K/L306V/S356G ++ 309/310
 Q22P/M87H/H154L/P322S/S356G ++ 311/312 Q22P/M87H/I198S/L199K/S356G + 313/314
 Q22P/L84G/H154L/L199K/I207L + 315/316 Q22P/M87H/I198S/L199S/I207L/S356G + 317/318
 Q22P/L84G/M87H/I198S/L199A/L306V/S356G + 319/320 Q22P/L84G/M87H/I198S/L199S/S356G +
 321/322 Q22P/L84G/M87H/I198S/L199S/L306V/S356G + 323/324 Q22P/L199A/S356G + 325/326
 Q22P/L84G/H154L/I198S/L199A/L306V + 327/328 L84G/H154L/I198S/L199K + 329/330
 Q22P/L84G/M87H/L199S/S356G + 331/332 Q22P/L84G/M87H/H154L/S356G + 333/334
 Q22P/I198S/L199K/I207L/G329C/S356G + 335/336 Q22P/M87H/H154V/I198S/L199K/S356G +
 337/338 Q22P/I198S/L199A/S356G + 339/340 Q22P/M87H/I198S/L199A/L306V/S356G + 341/342
 Q22P/I207L/S356G + 343/344 Q22P/L84G/S356G + 345/346 Q22P/L84G/H154V/I198S/L199K +
 .sup. 1289/1290.sup.b Q22P/S356G – .sup.a

Levels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 32, and defined as follows: “–” = production less than 100-fold; the reference polypeptide; “+” = production at least 100-fold and less than 300-fold; “++” = at least 300-fold, but less than 600-fold; “+++” = at least 600-fold, but less than 900-fold; and “++++” = greater than 900-fold increased production, as compared to the reference polypeptide. .sup.bDoes not contain diversity specific to conversion of rebaudioside D to rebaudioside M; useful as an engineered biocatalyst to convert stevioside to rebaudioside A.

Purified Protein Characterization Assay and Analysis for Glucosyl Transfer from ADP-Glucose to Stevioside or Rebaudioside D

(192) First, 2 μ L purified protein was diluted in 100 μ L total reaction volume of 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl₂, with 1 mM rebaudioside D (ChromaDex, >93% purity) and 2 mM ADP-glucose. The reaction was performed at 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1-18 h. The reaction described above was quenched with 0.5 volume/volume acetonitrile with 0.2% formic acid and precipitated by centrifugation.

(193) Glycosylated products were detected in the supernatant by LC-MS/MS following 1:50 dilution in water with the instrument and parameters described in Example 1, Table 1.1. Specific activities described in Table 6.4 for stevioside was determined as μ mol rebaudioside A product formed per min per mg purified protein and for rebaudioside D was determined as μ mol rebaudioside M product formed per min per mg purified protein from the linear portion of the reaction progress curve. The enzymes listed in Table

6.4 catalyzed the conversion of RebD to RebM to >99% conversion and catalyzed the conversion of stevioside to a mixture of RebA and RebI with >85% conversion in less than 18 h with 0.8 g/L stevioside or 1.3 g/L rebD, 2 molar excess of ADP-glucose, and 35-77 mg/L purified protein.

(194) TABLE-US-00012 TABLE 6.4 Round 3.3 Variants Specific Activity for Stevioside and RebD SEQ ID μmol μmol NO: Amino Acid Differences RebA/ RebM/ (nt/aa) (Relative to SEQ ID NO: 32) min-mg.sup.a min-mg.sup.b 229/230 Q22P/L84G/H154L/I198S/L199A/S356G ++ ++ 231/232 Q22P/L84G/H154L/L199A/S356G ++ +++ 233/234 Q22P/L84G/H154L/I198S/L199S/S356G ++ ++ 237/238 Q22P/L84G/H154V/I198S/L199A/L306V/ ++ ++ S356G 243/244

Q22P/L84G/M87H/H154L/I198S/L199S/ + + L306V/S356G 253/254 Q22P/L84G/H154V/L199S/S356G +++ + .sup.aLevels of increased specific activity were determined relative to the reference polypeptide of SEQ ID NO: 32, and defined as follows: “+” = activity at least that of the reference polypeptide, but less than 2-fold; “++” = at least 2-fold, but less than 4-fold; and “+++” = at least 4-fold increased activity as compared to the reference polypeptide. .sup.bLevels of increased activity were determined relative to the reference polypeptide of SEQ ID NO: 32, and defined as follows: “-” = activity less than 10-fold; “+” = activity at least 10-fold, but less than 15-fold; “++” = at least 15-fold, but less than 20-fold; and “+++” = at least 20-fold increased activity, as compared to the reference polypeptide.

Example 7

Transformation of Rebaudioside D to Rebaudioside M with AGT (SEQ ID NO: 232)

(195) A 250-mL shake-flask culture was grown for protein purification of the polypeptide SEQ ID NO: 232 as described in Example 1. Then, 2.4 mL of the 50% glycerol stock of was diluted in 60 mL total reaction volume in a 250-mL baffled flask with 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl.sub.2, 2 mM ADP-glucose, and 1 mM rebaudioside D (ChromaDex, >93% purity). The reaction was performed at 30° C. in an Innova® shaking incubator with 250 RPM shaking for 2 h and quenched to pH<4 with 0.12 mL formic acid. The reaction was precipitated by centrifugation at 10,000 RPM for 10 m at 4° C. 6 g XAD-4 resin (Sigma) were added to the supernatant and incubated in the shake flask for 2 h. The resin was filtered and eluted with 16.3 mL 50:24:26 water:ACN:EtOH by incubating 4 h and re-filtering. A second elution was performed with 10 mL 50:50 water:EtOH, which filtered and combined with the first elution. The eluent was concentrated to about 6 mL by rotary evaporation, filtered through WHATMAN® UNIPREP® syringeless filters, and fractionated by HPLC using the instrument and parameters described in Table 7.1. From the C18 column, fractions were manually collected at retention times 5.8-6.2 m. Fractions were pooled, concentrated by rotary evaporation, and lyophilized. The sample was then resuspended in 1.5 mL ethanol and incubated on a stir plate at 80° C. for 2 h, concentrated by rotary evaporation, and dried 14 h under vacuum at 40° C. The samples were resuspended in pyridine-d5 and used for 1H, COSY, and HSQC NMR spectra acquisition performed by Acorn NMR.

(196) TABLE-US-00013 TABLE 7.1 Semi-Preparative HPLC Fractionation of Steviol Glycosides Instrument Agilent HPLC 1200 series Column Higgins C18 250 × 10 mm, 5 μm (Higgins Analytical) Mobile phase Isocratic 68:32 A:B A: 0.1% formic acid in water B: 0.1% formic acid in methanol Flow rate 2.5 mL/m Run time 10 m Peak retention times Rebaudioside M 6.00 m Column temperature 40° C. Injection volume 100 μL UV detection 210 nm (for steviol glycosides) 254 nm (for organic contaminants)

(197) The isolated product was determined to be rebaudioside M based on identity of the spectra to that of an authentic rebaudioside M standard (Chromadex, purity 95.6% by HPLC). By .sup.1H NMR, the exchangeable protons in the sample were broader than those in the standard and the sample contained a minor methyl contaminant. Otherwise, the spectra of the sample and standard were identical. Except for minor impurities, the COSY and HSQC spectra of the sample and standard were identical. The presence of six anomeric protons evident from the .sup.1H and .sup.1H-.sup.13C HSQC spectra [$\delta_{\text{sub.H}}$ 6.44, $\delta_{\text{sub.H}}$ 5.85, $\delta_{\text{sub.H}}$ 5.524, $\delta_{\text{sub.H}}$ 5.518, $\delta_{\text{sub.H}}$ 5.49, $\delta_{\text{sub.H}}$ 5.34] confirmed the presence of six sugar units in the structure and were coincident with the standard, indicating β -anomeric conformation. The attachment of sugars at the C2' and C3' hydroxyl positions is supported by the relatively downfield chemical shift of H-2' ($\delta_{\text{sub.H}}$ (4.54) and H-3' ($\delta_{\text{sub.H}}$ 5.15) in sugar I, suggesting a 2,3-branched-D-glucotriosyl substituent at C-19 (See FIG. 2 for carbon numbering). Peak assignments are listed in Table 7.2 and were determined from .sup.1H, COSY, and HSQC spectra and compared to literature (See e.g., Prakash et al., Nat. Prod. Comm., 11:1523-6 [2013]).

(198) TABLE-US-00014 TABLE 7.2 NMR Spectra of Enzymatically Produced Rebaudioside M in

C.sub.5D.sub.5N. Position δ .sup.13C, ppm δ .sup.1H, ppm .sup.1H Multiplicity (J, Hz) 1 40.3 0.77 t(12.1) 1 40.3 1.79 m 2 19.6 1.37 m 2 19.6 2.27 m 3 38.4 1.04 m 3 38.4 2.32 d(13.0) 4 Null 5 57.3 1.08 d(13.9) 6 23.5 2.26 m 6 23.5 2.44 q(12.6) 7 42.6 1.45 m 7 42.6 1.83 m 8 Null 9 54.3 0.93 d(8.49) 10 Null 11 20.2 1.67 m 11 20.2 1.79 m 12 38.5 1.88 m 12 38.5 2.78 m 13 Null 14 43.3 2.05 m 14 43.3 2.77 m 15 46.5 1.9 d(17) 15 46.5 2.06 m 16 Null 17 104.9 4.92 s 17 104.9 5.72 s 18 28.2 1.32 s 19 Null 20 16.7 1.41 s 1' 94.9 6.44 d(8.2) 2' 77 4.55 m 3' 88.7 5.15 t(8.5) 4' 70.1 4.22 m 5' 78.6 4.15 m 6' 61.7 4.22 m 6' 61.7 4.35 m 1'' 96.3 5.52 d(7.8) 2'' 81.5 4.15 m 3'' 88 5.01 obscured by water 4'' 70.3 4.1 m 5'' 77.7 3.98 m 6'' 62.5 4.23 m 6'' 62.5 4.35 m 1''' 104.7 5.52 d(7.6) 2''' 75.8 4.2 m 3''' 78.6 4.17 m 4''' 73.2 4.01 m 5''' 77.5 3.76 ddd(3.1, 6.3, 9.5) 6''' 64 4.3 m 6''' 64 4.53 m 1'''' 104 5.49 d(8.0) 2'''' 75.6 4 m 3'''' 77.7 4.53 m 4'''' 71.3 4.19 m 5'''' 77.9 4.02 m 6'''' 62.1 4.24 m 6'''' 62.1 4.35 m 1''''' 104.2 5.86 d(7.5) 2''''' 75.5 4.24 m 3''''' 78.4 4.24 m 4''''' 73.6 4.14 m 5''''' 77.8 3.94 ddd(3.0, 6.6, 9.6) 6''''' 64 4.36 m 6''''' 64 4.65 d(9.1) 1'''''' 104.2 5.34 d(8.1) 2'''''' 75.5 3.98 m 3'''''' 78 4.4 m 4'''''' 71.1 4.15 m 5'''''' 78.1 3.88 m 6'''''' 62 4.14 m 6'''''' 62 4.35 m

Example 8

ADP-Glycosyltransferase Variants of SEQ ID NO: 232

(199) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 232 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO:231 (i.e., SEQ ID NO:232) was carried out by constructing a library in which surface residue mutations associated with improved activity in round 3 were recombined. This library was then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a fourth round ("Round 4") of 76 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(200) HTP Assay for Glucose Transfer from ADP-Glucose to Rebaudioside D

(201) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 232 variants with lysate loading of 2.5 μ L lysate in 100 μ L reactions and with substrate loading of 0.5 mM rebaudioside D (ChromaDex, >94% purity), from a 20 mM stock solution in 50% ethanol and co-substrate loading of 0.5 mM ADP-glucose (Sigma, >93% purity). The following reaction conditions were used: 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl.sub.2, 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 h. The reactions were quenched by diluting the assay 1:10 and then quenching with 50 μ L/well acetonitrile with 0.2% formic acid, centrifuged 10 m at 4° C., and the supernatants were diluted 1:10 in water and analyzed by HPLC-MS/MS as described in Example 1, Table 1.1.

(202) The engineered polypeptides are listed in Table 8.1. Although the parent and variant constructs contain an N-terminal histidine tag for affinity purification, the mutations were numbered relative to the untagged reference sequence for clarity. Shake-flask scale cultures were grown, lysed, and lyophilized to powder as described in Example 1 for variants with SEQ ID NOS: 348, 350, 352, 354, 356, 364, 408, and 428.

(203) TABLE-US-00015 TABLE 8.1 Round 4 Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 232) RebM.sup.a 347/348

L109R/S131P/R139P/S179V/R261P/Y396R/Y421V +++ 349/350

L106T/L109R/R139P/G347D/E417A/Y421V/R427A +++ 351/352

L109R/E112P/S179V/G204D/I233R/E417A/Y421V/R427L +++ 353/354

F64P/L109R/E112P/R139P/E417A ++ 355/356

L106S/L109R/E112P/S131P/Q159R/G204D/G347D/E417A/Y421V/ ++ R427L 357/358

L109R/E112P/R139P/S179V/E417R/R427A ++ 359/360

F64P/L106S/R139P/S179L/I233R/E417A/R427A/Q431D ++ 361/362

F64P/L106S/L109R/R139P/S179L/E417R/Y421V/R427L/Q431D ++ 363/364

L106S/L109R/E112P/S131P/Q159R/S179L/E417A/Y421V ++ 365/366

F64P/L109R/R139P/I233R/E417R/Y421V ++ 367/368

L106S/E112P/Q159R/G204D/I233R/E417R/Y421V/R427A/Q431D ++ 369/370

L106S/L109R/E112P/R261P/E417R/Q431D ++ 371/372 L106S/E417A ++ 373/374

L109R/E112P/Q159R/S179V/E417R/Y421V ++ 375/376 L106T/G347D/E417R/R427A/Q431D ++

377/378 F64P/L109R/E112P/R139P/Q159R/S179L/E417R/Q431D ++ 379/380

E112P/S131P/S179L/G347D/E417R/Y421V ++ 381/382

F64P/L106T/E112P/Q159R/S179L/E417R/Y421V ++ 383/384
 F64P/L106S/L109R/E112P/E417A/Y421V/R427A/Q431D + 385/386
 L106S/L109R/E112P/G347D/R427A + 387/388 F64P/R139P/G347D/E417R/Y421V/R427L/Q431D +
 389/390 F64P/L109R/Q159R/S179V/G204D/I233R/E417R/Y421V + 391/392
 F64P/L106T/G204D/E417A/Y421V/R427A + 393/394 S179V/E417R/Y421V + 395/396
 F64P/L106S/E417R/Y421V + 397/398 E112P/Q159R/E417R/Y421V + 399/400
 L109R/E112P/E417A/R427A/Q431D + 401/402 L106S/E112P/I233R/E417R + 403/404
 L109R/E112P/I233R/E417A/Q431D + 405/406 L109R/E112P/S131P/Q159R/E417R/Y421V + 407/408
 L109R/R139P/S179L/E417R/Y421V/R427A + 409/410 L106T/R139P/Y421V + 411/412
 L109R/E112P/E417A/Y421V/R427L + 413/414 E112P/E417R/Y421V + 415/416
 F64P/R139P/I233R/E417R/R427L/Q431D/K439P + 417/418
 L106T/E112P/Q159R/S179L/G204D/E417A/Y421V + 419/420 I233R/E417R/Y421V/R427L/Q431D +
 421/422 L106S/I233R/Y421V/R427A + 423/424 L109R/S179V/I233R/Y421V + 425/426
 F64P/L106T/E112P/R139P/Q159R/S179V/G204D/Y396R/E417A + 427/428
 L109R/G204D/E417R/Q431D + 429/430 F64P/L109R/E417R/Y421V + 431/432
 L109R/E112P/S179V/G347D/E417R + 433/434 F64P/L106S/L109R/I233R/G347D/R427A/Q431D +
 435/436 L106S/L109R/R139P/R427A/Q431D + 437/438
 L106S/R139P/Q159R/I233R/G347D/E417R/Y421V/R427A/Q431D + 439/440
 L106S/E112P/Y396R/E417R/Y421V + 441/442 L106S/L109R/E112P/G204D/G347D/Y421V/K439P +
 443/444 R139P/I233R/E417A + 445/446 L109R/E417R/Y421V + 447/448
 L109R/E417R/R427A/Q431D + 449/450 L109R/S131P/G204D + 451/452 G347D/E417R + 453/454
 L106T/L109R/R139P/E417R + 455/456 L106T/L109R/I233R/R427A/Q431D + 457/458
 L106S/L109R/E417R/Y421V/R427L + 459/460 L109R/E112P/G204D/I233R/E417R + 461/462
 Q159R/G347D/E417A/Y421V/Q431D + 463/464 L106T/L109R + 465/466
 L109R/E112P/S131P/Q159R/S179V/K439P + 467/468 F64P/L106S/E112P/G347D/E417A/Y421V +
 469/470 L109R/E112P/Q159R/E417R/R427L + 471/472 L109R/E112P/G204D/R427A + 473/474
 F64P/L106T/L109R/E112P/S131P/S179L/E417R/R427A/Q431D + 475/476 F64P/L106S/Q431D +
 477/478 F64P/E417R/Y421V/Q431D + 479/480 L106T/L109R/R139P/I233R/E417R/Y421V + 481/482
 E112P/S131P/S179V/G204D/E417R/Y421V/R427L + 483/484
 L106T/E112P/Q159R/S179V/I233R/E417A/Y421V/R427L/K439P + 485/486
 F64P/L106S/L109R/E112P/S131P/Q159R/I233R/Y421V/R427L/ + Q431D 487/488
 S131P/S179L/I233R/E417R/R427A + 489/490 F64P/L109R/E417A/Y421V + 491/492
 L109R/R139P/S179L/E417R/R427L + 493/494
 F64P/L106T/L109R/E417A/Y421V/R427L/Q431D/K439P + 495/496
 E112P/R139P/S179V/G204D/I233R/G347D/R427L + 497/498
 L106T/S131P/S179L/I233R/Y421V/R427L/Q431D + .sup.a

Levels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 232, and defined as follows: “+” = production at least 1.6-fold, but less than 2-fold; “++” = at least 2-fold, but less than 2.4-fold; and “+++” = at least 2.4-fold increased production, as compared to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from ADP-Glucose to Stevioside or Rebaudioside D

(204) A time course experiment was performed to characterize the activity of the engineered round 4 variants on stevioside and rebaudioside D. One g/L shake flask powder (SFP) was added to a 100 μ L total reaction volume containing 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl₂, 2.5% v/v ethanol with 1 mM stevioside (Chromadex, >94% purity) or rebaudioside D (ChromaDex, >93% purity) and 1 mM ADP-glucose. The reaction was performed at 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 0.5-2 h. The reaction was quenched by adding 50 μ L acetonitrile with 0.2% formic acid and precipitated by centrifugation. The supernatant was diluted 1:50 in water and analyzed for steviol glycosides by LC-MS/MS as described in Example 1, Table 1.1. All 8 variants had higher activities on both stevioside and rebaudioside D than SEQ ID NO: 232. The levels of rebaudioside A produced from stevioside by the variants relative to SEQ ID NO: 232 at the 1 hour time point are listed in Table 8.2.

(205) TABLE-US-00016 TABLE 8.2 Round 4 Variant SFP Activity on Stevioside SEQ ID Amino Acid Differences Increased NO: (nt/aa) (Relative to SEQ ID NO: 232) RebA.sup.a 355/356

L106S/L109R/E112P/S131P/Q159R/G204D/G347D/E417A/Y421V/R427L +++ 351/352
 L109R/E112P/S179V/G204D/I233R/E417A/Y421V/R427L ++ 427/428 L109R/G204D/E417R/Q431D
 ++ 347/348 L109R/S131P/R139P/S179V/R261P/Y396R/Y421V ++ 363/364
 L106S/L109R/E112P/S131P/Q159R/S179L/E417A/Y421V ++ 349/350
 L106T/L109R/R139P/G347D/E417A/Y421V/R427A + 407/408
 L109R/R139P/S179L/E417R/Y421V/R427A + 353/354 F64P/L109R/E112P/R139P/E417A +
 .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID
 NO: 232, at the 1 h time point and defined as follows: “+” = production at least 1.5-fold, but less than 2-
 fold; “++” = at least 2-fold, but less than 3-fold; and “+++” = at least 3-fold increased production, as
 compared to the reference polypeptide.

(206) The activities of the 8 engineered variants were clearly higher on rebaudioside D than the activity
 of SEQ ID NO: 232, but they were not well distinguished from each other in the time course. Therefore, a
 follow-up experiment was performed as follows: A dose response curve of 0.03-1 g/L shake flask powder
 (SFP) was added to a 100 μ L total reaction volume containing 50 mM Tris-HCl buffer, pH 7.5, 3 mM
 MgCl₂, 2.5% v/v ethanol with 1 mM rebaudioside D (ChromaDex, >93% purity) and 1 mM ADP-
 glucose. The reaction was performed at 30° C. in a Thermotron® titre-plate shaker with 300 RPM
 shaking for 0.5 h. The reaction was quenched by adding 50 μ L acetonitrile with 0.2% formic acid and
 precipitated by centrifugation. The supernatant was diluted 1:67 in water and analyzed for steviol
 glycosides by LC-MS/MS as described in Example 1, Table 1.1. The activities of the round 4 variants on
 rebaudioside D at 0.0625 g/L SFP loading are listed in Table 8.3.

(207) TABLE-US-00017 TABLE 8.3 Round 4 Variant SFP Activity on Rebaudioside D SEQ ID NO:
 Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 232) RebM.sup.a 349/350

L106T/L109R/R139P/G347D/E417A/Y421V/R427A + 407/408
 L109R/R139P/S179L/E417R/Y421V/R427A – 355/356
 L106S/L109R/E112P/S131P/Q159R/G204D/G347D/E417A/Y421V/ – R427L 353/354
 F64P/L109R/E112P/R139P/E417A + 347/348 L109R/S131P/R139P/S179V/R261P/Y396R/Y421V +++
 363/364 L106S/L109R/E112P/S131P/Q159R/S179L/E417A/Y421V – .sup.aLevels of increased
 production were measured as μ mol RebM/mg SFP-min at 0.0625 g/L SFP and 30 min and defined
 relative to SEQ ID NO: 232, as follows: “+” = production at least that of the reference polypeptide, but
 less than 2-fold; “++” = at least 2-fold, but less than 3-fold; “+++” = at least 3-fold, but less than 4-fold;
 and “++++” = at least 4-fold increased production, as compared to the reference polypeptide.

Example 9

ADP-Glycosyltransferase Variants of SEQ ID NO: 348

(208) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID
 NO: 348, for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed
 evolution of the GT encoded by SEQ ID NO:347 (i.e., SEQ ID NO:348) was carried out by constructing
 libraries in which mutations associated with improved activity in previous rounds were recombined and
 in which mutations identified from homologs in publically available databases were recombined. These
 libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below
 to provide a fifth round (“Round 5”) of engineered GT variant polypeptides, 6 from library 5.01 (Table
 9.1) and 18 from the remaining libraries in the round (Table 9.2), with glucosyltransferase activity toward
 ADP-glucose and steviol glycosides.

(209) HTP Assay for Glucose Transfer from ADP-Glucose to Steviol Glycosides

(210) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO:
 348 variants. Lysis buffer volume was increased to 400 μ L from the 250 μ L used in previous rounds, and
 the lysate was diluted 10-fold. For round 5.01, assays were conducted with 10 μ L lysate in 100 μ L
 reactions and with substrate loading of 1 mM stevioside (ChromaDex, >94% purity), from a 20 mM stock
 solution in 50% ethanol and co-substrate loading of 1 mM ADP-glucose (Sigma, >93% purity). The
 following reaction conditions were used: 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl₂, 40° C. in a
 Thermotron titre-plate shaker with 300 RPM shaking for 1 h. The reactions were quenched by adding 10
 μ L assay 90 μ L acetonitrile with 0.2% formic acid and centrifuged 10 m at 4° C. The supernatants were
 diluted 1:10 in water and analyzed by RapidFire SPE-MS/MS (Agilent) as described in Table 9.3. For the
 remaining round 5 libraries, the lysate was diluted 4-fold instead of 10-fold, 50 mM potassium phosphate

buffer pH 7 was used instead of Tris-HCl the temperature was 50° C., the reaction time was 2 h, and the assay was performed with both stevioside and rebaudioside D (Chromadex, >93%).

(211) The engineered polypeptides are listed in Table 9.1 and Table 9.2. Although the parent and variant constructs contain an N-terminal histidine tag for affinity purification, the mutations were numbered relative to the untagged reference sequence for clarity. Shake-flask scale cultures were grown, lysed, and lyophilized to powder as described in Example 1 for SEQ ID NO: 500.

(212) TABLE-US-00018 TABLE 9.1 Round 5.01 Variants and RebA Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 348) RebA.sup.a 499/500

L106S/E112P/G204D/G347D/R396Y/E417R/ +++ R427A/Q431D 501/502

G204D/G347D/R396Y/E417R/Q431D ++ 503/504 L106S/E112P/G204D/G347D/R396Y/E417R/ ++

R427A 505/506 E112P/G204D/G347D/R396Y/E417R/R427A/ + Q431D 507/508

L106S/E112P/G204D/G347D/R396Y/ + E417R 509/510 E112P/G204D/G347D/R396Y/E417R/ +

R427A .sup.aLevels of increased production were determined relative to the reference polypeptide of

SEQ ID NO: 348, and defined as follows: “+” = production at least 10-fold that of the reference

polypeptide, but less than 20-fold; “++” = at least 20-fold, but less than 30-fold; and “+++” = at least 30-

fold increased production, as compared to the reference polypeptide.

(213) TABLE-US-00019 TABLE 9.2 Additional Round 5 Variants and RebA and RebM Levels SEQ ID

NO: Amino Acid Differences Increased Increased (nt/aa) (Relative to SEQ ID NO: 348) RebA.sup.a

RebM.sup.b 511/512 N137G/D161L/V435R +++++ 513/514 K38R/A178V/W401L ++ +++++

515/516 R74W/R102K/N137G/D161L/ ++ + H259S/K289S 517/518 I14V/L100F ++ +++++ 519/520

D290E/A351G/W401L ++ +++ 521/522 I28M/I44V/V365I/A407E ++ +++ 523/524

K185R/D290E/W401L/I422M ++ ++ 525/526 D98P/I233W ++ +++ 527/528 T54P/V413L ++ +++

529/530 R102K/D161L/T250A/V435E ++ ++ 531/532 I92L/S118A ++ ++ 533/534

K38R/S118A/D290E/A351G/D375P/ ++ ++ W401L/I422M 535/536 N137G/D169G + ++ 537/538

K38R/D290E/A351G/W401L/I422M + + 539/540 V435Q/M438A + ++ 541/542

Q159M/D169S/R173G/D300Q/ + + Q424E/M438A 543/544 S118A/S156A/A178V/D290E/ + ++

D375P/W401L/I422M 545/546 A110G/K222R/T250R/H259P/ + + V435G .sup.aLevels of increased

RebA production were determined relative to the reference polypeptide of SEQ ID NO: 348, and defined

as follows: “+” = production at least that of the reference polypeptide, but less than 3-fold; “++” =

production at least 3-fold, but less than 6-fold; “+++” = at least 6-fold, but less than 9-fold; and “++++” =

at least 9-fold increased production, as compared to the reference polypeptide. .sup.bLevels of increased

RebM production were determined relative to the reference polypeptide of SEQ ID NO: 348, and defined

as follows: “+” = production at least that of the reference polypeptide, but less than 15-fold; “++” =

production at least 15-fold, but less than 30-fold; “+++” = at least 30-fold, but less than 45-fold; and

“++++” = at least 45-fold increased production, as compared to the reference polypeptide.

(214) TABLE-US-00020 TABLE 9.3 RapidFire SPE-MS/MS Conditions for Steviol Glycoside Detection.

Agilent RapidFire Conditions Buffer A 0.1% formic acid in LC/MS grade water; 1.5 mL/min flow rate

Buffer B 0.1% formic acid in LC/MS grade methanol; 0.5 mL/min flow rate Aqueous wash Water

Organic wash Acetonitrile SPE cartridge Agilent RapidFire cartridge A (C4) RF state 1 600 ms RF state 2

2500 ms RF state 3 0 RF state 4 5000 ms RF state 5 1000 ms Agilent Jet Stream source parameters

Drying gas 325° C. temperature Drying gas flow 7 L/min Nebulizer pressure 50 psi Sheath gas 300° C.

temperature Sheath gas flow 12 L/min Capillary voltage +4000 V Nozzle voltage +500 V Agilent 6470

Triple Quadrupole MRM parameters Frag- Compound Q1 Q3 Dwell mentor CE CAV Stevioside 827.4

665.3 50 150 50 5 RebA 989.5 827.5 50 300 60 5 RebD or RebI 1151.7 827.5 50 285 55 5 RebM 1313.7

827.5 50 350 70 5

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from ADP-Glucose to Stevioside or Rebaudioside D

(215) A shake flask powder loading dose response experiment was performed to characterize activity of the engineered round 5 variant SEQ ID NO: 500 relative to SEQ ID NO: 348 on stevioside and

rebaudioside D. Levels of 0.03-1 g/L shake flask powder (SFP) were added to a 100 µL total reaction

volume containing 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl.sub.2, 2.5% v/v ethanol with 1 mM

stevioside (Chromadex, >94% purity) or rebaudioside D (ChromaDex, >93% purity) and 1 mM ADP-

glucose. The reaction was performed at 40° C. in a thermocycler for 0.5 h, and the reaction was quenched

with 50 μ L acetonitrile with 0.2% formic acid and precipitated by centrifugation. The supernatant was diluted 1:67 in water and analyzed for steviol glycosides by LC-MS/MS as described in Example 1, Table 1.1. At 1 g/L loading, SEQ ID NO: 500 produced >1.2-fold as much rebaudioside A from stevioside as SEQ ID NO: 348, and >1.07-fold as much rebaudioside M from rebaudioside D. Subsequently, SEQ ID NO: 499 (i.e., 500) was re-cloned to omit the N-terminal histidine tag, expressed in high throughput, and the plates were lysed in 400 μ L. The resulting powder, containing the variant SEQ ID NO: 548, was assayed relative to the SEQ ID NO: 500 variant, with 2.5 μ L lysate added to a 100 μ L total reaction volume containing 50 mM potassium phosphate buffer, pH 7, 3 mM MgCl₂, 2.5% v/v ethanol with 1 mM stevioside (Chromadex, >94% purity) or rebaudioside D (ChromaDex, >93% purity) and 1 mM ADP-glucose. The reaction was performed at 50° C. in a Thermotron® shaking incubator at 300 RPM for 4 h, and the reaction was quenched by adding 10 μ L assay to 90 μ L acetonitrile with 0.2% formic acid and precipitated by centrifugation. The supernatant was diluted 1:10 in water and analyzed for steviol glycosides by RapidFire SPE-MS/MS as described in Table 9.3. Under these conditions, the SEQ ID NO: 548 variant produced nearly 1.5-fold more rebaudioside A from stevioside as the SEQ ID NO: 500 variant, and over 1.6-fold more rebaudioside M from rebaudioside D.

(216) Subsequently, shake flask powder was prepared containing the variant of SEQ ID NO: 548, and it was assayed relative to the SEQ ID NO: 348 variant. Levels of 0.25-10 g/L shake flask powder (SFP) were added to a 100 μ L total reaction volume containing 50 mM potassium phosphate buffer, pH 7, 3 mM MgCl₂, 2 mM ADP-glucose, and 1 g/L rebaudioside A 60 (60% rebaudioside A, >35% stevioside) or rebaudioside D (Chromadex, >93% purity). The reaction was performed at 40° C. in a thermocycler for 1 h, and the reaction was quenched by adding 10 μ L assay to 90 μ L acetonitrile with 0.2% formic acid and precipitated by centrifugation. The supernatant was diluted 1:10 in water and analyzed for steviol glycosides by LC-MS/MS as described in Example 1, Table 1.1. The SEQ ID NO: 548 variant reached >65% conversion of rebaudioside A 60 to a mixture of rebaudioside A and rebaudioside I and nearly complete conversion of rebaudioside D to rebaudioside M. At 0.25 g/L shake flask powder loading, the SEQ ID NO: 548 variant reached 1.4-fold the conversion of rebaudioside D as the SEQ ID NO: 348 variant, and 2.2-fold the conversion of rebaudioside A 60.

(217) Expression Analysis of AGT Enzymes

(218) The six rounds of engineered AGT polypeptides for β -1,3-glucosylation of steviol glycosides were analyzed by polyacrylamide gel-electrophoresis to determine relative protein expression levels. Samples were prepared with 1 $\times$ LDS loading buffer and 1 $\times$ reducing agent (Life Technologies). A 4-12% Bis-Tris acrylamide gel (Life Technologies) was loaded with 5 μ g per lane of lyophilized soluble crude lysate from shake flask scale cultures and run with MES running buffer for 25 min at 200 V, and bands were quantified using ImageJ analysis software. The relative expression levels are listed in Table 9.4. SEQ ID NOs: 547/548, 499/500, and 347/348 are significantly better expressed genes and/or significantly better folded/more stable proteins. Thus, these genes produced more protein than the wild-type gene.

(219) TABLE-US-00021 TABLE 9.4 Protein Levels of Engineered AGT Variants SEQ ID NO: Increased Protein.sup.a 548 +++ 500 ++ 348 ++ 232 + 32 - 8 - .sup.aLevels of increased protein were determined relative to the reference polypeptide of SEQ ID NO: 4 and defined as follows: “+” = band intensity at least that of the reference polypeptide, but less than 2-fold; “++” = band intensity at least 2-fold, but less than 4-fold; and “+++” = at least 4-fold increased band intensity, relative to the reference polypeptide.

Example 10

Transformation of Rebaudioside a to Rebaudioside I with AGT (SEQ ID NO: 500)

(220) In this Example, scale-up of the reaction transforming rebaudioside A to rebaudioside I using the SEQ ID NO: 500 variant, and the isolation and characterization of rebaudioside I are described. A reaction containing 5 g/L of the SEQ ID NO: 500 variant lyophilized shake flask powder, 50 mM potassium phosphate buffer pH 7, 10.3 mM magnesium chloride, 10 g/L rebaudioside A (>97% purity), and 10.3 mM ADP-glucose (Sigma) with 10 mL total volume was stirred on a stir plate at 300 RPM and 35° C. for 89 h. The reaction was transparent when it was begun, and by the end of the reaction it was a white emulsion. The assay was diluted 1:10 in water, and 10 μ L of diluted assay was added to 90 μ L acetonitrile with 0.2% formic acid and precipitated by centrifugation. The supernatant was diluted 1:10 in water and analyzed for steviol glycosides by LC-MS/MS as described in Example 1, Table 1.1. This analysis confirmed production of rebaudioside I at high conversion. Because of its low solubility,

rebaudioside I settled. Therefore, the isolation was conducted by removing the supernatant, resuspending the sediment with the minimal volume of water, and centrifuging. This wash step was repeated twice, and a brown sediment at the surface of the white sediment was scraped off. The material was lyophilized and analyzed by Acorn NMR. The material was dissolved in pyridine-d₅, and ¹H 1-D, ¹³C DEPT-135, ¹H-¹³C HSQC NMR experiments were performed with a Varian Inova 500 NMR spectrometer. The NMR spectra were completely consistent with literature reports for rebaudioside I (see Table 10.1). See FIG. 3 for the structure of rebaudioside I with carbons numbered.

(221) TABLE-US-00022 TABLE 10.1 NMR Characterization of Rebaudioside I Position δ ¹³C, ppm. ¹H, ppm ¹H Multiplicity (J, Hz) 1 40.4 0.73 t (13.4) 1 40.4 1.74 m 2 19.1 1.42 m 2 19.1 2.18 m 3 38.2 1.02 M 3 38.2 2.35 m 4 — — 5 57 1.03 m 6 21.9 1.88 m 6 21.9 2.32 m 7 41.5 1.29 m 7 41.5 1.31 m 8 — — 9 53.9 0.88 d (7.2) 10 — — 11 20.3 1.69 m 11 20.3 1.7 m 12 37 1.97 m 12 37 2.27 m 13 — — 14 44.1 1.79 m 14 44.1 2.59 d (11.9) 15 47.4 2.05 Brs 16 — — 17 104.5 5.02 s 17 104.5 5.66 s 18 28.1 1.22 s 19 — — 20 15.4 1.27 s 1' 95 6.14 d (8.2) 2' 72.3 4.17 m 3' 89.4 4.25 m 4' 69 4.23 m 5' 77.8 3.91 m 6' 61.4 4.33 m 6' 61.4 4.25 m 1'' 97.8 5.05 d (7.8) 2'' 80.4 4.32 m 3'' 87.5 4.17 m 4'' 69.9 3.95 m 5'' 77.3 3.79 m 6'' 62.3 4.16 m 6'' 62.3 4.49 m 1''' 104.4 5.55 d (7.7) 2''' 76 4.18 m 3''' 78.3 4.26 or 4.23 m 4''' 71.9 4.22 m 5''' 78.5 3.92 m 6''' 62.9 4.39 m 6''' 62.9 4.51 m 1'''' 104.5 5.35 d (7.9) 2'''' 75.1 or 75.3 4.03 m 3'''' 78.3 4.26 m 4'''' 71.2 or 71.4 4.12 m 5'''' 78.3 4.11 m 6'''' 62.1 4.24 m 6'''' 62.1 4.56 m 1''''' 104.8 5.26 d (7.9) 2''''' 75.1 or 75.3 4.03 M 3''''' 78.3 4.26 or 4.23 M 4''''' 71.2 or 71.4 4.12 M 5''''' 78.3 4 M 6''''' 62.2 4.24 M 6''''' 62.2 4.54 M ¹H, DEPT-135, and HSQC in pyridine-d₅ on a Varian Inova 500 NMR instrument. The spectra were compared to literature spectra (See e.g., Prakash et al., Molecules, 19(11): 17345-55 [2014]) in order to assign peaks. ¹³C signals are from DEPT-135. Some peaks were too close in shift to determine assignment; when this occurred both options are listed.

Example 11

ADP-Glycosyltransferase Variants of SEQ ID NO: 548

(222) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 548 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO:547 (i.e., SEQ ID NO:548) was carried out by constructing libraries in which mutations associated with improved activity in previous rounds were recombined. This library was then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a sixth round ("Round 6") of 66 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides (Table 11.1).

(223) HTP Assay for Glucose Transfer from ADP-Glucose to Steviol Glycosides

(224) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 548 variants. Lysis buffer volume was 400 μL, and the lysate was diluted 4-fold. The assay was conducted with 10 μL lysate in 100 μL reactions and with substrate loading of 1 mM stevioside (ChromaDex, >94% purity), from a 20 mM stock solution in 50% ethanol and co-substrate loading of 1 mM ADP-glucose (Sigma, >93% purity). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 7, 3 mM MgCl₂, 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. The reactions were quenched by adding 10 μL assay 90 μL acetonitrile with 0.2% formic acid and centrifuged 10 m at 4° C. The supernatants were diluted 1:10 in water and analyzed by RapidFire SPE-MS/MS as described in Table 9.3. The top 84 variants were retested with the same conditions using 1 mM stevioside (Chromadex, >94% purity), 1 mM rebaudioside A (>97% purity), or rebaudioside D (Chromadex, >93% purity). The resulting engineered GT variant polypeptides are listed in Table 11.1. Shake-flask scale cultures were grown, lysed, and lyophilized to powder as described in Example 1 for variants with SEQ ID NOS: 554, 562, 568, and 576.

(225) TABLE-US-00023 TABLE 11.1 Round 6 Variants and RebA, RebI, and RebM Levels SEQ ID NO: Amino Acid Differences Increased Increased Increased (nt/aa) (Relative to SEQ ID NO: 548) RebA.¹H, ppm RebI.¹H, ppm RebM.¹H, ppm 549/550 I28M/R74W/D161L/D290E/V365I +++ +++ + 551/552 I28M/I44V/I92L/D161L/K222R/D300Q/ +++ + + V413L 553/554 I28M/R74W/S156A/D161L/V365I/A407E +++ ++ + 555/556 I28M/R74W/S156A/Q159M/D161L/ +++ + + A178 V/D300Q/V365I/V435Q/M438A 557/558 I14V/I28M/I92L/L100F/R102K/Q159M/ +++ + + D161L/I233W/A351G/I422M 559/560 I14V/R74W/D161L/D375P/W401L +++ - + 561/562

I14V/I28M/K38R/L100F/R102K/ +++ + + S118A/D161L/D169G/A178V/I233W/
 T250R/A407E/I422M/M438A 563/564 I28M/K38R/I92L/D98P/L100F/R102K/ +++ + + S156A/D161L
 565/566 I14V/I28M/I92L/L100F/R102K/Q159M/ +++ - + D161L/D169S/I233W 567/568
 I28M/I44V/S118A/S156A/D161L/K222R/ +++ + - K289S/V435Q/M438A 569/570
 I28M/I92L/L100F/R102K/A110G/D161L/ +++ - + K185R/T250A/D300Q/D375P/V435Q 571/572
 I14V/I28M/I44V/D161L/D169S +++ + + 573/574 I28M/S156A/D161L/I233W/H259S/ +++ ++ +
 D300Q/V435R 575/576 I14V/I28M/T54P/D161L/K185R/V413L +++ ++ + 577/578
 I14V/I28M/K38R/L100F/R102K/A110G/ ++ + + Q159M/D161L/I233W/H259S/D290E/
 D300Q/A351G/V435Q 579/580 I28M/S156A/D161L/K185R/V435R/ ++ ++ + M438A 581/582
 I28M/I92L/D98P/L100F/A110G/S156A/ ++ - - D161L/W401L 583/584
 I28M/D98P/L100F/R102K/D161L/K185R/ ++ - + A351G/W401L/V435E/M438A 585/586
 I14V/I28M/L100F/N137G/S156A/D161L/ ++ + + K222R/H259S/K289S/V365I/W401L/ V435R/M438A
 587/588 I14V/I28M/I92L/L100F/D161L/K222R/ ++ - + I233W/K289S/D300Q 589/590
 R102K/N137G/Q159M/D161L/I422M/ ++ - + Q424E 591/592 I14V/I28M/L100F/S156A/D161L/T250R
 ++ + + 593/594 I14V/I28M/R74W/D98P/L100F/R102K/ ++ - - D161L/R173G/A178V/I233W/T250A/
 H259S/D290E/A407E 595/596 I14V/I28M/K38R/A110G/N137G/D161L/ ++ - - K222R/K289S/W401L
 597/598 I28M/T54P/I92L/Q159M/D161L/D290E ++ + + 599/600
 I14V/I28M/I44V/R74W/D98P/R102K/ ++ - - N137G/Q159M/D161L/K185R/K222R/ T250A 601/602
 I14V/I28M/R102K/S118A/N137G/D161L/ ++ - + K185R/K222R/T250R/H259P/W401L 603/604
 Q159M/D161L/K222R/D290E/D375P/ ++ - + A407E 605/606 I14V/A110G/S156A/D161L/D375P/ ++
 - + W401L/V435E/M438A 607/608 I14V/I92L/D98P/L100F/Q159M/D161L/ ++ - +
 H259P/V365I/I422M/Q424E/V435R 609/610 K38R/D161L/D300Q/M438A ++ - + 611/612
 R74W/S156A/D161L/R173G/A178V/ ++ - - Q424E/V435E 613/614
 R74W/D98P/L100F/R102K/S118A/ ++ - - S156A/Q159M/D161L/V435E/M438A 615/616
 I14V/Q159M/D161L/V365I/V435E/ ++ - - M438A 617/618 N137G/Q159M/D161L/K185R/D300Q/ ++
 - - A351G/V365I/V435Q 619/620 I14V/R74W/A110G/Q159M/D161L/ ++ - -
 D169G/R173G/T250A/H259P/D290E/ D375P/A407E/I422M 621/622
 I14V/D161L/K222R/T250R/H259S/K289 ++ - - S/D375P/W401L/V413L 623/624
 Q159M/D161L/W401L ++ - - 625/626 I44V/R74W/L100F/R102K/D161L/I233W/ ++ - -
 V365I/V435G/M438A 627/628 I14V/D161L/D300Q + - - 629/630 I28M/I44V/R74W/W401L + + -
 631/632 I14V/D161L/K222R/T250R/V435Q/ + - - M438A 633/634
 R74W/D98P/L100F/R102K/A110G/ + - - S118A/D161L/A178V/T250R/K289S/
 D290E/D300Q/V435E/M438A 635/636 S156A/Q159M/D161L/D169S + - - 637/638
 I14V/K38R/L100F/R102K/D161L/R173G/ + - - A178V/K222R/T250R/D375P/W401L/ V413L 639/640
 D98P/L100F/S156A/Q159M/D161L/ + - - A178V/H259S/K289S/D290E/A351G/ I422M 641/642
 I14V/K38R/R74W/A110G/S156A/D161L/ + - - R173G/A178V/K222R/D300Q 643/644
 D98P/L100F/S118A/Q159M/D161L/ + - - D300Q 645/646 I14V/I28M/I92L/L100F/R102K/I422M/ +
 +++ + Q424E/V435Q 647/648 I14V/I28M/Q159M/V365I/V435Q + +++ + 649/650
 I28M/I92L/S118A/Q159M/K222R/T250R/ + + + H259P/D300Q/A407E 651/652
 D161L/H259S/K289S/V435R/M438A + - - 653/654 I28M + ++ + 655/656
 I28M/T54P/Q159M/D290E/M438A + ++ + 657/658 I14V/I28M/Q159M/K289S/D290E/D300Q + ++ +
 659/660 I28M/I44V/K289S/D290E/A351G/I422M + +++ + 661/662 I28M/I44V/V435R/M438A + ++ +
 663/664 I14V/I28M/I44V/I92L + ++ + 665/666 I14V/I28M/A351G + ++ + 667/668
 I28M/I44V/D98P/L100F/R102K/S118A + ++ + 669/670 I14V/I28M/I44V/D375P + + + 671/672
 I14V/I28M/T54P/V365I + +++ + 673/674 I14V/I28M/K38R/R74W/R102K/S156A/ + ++ +
 Q159M/I233W/T250A/K289S/V413L/ I422M/Q424E/V435R/M438A 675/676 R74W/D375P/V435G +
 - - 677/678 S118A + + - 679/680 I28M/T54P/T250R/K439N - ++ + .sup.aLevels of increased
 production were determined relative to the reference polypeptide of SEQ ID NO: 548 and defined as
 follows: “-” = production less than that of the reference polypeptide; “+” = production at least that of the
 reference polypeptide, but less than 1.5-fold; “++” = at least 1.5-fold, but less than 2-fold; and “+++” = at
 least 2-fold increased production, as compared to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from ADP-Glucose to
 Stevioside or Rebaudioside D

(226) A shake flask powder loading response experiment was performed to characterize activity of the engineered round 6 variants on stevioside and rebaudioside D. Levels of 0.03-1 g/L shake flask powder (SFP) were added to a 100 µL total reaction volume containing 50 mM potassium phosphate buffer, pH 7, 3 mM MgCl₂, 2.5% v/v ethanol with 1 mM stevioside (Chromadex, >94% purity) or rebaudioside D (ChromaDex, >93% purity) and 1 mM ADP-glucose. The reaction was performed at 50° C. in a Thermotron titre-plate shaker at 300 RPM for 1 h, and the reaction was diluted with 150 µL water and then quenched by transferring 12.5 µL into 87.5 µL acetonitrile with 0.2% formic acid and precipitated by centrifugation. The supernatant was diluted 1:20 in water and analyzed for steviol glycosides by RapidFire SPE-MS/MS as described in Example 9, Table 9.3. The relative productivities of the enzymes are listed in Table 11.2.

(227) TABLE-US-00024 TABLE 11.2 Round 6 Variant SFP Activity on Stevioside and Rebaudioside D SEQ ID NO: Amino Acid Differences Increased Increased (nt/aa) (Relative to SEQ ID NO: 548)

RebA.sup.a RebM.sup.a 576 I14V/I28M/T54P/D161L/K185R/ +++ ++ V413L 568

I28M/I44V/S118A/S156A/D161L/ +++ ++ K222R/K289S/V435Q/M438A 554

I28M/R74W/S156A/D161L/V365I/ +++++ + A407E 562 I14V/I28M/K38R/R74W/L100F/ ++ ++

R102K/S118A/D161L/D169G/ A178V/I233W/T250R/A407E/ I422M/M438A .sup.a Levels of increased activity were measured as fold-change relative to SEQ ID NO: 548 in µmol RebA or RebM/mg SFP-min at 0.0625 g/L SFP and 60 min and defined as follows: “+” = activity at least that of the reference polypeptide, but less than 2-fold; “++” = at least 2-fold, but less than 3-fold; “+++” = at least 3-fold, but less than 4-fold; and “++++” = at least 4-fold increased activity, as compared to the reference polypeptide.

Example 12

ADP-Glycosyltransferase Variants of SEQ ID NO: 562

(228) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 562 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO:561 (i.e., SEQ ID NO:562) was carried out by constructing libraries in which mutations associated with improved activity in previous rounds were recombined. This library was then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a seventh round (“Round 7”) of 37 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides (Table 12.1).

(229) HTP Assay for Glucose Transfer from ADP-Glucose to Steviol Glycosides

(230) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 562 variants. Lysis buffer volume was 400 uL, and the lysate was diluted 4-fold. The assay was conducted with 10 µL lysate in 100 µL reactions and with substrate loading of 1 mM stevioside (ChromaDex, >94% purity), from a 20 mM stock solution in 50% ethanol and co-substrate loading of 1 mM ADP-glucose (Sigma, >93% purity). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 7, 3 mM MgCl₂, 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 h. The reactions were quenched by adding 10 µL assay 90 µL acetonitrile with 0.2% formic acid and centrifuged 10 m at 4° C. The supernatants were diluted 1:10 in water and analyzed by RapidFire SPE-MS/MS as described in Table 9.3. The top 84 variants were retested with the same conditions using 1 mM stevioside (Chromadex, >94% purity), 1 mM rebaudioside A (>97% purity), or rebaudioside D (Chromadex, >93% purity). The resulting engineered GT variant polypeptides are listed in Table 12.1.

(231) TABLE-US-00025 TABLE 12.1 Round 7 Variants and RebA, RebI, and RebM Levels SEQ ID NO: Amino Acid Differences Increased Increased Increased (nt/aa) (Relative to SEQ ID NO: 562) RebA.sup.a

RebI.sup.b RebM.sup.b 681/682 I75L/M87W/A110G/N137G/G169S/S191R/ +++ – –

A199K/K208A/Q209G/K289S/V435Q 683/684 M87W/G169S/S191R/A199K/I207L/ +++ – –

Q209G/W401L/V413L 685/686 I44V/M87W/N137G/Q159M/G169S/ +++ – – S191R/A199K/K208D

687/688 I44V/M87W/A199K/K208A +++ – – 689/690 M87W/I92L/Q159M/S191R/A199K/ +++ – –

K208A/Q209G/K289S/D290E 691/692 N137G/I198M/A199K/I207L/K208D/ ++ ++ + A426V/V435R

693/694 I75L/S76R/M87W/S191R/Q197R/A199K/ ++ – – Q209G/D300Q 695/696

I75L/S76R/M87W/S191R ++ ++ + 697/698 V19L/I75L/S76R/M87W/I92L/A199K/ ++ – –

I207L/K208A 699/700 N137G/A199K/K208A/Q209G/D290E/ ++ – – V435R 701/702

G169S/Q197R/A199K/I207L/Q209G/ ++ -- K222R/D300Q/V413L/V435R 703/704
 G169S/S191R/Q197R/A199K/I207L/ ++ -- K208D 705/706 I75L/A110G/Q197R/A199K/K208A/ ++ --
 -- D290E/D300Q/W401L/V413L 707/708 I92L/N137G/S191R/A199K/Q209G ++ -- 709/710
 M87W/I92L/A110G/G169S/A199K/ ++ -- I207L/Q209G/D290E/D300Q 711/712
 S76R/I92L/A199K/Q209G + -- 713/714 M87W/N137G/V435R + +++ + 715/716
 I44V/S76R/Q197R/A199K/K208D/A351G + -- 717/718 I75L/M87W/D300Q + +++ + 719/720
 V19L/I44V/A110G/S191R/I198M/A199K/ + -- + K208A/D300Q/V365I 721/722
 I44V/M87W/I92L/N137G/Q159M/A199K/ + -- Q209G 723/724
 I75L/S76R/M87W/I92L/D290E/D300Q + ++ -- 725/726 M87W/S191R/I198M/A199K/K222R/ + -- --
 P244L/K289S/D300Q/V435R 727/728 M87W/I92L/K208A/W401L + + -- 729/730
 Q197R/I198M/A199K/K208D/Q209G + -- -- 731/732 M87W/I92L/Q159M/G169S/S191R/ + -- --
 I198M/D290E/V413L/V435Q 733/734 S191R/I207L/K208A/K289S/D290E/ + + ++ V413L/V435Q
 735/736 N137G/V365I + ++ + 737/738 V19L/M87W/I92L/S191R/A199K/Q209G/ + -- -- V413L/V435Q
 739/740 I75L/S76R/M87W/I92L + -- -- 741/742 I75L/M87W/I92L/G169S/I207L/K208A/ + + --
 D300Q/V413L/V435Q 743/744 I44V/M87W/N137G/S191R/Q197R/ + -- -- Q209G/K289S/W401L
 745/746 I92L/Q197R/A199K/I207L/K208D/W401L + -- -- 747/748 I75L/M87W/A110G/S191R/Q197R/
 + -- -- I198M/I207L/K208D/K289S/D290E/ D300Q/W401L/V413L 749/750
 Q159M/Q197R/A199K/I207L/Q209G + -- -- 751/752 M87L/I92L/Q197R/I198M/A199K/K208D/ + -- --
 D300Q 753/754 M87W/V435R + -- -- .sup.aLevels of increased production were determined relative to
 the reference polypeptide of SEQ ID NO: 562, and defined as follows: “--” = production less than that of
 the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 3-fold;
 “++” = at least 3-fold, but less than 5-fold; and “+++” = at least 5-fold increased production, as compared
 to the reference polypeptide. .sup.bLevels of increased production were determined relative to the
 reference polypeptide of SEQ ID NO: 562, and defined as follows: “--” = production less than that of the
 reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.5-fold;
 “++” = at least 1.5-fold, but less than 2-fold; “+++” = at least 2-fold, but less than 2.5-fold; and “++++” =
 at least 2.5-fold increased production, as compared to the reference polypeptide.

Example 13

Synthesis, Optimization, and Assaying of Glycosyltransferase Enzymes with Glucosylation Activity

(232) In this Example, methods used in the synthesis, optimization and assaying of UGT enzymes with glucosylation activity are described.

(233) Gene Synthesis and Optimization

(234) Polynucleotide sequences encoding glycosyltransferases from *Oryza sativa Japonica*, *Solanum tuberosum*, *Lycium barbarum*, and *Solanum lycopersicum* were codon-optimized and synthesized as the genes of SEQ ID NOS: 755, 757, 759, 761, 763, 765, and 767. These synthetic genes were cloned into a pCK110900 vector system (See e.g., US Pat. Appln. Publn. No. 2006/0195947, which is hereby incorporated by reference) and subsequently expressed in *E. coli* W3110 (Δ fhuA). The *E. coli* strain W3110 expressed the UGT enzymes under the control of the lac promoter.

(235) Production of Shake Flask Powders (SFP)

(236) A shake-flask procedure was used to generate the glycosyltransferase polypeptide shake flask powders (SFP) for characterization assays used in the biocatalytic processes described herein. Shake flask powder (SFP) preparation of enzymes provides a more purified preparation (e.g., up to >30% of total protein) of the enzyme as compared to the cell lysate used in HTP assays and also allows for the use of more concentrated enzyme solutions. A single colony of *E. coli* containing a plasmid encoding an engineered polypeptide of interest was inoculated into 5 mL Luria Bertani broth containing 30 μ g/ml chloramphenicol and 1% glucose. Cells were grown overnight (at least 16 hours) in an incubator at 30° C. with shaking at 250 rpm. The culture was diluted into 250 mL Terrific Broth (12 g/L bacto-tryptone, 24 g/L yeast extract, 4 mL/L glycerol, 65 mM potassium phosphate, pH 7.0, 1 mM MgSO₄) containing 30 μ g/ml CAM, in a 1 L flask to an optical density of 600 nm (OD₆₀₀) of 0.2 and allowed to grow at 30° C.

(237) Expression of the glycosyltransferase gene was induced by addition of IPTG to a final concentration of 1 mM when the OD₆₀₀ of the culture was 0.6 to 0.8. Incubation was then continued overnight (at least 16 hours). Cells were harvested by centrifugation (5000 rpm, 15 mM, 4° C.) and the supernatant discarded. The cell pellet was resuspended in two volumes of 25 mM triethanolamine buffer,

pH 7.5, and passed through a MICROFLUIDIZER® high pressure homogenizer (Microfluidics), with standard *E. coli* lysis settings and maintained at 4° C. Cell debris was removed by centrifugation (10,000 rpm, 45 minutes, 4° C.). The cleared lysate supernatant was collected and frozen at -80° C. and then either His-affinity purified and dialyzed to produce purified protein or lyophilized to produce a dry shake-flask powder of crude protein.

(238) Assay for Rebaudioside A Glucosylation with Purified Proteins

(239) First, 50 µL purified protein was diluted in 200 µL total reaction volume consisting of 50 mM Tris-HCl buffer pH 7.5, 3 mM magnesium chloride, 1 mM rebaudioside A, and 0.5 mM uridine diphosphoglucose. The reaction was performed at 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 18 h. Boiled enzyme reaction was used as the negative control. Ten µL of the reaction was quenched with 90 µL acetonitrile with 0.2% formic acid and precipitated by centrifugation.

Glycosylated rebaudioside A products were detected in the supernatant by LC-MS/MS as described in Example 1, Table 1.1.

(240) Production of rebaudioside D from rebaudioside A was detected for SEQ ID NO: 756, 758, 762, and 768. Some enzymes also produced a regioisomer of rebaudioside D, likely the 6-linked molecule rebaudioside D2. Despite poor soluble expression, SEQ ID NO: 758 demonstrated high specific activity and good selectivity toward producing β-1,2-glucose linkages in the steviol glycoside substrates.

(241) Assay for Rebaudioside A Glucosylation with Shake Flask Powder

(242) Lyophilized shake flask powder was reconstituted to 20 mg/mL. Then, 10 µL of these stocks were diluted in 100 µL total reaction volume of 50 mM potassium phosphate (KPhos) buffer, pH 7, with 3 mM MgCl.sub.2, 1 mM rebaudioside A (>97% purity), and 2 mM uridine diphosphoglucose (UDP-glucose). The reaction was performed at 40° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 16-18 h. Activity was detected for SEQ ID NO: 758 over negative control. Low conversion (i.e., <10%), of rebaudioside A to rebaudioside D was observed in the LC-MS/MS analysis.

Example 14

GT Variants of SEQ ID NO: 758

(243) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 758 for improved glucosylation of steviol glycosides are described. Directed evolution of the GT encoded by SEQ ID NO: 757 (i.e., SEQ ID NO: 758) was carried out by constructing combinatorial libraries of variant genes in which positions associated with surface residues of the enzyme were subjected to mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a first round ("Round 1") of 10 engineered GT variant polypeptides with β-1,2-glucosyltransferase activity toward steviol glycosides.

(244) HTP Growth, Expression, and Lysate Preparation

(245) Cells were picked into 96-well plates and grown overnight in LB media containing 1% glucose and 30 µg/mL CAM, 30° C., 200 rpm, 85% humidity. Then, 20 µL of overnight growth were transferred to a deep-well plate containing 380 µL TB growth media containing 30 µg/mL CAM, induced with 1 mM IPTG, and incubated for 18 h at 30° C., 200 rpm, 85% humidity. Cell cultures were centrifuged at 4000 rpm, 4° C. for 10 m, and the media discarded. Cell pellets thus obtained were frozen at -80° C., and lysed in 250 µL lysis buffer (0.5 g/L lysozyme and 0.5 g/L PMBS in 20 mM Tris-HCl buffer, pH 7.5) with low-speed shaking for 2 h on titre-plate shaker at room temperature. The plates were then centrifuged at 4000 rpm and 4° C. for 20 min and the cleared lysate supernatants were used in the HTP assay reactions described below.

(246) HTP Assay for Rebaudioside A Glucosylation

(247) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 757 variants with lysate loading of 25 µL lysate in 100 µL reactions and with substrate loading of 1 mM rebaudioside A (Sigma, >96% purity), from a 20 mM stock solution in 50% ethanol, and co-substrate loading of 0.5 mM UDP-glucose (Sigma, >98% purity). The following reaction conditions were used: 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl.sub.2, 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. The reactions were quenched with 0.5 volume/volume acetonitrile with 0.2% formic acid and precipitated by centrifugation for 10 m at 4° C. Glycosylated products were detected in the supernatant by LC-MS/MS following 1:20 dilution in water with the instrument and parameters described in Example 1, Table 1.1. Glycosyltransferase variant polypeptides that produced rebaudioside

D from rebaudioside A at greater quantities than SEQ ID NO: 758 were identified. The engineered polypeptides are listed in Table 14.1. Shake-flask scale cultures were grown for lyophilized powder production as described in Example 1 for analysis of variants with the following amino acid mutations shown in Table 14.1, relative to SEQ ID NO: 758.

(248) TABLE-US-00026 TABLE 14.1 Round 1 Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 758) RebD.sup.a 769/770

R69H/Y173N/V243A/M383V/T399A +++ 771/772 Y173H/P175S/N191D/M365I/M383V/T399A +++ 773/774 I56T/N191D/L354I/M383V/T399A +++ 775/776

R69H/Y173N/P175S/V243A/M246K/L354I/M365I/M383V/T399A +++ 777/778 F70L/N225G/I413V ++ 779/780 F70L/N225G/E247G ++ 781/782 F70L/N225G/M246P/E409K/I413V ++ 783/784

F70L/Q115S/N225G/E409K ++ 785/786 H74T/K310D/G396E/N424S ++ 787/788 H74T/G396E +

.sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 4, and defined as follows: "-" = production at least that of the reference polypeptide, but less than 3-fold; "++" = at least 3-fold, but less than 5-fold; and "+++" = at least 5-fold increased production, as compared to the reference polypeptide.

Shake Flask Lysate Characterization Assay and Analysis for Rebaudioside A Glucosylation

(249) First, 250 mL shake flask cultures were grown, induced, and lysed. Cell debris was removed by centrifugation as described in Example 1, and the cleared lysate supernatant was collected. Then, 10 μ L of the lysate were diluted in 100 μ L total reaction volume of 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl.sub.2, 1 mM rebaudioside A (Sigma, >96% purity), and 2 mM UDP-glucose (Sigma, >98% purity). The reaction was performed at 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 0-18 h. The reaction described above was quenched with 0.5 volume/volume acetonitrile with 0.2% formic acid and precipitated by centrifugation. Glycosylated products were detected in the supernatant by LC-MS/MS following 1:20 dilution in water with the instrument and parameters described in Example 1, Table 1.1.

(250) TABLE-US-00027 TABLE 14.2 Shake-flask Characterization of Round 1 Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 758) RebD.sup.a 769/770 R69H/Y173N/V243A/M383V/T399A + 771/772 Y173H/P175S/N191D/M365I/M383V/T399A + 777/778 F70L/N225G/I413V - 779/780 F70L/N225G/E247G - 785/786 H74T/K310D/G396E/N424S - 787/788 H74T/G396E - .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 758, at 4 h and defined as follows: "-" = production less than the reference polypeptide; and "+" = production at least that of the reference polypeptide, but less than 1.5-fold increased production, as compared to the reference polypeptide.

(251) Variants corresponding to SEQ ID NOS:770 and 772 produced rebaudioside D from rebaudioside A at greater quantities than the variant of SEQ ID NO: 758. The variant of SEQ ID NO: 770 exhibited the highest activity on RebA. Thus, the encoding polynucleotide (SEQ ID NO: 769) was selected for further directed evolution.

Example 15

ADP-Glycosyltransferase Variants of SEQ ID NO: 770

(252) In this Example, experiments for evolution and screening of GT polypeptides derived from the SEQ ID NO: 770 variant for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 769 (i.e., SEQ ID NO: 770) was carried out by constructing libraries of variant genes. Libraries recombined beneficial mutations identified in Example 14 (Round 1), combinatorially incorporated diversity from homologs in publically available databases, or subjected certain structural features of the enzyme with saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a second round ("Round 2") of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. 10 engineered variants were identified from the recombined beneficial mutations (Table 15.1), 19 from saturation mutagenesis (Table 15.2), and 53 from surface residue and homolog diversity (Table 15.3).

(253) HTP Assay for Glucose Transfer from ADP-Glucose to Rebaudioside A

(254) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 769 variants with lysate loading of 25 μ L lysate in 100 μ L reactions and with substrate loading of 1 mM

rebaudioside A (Sigma, >96% purity) and co-substrate loading of 4 mM ADP-glucose (Sigma, >93% purity). The following reaction conditions were used: 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl₂, 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 18 h. The reactions were quenched with 0.5 volume/volume acetonitrile with 0.2% formic acid and precipitated by centrifugation for 10 min at 4° C. Glycosylated products were detected in the supernatant by LC-MS/MS following 1:10 dilution in water with the instrument and parameters described in Example 1, Table 1.1. Glycosyltransferase variant polypeptides that produced rebaudioside D from rebaudioside A with ADP-glucose, at greater quantities than SEQ ID NO: 770 were identified. The engineered polypeptides from the recombined beneficial mutations are listed in Table 15.1. The engineered polypeptides from the saturation mutagenesis libraries are listed in Table 15.2.

(255) TABLE-US-00028 TABLE 15.1 Combinatorial Round 2 AGT Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 770) RebD.sup.a 789/790 Y24E/S28N/F262Y/C264S ++++ 791/792 Y24E/S28N/F262Y/C264S/K423E +++ 793/794 S28N/F262Y/C264S +++ 795/796 F262Y/C264S ++ 797/798 C264S/I291V + 799/800 Y24E/C264S/A294V + 801/802 S28N + 803/804 S28N/K423E + 805/806 Y24E/S28N/F262Y + 807/808 Y24E/S28N + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 770, and defined as follows: “+” = production at least 1.2-fold, but less than 3-fold; “++” = at least 3-fold, but less than 6-fold; “+++” = at least 6-fold, but less than 9-fold; and “++++” = at least 9-fold increased production, as compared to the reference polypeptide.

(256) TABLE-US-00029 TABLE 15.2 Saturation Mutagenesis Round 2 Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 770) RebD.sup.a 809/810 Y24V ++++ 811/812 C264A ++++ 813/814 Y24L +++ 815/816 S28L ++ 817/818 Y269S ++ 819/820 H366Q ++ 821/822 H366T ++ 823/824 S28G + 825/826 N32C + 827/828 K325G + 829/830 Y269W + 831/832 N32S + 833/834 M351L + 835/836 H366L + 837/838 N32R + 839/840 C264G + 841/842 K325H + 843/844 F341V + 845/846 S28K + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 770 and defined as follows: “+” = production that of the reference polypeptide, but less than 2-fold; “++” = at least 2-fold, but less than 3-fold; “+++” = at least 3-fold, but less than 4-fold; and “++++” = at least 4-fold increased production, as compared to the reference polypeptide.

HTP Assay for Glucosylation of Rebaudioside A

(257) The remaining combinatorial round 2 libraries were screened as follows: Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 769 variants with lysate loading of 25 µL lysate in 100 µL reactions or with 10 µL 4-fold diluted lysate in 100 µL reactions with substrate loading of 1 mM rebaudioside A (Sigma, >96% purity) and co-substrate loading of 1 mM UDP-glucose (Sigma, >98% purity). The following reaction conditions were used: 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl₂, 40° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1-2 h. The reactions were either quenched with 0.5 volume/volume acetonitrile with 0.2% formic acid or 10 µL of the reaction was quenched with 90 µL acetonitrile with 0.2% formic acid and precipitated by centrifugation. Glycosylated products were detected in the supernatant by LC-MS/MS following 1:100 or 1:10 dilution in water with the instrument and parameters described in Example 1, Table 1.1.

Glycosyltransferase variant polypeptides that produced rebaudioside D from rebaudioside A at greater quantities than SEQ ID NO: 770 were identified. The engineered polypeptides are listed in Table 15.3.

(258) TABLE-US-00030 TABLE 15.3 Additional Combinatorial Round 2 Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 770) RebD.sup.a 847/848 I56T/F70L/N191D/M246K/L354I/M365I +++ 849/850 H69Q/F70L/N191D/M246K/L354I/M365I +++ 851/852 P175S/M246K/L354I +++ 853/854 P175S +++ 855/856 I56T/F70L/P175S/M246K/L354I ++ 857/858 I56T/H69Q/P175S/N191D/M246K/L354I ++ 859/860 I56T/F70L/P175S/N191D/L354I/M365I ++ 861/862 I56T/F70L/N191D/L354I ++ 863/864 P175S/N191D ++ 865/866 P175S/L354I ++ 867/868 I56T/H69Q/F70L/M246K/L354I ++ 869/870 F70L/N191D/M246K/L354I/M365I ++ 871/872 I56T/H69Q/F70L/P175S/L354I ++ 873/874 P175S/M246K/L354I/M365I ++ 875/876 P175S/N191D/M246K/L354I/M365I ++ 877/878 H69Q/P175S/N191D/M246K/L354I ++ 879/880 P175S/N191D/L354I ++ 881/882 F70L/P175S/N191D/M246K/L354I/M365I ++ 883/884 H69Q/F70L/L354I/M365I ++ 885/886 I56T/H69Q/M246K/L354I ++ 887/888 H69Q/P175S/L354I +

889/890 I56T/H69Q/F70L/N191D/M246K/L354I/ + M365I 891/892
 I56T/H69Q/F70L/P175S/N191D/M246K/ + L354I 893/894 I56T/P175S/M246K + 895/896
 I56T/P175S/L354I/M365I + 897/898 I56T/H69Q/P175S/M246K/L354I + 899/900
 I56T/N191D/M246K/L354I + 901/902 L354I/M365I + 903/904 M246K/L354I + 905/906
 I56T/P175S/L354I + 907/908 I56T/F70L/P175S/M246K/M365I + 909/910 I56T/M246K + 911/912
 I56T/H69Q/M246K/M365I + 913/914 I126F/C220L + 915/916 I56T/H69Q/P175S/M246K/L354I/M365I
 + 917/918 I126F/K403R + 919/920 I56T/M246K/L354I/M365I + 921/922 I56T/L354I + 923/924
 F70L/N191D/M246K/M365I + 925/926 H69Q/L354I + 927/928
 I56T/H69Q/F70L/P175S/N191D/M246K + 929/930 H69Q/M246K/L354I/M365I + 931/932 I126F +
 933/934 H69Q + 935/936 H69Q/F70L/P175S/N191D/M246K/L354I/ + M365I 937/938
 P175S/N191D/L354I/M365I + 939/940 F70L + 941/942 N191D/M246K/L354I + 943/944
 F70L/P175S/N191D/L354I/M365I + 945/946 I56T/F70L/P175S/N191D/M246K/L354I + 947/948
 H69Q/F70L/M246K + 949/950 L354I + 951/952 I56T/F70L/P175S/N191D/M246K + .sup.aLevels of
 increased production were determined relative to the reference polypeptide of SEQ ID NO: 770, and
 defined as follows: “+” = production at least 1.5-fold, but less than 3-fold; “++” = at least 3-fold, but less
 than 4.5-fold; and “+++” = at least 4.5-fold increased production, as compared to the reference
 polypeptide.

HTP Assay for Glucosylation of Rebaudioside I

(259) 88 variants from the round 2 saturation mutagenesis library were screened as follows: Assays were
 performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 769 variants with
 lysate loading of 25 µL lysate in 100 µL reactions with substrate loading of 1 mM rebaudioside I
 (prepared from rebaudioside A as described in example 10) and co-substrate loading of 1 mM UDP-
 glucose (Sigma, >98% purity). The following reaction conditions were used: 50 mM potassium phosphate
 buffer, pH 7, 3 mM MgCl₂, 40° C. in a Thermotron titre-plate shaker with 300 RPM shaking for 66
 h. The reactions were quenched by adding 10 µL of the reaction to 90 µL acetonitrile with 0.2% formic
 acid and precipitated by centrifugation. Glycosylated products were detected in the supernatant by LC-
 MS/MS following 1:10 dilution in water with the instrument and parameters described in Table 15.5.
 Glycosyltransferase variant polypeptides that produced rebaudioside M from rebaudioside I at greater
 quantities than SEQ ID NO: 770 were identified. The top two engineered polypeptides are SEQ ID NO:
 1292 and 1294, which have mutations F156R and G199H, respectively, relative to SEQ ID NO: 770.

(260) Shake-flask scale cultures were grown for SFP production as described in Example 1 for analysis of
 variants with the amino acid mutations shown in Table 15.4 (relative to SEQ ID NO: 770).

(261) SFP Characterization Assay and Analysis for Glucosyl Transfer from ADP-Glucose to
 Rebaudioside A

(262) Shake flask powders were reconstituted to provide 20 g/L powder. Then, 10 µL of these stocks were
 diluted in 100 µL total reaction volume of 50 mM Tris-HCl buffer, pH 7.5, with 3 mM MgCl₂, 1 mM
 rebaudioside A (Sigma, >96% purity), and 4 mM ADP-glucose (Sigma, >93% purity). The reaction was
 performed at 30° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 0-19 h. The reaction
 was quenched and precipitated as described in Example 14. Glycosylated products were detected in the
 supernatant by LC-MS/MS following 1:100 dilution in water with the instrument and parameters
 described in Example 1, Table 1.1.

(263) TABLE-US-00031 TABLE 15.4 Shake Flask Powder Characterization of Round 2 Variants and
 RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 770)
 RebD.sup.a 789/790 Y24E/S28N/F262Y/C264S +++ 791/792 Y24E/S28N/F262Y/C264S/K423E +++
 793/794 S28N/F262Y/C264S ++ 795/796 F262Y/C264S ++ .sup.aLevels of increased production were
 determined relative to the reference polypeptide of SEQ ID NO: 770, at 0.5 h and defined as follows: “-”
 = production less than the reference polypeptide; “+” = production at least that of the reference
 polypeptide, but less than 5-fold; “++” = at least 5-fold, but less than 10-fold; and “+++” = at least 10-
 fold increased production, as compared to the reference polypeptide.

(264) All variants in Table 15.4 (i.e., variants of SEQ ID NOS: 790, 792, 794, and 796) produced
 rebaudioside D from rebaudioside A with ADP-glucose, at greater quantities than SEQ ID NO: 770. Thus,
 these engineered ADP-glycosyltransferase enzymes provide new biocatalytic reagents for the 13-
 glucosylation of rebaudioside A to rebaudioside D. In these experiments, the variant of SEQ ID NO: 792,

had the highest initial activity on rebaudioside A with ADP-glucose as a co-substrate. Thus, the encoding polynucleotide (SEQ ID NO: 791) was selected for further directed evolution.

(265) TABLE-US-00032 TABLE 15.5 HPLC-MS/MS Analysis of Steviol Glycosides Instrument Agilent HPLC 1200 series, Sciex 4000 QTrap Column Poroshell 120 EC C18 50 × 3.0 mm, 2.7 μm with Poroshell 120 EC C18 5 × 3.0, 2.7 μm guard column (Agilent Technologies) Mobile phase Gradient (A: 0.1% formic acid in water, B: 0.1% formic acid in methanol) Time (m) % B 0 40 0.50 53 5.00 53 5.50 70 7.50 70 8.00 95 8.50 95 8.51 60 9.20 40 Flow rate 0.8 mL/m Run time 9.2 m Peak retention times Rebaudioside M: 4.37 m Rebaudioside I: 6.70 m Other glucosylated rebaudioside I product: 4.8 m Second glucosylated rebaudioside I product: 6.7 m Column temperature 40° C. Injection volume 10 μL MS detection MRM 990/828 (for steviol tetraglycosides, e.g., rebaudioside A), 1152/828 (for steviol pentaglycosides, e.g., rebaudioside D), 1314/828 (steviol hexaglycosides, e.g., rebaudioside M), 828/666 (for steviol triglycosides, e.g., stevioside), 666/504 (steviol diglycosides, e.g., rubusoside) MS conditions MODE: MRM; CUR: 30; IS: 4750; CAD: high; TEM: 550° C.; GS1: 50; G52: 50; DP: 150; EP: 10; CXP: 14; DT: 50 ms for each transition. For the first three transitions CE: 85; for the last two transitions CE: 60.

Example 16

ADP-Glycosyltransferase Variants of SEQ ID NO: 792

(266) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 792 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 791 (i.e., SEQ ID NO: 792) was carried out by constructing libraries of variant genes in which mutations associated with improved activity in previous rounds above were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a third round ("Round 3") of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(267) HTP Assay for Glucose Transfer from ADP-Glucose to Rebaudioside A

(268) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 792 variants with lysate loading of 25 μL lysate in 100 μL reactions and with substrate loading of 1 mM rebaudioside A (>97% purity) and co-substrate loading of 1 mM ADP-glucose (Sigma, >93% purity). The following reaction conditions were used: 50 mM KPhos buffer, pH 7, 3 mM MgCl₂, 40° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 5-6 h. Then, 10 μL of the reactions were quenched with 90 μL acetonitrile with 0.2% formic acid, centrifuged 10 m at 4° C., and, the supernatants were analyzed by RapidFire-MS/MS as described in Example 9, Table 9.3. Glycosyltransferase variant polypeptides that produced rebaudioside D from rebaudioside A with ADP-glucose, at greater quantities than SEQ ID NO: 792 were identified. The engineered polypeptides are listed in Table 16.1. Shake-flask scale cultures were grown for SFP production as described in Example 1 for analysis of variants shown in Table 16.1 relative to SEQ ID NO: 792.

(269) TABLE-US-00033 TABLE 16.1 Round 3 Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 792) RebD.sup.a 953/954

F70L/I126F/N191D/M246K/K325H/M351L/ ++++ L354I/H366Q/E423K 955/956

F70L/I126F/N191D/M246K/K325H/M351L/ ++++ H366Q/E423K 957/958

F70L/I126F/N191D/M246K/L354I/H366Q +++ 959/960 F70L/I126F/P175S/N191D/M246K/K325H/ +++ L354I/H366Q 961/962 F70L/I126F/M246K/L354I/H366Q +++ 963/964

F70L/I126F/N191D/M246K/K325H/M351L/ +++ L354I/E423K 965/966

F70L/I126F/M246K/P330Q/H366Q +++ 967/968 F70L/I126F/P175S/K325H/P330Q/M351L/ +++

L354I/H366Q 969/970 E24L/N32S/Y269S/A382G/I385V/D389E/ ++ L402I/I406M 971/972

E24L/N32S/P330Q ++ 973/974 E24L/N32S/S264A/Y269S/P330Q/L402V/ ++ K403R 975/976

E24L/N32S/S264A/P330Q ++ 977/978 I126F/T211E/T260V/E423K ++ 979/980 I126F/C220L/T260V

++ 981/982 E24L/N32S/Y269S/K403R ++ 983/984 E24L/N32S/Y269S ++ 985/986 I126F/C220L/I316V

++ 987/988 E24L/N32S/S264A/Y269S/A382G/K403R/ ++ I406M 989/990 C220L/T260V/E423K ++

991/992 I126F/T211E/C220L/G253D/I316V/I342L ++ 993/994

I126F/T211E/C220L/D275Q/V279L/L323V ++ 995/996 C220L/T260V + 997/998

I56V/C220L/T260V/E423K + 999/1000 E24L/S264A/Y269S/D389E/L402V/I406M + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 792, and

defined as follows: “+” = production at least 1.4-fold that of the reference polypeptide, but less than 2.8-fold; “++” = at least 2.8-fold, but less than 4.2-fold; “+++” = at least 4.2-fold, but less than 5.6-fold; and “++++” = at least 5.6-fold increased production, as compared to the reference polypeptide.

SFP Characterization Assay and Analysis for Glucosyl Transfer from ADP-Glucose to Rebaudioside A (270) Shake flask powders were reconstituted to provide 20 g/L powder. Then, 10 µL of these stocks were diluted in 100 µL total reaction volume of 50 mM KPhos buffer, pH 7, with 3 mM MgCl.sub.2, 1 mM rebaudioside A (>97% purity), and 2 mM ADP-glucose (Sigma, >93% purity).

(271) The reaction was performed at 40° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 0-21 h. The reaction was quenched and precipitated as described above. Glycosylated products were detected in the supernatant by LC-MS/MS following 1:10 dilution in water with the instrument and parameters described in Example 1, Table 1.1.

(272) TABLE-US-00034 TABLE 16.2 Shake Flask Powder Characterization of Round 3 Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 792) RebD.sup.a 953/954 F70L/I126F/N191D/M246K/K325H/M351L/ +++ L354I/H366Q/E423K 955/956 F70L/I126F/N191D/M246K/K325H/M351L/ ++ H366Q/E423K 989/990 C220L/T260V/E423K +++ .sup.a Levels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 792, at 2 h and defined as follows: “+” = production at least that of the reference polypeptide, but less than 2-fold; “++” = production at least 2-fold, but less than 3-fold; and “+++” = at least 3-fold increased production, as compared to the reference polypeptide.

(273) In these experiments, all of the variants in Table 16.2 (i.e., SEQ ID NOS: 954, 956, and 990) produced rebaudioside D from rebaudioside A with ADP-glucose, at greater quantities than SEQ ID NO: 792. The variant of SEQ ID NO: 954 had the highest activity on rebaudioside A with ADP-glucose as a co-substrate in these experiments. Thus, the encoding polynucleotide (SEQ ID NO: 953) was selected for further directed evolution.

Example 17

ADP-Glycosyltransferase Variants of SEQ ID NO: 954 for Steviol Glycoside Glucosylation, Including Glucosylation of Rebaudioside I to Rebaudioside M

(274) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 954 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 953 (i.e., SEQ ID NO: 954) was carried out by constructing libraries of variant genes in which mutations associated with improved activity in previous rounds above were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a fourth round (“Round 4”) of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(275) HTP Assay for Glucose Transfer from ADP-Glucose to Rebaudioside A

(276) Cells were lysed with 400 µL lysis buffer as described in Example 13. Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 954 variants with lysate loading of 20 µL lysate in 100 µL reactions and with substrate loading of 1 mM rebaudioside A (>97% purity) and co-substrate loading of 1 mM ADP-glucose (Sigma, >93% purity). The following reaction conditions were used: 50 mM KPhos buffer, pH 7, 3 mM MgCl.sub.2, 40° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. The reactions were quenched as described in Example 16, and the supernatants were analyzed by RapidFire-MS/MS as described in Example 9, Table 9.3.

Glycosyltransferase variant polypeptides that produced rebaudioside D from rebaudioside A with ADP-glucose, at greater quantities than SEQ ID NO: 954 were identified. The engineered polypeptides are listed in Table 17.1. Shake-flask scale cultures were grown for SFP production as described in Example 1 for analysis of variants with the following amino acid mutations shown in Table 17.1 (relative to SEQ ID NO: 150).

(277) TABLE-US-00035 TABLE 17.1 Round 4 Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 954) RebD.sup.a 1001/1002 P175S/T211E/S264A/V279L/I316V/L323V +++++ 1003/1004 T211E/I385V/D389E +++++ 1005/1006 I56V/T211E/S264A/I316V/D389E +++ 1007/1008 I56V/P175S/A197P/G253D/I385V/D389E +++ 1009/1010 A197P/T211E/L402I +++ 1011/1012 T211E/L402I/K403R +++ 1013/1014 P175S/T211E/V279L/L323V/P330Q/L402I/ ++ K403R/I406M 1015/1016

P175S/T211E/S264A/P330Q ++ 1017/1018 P175S/T211E/L323V/A382G/L402I/K403R/ ++ I406M 1019/1020 P175S/T211E/K403R/I406M ++ 1021/1022 I56V/P175S/A197P/T211E/P330Q/A382G/ + I385V 1023/1024 P175S/S264A/L323V/P330Q + 1025/1026 A197P/V279L/L323V + 1027/1028 N162R/C220L/W226V/N367W + 1029/1030 I56V/S264A/I385V/D389E + 1031/1032 E24L/N32S/F126A/E198P/M201G/N367W + 1033/1034 I56V/T211E/V279L/L323V/P330Q/L402I + 1035/1036 N32S/H97G/E198P/Q202G/W226V/T260V + 1037/1038 E24L/N32S/E198P/M201G/W226V + 1039/1040 I56V/P175S/S264A/A382G/I385V/D389E/ + L402V/I406M 1041/1042 E198P/M201G/N367W + 1043/1044 E24L/N32S/L146A/W226V + 1045/1046 D389E + 1047/1048 M201G/Q202G/N367W + 1049/1050 T211E/A382G/I406M + 1051/1052 E24L/N32S/E198P/M201G/C220L/W226V + 1053/1054 H97G/Q202G/N367W + 1055/1056 E24L/H97G/T260V/N367W + 1057/1058 P175S/S264A/I316V/L323V/P330Q/K403R/ + I406M 1059/1060 A197P/T211E/I316V/I342L/I406M + 1061/1062 N32S/Q202G/N367W + 1063/1064 A197P/T211E/I316V/A382G/D389E/L402I/ + K403R 1065/1066 N32S/W226V/N367W + 1067/1068 I56V/T211E/G253D/I316V/L323V + 1069/1070 E24L/N32S/F126A/E198P/Q202G/C220L/ + W226V/T260V/Y269S 1071/1072 E198P/Q202G/C220L/Y269S/N367W + 1073/1074 P175S/S264A/I316V/D389E/L402V/ + 1075/1076 E24L/N32S/H97G/N162R/Q202G + 1077/1078 I56V/A197P/S264A/V279L/P330Q/A382G/ + D389E/L402V/K403R/I406M .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 954, and defined as follows: “+” = production at least 1.3-fold, but less than 2.6-fold; “++” = at least 2.6-fold, but less than 3.9-fold; “+++” = at least 3.9-fold, but less than 5.2-fold; and “++++” = at least 5.2-fold increased production, as compared to the reference polypeptide.

SFP Characterization Assay and Analysis for Glucosyl Transfer from ADP-Glucose to Rebaudioside A (278) Shake flask powders were reconstituted to provide 50 g/L powder. Then, 1 µL of these stocks were diluted in 100 µL total reaction volume of 50 mM KPhos buffer, pH 7, with 3 mM MgCl₂, 1 mM rebaudioside A (>97% purity), and 2 mM ADP-glucose (Sigma, >93% purity). The reaction was performed at 40° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 2 h. The reaction was then diluted 1:5 in water, and 25 µL of the diluted reaction was quenched with 75 µL acetonitrile with 0.2% formic acid and precipitated by centrifugation. Glycosylated products were detected in the supernatant by RapidFire-MS/MS following 1:10 dilution in water with the instrument and parameters described in Example 9, Table 9.3.

(279) TABLE-US-00036 TABLE 17.2 Shake Flask Characterization of Round 4 Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 954) RebD.sup.a 1053/1054 H97G; Q202G/N367W +++ 1031/1032 E24L/N32S/F126A/E198P/M201G/N367W +++ 1027/1028 N162R/C220L/W226V/N367W ++ 1051/1052 E24L/N32S/E198P/M201G/C220L/W226V ++ 1057/1058 P175S/S264A/I316V/L323V/P330Q/K403R/ ++ I406M 1013/1014 P175S/T211E/V279L/L323V/P330Q/L402I/ + K403R/I406M 1001/1002

P175S/T211E/S264A/V279L/I316V/L323V + 1029/1030 I56V/S264A/I385V/D389E – .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 954, and defined as follows: “–” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.5-fold; “++” = at least 1.5-fold, but less than 2-fold; and “+++” = at least 2-fold increased production, as compared to the reference polypeptide.

(280) With the exception of SEQ ID NO: 1030, all other variants in Table 17.2 (SEQ ID NOS: 1054, 1032, 1028, 1052, 1058, 1014, and 1002) produced rebaudioside D from rebaudioside A with ADP-glucose, at greater quantities than SEQ ID NO: 954, in these experiments. The variant with mutations H97G, Q202G, and N367W (SEQ ID NO: 1054), relative to SEQ ID NO: 954, had the highest activity on rebaudioside A with ADP-glucose as a co-substrate in these experiments. Thus, the encoding polynucleotide (SEQ ID NO: 1053) was selected for further directed evolution.

(281) SFP Characterization Assay and Analysis for Glucosyl Transfer from UDP-Glucose to Rebaudioside I

(282) Shake flask powders were reconstituted to provide 50 g/L powder. Then, 2 µL of these stocks were diluted in 100 µL total reaction volume of 50 mM KPhos buffer, pH 7, with 3 mM MgCl₂, 1 mM rebaudioside A (>97% purity), and 2 mM UDP-glucose (Sigma, >98% purity). The reaction was performed at 40° C. in a Thermotron titre-plate shaker with 300 RPM shaking for 20 h. The reaction was

then quenched with 0.5 volume/volume acetonitrile with 0.2% formic acid and precipitated by centrifugation. Glycosylated products were detected in the supernatant by LC-MS/MS following 1:20 dilution in water with the instrument and parameters described in Example 15, Table 15.5.

(283) TABLE-US-00037 TABLE 17.3 Shake Flask Characterization of Round 4 Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 954)
RebM.sup.a 1001/1002 P175S/T211E/S264A/V279L/I316V/L323V ++ 1013/1032
E24L/N32S/F126A/E198P/M201G/N367W + 1057/1058 P175S/S264A/I316V/L323V/P330Q/K403R/ +
I406M 1013/1014 P175S/T211E/V279L/L323V/P330Q/L402I/ – K403R/I406M 1020/1030
I56V/S264A/I385V/D389E – 1051/1052 E24L/N32S/E198P/M201G/C220L/W226V – 1053/1054
H97G/Q202G/N367W – 1027/1028 N162R/C220L/W226V/N367W – .sup.aLevels of increased
production were determined relative to the reference polypeptide of SEQ ID NO: 954, and defined as
follows: “–” = production less than that of the reference polypeptide; “+” = production at least that of the
reference polypeptide, but less than 1.2-fold; and “++” = at least 1.2-fold increased production, as
compared to the reference polypeptide.

(284) SEQ ID NO: 954 and all variants in Table 17.3 (SEQ ID NOS: 1054, 1032, 1028, 1052, 1014, 1002, and 1030) produced rebaudioside M from rebaudioside I with UDP-glucose, at levels above the negative control. SEQ ID NOS: 1002, 1032, and 1058 produced rebaudioside M from rebaudioside I at levels equal to or above SEQ ID: 954 levels with UDP-glucose. The variant with mutations P175S, T211E, S264A, V279L, I316V, and L323V (SEQ ID NO: 1002), relative to SEQ ID NO: 954, had the highest activity on rebaudioside I with UDP-glucose as a co-substrate. Thus, the encoding polynucleotide (SEQ ID NO: 1001) was selected for further directed evolution for the transformation of rebaudioside I to rebaudioside M.

Example 18

Sucrose Synthase Variants of SEQ ID NO: 74

(285) In this Example, experiments for the evolution and screening of sucrose synthase (SuS) polypeptides derived from SEQ ID NO: 73 for improved production of ADP-glucose from sucrose and ADP are described. Directed evolution of the SuS encoded by SEQ ID NO: 73 (i.e., SEQ ID NO: 74) was carried out by constructing libraries of variant genes in which positions associated with certain structural features of the enzyme were subjected to saturation mutagenesis and diversity from homologs in publically available databases was recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a first round (“Round 1”) of 37 engineered SuS variant polypeptides with improved activity toward synthesizing ADP-glucose.

(286) HTP Assay for Glucose Transfer from Sucrose to ADP

(287) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 73 variants with lysate loading of 25 μ L lysate in 100 μ L reactions and with substrate loading of 30% w/v sucrose (Sigma) from a 60% stock solution in water and co-substrate loading of 2 mM ADP (Sigma, >95%). The following reaction conditions were used: 50 mM Tris-HCl buffer, pH 7.5, 3 mM MgCl₂, 30° C. in a Thermocycler for 2 h. The reactions were heat quenched at 95° C. for 10 minutes, and then analyzed by a colorimetric D-fructose dehydrogenase assay adapted from the literature (See e.g., Ameyama et al., J. Bacteriol., 145:814-823 [1981]; and Ameyama, Meth. Enzymol., 89:20-29 [1982]). Briefly, an overnight enzyme-coupled assay was conducted in 96-well plates with 20 μ L sample, diluted such that fructose concentration is <1 g/L, 20 μ L 100 mM potassium ferricyanide (Sigma P-8131), and 160 μ L 0.8 units/mL fructose dehydrogenase (Sigma F4892) dissolved in pH 4.6 McIlvaine buffer with 0.1% Triton X-100. This reaction quantitatively converts fructose to K₃Fe(CN)₆, which is then quantified colorimetrically by adding 67 μ L of the overnight reaction to 33 μ L of stop solution (0.3% w/v sodium dodecyl sulfate, Sigma L-4509, 8.1% v/v phosphoric acid, Sigma P-6560, and 0.5% w/v ferric sulfate, Sigma F-1135) and shaking for 20 minutes to allow for complete conversion of K₃Fe(CN)₆ to Prussian blue, the absorbance of which is read on a plate reader at a wavelength of 690 nm.

(288) Following the primary assay, 84 engineered sucrose synthase (SuS) variant polypeptides with higher fructose, and therefore higher stoichiometric ADP-glucose, formation activity than SEQ ID NO: 74 were screened in triplicate at a lower substrate load of 2% w/v sucrose (Sigma) and co-substrate load of 1 mM ADP (Sigma, >95%). The engineered polypeptides are listed in Table 18.1.

(289) TABLE-US-00039 TABLE 18.1 Round 1 SuS Variants and Fructose Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 74) Fructose.sup.a 1079/1080 A4E/Y33Q/L47H/A52D/V343H/F532S +++ 1081/1082 A4E/L47H/A52D/F532S +++ 1083/1084 A4E/I113Q/F532S ++ 1085/1086 V343S/F532S ++ 1087/1088 R615L/A789D ++ 1089/1090 Q7T ++ 1091/1092 R722Y ++ 1093/1094 R615E ++ 1095/1096 A4E/L9T/Q349H/F532S ++ 1097/1098 A4E/V343H ++ 1099/1100 A4E/P13R/I113Q ++ 1101/1102 R221H ++ 1103/1104 A4E/P13R/I113Q/V343H/F532S + 1105/1106 A4E/P13R/F532S + 1107/1108 R44K + 1109/1110 A4E/P13R/I113Q/F532S + 1111/1112 A4E/Y33Q/I113Q + 1113/1114 V695L + 1115/1116 Q8M + 1117/1118 G117D/R440T + 1119/1120 R440T + 1121/1122 H788E + 1123/1124 R478T + 1125/1126 R611V + 1127/1128 R615T + 1129/1130 Q95S + 1131/1132 F532R + 1133/1134 Q444K + 1135/1136 R440P + 1137/1138 R478V + 1139/1140 R615C + 1141/1142 F532T + 1143/1144 Q444T + 1145/1146 R136S + 1147/1148 R583Q + 1149/1150 R615V + 1151/1152 R221A + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 74, and defined as follows: “+” = production at least that of the reference polypeptide, but less than 1.5-fold; “++” = at least 1.5-fold, but less than 2.5-fold; and “+++” = at least 2.5-fold increased production, as compared to the reference polypeptide.

Example 19

Sucrose Synthase Variants of SEQ ID NO: 1080

(290) Directed evolution of the SuS encoded by SEQ ID NO: 1079 (i.e., SEQ ID NO: 1080) was continued by constructing libraries of variant genes in which mutations associated with improved activity above were recombined and libraries in which additional homolog diversity was recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described in Example 18, with the lower substrate load, to provide a second round (“Round 2”) of 34 engineered SuS variant polypeptides with activity towards the generation of ADP-glucose.

(291) Following primary screening, 42 variants were retested in duplicate and 56 variants were retested in triplicate under the same conditions with the following modifications: 50 mM Tris-HCl pH 7.5 was changed to 50 mM potassium phosphate buffer, pH 7 and the temperature was increased to 50° C. The fructose dehydrogenase assay described in Example 18, was used to quantify fructose production as a proxy for stoichiometric ADP-glucose. The resulting engineered polypeptides are listed in Table 19.1. Shake-flask scale cultures were grown for protein characterization as described in Example 1 for variants with the amino acid mutations shown in Table 19.1 (relative to SEQ ID NO: 1080).

(292) TABLE-US-00039 TABLE 19.1 Round 2 SuS Variants and Fructose Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 1080) Fructose.sup.a 1153/1154 R136S/R440P/Q444T/R478T/G603S ++ 1155/1156 V68A/R129E/S248A/I595V/V600I/I756V ++ 1157/1158 I113L/A225E/R266N/R415K ++ 1159/1160 V68A/R129E/S248A ++ 1161/1162 M75V/A105E/R154H/I215F/I264V/A345T ++ 1163/1164 A225E/I372V/E534H ++ 1165/1166 R85V/I170L/A225E/R266N/E534H ++ 1167/1168 Q95S/Q444T/R478V/G603S/M792S ++ 1169/1170 H47L/R221H ++ 1171/1172 I87E/T125E/N230D/I267V/W375Y/I464F/ ++ T708A 1173/1174 R440P/Q444T/R583Q/K724H/H788E + 1175/1176 Q95S/R478T/K724H + 1177/1178 R266N + 1179/1180 R136S/R440T/Q444T/R478V/R583Q/H788E + 1181/1182 R440T/R478V + 1183/1184 R306L/P358E/N703Y/Q776E + 1185/1186 R136S/Q444T/R478V/R583Q/H788E/M792S + 1187/1188 L98V/S250D + 1189/1190 Q95S/R440T/R478V/K724H/H788E/M792S + 1191/1192 Q95S/Q201E/R478V/R583Q/K724H/H788E + 1193/1194 R93V/D477K/A635S + 1195/1196 M75V/A105E/A345T/T410S/Q769R + 1197/1198 M75V/A105E/P530L + 1199/1200 M75V/A345T/P530L + 1201/1202 Q8M/R221H + 1203/1204 V466I + 1205/1206 Q95S/R136S/H788E + 1207/1208 P358E/L636Q/V737I + 1209/1210 R93V/R129E/S506P/R550H/I595V/A719C/ + I756V 1211/1212 V68A/G189R/I272L/V316I/D477K/A719C/ + I756V 1213/1214 V126L/V314L/N499H/D549E/G589E/R755G + 1215/1216 Q95S/Q444T/R478V/K724H/H788E + 1217/1218 Q95S/R385L/R478V/R583Q/H788E + 1219/1220 V68A/D146N/S248A/V387I/S506P/R550H + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 1080, and defined as follows: “+” = production at least that of the reference polypeptide, but less than 1.2-fold; and “++” = at least 1.2-fold increased production, as compared to the reference polypeptide.

Characterization of SuS Shake Flask Powders with a Coupled Assay for Glucosylation of Rebaudioside A Using ADP-glucose Produced by SuS from Sucrose and ADP

(293) First, SuS lyophilized shake flask powders were reconstituted in water and added for final concentrations ranging from 0.025-1 g/L of protein in total reaction volume. The reaction conditions were as follows: 50 mM potassium phosphate buffer, pH 7, 3 mM MgCl₂, with 1 mM rebaudioside A (>97% purity), 10 mM sucrose (Sigma), and 1 mM ADP (Sigma). The GT added to catalyze the glucosyl transfer from ADP-glucose to rebaudioside A was SEQ ID NO: 548 at 2 g/L final concentration in 100 µL total reaction volume. The reaction was performed at 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 h. The reaction described above was quenched by adding 10 µL assay to 90 µL acetonitrile with 0.2% formic acid and precipitated by centrifugation. Glycosylated products were detected in the supernatant by RapidFire SPE-MS/MS following 1:10 dilution in water with the instrument and parameters described in Example 9, Table 9.3. The relative levels of rebaudioside I produced from rebaudioside A in the coupled shake flask powder assays are listed in Table 19.2.

(294) TABLE-US-00040 TABLE 19.2 Round 2 Shake Flask Powder Characterization in Coupled Assay SEQ ID NO: Amino Acid Differences RebI (nt/aa) (Relative to SEQ ID NO: 1080) Levels.sup.a 1161/1162 M75V/A105E/R154H/I215F/I264V/A345T ++ 1153/1154

R136S/R440P/Q444T/R478T/G603S + 1165/1166 R85V/I170L/A225E/R266N/E534H + 1157/1158 I113L/A225E/R266N/R415K + 1163/1164 A225E/I372V/E534H - .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 1080, and defined as follows: “-” = production less than 0.8-fold; “+” = production at least 0.8-fold, but less than 1.2-fold; and “++” = at least 1.2-fold increased production, as compared to the reference polypeptide.

Example 20

Sucrose Synthase Variants of SEQ ID NO: 1158

(295) Directed evolution of the SuS encoded by SEQ ID NO: 1157 (i.e., SEQ ID NO: 1158) was continued by constructing libraries of variant genes in which mutations associated with improved activity above were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below, to provide a third round (“Round 3”) of 34 engineered SuS variant polypeptides with activity towards the generation of ADP-glucose.

(296) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(297) Libraries were screened using the following high-throughput (HTP) enzyme coupled assay: 10 µL SuS lysate and 2 g/L GT SEQ ID NO: 548 in 100 µL reaction volume with substrate loading of 1 mM rebaudioside A (>97% purity) and co-substrate loadings of 1 mM ADP (Sigma, >95%) and 10 mM sucrose (Sigma). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 7, 3 mM MgCl₂, 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 h. The reaction described above was quenched by adding 10 µL assay to 90 µL acetonitrile with 0.2% formic acid and precipitated by centrifugation. The supernatant was diluted 1:10 in water and steviol glycoside products were detected by RapidFire SPE-MS/MS with the instrument and parameters described in Example 9, Table 9.3. After analysis, 56 engineered SuS variant polypeptides that showed activity coupled with a GT on rebaudioside A were retested in triplicate with the same conditions and 2.5-fold reduced lysate load. The resulting engineered polypeptides are listed in Table 20.1.

(298) Shake-flask scale cultures were grown for protein characterization as described in Example 1 for variants with the amino acid mutations shown in Table 20.1 (relative to SEQ ID NO: 1158).

(299) TABLE-US-00041 TABLE 20.1 Round 3 Variants and RebI Levels SEQ ID NO: Amino Acid Differences RebI (nt/aa) (Relative to SEQ ID NO: 1158) Levels.sup.a 1221/1222

R85V/I87E/A105E/I215F/I267V/I756V ++++ 1223/1224

H47L/R85V/T125E/I372V/R583Q/A635S/I756V ++++ 1225/1226 H47L/R85V/I87E/R154H/I756V ++++ 1227/1228 R85V/I87E/R129E/W375Y/I756V/Q776E +++ 1229/1230

V68A/R93V/L98V/R136S/S248A/S250D/P358E/ +++ R440P/E534H/K724H 1231/1232

H47L/R85V/I87E/A105E/T125E/A635S +++ 1233/1234 H47L/R129E/I170L/A635S +++ 1235/1236

Q8M/R136S/S248A/R478V/H788E +++ 1237/1238 H47L/R85V/A105E/R129E/Q201E/N230D/ ++

I267V/R583Q 1239/1240 H47L/M75V/R85V/A105E/T125E/R129E/I170L/ ++ A635S 1241/1242

M75V/R85V/R129E/R154H/I264V/W375Y ++ 1243/1244 I87E/T125E/R129E/I170L/N230D/I756V ++

1245/1246 H47L/M75V/R85V/I264V/I267V/I372V/K415R/ ++ A635S 1247/1248

H47L/R85V/I170L/I756V ++ 1249/1250 Q8M/Q95S/L98V/R440P/R478V/E534H/V600I/ ++ H788E
 1251/1252 Q8M/V68A/L98V/R221H/S248A/S250D/R440P/ ++ D477K/E534H/I595V/K724H
 1253/1254 Q8M/V68A/Q95S/L98V/P358E/R478T/I595V/ ++ K724H/M792S 1255/1256
 Q95S/R440P/Q444T/K724H/H788E ++ 1257/1258 I87E/R154H/R306L/W375Y/I756V ++ 1259/1260
 I170L/I264V/I267V ++ 1261/1262 M75V/R85V/I87E/A105E/I264V/I267V/R583Q/ ++ T708A
 1263/1264 Q8M/R93V/Q95S/L98V/R136S/R221H/I595V/ ++ V600I/H788E 1265/1266
 V68A/R93V/Q95S/P358E/R440T/Q444T/ ++ R478V/E534H/I595V/G603S 1267/1268
 H47L/R154H/I372V/W375Y/R583Q/A635S/ ++ T708A/I756V 1269/1270
 H47L/M75V/R85V/A105E/W375Y/I756V/ ++ Q776E 1271/1272 R129E/I215F/I372V/I756V ++
 1273/1274 V68A/R93V/Q95S/Q444T/H788E ++ 1275/1276
 H47L/M75V/R85V/I87E/I170L/I372V/I756V ++ 1277/1278
 Q8M/R93V/Q95S/L113I/S250D/R440T/I595V/ + V600I/K724H/H788E 1279/1280
 H47L/M75V/R85V/I87E/R129E/W375Y/Q776E + 1281/1282 R93V/Q95S/L98V/E534H/M792S +
 1283/1284 Q8M/V68A/H788E + 1285/1286 H47L/T125E/R129E/W375Y/I756V/Q776E + 1287/1288
 R85V/T125E/I215F/W375Y/K415R/A635S/ + Q776E .sup.aLevels of increased production were
 determined relative to the reference polypeptide of SEQ ID NO: 1158, and defined as follows: “+” =
 production at least that of the reference polypeptide, but less than 1.5-fold; “++” = production at least 1.5-
 fold, but less than 2-fold; “+++” = production at least 2-fold, but less than 2.5-fold; and “++++” = at least
 2.5-fold increased production, as compared to the reference polypeptide.

Characterization of SuS Shake Flask Powders with a Coupled Assay for Glucosylation of Rebaudioside A Using ADP-glucose Produced by SuS from Sucrose and ADP

(300) First, SuS lyophilized shake flask powders were reconstituted in water and added for final concentrations ranging from 0.025-5 g/L of protein in total reaction volume. The reaction conditions were as follows: 50 mM potassium phosphate buffer, pH 7, 3 mM MgCl₂, with 1 g/L rebaudioside D, 10 mM sucrose (Sigma), and 1 mM ADP (Sigma). The GT added to catalyze the glucosyl transfer from ADP-glucose to rebaudioside A was SEQ ID NO: 561/562 at 1 g/L final concentration in 100 µL total reaction volume. The reaction was performed at 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 h. The reaction described above was quenched by adding 10 µL assay to 90 µL acetonitrile with 0.2% formic acid and precipitated by centrifugation. Glycosylated products were detected in the supernatant by RapidFire SPE-MS/MS following 1:10 dilution in water with the instrument and parameters described in Example 9, Table 9.3. The relative levels of rebaudioside M produced from rebaudioside D in the coupled shake flask powder assays are listed in Table 20.2.

(301) TABLE-US-00042 TABLE 20.2 Round 3 Shake Flask Powder Characterization in Coupled Assay
 SEQ ID NO: Amino Acid Differences RebM (nt/aa) (Relative to SEQ ID NO: 1158) Levels.sup.a

1225/1226 H47L/R85V/I87E/R154H/I756V ++ 1221/1222 R85V/I87E/A105E/I215F/I267V/I756V ++
 1229/1230 V68A/R93V/L98V/R136S/S248A/S250D/P358E/ + R440P/E534H/K724H 1227/1228
 R85V/I87E/R129E/W375Y/I756V/Q776E + 1223/1224

H47L/R85V/T125E/I372V/R583Q/A635S/I756V + .sup.aLevels of increased production were
 determined relative to the reference polypeptide of SEQ ID NO: 1158, and defined as follows: “+” =
 production at least that of the reference polypeptide, but less than 1.7-fold; and “++” = at least 1.7-fold
 increased production, as compared to the reference polypeptide.

Expression Analysis of SuS Enzymes

(302) The three rounds of engineered SuS polypeptides for ADP recycle with sucrose were analyzed by polyacrylamide gel-electrophoresis to determine relative protein expression levels. Samples were prepared with 1×LDS loading buffer and 1× reducing agent (Life Technologies). A 4-12% Bis-Tris acrylamide gel (Life Technologies) was loaded with 5 µg per lane of lyophilized soluble crude lysate from shake flask scale cultures and run with MES running buffer for 25 min at 200 V, and bands were quantified using ImageJ analysis software. The relative expression levels are listed in Table 20.3. SEQ ID NOs: 1079/1080, 1157/1158, and 1221/1222 are significantly better expressed genes and/or significantly better folded/more stable proteins. These produce more protein than the wild type gene.

(303) TABLE-US-00043 TABLE 20.3 Protein Levels of Engineered SuS Variants SEQ ID NO: (nt/aa)
 Increased Protein.sup.a 1079/1080 +++ 1157/1158 ++ 1221/1222 ++ .sup.aLevels of increased protein
 were determined relative to the reference polypeptide of SEQ ID NO: 74, and defined as follows: “+” =

band intensity at least that of the reference polypeptide, but less than 2-fold; “++” = band intensity at least 2-fold, but less than 3-fold; and “+++” = at least 3-fold increased band intensity, relative to the reference polypeptide.

Example 21

Transformation of Stevioside to Rebaudioside A with AGT and ACSuS

(304) In this Example, experiments conducted to produce rebaudioside A using AGT and ACSuS are described. The buffer used was 50 mM pH 7.0 potassium phosphate with 3 mM MgSO₄. Enzyme and sucrose stock solutions (400 g/L) were prepared in the buffer. “Reb A 60” was a ~1:2 mixture of stevioside and rebaudioside A respectively. To a vial under air was added 250 µL of 8 g/L β-1,3-GT SEQ ID NO: 1290 stock solution, 250 µL of 2 g/L SuS SEQ ID NO: 1158 stock solution and 500 µL of 50 g/L Reb A 60 and 2.6 g/L ADP stock solution in 400 g/L of sucrose. The final composition was 25 g/L Reb A 60, 200 g/L sucrose, 2 g/L SEQ ID NO: 1290, 0.5 g/L SEQ ID NO: 1158, and 1.3 g/L (3 mM) of ADP. The resulting clear homogenous solution was stirred under air (the reaction gradually turned cloudy with formation of precipitates). The progress of the reaction was followed by taking aliquots and quenched with 1:1 acetonitrile/water. After centrifuging at 4000 rpm/r.t./5 min, the clear homogeneous supernatant was analyzed by HPLC using the instrument and parameters listed in Table 21.1 and Table 21.2. After 48 h, the reaction mixture consisted of >90% Reb A (Table 21.3).

(305) TABLE-US-00044 TABLE 21.1 Analytical HPLC Method Steviol Glycosides Instrument Agilent HPLC 1100 series Column 3.5 × 50 mm 3.5 µm Waters XBridge Phenyl column Mobile phase Isocratic 65:35 A:B A: 0.1% formic acid in water B: 0.1% formic acid in methanol Flow rate 2.4 mL/min Run time 12 min Peak retention times Rebaudioside D: 3.7 min Rebaudioside M: 4.7 min Stevioside: 8.8 min Rebaudioside A and I: 9.9 min Column temperature 50° C. Injection volume 10 µL UV detection 210 nm

(306) TABLE-US-00045 TABLE 21.2 Analytical HPLC Method Steviol Glycosides Instrument Thermo Ultimate 3000 UPLC with CAD Column 3.2 × 250 mm 3.5 µm Restek Pinnacle II Amino column Mobile phase Isocratic 22:78 A:B A: water B: acetonitrile Flow rate 2.0 mL/min Run time 12 min Peak retention times Fructose: 1.7 min Glucose: 2.0 min Stevioside: 2.6 min Sucrose: 3.1 min Rebaudioside A: 4.2 min Rebaudioside I: 6.6 min Rebaudioside D: 10.0 min Rebaudioside M: 10.7 min Column temperature 50° C. Injection volume 10 µL UV detection 210 nm for organic compounds CAD for sugars

(307) TABLE-US-00046 TABLE 21.3 Conversion of Stevioside to Rebaudioside A Over Time t [h] % stevioside % Reb A

0	36	64	1	35	65	2	33	67	4	31	69	8	27	73	24	18	82	48	8	92	72	6	94
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Example 22

Transformation of Stevioside to Rebaudioside A and Rebaudioside A to Rebaudioside D with AGT and ACSuS

(308) In this Example, experiments conducted to produce rebaudioside A and rebaudioside D are described. To a vial under air was added 250 µL of 20 g/L β-1,3-GT SEQ ID NO: 1290 stock solution, 250 µL of 40 g/L β-1,2-GT SEQ ID NO: 954 stock solution and 500 µL of 20 g/L Reb A 60, 2.6 g/L ADP and 1 g/L SuS SEQ ID NO: 1158 stock solution in 400 g/L of sucrose. The buffer used was 50 mM pH 7.0 potassium phosphate with 3 mM MgSO₄. Enzyme and sucrose stock solution (400 g/L) was prepared in the buffer. The final composition was 10 g/L Reb A 60, 200 g/L sucrose, 5 g/L SEQ ID NO: 1290, 10 g/L SEQ ID NO: 954, 0.5 g/L SEQ ID NO: 1158, and 1.3 g/L (3 mM) of ADP. The resulting clear homogenous solution was stirred under air (the reaction gradually turned cloudy with formation of precipitates). The progress of the reaction was followed by taking aliquots and quenched with 1:1 acetonitrile/water. After centrifuging at 4000 rpm/r.t./5 min, the clear homogeneous supernatant was analyzed by HPLC using the instrument and parameters listed in Table 21.1 and Table 21.2. After 24 h, no stevioside remained and formation of 16% of Reb D was observed (Table 22.1).

(309) TABLE-US-00047 TABLE 22.1 Conversion of Stevioside and Rebaudioside A to Rebaudioside A and Rebaudioside D t [h] % stevioside % Reb A % Reb D

0	35	65	0	1	27	70	3	2	25	70	5	4	23	72	5	8	15	79	6	24	0	84	16
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Example 23

Transformation of Rebaudioside D to Rebaudioside M with AGT and ACSuS

(310) In this Example, experiments conducted to produce rebaudioside M from rebaudioside D using AGT and ACSuS variants are described. To a vial under air was added 100 mg of Reb D, 250 µL of 80 g/L β-1,3-GT SEQ ID NO: 548 stock solution, 250 µL of 10 g/L SuS SEQ ID NO: 1158 stock solution

and 500 μL of 2.6 g/L ADP stock solution in 400 g/L of sucrose. The buffer used was 50 mM pH 7.0 potassium phosphate with 3 mM MgSO_4 . Enzyme and sucrose stock solution (400 g/L) was prepared in the buffer. The final composition was 100 g/L Reb D, 200 g/L sucrose, 20 g/L SEQ ID NO: 548, 2.5 g/L SEQ ID NO: 1158, and 1.3 g/L (3 mM) of ADP. The resulting thick slurry was stirred under air. The progress of the reaction was followed by taking aliquots and quenched with 1:1 acetonitrile/water. After centrifuging at 4000 rpm/r.t./5 mM, the clear homogeneous supernatant was analyzed by HPLC using the instrument and parameters listed in Table 21.1 and Table 21.2. After 48 h, the reaction mixture consisted of >90% Reb M (Table 23.1).

(311) TABLE-US-00048 TABLE 23.1 Conversion of Rebaudioside D to Rebaudioside M Over Time. t [h] % Reb M

0	0	1	16	2	28	4	40	8	52	24	79	32	88	48	94
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Example 24

Transformation of Stevioside to Rebaudioside A, Rebaudioside A to Rebaudioside D, and Rebaudioside D to Rebaudioside M with AGT and ACSuS

(312) In this Example, experiments conducted to produce rebaudioside A from stevioside, rebaudioside D from rebaudioside A, and rebaudioside M from rebaudioside D are described. To a vial under air was added 250 μL of 20 g/L β -1,3-GT SEQ ID NO: 1290 stock solution, 250 μL of 40 g/L β -1,2-GT SEQ ID NO: 954 stock solution, and 500 μL of 50 g/L Reb A 60, 2.6 g/L ADP, 5 g/L β -1,3-GT SEQ ID NO: 548, and 1 g/L SuS SEQ ID NO: 1158 stock solution, in 400 g/L of sucrose. The buffer used was 50 mM pH 7.0 potassium phosphate with 3 mM MgSO_4 . Enzyme and sucrose stock solution (400 g/L) was prepared in the buffer. The final composition was 25 g/L Reb A 60, 200 g/L sucrose, 5 g/L SEQ ID NO: 1290, 10 g/L SEQ ID NO: 954, 2.5 g/L SEQ ID NO: 548, 0.5 g/L SEQ ID NO: 1158 and 1.3 g/L (3 mM) of ADP. The resulting clear homogeneous solution was stirred under air (the reaction gradually turned cloudy with formation of precipitates). The progress of the reaction was followed by taking aliquots and quenched with 1:1 acetonitrile/water. After centrifuging at 4000 rpm/r.t./5 mM, the clear homogeneous supernatant was analyzed by HPLC using the instrument and parameters listed in Table 21.1 and Table 21.2. After 24 h, <5% stevioside remained and formation of 9% of Reb M was observed (Table 24.1).

(313) TABLE-US-00049 TABLE 24.1 Conversion of Stevioside and Rebaudioside A to Glucosylated Products Over Time. t [h] % stevioside % Reb A % Reb D % Reb I % Reb M

0	35	65	0	0	0	1	29	71	0	0	0
2	26	73	0	1	0	4	2	97	0	1	0
8	8	91	0	1	0	24	1	70	0	18	9

Example 25

Immobilization of AGT and/or ACSuS

(314) In this Example, experiments conducted to immobilize an AGT (e.g., the variants of SEQ ID NOS: 1290, 954, and/or 548) and/or an ACSuS variant (e.g., SEQ ID NO: 1158) are described. To a vessel is added the enzyme solution, either alone or as a combination of enzymes (i.e., AGT and/or ACSuS enzymes), and the solid support. The solid support is cationic, anionic, hydrophobic, hydrophilic with or without the presence of covalent bond forming functional groups such as thiol, alcohol, amines, olefin, alkyl halide and/or epoxide. The solid support is either a discrete polymeric resin or amorphous (nano) clay or activated carbon. The magnetic particle is used when it is suitable for product isolation/enzyme recycling. The reaction is carried out in the presence or absence of glutaraldehyde. The progress of enzyme uptake by the solid support is followed by Bradford assay. Alternatively, the solid support is packed in a column and the enzyme solution is flowed through the column, with recycling if necessary, until the desired degree of enzyme capture is reached. In some embodiments, all enzymes of interest are immobilized on a same solid support in the same reaction vessel, while in some alternative embodiments, the enzymes are immobilized individually in separate vessels or a combination thereof. In some embodiments, the immobilized enzyme is either isolated via filtration or is used immediately by adding buffer, sucrose, ADP and substrate to the immobilization reaction mixture.

Example 26

Transformation of Stevioside to Rebaudioside A with Immobilized AGT and/or ACSuS and Recycling of Sugar Solution

(315) In this Example, transformation of stevioside to rebaudioside A using an immobilized AGT (e.g., SEQ ID NO: 1290) and ACSuS (e.g., SEQ ID NO: 1158), and sugar solution recycling are described. To a vessel is added immobilized β -1,3-GT (e.g., SEQ ID NO: 1290) and immobilized SuS (e.g., SEQ ID NO: 1158). Alternatively, in some embodiments, one of the enzymes is used in the immobilized form and the

other is used in solution form. After the addition of buffer, sucrose, ADP and substrate (i.e., stevioside or Reb A 60), the reaction is monitored until the desired conversion is reached. In some embodiments, the product and immobilized enzyme are isolated by filtration. In some embodiments, immobilized enzyme is further separated from the product via either centrifugation, particle size filtration or magnetic retrieval and is re-used. The sugar filtrate is returned to the vessel for the next iteration.

Example 27

Transformation of Rebaudioside A to Rebaudioside D with Immobilized AGT and/or ACSuS and Recycling of Sugar Solution

(316) In this Example, experiments conducted to produce rebaudioside D from rebaudioside D with immobilized AGT and ACSuS, and sugar solution recycling are described. To a vessel is added immobilized β -1,2-GT (e.g., SEQ ID NO: 954) and immobilized SuS (e.g., SEQ ID NO: 1158). Alternatively, in some embodiments, one of the enzymes is used in the immobilized form and the other is used in solution form. After the addition of buffer, sucrose, ADP and substrate (stevioside or Reb A 60), the reaction is monitored until the desired conversion is reached. In some embodiments, the product and immobilized enzyme is isolated by filtration. In some embodiments, immobilized enzyme is further separated from the product via either centrifugation, particle size filtration or magnetic retrieval and is re-used. The sugar filtrate is returned to the vessel for the next iteration.

Example 28

Transformation of Rebaudioside D to Rebaudioside M with Immobilized AGT and/or ACSuS and Recycling of Sugar Solution

(317) In this Example, experiments conducted to produce rebaudioside M from rebaudioside D with immobilized AGT and ACSuS, and sugar solution recycling are described. To a vessel is added immobilized β -1,3-GT (e.g., SEQ ID NO: 548) and immobilized SuS (e.g., SEQ ID NO: 1158). Alternatively, in some embodiments, one of the enzymes is used in the immobilized form and the other is used in solution form. After the addition of buffer, sucrose, ADP and Reb D, the reaction is monitored until the desired conversion is reached. In some embodiments, the product and immobilized enzyme are isolated by filtration. In some additional embodiments, immobilized enzyme is further separated from the product via either centrifugation, particle size filtration or magnetic retrieval and is re-used. The sugar filtrate is returned to the vessel for the next iteration.

Example 29

Transformation of Stevioside to Rebaudioside A and Rebaudioside A to Rebaudioside D with Immobilized AGT and ACSuS and Recycling of Sugar Solution

(318) In this Example, experiments conducted to produce rebaudioside A from stevioside, and rebaudioside D from rebaudioside A with immobilized AGT and ACSuS, and sugar solution recycling are described. To a vessel is added immobilized β -1,3-GT (e.g., SEQ ID NO: 1290) and β -1,2-GT (e.g., SEQ ID NO: 954) and immobilized SuS (e.g., SEQ ID NO: 1158). Alternatively, one or two of these enzymes is used in the immobilized form and the others are used in solution form. After the addition of buffer, sucrose, ADP and substrate (stevioside or Reb A 60), the reaction is monitored until the desired conversion is reached. In some embodiments, the product and immobilized enzyme are isolated by filtration. In some embodiments, immobilized enzyme is further separated from the product via either centrifugation, particle size filtration or magnetic retrieval and is re-used. The sugar filtrate is returned to the vessel for the next iteration.

Example 30

Transformation of Stevioside to Rebaudioside A, Rebaudioside A to Rebaudioside D and Rebaudioside D to Rebaudioside M with Immobilized AGT and ACSuS and Recycling of Sugar Solution

(319) To a vessel is added immobilized β -1,3-GT (e.g., SEQ ID NO: 1290), β -1,2-GT (e.g., SEQ ID NO: 954), β -1,3-GT (e.g., SEQ ID NO: 548), and immobilized SuS (e.g., SEQ ID NO: 1158). Alternatively, in some embodiments, one, two or three of these enzymes is/are used in the immobilized form and the others are used in solution form. After the addition of buffer, sucrose, ADP and substrate (stevioside or Reb A 60), the reaction is monitored until the desired conversion is reached. In some embodiments, the product and immobilized enzyme are isolated by filtration. In some embodiments, immobilized enzyme is further separated from the product via either centrifugation, particle size filtration or magnetic retrieval and is re-used. The sugar filtrate is returned to the vessel for the next iteration.

Example 31

Sucrose Synthase Variants of SEQ ID NO: 1222

(320) Directed evolution of the SuS encoded by SEQ ID NO: 1222 was continued by constructing libraries of variant genes in which mutations associated with improved activity in earlier rounds of evolution were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below, to provide a fourth round ("Round 4") of 49 engineered SuS variant polypeptides with activity toward the generation of ADP-glucose.

(321) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside D

(322) Libraries were screened using the following HTP enzyme-coupled assay. Pelleted *E. coli* cultures were lysed with 250 μ L of Tris-HCl, pH 7.5, with 1 mM magnesium sulfate and 0.5 mg/mL lysozyme and polymyxin B sulfate (PMBS) and cleared by centrifugation. Lysate was diluted 20 $\times$ into Tris-HCl, pH 7.5. Then, 10 μ L diluted SuS lysate and 2 g/L GT SEQ ID NO: 696 (Rd8BB) were combined in 100 μ L reaction volume with substrate loading of $\sim$ 1 mM rebaudioside D and co-substrate loadings of 1 mM ADP (Sigma, >95%) and 10 mM sucrose (Sigma). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 7, 3 mM MgCl₂, 50 $^{\circ}$ C., in a Thermotron[®] titre-plate shaker with 300 RPM shaking for 2 h. The reaction described above was quenched by adding 10 μ L assay mixture to 90 μ L acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 10 $\times$ in water and steviol glycoside products were detected by RapidFire SPE-MS/MS, with the instrument and parameters described in Table 31.1.

(323) TABLE-US-00050 TABLE 31.1 RapidFire SPE-MS/MS Conditions for Steviol Glycoside Detection. Agilent RapidFire Conditions Buffer A 0.1% formic acid in LC/MS grade water; 1.5 mL/min flow rate Buffer B 0.1% formic acid in LC/MS grade methanol; 0.8 mL/min flow rate Aqueous wash Water Organic wash Acetonitrile SPE cartridge Agilent RapidFire cartridge A (C4) RF state 1 600 ms RF state 2 2500 ms RF state 3 0 RF state 4 5000 ms RF state 5 1000 ms Agilent Jet Stream Source Parameters Drying gas temperature 300 $^{\circ}$ C. Drying gas flow 10 L/min Nebulizer pressure 45 psi Sheath gas temperature 350 $^{\circ}$ C. Sheath gas flow 11 L/min Capillary voltage +3500 V Nozzle voltage +2000 V Agilent 6470 Triple Quadrupole MRM Parameters Compound Q1 Q3 Dwell Fragmentor CE CAV Stevioside 827.4 665.3 50 150 50 5 RebA 989.5 827.5 50 350 60 5 RebD or RebI 1151.7 827.5 50 350 55 5 RebM 1313.7 827.5 50 350 70 5

(324) After analysis, 49 engineered SuS variant polypeptides that showed improved activity coupled with a GT on rebaudioside D were identified. The resulting engineered polypeptides are listed in Table 31.2. Shake-flask scale cultures were grown for protein characterization as described in below for variants with the amino acid mutations shown in Table 31.3.

(325) TABLE-US-00051 TABLE 31.2 SUS Round 4 Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 1222) RebM.sup.a 1295/1296 H47L/I372V/A635S/Q776E + 1297/1298 H47L/R93V/L98V/P358E/R583Q/A635S + 1299/1300 V68A/R93V/R154H/P358E/I372V/R440P/ + Q776E 1301/1302 I372V/W375Y/Q776E + 1303/1304 H47L/L98V/R129E/P358E/I372V/W375Y/ + H438Q/E534H 1305/1306 R154H/E534H/A635S/K724H + 1307/1308 R93V/R154H/S248A/K724H + 1309/1310 H47L/R93V/P358E/I372V/W375Y/Q776E + 1311/1312 V68A/R136S/K724H + 1313/1314 H47L/P358E/A635S/Q776E + 1315/1316 L98V/E534H/R583Q/A635S + 1317/1318 R93V/L98V/T125E/R154H/S250D/R440P + 1319/1320 R93V/L98V/R154H/A635S/Q776E +++ 1321/1322 R129E/R154H/S248A/S250D/P358E/W375Y/ +++ E534H/A635S 1323/1324 R93V/T125E/R154H/R440P/E534H ++ 1325/1326 R93V/L98V/E534H + 1327/1328 R93V/L98V/R154H/S250D/P358E/W375Y/ ++ E534H 1329/1330 H47L/L98V/I372V/W375Y ++ 1331/1332 H47L/R93V/L98V/R154H/I372V/W375Y/ +++ Q776E 1333/1334 H47L/R93V/R129E/R136S/R154H/S250D/ + I372V/E534H/A635S/K724H 1335/1336 R93V/R129E/S250D/P358E/I372V/W375Y/ + R583Q 1337/1338 V68A/W375Y/R440P/E534H/K724H/Q776E + 1339/1340 H47L/T125E/R154H +++ 1341/1342 R93V/L98V/T125E/R154H/S248A ++ 1343/1344 V68A/R154H/E534H/A635S ++ 1345/1346 H47L/R93V/R129E/R136S/W375Y/E534H/ ++ R583Q 1347/1348 V68A/R129E/R583Q/K724H ++ 1349/1350 H47L/R93V/L98V/R136S/R154H/E772G/ +++ Q776E 1351/1352 H47L/R136S/R583Q/Q776E ++ 1353/1354 R154H/S250D/P358E/W375Y/R583Q +++ 1355/1356 H47L/R129E/S248A/S250D/I372V/W375Y/ + E534H/K724H 1357/1358 W375Y/A635S + 1359/1360

H47L/P358E/R440P/K724H + 1361/1362 R129E/R136S/W375Y + 1363/1364
H47L/V68A/R93V/L98V/P358E/R440P + 1365/1366 L98V/R129E/W375Y/R583Q + 1367/1368
R93V/P358E/E534H/A635S + 1369/1370 E534H/R583Q + 1371/1372
H47L/V68A/R154H/I372V/W375Y + 1373/1374 R129E/P358E/I372V/A635S ++ 1375/1376
H47L/R129E/R136S/I372V/W375Y/E534H ++ 1377/1378 R93V/L98V/R129E/R154H/S248A ++
1379/1380 V68A/R129E/R440P +++ 1381/1382 R154H/P358E/W375Y/E534H/Q776E +++ 1383/1384
H47L/L98V/R129E/W375Y/E534H/A635S/ ++ K724H/Q776E 1385/1386 V68A/R154H/P358E/W375Y
++ 1387/1388 V68A/R93V/L98V/R129E/P358E/W375Y/ ++ K724H 1389/1390
R129E/R136S/R154H/S248A/S250D/I372V/ ++ W375Y/E534H 1391/1392

H47L/R93V/P358E/I372V/W375Y/R440P/ +++ K724H .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 1222, and are defined as follows: “+” = production at least 2-fold, but less than 3.05-fold; “++” = at least 3.05-fold, but less than 3.29-fold; and “+++” = at least 3.29-fold increased production, as compared to the reference polypeptide.

Production of Shake Flask Powders (SFP)

(326) A shake-flask procedure was used to generate the glycosyltransferase polypeptide shake flask powders (SFP) for characterization assays used in the biocatalytic processes described herein. Shake flask powder (SFP) preparation of enzymes provides a more purified preparation (e.g., up to >30% of total protein) of the enzyme, as compared to the cell lysate used in HTP assays, and also allows for the use of more concentrated enzyme solutions. A single colony of *E. coli* containing a plasmid encoding an engineered polypeptide of interest was inoculated into 5 mL Luria Bertani broth containing 30 µg/ml chloramphenicol and 1% glucose. Cells were grown overnight (at least 16 hours) in an incubator at 30° C. with shaking at 250 rpm. The culture was diluted into 250 mL Terrific Broth (12 g/L bacto-tryptone, 24 g/L yeast extract, 4 mL/L glycerol, 65 mM potassium phosphate, pH 7.0, 1 mM MgSO₄) containing 30 µg/ml CAM, in a 1 L flask to an optical density of 600 nm (OD₆₀₀) of 0.2, and allowed to grow at 30° C.

(327) Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to Rebaudioside D

(328) An experiment was performed to characterize the activity of the engineered round 4 SUS variants on sucrose and ADP to facilitate the formation of rebaudioside M from rebaudioside D. Shake flask powder (SFP) was added to a 100 µL total reaction volume at 0.125 g/L concentration containing 50 mM potassium phosphate buffer, pH 7, 3 mM magnesium chloride, 1 g/L rebaudioside D, 10 mM sucrose, 1 mM ADP, and 2 g/L GT SEQ ID NO: 734. The reaction was performed at 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 h. The reaction was quenched by adding 10 µL of the reaction mixture to 90 µL acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 10× in water and analyzed for steviol glycosides by RapidFire SPE-MS/MS, with the instrument and parameters described in Table 31.1. At least 3 of the variants had higher activities than SEQ ID NO: 1222. The levels of rebaudioside M produced from rebaudioside D by the variants relative to SEQ ID NO: 1222 are listed in Table 31.3. The variant with mutations H47L, R93V, P358E, I372V, W375Y, R440P, and K724H (SEQ ID NO: 1392), relative to SEQ ID NO: 1222, had the highest activity. Therefore, the encoding polynucleotide (SEQ ID NO: 1391) was selected for further directed evolution for the catalysis of the recycling reaction transferring a glucose from sucrose to ADP.

(329) TABLE-US-00052 TABLE 31.3 SUS Round 4 SFP Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 1222) RebM.sup.a 1391/1392

H47L/R93V/P358E/I372V/W375Y/R440P/K724H + 1349/1350
H47L/R93V/L98V/R136S/R154H/E772G/Q776E + 1381/1382 R154H/P358E/W375Y/E534H/Q776E + 1365/1366 L98V/R129E/W375Y/R583Q + 1379/1380 V68A/R129E/R440P + 1391/1392

H47L/R93V/P358E/I372V/W375Y/R440P/K724H + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 1222, and defined as follows: “+” = production at least 1.04-fold, but less than 1.2-fold greater productivity, as compared to the reference polypeptide.

Example 32

Sucrose Synthase Variants of SEQ ID NO: 1392

(330) Directed evolution of the SuS encoded by SEQ ID NO: 1391 was continued by constructing

libraries of variant genes in which mutations associated with improved activity in earlier rounds of evolution were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below, to provide a fifth round ("Round 5") of 86 engineered SuS variant polypeptides with activity toward the generation of ADP-glucose.

(331) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside D

(332) Libraries were screened using the following HTP enzyme-coupled assay. Pelleted *E. coli* cultures were lysed with 400 μ L of Tris-HCl, pH 7.5 with 1 mM magnesium sulfate and 0.5 mg/mL lysozyme and polymyxin B sulfate (PMBS) and cleared by centrifugation. Lysate was diluted 25-60 $\times$ into Tris-HCl, pH 7.5. 10 μ L diluted SuS lysate and 2 g/L GT SEQ ID NO: 4684 (Rd9BB) in 100 μ L reaction volume with substrate loading of $\sim$ 1 mM rebaudioside D and co-substrate loadings of 1 mM ADP (Sigma, >95%) and 10 mM sucrose (Sigma). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 7, 3 mM MgCl₂, 50 $^{\circ}$ C. in a Thermotron[®] titre-plate shaker with 300 RPM shaking for 1 h. The reaction described above was quenched by adding 10 μ L assay to 90 μ L acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 10 $\times$ in water and steviol glycoside products were detected by RapidFire SPE-MS/MS, with the instrument and parameters described in Table 31.1.

(333) After analysis, 86 engineered SuS variant polypeptides that showed improved activity coupled with a GT on rebaudioside D were identified. The resulting engineered polypeptides are listed in Table 32.1. Shake-flask scale cultures were grown for protein characterization as described in below for variants with the amino acid mutations shown in Table 32.2.

(334) TABLE-US-00053 TABLE 32.1 SUS Round 5 Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 1392) RebM.sup.a 1395/1396

G54D/A97V/A118N/N307E/G694N/L727E/ + A738E 1397/1398

A97V/A118N/H442N/G694N/L727E/A738E + 1399/1400 Y17D/F684H + 1401/1402

Y17D/Y434H/F684H + 1403/1404 Y17D/Y357K/Y434H/F519T/F684H ++ 1405/1406

Y17D/Y357K/P364R/Y434H/F519T/F684H ++ 1407/1408 R550I + 1409/1410

V68A/L98V/R129E/R136S ++

(335) TABLE-US-00054 TABLE 32.1 SUS Round 5 Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 1392) RebM.sup.a 1411/1412

R136S/G603D ++ 1413/1414 L98V/R136S/R154H/A635S ++ 1415/1416 V68A/R136S/E534H/A635S

++ 1417/1418 V68A/R154H ++ 1419/1420 V68A/R129E/R136S/R154H ++ 1421/1422

L98V/R129E/R136S/R154H/A635S +++ 1423/1424 V68A/L98V/R154H/E534H/A635S ++ 1425/1426

L98V/R154H + 1427/1428 R136S/E534H/A635S + 1429/1430 L98V/R154H/E534H ++ 1431/1432

V68A/L98V/R136S/R154H/A635S ++ 1433/1434 R129E/A635S + 1435/1436 R129E/R154H/A635S

+++ 1437/1438 V68A/L98V/R154H/E534H + 1439/1440 V68A/R129E/R136S + 1441/1442

V68A/R129E/R136S/R154H/I464F/A635S +++ 1443/1444 L98V/R129E/R154H +++ 1445/1446

R154H/A635S ++ 1447/1448 R129E/R154H/E534H ++ 1449/1450 R154H + 1451/1452

L98V/R136S/A635S ++ 1453/1454 V68A/R136S/R154H/E534H/A635S +++ 1455/1456

V68A/L98V/R129E/R154H/A635S +++ 1457/1458 V68A/R129E/R154H ++ 1459/1460

V68A/R154H/A635S ++ 1461/1462 V68A/L98V/R154H + 1463/1464 L98V/R136S/E534H/A635S ++

1465/1466 V68A/L98V/R136S/R154H/E534H/A635S +++ 1467/1468 L98V/R129E/E534H/A635S ++

1469/1470 V68A/L98V/R129E/R136S/R154H ++ 1471/1472 R129E/R136S/E534H + 1473/1474

V68A/R129E/R136S/E534H/A635S +++ 1475/1476 V68A/R129E/R154H/D765H +++ 1477/1478

L98V/R129E/R154H/E534H/A635S +++ 1479/1480 V68A/R136S/A635S ++ 1481/1482

V68A/L98V/R129E/R136S/R154H/E534H ++ 1483/1484 R129E/R136S/A635S +++ 1485/1486

R136S/R154H/A635S +++ 1487/1488 L98V/R129E/R136S/E534H + 1489/1490

L98V/R129E/R136S/R154H ++ 1491/1492 R136S/A635S ++ 1493/1494

R132C/R136S/R154H/E534H/A635S ++ 1495/1496 L98V/R129E/R136S/A635S +++ 1497/1498

R129E/R136S/R154H/A635S +++ 1499/1500 R129E/R136S + 1501/1502 V68A/R154H/E534H/A635S

+++ 1503/1504 V68A/L98V/R129E/R154H/E534H +++ 1505/1506 F160W + 1507/1508 S161Q +

1509/1510 F160M + 1511/1512 A635E + 1513/1514 A253G + 1515/1516 F160E + 1517/1518 F160A +

1519/1520 Q381S + 1521/1522 R550M + 1523/1524 F519A + 1525/1526 P785D + 1527/1528 R167E +

1529/1530 L563V + 1531/1532 A635D ++ 1533/1534 P285A + 1535/1536 R550Q + 1537/1538 F519L +

1539/1540 F519T + 1541/1542 A635R + 1543/1544 F519G + 1545/1546 G157A ++ 1547/1548 A122E + 1549/1550 S564A +++ 1551/1552 A122D + 1553/1554 R550S + 1555/1556 A253T + 1557/1558 F160S + 1559/1560 F160N + 1561/1562 F519S + 1563/1564 G157F + 1565/1566 A253V + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 1392, and are defined as follows: “+” = production at least 1.2-fold, but less than 1.54-fold; “++” = at least 1.54-fold, but less than 1.73-fold; and “+++” production at least 1.73-fold greater than that of the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to Rebaudioside D

(336) An experiment was performed to characterize the activity of the engineered round 4 SUS variants on sucrose and ADP to facilitate the formation of rebaudioside M from rebaudioside D. Shake flask powder (SFP) was added to a 100 μ L total reaction mixture volume at 0.05 g/L concentration containing 50 mM potassium phosphate buffer, pH 7, 3 mM magnesium chloride, 1 g/L rebaudioside D, 10 mM sucrose, 1 mM ADP, and 2 g/L GT SEQ ID NO: 4684. The reaction was performed at 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1-2 h. The reaction was solubilized by diluting 2.5 $\times$ into water, quenched by adding 10 μ L of the diluted reaction to 90 μ L acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 3.33 $\times$ in water and analyzed for steviol glycosides by RapidFire SPE-MS/MS, with the instrument and parameters described in Table 31.1. All of the variants in Table 32.2 had higher activities than SEQ ID NO: 1392. The levels of rebaudioside M produced from rebaudioside D by the variants relative to SEQ ID NO: 1392 are listed in Table 32.2. The variant with mutations V68A, L98V, R129E, R154H, and A635S (SEQ ID NO: 1455) relative to SEQ ID NO: 1392, was selected for further directed evolution for the catalysis of the recycling reaction transferring a glucose from sucrose to ADP.

(337) TABLE-US-00055 TABLE 32.2 SUS Round 5 SFP Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 1392) RebM.sup.a 1393/1394 R154Q/S161T + 1397/1398 A97V/A118N/H442N/G694N/L727E/A738E + 1405/1406

Y17D/Y357K/P364R/Y434H/F519T/F684H +++ 1407/1408 R550I + 1455/1456

V68A/L98V/R129E/R154H/A635S ++ 1477/1478 L98V/R129E/R154H/E534H/A635S + 1497/1498 R129E/R136S/R154H/A635S ++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 1392, and are defined as follows: “+” = production at least that of the reference polypeptide, but less than 1.12-fold; “++” = production at least 1.12-fold, but less than 1.20-fold; and “+++” = production at least 1.20-fold greater, as compared to the reference polypeptide.

Example 33

Sucrose Synthase Variants of SEQ ID NO: 1456

(338) Directed evolution of the SuS encoded by SEQ ID NO: 1455 was continued by constructing libraries of variant genes in which mutations associated with improved activity in earlier rounds of evolution were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below, to provide a sixth round (“Round 6”) of 16 engineered SuS variant polypeptides with activity toward the generation of ADP-glucose.

(339) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(340) Libraries were screened using the following HTP enzyme-coupled assay. Pelleted *E. coli* cultures were lysed with 400 μ L of Tris-HCl, pH 7.5, with 1 mM magnesium sulfate and 0.5 mg/mL lysozyme and polymyxin B sulfate (PMBS) and cleared by centrifugation. Lysate was diluted 30 $\times$ into Tris-HCl, pH 7.5. Then, 10 μ L diluted SuS lysate and 1 g/L GT SEQ ID NO: 2814 (Rd8BB) were combined in 100 μ L reaction volume with substrate loading of 4.5 mM rebaudioside A 97 and co-substrate loadings of 0.25 mM ADP (Sigma, >95%) and 10 mM sucrose (Sigma). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6.5, 50° C., in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 h. The reaction described above was solubilized by adding 10 μ L assay mixture to 90 μ L water, quenched by adding 10 μ L solubilized assay mixture to 90 μ L acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 7.3 $\times$ in water, and steviol glycoside products were detected by RapidFire SPE-MS/MS, with the instrument and parameters described in Table 31.1.

(341) After analysis, 16 engineered SuS variant polypeptides that showed improved activity coupled with

a GT on rebaudioside A were identified. The resulting engineered polypeptides are listed in Table 33.1. Shake-flask scale cultures were grown for protein characterization as described in Example 31, for variants with the amino acid mutations shown in Table 33.2.

(342) TABLE-US-00056 TABLE 33.1 SUS Round 6 Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 1456) RebD.sup.a 1567/1568

Y17D/G54D/A524T/R550I/L727E + 1569/1570 Y17D/Y434H/A524T/F684H + 1571/1572

Y17D/A738E + 1573/1574 Y17D/G54D/A97V/R136P/E329Q/R550I/ + F684H/A738E 1575/1576

G54D/A97V/Y434H/A524T/R550I/F684H/ + L727E 1577/1578 Y17D/S161T/Y434H/A524T/R766H ++

1579/1580 E329Q/R550I/F684H/L727E/A738E + 1581/1582 Y17D/G54D/S161T/F519T/L727E/A738E

++ 1583/1584 Y17D/Y434H/A738E +++ 1585/1586 G54D/A97V/S161T/Y434H/H442N +++ 1587/1588

G54D/R136P/H442N/R550I ++ 1589/1590 Y17D/G54D/A97V/E329Q/A524T/F684H ++ 1591/1592

G54D/Y434H/A524T/A738E ++ 1593/1594 Y17D/H442N/A524T/R550I/F684H/E721K + 1595/1596

Y17D/L727E +++ 1597/1598 A97V/R136P/F519T/R550I/L727E/A738E + .sup.aLevels of increased

production were determined relative to the reference polypeptide of SEQ ID NO: 1456, and defined as follows: “+” = production at least 1.4-fold, but less than 1.5-fold; “++” = greater than 1.5-fold, but less than 1.54-fold; and “+++” = greater than 1.54-fold increased production, relative to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to Rebaudioside A

(343) An experiment was performed to characterize the activity of the engineered round 6 SUS variants on sucrose and ADP to facilitate the formation of rebaudioside D from rebaudioside A. Shake flask powder (SFP) was added to a 100 μ L total reaction volume at 0.01 g/L concentration containing 50 mM potassium phosphate buffer, pH 6.5, 4.5 mM rebaudioside A 97, 30 mM sucrose, 0.25 mM ADP, and 1 g/L GT SEQ ID NO: 2814. The reaction was performed at 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1-2 h. The reaction was solubilized by diluting 10 $\times$ into water, quenched by adding 10 μ L of the diluted reaction to 90 μ L acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 4.4 $\times$ in water and analyzed for steviol glycosides by RapidFire SPE-MS/MS, with the instrument and parameters described in Table 31.1. At least 4 of the variants had higher activities than SEQ ID NO: 1456. The levels of rebaudioside D produced from rebaudioside A by the variants relative to SEQ ID NO: 1456 are listed in Table 33.2. The variant with mutations Y17D, G54D, S161T, F519T, L727E, and A738E (SEQ ID NO: 1582) relative to SEQ ID NO: 1456, was selected for further directed evolution for the catalysis of the recycling reaction transferring a glucose from sucrose to ADP.

(344) TABLE-US-00057 TABLE 33.2 SUS Round 6 SFP Variants and RebD Levels SEQ ID NO: Amino Acid Differences (Relative to Increased (nt/aa) SEQ ID NO: 1456) RebD.sup.a 1577/1578

Y17D/S161T/Y434H/A524T/R766H + 1581/1582 Y17D/G54D/S161T/F519T/L727E/A738E +

1583/1584 Y17D/Y434H/A738E + 1595/1596 Y17D/L727E + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 1456, and defined as follows: “+” = production between 1.15- and 1.36-fold greater than that of the reference polypeptide.

Example 34

Sucrose Synthase Variants of SEQ ID NO: 1582

(345) Directed evolution of the sucrose synthase encoded by SEQ ID NO: 1581 was continued by constructing libraries of variant genes in which mutations associated with improved activity in earlier rounds of evolution were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below, to provide a seventh round (“Round 7”) of 87 engineered SuS variant polypeptides with activity toward the generation of ADP-glucose.

(346) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(347) Combinatorial libraries were screened using the following HTP enzyme-coupled assay. Pelleted *E. coli* cultures were lysed with 400 μ L of Tris-HCl, pH 7.5 with 1 mM magnesium sulfate and 0.5 mg/mL lysozyme and polymyxin B sulfate (PMBS), and cleared by centrifugation. Lysate was diluted ~90 $\times$ into Tris-HCl, pH 7.5. Then, 10 μ L diluted SuS lysate and 1 g/L GT SEQ ID NO: 2884 (Rd9BB) were combined in 100 μ L reaction volume with substrate loading of 4.5-7.5 mM rebaudioside A 97 and co-substrate loadings of 0.2-0.25 mM ADP (Sigma, >95%) and 30 mM sucrose (Sigma). The following

reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 55° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 2 h. The reaction described above was solubilized by adding 10 µL assay to 90-190 µL water, quenched by adding 10 µL solubilized assay to 90 µL acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 4.4-6.7× in water and steviol glycoside products were detected by RapidFire SPE-MS/MS, with the instrument and parameters described in Table 31.1.

(348) After analysis, 87 engineered SuS variant polypeptides that showed improved activity coupled with a GT on rebaudioside A were identified. The resulting engineered polypeptides are listed in Table 34.1. Shake-flask scale cultures were grown for protein characterization as described in Example 31 for variants with the amino acid mutations shown in Table 34.2.

(349) TABLE-US-00058 TABLE 34.1 SUS Round 7 Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 1582) RebD.sup.a 1599/1600 L562I ++
1601/1602 V270L/F452Y/P517A/1700F/L750M + 1603/1604 P517A/L562I/L750M + 1605/1606 V270L/I367V/F452Y/P517A/E613Q/I700F/ ++ L750M 1607/1608 V270L/R570H + 1609/1610 P517A + 1611/1612 V270L/I322V/P517A + 1613/1614 L562I + 1615/1616 P517A/T640N + 1617/1618 R71Q/R222L/R236S ++ 1619/1620 R14D/R137K/R356H/R745L + 1621/1622 R14D/R570H/R745L +++
1623/1624 R236S ++ 1625/1626 R356H/R570H +++ 1627/1628 R71Q/R158T/R222L/R356H + 1629/1630 R14D/R570H + 1631/1632 R137K/R319K/R570H + 1633/1634 R71Q/R319K/R356H/R606S + 1635/1636 S26E/A164E/A165E/A213E/R586E +++ 1637/1638 G117E/A608E ++ 1639/1640 A213E/H365E/P517E/A707E ++ 1641/1642 G117E/R158E/A213E/G332E/A608E + 1643/1644 S26E/G117E/H365E + 1645/1646 G117E/A164E/A707E + 1647/1648 A165E/Q311E ++ 1649/1650 G117E/Q311E/G332E ++ 1651/1652 G117E/A213E/H365E/P517E + 1653/1654 A253G/T519L + 1655/1656 T519L/L563V ++ 1657/1658 A253T/T519L + 1659/1660 A253V/T519G ++ 1661/1662 A253V/T519L + 1663/1664 A253T/T519L/L563V + 1665/1666 A253T/T519L/S635E ++ 1667/1668 A253T/L563V/S635R ++ 1669/1670 G157A/A253T/T519L ++ 1671/1672 A253T/T519L/S635D +++
1673/1674 A122E/R550Q + 1675/1676 A122D/R167E + 1677/1678 F160W/L282M/Q381S/R550M + 1679/1680 A122E/F160I/R167E/L282M/Q381H/R550Q/ ++ L636Q 1681/1682 A122D/F160M/L282M/R550S + 1683/1684 F160M/T161Q/L282M/R550Q ++ 1685/1686 A122D/R167E + 1687/1688 L282M/R550Q ++ 1689/1690 F160W/T161Q/L282M/R550M +++
1691/1692 F160M/L282M/R550M/L636Q +++ 1693/1694 A122D/L282M/Q381S/R550M +++ 1695/1696 A122D/R550M +++ 1697/1698 A122D +++ 1699/1700 A122E/L282M/R550Q +++
1701/1702 T161Q/L282M/R550Q/L636Q +++ 1703/1704 A122E/F160M/T161Q/L282M/Q381S/R550M/ +++ L636Q 1705/1706 R167E/R550Q ++ 1707/1708 A122D/L282M/R550Q + 1709/1710 R167E/L282M/L636Q + 1711/1712 L282M/Q381S/R550S +++
1713/1714 Q381S/R550S + 1715/1716 T161Q/Q381S/R550Q + 1717/1718 A122E/R167E/R550M +++
1719/1720 R550Q ++ 1721/1722 F160M/T161Q/L282M/Q381S/R550M ++ 1723/1724 A122E/F160W/Q381S/R550M/L636Q + 1725/1726 F160M/L282M/Q381S/R550M ++ 1727/1728 A122D/F160W/T161Q/L282M/R550Q + 1729/1730 F160M/T161Q +++ 1731/1732 Q381S/R550Q + 1733/1734 A122E/F160W/Q381S/R550Q ++ 1735/1736 A122D/F160W/T161Q/R167E/R550M ++
1737/1738 F160W/T161Q/L282M/R550Q +++ 1739/1740 A122E/F160W/T161Q/R167E/R550S ++ 1741/1742 A122D/F160W/R550M + 1743/1744 F160W/T161Q/R550Q/L636Q/A735V + 1745/1746 F160W/R167E/L282M/Q381S/L636Q ++ 1747/1748 A122E/F160M/R550Q/L636Q + 1749/1750 A122E/F160W/T161Q/L282M/Q381S/R550S + 1751/1752 A122E/F160W/L282M/Q381S + 1753/1754 A122D/F160W/L282M/Q381S/R550M + 1755/1756 A122E/F160W/T161Q/R550M/L636Q + 1757/1758 A122E/T161Q/R550Q + 1759/1760 A122E/L282M/R550M/L636Q ++ 1761/1762 A122E/Q381S/E706K ++ 1763/1764 F160W/Q381S/R550Q/L636Q/A681V +++ 1765/1766 R550M/L636Q + 1767/1768 A122E/L282M/R550S + 1769/1770 A122D/F160W/R550Q/L636Q + 1771/1772 F160M/L282M + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 1582, and are defined as follows: “+” = production at least that of the reference polypeptide, but less than 1.42-fold; “++” = at least 1.42-fold, but less than 1.60-fold; and “+++” = at least 1.60-fold increased production, relative to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to Rebaudioside A

(350) An experiment was performed to characterize the activity of the engineered round 7 SUS variants on sucrose and ADP to facilitate the formation of rebaudioside D from rebaudioside A. Shake flask powder (SFP) was added to a 100 μ L total reaction volume at 0.02 g/L concentration containing 50 mM potassium phosphate buffer, pH 6, 7.5 mM rebaudioside A 97, 30 mM sucrose, 0.2 mM ADP, and 1 g/L GT SEQ ID NO: 2884. The reaction was performed at 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 h. The reaction was solubilized by diluting 20 $\times$ into water, quenched by adding 10 μ L of the diluted reaction to 90 μ L acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 4.4 $\times$ in water and analyzed for steviol glycosides by RapidFire SPE-MS/MS, with the instrument and parameters described in Table 31.1. At least 3 of the variants had higher activities than SEQ ID NO: 1582. The levels of rebaudioside D produced from rebaudioside A by the variants relative to SEQ ID NO: 1582 are listed in Table 34.2. The variant with mutations F160W, Q381S, R550Q, L636Q, and A681V (SEQ ID NO: 1764), relative to SEQ ID NO: 1582, was selected for further directed evolution for the catalysis of the recycling reaction transferring a glucose from sucrose to ADP.

(351) TABLE-US-00059 TABLE 34.2 SUS Round 7 SFP Variants and RebD Levels SEQ ID NO: Amino Acid Differences (Relative Increased (nt/aa) to SEQ ID NO: 1582) RebD.sup.a 1599/1600 L562I + 1615/1616 P517A/T640N + 1763/1764 F160W/Q381S/R550Q/L636Q/A681V + .sup.a Levels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 1582, and are defined as follows: “+” = production between 1.1- and 1.25-fold greater activity, as compared to the reference polypeptide.

Example 35

Sucrose Synthase Variants of SEQ ID NO: 1764

(352) Directed evolution of the sucrose synthase encoded by SEQ ID NO: 1763 was continued by constructing libraries of variant genes in which mutations associated with improved activity in earlier rounds of evolution were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below, to provide an eighth and ninth round (“Round 8” and “Round 9”) of 24 engineered SuS variant polypeptides with activity toward the generation of ADP-glucose.

(353) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A for Round 8

(354) Combinatorial libraries were screened using the following HTP enzyme-coupled assay. Pelleted *E. coli* cultures were lysed with 400 μ L of Tris-HCl, pH 7.5 with 1 mM magnesium sulfate and 0.5 mg/mL lysozyme and polymyxin B sulfate (PMBS), and cleared by centrifugation. Lysate was diluted 15 $\times$ into Tris-HCl, pH 7.5. Then, 10 μ L diluted SuS lysate and 1 g/L GT SEQ ID NO: 2884 (Rd9BB) were combined in 100 μ L reaction volume with substrate loading of 15 mM rebaudioside A 97 and co-substrate loadings of 0.2 mM ADP (Sigma, >93%) and 45 mM sucrose (cane sugar). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 55° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 2 h. The reaction described above was solubilized by adding 10 μ L assay to 390 μ L water, quenched by adding 20 μ L solubilized assay to 180 μ L acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 3.3 $\times$ in water and steviol glycoside products were detected by RapidFire SPE-MS/MS, with the instrument and parameters described in Table 31.1. No substantially improved combinatorial variants were identified relative to SEQ ID NO: 1764, so another set of combinatorial and saturation mutagenesis libraries were generated using the same backbone, and this set was called “Round 9”.

(355) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A for Round 9

(356) Combinatorial libraries were screened using the following HTP enzyme-coupled assay. Pelleted *E. coli* cultures were lysed with 400 μ L of Tris-HCl, pH 7.5 with 1 mM magnesium sulfate and 0.5 mg/mL lysozyme and polymyxin B sulfate (PMBS), and cleared by centrifugation. Undiluted lysate (saturation mutagenesis library) or 32 $\times$ (Round 9 combinatorial libraries) were added to Tris-HCl, pH 7.5. Round 9 library lysates were pre-incubated at 62° C. for 1 h (combinatorial libraries) or 3.75 h (saturation mutagenesis library) to thermally challenge the enzymes. Then, 10 μ L diluted SuS lysate and 1 g/L GT SEQ ID NO: 3244 (Rd12B) were combined in 100 μ L reaction volume with substrate loading of 15 mM rebaudioside A 97, co-substrate loadings of 0.2 mM ADP (Sigma, >95%), 45 mM sucrose (cane sugar), and 9 mM fructose (Sigma). The following reaction conditions were used: 50 mM potassium phosphate

buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 2-3 h. The reaction described above was solubilized by adding 10 µL assay to 390 µL water, quenched by adding 20 µL solubilized assay to 180 µL acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 3.3× in water and steviol glycoside products were detected by RapidFire SPE-MS/MS, with the instrument and parameters described in Table 31.1. After analysis, 24 engineered SuS variant polypeptides that showed improved activity coupled with a GT on rebaudioside A were identified. The resulting engineered polypeptides are listed in Table 35.1. Shake-flask scale cultures were grown for protein characterization as described in Example 31 for variants with the amino acid mutations shown in Table 35.2.

(357) TABLE-US-00060 TABLE 35.1 SUS Round 9 Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 1764) RebD.sup.a 1773/1774 E536L +++ 1775/1776 H705P ++ 1777/1778 L580M + 1779/1780 H699F +++ 1781/1782 N347R/S532Y +++ 1783/1784 H705M + 1785/1786 A548P ++ 1787/1788 V681A +++ 1789/1790 L407I/R570H/V681A + 1791/1792 V270L/V681A ++ 1793/1794 G117E/A122D/V270L/L540M/V681A + 1795/1796 L407T/V681A ++ 1797/1798 E536L/H705M + 1799/1800 A548P/L580M + 1801/1802 N347R/E536L/A548P/H705P ++ 1803/1804 G181N/A548P/H705P +++ 1805/1806 A63I/E536L + 1807/1808 A548P/H705P + 1809/1810 L580M + 1811/1812 E536L/A548P/H699F + 1813/1814 G181N/E536L/A548P/H705M + 1815/1816 G181N/E536L/A548P ++ 1817/1818 E536L/A548P + 1819/1820 E536L ++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 1764, and are defined as follows: “+” = production at least 1.5-fold, but less than 2.4-fold; “++” = at least 2.4-fold, but less than 3.0-fold; and “+++” = at least 3.0-fold increased production, relative to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to Rebaudioside A

(358) An experiment was performed to characterize the activity of the engineered round 9 SUS variants on sucrose and ADP to facilitate the formation of rebaudioside D from rebaudioside A. Shake flask powder (SFP) was added to a 100 µL total reaction volume at 0.006-0.2 g/L concentration containing 50 mM potassium phosphate buffer, pH 6, 15 mM rebaudioside A (>97% purity), 45 mM sucrose, 9 mM fructose, 0.2 mM ADP, and 1 g/L GT SEQ ID NO: 3244. The reaction was performed at 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 2 h without pre-incubation, or the SFP was pre-incubated at 10× final concentration in pH 6 potassium phosphate buffer at 62° C. for 1 h prior to 60° C. reaction for 2 h. The reaction was solubilized by diluting 40× into water, quenched by diluting 10× into acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 3.3× in water and analyzed for steviol glycosides by RapidFire SPE-MS/MS, with the instrument and parameters described in Table 31.1. All 8 of the variants listed in Table 35.2 had higher activities than SEQ ID NO: 1764. The variant with the mutations G181N, A548P, H705P (SEQ ID NO: 1804), relative to SEQ ID NO: 1764, had the most beneficial mutations and was selected for further directed evolution for the catalysis of the recycling reaction transferring a glucosyl from sucrose to ADP.

(359) TABLE-US-00061 TABLE 35.2 SUS Round 9 SFP Variants and RebD Levels SEQ ID Increased Increased NO: Amino Acid Differences RebD, RebD, (nt/aa) (Relative to SEQ ID NO: 1764) 50° C..sup.a 60° C..sup.a 1787/1788 V681A ++ +++ 1791/1792 V270L/V681A ++ +++ 1793/1794 G117E/A122D/V270L/L540M/V681A ++ 1795/1796 L407T/V681A + +++ 1803/1804 G181N/A548P/H705P ++ +++ 1807/1808 A548P/H705P ++ +++ 1809/1810 L580M + + 1815/1816 G181N/E536L/A548P + +++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ IDNO: 1764, and are defined as follows: “+” = production at least 1.1-fold, but less than 1.7-fold; “++” = production at least 1.7-fold, but less than 2.3-fold; and “+++” = production at least 2.3-fold greater, as compared to the reference polypeptide.

Example 36

Sucrose Synthase Variants of SEQ ID NO: 1804

(360) Directed evolution of the sucrose synthase encoded by SEQ ID NO: 1803 was continued by constructing libraries of variant genes in which mutations associated with improved activity in earlier rounds of evolution were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP)

assay described below, to provide a tenth round ("Round 10") of 82 engineered SuS variant polypeptides with increased activity toward the generation of ADP-glucose.

(361) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(362) Libraries were screened using the following HTP enzyme-coupled assay. Pelleted *E. coli* cultures were lysed with 400 μ L of Tris-HCl, pH 7.5, with 1 mM magnesium sulfate and 0.5 mg/mL lysozyme and polymyxin B sulfate (PMBS) and cleared by centrifugation. Lysate was diluted 20 $\times$ into potassium phosphate buffer, pH 6.0, and pre-incubated for 15 minutes at 64 $^{\circ}$ C. (combinatorial libraries) or 65 $^{\circ}$ C. (saturation mutagenesis library). Then, 10 μ L diluted SuS lysate and 0.5-1 g/L GT SEQ ID NO: 3696 or 3502, respectively, were used in 100 μ L reaction volume with 15 mM rebaudioside A (>97% purity), 0.2 mM ADP (Sigma, >93%), 45 mM sucrose (cane sugar), and 9 mM fructose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60 $^{\circ}$ C. in a Thermotron $^{\circ}$ titre-plate shaker with 300 RPM shaking for 3 h. The reaction described above was solubilized by diluting 40 $\times$ in water, quenched by diluting 10 $\times$ in acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 3.3 $\times$ in water and steviol glycoside products were detected by RapidFire SPE-MS/MS, with the instrument and parameters described in Table 31.1. After analysis, the engineered SuS variant polypeptides that showed improved activity coupled with a GT on rebaudioside A were identified and are listed in Table 36.1. Shake-flask scale cultures were grown for protein characterization as described in Example 31 for variants with the amino acid mutations shown in Table 36.2.

(363) TABLE-US-00062 TABLE 36.1 SUS Round 10 Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 1804) RebD.sup.a 1821/1822

N347R/Q550I/V681A + 1823/1824 N347R/S532Y/Q550I/T640N/V681A/H699F + 1825/1826

N347R/V681A + 1827/1828 N347R/Y434H/P517A/L562I/T640N/V681A + 1829/1830

N347R/P517A/S532Y/V681A + 1831/1832 N347R/Y434H/V681A + 1833/1834 E536L/L580M/V681A

+ 1835/1836 N347R/Q550I/L580M/V681A + 1837/1838 N347R/E536L/L562I/V681A + 1839/1840

P517A/V681A + 1841/1842 S532Y/V681A + 1843/1844 Y434H/P517A/S532Y/V681A + 1845/1846

N347R/Y434H/S532Y/L562I/T640N/V681A + 1847/1848 L562I/V681A + 1849/1850

N347R/Y434H/Q550I/L562I/V681A + 1851/1852 A389G ++ 1853/1854 S539A +++ 1855/1856 L433K

++ 1857/1858 R71Q +++ 1859/1860 E87L ++ 1861/1862 P13K ++ 1863/1864 D765A ++ 1865/1866

Q37R ++ 1867/1868 R90H ++ 1869/1870 D60L + 1871/1872 S531T ++ 1873/1874 Q37G + 1875/1876

S539R +++ 1877/1878 D17R +++ 1879/1880 P535S + 1881/1882 R606H ++ 1883/1884 V98L +

1885/1886 L388K ++ 1887/1888 P57W +++ 1889/1890 P535H + 1891/1892 P13H + 1893/1894 E727K

+ 1895/1896 V85H + 1897/1898 R606A ++ 1899/1900 A789R + 1901/1902 A789N + 1903/1904 A18G +

1905/1906 P13N + 1907/1908 K415H + 1909/1910 S531R +++ 1911/1912 R711K +++ 1913/1914

R606Q ++ 1915/1916 R606L + 1917/1918 R606M ++ 1919/1920 R606I + 1921/1922 D52P +++

1923/1924 D52W + 1925/1926 S30H + 1927/1928 E87H ++ 1929/1930 A118N ++ 1931/1932 Q769K

+++ 1933/1934 E129G +++ 1935/1936 D52R ++ 1937/1938 E99I + 1939/1940 E129A + 1941/1942

L433P ++ 1943/1944 E748T + 1945/1946 A164T + 1947/1948 A707G ++ 1949/1950 P57R ++

1951/1952 A608P ++ 1953/1954 A118G + 1955/1956 P535A + 1957/1958 D180P + 1959/1960 Q769R +

1961/1962 N183P +++ 1963/1964 R71G ++ 1965/1966 R606V ++ 1967/1968 E129T +++ 1969/1970

A164S ++ 1971/1972 L388R ++ 1973/1974 H365W + 1975/1976 D52G +++ 1977/1978 G589S +++

1979/1980 E738S +++ 1981/1982 D765S +++ 1983/1984 S531A ++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 1804, and are defined as follows:

"+" = production at least 1.3-fold, but less than 2-fold; "++" = at least 2-fold, but less than 2.36-fold; and

"+++" = at least 2.36-fold increased production, relative to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to Rebaudioside A

(364) An experiment was performed to characterize the activity of the engineered round 10 SUS variants on sucrose and ADP to facilitate the formation of rebaudioside D from rebaudioside A. Shake flask powder (SFP) was made up to 0.06-2 g/L in potassium phosphate buffer, pH 6, and an aliquot was pre-incubated at 64 $^{\circ}$ C. in a thermocycler for 15 minutes. Then, 10 μ L of these SFP dilutions, either pre-incubated or not pre-incubated, were added to a 100 μ L total reaction volume containing 50 mM potassium phosphate buffer, pH 6, 15 mM rebaudioside A (>97% purity), 45 mM sucrose, 9 mM fructose, 0.2 mM ADP, and 1 g/L GT SEQ ID NO: 3696. The reactions were performed at 55 $^{\circ}$ C. for SFP samples

that were not pre-incubated at 60° C. for pre-incubated SFP in a Thermotron® titre-plate shaker with 300 RPM shaking for 3 h. The reaction was solubilized by diluting 40× into water, quenched by diluting 10× into acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 3.3× in water and analyzed for steviol glycosides by RapidFire SPE-MS/MS, with the instrument and parameters described in Table 31.1. All 7 of the variants listed in Table 36.2 had higher activities than SEQ ID NO: 1804 under the pre-incubated condition, and all but one also had higher activity under the 55° C. condition. The variant with the mutations P517A and V681A (SEQ ID NO: 1840) was the most improved under both conditions relative to SEQ ID NO: 1804, so it was selected for further directed evolution for the catalysis of the recycling reaction transferring a glucose from sucrose to ADP.

(365) TABLE-US-00063 TABLE 36.2 SUS Round 10 SFP Variants and RebD Levels SEQ ID Amino Acid Differences Increased Increased NO: (Relative to RebD RebD (nt/aa) SEQ ID NO: 1804) 55° C..sup.a 60° C..sup.a 1821/1822 N347R/Q550I/V681A + ++ 1823/1824 N347R/S532Y/Q550I/ - + T640N/V681A/H699F 1827/1828 N347R/Y434H/P517A/L562I/ + ++ T640N/V681A 1831/1832 N347R/Y434H/V681A + ++ 1837/1838 N347R/E536L/L562I/V681A + ++ 1839/1840 P517A/V681A ++ +++ 1845/1846 N347R/Y434H/S532Y/ + ++ L562I/T640N/V681A .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 1804, and are defined as follows: “-” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.5-fold; “++” = production at least 1.5-fold, but less than 2-fold; and “+++” = production at least 2-fold greater, as compared to the reference polypeptide.

Example 37

Sucrose Synthase Variants of SEQ ID NO: 1840

(366) Directed evolution of the sucrose synthase encoded by SEQ ID NO: 1839 was continued by constructing libraries of variant genes in which mutations associated with improved activity in earlier rounds of evolution were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below, to provide an eleventh round (“Round 11”) of 167 engineered SuS variant polypeptides with increased activity toward the generation of ADP-glucose.

(367) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(368) Libraries were screened using the following HTP enzyme-coupled assay. Pelleted *E. coli* cultures were lysed with 400 µL of Tris-HCl, pH 7.5, with 1 mM magnesium sulfate and 0.5 mg/mL lysozyme and polymyxin B sulfate (PMBS), and cleared by centrifugation. Lysate was diluted 10-20× into potassium phosphate buffer, pH 6.0, and pre-incubated for 15 minutes at 66° C. Then, 10 µL diluted, pre-incubated SuS lysate and 0.5 g/L GT SEQ ID NO: 3696 or 3956, were used in 100 µL reaction volume with 15 mM rebaudioside A (>97% purity), 0.2 mM ADP (Amresco, >93%), 45 mM sucrose (cane sugar), and 9 mM fructose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C., in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. The reaction described above was solubilized by diluting 40× in water, quenched by diluting 5-10× in acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted in water to ~10 µM steviol glycosides and steviol glycoside products were detected by RapidFire SPE-MS/MS with the instrument and parameters described in Table 31.1. After analysis, the engineered SuS variant polypeptides that showed improved activity coupled with a GT on rebaudioside A were identified and are listed in Tables 37.1 and 37.2. Shake-flask scale cultures were grown for protein characterization as described in Example 31 for variants with the amino acid mutations shown in Table 37.3.

(369) TABLE-US-00064 TABLE 37.1 SUS Round 11 Combinatorial Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 1840) RebD.sup.a 1985/1986 D17R/E129T/G589S + 1987/1988 D17R/G589S/E738S ++ 1989/1990 D17R/E129T/G589S/E738S + 1991/1992 D17R/E129T/G589S/E738S/D765S +++ 1993/1994 D52P/E129T/G589S/E738S/D765S +++ 1995/1996 D52P/E129T/E738S + 1997/1998 D52P/A118N/E129T/D765S + 1999/2000 D17R/D52P/E87H/G589S/E738S/D765S +++ 2001/2002 E87H/E129T/L388K/G589S + 2003/2004 D17R/D52G/E129T/R653H/E738S/D765S + 2005/2006 D17R/E738S/D765S + 2007/2008 D17R/D52G/E87H/E129T/L388K/G589S/E738S ++ 2009/2010 D17R/L388K/G589S/E738S ++ 2011/2012 D17R/D52P/E129T/E738S ++ 2013/2014 D17R/D52P/E129T/G589S/D765S ++ 2015/2016 D17R/D52P/E87H/E129G/L388K/G589S +++ 2017/2018 D52G/A118N/L388K/G589S/E738S +++

2019/2020 D52P/E87H/A118N/E129T/L388K/D765S ++ 2021/2022
 D52G/G84A/E129T/L388K/E738S/D765S + 2023/2024
 D17R/D52P/A118N/E129T/S265T/G589S/D765S ++ 2025/2026
 D17R/D52P/A118N/L388K/G589S/E738S/D765S +++ 2027/2028 D17R/D52G/E129G/G589S/E738S +
 2029/2030 D17R/D52G/E129T/L388K/G589S/E738S +++ 2031/2032
 D17R/E87H/A118N/L388K/E738S + 2033/2034 D52G/L388K/E738S/D765S + 2035/2036
 D17R/D52G/A118N/L388K/E738S/D765S + 2037/2038 D17R/D52P/E129G/E738S/D765S + 2039/2040
 D17R/D52P/E87H/A118N/E129T/G589S/E738S/D765S + 2041/2042
 D52P/E87H/A118N/L388K/G589S/E738S ++ 2043/2044 G84A/E129T/G589S/E738S/D765S +
 2045/2046 D17R/D52G/G589S/D765S + 2047/2048 D17R/D52P/G589S/E738S/D765S +++ 2049/2050
 D17R/D52G/E87H/E129T/E738S + 2051/2052 D17R/D52G/E87H/A118N/E129T/G589S/D765S ++
 2053/2054 D17R/D52P/A118N/E129T/G589S/E738S/D765S ++ 2055/2056 D52G/E87H/E738S +
 2057/2058 D17R/D52G/E87H/A118N/E129G/G589S/E738S + 2059/2060
 D17R/D52P/E129G/L388K/G589S/E738S/D765S ++ 2061/2062
 D17R/D52P/E87H/A118N/E129T/L388K/G589S/E738S/D765S + 2063/2064
 D17R/D52P/L388K/G589S/E738S/D765S ++ 2065/2066
 D17R/D52P/E87H/E129T/L388K/G589S/E738S + 2067/2068 D52G/E87H/E129T/L388K/E738S/D765S
 + 2069/2070 D17R/D52G/E87H/L388K/G589S/D765S + 2071/2072
 D17R/D52P/A118N/L388K/G589S/E738S ++ 2073/2074 D17R/G589S/D765S + 2075/2076
 D52G/E87H/A118N/G589S/E738S/D765S ++ 2077/2078 D52G/E738S/D765S + 2079/2080
 D52P/G589S/E738S/D765S ++ 2081/2082 D52P/E87H/E129G/D765S +++ 2083/2084
 D17R/E129T/E738S/D765S ++ 2085/2086 D17R/D52P/L388K/G589S/E738S +++ 2087/2088
 D17R/A118N/E129T/L388K/E738S/D765S + 2089/2090 D52P/A118N/E129T/G589S/E738S ++
 2091/2092 D52G/A118N/L388K/E738S/D765S + 2093/2094 G589S/E738S/D765S ++ 2095/2096
 D17R/D52G/A118N/E129T/L388K/G589S/E738S/D765S ++ 2097/2098 D52G/E129G/G589S/D765S +
 2099/2100 D52P/G589S/E738S +++ 2101/2102 D52G/G589S/E738S/D765S +++ 2103/2104
 D52P/E87H/G589S/E738S +++ 2105/2106 D17R/D52P/A118N/E129T/E738S/D765S + 2107/2108
 D17R/D52G/E129T/G589S + 2109/2110 D17R/E87H/E129T/L388K/E738S + 2111/2112

E87H/A118N/E129T/D765S + .sup.aLevels of increased production were determined relative to the
 reference polypeptide of SEQ ID NO: 1840, and defined as follows: “+” = production at least 11-fold, but
 less than 14.9-fold; “++” = at least 14.9-fold, but less than 18.3-fold; and “+++” = at least 18.3-fold
 increased production, relative to the reference polypeptide.

(370) TABLE-US-00065 TABLE 37.2 SUS Round 11 Saturation Mutagenesis Variants and RebD Levels
 Amino Acid Differences SEQ ID NO: (Relative to SEQ ID Increased (nt/aa) NO: 1840) RebD.sup.a
 2113/2114 R14K ++ 2115/2116 I480V ++ 2117/2118 M263Y ++ 2119/2120 D218V + 2121/2122 H154A
 +++ 2123/2124 G360R + 2125/2126 E534W/E739K +++ 2127/2128 G603H + 2129/2130 E59N/D72N
 ++ 2131/2132 Q33S + 2133/2134 V134P ++ 2135/2136 P81I +++ 2137/2138 S393H ++ 2139/2140
 V134A ++ 2141/2142 E390M + 2143/2144 P364S + 2145/2146 Y434R + 2147/2148 W212Y +
 2149/2150 S15I + 2151/2152 V316T + 2153/2154 P530F +++ 2155/2156 H92G + 2157/2158 S175G
 +++ 2159/2160 G652L +++ 2161/2162 E534R +++ 2163/2164 G603S +++ 2165/2166 S26I + 2167/2168
 H154R + 2169/2170 E105S + 2171/2172 E59R + 2173/2174 D130Y +++ 2175/2176 F542W ++
 2177/2178 S26E + 2179/2180 S15P + 2181/2182 W79Y ++ 2183/2184 P81L +++ 2185/2186 A165L +
 2187/2188 P81G ++ 2189/2190 I362E + 2191/2192 G603A ++ 2193/2194 E534W +++ 2195/2196
 G652S ++ 2197/2198 D218T ++ 2199/2200 D218A ++ 2201/2202 L20M + 2203/2204 R319S +
 2205/2206 S15A + 2207/2208 E59W + 2209/2210 E59S + 2211/2212 G652T +++ 2213/2214 L241T ++
 2215/2216 D218N ++ 2217/2218 H154E + 2219/2220 Y434G ++ 2221/2222 Q46V + 2223/2224 Q46R +
 2225/2226 W79H ++ 2227/2228 G652K +++ 2229/2230 Q33H/H154C +++ 2231/2232 M263S +
 2233/2234 A165I + 2235/2236 V93T + 2237/2238 G603E +++ 2239/2240 Q349R ++ 2241/2242 I480P
 ++ 2243/2244 E534L +++ 2245/2246 E534G +++ 2247/2248 E59C + 2249/2250 A213V + 2251/2252
 Q33P + 2253/2254 Q46T + 2255/2256 E534K +++ 2257/2258 E59A ++ 2259/2260 Q46I + 2261/2262
 L58M + 2263/2264 E534T +++ 2265/2266 G50R + 2267/2268 G652R +++ 2269/2270 S256G ++
 2271/2272 A97V/H154S ++ 2273/2274 V316H ++ 2275/2276 V24S + 2277/2278 G360E + 2279/2280
 E498Q + 2281/2282 G360D + 2283/2284 Q349T ++ 2285/2286 S175T + 2287/2288 G603Q ++

2289/2290 A18V/1362A ++ 2291/2292 H154S + 2293/2294 D104T + 2295/2296 S26A ++ 2297/2298 Q46G + 2299/2300 Q33L + 2301/2302 D218S ++ 2303/2304 A165T + 2305/2306 Q349D + 2307/2308 D54M + 2309/2310 S26T + 2311/2312 E498L + 2313/2314 M185L + 2315/2316 D218Q ++ 2317/2318 E59V ++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 1840, and defined as follows: “+” = production at least 1.2-fold, but less than 1.62-fold; “++” = at least 1.62-fold, but less than 2.4-fold; and “+++” = at least 2.4-fold increased production, relative to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to Rebaudioside A

(371) An experiment was performed to characterize the activity of the engineered round 11 SUS variants on sucrose and ADP to facilitate the formation of rebaudioside D from rebaudioside A. Shake flask powder (SFP) was made up to 0.03-1 g/L in potassium phosphate buffer, pH 6, and an aliquot was pre-incubated at 66° C. in a thermocycler for 15 minutes. Then, 10 µL of these SFP dilutions, either pre-incubated or not pre-incubated, were added to a 100 µL total reaction volume containing 50 mM potassium phosphate buffer, pH 6, 15 mM rebaudioside A (>97% purity), 45 mM sucrose, 9 mM fructose, 0.2 mM ADP, and 0.5 g/L GT SEQ ID NO: 3956. The reactions were performed at 55° C. for SFP samples that were not pre-incubated and at 60° C. for pre-incubated SFP in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. The reaction was solubilized by diluting 40× into water, quenched by diluting 10× into acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 3.3× in water and analyzed for steviol glycosides by RapidFire SPE-MS/MS with the instrument and parameters described in Table 31.1. All 8 of the variants listed in Table 37.3 had higher activities than SEQ ID NO: 1840 under the pre-incubated condition, and all but three also had higher activity under the 55° C. condition. The variant with the mutations D17R, D52P, L388K, G589S, E738S, and D765S (SEQ ID NO: 2064), which was improved under both conditions relative to SEQ ID NO: 1840, was selected for further directed evolution for the catalysis of the recycling reaction transferring a glucose from sucrose to ADP.

(372) TABLE-US-00066 TABLE 37.3 SUS Round 11 SFP Variants and RebD Levels Increased Increased SEQ ID NO: Amino Acid Differences RebD, RebD, (nt/aa) (Relative to SEQ ID NO: 1840) 55° C..sup.a 60° C..sup.a 1999/2000 D17R/D52P/E87H/G589S/E738S/D765S – ++ 2017/2018

D52G/A118N/L388K/G589S/E738S – ++ 2025/2026 D17R/D52P/A118N/L388K/G589S/E738S/D765S – ++ 2047/2048 D17R/D52P/G589S/E738S/D765S + +++ 2059/2060

D17R/D52P/E129G/L388K/G589S/E738S/D765S + +++ 2063/2064

D17R/D52P/L388K/G589S/E738S/D765S + +++ 2099/2100 D52P/G589S/E738S + ++ 2103/2104

D52P/E87H/G589S/E738S + ++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 1840, and defined as follows: “–” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 2-fold; “++” = production at least 2-fold, but less than 3.2-fold; and “+++” = production at least 3.2-fold greater than that of the reference polypeptide.

Example 38

Sucrose Synthase Variants of SEQ ID NO: 2064

(373) Directed evolution of the sucrose synthase encoded by SEQ ID NO: 2063, was continued by constructing libraries of variant genes in which mutations associated with improved activity in earlier rounds of evolution were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below, to provide a twelfth round (“Round 12”) of 92 engineered SuS variant polypeptides with increased activity toward the generation of ADP-glucose.

(374) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(375) Libraries were screened using the following HTP enzyme-coupled assay. Pelleted *E. coli* cultures were lysed with 400 µL of Tris-HCl, pH 7.5, with 1 mM magnesium sulfate and 0.5 mg/mL lysozyme and polymyxin B sulfate (PMBS), and cleared by centrifugation. Lysate was diluted 10× into potassium phosphate buffer, pH 6.0, and pre-incubated for 15 minutes at 68° C. Then, 10 µL diluted, pre-incubated SuS lysate and 0.5 g/L GT SEQ ID NO: 3956 or 4256 were used in 100 µL reaction volume with 15 mM rebaudioside A (>97% purity), 0.2 mM ADP (Amresco, >93%), 37.5 mM sucrose (cane sugar), and 9 mM

fructose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. The reaction described above was solubilized by diluting 40× in water, quenched by diluting 5× in acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 7.5× in water and steviol glycoside products were detected by RapidFire SPE-MS/MS with the instrument and parameters described in Table 31.1. After analysis, the engineered SuS variant polypeptides that showed improved activity coupled with a GT on rebaudioside A were identified and are listed in Tables 38.1 and 38.2. Shake-flask scale cultures were grown for protein characterization as described in Example 31 for variants with the amino acid mutations shown in Table 38.3.

(376) TABLE-US-00067 TABLE 38.1 SUS Round 12 Combinatorial Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 2064) RebD.sup.a 2319/2320 P57W/S531R/S532Y/E536L/L562I ++ 2321/2322 E129T/Q550I/L562I + 2323/2324 R71Q/E129T/D180P/Y434H/S532Y/E536L/S539R/R711K/A789N + 2325/2326 E87H/N347R/E536L/S539R/Q550I + 2327/2328 P57W/N347R/Y434H/S531R/S532Y/S539R/A789N + 2329/2330 P57W/E129G/E536L/R606M/A789N + 2331/2332 R71Q/E129T/R606M + 2333/2334 N347R/S531R/Q550I/R711K ++ 2335/2336 P57W/E87H/N347R/L562I + 2337/2338 N347R/E536L/L562I/R606M/E612A + 2339/2340 E129T/E536L/S539R/L562I + 2341/2342 S531R/S532Y/E536L/Q550I/L562I/R606M/A789N + 2343/2344 E129T/D180P/Y434H/L562I/R711K/A789N + 2345/2346 P57W/R90H/E129G/L562I + 2347/2348 P57W/R71Q/E129G/S531R/S532Y/E536L/S539R/R606H ++ 2349/2350 R90H/E129T/S539R/Q550I/R606H + 2351/2352 R71Q/E129T/S531R + 2353/2354 P57W/L562I/R606M/R711K ++ 2355/2356 Y434H/S531R/S539R/Q550I/L562I/R711K + 2357/2358 P57W/E129G/S531R/S539R/L562I/A789N + 2359/2360 P57W/E129T/N347R/E536L/Q550I/L562I/R711K/A789N + 2361/2362 P57W/E129T/R606M + 2363/2364 E129T/D180P/R606M/R711K/A789N + 2365/2366 P57W/R90H/E129T/L562I/R711K ++ 2367/2368 E129G/Q550I + 2369/2370 P57W/R71Q/L562I/R606M/R711K/A789N + 2371/2372 E87H/G189D/S532Y/E536L/L562I/R711K/A789N + 2373/2374 L562I/R711K ++ 2375/2376 P57W/R71Q/E129T/D180P/Y434H/E536L/L562I ++ 2377/2378 P57W/R71Q/E87H/N347R/Y434H/L562I/R606H +++ 2379/2380 R71Q/E536L/S539R/L562I +++ 2381/2382 P57W/E129T/N347R/Q550I/R711K ++ 2383/2384 P57W/L96Q/E129T/D180P/S531R/S532Y/Q550I/L562I ++ 2385/2386 E87H/N347R/S531R/R606M/A789N + 2387/2388 D180P/S532Y + 2389/2390 S531R/S532Y/E536L/S539R/L562I/R711K/A789N +++ 2391/2392 R71Q/N347R/S532Y/Q550I/L562I/R711K +++ 2393/2394 P57W/D180P/L562I/R606H/E612A + 2395/2396 P57W/R71Q/A789N + 2397/2398 P57W/Y434H/Q550I/L562I/R606H/E612A/A789N ++ 2399/2400 P57W/E87H/D180P/S531R/S532Y/L562I/R606M/E612A/R711K ++ 2401/2402 S531R/S532Y/L562I/R606H/R711K ++ 2403/2404 E129G/L562I/R606M/R711K +++ 2405/2406 P57W/R71Q/E129T/Y434H/S531R/E536L/L562I +++ 2407/2408 N347R/Q550I/L562I/R606H +++ 2409/2410 E129T/N347R/L562I ++ 2411/2412 P57W/D180P/L562I ++ 2413/2414 E129T/S539R/L562I/A789N ++ 2415/2416 N347R/E536L/S539R/Q550I/R711K/A789N ++ 2417/2418 D180P/Q550I/R606M +++ 2419/2420 Y434H/Q550I +++ 2421/2422 P57W/E129T/N347R/S531R/S532Y/S539R/L562I/R711K/A747V + 2423/2424 R71Q/N347R/E536L/L562I/E612A/A789N ++ 2425/2426 R71Q/E129T/D180P/N347R/S531R/S539R/Q550I + 2427/2428 S532Y/S539R/Q550I +++ 2429/2430 E129G/Q550I/L562I +++ 2431/2432 P57W/L562I/R711K + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 2064, and defined as follows: “+” = production at least 15-fold, but less than 38.25-fold; “++” = at least 38.25-fold, but less than 50.3-fold increased production; and “+++” = at least 50.3-fold increased production, relative to the reference polypeptide.

(377) TABLE-US-00068 TABLE 38.2 SUS Round 12 Saturation Mutagenesis Variants and RebD Levels Amino Acid Differences SEQ ID NO: (Relative to SEQ ID Increased (nt/aa) NO: 2064) RebD.sup.a 2433/2434 V641L + 2435/2436 L688H + 2437/2438 Q487K +++ 2439/2440 A41K ++ 2441/2442 F684M ++ 2443/2444 L186V + 2445/2446 I91G + 2447/2448 G112Q ++ 2449/2450 V89M + 2451/2452 Q487T

++ 2453/2454 F684G + 2455/2456 F684T + 2457/2458 A25T/G112W ++ 2459/2460 P226V + 2461/2462 L764R +++ 2463/2464 L688G ++ 2465/2466 L688A +++ 2467/2468 V763L + 2469/2470 F684H ++ 2471/2472 S330A + 2473/2474 R21Q + 2475/2476 G112R + 2477/2478 D259G + 2479/2480 V200A + 2481/2482 Q487V + 2483/2484 Q487R ++ 2485/2486 L688Q +++ 2487/2488 P318A + 2489/2490 G485A ++ 2491/2492 G485S ++ 2493/2494 V89L + 2495/2496 I91C ++ 2497/2498 L688F +++ 2499/2500 Q487I +++ 2501/2502 S674A + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 2064, and defined as follows: “+” = production at least 2-fold, but less than 3.7-fold; “++” = at least 3.7-fold, but less than 6.4-fold; and “+++” = at least 6.4-fold increased production, relative to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to Rebaudioside A

(378) An experiment was performed to characterize the activity of the engineered round 12 SUS variants on sucrose and ADP to facilitate the formation of rebaudioside D from rebaudioside A. Shake flask powder (SFP) was made up to 0.03-1 g/L in potassium phosphate buffer, pH 6, and an aliquot was pre-incubated at 68° C. in a thermocycler for 15 minutes. Then, 10 µL of these SFP dilutions, either pre-incubated or not pre-incubated, were added to a 100 µL total reaction volume containing 50 mM potassium phosphate buffer, pH 6, 15 mM rebaudioside A (>97% purity), 37.5 mM sucrose, 9 mM fructose, 0.2 mM ADP, and 0.5 g/L GT SEQ ID NO: 4256. The reactions were performed at 55° C. for SFP samples that were not pre-incubated and at 60° C. for pre-incubated SFP in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. The reaction was solubilized by diluting 40× into water, quenched by diluting 10× into acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 3.3× in water and analyzed for steviol glycosides by RapidFire SPE-MS/MS with the instrument and parameters described in Table 31.1. A further thermostability characterization was conducted with the clarified shake flask lysates prior to lyophilization as follows: lysates were diluted 400× in buffer and incubated in a thermocycler at a gradient of 55-70° C. for 16-18 h. To determine the % activity remaining, the pre-incubated lysates were assayed as described above with either stevioside or rebaudioside D and 4 h incubation at 55° C. The percent activity remaining was expressed as activity at high temperature divided by activity at lowest pre-incubated temperature. All 7 of the variants listed in Table 38.3 had higher activities than SEQ ID NO: 2064 under the pre-incubated conditions, and two also were less deleterious under the 55° C. condition. The variant with the mutations P57W, L562I, and R711K (SEQ ID NO: 2432), which was improved under all conditions relative to SEQ ID NO: 2064 and contained the top mutation from the round 10 saturation mutagenesis screen, was selected for further directed evolution for the catalysis of the recycling reaction transferring a glucose from sucrose to ADP.

(379) TABLE-US-00069 TABLE 38.3 SUS Round 12 SFP Variants and RebD Levels Increased Increased SEQ ID NO: Amino Acid Differences RebD, RebD, % Activity (nt/aa) (Relative to SEQ ID NO: 2064) 55° C..sup.a 60° C..sup.a Remaining.sup.b 2333/2334 N347R/S531R/Q550I/R711K – ++ ++ 2373/2374 L562I/R711K – +++ + 2389/2390 S531R/S532Y/E536L/S539R/L562I/ – ++ ++ R711K/A789N 2405/2406 P57W/R71Q/E129T/Y434H/S531R/ – ++ +++ E536L/L562I 2427/2428 S532Y/S539R/Q550I + +++ ++ 2429/2430 E129G/Q550I/L562I – ++ ++ 2431/2432 P57W/L562I/R711K + +++ ++

.sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 2064, and defined as follows: “–” = production less than 0.8-fold that of the reference polypeptide; “+” = production at least 0.8-fold, but less than 1.2-fold; “++” = production at least 1.2-fold, but less than 4.7-fold; and “+++” = production at least 4.7-fold increased, relative to that of the reference polypeptide. .sup.bThe percent of activity remaining for each variant was determined following 16-18 h pre-incubation at 65° C. relative to the production of each variant following pre-incubation at 55° C. and is defined as follows: “–” = less than 55% of activity remained following pre-incubation at 55° C.; “+” = at least 55% activity, but less than 65% activity remained; “++” = at least 65% activity, but less than 75% activity remained; and “+++” = at least 75% activity remained.

Example 39

Sucrose Synthase Variants of SEQ ID NO: 2432

(380) Directed evolution of the sucrose synthase encoded by SEQ ID NO: 2431, was continued by constructing libraries of variant genes in which mutations associated with improved activity in earlier rounds of evolution were recombined and in which certain structural features were subjected to saturation

mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below, to provide a thirteenth round ("Round 13") of 46 engineered SuS variant polypeptides with increased activity toward the generation of ADP-glucose.

(381) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(382) Libraries were screened using the following HTP enzyme-coupled assay. Pelleted *E. coli* cultures were lysed with 400 μ L of Tris-HCl, pH 7.5, with 1 mM magnesium sulfate and 0.5 mg/mL lysozyme and polymyxin B sulfate (PMBS) and cleared by centrifugation. For the combinatorial library, lysate was diluted 10 $\times$ into potassium phosphate buffer, pH 6.0, and pre-incubated for 15 minutes at 73 $^{\circ}$ C. For the saturation mutagenesis library, lysate was diluted 20 $\times$ into potassium phosphate buffer, pH 6.0, and pre-incubated for 17.5 hours at 62 $^{\circ}$ C. Then, 10 μ L diluted, pre-incubated SuS lysate and 0.5 g/L GT SEQ ID NO: 4256 or 4550 were used in 100 μ L reaction volume with 15 mM rebaudioside A (>97% purity), 0.2 mM ADP (Amresco, >93%), 37.5 mM sucrose (cane sugar), and 9 mM fructose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60 $^{\circ}$ C. in a Thermotron $^{\circ}$ titre-plate shaker with 300 RPM shaking for 3-4 h. The reaction described above was solubilized by diluting 40 $\times$ in water, quenched by diluting 5 $\times$ in acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 7.5 $\times$ in water and steviol glycoside products were detected by RapidFire SPE-MS/MS with the instrument and parameters described in Table 31.1. After analysis, the engineered SuS variant polypeptides that showed improved activity coupled with a GT on rebaudioside A were identified and are listed in Tables 39.1 and 39.2. Shake-flask scale cultures were grown for protein characterization as described in Example 31 for variants with the amino acid mutations shown in Table 39.3.

(383) TABLE-US-00070 TABLE 39.1 SUS Round 13 Combinatorial Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 2432) RebD.sup.a 2503/2504 Q33H/H154A/I480P/P530F/E534W/G603Q/R606M + 2505/2506

L58M/E59A/W79H/S175G/I480P/E534W/Q550I/G652R + 2507/2508

Q33H/L58M/E59A/P81G/D130Y/I480V/P530F/E534W/Q550I/G652K ++ 2509/2510

Q33H/L47P/E59A/P81L/S175G/P530F/E534W/Q550I/R606M +++ 2511/2512

Q33H/D130Y/P530F/E534W/Q550I + 2513/2514 Q33H/L58M/E59A/I480P/P530F/E534W/Q550I ++

2515/2516 E59A/H154A/P530F/E534W/Q550I + 2517/2518

Q33H/W79H/P81L/S175G/P530F/E534W/G603Q + 2519/2520

Q33H/L58M/H154A/I480P/E534W/Q550I/G603A/R606M ++ 2521/2522

Q33H/P81G/D130Y/I480V/P530F/E534W/Q550I ++ 2523/2524

Q33H/W79H/P81G/S175G/P530F/E534W + 2525/2526 P81G/I480V/P530F/E534W/Q550I + 2527/2528

Q33H/E59A/I480P/P530F/E534W/Q550I/R606M +++ 2529/2530 Q33H/H154A/E534W + 2531/2532

W79H/P81I/I480P/P530F/E534W/Q550I/G603E/R606M/G652R ++ 2533/2534

Q33H/W79H/H154A/I480P/P530F/E534W/Q550I ++ 2535/2536 Q33H/P530F/E534W/Q550I +

2537/2538 D130Y/I480V/P530F/E534W/Q550I/G603Q/R606M + 2539/2540

Q33H/P81I/S175G/P530F/E534W/F542W/Q550I/G652R +++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 2432, and defined as follows: "+" = production at least 1.2-fold, but less than 5.2-fold; "++" = at least 5.2-fold, but less than 30-fold; and "+++" = at least 30-fold increased production, relative to the reference polypeptide.

(384) TABLE-US-00071 TABLE 39.2 SUS Round 13 Saturation Mutagenesis Variants and RebD Levels Amino Acid Differences SEQ ID NO: (Relative to SEQ ID Increased (nt/aa) NO: 2432) RebD.sup.a

2541/2542 L561I +++ 2543/2544 D42H + 2545/2546 V267I + 2547/2548 T410S ++ 2549/2550 F70R ++

2551/2552 F70V + 2553/2554 T199A ++ 2555/2556 D42S ++ 2557/2558 A380T + 2559/2560 G758R +

2561/2562 A25G + 2563/2564 A25E ++ 2565/2566 D42T +++ 2567/2568 F70S + 2569/2570 S265A +

2571/2572 F77W ++ 2573/2574 M75T + 2575/2576 F77L ++ 2577/2578 M75W + 2579/2580 F70H +

2581/2582 A25L + 2583/2584 L561V +++ 2585/2586 Y106W +++ 2587/2588 F70N +++ 2589/2590

A642V + 2591/2592 G758Q ++ 2593/2594 S265Q + .sup.aLevels of increased production were

determined relative to the reference polypeptide of SEQ ID NO: 2432, and defined as follows: "+" = production at least that of the reference polypeptide, but less than 1.08-fold; "++" = at least 1.08-fold, but less than 1.16-fold; and "+++" = at least 1.16-fold increased production, relative to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to Rebaudioside A

(385) An experiment was performed to characterize the activity of the engineered round 13 SUS variants on sucrose and ADP to facilitate the formation of rebaudioside D from rebaudioside A. Shake flask powder (SFP) was made up to 0.03-1 g/L in potassium phosphate buffer, pH 6, and an aliquot was pre-incubated at 73° C. in a thermocycler for 15 minutes. Then, 10 µL of these SFP dilutions, either pre-incubated or not pre-incubated, were added to a 100 µL total reaction volume containing 50 mM potassium phosphate buffer, pH 6, 15 mM rebaudioside A (>97% purity), 37.5 mM sucrose, 9 mM fructose, 0.2 mM ADP, and 0.5 g/L GT SEQ ID NO: 4550. The reactions were performed at 55° C. for SFP samples that were not pre-incubated and at 60° C. for pre-incubated SFP in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. The reaction was solubilized by diluting 40× into water, quenched by diluting 5× into acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 3.3× in water and analyzed for steviol glycosides by RapidFire SPE-MS/MS with the instrument and parameters described in Table 31.1. A further thermostability characterization was conducted with the clarified shake flask lysates prior to lyophilization as follows: lysates were diluted 400× in buffer and incubated in a thermocycler at a gradient of 62-78° C. for 16.7 h. To determine the percent activity remaining, the pre-incubated lysates were assayed as described above with either stevioside or rebaudioside D and 4 h incubation at 60° C. The percent activity remaining was expressed as activity at high temperature divided by activity at lowest pre-incubated temperature. All 8 of the variants listed in Table 39.3 had higher activities than SEQ ID NO: 2432 under at least one condition, and five were improved under all conditions. The variant with the mutations Q33H, L47P, E59A, P81L, S175G, P530F, E534W, Q550I, and R606M (SEQ ID NO: 2510), which was improved under all conditions relative to SEQ ID NO: 2432, was selected as the best enzyme for the catalysis of the recycling reaction transferring a glucose from sucrose to ADP.

(386) TABLE-US-00072 TABLE 39.3 SUS Round 13 SFP Variants and RebD Levels Increased Increased SEQ ID NO: Amino Acid Differences RebD, RebD, % Activity (nt/aa) (Relative to SEQ ID NO: 2432) 55° C..sup.a 60° C..sup.a Remaining.sup.b 2507/2508 Q33H/L58M/E59A/P81G/D130Y/I48 - +++ +++ 0V/P530F/E534W/Q550I/G652K 2509/2510 Q33H/L47P/E59A/P81L/S175G/P530 + ++ ++ F/E534W/Q550I/R606M 2513/2514 Q33H/L58M/E59A/I480P/P530F/E53 + + + 4W/Q550I 2519/2520 Q33H/L58M/H154A/I480P/E534W/ + + Q550I/G603A/R606M 2521/2522 Q33H/P81G/D130Y/I480V/P530F/E5 - + ++ 34W/Q550I 2527/2528 Q33H/E59A/I480P/P530F/E534W/Q5 + ++ + 50I/R606M 2533/2534 Q33H/VV79H/H154A/I480P/P530F/E5 + + +++ 34W/Q550I 2539/2540 Q33H/P81I/S175G/P530F/E534W/F5 + +++ ++ 42W/Q550I/G652R .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 2432, and defined as follows: “-” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.6-fold; “++” = production at least 1.6-fold, but less than 3.7-fold; and “+++” = production at least 3.7-fold increased, relative to that of the reference polypeptide. .sup.bThe percent of activity remaining for each variant was determined following 16.7 h pre-incubation at 66.5° C., relative to the production of each variant following pre-incubation at 62° C., and are defined as follows: “-” = less than 40% of activity remained following pre-incubation at 62° C.; “+” = at least 40% activity, but less than 60% activity remained; “++” = at least 60% activity, but less than 70% activity remained; and “+++” = at least 70% activity remained.

Example 40

Sucrose Synthase Variants of SEQ ID NO: 2510

(387) Directed evolution of the sucrose synthase encoded by SEQ ID NO: 2509 was continued by constructing libraries of variant genes in which mutations associated with improved production in earlier rounds of evolution were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below, to provide a fourteenth round (“Round 14”) of 164 engineered SuS variant polypeptides with increased activity toward the generation of ADP-glucose.

(388) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A 60 to Form Rebaudioside M

(389) Libraries were screened using the following HTP enzyme-coupled assay. Pelleted *E. coli* cultures were lysed with 400 μ L of Tris-HCl, pH 7.5, with 1 mM magnesium sulfate and 0.5 mg/mL lysozyme and polymyxin B sulfate (PMBS) and cleared by centrifugation. Lysate was diluted 35 $\times$ into potassium phosphate buffer, pH 6.0, with 14.5 g/L RebA60 and pre-incubated for 15 minutes at 73 $^{\circ}$ C. Then, 10 μ L diluted pre-incubated SuS lysate, 0.08 g/L β 1,2GT SFP SEQ ID NO: 4550, and 0.2 g/L β 1,3GT SFP SEQ ID NO: 6864 were used in 100 μ L reaction volume with 20 g/L RebA60, 0.1 g/L ADP (Amresco, ultra pure grade), 40 g/L sucrose (cane sugar), and 9.6 g/L fructose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60 $^{\circ}$ C. in a Thermotron $^{\circ}$ titre-plate shaker with 300 RPM shaking for 16-18 h. The reaction described above was solubilized by diluting 40 $\times$ in water, quenched by diluting 5 $\times$ in acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 15 $\times$ in water and steviol glycoside products were detected by RapidFire SPE-MS/MS with the instrument and parameters described in Table 31.1. After analysis, the engineered SuS variant polypeptides that showed improved activity coupled with a GT on rebaudioside A were identified and are listed in Tables 40.1 and 40.2. Shake-flask scale cultures were grown for protein characterization as described in Example 1 for variants with the amino acid mutations shown in Table 40.3.

(390) TABLE-US-00073 TABLE 40.1 SUS Round 14 Combinatorial Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 2510) RebM.sup.a 7437/7438 D259G/Q487I/F684M/L688G + 7439/7440 A41K/R71Q/S532Y/F684H +++ 7441/7442 A41K/G485A/Q487K +++ 7443/7444 A41K/I91C/G485S + 7445/7446 S532Y ++ 7447/7448 G112W + 7449/7450 A41K/G485A/S532Y/L688G + 7451/7452 F684H ++ 7453/7454 A41K/G112Q/Q487I/F684H +++ 7455/7456 G112Q ++ 7457/7458 G485A/F684H/L688G + 7459/7460 A41K/R71Q + 7461/7462 A41K/D259G/S532Y + 7463/7464 A41K/G485S/Q487I/F684M/L688G + 7465/7466 A41K/G112W + 7467/7468 G112Q/D259G/S532Y/F684M/L688G + 7469/7470 A41K/S532Y ++ 7471/7472 A41K/R71Q/G112Q/D259G/G485A/L688Q ++ 7473/7474 G485A/S532Y +++ 7475/7476 R71Q/F684M/L688A + 7477/7478 A41K/D259G/G485A/Q487R/F684H/L688A ++ 7479/7480 S532Y/F684H/L688Q + 7481/7482 A41K ++ 7483/7484 G485S/Q487I/F684H/L688Q + 7485/7486 A41K/G485A/Q487R ++ 7487/7488 G112Q/G485S/F684H/L688G + 7489/7490 D259G/S532Y + 7491/7492 A41K/G112Q/F684H/L688A + 7493/7494 A41K/G485A ++ 7495/7496 A41K/R71Q/Q487R + 7497/7498 A41K/R71Q/F684H + 7499/7500 A41K/R71Q/G485A ++ 7501/7502 A41K/D259G/G485A/Q487I + 7503/7504 A41K/D259G/G485S + 7505/7506 A41K/G112Q/G485S/F684H ++ 7507/7508 R71Q/G485S/F684M/L688Q + 7509/7510 G485A/Q487R + 7511/7512 A41K/R71Q/G112Q +++ 7513/7514 G485S/Q487I/F684M + 7515/7516 G485S +++ 7517/7518 A41K/G84A/D259G/G485A/Q487I + 7519/7520 G485S/Q487K/S532Y +++ 7521/7522 R71Q/S532Y ++ 7523/7524 A41K/Q487R/F684M ++ 7525/7526 G112Q/G485S/F684M/L688G + 7527/7528 G112Q/D259G/F684M/L688Q + 7529/7530 A41K/R71Q/D259G/G485A/S532Y ++ 7531/7532 A41K/G112Q/S532Y ++ 7533/7534 A41K/F684M/L688A + 7535/7536 G485A ++ 7537/7538 G485S/Q487R/S532Y/F684H ++ 7539/7540 D259G/G485S/Q487K/F684H +++ 7541/7542 R71Q/G112W/G485S/L688A ++ 7543/7544 R71Q/G112Q/D259G/G485A/Q487R/F684M + 7545/7546 A41K/R71Q/S532Y +++ 7547/7548 G112W/D259G +++ 7549/7550 A41K/G485A/S532Y +++ 7551/7552 A41K/I91C/G112Q/G485S/Q487K/S532Y/F684H + 7553/7554 A41K/Q487I/F684M ++ 7555/7556 A41K/R71Q/G485A/S532Y/F684H +++ 7557/7558 A41K/G112Q/D259G/Q487R/S532Y/F684H +++ 7559/7560 A41K/G112Q/D259G/G485S/Q487R ++ 7561/7562 D259G/G485S/S532Y ++ 7563/7564 A41K/F684H + 7565/7566 G485S/F684M/L688G + 7567/7568 A41K/D259G/G485A/Q487R/S532Y/F684H + 7569/7570 A41K/G485S/F684M/L688A ++ 7571/7572 A41K/I91C/G112Q/G485S/S532Y/F684H + 7573/7574 A41K/R71Q/Q487R/S532Y/F684H +++ 7575/7576 G112Q/G485S/F684H +++ 7577/7578 A41K/I91C/G112Q/G485S + 7579/7580 G485A/F684M + 7581/7582 R44C/G112W/F684H/L688A + 7583/7584 P226V/Q487I/F684M/L688A + 7585/7586 A41K/R71Q/G112W/D259G/G485S/Q487R/F684H/L688Q ++ 7587/7588 F684M/L688Q + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 2510, and defined as follows: “+” = production at least 1.5-fold, but less than 2.9-fold; “++” = at least 2.9-fold, but less than 4.2-fold; and “+++” = at least 4.2-fold increased production, relative to the reference polypeptide.

(391) TABLE-US-00074 TABLE 40.2 SUS Round 14 Saturation Mutagenesis Variants and RebM Levels

Amino Acid Differences SEQ ID NO: (Relative to SEQ ID Increased (nt/aa) NO: 2510) RebM.sup.a
 7589/7590 R136A ++ 7591/7592 R615L ++ 7593/7594 H343N + 7595/7596 E45C + 7597/7598 D12C +
 7599/7600 D72E + 7601/7602 D12V +++ 7603/7604 R478I + 7605/7606 D12S +++ 7607/7608 P48V +
 7609/7610 L176M ++ 7611/7612 H343A + 7613/7614 E45G + 7615/7616 R136Q ++ 7617/7618 D12Q
 ++ 7619/7620 T205S + 7621/7622 Q95V ++ 7623/7624 H788K ++ 7625/7626 E358P + 7627/7628
 D29G + 7629/7630 Q7K/D12L ++ 7631/7632 E100K +++ 7633/7634 L630M +++ 7635/7636 H724S ++
 7637/7638 P47I +++ 7639/7640 Q95N + 7641/7642 I55D ++ 7643/7644 P47V + 7645/7646 R615K +
 7647/7648 R136P +++ 7649/7650 R478H ++ 7651/7652 E45V ++ 7653/7654 D12N +++ 7655/7656
 R44V + 7657/7658 L176R + 7659/7660 R361A + 7661/7662 R44L + 7663/7664 R361L + 7665/7666
 D29P +++ 7667/7668 R136K + 7669/7670 P440R ++ 7671/7672 L176V + 7673/7674 P47C + 7675/7676
 E45A +++ 7677/7678 L116A ++ 7679/7680 R303V + 7681/7682 Q27R + 7683/7684 R611A +++
 7685/7686 Q95T +++ 7687/7688 E45L ++ 7689/7690 R615C + 7691/7692 D29A ++ 7693/7694 P47N +
 7695/7696 R178H + 7697/7698 H724G ++ 7699/7700 R136F +++ 7701/7702 P47T + 7703/7704 P47L
 +++ 7705/7706 Q95L + 7707/7708 Q95D ++ 7709/7710 D29L +++ 7711/7712 R361T + 7713/7714
 L176T + 7715/7716 E100P ++ 7717/7718 E100Q + 7719/7720 S675C +++ 7721/7722 T205R/G485S +
 7723/7724 V756C ++ 7725/7726 H724K ++ 7727/7728 P47D + 7729/7730 P207K + 7731/7732 Q198A
 + 7733/7734 Y51P + 7735/7736 P48A + 7737/7738 Q198R ++ 7739/7740 I55S + 7741/7742 R136N ++
 7743/7744 Y51S ++ 7745/7746 Q201S + 7747/7748 R139K ++ 7749/7750 E358S +++ 7751/7752 E317I
 + 7753/7754 R208K + 7755/7756 R615M +++ 7757/7758 K280G + 7759/7760 E45S ++ 7761/7762
 L116F +++ 7763/7764 R478Q + .sup.aLevels of increased production were determined relative to the
 reference polypeptide of SEQ ID NO: 2510, and defined as follows: “+” = production at least 1.5-fold,
 but less than 1.92-fold; “++” = at least 1.92-fold, but less than 2.6-fold; and “+++” = at least 2.6-fold
 increased production, relative to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to
 Rebaudioside A 60 to form Rebaudioside M

(392) An experiment was performed to characterize the activity of the engineered round 14 SUS variants
 on sucrose and ADP to facilitate the formation of rebaudioside D from rebaudioside A. Shake flask
 powder (SFP) was made up to 0.03-1 g/L in 14.5 g/L RebA60 in potassium phosphate buffer, pH 6, and
 an aliquot was pre-incubated at 73° C. in a thermocycler for 15 minutes. Then, 10 µL diluted, pre-
 incubated SuS lysate, 0.08 g/L β1,2GT SFP SEQ ID NO: 4550, and 0.2 g/L β1,3GT SFP SEQ ID NO:
 6864, were used in 100 µL reaction volume with 20 g/L RebA60, 0.1 g/L ADP (Amresco, ultra pure
 grade), 40 g/L sucrose (cane sugar), and 9.6 g/L fructose. The following reaction conditions were used: 50
 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking
 for 16-18 h. The reaction described above was solubilized by diluting 40× in water, quenched by diluting
 5× in acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted
 15× in water and steviol glycoside products were detected by RapidFire SPE-MS/MS with the instrument
 and parameters described in Table 31.1. All 8 of the variants listed in Table 40.3 had higher activities than
 SEQ ID NO: 2510. The variant with the mutations A41K, G112Q, G485S, and F684H (SEQ ID NO:
 7506), which was most improved relative to SEQ ID NO: 2510, was selected as the best enzyme for the
 catalysis of the recycling reaction transferring a glucose from sucrose to ADP.

(393) TABLE-US-00075 TABLE 40.3 SUS Round 14 SFP Variants and RebM Levels SEQ ID NO:
 Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 2510) RebM.sup.a 7439/7440
 A41K/R71Q/S532Y/F684H +++ 7453/7454 A41K/G112Q/Q4871/F684H ++ 7493/7494 A41K/G485A +
 7505/7506 A41K/G112Q/G485S/F684H +++ 7515/7516 G485S + 7549/7550 A41K/G485A/S532Y ++
 7555/7556 A41K/R71Q/G485A/S532Y/F684H + 7575/7576 G112Q/G485S/F684H + .sup.aLevels of
 increased production were determined relative to the reference polypeptide of SEQ ID NO: 2510, and
 defined as follows: “+” = production at least 2.1-fold, but less than 2.85-fold; “++” = production at least
 2.85-fold, but less than 3.1-fold; and “+++” = production at least 3.1-fold increased, relative to that of the
 reference polypeptide.

Example 41

Sucrose Synthase Variants of SEQ ID NO: 7506

(394) Directed evolution of the sucrose synthase encoded by SEQ ID NO: 7505 was continued by
 constructing libraries of variant genes in which mutations associated with improved production in earlier

rounds of evolution were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below, to provide a fifteenth round ("Round 15") of 56 engineered SuS variant polypeptides with increased activity toward the generation of ADP-glucose.

(395) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A 60 to Form Rebaudioside M

(396) Libraries were screened using the following HTP enzyme-coupled assay. Pelleted *E. coli* cultures were lysed with 400 μ L of Tris-HCl, pH 7.5 with 1 mM magnesium sulfate and 0.5 mg/mL lysozyme and polymyxin B sulfate (PMBS) and cleared by centrifugation. Lysate was diluted 50 $\times$ into potassium phosphate buffer, pH 6.0, with 14.5 g/L RebA60 and pre-incubated for 15 minutes at 73 $^{\circ}$ C. Then, 10 μ L diluted, pre-incubated SuS lysate, 0.08 g/L β 1,2GT SFP SEQ ID NO: 4550, and 0.2 g/L β 1,3GT SFP SEQ ID NO: 6864, were used in 100 μ L reaction volume with 20 g/L RebA60, 0.1 g/L ADP (Amresco, ultra pure grade), 40 g/L sucrose (cane sugar), and 9.6 g/L fructose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60 $^{\circ}$ C. in a Thermotron $^{\circ}$ titre-plate shaker with 300 RPM shaking for 16-18 h. The reaction described above was solubilized by diluting 20 $\times$ in water, quenched by diluting 5 $\times$ in acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 20 $\times$ in water and steviol glycoside products were detected by RapidFire SPE-MS/MS with the instrument and parameters described in Table 31.1. After analysis, the engineered SuS variant polypeptides that showed improved activity coupled with a GT on rebaudioside A were identified and are listed in Table 41.1. Shake-flask scale cultures were grown for protein characterization as described in Example 1 for variants with the amino acid mutations shown in Table 41.2.

(397) TABLE-US-00076 TABLE 41.1 SUS Round 15 Combinatorial Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 7506) RebM.sup.a 8369/8370 D12N/E45A/P47I/Y51P/R136P/R139K/L630M/G758R ++ 8371/8372 D12N/E45A/Y51P/R136P/R139K/L630M/S675C/V756C/G758Q +++ 8373/8374 D12S/E45A/Y51P/L630M/V756C ++ 8375/8376 R136P/R139K/S142N ++ 8377/8378 D12N/R136Q/S142N +++ 8379/8380 D12V/E45A/P47I/R136Q/R139K/S142N/S675C/G758Q +++ 8381/8382 D12V/E45A/P47L +++ 8383/8384 D12V/R136Q/R139K/S142N/V756C/G758Q +++ 8385/8386 D12S/L630M/V756C ++ 8387/8388 D12N/P48A/Y51P/R136Q/R139K/G758Q +++ 8389/8390 D12V/E45A/P47I/P48A/Y51S/R136Q/S142N/L630M +++ 8391/8392 P47I/Y51S/R136Q/V756C/G758Q ++ 8393/8394 T199A/S532Y/S539R/L561I/G652R/H724S + 8395/8396 S532Y/L561I/H724K + 8397/8398 Q198R/I480P/L561I ++ 8399/8400 Q198R/T199A/L561V/H724G +++ 8401/8402 D42T/Q198R/S532Y/L561I/H724K ++ 8403/8404 Y106W/T199A/S539R/L561I/G652R ++ 8405/8406 I480V/L561V/G652R ++ 8407/8408 Q198R/I480V/H724K ++ 8409/8410 D42T/Q198R/S532Y/L561V ++ 8411/8412 D42T/D259G/I480V/L561V + 8413/8414 Q198R/T199A/I480P/L561I ++ 8415/8416 S532Y/S539R/L561V + 8417/8418 D42T/Q198R/T199A/I480V/S532Y/S539R/L561V +++ 8419/8420 D42T/I480V/L561I/H724K +++ 8421/8422 D42T/I480P/L561I ++ 8423/8424 D42T/L561I ++ 8425/8426 D259G/I480V + 8427/8428 I480V/L561V/G652R/H724K/L764R + 8429/8430 I480V/H724K ++ 8431/8432 Q198R/T199A/I480P/L561V/H724K ++ 8433/8434 D42T/D259G/I480V/G652K + 8435/8436 D42T/T199A/I480V/S532Y/L561V +++ 8437/8438 A25E/A517P + 8439/8440 H154A/R361T/P440R/A517P/G603E/H788K + 8441/8442 Q95T/G603E + 8443/8444 A25E/E100K/H154A/R208K/P440R/A517P/P705M/H788K + 8445/8446 A25E/D29P/R208K/P440R + 8447/8448 K280G/P440R/A517P + 8449/8450 A517P + 8451/8452 I55D/A517P/H788K ++ 8453/8454 I55D/A517P + 8455/8456 I55D + 8457/8458 Q8R/A25E/I55D/Q95T/R208K/E358S/P440V/A517P/H788K + 8459/8460 D29P/R208K/R361T/A517P/H788K + 8461/8462 G603E + 8463/8464 I55D/T410S/P440R/G603E/H788K + 8465/8466 A642V + 8467/8468 F77L/L176M/Q487K/R615M/A642V + 8469/8470 F70N/A642V + 8471/8472 L116F + 8473/8474 A380T + 8475/8476 L176T + 8477/8478 F70N + 8479/8480 V267I/R611Q/A642V + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 7506, and defined as follows: "+" = activity at least 1.07-fold, but less than 1.95-fold; "++" = at least 1.95-fold, but less than 2.4-fold; and "+++" = at least 2.4-fold increased production, relative to the reference

polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to Rebaudioside A 60 to form Rebaudioside M

(398) An experiment was performed to characterize the activity of the engineered round 15 SUS variants on sucrose and ADP to facilitate the formation of rebaudioside D from rebaudioside A. Shake flask powder (SFP) was made up to 0.03-1 g/L in 14.5 g/L RebA60 in potassium phosphate buffer, pH 6, and an aliquot was pre-incubated at 73° C. in a thermocycler for 15 minutes. 10 µL diluted, pre-incubated SuS lysate, 0.08 g/L β1,2GT SFP SEQ ID NO: 4550, and 0.2 g/L β1,3GT SFP SEQ ID NO: 6864, were used in 100 µL reaction volume with 20 g/L RebA60, 0.1 g/L ADP (Amresco, ultra pure grade), 40 g/L sucrose (cane sugar), and 9.6 g/L fructose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 16-18 h. The reaction described above was solubilized by diluting 20× in water, quenched by diluting 5× in acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 20× in water and steviol glycoside products were detected by RapidFire SPE-MS/MS with the instrument and parameters described in Table 31.1. All 8 of the variants listed in Table 41.2 had higher activities than SEQ ID NO: 7506. The variant with the mutations D42T, I480V, L561I, and H724K (SEQ ID NO: 8420), which was most improved relative to SEQ ID NO: 7506, was selected as the best enzyme for the catalysis of the recycling reaction transferring a glucose from sucrose to ADP.

(399) TABLE-US-00077 TABLE 41.2 SUS Round 15 SFP Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 7506) RebM.sup.a 8371/8372

D12N/E45A/Y51P/R136P/R139K/L630M/S675C/V756C/G758Q + 8383/8384

D12V/R136Q/R139K/S142N/V756C/G758Q + 8389/8390

D12V/E45A/P471/P48A/Y51S/R136Q/S142N/L630M ++ 8399/8400 Q198R/T199A/L561V/H724G +

8419/8420 D42T/I480V/L561I/H724K +++ 8439/8440 H154A/R361T/P440R/A517P/G603E/H788K

+++ 8451/8452 I55D/A517P/H788K ++ 8475/8476 L176T + .sup.a Levels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 7506, and defined as follows: “+” = production at least 2.1-fold, but less than 2.2-fold; “++” = production at least 2.2-fold, but less than 2.3-fold; and “+++” = production at least 2.3-fold increased, relative to that of the reference polypeptide.

Example 42

Sucrose Synthase Variants of SEQ ID NO: 8420

(400) Directed evolution of the sucrose synthase encoded by SEQ ID NO: 8419 was continued by constructing libraries of variant genes in which mutations associated with improved production in earlier rounds of evolution were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below, to provide a sixteenth round (“Round 16”) of 155 engineered SuS variant polypeptides with increased activity toward the generation of ADP-glucose.

(401) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A 60 to Form Rebaudioside M

(402) Libraries were screened using the following HTP enzyme-coupled assay. Pelleted *E. coli* cultures were lysed with 400 µL of Tris-HCl, pH 7.5 with 1 mM magnesium sulfate and 0.5 mg/mL lysozyme and polymyxin B sulfate (PMBS) and cleared by centrifugation. Lysate was diluted 30× into potassium phosphate buffer, pH 6.0, with 14.5 g/L RebA60 and pre-incubated for 1 hour at 75° C. Then, 10 µL diluted, pre-incubated SuS lysate, 0.08 g/L β1,2GT SFP SEQ ID NO: 7784, and 0.2 g/L β1,3GT SFP SEQ ID NO: 8088, were used in 100 µL reaction volume with 20 g/L RebA60, 0.025 g/L ADP (Amresco, ultra pure grade), 40 g/L sucrose (cane sugar), and 9.6 g/L fructose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 16-18 h. The reaction described above was solubilized by diluting 20× in water, quenched by diluting 5× in acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 20× in water and steviol glycoside products were detected by RapidFire SPE-MS/MS with the instrument and parameters described in Table 31.1. After analysis, the engineered SuS variant polypeptides that showed improved activity coupled with a GT on rebaudioside A were identified and are listed in Table 42.1. Shake-flask scale cultures were grown for protein characterization as described in Example 1 for variants with the amino acid mutations shown in Table 42.2.

(403) TABLE-US-00078 TABLE 42.1 SUS Round 16 Combinatorial Variants and RebM Levels SEQ ID

N/Amino Acid Position Differences Increased (nt/aa) (Relative to SEQ ID NO: 8420) RebM.sup.a 8797/8798
D12S/E45A/L176T/G603Q/L630M/A642V/V756C + 8799/8800 D12N/T199A/R208K/A517P/A642V +
8801/8802 D12N/E45A/R136Q/R208K/A517P/L630M/V756C +++ 8803/8804
D12S/R139K/G603Q/V756C + 8805/8806 R136Q/R139K/G603Q + 8807/8808
R136Q/R139K/A517P/G603Q/V756C +++ 8809/8810 D12S/R136Q/T199A/R208K/L630M/A642V ++
8811/8812 D12S/R136Q/R139K/R208K/A517P ++ 8813/8814
D12N/E45A/R136Q/R139K/A642V/V756C ++ 8815/8816 D12S/R139K/L176T/L630M/V756C +
8817/8818 D12S/R136Q/A517P/G603Q/L630M/A642V ++ 8819/8820
D12S/R136Q/R139K/L176T/V756C + 8821/8822 D12N/E45A/R136Q/A517P/L630M/A642V/V756C
+++ 8823/8824 D12N/E45A/R136Q/G603E/V756C + 8825/8826
D12S/E45A/R136Q/R139K/R208K/G603E/L630M/V756C ++ 8827/8828
D12S/R136Q/L176T/R208K/A517P/G603E/L630M/V756C +++ 8829/8830
D12N/L176T/A517P/V756C + 8831/8832 D12S/R136Q/R208K/G603Q/A642V + 8833/8834
D12N/R136Q/R139K/G603Q/A642V/V756C +++ 8835/8836
D12S/R136Q/R139K/G603Q/L630M/V756C +++ 8837/8838
D12S/R136Q/R139K/L176T/A517P/G603Q/L630M +++ 8839/8840 D12N/R136Q/R139K/L630M +
8841/8842 D12S/R136Q/R139K/A517P/L630M ++ 8843/8844 D12N/R136Q/T199A/A517P/V756C +
8845/8846 D12S/R136Q/R139K/T199A/R208K/A517P/G603E/V756C +++ 8847/8848
D12N/E45A/R136Q/R139K/T199A/A517P/G603E ++ 8849/8850 D12N/A517P/G603E/V756C +
8851/8852 D12N/R136Q/G603Q/L630M/V756C ++ 8853/8854 D12N/R136Q/R139K/A642V/V756C +
8855/8856 D12S/R136Q/R139K/L630M/A642V/V756C +++ 8857/8858
D12N/E45A/R139K/T199A/R208K/G603E + 8859/8860 T199A/A517P/G603Q/L630M/V756C +
8861/8862 R136Q/R139K/L176T/A517P/L630M/A642V/V756C +++ 8863/8864
R136Q/R139K/L176T/T199A/R208K/A517P/L630M/A642V ++ 8865/8866
D12S/R136Q/L176T/A517P/A642V + 8867/8868 D12N/R139K/G603Q/V756C + 8869/8870
D12N/R136Q/G603E/A642V ++ 8871/8872 D12N/E45A/R136Q/R139K/A517P/L630M/A642V/V756C
+++ 8873/8874 D12S/R136Q/L630M + 8875/8876 D12N/R136Q/R139K/A517P/V756C ++ 8877/8878
R136Q/R139K/A517P/G603E/A642V/V756C +++ 8879/8880
D12S/R136Q/R139K/G603Q/L630M/A642V ++ 8881/8882 D12N/G603Q/L630M/V756C + 8883/8884
D12S/A517P/G603Q/L630M/A642V/V756C ++ 8885/8886 D12N/E45A/R136Q/R139K/G603E/V756C
++ 8887/8888 D12N/R136Q/R139K/A517P/L630M/A642V ++ 8889/8890
D12S/R136Q/R139K/L630M/V756C ++ 8891/8892 D12N/Q95T/A517P/L630M/V756C + 8893/8894
D12S/Q95T/R139K/A517P/L630M/V756C ++ 8895/8896
D12N/E45A/T199A/R208K/A517P/G603E/L630M/V756C + 8897/8898
D12N/R136Q/L176T/G603Q/V756C + 8899/8900 D12S/R136Q/G603Q/L630M/A642V/V756C ++
8901/8902 D12N/T199A/L630M/A642V/V756C + 8903/8904 D12N/L630M/A642V/V756C +
8905/8906 D12S/R136Q/A517P/G603Q/L630M/V756C ++ 8907/8908
D12N/E45A/R136Q/R139K/A517P/G603E/V756C +++ 8909/8910
D12S/R136Q/R139K/A517P/G603Q/L630M/A642V/V756C +++ 8911/8912
D12N/Q95T/R136Q/R139K/A517P/G603Q/V756C ++ 8913/8914 D12N/R208K/L630M/V756C +
8915/8916 D12N/R136Q/R139K/G603E/L630M +++ 8917/8918 D12N/R136Q/A517P/L630M +++
8919/8920 R136Q/R139K/A517P/G603Q/L630M/A642V/V756C +++ 8921/8922
D12S/R136Q/A517P/A642V/V756C +++ 8923/8924 R139K/A517P/L630M/V756C ++ 8925/8926
D12S/R139K/A517P/L630M/A642V/V756C ++ 8927/8928 A517P/G603E/L630M/A642V/V756C +
8929/8930 D12N/L176T/A517P/G603E/L630M + 8931/8932 D12S/A517P/A642V/V756C ++
8933/8934 D12S/E45A/Q95T/R136Q/V756C ++ 8935/8936
D12N/R139K/A517P/G603Q/A642V/V756C +++ 8937/8938 R136Q/A517P/V756C + 8939/8940
D12N/G603Q/A642V/V756C ++ 8941/8942 D12S/T199A/V756C + 8943/8944
D12S/E45A/R139K/V756C + 8945/8946 D12S/L176T/G603Q/L630M/V756C ++ 8947/8948
D12S/R139K/R208K/A642V + 8949/8950
D12N/E45A/Q95T/R136Q/R139K/T199A/A517P/L630M/V756C +++ 8951/8952
D12S/R136Q/R139K/A517P/G603E/V756C +++ 8953/8954
D12N/R136Q/T199A/R208K/A517P/G603E/A642V/V756C +++ 8955/8956

R136Q/R139K/T199A/A517P/L630M/V756C +++ 8957/8958
R136Q/R139K/T199A/A517P/G603E/V756C +++ 8959/8960 D12S/L630M/V756C + 8961/8962
D12S/G603Q/L630M/V756C ++ 8963/8964 R136Q/R139K/A517P/V756C ++ 8965/8966
D12N/Q95T/R139K/L630M/A642V + 8967/8968 D12N/R136Q + 8969/8970
D12N/R139K/T199A/A517P/L630M/V756C +++ 8971/8972 R139K/A642V/V756C + 8973/8974
D12S/R136Q/R139K/A517P/G603E/L630M/V756C +++ 8975/8976
D12S/R139K/L176T/A517P/G603Q/L630M/V756C +++ 8977/8978
D12N/A517P/L630M/A642V/V756C + 8979/8980 D12S/R136Q/A517P/L630M/A642V + 8981/8982
D12N/R139K/L630M/A642V/V756C ++ 8983/8984
D12N/R136Q/R139K/L176T/A517P/G603Q/V756C +++ 8985/8986
R136Q/R139K/L630M/A642V/V756C ++ 8987/8988 D12S/R139K/A517P/A642V + 8989/8990
D12N/R136Q/G603Q/V756C ++ 8991/8992 D12N/Q95T/T199A/A517P/A642V + 8993/8994
D12N/R139K/G603E/A642V/V756C ++ 8995/8996 R139K/A517P/G603Q/V756C + 8997/8998
D12N/R136Q/R139K/L176T/L630M/V756C +++ 8999/9000
D12S/E45A/R208K/A517P/G603E/A642V/V756C + 9001/9002 D12N/R136Q/G603E/A642V/V756C
++ 9003/9004 D12N/A517P/G603Q/L630M/V756C + 9005/9006 R139K/T199A/A517P/A642V +
9007/9008 G603Q/V756C + 9009/9010 D12N/R139K/L630M/V756C ++ 9011/9012
D12S/R136Q/A517P/V756C ++ 9013/9014 D12S/E45A/R136Q/L176T/A517P/G603E/L630M/A642V +
9015/9016 R136Q/A642V + 9017/9018 D12S/A517P/L630M/A642V/V756C ++ 9019/9020
D12N/R136Q/R139K/L176T/A642V ++ 9021/9022 D12N/R136Q/R208K + 9023/9024
D12S/R136Q/G603E/V756C ++ 9025/9026 D12N/R139K/T199A/R208K/A642V + 9027/9028
D12N/R208K/A517P/G603E/H623N/L630M/A642V + 9029/9030 A517P/L630M/V756C + 9031/9032
D12S/R136Q/A517P/A642V + 9033/9034 D12N/E45A/R136Q/L630M/A642V + 9035/9036
D12S/R136Q/R139K/A517P/G603E/L630M/A642V/V756C +++ 9037/9038 D12N/G603Q/V756C +
9039/9040 D12S/Q95T/L630M/V756C + 9041/9042 R136Q/A642V/V756C + 9043/9044
D12N/R139K/L630M/A642V + 9045/9046 D12N/L176T/A517P/L630M/A642V/V756C ++ 9047/9048
R136Q/V756C + 9049/9050 D12S/E45A/R136Q/R139K/A517P/V756C ++ 9051/9052
D12N/G603E/A642V/V756C + 9053/9054 R136Q/G603Q/V756C + 9055/9056
D12N/R136Q/R139K/A517P ++ 9057/9058 R208K/A517P/L630M/A642V/V756C + 9059/9060
R136Q/L630M/A642V + 9061/9062
D12N/R136Q/R139K/T199A/R208K/A517P/L630M/A642V/V756C ++ 9063/9064
R139K/R208K/A517P/L630M/A642V/V756C + 9065/9066 R136Q/L630M/V756C + 9067/9068
D12S/E45A/R139K/A517P/V756C + 9069/9070 R136Q/R139K/R208K/G603Q/L630M/V756C ++
9071/9072 D12N/T199A/A642V/V756C + 9073/9074 D12S/R208K/G603Q/L630M/V756C +
9075/9076 D12N/Q95T/R139K/A517P/A642V + 9077/9078 D12S/L176T/G603E/V756C + 9079/9080
D12N/R136Q/R139K/A517P/L630M/A642V/V756C ++ 9081/9082
D12S/R136Q/R139K/A517P/G603Q/V756C ++ 9083/9084 D12S/R136Q/R139K/A517P/L630M/A642V
++ 9085/9086 D12S/R136Q/L630M/V756C + 9087/9088 D12N/R139K/R208K/A517P + 9089/9090
D12N/E45A/R139K/A517P/V756C + 9091/9092
D12S/E45A/R136Q/A517P/G603Q/A642V/V756C/A789V + 9093/9094
D12S/E45A/R139K/L176T/R208K/A517P/G603E/L630M + 9095/9096
D12N/R136Q/G603Q/A642V/V756C ++ 9097/9098 R139K/T199A/R208K/A517P/L630M/V756C +
9099/9100 G603E/L630M/V756C + 9101/9102 A25E/L176M/Q198R/S532Y/S539R + 9103/9104
H154A/S532Y/G652R/H788K + 9105/9106 L116F/S142N/Q198R/Y434H/P440R + .sup.aLevels of
increased production were determined relative to the reference polypeptide of SEQ ID NO: 8420, and
defined as follows: “+” = production at least 7.8-fold, but less than 16-fold; “++” = at least 16-fold, but
less than 22.3-fold; and “+++” = at least 22.3-fold increased production, relative to the reference
polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to
Rebaudioside A 60 to form Rebaudioside M

(404) An experiment was performed to characterize the activity of the engineered round 16 SUS variants
on sucrose and ADP to facilitate the formation of rebaudioside M from rebaudioside A. Shake flask
powder (SFP) was made up to 0.03-1 g/L in 14.5 g/L RebA60 in potassium phosphate buffer, pH 6, and

an aliquot was pre-incubated at 75° C. for 1 hour. 10 µL diluted, either pre-incubated or not pre-incubated SuS lysate, 0.08 g/L β1,2GT SFP SEQ ID NO: 7784, and 0.2 g/L β1,3GT SFP SEQ ID NO: 8088, were used in 100 µL reaction volume with 20 g/L RebA60, 0.025 g/L ADP (Amresco, ultra pure grade), 40 g/L sucrose (cane sugar), and 9.6 g/L fructose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 16-18 h. The reaction described above was solubilized by diluting 20× in water, quenched by diluting 5× in acetonitrile with 0.2% formic acid, and precipitated by centrifugation. The supernatant was diluted 20× in water and steviol glycoside products were detected by RapidFire SPE-MS/MS with the instrument and parameters described in Table 31.1. All 8 of the variants listed in Table 42.2 had higher production following preincubation than SEQ ID NO: 8420. The variant with the mutations D125, R136Q, R139K, A517P, G603Q, L630M, A642V, and V756C (SEQ ID NO: 8910), which was most improved relative to SEQ ID NO: 8420, was selected as the best enzyme for the catalysis of the recycling reaction transferring a glucose from sucrose to ADP.

(405) TABLE-US-00079 TABLE 42.2 SUS Round 16 SFP Variants and RebM Levels Increased RebM Increased RebM SEQ ID NO: Amino Acid Differences without with 75° C. (nt/aa) (Relative to SEQ ID NO: 8420) preincubation.sup.a preincubation.sup.a 8827/8828 D12S/R136Q/L176T/R208K/A517P/G60 + ++ 3E/L630M/V756C 8909/8910 D12S/R136Q/R139K/A517P/G603Q/L63 + +++ 0M/A642V/V756C 8919/8920 R136Q/R139K/A517P/G603Q/L630M/A + +++ 642V/V756C 8953/8954

D12N/R136Q/T199A/R208K/A517P/G60 – ++ 3E/A642V/V756C 9035/9036

D12S/R136Q/R139K/A517P/G603E/L63 – +++ 0M/A642V/V756C 9079/9080

D12N/R136Q/R139K/A517P/L630M/A6 + +++ 42V/V756C 9101/9102

A25E/L176M/Q198R/S532Y/S539R – ++ 9105/9106 L116F/S142N/Q198R/Y434H/P440R – ++

.sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 8420, and defined as follows: “–” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 3-fold; “++” = production at least 3-fold, but less than 13-fold; and “+++” = production at least 13-fold increased, relative to that of the reference polypeptide.

Example 43

Beta-1,2-ADP-Glycosyltransferase Variants of SEQ ID NO: 1054

(406) In this Example, experiments for evolution and screening of GT polypeptides (β1,2GT) derived from SEQ ID NO: 1054, for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 1053 was carried out by constructing libraries in which mutations associated with improved activity in previous rounds were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a fifth round (“Round 5”) of 26 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(407) HTP Assay for Glucose Transfer from ADP-Glucose to Rebaudioside A

(408) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 1053 variants. Lysis buffer volume was 400 µL, and the lysate was diluted 4-fold. Assays were conducted with 10 µL diluted lysate in 100 µL reaction volume with substrate loading of 1 mM rebaudioside A (>97% purity) and co-substrate loading of 1 mM ADP-glucose (Sigma, >93% purity). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 7, 3 mM MgCl₂.sub.2, 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 h. The reactions were diluted 1:5 in water and then quenched by adding 25 µL of the diluted assay to 75 µL acetonitrile with 0.2% formic acid. The resulting mixture was precipitated by centrifugation at 4° C. for 10 m. The supernatants were diluted 1:5 in water and analyzed by RapidFire SPE-MS/MS as described in Table 31.1. Glycosyltransferase variant polypeptides that produced rebaudioside D from rebaudioside A with ADP-glucose were identified, and the engineered polypeptides are listed in Table 43.1. Shake-flask scale cultures were grown, lysed, and lyophilized to powder as described in Example 31 for the variants listed in Table 43.2.

(409) TABLE-US-00080 TABLE 43.1 β1,2GT Round 5 Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 1054) RebD.sup.a 2595/2596

E24L/N162R/P175S/M201G/D275Q/1316V + 2597/2598 E24L/W226V/P330Q/L351M/K403R + 2599/2600 E24L/N162R/E198P/M201G/T211E/W226V/L323V/L351M + 2601/2602

E24L/P175S/M201G/D275Q/I316V/L351M + 2603/2604
E24L/M201G/W226V/G253D/L402I/K403R/I406M ++ 2605/2606
P175S/E198P/M201G/T211E/W226V/T260V/S264A/L323V/ +++ L402I/I406M 2607/2608
E24L/P175S/M201G/I316V + 2609/2610 E24L/P175S/T211E/I316V/P330Q + 2611/2612
E24L/P175S/T211E/G253D/I316V ++ 2613/2614 E24L/E198P/M201G/T211E ++ 2615/2616
E24L/P175S/W226V/L323V/L351M + 2617/2618 E24L/E198P/M201G + 2619/2620
E24L/M201G/T211E/G253D/L323V/L351M/Q366H/D389E/ + L402I/K403R 2621/2622
E24L/P175S/T211E/C220L/T260V/D275Q/P330Q/D389E + 2623/2624
E24L/E198P/M201G/T211E/C220L/T260V +++ 2625/2626
P175S/E198P/W226V/T260V/L351M/L402I/K403R/I406M ++ 2627/2628
E24L/E198P/M201G/W226V/P330Q/D389E ++ 2629/2630
E24L/N32S/P175S/W226V/G253D/D275Q/I316V + 2631/2632
E24L/N32S/P175S/T211E/T260V/P330Q/K403R/I406M + 2633/2634
E24L/N162R/M201G/G253D/S264A/L351M/L402I/I406M ++ 2635/2636 E24L/E198P/M201G/L351M
++ 2637/2638 E24L/N162R/E198P/M201G/W226V/L351M +++ 2639/2640
E24L/N162R/P175S/E198P/T211E/W226V/D275Q/I316V/ + L323V/D389E 2641/2642
E24L/P175S/E198P/T211E/W226V/G253D/I316V +++ 2643/2644
E24L/E198P/M201G/C220L/D275Q/D389E/L402I/I406M ++ 2645/2646
E24L/N32S/M201G/W226V/T260V/S264A/P330Q/ +++ L402I/I406M Levels of increased production
were determined relative to the reference polypeptide of SEQ ID NO: 1054, and defined as follows: “+” =
production at least 4.1-fold, but less than 4.58-fold; “++” = at least 4.58-fold, but less than 5.17-fold; and
“+++” at least 5.17-fold increased production, as compared to the reference polypeptide.
Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from ADP-Glucose to
Rebaudioside A

(410) A shake flask powder loading dose response experiment was performed to characterize activity of
the engineered round 5 variants on rebaudioside A. Levels of 0.006-0.2 g/L shake flask powder (SFP)
were added to a 100 μ L total reaction volume containing 50 mM potassium phosphate buffer, pH 7, 3 mM
MgCl₂, 1 mM rebaudioside A, and 1 mM ADP-glucose. The reaction was performed at 50° C. in a
Thermotron® titre-plate shaker at 300 RPM for 1 h. The reaction was diluted 1:5 in water and then
quenched by adding 25 μ L of the diluted assay to 75 μ L acetonitrile with 0.2% formic acid and
precipitated by centrifugation at 4° C. for 10 m. The supernatant was diluted 1:5 in water and analyzed for
steviol glycosides by RapidFire SPE-MS/MS. The activities of the round 5 variants on rebaudioside A at
0.05 g/L SFP loading are listed in Table 43.2. All 6 of the variants listed in Table 43.2 had higher
activities than SEQ ID NO: 1054. The variant with the mutations E24L, N162R, E198P, M201G, T211E,
W226V, L323V, and L351M (SEQ ID NO: 2600) and its encoding polynucleotide (SEQ ID NO: 2599)
were selected for further directed evolution for the glucosylation of rebaudioside A.

(411) TABLE-US-00081 TABLE 43.2 β 1,2GT Round 5 SFP Variants and RebD Levels SEQ ID NO:
Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 1054) RebD.sup.a 2595/2596
E24L/N162R/P175S/M201G/D275Q/I316V ++ 2599/2600
E24L/N162R/E198P/M201G/T211E/W226V/L323V/L351M +++ 2605/2606
P175S/E198P/M201G/T211E/W226V/T260V/S264A/L323V/ +++ L402I/I406M 2623/2624
E24L/E198P/M201G/T211E/C220L/T260V + 2633/2634
E24L/N162R/M201G/G253D/S264A/L351M/L402I/I406M + 2637/2638
E24L/N162R/E198P/M201G/W226V/L351M ++ .sup.aLevels of increased production were determined
relative to the reference polypeptide of SEQ ID NO: 1054, and defined as follows: “+” = production at
least 8.75-fold, but less than 10-fold; “++” = at least 10-fold, but less than 11-fold; and “+++” at least 11-
fold increased production, relative to the reference polypeptide.

Example 44

Glycosyltransferase Variants of SEQ ID NO: 1002 for Glucosylation of Rebaudioside I to Rebaudioside
M

(412) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID
NO: 1002 for improved glucosylation of steviol glycosides are described. Directed evolution of the GT
encoded by SEQ ID NO: 1001 was carried out by constructing libraries of variant genes in which

mutations associated with improved glycosyltransferase activity toward rebaudioside I in previous rounds were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a round (Round 5 Rebl) of engineered GT variant polypeptides with glucosyltransferase activity toward UDP-glucose and rebaudioside I.

(413) HTP Assay for Glucose Transfer from UDP-Glucose to Rebaudioside I

(414) Cells were lysed with 250 μ L lysis buffer. Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 1002 variants with lysate loading of 25 μ L lysate in 100 μ L reactions and with substrate loading of 1 mM rebaudioside I and co-substrate loading of 1 mM UDP-glucose (Sigma, >98% purity). The following reaction conditions were used: 50 mM KPhos buffer, pH 7, 3 mM MgCl₂, 40° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 18 h. The reactions were quenched by adding 10 μ L of the assay to 90 μ L acetonitrile with 0.2% formic acid. The resulting mixture was precipitated by centrifugation at 4° C. for 10 m. The supernatants were diluted 1:5 in water and analyzed by LC-MS/MS as described in Table 44.1. Glycosyltransferase variant polypeptides that produced rebaudioside M from rebaudioside I with UDP-glucose, at greater quantities than SEQ ID NO: 1002 were identified. The engineered polypeptides are listed in Table 44.2.

(415) TABLE-US-00082 TABLE 44.1 HPLC-MS/MS Analysis of Steviol Glycosides for Rebaudioside I Products. Instrument Agilent HPLC 1200 series, Sciex 4000 QTrap Column Poroshell 120 EC C18 50 $\times$ 3.0 mm, 2.7 μ m with Poroshell 120 EC C18 5 $\times$ 3.0, 2.7 μ m guard column (Agilent Technologies) Mobile phase Gradient (A: 0.1% formic acid in water, B: 0.1% formic acid in methanol) Time (m) % B 0 40 0.50 53 5.00 53 5.50 70 7.5 70 8.00 95 8.50 95 8.51 60 9.20 40 Flow rate 0.8 mL/m Run time 9.2 m Peak retention times Rebaudioside M: 4.37 m Rebaudioside I: 6.70 m Other glucosylated rebaudioside I product: 4.8 m Second glucosylated rebaudioside I product: 6.7 m Column temperature 40° C. Injection volume 10 μ L MS detection MRM 990/828 (for steviol tetraglycosides, e.g., rebaudioside A), 1152/828 (for steviol pentaglycosides, e.g., rebaudioside D), 1314/828 (steviol hexaglycosides, e.g., rebaudioside M), 828/666 (for steviol triglycosides, e.g., stevioside), 666/504 (steviol diglycosides, e.g., rubusoside) MS conditions MODE: MRM; CUR: 30; IS: 4750; CAD: high; TEM: 550° C.; GS1: 50; GS2: 50; DP: 150; EP: 10; CXP: 14; DT: 50 ms for each transition. For the first three transitions CE: 85; for the last two transitions CE: 60.

(416) TABLE-US-00083 TABLE 44.2 β 1,2GT Round 5 Rebl Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 1002) RebM.sup.a 2647/2648 F156R + 2649/2650 F156R/K161S + 2651/2652 F156R/G199H/N200A + 2653/2654 F156R/K161S/N162G + 2663/2664 F156R/K161S/N162G ++ 2665/2666 F156R/N162G/G199H ++ 2667/2668 W21Y/L127H/P129A/K161S/N162G +++ 2669/2670 L127Q/P129A/K161S/G199H/N200A +++ 2671/2672 L127H/P129A/N162T + 2673/2674 F156R/K161S/N162G + 2675/2676 L127Q/P129A/K161S/N162G/G199H +++ 2677/2678 F156R/K161S/N162G ++ 2679/2680 W21Y/L127H/P129A/N162T/G199H/N200A +++ 2681/2682 F156R/K161S/N162T ++ 2683/2684 W21Y/L127H/P129A/K161S ++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 1002, and defined as follows: “+” = production at least 2.5-fold, but less than 3.13-fold; “++” = at least 3.13-fold, but less than 3.28-fold; and “+++” at least 3.28-fold increased production, as compared to the reference polypeptide.

Example 45

Beta-1,2-ADP-Glycosyltransferase Variants of SEQ ID NO: 2600

(417) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 2600 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 2599 was carried out by constructing libraries of variant genes. Libraries recombined mutations associated with the surface residues of the enzyme and beneficial mutations associated with improved activity in previous rounds, combinatorially incorporated diversity from homologs in publicly available databases, or subjected certain structural features of the enzyme with saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a sixth round (“Round 6”) of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. Twenty-four engineered variants were identified from the recombined beneficial mutations and homolog diversity (Table 45.1), and 21 were identified from saturation mutagenesis (Table 45.2).

(418) HTP Assay for Glucosyl Transfer from ADP-Glucose to Rebaudioside A

(419) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 2600 variants. Lysis buffer volume was 400 μ L, and the lysate was diluted 40-fold. Assays were conducted with 10 μ L diluted lysate in 100 μ L reaction volume with substrate loading of 1 mM rebaudioside A (>97% purity) and co-substrate loading of 1 mM ADP-glucose (Sigma, >93% purity). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 7, 3 mM MgCl₂, 45° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 h. The reactions were diluted 1:5 in water and then quenched by adding 25 μ L of the diluted assay to 75 μ L acetonitrile with 0.2% formic acid. The resulting mixture was precipitated by centrifugation at 4° C. for 10 m. The supernatants were diluted 1:5 in water and analyzed by RapidFire SPE-MS/MS. The engineered variants that produced rebaudioside D from rebaudioside A with ADP-glucose and that were identified from the combinatorial libraries are listed in Table 45.1. The top 84 variants from the saturation mutagenesis library were retested in triplicate as described above at 50° C. The resulting engineered GT variant polypeptides are listed in Table 45.2. Shake-flask scale cultures were grown, lysed, and lyophilized to powder for variants listed in Table 45.3.

(420) TABLE-US-00084 TABLE 45.1 β 1,2GT Round 6 Combinatorial Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 2600) RebD.sup.a 2685/2686 A12S/R15K/I57K/E318D/L402I/A435V + 2687/2688 I57K/T260V/E400Q/L402I/D404S + 2689/2690 I57K/L402I + 2691/2692 A12S/P175S/E400Q/L402I/I406M ++ 2693/2694 I57K/E400Q/L402I + 2695/2696 I57K/P175S/C220L/T260V/S264A/L402I ++ 2697/2698 A12S/I57K/P175S/K451N + 2699/2700 H7E/A12S/E400Q/A435V + 2701/2702 A12S/E318D/L402I/D404S/K451N ++ 2703/2704 A12S/R15K/I57K/V71I/P175S/T260V/E400Q/L402I +++ 2705/2706 A12S/E318D/L402I/D404S/I406M/K451N +++ 2707/2708 A12S/P175S/T260V/S264A/E318D/E400Q +++ 2709/2710 A12S/R15K/E318D/E400Q/L402I/I406M + 2711/2712 T260V + 2713/2714 I57K/P175S/D404G ++ 2715/2716 E400Q/L402I ++ 2717/2718 H2-/H7E/A12S/R15K/P175S/T260V/E318D +++ 2719/2720 R15K/P175S/E318D/E400Q/L402I ++ 2721/2722 I57K/C220L/T260V/E400Q/L402I/I406M +++ 2723/2724 A12S/R15K/I57K/C220L/T254K/T260V/E318D/L402I + 2725/2726 H7E/E318D/K451N + 2727/2728 L160V/F186M/R195P ++ 2729/2730 R195P + 2731/2732 L152V/E192D/R195P + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 2600, and defined as follows: “+” = production at least 1.4-fold, but less than 1.64-fold; “++” = at least 1.64-fold, but less than 1.84-fold; and “+++” at least 1.84-fold increased production, as compared to the reference polypeptide.

(421) TABLE-US-00085 TABLE 45.2 β 1,2GT Round 6 Saturation Mutagenesis Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 2600) RebD.sup.a 2733/2734 T240A +++ 2735/2736 I239F + 2737/2738 R331S + 2739/2740 S148A ++ 2741/2742 T240P +++ 2743/2744 I239E + 2745/2746 A327V +++ 2747/2748 V323L + 2749/2750 V135A ++ 2751/2752 F326M ++ 2753/2754 L152V +++ 2755/2756 F356G ++ 2757/2758 F186V + 2759/2760 N32R ++ 2761/2762 D237T + 2763/2764 H325R + 2765/2766 R331H + 2767/2768 P330A ++ 2769/2770 H325G + 2771/2772 I239Y + 2773/2774 R331C + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 2600, and defined as follows: “+” = production at least 1.18-fold, but less than 1.4-fold; “++” = at least 1.4-fold, but less than 1.6-fold; and “+++” at least 1.6-fold increased production, as compared to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from ADP-Glucose to Rebaudioside A

(422) A shake flask powder loading dose response experiment was performed to characterize activity of the engineered round 6 variants on rebaudioside A. The experiment was performed as described in Example 40, with levels of 0.003-0.1 g/L SFP. The conversions of the round 6 variants on rebaudioside A at 0.025 g/L SFP loading are listed in Table 45.3. All 5 of the variants listed in Table 45.3 had higher activities than SEQ ID NO: 2600. The variant with the mutations H2-, H7E, A12S, R15K, P175S, T260V, and E318D (SEQ ID NO: 2718) and its encoding polynucleotide (SEQ ID NO: 2717) were selected for further directed evolution for the glucosylation of rebaudioside A.

(423) TABLE-US-00086 TABLE 45.3 β 1,2GT Round 6 Shake Flask Powder Variants and RebD Levels

SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 2600) RebD.sup.a
2687/2688 I57K/T260V/E400Q/L402I/D404S + 2695/2696 I57K/P175S/C220L/T260V/S264A/L402I +
2703/2704 A12S/R15K/I57K/V71I/P175S/T260V/E400Q/L402I + 2717/2718

H2-/H7E/A12S/R15K/P175S/T260V/E318D ++ 2727/2728 L160V/F186M/R195P + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 2600, and defined as follows: “+” = production at least 1.1-fold, but less than 1.6-fold; and “++” = at least 1.6-fold increased production, as compared to the reference polypeptide.

Example 46

Beta-1,2-ADP-Glycosyltransferase Variants of SEQ ID NO: 2718

(424) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 2718 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 2717 was carried out by constructing combinatorial libraries of variant genes in which beneficial mutations associated with improved activity or expression in previous rounds were recombined. Another library subjected certain structural features of the enzyme to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a seventh round (“Round 7”) of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. Twenty-five engineered variants were identified from the recombined beneficial mutations (Table 46.1) and 29 engineered variants were identified from saturation mutagenesis (Table 46.2).

(425) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(426) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 2717 variants. Lysis buffer volume was 400 μ L, and the lysate was diluted 10-fold. For screening the library in which beneficial mutations from previous rounds were recombined, assays were conducted with 10 μ L diluted lysate and 0.2 g/L SUS SFP SEQ ID NO: 1392 in 100 μ L reaction volume with substrate loading of 7.5 mM rebaudioside A and co-substrate loadings of 1 mM ADP (Sigma, >95%) and 15 mM sucrose (Sigma). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6.5, 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 2 h. The reactions were diluted 1:10 in water and then quenched by adding 10 μ L of the diluted assay to 90 μ L acetonitrile with 0.2% formic acid. The resulting mixture was precipitated by centrifugation at 4° C. for 10 m. The supernatants were diluted 1:10 in water and analyzed by RapidFire SPE-MS/MS as described in Table 31.1. For the remaining two round 7 libraries, screening was performed as described above, with the exceptions that 0.2 g/L SUS SFP SEQ ID NO: 1456 SFP and 30 mM sucrose were used. The engineered variants with glucosyltransferase activity coupled with a SuS on rebaudioside A from the recombined beneficial mutations are listed in Table 46.1. The top variants from the saturation mutagenesis library were retested in triplicate, using the procedure described above with the exceptions that 0.8 mM ADP was used, and the reactions were diluted 1:20 in water and then quenched by adding 20 μ L of the diluted assay to 80 μ L acetonitrile with 0.2% formic acid. The resulting engineered GT variant polypeptides are listed in Table 46.2. Shake-flask scale cultures were grown, lysed, and lyophilized to powder for variants listed in Table 46.3.

(427) TABLE-US-00087 TABLE 46.1 β 1,2GT Round 7 Combinatorial Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 2718) RebD.sup.a 2775/2776 F185M/R194P/D236T/T239A/F325M/A326V ++ 2777/2778

K14R/M184A/F185M/R194P/D388E/E399Q/L401I +++ 2779/2780 K14R/R194P ++ 2781/2782

K14R/M184A/F185M/R194P/F234Y/F325M/A326V/P329A +++ 2783/2784 K14R/A326V/R330H +++

2785/2786 K14R/N31R/M184A/D274Q/V322L/A326V/P329A/R330H + 2787/2788 K14R/F355G ++

2789/2790 K14R/I56K/M184A/F185M/R194P/I238M/T239A/D274Q/P329A/ +++ E399Q 2791/2792

K14R/F185M/D236T/I238M/T239A/D274Q/V322L/A326V/P329Q/ ++ F355G/E399Q/L401I

2793/2794 K14R/F185M/R194P/I238M/E399Q + 2795/2796 K14R/L23Q/D274Q/A326V/P329A +

2797/2798 N31R/I56K/I315V/P329A/R330H + 2799/2800

K14R/I56K/M184A/R194P/F234Y/I315V/A326V/P329Q/E399Q/ ++ L401I 2801/2802

K14R/V322L/A326V/R330H + 2803/2804

S147A/D236T/I238M/D243G/I315V/P329A/R330H/E399Q/L401I + 2805/2806

K14R/M184A/R194P/F355G/E399Q ++ 2807/2808

K14R/I56K/G252D/D274Q/I315V/A326V/P329A/R330H/L401I + 2809/2810
L23Q/N31R/S147A/M184A/F185M/I238M/G252D/F325M/P329A/ ++ R330H/D388E/L401I 2811/2812
K14R/F355G/E399Q + 2813/2814 K14R/I56K/R194P/I238M/I315V/F325M/A326V/P329A/R330H/ ++
E399Q 2815/2816 I238M + 2817/2818 E65D/I238T/N240S + 2819/2820 E65D + 2821/2822
E65D/Q114E/H132R/I238M + 2823/2824 S223T/I412S + .sup.aLevels of increased production were
determined relative to the reference polypeptide of SEQ ID NO: 2718, and defined as follows: “+” =
production at least 1.05-fold, but less than 2.5-fold; “++” = at least 2.5-fold, but less than 3-fold; and
“+++” at least 3-fold increased production, as compared to the reference polypeptide.

(428) TABLE-US-00088 TABLE 46.2 β 1,2GT Round 7 Saturation Mutagenesis Variants and RebD
Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 2718)
RebD.sup.a 2825/2826 N391R ++ 2827/2828 K309E ++ 2829/2830 E65N +++ 2831/2832 M104L +
2833/2834 K422R + 2835/2836 I56T + 2837/2838 K58R + 2839/2840 S11G +++ 2841/2842 L45V ++
2843/2844 N138G ++ 2845/2846 N135L ++ 2847/2848 N138K +++ 2849/2850 H132Q ++ 2851/2852
K309H + 2853/2854 E430L ++ 2855/2856 S11Q +++ 2857/2858 E165P + 2859/2860 Y449F +
2861/2862 N286R +++ 2863/2864 Q114R + 2865/2866 E256P + 2867/2868 E430V + 2869/2870 I55L +
2871/2872 H132S + 2873/2874 I238G + 2875/2876 L113V + 2877/2878 E273R + 2879/2880 E65S ++
2881/2882 L45F ++ .sup.aLevels of increased production were determined relative to the reference
polypeptide of SEQ ID NO: 2718, and defined as follows: “+” = production at least that of the reference
polypeptide, but less than 1.3-fold; “++” = at least 1.3-fold, but less than 1.7-fold; and “+++” at least 1.7-
fold increased production, as compared to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from ADP-Glucose to
Rebaudioside A

(429) A shake flask powder loading dose response experiment was performed to characterize activity of
the engineered round 7 variants on rebaudioside A. Levels of 0.002-0.2 g/L shake flask powder (SFP)
were added to a 100 μ L total reaction volume containing 50 mM potassium phosphate buffer, pH 6.5, 2
mM rebaudioside A, and 2 mM ADP-glucose. The reaction was performed at 50° C. in a Thermotron®
titre-plate shaker at 300 RPM for 1 h. The reaction was diluted 1:5 in water and then quenched by adding
25 μ L of the diluted assay to 75 μ L acetonitrile with 0.2% formic acid and precipitated by centrifugation
at 4° C. for 10 m. The supernatant was diluted 1:10 in water and analyzed for steviol glycosides by
RapidFire SPE-MS/MS. The conversions of the round 7 variants on rebaudioside A at 0.025 g/L SFP
loading are shown in Table 46.3. All 7 of the variants listed in Table 46.3 had higher activities than SEQ
ID NO: 2718. The variant with the mutations K14R, I56K, R194P, I238M, I315V, F325M, A326V,
P329A, R330H, and E399Q (SEQ ID NO: 2814) and its encoding polynucleotide (SEQ ID NO: 2813)
were selected for further directed evolution for the glucosylation of rebaudioside A.

(430) TABLE-US-00089 TABLE 46.3 β 1,2GT Round 7 Shake Flask Powder Variants and RebD Levels
SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 2718) RebD.sup.a

2777/2778 K14R/M184A/F185M/R194P/D388E/E399Q/L401I ++ 2793/2794

K14R/F185M/R194P/I238M/E399Q ++ 2799/2800

K14R/I56K/M184A/R194P/F234Y/I315V/A326V/P329Q/E399Q/ + L401I 2805/2806

K14R/M184A/R194P/F355G/E399Q + 2809/2810

L23Q/N31R/S147A/M184A/F185M/I238M/G252D/F325M/P329A/ +++ R330H/D388E/L401I

2811/2812 K14R/F355G/E399Q ++ 2813/2814

K14R/I56K/R194P/I238M/I315V/F325M/A326V/P329A/R330H/ +++ E399Q .sup.aLevels of increased
production were determined relative to the reference polypeptide of SEQ ID NO: 2718, and defined as
follows: “+” = production at least 1.6-fold, but less than 2.3-fold; “++” = at least 2.3-fold, but less than
2.5-fold; and “+++” at least 2.5-fold production, as compared to the reference polypeptide.

Example 47

Beta-1,2-ADP-Glycosyltransferase Variants of SEQ ID NO: 2814

(431) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID
NO: 2814 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed
evolution of the GT encoded by SEQ ID NO: 2813 was carried out by constructing combinatorial
libraries of variant genes in which beneficial mutations associated with improved activity in previous
rounds were recombined and in which certain structural features of the enzyme were subjected with

saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide an eighth round (“Round 8”) of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. Sixteen engineered variants were identified from the recombined beneficial mutations (Table 47.1), and 18 were identified from saturation mutagenesis (Table 47.2).

(432) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(433) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 2813 variants. Lysis buffer volume was 400 μ L, and the lysate was diluted 20-fold. For screening the library in which beneficial mutations from previous rounds were recombined, assays were conducted with 10 μ L diluted lysate and 0.2 g/L SUS SFP SEQ ID NO: 1456 in 100 μ L reaction volume with substrate loading of 7.5 mM rebaudioside A and co-substrate loadings of 0.4 mM ADP (Sigma, >95%) and 30 mM sucrose (Sigma). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6.5, 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 h. The reactions were diluted 1:20 in water and then quenched by adding 20 μ L of the diluted assay to 80 μ L acetonitrile with 0.2% formic acid. The resulting mixture was precipitated by centrifugation at 4° C. for 10 m. The supernatants were diluted 1:10 in water and analyzed by RapidFire SPE-MS/MS. The resulting engineered variants with glucosyltransferase activity coupled with a SuS on rebaudioside A are listed in Table 47.1. For the round 8 saturation mutagenesis library, screening was performed as described above with the exceptions that 0.2 g/L SUS SFP SEQ ID NO: 1582, 0.2 mM ADP, and the following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 55° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 2 h. The top 56 variants from this library were retested in triplicate using the same assay conditions with the exception that 0.15 g/L SUS SFP SEQ ID NO: 1582 was used. The resulting engineered GT variant polypeptides are listed in Table 47.2. Shake-flask scale cultures were grown, lysed, and lyophilized to powder for variants listed in Table 47.3.

(434) TABLE-US-00090 TABLE 47.1 β 1,2GT Round 8 Combinatorial Variants and RebD Levels SEQ ID Amino Acid Differences Increased NO: (nt/aa) (Relative to SEQ ID NO: 2814) RebD.sup.a 2883/2884 N31R/D388E ++ 2885/2886 M238Y/H324G/A329Q/F355G ++ 2887/2888

N31R/V134A/F185M/G252D/D274Q/H324G/D388E +++ 2889/2890

L23Q/N31R/F185M/H324G/L401I + 2891/2892 N31R/V134A/G252D/H324G + 2893/2894

N31R/V134A/M238E/G252D/V322L/H324G/D388E ++ 2895/2896 H324G/A329Q/F355G/L401I + 2897/2898 V134A/M184A/F185M/F234Y/D236T/T239A/D274Q/H324G/D388E +++ 2899/2900

N31R/V322L ++ 2901/2902 D236T + 2903/2904 F234Y/D236T/M238E/V322L/H324G/F355G +++

2905/2906 V322L/H324G + 2907/2908 M184A/F185M/V322L + 2909/2910 N31R/D236T + 2911/2912

L23Q/N31R/F185M/F355G + 2913/2914 N31R/M184A/F185M/M238Y/T239A/V322L + .sup.aLevels

of increased production were determined relative to the reference polypeptide of SEQ ID NO: 2814, and defined as follows: “+” = production at least 1.3-fold that of the reference polypeptide, but less than 1.4-fold; “++” = at least 1.4-fold but less than 1.7-fold; and “+++” = greater than 1.7-fold increased production, as compared to the reference polypeptide.

(435) TABLE-US-00091 TABLE 47.2 β 1,2GT Round 8 Saturation Mutagenesis Variants and RebD Levels Amino Acid Differences SEQ ID NO: (Relative to SEQ ID Increased (nt/aa) NO: 2814)

RebD.sup.a 2915/2916 I427R + 2917/2918 S426R +++ 2919/2920 V164H ++ 2921/2922 I440R ++

2923/2924 M325L +++ 2925/2926 V176R +++ 2927/2928 V164R + 2929/2930 S446R ++ 2931/2932

T400V ++ 2933/2934 I122L + 2935/2936 S426A +++ 2937/2938 V164M + 2939/2940 V176K ++

2941/2942 V176N + 2943/2944 V176L + 2945/2946 E177A + 2947/2948 G316R + 2949/2950 K425R +

.sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID

NO: 2814, and defined as follows: “+” = production at least that of the reference polypeptide, but less than 1.25-fold; “++” = at least 1.25-fold, but less than 1.35-fold; and “+++” = at least 1.35-fold increased production, as compared to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP to Rebaudioside A

(436) A shake flask powder loading dose response experiment was performed to characterize activity of the engineered round 8 variants on rebaudioside A. Levels of 0.002-0.2 g/L shake flask powder (SFP) were added to a 100 μ L total reaction volume containing 50 mM potassium phosphate buffer, pH 6.5, 8

mM rebaudioside A, 0.2 mM ADP, 30 mM sucrose, and 0.2 g/L SUS SFP SEQ ID NO: 1456. The reaction was performed at 50° C. in a Thermotron® titre-plate shaker at 300 RPM for 1 h. The reaction was diluted 1:20 in water and then quenched by adding 20 µL of the diluted assay to 80 µL acetonitrile with 0.2% formic acid and precipitated by centrifugation at 4° C. for 10 m. The supernatant was diluted 1:10 in water and analyzed for steviol glycosides by RapidFire SPE-MS/MS. The conversion of the round 8 variants on rebaudioside A at 0.1 g/L SFP loading are shown in Table 47.3. All 8 of the variants listed in Table 47.3 had higher activities than SEQ ID NO: 2814. The variant with the mutations N31R and D388E (SEQ ID NO: 2884) and its encoding polynucleotide (SEQ ID NO: 2883) were selected for further directed evolution for the glucosylation of rebaudioside A.

(437) TABLE-US-00092 TABLE 47.3 β1,2GT Round 8 Shake Flask Variants and RebD Levels SEQ ID Amino Acid Differences Increased NO: (nt/aa) (Relative to SEQ ID NO: 2814) RebD.sup.a 2883/2884 N31R/D388E +++ 2887/2888 N31R/V134A/F185M/G252D/D274Q/H324G/D388E + 2893/2894 N31R/V134A/M238E/G252D/V322L/H324G/D388E + 2897/2898 V134A/M184A/F185M/F234Y/D236T/T239A/D274Q/H324G/D388E + 2899/2900 N31R/V322L ++ 2903/2904 F234Y/D236T/M238E/V322L/H324G/F355G ++ 2909/2910 N31R/D236T + 2913/2914 N31R/M184A/F185M/M238Y/T239A/V322L +++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 2814, and defined as follows: “+” = production at least 1.2-fold, less than 1.7-fold; “++” = at least 1.7-fold, but less than 1.9-fold; and “+++” = at least 1.9-fold increased production, as compared to the reference polypeptide.

Example 48

Beta-1,2-ADP-Glycosyltransferase Variants of SEQ ID NO: 2884

(438) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 2884 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 2883 was carried out by constructing combinatorial libraries of variant genes in which beneficial mutations associated with improved activity in previous rounds were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a ninth round (“Round 9”) of 33 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(439) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(440) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 2883 variants. Lysis buffer volume was 400 µL, and the lysate was diluted 20-fold. Assays were conducted with 10 µL diluted lysate and 0.15 g/L SUS SFP SEQ ID NO: 1582 in 100 µL reaction volume with substrate loading of 7.5 mM rebaudioside A and co-substrate loadings of 0.2 mM ADP (Sigma, >95%) and 24 mM sucrose (Sigma). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 55° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 2 h. The reactions were diluted, quenched, and analyzed as described above. The resulting engineered variants with glucosyltransferase activity coupled with a SuS on rebaudioside A are listed in Table 48.1. Shake-flask scale cultures were grown, lysed, and lyophilized to powder for variants listed in Table 48.2.

(441) TABLE-US-00093 TABLE 48.1 β1,2GT Round 9 Combinatorial Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 2884) RebD.sup.a 2951/2952 S11Q/K58R/E65N/M104L/P107G/N138G/N286R + 2953/2954

S11G/K58R/H132Q/E165P/N286R/N391R/K422S/E430L + 2955/2956

N135L/N138G/E165P/K309H/E430L + 2957/2958

S11Q/E65N/M104L/P107G/N138G/E165P/N286R/N391R/E430L ++ 2959/2960

P107G/Q114R/H132Q/N138G + 2961/2962 S11Q/K58R + 2963/2964

K58R/E65N/E165P/K309H/E430L ++ 2965/2966 S11Q + 2967/2968 S11Q/K58R/E65N/N135L/N138G

++ 2969/2970 K58R/M104L/Q114R/E165P/N391R + 2971/2972 S11Q/N138K/E165P/5223T/K309H +

2973/2974 S11Q/E65N/P107G/N135L/E165P/N391R/E430L + 2975/2976 S11Q/P107G/N138G +

2977/2978 S11Q/E65N/N135L/N138G + 2979/2980 S11G/E65N/H132Q/N391R/E430L + 2981/2982

S11Q/M104L/H132Q/N138K/N391R + 2983/2984 S11Q/Q114R/5223T/G252D/N286R/N391R ++

2985/2986 S11G/M104L/H132Q/N138G/K309H/N391R ++ 2987/2988

S11G/K58R/M104L/P107G/Q114R/N138K/S223T/N391R +++ 2989/2990

S11G/L45V/K58R/H132Q/N138G/N286R ++ 2991/2992 S11G/P107G/Q114R/5223T/K309H/E430L

+++ 2993/2994 S11Q/H132Q/N135L/N138K/5223T/N286R +++ 2995/2996 S223T/K309E ++
 2997/2998 S11Q/H132Q/N138G/5223T/N286R/N391R +++ 2999/3000 K58R/E65N/N138K ++
 3001/3002 P107G/E430L + 3003/3004 S11Q/K58R/N138G/K309E +++ 3005/3006
 S11G/K58R/Q114R/N286R/K309H/N391R + 3007/3008 K58R/E65N/N138G/E165P ++ 3009/3010
 S11Q/E65N/H132Q/N135L/G252D + 3011/3012 S11Q/Q114R/N135L/N138G/5223T ++ 3013/3014
 S11Q/K58R/E65N/P107G/N135L/5223T/N286R/N391R/E430L + 3015/3016
 S11Q/E65N/H132Q/N135L/N138G/5223T/N391R +++ .sup.aLevels of increased production were
 determined relative to the reference polypeptide of SEQ ID NO: 2884, and defined as follows: “+” =
 production at least 1.5-fold, but less than 2.1-fold; “++” = at least 2.1-fold, but less than 2.5-fold; and
 “+++” = at least 2.5-fold increased production, relative to reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP to Rebaudioside A

(442) A shake flask powder loading dose response experiment was performed to characterize activity of the engineered round 9 variants on rebaudioside A. Levels of 0.002-0.2 g/L shake flask powder (SFP) were added to a 100 μ L total reaction volume containing 50 mM potassium phosphate buffer, pH 6, 8 mM rebaudioside A, 0.4 mM ADP, 24 mM sucrose, and 0.15 g/L SUS SFP SEQ ID NO: 1582. The reaction was performed at 55° C. in a Thermotron® titre-plate shaker at 300 RPM for 1 h. The reactions were diluted, quenched, and analyzed as described above. The conversion of the round 9 variants on rebaudioside A at 0.05 g/L SFP loading are shown in Table 48.2. All 6 of the variants listed in Table 48.2 had higher activities than SEQ ID NO: 2884. The variant with the mutations S11Q, E65N, H132Q, N135L, N138G, S223T, and N391R (SEQ ID NO: 3016) and its encoding polynucleotide (SEQ ID NO: 3015) were most improved and were selected for further directed evolution for the glucosylation of rebaudioside A.

(443) TABLE-US-00094 TABLE 48.2 β 1,2GT Round 9 Shake Flask Powder Variants and RebD Levels
 SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 2884) RebD.sup.a
 3015/3016 S11Q/E65N/H132Q/N135L/N138G/S223T/ +++ N391R 2983/2984
 S11Q/Q114R/5223T/G252D/N286R/N391R ++ 2997/2998 S11Q/H132Q/N138G/5223T/N286R/N391R
 + 2987/2988 S11G/K58R/M104L/P107G/Q114R/N138K/ ++ S223T/N391R 3011/3012
 S11Q/Q114R/N135L/N138G/5223T + 2975/2976 S11Q/P107G/N138G + .sup.aLevels of increased
 production were determined relative to the reference polypeptide of SEQ ID NO: 2884, and defined as
 follows: “+” = production at least 1.07-fold, but less than 1.2-fold; “++” = at least 1.2-fold, but less than
 1.85-fold; and “+++” = at least 1.85-fold increased production, as compared to the reference polypeptide.
 Example 49

Beta-1,2-ADP-Glycosyltransferase Variants of SEQ ID NO: 3016

(444) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 3016 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 3015 was carried out by constructing libraries of variant genes. Libraries recombined mutations associated with improved activity in previous rounds and subjected certain structural features of the enzyme with saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a tenth round (“Round 10”) of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. Forty engineered variants were identified from the recombined beneficial mutations (Table 49.1), and 40 were identified from saturation mutagenesis (Table 49.2).

(445) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(446) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 3015 variants. Lysis buffer volume was 400 μ L, and the lysate was diluted 20-fold. For screening the library in which beneficial mutations from previous rounds were recombined, assays were conducted with 10 μ L diluted lysate and 0.15 g/L SUS SFP SEQ ID NO: 1764 in 100 μ L reaction volume with substrate loading of 8 mM rebaudioside A and co-substrate loadings of 0.2 mM ADP (Sigma, >95%) and 24 mM sucrose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 55° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 h. The reactions were diluted 1:20 in water and then quenched by adding 20 μ L of the diluted assay to 80 μ L acetonitrile with 0.2% formic acid. The resulting mixture was precipitated by centrifugation at 4° C. for 10 m. The supernatants were

diluted 1:10 in water and analyzed by RapidFire SPE-MS/MS. The resulting engineered variants with glucosyltransferase activity coupled with a SuS on rebaudioside A are listed in Table 49.1. For the round 10 saturation mutagenesis library, screening was performed as described above with the exceptions that the lysate was diluted 10-fold, and the following reaction conditions were used: 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 2 h. The resulting engineered GT variant polypeptides are listed in Table 49.2. Shake-flask scale cultures were grown, lysed, and lyophilized to powder for variants listed in Table 49.3.

(447) TABLE-US-00095 TABLE 49.1 β 1,2GT Round 10 Combinatorial Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 3016) RebD.sup.a 3017/3018 K58R/I122L/I440R + 3019/3020 K58R/I122L/V164H/D236T/S446R + 3021/3022 K58R/P107G/I122L/V322L/T400V/K425R/I427R/I440R/S446R ++ 3023/3024 K58R/P107G/D236T/T400V + 3025/3026 I122L/V164H/V176R/H324G/T400V +++ 3027/3028 I122L/V176R/D236T/T400V ++ 3029/3030 P107G/V164H/D236T/T400V/S446R ++ 3031/3032 V164H ++ 3033/3034 K58R/V176R/D236T +++ 3035/3036 V176R/T400V +++ 3037/3038 K58R/P107G/I122L/D236T/H324G/T400V/K425R/S446R ++ 3039/3040 V164H/V322L/H324G/M325L/T400V/K425R/I440R/S446R + 3041/3042 I122L/T400V +++ 3043/3044 K58R/V164H/V176R +++ 3045/3046 P107G/I440R + 3047/3048 P107G/I122L/D236T/K425R/S426A/S446R +++ 3049/3050 P107G/V176R/V322L/M325L/I440R/S446R ++ 3051/3052 I122L/V164H/T400V/I440R +++ 3053/3054 K58R/P107G/I122L/V176R/D236T/H324G/M325L/T400V/S426R/ + I427R 3055/3056 I122L/V164H/T400V/I440R/S446R + 3057/3058 K58R/P107G/T400V + 3059/3060 H324G/T400V/K425R/S426R/I440R/S446R ++ 3061/3062 I122L/I440R/S446R +++ 3063/3064 V164H/S271G/K425R/S426R ++ 3065/3066 V164H/T400V ++ 3067/3068 P107G/D236T ++ 3069/3070 K58R/P107G/V164H/T400V ++ 3071/3072 K58R/I122L/M325L + 3073/3074 K58R/I122L/V322L/M325L ++ 3075/3076 K58R/I122L/V176R/D236T/T400V/S446R + 3077/3078 K58R/V164H/H324G/K425R/I427R + 3079/3080 P107G/V176R/T400V/K425R/I427R/I440R + 3083/3084 K58R/D236T + 3085/3086 I122L/K425R/S426R/S446R + 3087/3088 I122L/V176R/H324G/T400V/I440R + 3089/3090 K58R/I122L/V176R/V322L/H324G/S426A/I427R + 3093/3094 I139V/G252D + 3095/3096 G8S/K448A + 3097/3098 I139V + 3099/3100 P189R + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 3016, and defined as follows: “+” = production at least 1.2-fold, but less than 1.43-fold; “++” = at least 1.43-fold, but less than 1.67-fold; and “+++” = at least 1.67-fold increased production, as compared to the reference polypeptide.

(448) TABLE-US-00096 TABLE 49.2 β 1,2GT Round 10 Saturation Mutagenesis Variants and RebD Levels Amino Acid Differences SEQ ID NO: (Relative to SEQ ID Increased (nt/aa) NO: 3016) RebD.sup.a 3101/3102 N240E +++ 3103/3104 A242Y + 3105/3106 S9C + 3107/3108 E210V +++ 3109/3110 E408P ++ 3111/3112 G200T ++ 3113/3114 N115R + 3115/3116 N172R ++ 3117/3118 A242I ++ 3119/3120 L116V + 3121/3122 G415R +++ 3123/3124 K245M + 3125/3126 G200A +++ 3127/3128 E411Q + 3129/3130 A213G + 3131/3132 G415K ++ 3133/3134 T409L +++ 3135/3136 N423R +++ 3137/3138 N172S + 3139/3140 N65S + 3141/3142 K106A ++ 3143/3144 Q178K + 3145/3146 N423A ++ 3147/3148 N447R + 3149/3150 D255P +++ 3151/3152 N240C ++ 3153/3154 S9M ++ 3155/3156 H324R/N423R +++ 3157/3158 N447L + 3159/3160 K416R + 3161/3162 V385R + 3163/3164 N115D + 3165/3166 N240L ++ 3167/3168 E210L ++ 3169/3170 G415A ++ 3171/3172 G200V + 3173/3174 N240P + 3175/3176 N240V + 3177/3178 I412R + 3179/3180 G415H + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 3016, and defined as follows: “+” = production at least 1.1-fold that of the reference polypeptide, but less than 1.38-fold; “++” = at least 1.38-fold but less than 1.76-fold; and “+++” = at least 1.76-fold production, relative to reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP to Rebaudioside A

(449) A shake flask powder loading dose response experiment was performed to characterize activity of the engineered round 10 variants on rebaudioside A. Levels of 0.002-0.2 g/L shake flask powder (SFP) were added to a 100 μ L total reaction volume containing 50 mM potassium phosphate buffer, pH 6, 8 mM

rebaudioside A, 0.4 mM ADP, 24 mM sucrose, and 0.15 g/L SUS SFP SEQ ID NO: 1764. The reaction was performed at 60° C. in a Thermotron® titre-plate shaker at 300 RPM for 1 h. The reaction was diluted 1:20 in water and then quenched by adding 20 µL of the diluted assay to 80 µL acetonitrile with 0.2% formic acid and precipitated by centrifugation at 4° C. for 10 m. The supernatant was diluted 1:10 in water and analyzed for steviol glycosides by RapidFire SPE-MS/MS. The activities of the round 10 variants on rebaudioside A at 0.25 g/L SFP loading are shown in Table 49.3. All 5 of the variants listed in Table 49.3 had higher activities than SEQ ID NO: 3016. The variant with the mutations K58R, I122L, V176R, T400V, K425R, S426A, I427R, and S446R (SEQ ID NO: 3082) and its encoding polynucleotide (SEQ ID NO: 3081) were most improved and were selected for further directed evolution for the glucosylation of rebaudioside A.

(450) TABLE-US-00097 TABLE 49.3 β 1,2GT Round 10 Shake Flask Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 3016) RebD.sup.a 3061/3062 I122L/I440R/S446R + 3081/3082 K58R/I122L/V176R/T400V/K425R/S426A/ ++ I427R/S446R 3089/3090 K58R/I122L/V176R/V322L/H324G/S426A/ + I427R 3091/3092 K448A + 3099/3100 P189R + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 3016, and defined as follows: “+” = production at least that of the reference polypeptide, but less than 1.3-fold; and “++” = at least 1.3-fold increased production, as compared to the reference polypeptide.

Example 50

Beta-1,2-ADP-Glycosyltransferase Variants of SEQ ID NO: 3082

(451) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 3082 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 3081 was carried out by constructing combinatorial libraries of variant genes. Libraries recombined mutations associated with improved activity in previous rounds and subjected certain structural features of the enzyme with saturation mutagenesis. These libraries were then plated, grown, and screened using the HTP assay described below, to provide an eleventh round (“Round 11”) of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. Fifty engineered variants were identified from the recombined beneficial mutations (Table 50.1), and 53 were identified from saturation mutagenesis (Table 50.2).

(452) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(453) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 3081 variants. Lysis buffer volume was 400 µL, and the lysate was diluted 20-fold. For screening the library in which beneficial mutations from previous rounds were recombined, assays were conducted with 10 µL diluted lysate and 0.15 g/L SUS SFP SEQ ID NO: 1764 in 100 µL reaction volume with substrate loading of 8 mM rebaudioside A and co-substrate loadings of 0.2 mM ADP (Sigma, >95%) and 24 mM sucrose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 2 h. The reactions were diluted, quenched, and analyzed as described above. The resulting engineered variants with glucosyltransferase activity coupled with a SuS on rebaudioside A are listed in Table 50.1. For the round 11 saturation mutagenesis library, screening was performed as described above with the exception that a substrate loading of 10 mM rebaudioside A was used. The resulting engineered GT variant polypeptides are listed in Table 50.2.

Shake-flask scale cultures were grown, lysed, and lyophilized to powder for several variants.

(454) TABLE-US-00098 TABLE 50.1 β 1,2GT Round 11 Combinatorial Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 3082) RebD.sup.a 3181/3182 V164H/G200A/D236T +++ 3183/3184 G8S/I139V/G200A/D236T/N240E/D255P/N423R +++ 3185/3186 P107G/D236T/N240E/M325L/I440R + 3187/3188 V164H/V322L/M325L/K416R/N423R +++ 3189/3190 G8S/P107G/V164H/D255P ++ 3191/3192

P107G/I139V/G200A/N240E/V322L/H324G/M325L/K448A + 3193/3194 G8S ++ 3195/3196

V164H/P189R/D236T/N240E ++ 3197/3198 V164H/D236T/N423R + 3199/3200

V164H/D236T/N240E/I440R + 3201/3202 G8S/P107G/I139V/V164H/G415A/K416R/I440R/K448A +

3203/3204 G8S/V164H/N240E/N423R +++ 3205/3206 P107G/N423R ++ 3207/3208

T12S/V164H/I440R ++ 3209/3210 V164H/D236T + 3211/3212

G8S/V164H/G200A/D236T/N240E/V322L/I440R/K448A ++ 3213/3214
V164H/P189R/N240E/G252D/G415A/N423R + 3215/3216
V164H/G200A/D236T/D255P/V322L/H324G/N423R/I440R +++ 3217/3218 I139V + 3219/3220
G8S/P107G/I139V/D255P/V322L/M325L/G415A/K416R/I440R/ +++ K448A 3221/3222 V164H/P189R
+ 3223/3224 G8S/V164H/P189R/G200A/V322L/H324G/M325L/K416R/K448A + 3225/3226
G8S/D236T/N240E/G252D/K448A + 3227/3228 G8S/I139V/D255P/G415A/K416R ++ 3229/3230
I139V/V164H/D236T/N240E + 3231/3232 P107G/N240E/G252D/N423R/K448A ++ 3233/3234
G8S/I139V/V164H/D236T/K416R/N423R/I440R + 3235/3236 N240E/G252D/D255P/G415A/K448A
+++ 3237/3238 N240E/G252D/D255P/V322L/G415A/K416R ++ 3239/3240 G8S/I139V + 3241/3242
V164H/G200A/D236T/N240E/H324G/K416R/I440R + 3243/3244
G8S/G252D/D255P/V322L/M325L/K448A + 3245/3246 G200A/D236T/V322L/M325L/K416R +
3247/3248 G200A/V322L/M325L/G415A/K448A ++ 3249/3250 G415A/K416R/K448A ++ 3251/3252
G8S/V164H/K448A + 3253/3254 I139V/D255P/M325L/G415A/I440R ++ 3255/3256 P107G/I139V +
3257/3258 G8S/V164H/G252D/D255P/K448A + 3259/3260 G8S/V164H + 3261/3262
D236T/G415A/K416R + 3263/3264 G8S/P107G/V164H/N423R/I440R + 3265/3266
V164H/K416R/K448A ++ 3267/3268 V164H/V322L/M325L ++ 3269/3270
G8S/I139V/P189R/N240E/M325L/K416R + 3271/3272 G8S/N240E/N423R +++ 3273/3274
D255P/N423R +++ 3275/3276 M325L + 3277/3278 N240E + 3279/3280 H2- +++ .sup.aLevels of
increased production were determined relative to the reference polypeptide of SEQ ID NO: 3082, and
defined as follows: “+” = production at least 1.25-fold, but less than 1.44-fold; “++” = at least 1.44-fold,
but less than 1.62-fold; and “+++” = at least 1.62-fold increased production, as compared to the reference
polypeptide.

(455) TABLE-US-00099 TABLE 50.2 β 1,2GT Round 11 Saturation Mutagenesis Variants and RebD
Levels Amino Acid Differences SEQ ID NO: (Relative to SEQ ID Increased (nt/aa) NO: 3082)
RebD.sup.a 3385/3386 H73T + 3387/3388 N286L + 3389/3390 F355A ++ 3391/3392 P75H + 3393/3394
D389F ++ 3395/3396 E299L ++ 3397/3398 H2T + 3399/3400 L113I +++ 3401/3402 H73A +++
3403/3404 K309R ++ 3405/3406 P237M + 3407/3408 R446H + 3409/3410 H2N ++ 3411/3412 P237V
++ 3413/3414 T406N +++ 3415/3416 H73P + 3417/3418 E299R + 3419/3420 E256I + 3421/3422 D323P
+ 3423/3424 P189S + 3425/3426 T406F + 3427/3428 H2S + 3429/3430 T406Q ++ 3431/3432 E299A +
3433/3434 K34R ++ 3435/3436 N286S +++ 3437/3438 D305G + 3439/3440 E313D + 3441/3442 E256S
++ 3443/3444 H73V + 3445/3446 E256T + 3447/3448 E186G +++ 3449/3450 K422R ++ 3451/3452
G8S + 3453/3454 E186I + 3455/3456 T406G + 3457/3458 R446P ++ 3459/3460 T239A ++ 3461/3462
K422S + 3463/3464 E299V + 3465/3466 P237L ++ 3467/3468 L312V ++ 3469/3470 Q114V +++
3471/3472 H3I + 3473/3474 E221K + 3475/3476 L72Y ++ 3477/3478 L312S +++ 3479/3480 Q235M
+++ 3481/3482 E256L +++ 3483/3484 D389G + 3485/3486 K422C + 3487/3488 L312T +++ 3489/3490
E438T +++ .sup.aLevels of increased production were determined relative to the reference polypeptide of
SEQ ID NO: 3082, and defined as follows: “+” = production at least 1.15-fold, but less than 1.2-fold;
“++” = at least 1.2-fold, but less than 1.27-fold; and “+++” = at least 1.27-fold increased production, as
compared to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP to
Rebaudioside A

(456) A shake flask powder loading dose response experiment was performed to characterize activity of
the engineered round 11 variants on rebaudioside A. Levels of 0.006-0.2 g/L shake flask powder (SFP)
were added to a 100 μ L total reaction volume containing 50 mM potassium phosphate buffer, pH 6, 10
mM rebaudioside A, 0.2 mM ADP, 24 mM sucrose, and 0.15 g/L SUS SFP SEQ ID NO: 1764. The
reaction was performed at 60° C. in a Thermotron® titre-plate shaker at 300 RPM for 2 h. The reactions
were diluted, quenched, and analyzed as described above. The round 11 variant that was most improved
on rebaudioside A at 0.025 g/L SFP loading was SEQ ID NO: 3244, which had mutations G8S, G252D,
D255P, V322L, M325L, and K448A relative to SEQ ID NO: 1764. This variant, SEQ ID NO: 3244, was
used for further directed evolution.

(457) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside I

(458) Directed evolution of the GT encoded by SEQ ID NO: 3081 was carried out by constructing
libraries of variant genes in which mutations associated with improved glycosyltransferase activity

toward rebaudioside I in previous rounds were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a round (“Round 11.04”) of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and rebaudioside I. Cells were lysed with 400 μ L lysis buffer. Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 3081 variants with lysate loading of 25 μ L lysate in 100 μ L reactions and with substrate loading of 1 mM rebaudioside I and co-substrate loading of 1 mM ADP-glucose. The following reaction conditions were used: 50 mM KPhos buffer, pH 6, 40° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 18 h. The reactions were diluted 1:4 in water and then quenched by adding 25 μ L of the diluted assay to 75 μ L acetonitrile with 0.2% formic acid. The resulting mixture was precipitated by centrifugation at 4° C. for 10 m. The supernatants were diluted 1:5 in water and analyzed by LC-MS/MS as described in Table 44.1. The resulting 52 engineered variants with glucosyltransferase activity on rebaudioside I are listed in Table 50.3.

(459) TABLE-US-00100 TABLE 50.3 β 1,2GT Round 11 Combinatorial Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 3082) RebM.sup.a

3281/3282 L126Q/P128A/F155R/R161G/N199A + 3283/3284 W20Y/P128A/K160S/R161T/G198P + 3285/3286 W20Y/L126Q/P128A/K160S/R161T/G198H + 3287/3288 L126Q/P128A/K160S/N199A/P369N + 3289/3290 P128A/K160S/R161G/P369N + 3291/3292 P128A/F155R/K160S/G198P + 3293/3294 L126H/P128A/K160S/G198H/P369N + 3295/3296 P128A/G198H/N199A/P369N + 3297/3298 N199A/P369N ++ 3299/3300 L126Q/G198P/P369N + 3301/3302 W20Y/L126H/K160S + 3303/3304 L126Q/N199A/P369N + 3305/3306 W20Y/F155R/R161T + 3307/3308 W20Y/L126H/F155R/K160S/G198P + 3309/3310 F155R/R161G/G198H/N199A + 3311/3312 W20Y/L126H/P128A/R161T + 3313/3314 L126H/K160S/G198P/P369N + 3315/3316 L126Q/P128A/K160S/P369N + 3317/3318 V121F/L126H/P128A/R161T/P369N ++ 3319/3320 W20Y/P128A/F155R/N199A + 3321/3322 P128A/F155R/K160S/R161T/N199A + 3323/3324 W20Y/F155R/K160S/R161G + 3325/3326 L126H/K160S/R161G/N199A/P369N ++ 3327/3328 W20Y/L126Q/P128A/R161T/G198H/N199A + 3329/3330 L126H/A196V/G198P/P369N + 3331/3332 P128A/F155R/K160S/R161T ++ 3333/3334 W20Y/P128A/F155R/K160S/R161T/G198H + 3335/3336 W20Y/L126H/P128A/F155R/R161T ++ 3337/3338 P128A/N199A/P369N ++ 3339/3340 R161T/G198P/P369N ++ 3341/3342 W20Y/L126H/K160S/R161T ++ 3343/3344 L126H/P128A/R161G/P369N ++ 3345/3346 L126Q/P128A/F155R/K160S/R161T/N199A ++ 3347/3348 K160S/R161G/P369N + 3349/3350 K160S/R161T/P369N ++ 3351/3352 W20Y/P128A/F155R/R161T/N199A ++ 3353/3354 L126H/P128A/F155R/K160S/R161T + 3355/3356 W20Y/L126Q/R161G + 3357/3358 F155R/N199A ++ 3359/3360 W20Y/L126Q/P128A/K160S/R161T +++ 3361/3362 L126H/P128A/R161G/N199A/P369N +++ 3363/3364 P128A/F155R/N199A +++ 3365/3366 R161T/P369N +++ 3367/3368 L126Q/P128A/K160S/R161T/P369N +++ 3369/3370 W20Y/L126Q/F155R/R161G +++ 3371/3372 L126Q/P369N +++ 3373/3374 W20Y/F155R/R161T/N199A +++ 3375/3376 W20Y/K160S/R161T/G198H +++ 3377/3378 W20Y/F155R/K160S +++ 3379/3380 L126H/P128A/P369N ++ 3381/3382 P128A/P369N ++ 3383/3384 P128A/R161G/N199A/P369N ++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 3082, and defined as follows: “+” = production at least 3.24-fold, but less than 4.14-fold; “++” = at least 4.14-fold, but less than 6-fold; and “+++” = greater than 6-fold increased production, as compared to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP to Rebaudioside I

(460) A shake flask powder loading dose response experiment was performed to characterize activity of the engineered round 11.04 variants on rebaudioside I. Levels of 0.16-5 g/L shake flask powder (SFP) were added to a 100 μ L total reaction volume containing 50 mM potassium phosphate buffer, pH 6, 1 mM rebaudioside 1, and 1 mM ADP-glucose. The reaction was performed at 50° C. in a Thermotron® titre-plate shaker at 300 RPM for 4 h. The reactions were diluted, quenched, and analyzed as described above. The production levels of rebaudioside M by the round 11.04 variants at 0.3 g/L SFP loading are shown in Table 50.4. All 5 of the variants listed in Table 50.4 had higher activities than SEQ ID NO: 3082 on RebI. The variant with the mutations L126Q, P128A, F155R, K160S, R161T, and N199A (SEQ ID NO: 3346) and its encoding polynucleotide (SEQ ID NO: 3345) were most improved and were selected for further

directed evolution for the glucosylation of rebaudioside I.

(461) TABLE-US-00101 TABLE 50.4 β 1,2GT Round 11 Shake Flask Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 3082) RebM.sup.a 3295/3296 P128A/G198H/N199A/P369N + 3345/3346 L126Q/P128A/F155R/K160S/R161T/N199A ++ 3377/3378 W20Y/F155R/K160S ++ 3381/3382 P128A/P369N ++ 3379/3380 L126H/P128A/P369N ++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 3082, and defined as follows: “+” = production at least 1.15-fold, but less than 1.8-fold; and “++” = at least 1.8-fold increased production, as compared to the reference polypeptide.

Example 51

Beta-1,2-ADP-Glycosyltransferase Variants of SEQ ID NO: 3244

(462) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 3244 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 3243 was carried out by constructing combinatorial libraries of variant genes. Libraries recombined mutations associated with improved activity in previous rounds and subjected certain structural features of the enzyme with saturation mutagenesis. These libraries were then plated, grown, and screened using the HTP assay described below to provide a twelfth round (“Round 12”) of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. Fifty engineered variants were identified from the recombined beneficial mutations (Table 51.1), and 31 were identified from saturation mutagenesis (Table 51.2).

(463) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(464) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 3243 variants. Lysis buffer volume was 400 μ L, and the lysate was diluted 40-fold. For screening the library in which beneficial mutations from previous rounds were recombined, diluted lysates were preincubated at 62° C. for 0.5 h in an Eppendorf thermocycler. Assays were then conducted with 10 μ L diluted lysate and 0.15 g/L SUS SFP SEQ ID NO: 1764 in 100 μ L reaction volume with substrate loading of 15 mM rebaudioside A and co-substrate loadings of 0.2 mM ADP (Sigma, >95%) and 37.5 mM sucrose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4.5 h. The reactions were diluted, quenched, and analyzed as described above. The resulting engineered variants with glucosyltransferase activity coupled with SUS SEQ ID NO: 1764 on rebaudioside A are listed in Table 51.1. For the round 12 saturation mutagenesis library, screening was performed as described above, with the exception that the diluted lysates were preincubated at 65° C. for 0.5 h. The resulting engineered GT variant polypeptides are listed in Table 51.2. Shake-flask scale cultures were grown, lysed, and lyophilized to powder for variants listed in Table 51.3.

(465) TABLE-US-00102 TABLE 51.1 β 1,2GT Round 12 Combinatorial Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 3244) RebD.sup.a 3491/3492

K106A/N172R/G200A/E210V/N240E/ ++ A242I/E408P/G415A/K416R/N423R 3493/3494

N240E/A242I/G415A/N423R ++ 3495/3496 N240E/A242I +++ 3497/3498

V164H/N240E/A242I/G415R/N423R + 3499/3500 G200A/N240E/A242I/G407S ++ 3501/3502

K106A/V164H/G200A/E210V/G415A +++ 3503/3504 N240E +++ 3505/3506

V164H/G200A/E408P/G415A/K416R ++ 3507/3508 T409L/G415R + 3509/3510 G200A/E408P/T409L

++ 3511/3512 E210V/N240E +++ 3513/3514 V164H/N240E/A242I/E408P + 3515/3516

K106A/V164H/G200A/N240E/E408P/ ++ T409L/K416R 3517/3518

G200A/A242I/G415A/K416R/N423R + 3519/3520 K106A/V164H/E210V/N240E/E408P/ +

T409L/K416R 3521/3522 N172R/N240E + 3523/3524 N172R/N240E/A242I/G415R + 3525/3526

K106A/N240E/A242I/T409L + 3527/3528 K106A/V164H/E210V/E408P/T409L + 3529/3530

K106A/E210V/N240E/E408P/G415R ++ 3531/3532 K106A/N172R/G200A/E210V/A242I +++

3533/3534 N240E/E408P +++ 3535/3536 N172R/G200A/E408P/K416R + 3537/3538

K106A/N172R/E408P/T409L ++ 3539/3540 V164H/N172R/N240E/A242I/G415R ++ 3541/3542

V164H/N172R/A242I/E408P + 3543/3544 N172R/G200A/N240E + 3545/3546

N172R/G200A/E210V/N240E/E408P + 3547/3548 K106A/N172R/T409L/N423R ++ 3549/3550

N240E/A242I/E408P/K416R + 3551/3552 K106A/N240E/A242I ++ 3553/3554 N172R/E210V/G415A

++ 3555/3556 V164H/N172R/G200A/E210V/A242I/T409L + 3557/3558 K106A/A242I/E408P/T409L

+++ 3559/3560 K106A/N172R/N240E/A242I/T409L/G415R ++ 3561/3562 N240E/G415R ++
 3563/3564 K106A/V164H/N172R/A242I/T409L/K416R + 3565/3566
 G200A/E210V/N240E/A242I/T409L + 3567/3568 V164H/N423R ++ 3569/3570
 K106A/V164H/G200A/E210V/A242I +++ 3571/3572 K106A/V164H/N240E/G415A ++ 3573/3574
 K106A/G200A/E210V/T409L +++ 3575/3576 K106A/V164H/T409L/G415A/N423R + 3577/3578
 K106A/V164H/N172R/G200A/T409L + 3579/3580 N172R/N240E/T409L + 3581/3582
 K106A/N172R/A242I/K416R/N423R + 3583/3584 K106A/E210V/N240E + 3585/3586
 K106A/N172R/A242I + 3587/3588 K106A/N172R/N240E +++ 3589/3590
 K106A/V164H/N172R/E210V/N240E/G415R + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 3244, and defined as follows: “+” = production at least 1.8-fold, but less than 1.9-fold; “++” = at least 1.9-fold, but less than 2.07-fold; and “+++” = at least 2.07-fold increased production, as compared to the reference polypeptide.

(466) TABLE-US-00103 TABLE 51.2 β 1,2GT Round 12 Saturation Mutagenesis Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 3244)
 RebD.sup.a 3591/3592 C46T + 3593/3594 L49M + 3595/3596 L49P +++ 3597/3598 V134S + 3599/3600
 K143P +++ 3601/3602 V134A +++ 3603/3604 L49A ++ 3605/3606 V278L + 3607/3608 V179T +
 3609/3610 P232T + 3611/3612 C46V +++ 3613/3614 S105A ++ 3615/3616 L49Q ++ 3617/3618 L42I +
 3619/3620 V179D ++ 3621/3622 I290L ++ 3623/3624 Q35D ++ 3625/3626 L42F ++ 3627/3628 V134C
 ++ 3629/3630 L49S ++ 3631/3632 Q441R + 3633/3634 L42V + 3635/3636 A381G + 3637/3638 L401V
 + 3639/3640 V278I + 3641/3642 R14K + 3643/3644 K373R + 3645/3646 Q441I + 3647/3648 S336A
 +++ 3649/3650 M181L +++ 3651/3652 V179A + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 3244, and defined as follows: “+” = production at least that of the reference polypeptide, but less than 1.29-fold; “++” = at least 1.29-fold, but less than 1.4-fold; and “+++” = production at least 1.4-fold increased production, relative to the reference polypeptide.
 Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP to Rebaudioside A

(467) A shake flask powder loading dose response experiment was performed to characterize activity of the engineered round 12 variants on rebaudioside A. Levels of 0.006-0.2 g/L shake flask powder (SFP) were preincubated at 62° C. for 0.5 h and then added to a 100 μ L total reaction volume containing 50 mM potassium phosphate buffer, pH 6, 15 mM rebaudioside A, 0.2 mM ADP, 37.5 mM sucrose, and 0.15 g/L SUS SFP SEQ ID NO: 1764. The reaction was performed at 50° C. and 60° C. in a Thermotron® titre-plate shaker at 300 RPM for 4.5 h. The reactions were diluted, quenched, and analyzed as described above. The conversion of the round 12 variants on rebaudioside A at 0.025 g/L SFP loading are listed in Table 51.3. All 7 of the variants listed in Table 51.3 had higher activities than SEQ ID NO: 3244 at 60° C., and 4 did not lose significant activity at 50° C. The variant with the mutations K106A, V164H, G200A, E210V, and G415A (SEQ ID NO: 3502) and its encoding polynucleotide (SEQ ID NO: 3501) were selected for further directed evolution for the glucosylation of rebaudioside A.

(468) TABLE-US-00104 TABLE 51.3 β 1,2GT Round 12 Shake Flask Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased RebD, (nt/aa) (Relative to SEQ ID NO: 3244) 60° C..sup.a
 3501/3502 K106A/V164H/G200A/E210V/G415A +++ 3515/3516
 K106A/V164H/G200A/N240E/E408P/ + T409L/K416R 3531/3532
 K106A/N172R/G200A/E210V/A242I ++ 3539/3540 V164H/N172R/N240E/A242I/G415R ++
 3573/3574 K106A/G200A/E210V/T409L +++ 3581/3582 K106A/N172R/A242I/K416R/N423R ++
 3589/3590 K106A/V164H/N172R/E210V/N240E/G415R ++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 3244, and defined as follows: “+” = production at least 1.24-fold, but less than 1.4-fold; “++” = at least 1.4-fold, but less than 1.5-fold; and “+++” = at least 1.5-fold increased production, relative to the reference polypeptide.

Example 52

Beta-1,2-ADP-Glycosyltransferase Variants of SEQ ID NO: 3346 for Rebaudioside M Production From Rebaudioside I

(469) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 3346 for improved glucosylation of steviol glycosides are described. Directed evolution of the GT encoded by SEQ ID NO: 3345 was carried out by constructing a library that subjected certain structural

the surface of the enzyme to saturation mutagenesis. The library was then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a round (“Round 12 Rebl”) of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and rebaudioside I.

(470) HTP Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside M

(471) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 3345 variants. Lysis buffer volume was 400 μ L, and assays were then conducted with 25 μ L lysate and 0.1 g/L SUS SFP SEQ ID NO: 1804 in 100 μ L reaction volume with substrate loading of 1 mM rebaudioside I and co-substrate loadings of 1 mM ADP (Sigma, >95%) and 15 mM sucrose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. The reactions were diluted, quenched, and analyzed as described in Example 50. The resulting engineered variants with glucosyltransferase activity coupled with a SuS on rebaudioside I are shown in Table 52.1.

(472) TABLE-US-00105 TABLE 52.1 β 1,2GT Round 12 Rebl Saturation Mutagenesis Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 3346) RebM.sup.a 3759/3760 G200T + 3761/3762 F152L + 3763/3764 P197D + 3765/3766 P197Q ++ 3767/3768 I202W ++ 3769/3770 K143H +++ 3771/3772 K187S ++ 3773/3774 L195R + 3775/3776 L195Y ++ 3777/3778 G201W ++ 3779/3780 Y268Q + 3781/3782 P91M + 3783/3784 G200P ++ 3785/3786 P197W + 3787/3788 L21M + 3789/3790 G200E ++ 3791/3792 F125C +++ 3793/3794 Q127G ++ 3795/3796 G201R +++ 3797/3798 F264S + 3799/3800 K143P/A150S + 3801/3802 L21P +++ 3803/3804 A130T/K187Q +++ 3805/3806 I202A + 3807/3808 E186V + 3809/3810 G201A ++ 3811/3812 L195S + 3813/3814 F264T + 3815/3816 G200R +++ 3817/3818 F156Q + 3819/3820 M364W + 3821/3822 L195G ++ 3823/3824 E186N ++ 3825/3826 L145W + 3827/3828 Q365L +++ 3829/3830 G201S ++ 3831/3832 Y268W ++ 3833/3834 L195T + 3835/3836 F125M + 3837/3838 M364S + 3839/3840 Y268F ++ 3841/3842 G415D + 3843/3844 F125V + 3845/3846 L195F +++ 3847/3848 F156P ++ 3849/3850 P197L + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 3346, and defined as follows: “+” = production at least 1.07-fold, but less than 1.19-fold; “++” = at least 1.19-fold, but less than 1.33-fold; and “+++” = at least 1.33-fold increased production, as compared to the reference polypeptide.

Example 53

Beta-1,2-ADP-Glycosyltransferase Variants of SEQ ID NO: 3502

(473) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 3502 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 3501 was carried out by constructing combinatorial libraries of variant genes. Libraries recombined mutations associated with improved activity in previous rounds and subjected certain structural features of the enzyme with saturation mutagenesis. These libraries were then plated, grown, and screened using the HTP assay described below to provide a thirteenth round (“Round 13”) of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. 53 engineered variants were identified from the recombined beneficial mutations (Table 53.1), and 24 were identified from saturation mutagenesis (Table 53.2).

(474) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(475) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 3501 variants. Lysis buffer volume was 400 μ L, and the lysate was diluted 40-fold. For screening the library in which beneficial mutations from previous rounds were recombined, diluted lysates were preincubated at 69° C. for 0.8 h in an Eppendorf thermocycler. Assays were then conducted with 10 μ L diluted lysate and 0.15 g/L SUS SFP SEQ ID NO: 1764 in 100 μ L reaction volume with substrate loading of 15 mM rebaudioside A and co-substrate loadings of 0.2 mM ADP (Sigma, >95%) and 37.5 mM sucrose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. The reactions were diluted, quenched, and analyzed as described above. The resulting engineered variants with glucosyltransferase activity coupled with SUS SEQ ID NO: 1764 on rebaudioside A are listed in Table 53.1. For the round 13 saturation mutagenesis library, screening was performed as described above with the exceptions that the diluted lysates were preincubated at 68° C. for 0.5 h, 0.1 g/L SUS SFP SEQ ID NO: 1804 was used, and either 0.1 or 0.2 mM ADP was used. The resulting engineered GT variant polypeptides are listed in Table

53.2. Shake-flask scale cultures were grown, lysed, and lyophilized to powder for the variants listed in Table 53.3.

(476) TABLE-US-00106 TABLE 53.1 β 1,2GT Round 13 Combinatorial Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 3502) RebD.sup.a 3653/3654 P237L + 3655/3656 E186G/P237V/N286S + 3657/3658 H2-/H5N/L113I/E186G/P237V/T406Q +++ 3659/3660 H5N/E186G/P237V/K245T/E256I/T406Q + 3661/3662 E186G/P237L ++ 3663/3664 K34R/L113I/E186G/P237L/T406N + 3665/3666 H5N/K34R/H73A/L113I/T406N + 3667/3668 H5N/E186G/P237V/T406Q ++ 3669/3670 H73A/P237V + 3671/3672 H5N/L113I/P237V/T406N + 3673/3674 E186G ++ 3675/3676 H2-/H73A/E186G/P237V + 3677/3678 H73A/E186G/P237L/T406Q +++ 3679/3680 H5N/E256L/T406Q ++ 3681/3682 L113I/E186G/T406Q + 3683/3684 E186G/T406N + 3685/3686 E186G/T406Q + 3687/3688 H5N/H73A/E186G/T406N ++ 3689/3690 P189S/N333H/F355A/S421Q + 3691/3692 H2S/N286Q/F355A/K416R/K422R ++ 3693/3694 H2S/P189S/N286Q + 3695/3696 H73P/N172R/N240E/A242I/E408P + 3697/3698 H73P/Q235M/N240E/E408P + 3699/3700 H73P/N172R/Q235M/T239A/N240E ++ 3701/3702 H73P/N172R/T239A/N240E + 3703/3704 H73P/T239A/A242I ++ 3705/3706 T239A/N240E + 3707/3708 N172R/Q235M/N240E + 3709/3710 L72Y/N172R/N240E/A242I + 3711/3712 T239A/N240E/A242I/E256S/D323P + 3713/3714 N240E/A242I/E256S/E438T + 3715/3716 N172R/E188G/D323P ++ 3717/3718 H73P/Q235M/N240E/D323P ++ 3719/3720 A242I/E408P + 3721/3722 T239A/N240E/E408P +++ 3723/3724 H73P/N172R + 3725/3726 H73P/N172R/T239A/N240E/A242I +++ 3727/3728 T239A/A242I/E408P + 3729/3730 L72Y/H73P/N172R/Q235M/N240E/A242I/E438T ++ 3731/3732 Q235M/T239A/N240E ++ 3733/3734 N172R + 3735/3736 Q235M/N240E/A242I +++ 3737/3738 N172R/T239A/N240E/A242I/D323P/E408P ++ 3739/3740 H73P/N172R/Q235M/T239A/A242I + 3741/3742 N172R/Q235M ++ 3743/3744 T239A/N240E/A242I/D323P/E408P ++ 3745/3746 H73P/N172R/N240E/E408P +++ 3747/3748 L72Y/N172R/T239A/N240E/A242I/E408P +++ 3749/3750 N172R/Q235M/T239A/E408P + 3751/3752 N172R/N240E/E299A/D323P ++ 3753/3754 N172R/Q235M/N240E/A242I/E438T +++ 3755/3756 N172R/N240E +++ 3757/3758 N172R/Q235M/T239A/N240E/A242I +++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 3502, and defined as follows: “+” = production at least 1.58-fold, but less than 1.79-fold; “++” = at least 1.79-fold, but less than 2.07-fold; and “+++” = at least 2.07-fold increased production, as compared to the reference polypeptide.

(477) TABLE-US-00107 TABLE 53.2 β 1,2GT Round 13 Saturation Mutagenesis Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 3502) RebD.sup.a 3851/3852 E186R ++ 3853/3854 K187T ++ 3855/3856 N199Y + 3857/3858 Q132H ++ 3859/3860 S153G + 3861/3862 Q132T + 3863/3864 A196S + 3865/3866 S153A + 3867/3868 Q127V +++ 3869/3870 N199A + 3871/3872 I144V + 3873/3874 E186A ++ 3875/3876 K187A + 3877/3878 Q127L + 3879/3880 Q132K + 3881/3882 G96P ++ 3883/3884 K187R +++ 3885/3886 N199S ++ 3887/3888 E186G +++ 3889/3890 G96A ++ 3891/3892 A200S + 3893/3894 F156W + 3895/3896 Q127I +++ 3897/3898 F155M +++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 3502, and defined as follows: “+” = production at least that of the reference polypeptide, but less than 1.15-fold; “++” = at least 1.15-fold, but less than 1.18-fold; and “+++” = at least 1.18-fold increased production, as compared to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP to Rebaudioside A

(478) A shake flask powder loading dose response experiment was performed to characterize activity of the engineered round 13 variants on rebaudioside A. Levels of 0.006-0.2 g/L shake flask powder (SFP) were preincubated at 69° C. for 50 mM and then added to a 100 μ L total reaction volume containing 50 mM potassium phosphate buffer, pH 6, 15 mM rebaudioside A, 0.2 mM ADP, 37.5 mM sucrose, and 0.1 g/L SUS SFP SEQ ID NO: 1804. The reaction was performed at 60° C. in a Thermotron titre-plate shaker at 300 RPM for 4 h. The reactions were diluted, quenched, and analyzed as described above. The conversion of the round 13 variants on rebaudioside A at 0.025 g/L SFP loading are listed in Table 53.3. All 10 of the variants listed in Table 53.3 had higher activities than SEQ ID NO: 3502. The variant with the mutations H73P, N172R, N240E, A242I, and E408P (SEQ ID NO: 3696) and its encoding

polynucleotide (SEQ ID NO: 3695) were the most improved and were selected for further directed evolution for the glucosylation of rebaudioside A.

(479) TABLE-US-00108 TABLE 53.3 β 1,2GT Round 13 Shake Flask Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 3502) RebD.sup.a 3695/3696 H73P/N172R/N240E/A242I/E408P +++ 3689/3690 P189S/N333H/F355A/S421Q +++ 3675/3676 H2-/H73A/E186G/P237V ++ 3745/3746 H73P/N172R/N240E/E408P ++ 3691/3692

H2S/N286Q/F355A/K416R/K422R ++ 3657/3658 H2-/H5N/L113I/E186G/P237V/T406Q + 3699/3700 H73P/N172R/Q235M/T239A/N240E + 3703/3704 H73P/T239A/A242I + 3693/3694 H2S/P189S/N286Q + 3677/3678 H73A/E186G/P237L/T406Q + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 3502 and defined as follows: “+” = production at least 1.79-fold that of the reference polypeptide, but less than 3-fold; “++” = at least 3-fold increased production but less than 3.6-fold; and “+++” = at least 3.6-fold increased production relative to reference polypeptide.

Example 54

Beta-1,2-ADP-Glycosyltransferase Variants of SEQ ID NO: 3696

(480) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 3696 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 3695 was carried out by constructing combinatorial libraries of variant genes. Libraries recombined mutations associated with improved activity in previous rounds and subjected certain structural features of the enzyme with saturation mutagenesis. These libraries were then plated, grown, and screened using the HTP assay described below to provide a fourteenth round (“Round 14”) of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. Seventy-five engineered variants were identified from the recombined beneficial mutations (Table 54.1), and 49 were identified from saturation mutagenesis (Table 54.2).

(481) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(482) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 3695 variants. Lysis buffer volume was 400 μ L, and the lysate was diluted 40-fold. For screening the library in which beneficial mutations from previous rounds were recombined, diluted lysates were preincubated at 72° C. for 0.5 h in an Eppendorf thermocycler. Assays were then conducted with 10 μ L diluted lysate and 0.1 g/L sucrose synthetase (SUS) SFP SEQ ID NO: 1804 in 100 μ L reaction volume with substrate loading of 20 mM rebaudioside A and co-substrate loadings of 0.2 mM ADP (Sigma, >95%) and 50 mM sucrose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. The reactions were diluted, quenched, and analyzed as described above. The resulting engineered variants with glucosyltransferase activity coupled with a SuS on rebaudioside A are listed in Table 54.1. For the round 14 saturation mutagenesis library, screening was performed as described above with the exceptions that the lysate was diluted 80-fold and was preincubated at 70° C. for 0.5 h. The resulting engineered GT variant polypeptides are listed in Table 54.2. Shake-flask scale cultures were grown, lysed, and lyophilized to powder for variants listed in Table 54.3.

(483) TABLE-US-00109 TABLE 54.1 β 1,2GT Round 14 Combinatorial Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 3696) RebD.sup.a 3899/3900 I26V/L42V/C46V/V134A + 3901/3902 L42V/C46V/L49P/V134C/Q235R/F355A/L401V ++ 3903/3904 L42I/C46V/L49S/V134C/E186G + 3905/3906 I26V/L49Q/V134A + 3907/3908 C46V/L49Q/V134T + 3909/3910 I26V/V134T/L401V +++ 3911/3912 I26V/L42V/V134C/L401V + 3913/3914 L42V/C46V/L49A/V134A/L401V + 3915/3916 I26V/L42I/L49Q/V134C/E186G/F355A +++ 3917/3918 I26V/L42I/L49Q/V134C/L401V ++ 3919/3920 I26V/L42V/L49S/V134C + 3921/3922 L42V/L49A/V134C/E186G/F355A +++ 3923/3924 I26V/L42V/C46V/L49A/V134C/E186G + 3925/3926 L42V/C46V/L49S/T97P/V134C/L401V ++ 3927/3928 L42V/C46V/L49A/V134T/E186G/F355A + 3929/3930 I26V/V134A/E186G/F355A/L401V + 3931/3932 L42I/C46V/L49S/E186G + 3933/3934 I26V/L42V/C46V/L49A ++ 3935/3936 I26V/F355A/L401V + 3937/3938 L42V/V134C/E186G ++ 3939/3940 L42I/C46V/L49A/V134A ++ 3941/3942 L42V/L49S/V134A/E186G/L401V ++ 3943/3944 E186G/F355A + 3945/3946 I26V/L42V ++ 3947/3948

V134C/L401V + 3949/3950 L42V/L49S/V134C/F355A/L401V + 3951/3952 I26V/V134A +++
 3953/3954 I26V/V134C/F355A + 3955/3956 I26V/L42V/C46V/L49A/V134A +++ 3957/3958
 L49S/V134C/F355A + 3959/3960 L42V/E186G/F355A ++ 3961/3962 H2-/K143P/P232T/K373R/Q441R
 + 3963/3964 H2-/K143P/K373R/Q441R + 3965/3966 H5N/P237L/K373R/K422R + 3967/3968
 S105A/K373R/Q441R + 3969/3970 H5N/K143P/P189S/P232T/P237V/Q441R + 3971/3972
 K143P/P232T + 3973/3974 K143P/P189S/P232T/K422R/Q441R + 3975/3976
 H5N/K143P/P232T/K373R/K422R/Q441R + 3977/3978 S105A/K143P/P232T/P237V/K373R/K416R/ +
 K422R/Q441R 3979/3980 K143P/P237V + 3981/3982 H2-/H5N/Q35D/S105A/K143P/P237V/K373R/ +
 K416R/K422R 3983/3984 P189S/P232T/K373R/K416R/K422R/Q441R ++ 3985/3986
 S105A/P237L/K373R/K422R/Q441R +++ 3987/3988 S105A/K373R +++ 3989/3990
 H5N/S105A/P237V/K416R/K422R/Q441R + 3991/3992 H2-/H5N/Q35D/P232T/V278L/K373R/K416R
 ++ 3993/3994 S105A/P189S/P237V ++ 3995/3996 H2-/H5N/S105A/K143P/P232T/K373R/ +
 K416R/K422R 3997/3998 H2-/H5N/V278L +++ 3999/4000 P232T/P237L ++ 4001/4002 P237L/K422R
 + 4003/4004 H5N/K143P/K373R +++ 4005/4006 K143P +++ 4007/4008
 Q35D/S105A/P189S/P232T/K373R/K416R + 4009/4010 H2-/H5N/Q35D/K143P/P232T/P237L/K416R/
 + K422R/Q441R 4011/4012 S105A/P237L/V278L/K373R/K416R/K422R ++ 4013/4014
 H5N/S105A/P232T/P237L/K373R/K416R/Q441R ++ 4015/4016
 H5N/K143P/P232T/P237L/K416R/K422R +++ 4017/4018 S105A/K143P/V278L/K373R ++ 4019/4020
 K143P/P237L/K422R/Q441R +++ 4021/4022 H5N/P232T/K416R/K422R + 4023/4024
 H5N/Q35D/P232T/K373R/K416R/K422R ++ 4025/4026 P189S/P237L/V278I/K416R/Q441R ++
 4027/4028 H5N/K143P/P237L/V278L/K373R/K416R/K422R ++ 4029/4030
 K143P/P189S/P237L/K373R/K416R/K422R +++ 4031/4032 K143P/P189S +++ 4033/4034
 H5N/S105A/P237L/V278I/K422R/Q441R + 4035/4036 H5N/K373R/K416R ++ 4037/4038
 P237V/K373R/Q441R ++ 4039/4040 Q35D + 4041/4042 H2-/S105A/K143P/P232T/P237L/V278L/ ++
 K373R/Q441R 4043/4044 H5N/P189S/P237V/V278I/K373R/K416R/ + K422R/Q441R 4045/4046
 H5N/K143P/P232T/P237L/V278L ++ 4047/4048 P237V/K373R + .sup.aLevels of increased production
 were determined relative to the reference polypeptide of SEQ ID NO: 3696, and defined as follows: “+” =
 production at least 1.5-fold, but less than 1.75-fold; “++” = at least 1.75-fold, but less than 2-fold; and
 “+++” = at least 2-fold increased production, as compared to the reference polypeptide.

(484) TABLE-US-00110 TABLE 54.2 β 1,2GT Round 14 Saturation Mutagenesis Variants and RebD
 Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 3696)
 RebD.sup.a 4049/4050 A426L + 4051/4052 S352A ++ 4053/4054 T239F + 4055/4056 R176Q +++
 4057/4058 Q235R +++ 4059/4060 F158W + 4061/4062 H330C ++ 4063/4064 T239Y ++ 4065/4066
 D243Y +++ 4067/4068 T239P ++ 4069/4070 H164E +++ 4071/4072 D249H + 4073/4074 D243R +
 4075/4076 H3N + 4077/4078 R427S + 4079/4080 H164F + 4081/4082 I50P + 4083/4084 S8R +
 4085/4086 T239G + 4087/4088 T239M ++ 4089/4090 S8V ++ 4091/4092 D249G ++ 4093/4094 D243G
 ++ 4095/4096 P237A + 4097/4098 D243H ++ 4099/4100 D243S ++ 4101/4102 H164L + 4103/4104
 P237L + 4105/4106 D193F + 4107/4108 R176P +++ 4109/4110 I50V +++ 4111/4112 R176T +
 4113/4114 H164A + 4115/4116 A101L ++ 4117/4118 I248T + 4119/4120 T223A ++ 4121/4122
 T223L/D243G + 4123/4124 Q137R ++ 4125/4126 R427T ++ 4127/4128 E61Y + 4129/4130 E240V +++
 4131/4132 E61N +++ 4133/4134 M364Q + 4135/4136 Q301A ++ 4137/4138 K62A ++ 4139/4140
 R161L + 4141/4142 D323T + 4143/4144 D244N +++ 4145/4146 T239Q + .sup.aLevels of increased
 production were determined relative to the reference polypeptide of SEQ ID NO: 3696, and defined as
 follows: “+” = production at least 1.1-fold, but less than 1.18-fold; “++” = at least 1.18-fold, but less than
 1.24-fold; and “+++” = at least 1.24-fold increased production, as compared to the reference polypeptide.
 Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP to
 Rebaudioside A

(485) A shake flask powder loading dose response experiment was performed to characterize activity of
 the engineered round 14 variants on rebaudioside A. Levels of 0.006-0.2 g/L shake flask powder (SFP)
 were preincubated at 72° C. for 0.5 h and then added to a 100 μ L total reaction volume containing 50 mM
 potassium phosphate buffer, pH 6, 20 mM rebaudioside A, 0.2 mM ADP, 50 mM sucrose, and 0.1 g/L
 SUS SFP SEQ ID NO: 1804. The reaction was performed at 60° C. with preincubated SFP or at 55° C.
 with SFP that had not undergone preincubation in a Thermotron® titre-plate shaker at 300 RPM for 4 h.

The reactions were diluted, quenched, and analyzed as described above. The production levels of rebaudioside D by the round 14 variants at 0.05 g/L SFP loading are shown in Table 54.3. Three out of the 8 variants listed in Table 54.3 had higher activities under both conditions than SEQ ID NO: 3696. The variant with the mutations I26V, L42V, C46V, L49A, and V134A (SEQ ID NO: 3956) and its encoding polynucleotide (SEQ ID NO: 3955) were the most improved taking into account both conditions and were selected for further directed evolution for the glucosylation of rebaudioside A.

(486) TABLE-US-00111 TABLE 54.3 β 1,2GT Round 14 Shake Flask Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased RebD, (nt/aa) (Relative to SEQ ID NO: 3696) 60° C..sup.a
3899/3900 I26V/L42V/C46V/V134A ++ 3921/3922 L42V/L49A/V134C/E186G/F355A ++ 3955/3956
I26V/L42V/C46V/L49A/V134A ++ 3961/3962 H2-/K143P/P232T/K373R/Q441R + 3969/3970
H5N/K143P/P189S/P232T/P237V/Q441R + 3987/3988 S105A/K373R + 3997/3998 H2-/H5N/V278L +
4041/4042 H2-/S105A/K143P/P232T/P237L/ + V278L/K373R/Q441R .sup.a Levels of increased
production were determined relative to the reference polypeptide of SEQ ID NO: 3696, and defined as
follows: “+” = production at least 1.13-fold, but less than 1.4-fold; and “++” = at least 1.4-fold increased
production, as compared to the reference polypeptide.

Example 55

Beta-1,2-ADP-Glycosyltransferase Variants of SEQ ID NO: 3956

(487) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 3956 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 3955 was carried out by constructing combinatorial libraries of variant genes. Libraries recombined mutations associated with improved activity in previous rounds and subjected certain structural features of the enzyme with saturation mutagenesis. These libraries were then plated, grown, and screened using the HTP assay described below, to provide a fifteenth round (“Round 15”) of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. Sixty-two engineered variants were identified from the recombined beneficial mutations (Table 55.1), and 113 were identified from saturation mutagenesis (Table 55.2).

(488) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(489) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 3956 variants. Lysis buffer volume was 400 μ L, and the lysate was diluted 40-fold. For screening the library in which beneficial mutations from previous rounds were recombined, diluted lysates were preincubated at 76° C. for 0.5 h in an Eppendorf thermocycler. Assays were then conducted with 10 μ L diluted lysate and 0.1 g/L SUS SFP SEQ ID NO: 1840 in 100 μ L reaction volume with substrate loading of 20 mM rebaudioside A and co-substrate loadings of 0.2 mM ADP (Sigma, >95%) and 50 mM sucrose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. The reactions were diluted, quenched, and analyzed as described above. The resulting engineered variants with glucosyltransferase activity coupled with SUS SEQ ID NO: 1840 on rebaudioside A are listed in Table 55.1. For the round 15 saturation mutagenesis library, screening was performed as described above with the exceptions that the diluted lysates were preincubated at 65° C. for 16 h and 0.1 g/L SUS SFP SEQ ID NO: 2064 was used. The resulting engineered GT variant polypeptides are listed in Table 55.2. Shake-flask scale cultures were grown, lysed, and lyophilized to powder for the variants listed in Table 55.3.

(490) TABLE-US-00112 TABLE 55.1 β 1,2GT Round 15 Combinatorial Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 3956) RebD.sup.a 4147/4148
A49S/S153A/F155M ++ 4149/4150 S153A/N199S/T406Q + 4151/4152 G96A/S153A/F155M/F156W +
4153/4154 G96P/Q127I/S153A/F155M + 4155/4156 A49S/G96P/Q132K/F155M/P237L/V278L +
4157/4158 Q132H/F155M + 4159/4160 A49S/G96P/F155M/N199S/A200S/T406Q + 4161/4162
G96P/Q127I/Q132H/S153A/V278L ++ 4163/4164 G96P/S153A/F155M/N199S/A200S/P237L +
4165/4166 S153A/F155M/A196S/N199S/P237L + 4167/4168 Q127I/Q132H/F155M/F156W/T406Q +
4169/4170 G96A/Q132K/S153A/F155M/F156W + 4171/4172
G96A/Q132K/S153A/F155M/F156W/A200S + 4173/4174 G96P/S153A/T406Q + 4175/4176
Q127I/T406Q + 4177/4178 Q132K/P237L + 4179/4180 A49S/G96A/Q127I/Q132K/F156W/A196S +++
4181/4182 A49S/G96P/Q127I/S153A/V278L +++ 4183/4184

Q127I/S153A/P237L/N199S/A200S/P237L ++ 4185/4186 Q127I/Q132H/S153A/F156W +++
 4187/4188 S153A/P237L ++ 4189/4190 Q132K/S153A/F155M +++ 4191/4192
 A49S/G96P/Q132H/F155M ++ 4193/4194 S153A/F155M/F156W/P237L +++ 4195/4196
 G96A/Q132H/S153A/F156W +++ 4197/4198 G96P/Q132H/S153A/F155M/F156W ++ 4199/4200
 A49S/G96A/Q132H/S153A/N199S/A200S + 4201/4202 G96P/Q132K/V278L +++ 4203/4204
 Q127I/Q132H + 4205/4206 G96P/Q132H/A196S/N199S + 4207/4208 G96P/Q132K/S153A/F155M ++
 4209/4210 Q132H/S153A/P237L/T406Q ++ 4211/4212 F155M/N199S ++ 4213/4214
 A49S/G96P/Q132H/S153A/F155M/F156W/P237L/V278L/T406Q ++ 4215/4216
 H5N/P73A/I144V/K187R/K373R/K422R ++ 4217/4218 H5N/I144V/V179T/M181L/K373R/K422R ++
 4219/4220 V179T/M181L/E186A + 4221/4222 K143P/V179T/M181L/E186G/K187R/K422R/N423R +
 4223/4224 V179D/K187T/K373R/K422R ++ 4225/4226
 H5N/P73A/I144V/V179D/E186A/K373R/N423R +++ 4227/4228
 I144V/V179T/M181L/E186A/K187R/K373R +++ 4229/4230 V179D/M181L/E186G/K187R/N423R ++
 4231/4232 P73A/L113I/V179T/M181L/E186A/K187R ++ 4233/4234 K373R/N423R + 4235/4236
 V179T/E186A/K187T + 4237/4238 M181L/E186G/K187R/K422R/N423R + 4239/4240
 L113I/I144V/E186G/N423R + 4241/4242 I144V/V179T/E186A/K187T/K373R ++ 4243/4244
 K143P/I144V/V179T/M181L/E186G/K187R/K422R + 4245/4246 L113I/K373R/K422R + 4247/4248
 P73A/L113I/K143P/V179T/M181L/K422R ++ 4249/4250 L113I/I144V/K373R ++ 4251/4252
 P73A/M181L/E186A/K187T +++ 4253/4254 H5N/K143P/I144V/V179T/M181L/E186G/K187R/K373R
 + 4255/4256 P73A/K143P/I144V/V179D/E186G/K187T/K373R/N423R +++ 4257/4258
 H5N/I144V/K373R/K422R +++ 4259/4260 L113I/K143P/V179D/E186G/K187R + 4261/4262
 P73A/V179D/M181L/E186G/K187R/K373R ++ 4263/4264 L113I/M181L/E186A/K373R/K422R +
 4265/4266 H5N/P73A/K143P/I144V/V179D/E186G/K187R/K422R + 4267/4268
 P73A/V179D/M181L/E186A/K373R/K422R ++ 4269/4270

H5N/P73A/L113I/E186G/K187R/K373R/N423R + .sup.a Levels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 3956, and defined as follows: “+” = production at least 1.75-fold, but less than 2-fold; “++” = at least 2-fold, but less than 2.38-fold; and “+++” = at least 2.38-fold increased production, as compared to the reference polypeptide.

(491) TABLE-US-00113 TABLE 55.2 β 1,2GT Round 15 Saturation Mutagenesis Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 3956)
 RebD.sup.a 4271/4272 T12L + 4273/4274 L222F ++ 4275/4276 T220G +++ 4277/4278 M238I +
 4279/4280 K117N + 4281/4282 E388S + 4283/4284 K57H + 4285/4286 F41H + 4287/4288 S66L +
 4289/4290 Y449L +++ 4291/4292 P73S ++ 4293/4294 L74M/M238V + 4295/4296 D236L ++ 4297/4298
 H83R + 4299/4300 N65R + 4301/4302 P408T ++ 4303/4304 E388A + 4305/4306 K214T + 4307/4308
 M238G +++ 4309/4310 Q132G ++ 4311/4312 P82S + 4313/4314 D387P ++ 4315/4316 Y449H ++
 4317/4318 M238A ++ 4319/4320 K450V ++ 4321/4322 K309T +++ 4323/4324 K57G ++ 4325/4326
 D389G ++ 4327/4328 K103Q + 4329/4330 R391A +++ 4331/4332 Y44H + 4333/4334 K160Q +
 4335/4336 T406S + 4337/4338 R391L + 4339/4340 T406G ++ 4341/4342 G138K + 4343/4344 T220S
 ++ 4345/4346 D236T + 4347/4348 M238S + 4349/4350 Y44S/K187T ++ 4351/4352 N65Q ++
 4353/4354 N286G + 4355/4356 R391V + 4357/4358 R418H + 4359/4360 H85W + 4361/4362 L135E +
 4363/4364 G138T + 4365/4366 P140G + 4367/4368 L45Y +++ 4369/4370 Q11L + 4371/4372 R418A
 +++ 4373/4374 K214H + 4375/4376 T223D +++ 4377/4378 V70W + 4379/4380 M238R +++ 4381/4382
 H393P ++ 4383/4384 N65G +++ 4385/4386 N65C ++ 4387/4388 I55G + 4389/4390 E299V + 4391/4392
 D236V ++ 4393/4394 K103H + 4395/4396 R182I ++ 4397/4398 P408L + 4399/4400 K57P ++
 4401/4402 D389V ++ 4403/4404 Q114I ++ 4405/4406 K111F + 4407/4408 Y449S + 4409/4410 K309R
 ++ 4411/4412 R182L +++ 4413/4414 I55L + 4415/4416 K160T + 4417/4418 K117S + 4419/4420 T12I +
 4421/4422 H393T + 4423/4424 R418M ++ 4425/4426 Y449R + 4427/4428 K422L ++ 4429/4430
 K226Q + 4431/4432 E430P + 4433/4434 I55A ++ 4435/4436 K56R ++ 4437/4438 K450N + 4439/4440
 N65P + 4441/4442 L113S +++ 4443/4444 P162L +++ 4445/4446 K160D +++ 4447/4448 D387R +++
 4449/4450 E256R +++ 4451/4452 P73H +++ 4453/4454 P167L + 4455/4456 P140C +++ 4457/4458
 I412G +++ 4459/4460 L159N ++ 4461/4462 I412R ++ 4463/4464 R391S +++ 4465/4466 Q114G +
 4467/4468 M238T ++ 4469/4470 L72V +++ 4471/4472 G138S ++ 4473/4474 Q11D ++ 4475/4476
 R182Y + 4477/4478 D389A + 4479/4480 K309S + 4481/4482 E429D ++ 4483/4484 R182Q +

4485/4486 D387Q + 4487/4488 K309P ++ 4489/4490 D389L + 4491/4492 R182V + 4493/4494 Y449G + 4495/4496 R182T +++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 3956, and defined as follows: “+” = production at least 1.15-fold, but less than 1.22-fold; “++” = at least 1.22-fold, but less than 1.35-fold; and “+++” = at least 1.35-fold increased production, as compared to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP to Rebaudioside A

(492) A shake flask powder loading dose response experiment was performed to characterize activity of the engineered round 15 variants on rebaudioside A. Levels of 0.006-0.2 g/L shake flask powder (SFP) were preincubated at 76° C. for 0.5 h and then added to a 100 µL total reaction volume containing 50 mM potassium phosphate buffer, pH 6, 20 mM rebaudioside A, 0.2 mM ADP, 50 mM sucrose, and 0.1 g/L SUS SFP SEQ ID NO: 2064. The reaction was performed at 60° C. with preincubated SFP or at 55° C. without preincubation in a Thermotron® titre-plate shaker at 300 RPM for 4 h. The reactions were diluted, quenched, and analyzed as described above. The production of rebaudioside D by the round 15 variants at 0.025 g/L SFP loading are listed in Table 55.3. Five out of the 9 variants listed in Table 55.3 had higher activities under both conditions than SEQ ID NO: 3956. The variant with the mutations P73A, K143P, I144V, V179D, E186G, K187T, K373R, and N423R (SEQ ID NO: 4256) and its encoding polynucleotide (SEQ ID NO: 4255) were the most improved taking into account both conditions and were selected for further directed evolution for the glucosylation of rebaudioside A.

(493) TABLE-US-00114 TABLE 55.3 β1,2GT Round 15 Shake Flask Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased RebD, (nt/aa) (Relative to SEQ ID NO: 3956) 60° C.sup.a.
4149/4150 S153A/N199S/T406Q ++ 4151/4152 G96A/S153A/F155M/F156W + 4193/4194
S153A/F155M/F156W/P237L + 4217/4218 H5N/I144V/V179T/M181L/K373R/K422R ++ 4225/4226
H5N/P73A/I144V/V179D/E186A/K373R/N423R +++ 4251/4252 P73A/M181L/E186A/K187T +
4255/4256 P73A/K143P/I144V/V179D/E186G/K187T/K373R/N423R +++ 4267/4268
P73A/V179D/M181L/E186A/K373R/K422R + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 3956, and defined as follows: “+” = production at least 1.4-fold, but less than 1.75-fold; “++” = at least 1.75-fold, but less than 1.9-fold; and “+++” = at least 1.9-fold increased production, as compared to the reference polypeptide.

Example 56

Beta-1,2-ADP-Glycosyltransferase Variants of SEQ ID NO: 4256

(494) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 4256 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 4255 was carried out by constructing combinatorial libraries of variant genes. Libraries recombined mutations associated with improved activity in previous rounds and subjected certain structural features of the enzyme with saturation mutagenesis. These libraries were then plated, grown, and screened using the HTP assay described below to provide a sixteenth round (“Round 16”) of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. Twenty-seven engineered variants were identified from the recombined beneficial mutations (Table 56.1), and 66 were identified from saturation mutagenesis (Table 56.2).

(495) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A

(496) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 4255 variants. Lysis buffer volume was 400 µL, and the lysate was diluted 40-fold. For screening the library in which beneficial mutations from previous rounds were recombined, diluted lysates were preincubated at 79° C. for 0.5 h in an Eppendorf thermocycler. Assays were then conducted with 10 µL diluted lysate and 0.05 g/L sucrose synthetase (SUS) SFP SEQ ID NO: 2064 in 100 µL reaction volume with substrate loading of 20 mM rebaudioside A and co-substrate loadings of 0.2 mM ADP (Sigma, >95%) and 50 mM sucrose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. The reactions were diluted, quenched, and analyzed as described above. The resulting engineered variants with glucosyltransferase activity coupled with SUS SEQ ID NO: 2064 on rebaudioside A are listed in Table 56.1. For the round 16 saturation mutagenesis library, screening was performed as described above with

the exception that 0.05 g/L sucrose synthetase SFP SEQ ID NO: 2510 was used. The resulting engineered GT variant polypeptides are listed in Table 56.2. Shake-flask scale cultures were grown, lysed, and lyophilized to powder for variants listed in Table 56.3.

(497) TABLE-US-00115 TABLE 56.1 β 1,2GT Round 16 Combinatorial Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 4256) RebD.sup.a 4497/4498 R427S + 4499/4500 S153A/F155M/R427T + 4501/4502 S153A + 4503/4504 F155M/F156W + 4505/4506 K62A/S153A/F155M/F156W/L159V +++ 4507/4508 S153A/F156W/L159V + 4509/4510 S153A/F155M/P237L/M238T/T239F/T406Q +++ 4511/4512 S153A/F156W/N199S/P237L/R427T ++ 4513/4514 S153A/L159V/P237L/M238G/5352A ++ 4515/4516 F156W/N199S/R427T +++ 4517/4518 F156W/N199S + 4519/4520 S153A/F156W/R427T +++ 4521/4522 F155M/F156W/R176Q/M181L/N199S ++ 4523/4524 F155M/F156W/R176Q/M238T/R427S + 4525/4526 K62A/S153A/F155M/N199S/T406Q ++ 4527/4528 S153A/R176Q/M181L/R427S ++ 4529/4530 S153A/F155M/M238T/T239F + 4531/4532 F155M/M181L/N199S/M238G/T406Q +++ 4533/4534 S153A/F155M/F156W/ + L159V/N199S/M238G/T406Q 4535/4536 K62A/S153A/F155M/F156W/L159V/R427T + 4537/4538 S153A/F155M/F156W ++ 4539/4540 H164E/I375L/N433D + 4541/4542 I375L ++ 4543/4544 I50V/Q137R/P189G/I375L ++ 4545/4546 T223A/I375L + 4547/4548 Q137R/H164E/I375L + 4549/4550 S8V/I375L + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 4256, and defined as follows: “+” = production at least 1.75-fold, but less than 2.08-fold; “++” = at least 2.08-fold, but less than 2.33-fold; and “+++” = at least 2.33-fold increased production, as compared to the reference polypeptide.

(498) TABLE-US-00116 TABLE 56.2 β 1,2GT Round 16 Saturation Mutagenesis Variants and RebD Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 4256) RebD.sup.a 4551/4552 Y449K +++ 4553/4554 A106S + 4555/4556 N445R ++ 4557/4558 E192L +++ 4559/4560 F280W ++ 4561/4562 S110G + 4563/4564 L116I + 4565/4566 N115A + 4567/4568 A106K + 4569/4570 Q132R +++ 4571/4572 K416R ++ 4573/4574 Y449Q + 4575/4576 V385S ++ 4577/4578 S9G + 4579/4580 A106R ++ 4581/4582 T220S + 4583/4584 H99R + 4585/4586 T220Q ++ 4587/4588 N445K ++ 4589/4590 N302P + 4591/4592 G395Q +++ 4593/4594 N65F ++ 4595/4596 V385P +++ 4597/4598 E192A + 4599/4600 A200S ++ 4601/4602 E131C +++ 4603/4604 R423L +++ 4605/4606 N447L ++ 4607/4608 E136R ++ 4609/4610 Y449R ++ 4611/4612 N302S + 4613/4614 E131S ++ 4615/4616 N115T + 4617/4618 P335R + 4619/4620 E192P + 4621/4622 Q399K + 4623/4624 M238L + 4625/4626 K245P ++ 4627/4628 D190T +++ 4629/4630 Q132T + 4631/4632 T12S + 4633/4634 I412R +++ 4635/4636 E304I + 4637/4638 R272H + 4639/4640 M238I + 4641/4642 A170E +++ 4643/4644 P335K ++ 4645/4646 T220M +++ 4647/4648 H68I + 4649/4650 E136S + 4651/4652 G10D ++ 4653/4654 N257H +++ 4655/4656 Q178L + 4657/4658 P194F +++ 4659/4660 V385A + 4661/4662 Y449L + 4663/4664 V385C + 4665/4666 E136G + 4667/4668 I242L ++ 4669/4670 P408E + 4671/4672 E136D + 4673/4674 G7H + 4675/4676 N445H + 4677/4678 K402R + 4679/4680 E53N ++ 4681/4682 Q132G +++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 4256, and defined as follows: “+” = production at least 1.07-fold, but less than 1.27-fold; “++” = at least 1.27-fold, but less than 1.42 fold; and “+++” = at least 1.42-fold increased production, as compared to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP to Rebaudioside A

(499) A shake flask powder loading dose response experiment was performed to characterize activity of the engineered round 16 variants on rebaudioside A. Levels of 0.003-0.1 g/L shake flask powder (SFP) were preincubated at 79° C. for 0.5 h and then added to a 100 μ L total reaction volume containing 50 mM potassium phosphate buffer, pH 6, 20 mM rebaudioside A, 0.2 mM ADP, 50 mM sucrose, and 0.05 g/L SUS SFP SEQ ID NO: 2432. The reaction was performed at 60° C. in a Thermotron® titre-plate shaker at 300 RPM for 4 h. The reactions were diluted, quenched, and analyzed as described above. The production levels of rebaudioside D by the round 16 variants at 0.0125 g/L SFP loading are shown in Table 56.3. SEQ ID NO: 4550 was the most improved and was selected as the best enzyme for the catalysis of glycosyltransfer from ADP-glucose to rebaudioside A to form rebaudioside D.

(500) TABLE-US-00117 TABLE 56.3 β 1,2GT Round 16 Shake Flask Variants and RebD Levels SEQ ID

NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 4256) RebD.sup.a 4549/4550 S8V/I375L +++ 4539/4540 H164E/I375L/N433D +++ 4497/4498 R427S ++ 4545/4546 T223A/I375L ++ 4515/4516 F156W/N199S/R427T + 4531/4532 F155M/M181L/N199S/M238G/T406Q + 4543/4544 I50V/Q137R/P189G/I375L + 4501/4502 S153A + .sup.a Levels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 4256, and defined as follows: “+” = production at least that of the reference polypeptide, but less than 1.3-fold; “++” = at least 1.3-fold, but less than 1.6-fold; and “+++” = at least 1.6-fold increased production, as compared to the reference polypeptide.

Example 57

Beta-1,2-ADP-Glycosyltransferase Variants of SEQ ID NO: 4550

(501) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 4550 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 4549 was carried out by constructing combinatorial libraries of variant genes. Libraries recombined mutations associated with improved production identified during the development of the present invention. These libraries were then plated, grown, and screened using the HTP assay described below to provide a seventeenth round (“Round 17”) of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

Seventy-three engineered variants were identified from the recombined beneficial mutations (Table 57.1).

(502) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A 60

(503) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 4549 variants. Lysis buffer volume was 400 μ L, and the lysate was diluted 50 or 100-fold into 20 g/L rebaudioside A 60% and pre-incubated for 2 h at 75° C. Assays were then conducted with 10 μ L diluted lysate, 0.1 g/L SUS SFP SEQ ID NO: 2510, and 0.2 g/L β -1,3-glycosyltransferase (β 1,3GT) SFP SEQ ID NO: 6864 in 100 μ L reaction volume with substrate loading of 20 g/L rebaudioside A 60% (RebA60) and co-substrate loadings of 0.1 g/L ADP (Amresco, ultra pure grade) and 40 g/L sucrose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 16-18 h. The reactions were solubilized by 40 $\times$ dilution into water, quenched by 5 $\times$ dilution into acetonitrile with 0.2% formic acid, precipitated by centrifugation, and diluted 10 $\times$ into water for analysis as described above. The resulting engineered variants with glucosyltransferase activity coupled with a SuS on rebaudioside A are listed in Table 57.1. Shake-flask scale cultures were grown, lysed, and lyophilized to powder for variants listed in Table 57.2.

(504) TABLE-US-00118 TABLE 57.1 β 1,2GT Round 17 Combinatorial Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 4550) RebM.sup.a

7215/7216 R427S ++ 7217/7218 N65G/P143K/Q235R/R427S + 7219/7220 N65G/Q114I/K422L/R427S + 7221/7222 R427S/E429D ++ 7223/7224 R418A/R427S + 7225/7226 N65G/Q1141/R427S ++ 7227/7228 N65G/P143K + 7229/7230 Y44H/A73H/Q127I/L135E ++ 7231/7232

Q11L/A73S/Q132K/L135E/P408A + 7233/7234 Q11L/F41H/Q132K/V278L +++ 7235/7236

Q11L/Q127I/Q132K/L135E/T406S +++ 7237/7238 Q132K/L135E/R182I ++ 7239/7240 L45Y/T406G

++ 7241/7242 R182L + 7243/7244 L45Y/A73H/Q132K/L135E/R182L +++ 7245/7246

L45Y/A73H/Q127I/Q132K/L135E/T406G/P408A +++ 7247/7248

Q11L/F41H/Y44H/Q127I/Q132K/T406G + 7249/7250 Q11L/R182L/T406S + 7251/7252

L45Y/Q127I/Q132K/L135E/R182T/T406G/P408A + 7253/7254

Q11L/A73H/Q127I/Q132K/T406G/P408A +++ 7255/7256 A73H/Q127I ++ 7257/7258

Y44H/L45Y/Q127I/Q132K/L135E/G138T/R182L/T406S/P408A + 7259/7260

Q11L/F41H/L45Y/A73H/Q127I/V278L/T406G/P408A + 7261/7262 Q11L/Q127I/Q132K/L135E +++

7263/7264 Q11L/L45Y/Q127I/Q132K/L135E/G138T/R182T ++ 7265/7266

Y44H/L45Y/Q127I/Q132K/V278L +++ 7267/7268 Y44H/A73S/Q132K/L135E/T406G/P408A ++

7269/7270 Q132K/L135E/T406G + 7271/7272 F41H/A73H/Q132K/L135E/T406G + 7273/7274

Q11L/Y44H/A73S/Q127I/Q132K/L135E/T406G/P408A + 7275/7276 Q11L/Y44H/Q127I + 7277/7278

Q11L/Y44H/Q132K/V278L/T406G + 7279/7280 Q11L/Y44H/A73H/Q127I/Q132K/L135E ++

7281/7282 Q11L/Q127I/Q132K/R182T + 7283/7284 Q11L/A73H/Q127I/Q132K/V278L/T406G/P408A

+++ 7285/7286 F41H/Y44H/L45Y/A73H/Q127I/Q132K/V278L +++ 7287/7288

F41H/L45Y/Q127I/Q132K/L135E/V278L/T406S +++ 7289/7290 L45Y/A73H/Q132K +++ 7291/7292

Q11L/L45Y/T406G ++ 7293/7294 Q11L/A73H/Q132K ++ 7295/7296 L45Y/Q127I/Q132K +++

7297/7298 T406G + 7299/7300 L45Y/Q132K/P408A ++ 7301/7302 A73H/Q127I/Q132K ++ 7303/7304 Y44H/L45Y/A73H/Q132K/L135E/T406G ++ 7305/7306 Q11L/Q127I/Q132K/G138T/R182L +++ 7307/7308 Q11L/A73S/Q127I/Q132K/L135E/R182L/V278L +++ 7309/7310 Q11L/F41H/L45Y/Q127I/V278L/T406G ++ 7311/7312 L45Y/Q127I/Q132K/L135E/R182L ++ 7313/7314 Y44H/A73S/Q127I/L135E/R182L/V278L/T406G/P408A + 7315/7316 Q127I ++ 7317/7318 Q132K/L135E ++ 7319/7320 L45Y/Q127I/L135E ++ 7321/7322 Q127I/Q132K/L135E/R182I/T406G + 7323/7324 Q132K/T406S ++ 7325/7326 Q11L/L45Y/A73H/Q127I/L135E + 7327/7328 Q11L/F41H/Y44H/Q127I/V278L/T406S + 7329/7330 L45Y/A73S/V278L/T406G/P408A ++ 7331/7332 Q127I/Q132K/L135E/R182L + 7333/7334 Q11L/Y44H/L45Y/Q127I + 7335/7336 Q11L/Q127I/Q132K/L135E/R182L + 7337/7338 Q11L/Q132K/L135E/T406G/ ++ 7339/7340 H164E/T220G/K309S/Y449H + 7341/7342 H164E/T220G/Y449L + 7343/7344 K309P/Y449L + 7345/7346 K57G + 7347/7348 H164E + 7349/7350 Y449L + 7351/7352 K309R + 7353/7354 T220G/Y449H + 7355/7356 K56R/K309P/Y449L + 7357/7358 H164E/Y449H + 7359/7360 K309T/Y449H + .sup.a

Levels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 4550, and defined as follows: “+” = production at least 1.2-fold, but less than 1.52-fold; “++” = at least 1.52-fold, but less than 2.1-fold; and “+++” = at least 2.1-fold increased production, as compared to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP to Rebaudioside A 60

(505) A shake flask powder loading dose response experiment was performed to characterize activity of the engineered round 17 variants on rebaudioside A 60%. Levels of 0.0003-0.04 g/L shake flask powder (SFP) were assayed in 100 μ L total reaction volume containing 50 mM potassium phosphate buffer, pH 6, 20 g/L RebA60, 0.1 g/L ADP, 40 g/L sucrose, 0.1 g/L SUS SFP SEQ ID NO: 2510, and 0.2 g/L β -1,3-glycosyltransferase (β 1,3GT) SFP SEQ ID NO: 6864. The reaction was performed at 60° C. in a Thermotron® titre-plate shaker at 300 RPM for 16-18 h. The reactions were diluted, quenched, and analyzed as described above. The results for the production of rebaudioside M in the one-pot reaction by the round 17 variants at 0.005 g/L SFP loading are shown in Table 57.2. SEQ ID NO: 7324 was identified as being the most improved and was selected as the best enzyme for the catalysis of glycosyltransfer from ADP-glucose to rebaudioside A to form rebaudioside D.

(506) TABLE-US-00119 TABLE 57.2 β 1,2GT Round 17 Shake Flask Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 4550) RebM.sup.a 7215/7216 R427S – 7225/7226 N65G/Q114I/R427S – 7227/7228 N65G/P143K + 7265/7266 Y44H/L45Y/Q127I/Q132K/V278L – 7273/7274 Q11L/Y44H/A73S/Q127I/ – Q132K/L135E/T406G/P408A 7285/7286 F41H/Y44H/L45Y/A73H/Q127I/Q132K/V278L – 7323/7324 Q132K/T406S ++ 7343/7344 K309P/Y449L + .sup.a

Levels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 4550, and defined as follows: “–” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.15-fold; and “++” = at least 1.15-fold increased production, as compared to the reference polypeptide.

Example 58

Beta-1,2-ADP-Glycosyltransferase Variants of SEQ ID NO: 7324

(507) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 7324 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 7323 was carried out by constructing combinatorial libraries of variant genes. Libraries recombined mutations associated with improved production identified during the development of the present invention and subjected certain structural features of the enzyme with saturation mutagenesis. These libraries were then plated, grown, and screened using the HTP assay described below to provide an eighteenth round (“Round 18”) of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. Ninety engineered variants were identified from the recombined beneficial mutations (Table 58.1), and 124 were identified from saturation mutagenesis (Table 58.2).

(508) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A 60

(509) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO:

7323 variants. Lysate volume was 400 μ L, and the lysate was diluted 80-fold into 20 g/L rebaudioside A 60% and pre-incubated for 1.5 h at 75° C. Assays were then conducted with 10 μ L diluted lysate, 0.05 g/L SUS SFP SEQ ID NO: 7506, and 0.1 g/L β -1,3-glycosyltransferase (β 1,3GT) SFP SEQ ID NO: 7388, in 100 μ L reaction volume with substrate loading of 20 g/L rebaudioside A 60% (RebA60) and co-substrate loadings of 0.05 g/L ADP (Amresco, ultra pure grade) and 40 g/L sucrose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 16-18 h. The reactions were solubilized by 40 $\times$ dilution into water, quenched by 5 $\times$ dilution into acetonitrile with 0.2% formic acid, precipitated by centrifugation, and diluted 10 $\times$ into water for analysis as described above. The resulting engineered variants with glucosyltransferase activity coupled with a SuS on RebA60 are listed in Table 58.1. For the round 18 saturation mutagenesis library, screening was performed as described previously, with the exception that the lysate was diluted 100-fold into 50 mM Kphos pH6 and pre-incubated for 1 h at 75° C. The resulting engineered GT variant polypeptides are listed in Table 58.2. Shake-flask scale cultures were grown, lysed, and lyophilized to powder for variants listed in Table 58.3.

(510) TABLE-US-00120 TABLE 58.1 β 1,2GT Round 18 Combinatorial Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 7324) RebM.sup.a

7765/7766 L116I/D190T/S406G/P408A ++ 7767/7768 L116I/M238L/K245P + 7769/7770 D190T/P194F/I412R + 7771/7772 D190T/P194F/P335R/K416R + 7773/7774 D190T/P194F ++ 7775/7776 D190T/E192L/P194F +++ 7777/7778 A170E/P194F/P335R/K416R + 7779/7780 E192L/P194F + 7781/7782 A170E/P335K/K416R ++ 7783/7784 D190T/P194F/M238L/K245P ++ 7785/7786 P194F ++ 7787/7788 A170E/E192L/P194F/P335K + 7789/7790 A170E/P194F + 7791/7792 A170E/D190T/E192L/P194F ++ 7793/7794 D190T + 7795/7796 N115A/D190T/P194F ++ 7797/7798 A170E/M238L + 7799/7800 A170E/E192L + 7801/7802 N115T/L116I/V278L + 7803/7804 D190T/K245P/I412R + 7805/7806 A170E/P335K ++ 7807/7808 L116I/A170E/D190T ++ 7809/7810 L116I/K416R + 7811/7812 N115A/A170E/D190T/M238L/I412R + 7813/7814 N115A/K245P/V278L +++ 7815/7816 E192L/P194F/I242L/S406G/P408A +++ 7817/7818 L116I/M238L + 7819/7820 A170E/E192L/P194F/V278L ++ 7821/7822 A170E ++ 7823/7824 V278L ++ 7825/7826 R272H/V385P ++ 7827/7828 G10D/Q127I/K132R/V385P + 7829/7830 G10D/A73H + 7831/7832 G10D/E53N/R423L ++ 7833/7834 E53N/K132G/E136R/N302P/R423L/R427S + 7835/7836 G10D/K132R/N302P/V385P +++ 7837/7838 G10D + 7839/7840 K132G + 7841/7842 A73H/V385S/R427S +++ 7843/7844 A73H +++ 7845/7846 V385S/R427S + 7847/7848 N302P/V385S + 7849/7850 G10D/E53N/R272H/R423L/R427S + 7851/7852 Q127I + 7853/7854 E53N/Q127I/K132R/V385S ++ 7855/7856 N257H/V385S ++ 7857/7858 N302P/V385P +++ 7859/7860 E53N/A73H/A200S/R423L ++ 7861/7862 A73H/K132R/E136R/V385S/R427S ++ 7863/7864 V385S +++ 7865/7866 G10D/A73H/Q127I +++ 7867/7868 G7V/G10D/A73H/Q127I/K132G/V385P/R423L ++ 7869/7870 R423L/R427S +++ 7871/7872 A200S +++ 7873/7874 A73H/V385S +++ 7875/7876 E53N/Q127I/E136R/V385L +++ 7877/7878 R272H/N302P +++ 7879/7880 E53N/K132G/N302P/V385S/R423L/R427S +++ 7881/7882 A200S/V385S +++ 7883/7884 G10D/E53N/A73H/H99R ++ 7885/7886 A73H/Q127I/R427S +++ 7887/7888 T220M/N445K + 7889/7890 N65G/T220Q/N445R/Y449K + 7891/7892 Y449L + 7893/7894 Q399K/S406G + 7895/7896 Q399K/S406G/Y449L +++ 7897/7898 P143K/K309P/N445K/N447L/Y449L + 7899/7900 N65G/K402R/S 406G/N445R/Y449R ++ 7901/7902 P143K/K309P/K402R + 7903/7904 N65G/Q399K/S406G/N447L/Y449R + 7905/7906 T220Q/N445K/N447L + 7907/7908 N445K/Y449L + 7909/7910 K309P/Q399K/Y449L ++ 7911/7912 N445R/Y449L + 7913/7914 S406G/N445R/Y449K + 7915/7916 N445R/N447L/Y449L ++ 7917/7918 N65G/N445K/N447L/Y449R ++ 7919/7920 S406G/N445R/N447L/Y449L ++ 7921/7922 S406G/N445K/Y449R +++ 7923/7924 N65G/A106R/N445K/N447L/Y449L ++ 7925/7926 P143K/T220Q/N445K/N447L + 7927/7928 K402R/N445R/Y449L + 7929/7930 N65G/T220Q/K309P/N445R/N447L + 7931/7932 A106R/T220Q/Q399K/K402R/S406G + 7933/7934 N65G/N447L/Y449L ++ 7935/7936 N447L ++ 7937/7938 K309P/N445K/N447L/Y449L + 7939/7940 N65G/T220Q/N445R/Y449R + 7941/7942 T220Q/Q399K + 7943/7944 N65G/P143K/T220Q + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 7324, and defined as follows: “+” = production at least that of the reference polypeptide, but less than 1.16-fold; “++” = at least 1.16-fold, but

less than 1.23-fold; and “+++” = at least 1.23-fold increased production, as compared to the reference polypeptide.

(511) TABLE-US-00121 TABLE 58.2 β 1,2GT Round 18 Saturation Mutagenesis Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 7324)

RebM.sup.a 8091/8092 N51Q + 8093/8094 A434C + 8095/8096 S64R ++ 8097/8098 V46Q + 8099/8100 V46L ++ 8101/8102 Q11L/Y44H +++ 8103/8104 V46C + 8105/8106 V46T ++ 8107/8108 A49K ++ 8109/8110 D119T/R172H ++ 8111/8112 V376N + 8113/8114 G410R + 8115/8116 A130S + 8117/8118 A49N/S406G/P408A + 8119/8120 A398R +++ 8121/8122 G356T ++ 8123/8124 I420F ++ 8125/8126 D274Q + 8127/8128 Q11L/E71P +++ 8129/8130 A381C ++ 8131/8132 P118S ++ 8133/8134 D274T +++ 8135/8136 S64G +++ 8137/8138 A398H ++ 8139/8140 V133H ++ 8141/8142 D119G ++ 8143/8144 P118A ++ 8145/8146 A49G ++ 8147/8148 K100R + 8149/8150 G338S + 8151/8152 N157E + 8153/8154 V259T + 8155/8156 P118L ++ 8157/8158 A141Q ++ 8159/8160 D179I ++ 8161/8162 I43T +++ 8163/8164 S406G/P408A +++ 8165/8166 P169E +++ 8167/8168 Q441L ++ 8169/8170 Q11L/V287S +++ 8171/8172 N108G/R172H +++ 8173/8174 V376S +++ 8175/8176 G356S + 8177/8178 E71T +++ 8179/8180 I397L/S406G/P408A ++ 8181/8182 I420L + 8183/8184 Q11L/E71T + 8185/8186 N51M + 8187/8188 D119S + 8189/8190 A398K + 8191/8192 N333S/S406G/P408A + 8193/8194 V287L + 8195/8196 A398L + 8197/8198 E437G + 8199/8200 R172H/I420L +++ 8201/8202 R172H/E437D + 8203/8204 V42A +++ 8205/8206 V287A ++ 8207/8208 A141G +++ 8209/8210 S47N +++ 8211/8212 P118A/S406G/P408A + 8213/8214 Q11L/Q441S + 8215/8216 P118V +++ 8217/8218 S64L +++ 8219/8220 V417C +++ 8221/8222 M275L +++ 8223/8224 V376G +++ 8225/8226 E437V +++ 8227/8228 E396A +++ 8229/8230 E76H +++ 8231/8232 V133R ++ 8233/8234 T97V + 8235/8236 H2T ++ 8237/8238 I112T + 8239/8240 I112N + 8241/8242 N157T + 8243/8244 F109V ++ 8245/8246 S64A + 8247/8248 F109L + 8249/8250 L401T + 8251/8252 V133S + 8253/8254 V287S + 8255/8256 D119A + 8257/8258 V287K + 8259/8260 P118I + 8261/8262 I43L + 8263/8264 R394H/I420W + 8265/8266 N108T + 8267/8268 V376R + 8269/8270 N288P + 8271/8272 T48S/A398V + 8273/8274 V385A + 8275/8276 I112Q + 8277/8278 M181H ++ 8279/8280 V133L + 8281/8282 A134S ++ 8283/8284 N157C + 8285/8286 N51V + 8287/8288 N333S/A398T + 8289/8290 M181T + 8291/8292 T97L ++ 8293/8294 E396S ++ 8295/8296 H4F + 8297/8298 A141S ++ 8299/8300 I112V +++ 8301/8302 V42D/G138C ++ 8303/8304 S64V + 8305/8306 E76L + 8307/8308 K117R/N157T/Q301R + 8309/8310 V376M + 8311/8312 S64Q + 8313/8314 V417S ++ 8315/8316 V42A/A141P ++ 8317/8318 A398M + 8319/8320 A37G + 8321/8322 I112R ++ 8323/8324 I43M ++ 8325/8326 F109Y ++ 8327/8328 V357T + 8329/8330 D179R ++ 8331/8332 I43A + 8333/8334 G40Q + 8335/8336 V42G +++ 8337/8338 S64F +++

.sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 7324, and defined as follows: “+” = production at least 1.1-fold, but less than 1.22-fold; “++” = at least 1.22-fold, but less than 1.3-fold; and “+++” = at least 1.3-fold increased production, as compared to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP to Rebaudioside A 60

(512) A shake flask powder loading dose response experiment was performed to characterize activity of the engineered round 18 variants on rebaudioside A 60%. Levels of 0.0013-0.04 g/L shake flask powder (SFP) were assayed in 100 μ L total reaction volume containing 50 mM potassium phosphate buffer, pH 6, 20 g/L RebA60, 0.05 g/L ADP, 40 g/L sucrose, 0.05 g/L SUS SFP SEQ ID NO: 7506, and 0.15 g/L β -1,3-glycosyltransferase (β 1,3GT) SFP SEQ ID NO: 7388. The reaction was performed at 60° C. in a Thermotron® titre-plate shaker at 300 RPM for 16-18 h. The reactions were diluted, quenched, and analyzed as described above. The production levels of rebaudioside M in the one-pot reaction by the round 18 variants at 0.005 g/L SFP loading are shown in Table 58.3. The variant with SEQ ID NO: 7784 was the most improved variant. Thus, it was selected as the best enzyme for the catalysis of glycosyltransfer from ADP-glucose to rebaudioside A to form rebaudioside D.

(513) TABLE-US-00122 TABLE 58.3 β 1,2GT Round 18 Shake Flask Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 7324) RebM.sup.a 7783/7784 D190T/P194F/M238L/K245P ++ 7795/7796 N115A/D190T/P194F ++ 7805/7806 A170E/P335K – 7813/7814 N115A/K245P/V278L + 7841/7842 A73H/V385S/R427S – 7869/7870 R423L/R427S – 7885/7886 A73H/Q127I/R427S – 7895/7896 Q399K/S406G/Y449L + 7903/7904

N65G/Q399K/S406G/N447L/Y449L ++ 7933/7934 N65G/N447L/Y449L + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 7324, and defined as follows: “-” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.08-fold; and “++” = at least 1.08-fold increased production, as compared to the reference polypeptide.

Example 59

Beta-1,2-ADP-Glycosyltransferase Variants of SEQ ID NO: 7784

(514) In this Example, experiments for evolution and screening of GT polypeptides derived from SEQ ID NO: 7784 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 7783 was carried out by constructing libraries of variant genes. Libraries recombined mutations associated with improved production identified during the development of the present invention and subjected certain structural features of the enzyme with saturation mutagenesis. These libraries were then plated, grown, and screened using the HTP assay described below to provide an nineteenth round (“Round 19”) of engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. Eighty-seven engineered variants were identified from the combinatorial libraries (Table 59.1), and fifty-eight were identified from the saturation mutagenesis libraries (Table 59.2).

(515) HTP Coupled Assay for Glucose Transfer from Sucrose to ADP to Rebaudioside A 60

(516) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 7783 variants. Lysis buffer volume was 400 μ L, and the lysate was diluted 100-fold into 50 mM potassium phosphate, pH 6.0, and pre-incubated for 1 h at 75° C. Assays were then conducted with 10 μ L diluted lysate, 0.04 g/L SUS SFP SEQ ID NO: 8420, and 0.1 g/L β -1,3-glycosyltransferase (β 1,3GT) SFP SEQ ID NO: 8088, in 100 μ L reaction volume with substrate loading of 20 g/L rebaudioside A 60% (RebA60) and co-substrate loadings of 0.025 g/L ADP (Amresco, ultra pure grade) and 40 g/L sucrose. The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 16-18 h. The reactions were solubilized by 20 $\times$ dilution into water, quenched by 5 $\times$ dilution into acetonitrile with 0.2% formic acid, precipitated by centrifugation, and diluted 20 $\times$ into water for analysis as described above. The resulting engineered variants with glucosyltransferase activity coupled with a SuS on RebA60 are listed in Table 59.1 and 59.2. Shake-flask scale cultures were grown, lysed, and lyophilized to powder for variants listed in Table 59.3.

(517) TABLE-US-00123 TABLE 59.1 β 1,2GT Round 19 Combinatorial Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 7784) RebM.sup.a

8623/8624 S64R + 8625/8626 I112T +++ 8627/8628 Q11L/S64R/F109V ++ 8629/8630 S64R/N65G/F109V + 8631/8632 Q11L ++ 8633/8634 S64R/N65G/I112T ++ 8635/8636 S64R/N65G/I112N ++ 8637/8638 F109V/I112T + 8639/8640 F109V/S406G + 8641/8642 F109L/I112T + 8643/8644 F109V ++ 8645/8646 F109V/A134S + 8647/8648 I112N/N445K + 8649/8650 S64R/N445K ++ 8651/8652 S64R/A134S + 8653/8654 F109V/N115A/P118V +++ 8655/8656 N65G/I112N/N445K + 8657/8658 S64R/I112N + 8659/8660 Q11L/N445K + 8661/8662 N65G/I112N + 8663/8664 V42G/I43M + 8665/8666 V46T/N51Q + 8667/8668 A141P + 8669/8670 A73H/A141P +++ 8671/8672 A141G ++ 8673/8674 A73H/N302P ++ 8675/8676 E71T/A73H/A141S ++ 8677/8678 E71T ++ 8679/8680 E71T/N302P + 8681/8682 A73H ++ 8683/8684 A73H/A141G +++ 8685/8686 E71T/A141P +++ 8687/8688 A141S/N302P +++ 8689/8690 E71T/A73H/A141G ++ 8691/8692 A73H/L116I/A141P + 8693/8694 N302P + 8695/8696 A73H/A141S + 8697/8698 A141S + 8699/8700 V46L/S47N/N51Q ++ 8701/8702 V42A/Y44H/E71T/A73H/L116I ++ 8703/8704 A141G/N302P + 8705/8706 S47N/A49G + 8707/8708 S47N/N51V + 8709/8710 E71T/A73H + 8711/8712 I43T/A73H/A141P + 8713/8714 V287L/N288P ++ 8715/8716 V287S/A398T + 8717/8718 V287S/Q399K +++ 8719/8720 Q127I/P169E/R172H + 8721/8722 V376M/A398M + 8723/8724 Q399K +++ 8725/8726 V287S/V376M + 8727/8728 A398R ++ 8729/8730 V287M ++ 8731/8732 A398M/Q399K/I420F ++ 8733/8734 N288P/Q399K +++ 8735/8736 Q127I/P169E/A398L/Q399K +++ 8737/8738 V287L/Q399K +++ 8739/8740 A398M ++ 8741/8742 I420F + 8743/8744 A398T/Q399K + 8745/8746 Q127I/V287S + 8747/8748 P169E/A398T ++ 8749/8750 P169E/R172H/V287S + 8751/8752 Q127I + 8753/8754 V376M/Q399K + 8755/8756 N288P/A398K + 8757/8758 Q127I/P169E/V287S ++ 8759/8760

Q127I/V376M +++ 8761/8762 V376M + 8763/8764 Q127I/P169E/V376M/A398M/Q399K + 8765/8766 A398M/Q399K ++ 8767/8768 A398L/Q399K + 8769/8770 P169E/R172H/A398M/Q399K ++ 8771/8772 V376M/A398L ++ 8773/8774 P169E/R172H/N288P + 8775/8776 V287S +++ 8777/8778 L116I/V287S + 8779/8780 N288P +++ 8781/8782 P169E/A398M/Q399K ++ 8783/8784 A398R/Q399K ++ 8785/8786 N288P/V376S/A398L + 8787/8788 A398L +++ 8789/8790 P169E/V287S +++ 8791/8792 V287S/I420F/R423L/R427S +++ 8793/8794 A398L/R427S ++ 8795/8796

P169E/R172H/N288P/A398L/Q399K/ + I420F/R423L/R427S .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 7784, and defined as follows: “+” = production at least 1.1-fold, but less than 1.18-fold; “++” = at least 1.18-fold, but less than 1.24-fold; and “+++” = at least 1.24-fold increased production, as compared to the reference polypeptide.

(518) TABLE-US-00124 TABLE 59.2 β 1,2GT Round 19 Saturation Mutagenesis Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 7784)

RebM.sup.a 9107/9108 Q127A + 9109/9110 L16T ++ 9111/9112 I331V ++ 9113/9114 V15L/R394H ++ 9115/9116 Y123A ++ 9117/9118 I202Y + 9119/9120 A22G + 9121/9122 P143G ++ 9123/9124 L16A +++ 9125/9126 L116I/P143R + 9127/9128 G198Q +++ 9129/9130 F125L +++ 9131/9132 V144S +++ 9133/9134 F156V + 9135/9136 Y268W +++ 9137/9138 Q127G +++ 9139/9140 G201K +++ 9141/9142 I331C + 9143/9144 L116I/M350L ++ 9145/9146 L36M + 9147/9148 G201A + 9149/9150 L116I ++ 9151/9152 N199S ++ 9153/9154 H93A + 9155/9156 Q127V ++ 9157/9158 V144G + 9159/9160 L16G + 9161/9162 P197H + 9163/9164 G201N + 9165/9166 H324R + 9167/9168 N199Y + 9169/9170 Y123L +++ 9171/9172 G201L + 9173/9174 Y123V ++ 9175/9176 L13Q + 9177/9178 I202A + 9179/9180 G10D/V144L +++ 9181/9182 Y123S +++ 9183/9184 V144Q + 9185/9186 G186N ++ 9187/9188 Y123G + 9189/9190 L116I/Y123S ++ 9191/9192 G96M + 9193/9194 M350L + 9195/9196 V287A ++ 9197/9198 T187Y + 9199/9200 M203T ++ 9201/9202 R14Q + 9203/9204 N199R ++ 9205/9206 G89A ++ 9207/9208 V15A ++ 9209/9210 G10D/N199G +++ 9211/9212 L116I/Y123N + 9213/9214 A149S ++ 9215/9216 T187G + 9217/9218 N199G +++ 9219/9220 N199P ++ 9221/9222 G10D +++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 7784, and defined as follows: “+” = production at least 1.1-fold, but less than 1.18-fold; “++” = at least 1.18-fold, but less than 1.27-fold; and “+++” = at least 1.27-fold increased production, as compared to the reference polypeptide.

Shake Flask Powder Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP to Rebaudioside A 60

(519) A shake flask powder loading dose response experiment was performed to characterize activity of the engineered round 19 variants on rebaudioside A 60%. Levels of 0.0013-0.04 g/L shake flask powder (SFP) were assayed in 100 μ L total reaction volume containing 50 mM potassium phosphate buffer, pH 6, 20 g/L RebA60 or RebA97, 0.025 g/L ADP, 20 (single substrate) or 40 g/L (one-pot) sucrose, 0.04 g/L SUS SFP SEQ ID NO: 8420, and, for the one-pot reaction only, 0.15 g/L β -1,3-glycosyltransferase (β 1,3GT) SFP SEQ ID NO: 8088. The reaction was performed at 60° C. in a Thermotron® titre-plate shaker at 300 RPM for 16-18 h. The reactions were diluted, quenched, and analyzed as described above. The production levels of rebaudioside D in the single substrate and rebaudioside M in the one-pot reactions by the round 19 variants at 0.0025 and 0.005 g/L SFP loading, respectively, are shown in Table 59.3. The variant with SEQ ID NO: 9180 was the most improved variant. Thus, it was selected as the best enzyme for the catalysis of glycosyltransfer from ADP-glucose to rebaudioside A to form rebaudioside D.

(520) TABLE-US-00125 TABLE 59.3 β 1,2GT Round 19 Shake Flask Variants and RebD and RebM Levels SEQ ID Amino Acid Differ- NO: ences (Relative to Increased Increased (nt/aa) SEQ ID NO: 7784) RebD.sup.a RebM.sup.a 8635/8636 S64R/N65G/I112N - + 8651/8652 S64R/A134S ++ ++ 8685/8686 E71T/A141P + + 8765/8766 A398M/Q399K + + 8789/8790 P169E/V287S - + 9131/9132 V144S + ++ 9169/9170 Y123L - - 9179/9180 G10D/V144L + ++ 9181/9182 Y123S - - .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 7784, and defined as follows: “-” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.1-fold; and “++” = at least 1.1-fold increased production, as compared to the reference polypeptide.

Example 60

Beta-1,3-ADP-Glycosyltransferase Variants of SEQ ID NO: 696

(521) In this Example, experiments for evolution and screening of β 1,3-glycosyltransferase (GT) polypeptides derived from SEQ ID NO: 696 for improved glucosylation of steviol glycosides using ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 695 (i.e., SEQ ID NO: 696) was carried out by constructing libraries of variant genes in which mutations associated with improved activity in previous rounds above were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide an eighth round ("Round 8") of 65 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(522) HTP Assay for Glucose Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside

(523) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 695 variants. Pellets were lysed with 400 μ L lysis buffer with 0.5 mg/mL lysozyme and 0.5 mg/mL PMBS in 25 mM Tris-HCl pH 7.5 for 1.5 h and cleared by centrifugation. Assays were conducted with 10 μ L lysate in 100 μ L reactions and with 5 mM rebaudioside D substrate, 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.1 g/L SUS SFP SEQ ID NO: 1222, and 15 mM sucrose (cane sugar). The following reaction conditions were used: 50 mM KPhos buffer, pH 7, 3 mM MgCl.sub.2, 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4-5 h. Then, reactions were solubilized by 20 $\times$ dilution in water, quenched with 5 $\times$ dilution in acetonitrile with 0.2% formic acid, centrifuged 10 m at 4° C., and supernatants were diluted 5 $\times$ in water for analysis. Samples were analyzed by RapidFire-MS/MS as described in Example 31, Table 31.1. Glycosyltransferase variant polypeptides that produced rebaudioside M from rebaudioside D with with in situ synthesized ADP-glucose at greater quantities than SEQ ID NO: 696 were identified. Most were retested in triplicate on stevioside (95% purity). The engineered polypeptides are listed in Table 60.1. Shake-flask scale cultures were grown for SFP production as described in Example 31 for analysis of variants shown in Table 60.2 relative to SEQ ID NO: 696.

(524) TABLE-US-00126 TABLE 60.1 β 1,3GT Round 8 Variants and RebM and RebA Levels SEQ ID NO: Amino Acid Differences Increased Increased (nt/aa) (Relative to SEQ ID NO: 696) RebM.sup.a

RebA.sup.a 4683/4684 T262S/W401L ++ ND 4685/4686 I189L/M206K/K208A/V365I ++ ND

4687/4688 K185R/K208A/E230S/S252N/S255N/D290E/V365I ++ ND 4689/4690

V365I/W401L/V413L/V435Q ++ ND 4691/4692 K208A/V365I/V435Q +++ ND 4693/4694

S304P/P322S/V365I/W401L ++ ND 4695/4696 L249N + + 4697/4698 L249M + + 4699/4700 Q71R + +

4701/4702 L249S + + 4703/4704 I243L + + 4705/4706 K51A + + 4707/4708 K200N + + 4709/4710

L249Y + + 4711/4712 R308F ++ + 4713/4714 P339D + + 4715/4716 L249E + + 4717/4718 L78F + -

4719/4720 L78R + - 4721/4722 L249I ++ ++ 4723/4724 R88K + - 4725/4726 L78Q + - 4727/4728

L78I + - 4729/4730 C366A + + 4731/4732 F352Q ++ + 4733/4734 L245G + + 4735/4736 L78G +++ -

4737/4738 I25V + + 4739/4740 L78K ++ - 4741/4742 L78M + - 4743/4744 S374T + + 4745/4746

L249T + + 4747/4748 L78P ++ - 4749/4750 K200S + + 4751/4752 N56L + + 4753/4754 R308Y ++ -

4755/4756 H259Y + - 4757/4758 H259S +++ ND 4759/4760 T80L +++ ND 4761/4762 C366A ++ ND

4763/4764 P22L + - 4765/4766 I243M ++ + 4767/4768 I25L +++ + 4769/4770 L249H ++ - 4771/4772

I243C + + 4773/4774 S279G +++ +++ 4775/4776 H259G ++ + 4777/4778 L249P ++ ND 4779/4780

I243Y + - 4781/4782 L78P +++ ND 4783/4784 L362T ++ +++ 4785/4786 R88V +++ - 4787/4788 L78E

+++ - 4789/4790 H81C + - 4791/4792 L78G +++ ND 4793/4794 I243V ++ ND 4795/4796 L157G + -

4797/4798 I243V ++ - 4799/4800 R88I +++ - 4801/4802 S364G + + 4803/4804 S282T ++ + 4805/4806

S284T + + 4807/4808 V338C +++ ++ 4809/4810 S364G ++ ND 4811/4812 L157Q + - .sup.aLevels of

increased production were determined relative to the reference polypeptide of SEQ ID NO: 696, and defined as follows: "-" = production less than that of the reference polypeptide; "+" = production at least that of the reference polypeptide, but less than 1.24-fold; "++" = at least 1.24-fold, but less than 1.49-fold; and "+++" = at least 1.49-fold increased production, as compared to the reference polypeptide. ND = not determined.

SFP Characterization Assay and Analysis for Glucosyl Transfer from ADP-Glucose to Stevioside or Rebaudioside D

(525) Shake flask powders (SFP) were reconstituted to a concentration of 20 g/L and diluted to 0.002-5 g/L SFP in 100 μ L total reaction volumes of 50 mM potassium phosphate buffer, pH 7, with 3 mM MgCl.sub.2, 1 mM stevioside (95% purity) or rebaudioside D, and 1 mM ADP-glucose (Sigma, >93%

purify). The reactions were incubated at 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 hour. The reactions were solubilized by 5× dilution with water, quenched by 4× dilution in acetonitrile with 0.2% formic acid, cleared by centrifugation, and diluted 5× with water for analysis. Glycosylated products were detected in by SPE-QQQ as described in Example 31, Table 31.1.

(526) TABLE-US-00127 TABLE 60.2 β 1,3GT Round 8 Shake Flask Powder Variants and RebM and RebA Levels SEQ ID NO: Amino Acid Differences Increased Increased (nt/aa) (Relative to SEQ ID NO: 696) RebM.sup.a RebA.sup.a 4683/4684 T262S/W401L + + 4685/4686 I189L/M206K/K208A/V365I + + 4693/4694 S304P/P322S/V365I/W401L – + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 696, and defined as follows: “–” = production less than that of the reference polypeptide; and “+” = production at least that of the reference polypeptide, but less than 1.35-fold increased production, as compared reference polypeptide.

(527) In these experiments, all of the variants in Table 60.2 (i.e., SEQ ID NOS: 4684, 4686, and 4694) produced rebaudioside A from stevioside and/or rebaudioside M from rebaudioside D with ADP-glucose at greater quantities than SEQ ID NO: 696. The variant SEQ ID NO: 4684 was selected for further directed evolution.

Example 61

Beta-1,3-ADP-Glycosyltransferase Variants of SEQ ID NO: 4684

(528) In this Example, experiments for evolution and screening of β 1,3-glycosyltransferase (β 1,3GT) polypeptides derived from SEQ ID NO: 4684 for improved glucosylation of steviol glycosides using in situ synthesized ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 4684 (i.e., SEQ ID NO: 4683) was carried out by constructing a library of variant genes in which mutations associated with improved activity in previous rounds above were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a ninth round (“Round 9”) of 31 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(529) HTP Assay for Glucose Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside

(530) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 4684 variants. Pellets were lysed, and lysate was cleared as described in Example 60. Assays were conducted with 10 μ L lysate in 100 μ L reactions and with 5 mM stevioside (>95% purity) substrate, 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.1 g/L SUS SFP SEQ ID NO: 1222, and 15 mM sucrose (cane sugar). The following reaction conditions were used: 50 mM KPhos buffer, pH 7, 3 mM MgCl.sub.2, 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4-5 h. Reactions were solubilized, quenched, and diluted as described in Example 60. Samples were analyzed by RapidFire-MS/MS as described in Example 31, Table 31.1. Glycosyltransferase variant polypeptides that produced rebaudioside A from stevioside with in situ synthesized ADP-glucose at greater quantities than SEQ ID NO: 4684 were identified. The engineered polypeptides are listed in Table 61.1. Shake-flask scale cultures were grown, lysed, cleared, and lyophilized to powder as described in Example 31 for analysis of variants shown in Table 61.2 relative to SEQ ID NO: 4684.

(531) TABLE-US-00128 TABLE 61.1 β 1,3GT Round 9 Variants and RebA Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 4684) RebA.sup.a 4813/4814

N56H/L249M/S284T/W353Y + 4815/4816 K51A/I243L/L249E/R308F/W353Y ++ 4817/4818 K51A/N56H/L249M/L362M/C366S ++ 4819/4820 K51A/N56H/L249Y/W353Y + 4821/4822 I243L/R308F/W353Y +++ 4823/4824 K51A/L249Y/S284T/R308F/L362M/C366V +++ 4825/4826 K51A/N56H/L249M/L362M/S364G +++ 4827/4828 K51A/N56H/I243L/L249E/R308F/ +++ L362M/S364G 4829/4830 N56H/I243L/S364G/C366V ++ 4831/4832 N56H/L249M/W353Y ++ 4833/4834 N56H/L249Y +++ 4835/4836 R308F/C366A ++ 4837/4838 N56H/I243L/L249E/S282T/S364G/C366V +++ 4839/4840 K51A/L249M + 4841/4842 N56H + 4843/4844 K51A/L249M/S282T/W353Y/C366S ++ 4845/4846 K51A/L249E/W353Y/L362M/S364G + 4847/4848 N56H/S284T/C366V + 4849/4850 I243L/S282T/W353Y/L362M/S364G + 4851/4852 K51A/W353Y/L362M + 4853/4854 K51A/N56H + 4855/4856 I243L/L249E/S282T/S284T/L362M/ ++ S364G/C366S 4857/4858 S282T + 4859/4860 I243L/L249M/R308F/W353Y/C366A + 4861/4862 K51A/N56H/I243L/L249E/S282T/ + W353Y/L362M/C366S 4863/4864 I243L/S282T/L362M/S364G/C366V ++ 4865/4866 K51A/L249M/S282T/S284T/S364G + 4867/4868

L249Y/W353Y/L362M/C366S + 4869/4870 L362M/C366A + 4871/4872

I243L/L249Y/W353Y/L362M/C366S + 4873/4874 K51A/I243L/L249E/A348S/L362M/C366V ++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 4684, and defined as follows: “+” = production at least 1.2-fold, but less than 1.42-fold; “++” = at least 1.42-fold, but less than 1.59-fold; and “+++” = at least 1.59-fold increased production, as compared to the reference polypeptide.

SFP Characterization Assay and Analysis for Glucosyl Transfer from ADP-Glucose to Stevioside or Rebaudioside D

(532) Shake flask powders (SFP) were reconstituted to a concentration of 20 g/L and diluted to 0.002-5 g/L SFP in 100 μ L total reaction volumes of 50 mM potassium phosphate buffer, pH 7, with 3 mM MgCl₂, 1 mM stevioside (95% purity) or rebaudioside D, and 1 mM ADP-glucose (Sigma, >93% purity). The reactions were incubated at 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 hour. The reactions were solubilized by 5 $\times$ dilution with water, quenched by 4 $\times$ dilution with acetonitrile and 0.2% formic acid, cleared by centrifugation, and diluted 5 $\times$ with water for analysis. Glycosylated products were detected in by SPE-QQQ as described in Example 31, Table 31.1.

(533) TABLE-US-00129 TABLE 61.2 β 1,3GT Round 9 Shake Flask Powder Variants and RebM and RebA Levels SEQ ID Amino Acid Differ- NO: ences (Relative to Increased Increased (nt/aa) SEQ ID NO: 4684) RebA.sup.a RebM.sup.a 4837/4838 N56H; I243L; L249E; ++ +++ S282T; S364G; C366V; 4821/4822 I243L; R308F; W353Y; + ++ 4825/4826 K51A; N56H; L249M; + ++ L362M; S364G; 4823/4824 K51A; L249Y; S284T; + ++ R308F; L362M; C366V; 4833/4834 N56H; L249Y; + + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 4684, and defined as follows: “+” = production at least that of the reference polypeptide, but less than 1.4-fold; “++” = at least 1.4-fold, but less than 2-fold; and “+++” = at least 2-fold increased production, as compared to the reference polypeptide.

(534) In these experiments, all of the variants in Table 61.2 (i.e., SEQ ID NOS: 4822, 4824, 4826, 4834, and 4838) produced rebaudioside A from stevioside and rebaudioside M from rebaudioside D with ADP-glucose at greater quantities than SEQ ID NO: 4684. The variant of SEQ ID NO: 4838 had the highest activity on both stevioside and rebaudioside D with ADP-glucose as a co-substrate in these experiments. Thus, the encoding polynucleotide (SEQ ID NO: 4837) was selected for further directed evolution.

Example 62

Beta-1,3-ADP-Glycosyltransferase Variants of SEQ ID NO: 4838

(535) In this Example, experiments for evolution and screening of β 1,3-glycosyltransferase (β 1,3GT) polypeptides derived from SEQ ID NO: 4838 for improved glucosylation of steviol glycosides using in situ synthesized ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 4837 (i.e., SEQ ID NO: 4838) was carried out by constructing libraries of variant genes in which mutations associated with improved production identified during the development of the present invention were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a tenth round (“Round 10”) of 123 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(536) HTP Assay for Glucose Transfer from Sucrose to ADP and then from ADP-Glucose to Rebaudioside D

(537) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 4838 variants. Pellets were lysed, and the lysate was cleared as described in Example 60, and then diluted 4 $\times$ into potassium phosphate buffer, pH 6.5. Assays were conducted with 10 μ L lysate in 100 μ L reactions and with 5 mM stevioside (>95% purity) substrate, 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.2 g/L SUS SFP SEQ ID NO: 1222, and 15 mM sucrose (cane sugar). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6.5, 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 h. Reactions were solubilized, quenched, and diluted as described in Example 60. Samples were analyzed by RapidFire-MS/MS as described in Example 1, Table 1.1. Selected variants were retested in a similar assay with 10 $\times$ lysate dilution, 4 mM stevioside or rebaudioside D, 0.375 mM ADP, 0.5 g/L sucrose synthase SEQ ID NO: 1392, 30 mM sucrose, and 2 h incubation.

Glycosyltransferase variant polypeptides that produced rebaudioside M from rebaudioside D with in situ

synthesized ADP-glucose at greater quantities than SEQ ID NO: 4838 were identified. The engineered polypeptides are listed in Table 62.1. Shake-flask scale cultures were grown, lysed, cleared, and lyophilized to powder as described in Example 1 for analysis of variants shown in Table 62.2 relative to SEQ ID NO: 4838.

(538) TABLE-US-00130 TABLE 62.1 β 1,3GT Round 10 Variants and RebM and RebA Levels SEQ ID NO: Amino Acid Differences Increased Increased (nt/aa) (Relative to SEQ ID NO: 4838) RebA.sup.a RebM.sup.a 4875/4876 K51A/H56L/L362M +++ +++ 4877/4878 Q177S ++ ++ 4879/4880 K51A ++ ++ 4881/4882 H56L/L362M/V366A ++ +++ 4883/4884 H56L/L243M/E249Y +++ +++ 4885/4886 K208E ++ ++ 4887/4888 A110S ++ +++ 4889/4890 K439D ++ + 4891/4892 H56L/L243M/E249M +++ +++ 4893/4894 K208I ++ ++ 4895/4896 Q265A ++ +++ 4897/4898 K208W/V320I +++ ++ 4899/4900 K208L ++ ++ 4901/4902 K200S/L243M/E249Y/H259G ++ +++ 4903/4904 K336E ++ +++ 4905/4906 K208T ++ +++ 4907/4908 K222P +++ +++ 4909/4910 H259N ++ ++ 4911/4912 K289H ++ +++ 4913/4914 R234T ++ ++ 4915/4916 W353Y/L362M - + 4917/4918 L174S + + 4919/4920 H259S ++ ++ 4921/4922 I25V/L362M/V366S - + 4923/4924 E226R + + 4925/4926 K171V + + 4927/4928 R234E + ++ 4929/4930 R88I - +++ 4931/4932 L256A + + 4933/4934 E175N + ++ 4935/4936 V428K + + 4937/4938 E226T + + 4939/4940 K439Q ++ + 4941/4942 E405Q + + 4943/4944 R88C/R173S - +++ 4945/4946 I198S - +++ 4947/4948 K208G + + 4949/4950 L163K + + 4951/4952 I198D/F313S - + 4953/4954 R234A + + 4955/4956 K171D + + 4957/4958 Q177P + + 4959/4960 I198E - +++ 4961/4962 A85E - ++ 4963/4964 Q209E/R234Q + + 4965/4966 K51A/H56L/K200S/L243M/E249I/H259G/V338C ++ +++ 4967/4968 K208N + ++ 4969/4970 I25L/H56L/W353Y + ++ 4971/4972 I198D/F313S/V428S - ++ 4973/4974 R411T + +++ 4975/4976 R234H/N408D + + 4977/4978 I198D - +++ 4979/4980 P322V + + 4981/4982 Q177K + + 4983/4984 Q209A + + 4985/4986

K51A/Q71R/E249M/S279G/S284T/L362M/V366S + +++ 4987/4988 K171E + ++ 4989/4990 K289T + + 4991/4992 K171P + + 4993/4994 L243M/V338C/L362M/V366S + ++ 4995/4996 P272D + + 4997/4998 Q159N + ++ 4999/5000 H259T + + 5001/5002 P70S/I198D/H259E/F313S - ++ 5003/5004 I25L + ++ 5005/5006 I25V/L243M/E249Y/H259G/V366A + +++ 5007/5008 S253A + + 5009/5010 I25V/L243M/E249I/L362M + +++ 5011/5012 L174E + ++ 5013/5014 K208V + + 5015/5016 I198T - + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 4838, and defined as follows: “-” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.34-fold; “++” = at least 1.34-fold, but less than 1.54-fold; and “+++” = at least 1.54-fold increased production, as compared to the reference polypeptide.

SFP Characterization Assay and Analysis for Glucosyl Transfer from ADP-Glucose to Stevioside or Rebaudioside D

(539) Shake flask powders (SFP) were reconstituted to a concentration of 20 g/L and diluted to 0.006-0.2 g/L SFP in 100 μ L total reaction volumes of 50 mM potassium phosphate buffer, pH 6.5, 1 mM stevioside (95% purity) or rebaudioside D, and 1 mM ADP-glucose (Sigma, >93% purity). The reactions were incubated at 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 hour. The reactions were solubilized by 5 $\times$ dilution with water, quenched by 4 $\times$ dilution with acetonitrile and 0.2% formic acid, cleared by centrifugation, and diluted 5 $\times$ with water for analysis. Glycosylated products were detected in by SPE-QQQ as described in Example 31, Table 31.1.

(540) TABLE-US-00131 TABLE 62.2 β 1,3GT Round 10 Shake Flask Powder Variants and RebA and RebM Levels SEQ ID Amino Acid Differences NO: (Relative to Increased Increased (nt/aa) SEQ ID NO: 4838) RebA.sup.a RebM.sup.a 4875/4876 K51A/H56L/L362M + + 4879/4880 L243M/V338C/L362M/V366S - + 4881/4882 K200S/L243M/E249Y/H259G - - 4909/4910 I198D/F313S - ++ 4911/4912 P70S/I198D/H259E/F313S - - .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 4838, and defined as follows: “-” = production less than that of the reference polypeptide; and “+” = production at least that of the reference polypeptide, but less than 1.3-fold increased production, as compared to the reference polypeptide.

(541) In these experiments, one variant in Table 62.2 (i.e., SEQ ID NO: 4876) produced rebaudioside A from stevioside and rebaudioside M from rebaudioside D with ADP-glucose at greater quantities than SEQ ID NO: 4838. Thus, the encoding polynucleotide (SEQ ID NO: 4875) was selected for further directed evolution.

Example 63

Beta-1,3-ADP-Glycosyltransferase Variants of SEQ ID NO: 4876

(542) In this Example, experiments for evolution and screening of β 1,3-glycosyltransferase (β 1,3GT) polypeptides derived from SEQ ID NO: 4876 for improved glucosylation of steviol glycosides using in situ synthesized ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 4875 (i.e., SEQ ID NO: 4876) was carried out by constructing libraries of variant genes in which mutations associated with improved production identified during the development of the present invention were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide an eleventh round ("Round 11") of 122 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(543) HTP Assay for Glucose Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(544) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 4875 variants. Pellets were lysed, and lysate was cleared as described in Example 60, and then diluted 10 $\times$ into potassium phosphate buffer, pH 6.5. Assays of combinatorial libraries were conducted with 10 μ L lysate in 100 μ L reactions and with 4 mM stevioside (>95% purity) or rebaudioside D substrate, 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.2 g/L SUS SFP SEQ ID NO: 1456, and 24 mM sucrose (cane sugar). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6.5, 50 $^{\circ}$ C. in a Thermotron $^{\circ}$ titre-plate shaker with 300 RPM shaking for 1 h. Reactions were solubilized, quenched, and diluted as described in Example 60. Samples were analyzed by RapidFire-MS/MS as described in Example 1, Table 1.1. For the saturation mutagenesis library, the same assay was performed but with 5 $\times$ diluted lysate, pH 6 potassium phosphate buffer, and 1-2 incubation at 55 $^{\circ}$ C. Selected variants were also retested in triplicate under these conditions. Glycosyltransferase variant polypeptides that produced rebaudioside A/rebaudioside M from stevioside/rebaudioside D with in situ synthesized ADP-glucose at greater quantities than SEQ ID NO: 4876 were identified. The engineered polypeptides are listed in Table 63.1. Shake-flask scale cultures were grown, lysed, cleared, and lyophilized to powder as described in Example 1 for analysis of variants shown in Table 63.2 relative to SEQ ID NO: 4876.

(545) TABLE-US-00132 TABLE 63.1 β 1,3GT Round 11 Variants and RebM and RebA Levels SEQ ID NO: Amino Acid Differences Increased Increased (nt/aa) (Relative to SEQ ID NO: 4876) RebM.sup.a RebA.sup.a

5017/5018	I25L/Q209E/L243M/H259G	++	++					
5019/5020	Q209E/E249M/V338C	++	++					
5021/5022	I198D/K200S/Q209E/L243M/	++	-					
	E249Y/K289T	5023/5024	I25L/E249I ++ ++					
5025/5026	I25L/Q209E/E249I/H259G/V366A	++ ++	5027/5028	K200S/Q209E/V366A ++ +				
5029/5030	I25L/Q209E/H259G	++ ++	5031/5032	I25L/H259G/S279G/K289T ++ ++				
5033/5034	I25L/S279G/S284T/K289T	++ ++	5035/5036	I198D ++ -				
5037/5038	I25L/Q209E/S279G/V366A	++ ++	5039/5040	S279G ++ ++				
5041/5042	I25L/H259G/S279G	++ ++	5043/5044	Q209E/E249Y/S279G/S284T/K289T				
++ ++	5045/5046	I25L/Q209E/K289T/V366A	++ ++	5047/5048	V366A ++ ++			
5049/5050	E249M/H259G	++ ++	5051/5052	Q209E/E249I/H259G/K289T	++ ++			
5053/5054	K289T ++ ++	5055/5056	Q209E/S279G/K289T	++ ++	5057/5058	I25L/Q209E/L243M/E249Y		
++ ++	5059/5060	I25L/K200S/L243M/E249M	++ ++	5061/5062	I25L ++ +			
5063/5064	Q209E ++ ++	5065/5066	Q209E/E249Y/H259G/S279G/	+++ +++	V338C/V366A	5067/5068	I25L/K200S/Q209E/V338C	++ ++
5069/5070	H259G/V366A ++ +	5071/5072	I25H/Q209E/L243M/E249Y	++ +	5073/5074	Q209E/V366A	+++ ++	
5075/5076	I25L/I198D/Q209E	+++ -	5077/5078	V338C ++ ++	5079/5080	I25L/K289T ++ ++		
5081/5082	I25L/S279G/S284T	++ ++	5083/5084	S279G/V338C ++ ++	5085/5086	I198D/E249M/V338C	++ -	
5087/5088	I25L/Q209E/L243M	++ ++	5089/5090	K53R ++ +	5091/5092	R308T + ND	5093/5094	G364S ++ ++
5095/5096	R308C + ND	5097/5098	T54P + ND	5099/5100	R308L ++ ++	5101/5102	S55W + ND	
5103/5104	K53C ++ +	5105/5106	K53L + ND	5107/5108	K336Y + ND	5109/5110	K247L ++ ++	
5111/5112	K53V ++ ++	5113/5114	Q341V + ND	5115/5116	K247A ++ +	5117/5118	S55T ++ ++	
5119/5120	E342R + ND	5121/5122	K336M ++ +	5123/5124	T54V ++ ++	5125/5126	L107A ++ +	
5127/5128	S252Q ++ +	5129/5130	S254G ++ +	5131/5132	A426T + ND	5133/5134	R411S + ND	
5135/5136	S252E ++ ++	5137/5138	K392H ++ ++	5139/5140	S194Q + ND	5141/5142	V34I + ND	
5143/5144	L107T ++ +	5145/5146	S253G ++ +	5147/5148	L107C ++ +	5149/5150	P322M + ND	
5151/5152	K392Q ++ +	5153/5154	G297A + ND	5155/5156	F327L ++ +	5157/5158	T61S ++ ++	

5159/5160 L174M + ND 5161/5162 T8D + ND 5163/5164 R411E + ND 5165/5166 M414L + ND 5167/5168 R412T + ND 5169/5170 S388V + ND 5171/5172 P322T ++ + 5173/5174 S111G + + 5175/5176 S446F + ND 5177/5178 K432T + ND 5179/5180 S253P + + 5181/5182 L141M + ND 5183/5184 K430R + ND 5185/5186 V9G + ND 5187/5188 S254M ++ ++ 5189/5190 S252A + ND 5191/5192 P322R ++ ND 5193/5194 A238K + ND 5195/5196 E321D ++ ND 5197/5198 L256D + + 5199/5200 L256T + ND 5201/5202 M108C + ND 5203/5204 S111C + ND 5205/5206 L256W + ND 5207/5208 R234H/G297A + ND 5209/5210 S254P + + 5211/5212 S449L + + 5213/5214 T8L + ND 5215/5216 R412H + ND 5217/5218 S388T + + 5219/5220 D69S + + 5221/5222 P79S + - 5223/5224 E201T + + 5225/5226 I91R + + 5227/5228 P70K + - 5229/5230 D69T ++ + 5231/5232 W87R ++ - 5233/5234 I91T + + 5235/5236 P79V + - 5237/5238 D69N ++ + 5239/5240 P158Q + + 5241/5242 I91Q + - 5243/5244 W87M ++ - 5245/5246 E73S ++ + 5247/5248 K190R ++ ++ 5249/5250 A153S ++ ++ 5251/5252 E201P + ++ 5253/5254 P79G + - 5255/5256 W87L ++ - 5257/5258 D69Q + + 5259/5260 K205P - ++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 4876, and defined as follows: “-” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.2-fold; “++” = at least 1.2-fold, but less than 2-fold; and “+++” = at least 2-fold increased production, as compared to the reference polypeptide. ND = not determined.

SFP Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(546) Shake flask powders (SFP) were reconstituted to a concentration of 20 g/L and diluted to 0.006-0.2 g/L SFP in 100 μ L total reaction volumes of 50 mM potassium phosphate buffer, pH 6.5, 4 mM stevioside (95% purity) or rebaudioside D, 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.2 g/L SUS SFP SEQ ID NO: 1456, and 24 mM sucrose (cane sugar). The reactions were incubated at 50° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 hour. The reactions were solubilized by 20 $\times$ dilution with water, quenched by 5 $\times$ dilution in acetonitrile with 0.2% formic acid, cleared by centrifugation, and diluted 5 $\times$ with water for analysis. Glycosylated products were detected in by SPE-QQQ as described in Example 31, Table 31.1.

(547) TABLE-US-00133 TABLE 63.2 β 1,3GT Round 11 Shake Flask Powder Variants and RebA and RebM Levels SEQ ID NO: Amino Acid Differences Increased Increased (nt/aa) (Relative to SEQ ID NO: 4876) RebA.sup.a RebM.sup.a 5075/5076 I25L/I198D/Q209E - +++ 5041/5042 I25L/H259G/S279G ++ ++ 5025/5026 I25L/Q209E/E249I/H259G/V366A + ++ 5065/5066 Q209E/E249Y/H259G/S279G/ ++ ++ V338C/V366A 5073/5074 Q209E/V366A + ++ 5043/5044 Q209E/E249Y/S279G/S284T/K289T + + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 4876, and defined as follows: “-” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.3-fold; “++” = at least 1.3-fold, but less than 1.8-fold; and “+++” = at least 1.8-fold increased production, as compared to the reference polypeptide.

(548) In these experiments, the six variants in Table 63.2 (i.e., SEQ ID NO: 5076, 5042, 5026, 5066, 5074, and 5044) produced rebaudioside A from stevioside and/or rebaudioside M from rebaudioside D with ADP-glucose at greater quantities than SEQ ID NO: 4876. SEQ ID NO: 5066 performed best in the primary screen (Table 63.1) and had the most beneficial mutations. Thus, the encoding polynucleotide (SEQ ID NO: 5065) was selected for further directed evolution. The variant with SEQ ID NO: 5076 was used for process development due to its high RebD to RebM activity.

Example 64

Beta-1,3-ADP-Glycosyltransferase Variants of SEQ ID NO: 5066

(549) In this Example, experiments for evolution and screening of β 1,3-glycosyltransferase (β 1,3GT) polypeptides derived from SEQ ID NO: 5066 for improved glucosylation of steviol glycosides using in situ synthesized ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 5065 (i.e., SEQ ID NO: 5066) was carried out by constructing libraries of variant genes in which mutations associated with improved production identified during the development of the present invention were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a twelfth round (“Round 12”) of 40 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(550) HTP Assay for Glucose Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(551) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 5066 variants. Pellets were lysed, and the lysate was cleared as described in Example 60, and then diluted 10× into potassium phosphate buffer, pH 6. Assays of combinatorial libraries were conducted with 10 µL lysate in 100 µL reactions and with 4 mM stevioside (>95% purity) or rebaudioside D substrate, 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.15 g/L SUS SFP SEQ ID NO: 1582, and 24 mM sucrose (cane sugar). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 55° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 2 h. Reactions were solubilized, quenched, and diluted as described in Example 63. Samples were analyzed by RapidFire-MS/MS as described in Example 31, Table 31.1. Glycosyltransferase variant polypeptides that produced rebaudioside A/rebaudioside M from stevioside/rebaudioside D with in situ synthesized ADP-glucose at greater quantities than SEQ ID NO: 5066 were identified. The engineered polypeptides are listed in Table 64.1. Shake-flask scale cultures were grown, lysed, cleared, and lyophilized to powder as described in Example 31 for analysis of variants shown in Table 64.2 relative to SEQ ID NO: 5066.

(552) TABLE-US-00134 TABLE 64.1 β1,3GT Round 12 Variants and RebA and RebM Levels SEQ ID NO: Amino Acid Differences Increased Increased (nt/aa) (Relative to SEQ ID NO: 5066) RebA.sup.a RebM.sup.a 5261/5262 I198E/G259E – +++ 5263/5264 I198D/R234Q + ++ 5265/5266 Q159N/G259E/R411T + + 5267/5268 R88I/Q159N/I198D – ++ 5269/5270 G259E + ++ 5271/5272 R88I/A110D/R234T – ++ 5273/5274 I198D + +++ 5275/5276 I198E/K200S + ++ 5277/5278 R88I – ++ 5279/5280 R88I/I198D/G259E – +++ 5281/5282 R88I/I198D/K200S/G259E + ++ 5283/5284 R88I/A110D – ++ 5285/5286 R88I/G259N – ++ 5287/5288 R88I/A110D/Q159N/I198D/R234T/G259N/Q265A – ++ 5289/5290 R88I/A110D/Q159N/I198D/G259E + +++ 5291/5292 R88I/G259E/Q265A – ++ 5293/5294 Q159N/I198E/R411T – ++ 5295/5296 A110D/G259N/Q265A/R411T + + 5297/5298 R88I/A110D/Q159N/I198E/K200S – +++ 5299/5300 R88I/A110D/I198S – +++ 5301/5302 R88I/A110D/I198D – +++ 5303/5304 R234E + – 5305/5306 R88I/I198S – ++ 5307/5308 I198D/K200S/R234E + – 5309/5310 V320I/V428K + ++ 5311/5312 K208E/V320I/V428K + ++ 5313/5314 K171E/K208E/V320I + +++ 5315/5316 K171P/Q177P + ++ 5317/5318 L174E/V320I + ++ 5319/5320 K171P/L174E/E175N/Q177P/K208W/V320I/V428K/K439D + – 5321/5322 V428K + – 5325/5326 L174E/E175N/Q177P/K208E/V320I/V428K – ++ 5327/5328 K171E/K208E/V428K + ++ 5329/5330 K171E/Q177P/E226T/V428K/K439D + – 5331/5332 K171P/E175N/Q177P/K208L/V320I/V428K + + 5333/5334 K208E/V428K – ++ 5335/5336 K171E/K208E/V320I/V428K + +++ 5337/5338 L174E/E175N/V428K + – 5339/5340 K208E + + 5341/5342 K208L/V320I/R331C/V428K + – .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 5066, and defined as follows: “–” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.35-fold; “++” = at least 1.35-fold, but less than 1.8-fold; and “+++” = at least 1.8-fold increased production, as compared to the reference polypeptide.

SFP Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(553) Shake flask powders (SFP) were reconstituted to a concentration of 20 g/L and diluted to 0.006-0.2 g/L SFP in 100 µL total reaction volumes of 50 mM potassium phosphate buffer, pH 6, 4 mM stevioside (95% purity) or rebaudioside D, 0.4 mM ADP (Sigma, >93% purity) co-substrate, 0.15 g/L SUS SFP SEQ ID NO: 1582, and 24 mM sucrose (cane sugar). The reactions were incubated at 55° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 hour. The reactions were solubilized by 20× dilution with water, quenched by 5× dilution in acetonitrile with 0.2% formic acid, cleared by centrifugation, and diluted 5× with water for analysis. Glycosylated products were detected in by SPE-QQQ as described in Example 31, Table 31.1.

(554) TABLE-US-00135 TABLE 64.2 β1,3GT Round 12 Shake Flask Powder Variants and RebA and RebM Levels SEQ ID Amino Acid Differences Increased Increased NO: (nt/aa) (Relative to SEQ ID NO: 5066) RebA.sup.a RebM.sup.a 5279/5280 R88I/I198D/G259E – ++ 5289/5290 R88I/A110D/Q159N/I198D/G259E – +++ 5301/5302 R88I/A110D/I198D – ++ 5313/5314 K171E/K208E/V320I + + 5323/5324 V320I/V428K + + .sup.aLevels of increased production were

determined relative to the reference polypeptide of SEQ ID NO: 5066, and defined as follows: “-” = production less than 0.95-fold; “+” = production at least 0.95-fold, but less than 1.2-fold; “++” = at least 1.2-fold, but less than 1.9-fold; and “+++” = at least 1.9-fold increased production, as compared to the reference polypeptide.

(555) In these experiments, the six variants in Table 64.2 (i.e., SEQ ID NO: 5280, 5290, 5302, 5314, 5324 produced rebaudioside A from stevioside and/or rebaudioside M from rebaudioside D with ADP-glucose at greater quantities than SEQ ID NO: 5066. SEQ ID NO: 5290 performed best on rebaudioside D and had the most beneficial mutations. Thus, the encoding polynucleotide (SEQ ID NO: 5289) was selected for further directed evolution.

Example 65

Beta-1,3-ADP-Glycosyltransferase Variants of SEQ ID NO: 5290

(556) In this Example, experiments for evolution and screening of β 1,3-glycosyltransferase (β 1,3GT) polypeptides derived from SEQ ID NO: 5290 for improved glucosylation of steviol glycosides using in situ synthesized ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 5289 (i.e., SEQ ID NO: 5290) was carried out by constructing libraries of variant genes in which mutations associated with improved production identified during the development of the present invention were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a thirteenth round (“Round 13”) of 100 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(557) HTP Assay for Glucose Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(558) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 5289 variants. Pellets were lysed, and lysate was cleared as described in Example 60, and then diluted 5× into potassium phosphate buffer, pH 6. Assays were conducted with 10 μ L lysate in 100 μ L reactions and with 4 mM stevioside (>95% purity) or rebaudioside D substrate, 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.15 g/L SUS SFP SEQ ID NO: 1764, and 24 mM sucrose (cane sugar). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 55° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 2 h. Reactions were solubilized, quenched, and diluted as described in Example 33. Samples were analyzed by RapidFire-MS/MS as described in Example 31, Table 31.1.

Glycosyltransferase variant polypeptides that produced rebaudioside A/rebaudioside M from stevioside/rebaudioside D with in situ synthesized ADP-glucose at greater quantities than SEQ ID NO: 5290 were identified. The engineered polypeptides are listed in Table 65.1. Shake-flask scale cultures were grown, lysed, cleared, and lyophilized to powder as described in Example 31 for analysis of variants shown in Table 65.2 relative to SEQ ID NO: 5290.

(559) TABLE-US-00136 TABLE 65.1 β 1,3GT Round 13 Variants and RebA and RebM Levels SEQ ID NO: Amino Acid Differences Increased Increased (nt/aa) (Relative to SEQ ID NO: 5290) RebA.sup.a

RebM.sup.a 5343/5344 R308L + + 5345/5346 K53R/T54V/S55T/A153S/E201P + - 5349/5350

K53V/K171P/R308L/K392H + + 5351/5352 I91T/E201P/K439D/Y444H + + 5353/5354

R308L/F327L/K439D + + - 5355/5356 E73S/W87M/E201P - + + 5357/5358 K53V/T54V/R308L/K392H + - 5359/5360 K247A/Y249I/S252E/S254M/V320I/ + - V428K 5361/5362

I25L/K247L/Y249I/S252E/G364S/ + - V428K 5363/5364 V428K + + 5365/5366 S253G/V320I + +

5367/5368 Y249I/S252E/S253G/S254M + + 5369/5370 S252E/S253G/S254M + + 5371/5372

I25L/T61S/V428K + + 5373/5374 K190R + + 5375/5376 T61S/K208W/S252E/S254M/V428K + -

5377/5378 L107A/K247A/Y249I/S252E/S254M/ + + G364S 5379/5380

L107A/K208N/V320I/G364S/V428K + + 5381/5382 L107A + + 5383/5384

K190R/K208N/K247L/S252E/V428K + + 5385/5386 I25L/K208E/K247L/S252E/S253G/ + +

G364S/V428K 5387/5388 V320I + + 5389/5390 V320I/V428K + + 5391/5392

I25L/T61S/K208N/S252E/S253G + - 5393/5394 L107A/K247A/S252E + - 5395/5396

T61S/L107T/K208N/S252E/S253G/ + + + S254M/G364S/V428K 5397/5398

I25L/L107T/K208E/V320I/V428K + + 5399/5400 V320I/G364S/V428K + + 5401/5402

L107A/K247A/V428K - + 5403/5404 Y249I/S252E/S254M/V428K + - 5405/5406 V435Q + + +

5407/5408 T172H + + + 5409/5410 V435R + + + 5411/5412 A427L + + 5413/5414 R296H ND +

5415/5416 K289S ND + 5417/5418 Q177P ++ + 5419/5420 G404T ND + 5421/5422 A438Q ND + 5423/5424 D431E ND + 5425/5426 V428E/D434N ND + 5427/5428 T54A ND + 5429/5430 K171T ++ + 5431/5432 W233Q/Q269R ND + 5433/5434 D431R ND + 5435/5436 E2L ++ + 5437/5438 D347K ++ + 5439/5440 Q424W ND + 5441/5442 S106T ND + 5443/5444 G169L ++ + 5445/5446 W233Q ND + 5447/5448 R417P ++ + 5449/5450 G169C ND + 5451/5452 V179A ++ + 5453/5454 S317R ND + 5455/5456 V428I ++ + 5457/5458 W233M ++ + 5459/5460 A251L ND + 5461/5462 V435K ND + 5463/5464 N408R ND + 5465/5466 W233C ++ + 5467/5468 D347H ++ + 5469/5470 K171S ++ + 5471/5472 R296Q ++ + 5473/5474 D300G ND + 5475/5476 G169V ND + 5477/5478 E2N ++ + 5479/5480 V435M ND + 5481/5482 G169Q ND + 5483/5484 W233K ++ + 5485/5486 E99P ND + 5487/5488 V428S ND + 5489/5490 V179S ND + 5491/5492 D347P ++ + 5493/5494 R296A ND + 5495/5496 Q424A ++ + 5497/5498 W233V ++ + 5499/5500 G169E ++ + 5501/5502 V435T ND + 5503/5504 K222A ND + 5505/5506 V428R ++ + 5507/5508 V428Q ND + 5509/5510 W233L ++ + 5511/5512 N159L ND + 5513/5514 V428F ND + 5515/5516 R38Q - + 5517/5518 F264A ++ + 5519/5520 D347R ++ + 5521/5522 R109S ND + 5523/5524 T172N ++ + 5525/5526 V435C ND + 5527/5528 K171V ++ + 5529/5530 A427R ++ + 5531/5532 F64S ++ + 5533/5534 E318T - + 5535/5536 W233G ++ + 5537/5538 E259T ND + 5539/5540 T7P ++ + 5541/5542 N68K ND + .sup.a

Levels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 5290, and defined as follows: “-” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.3-fold; “++” = at least 1.3-fold increased production, as compared to the reference polypeptide.

SFP Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(560) Shake flask powders (SFP) were reconstituted to a concentration of 20 g/L and diluted to 0.006-0.2 g/L SFP in 100 μ L total reaction volumes of 50 mM potassium phosphate buffer, pH 6, 4 mM stevioside (95% purity), rebaudioside D, or rebaudioside A (97% purity), 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.15 g/L SUS SFP SEQ ID NO: 1764, and 24 mM sucrose (cane sugar). The reactions were incubated at 50 and 55° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 hour. The reactions were solubilized by 20 $\times$ dilution with water, quenched by 5 $\times$ dilution in acetonitrile with 0.2% formic acid, cleared by centrifugation, and diluted 5 $\times$ with water for analysis. Glycosylated products were detected in by SPE-QQQ as described in Example 31, Table 31.1.

(561) TABLE-US-00137 TABLE 65.2 β 1,3GT Round 13 Shake Flask Powder Variants and RebA and RebM Levels SEQ ID Amino Acid NO: Differences (Relative Increased Increased (nt/aa) to SEQ ID NO: 5290) RebA.sup.a RebM.sup.a 5347/5348 W87M ++ 5355/5356 E73S/W87M/E201P ++ 5363/5364 V428K ++ 5371/5372 I25L/T61S/V428K ++ 5385/5386 I25L/K208E/K247L/S252E/ + + S253G/G364S/V428K .sup.a Levels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 5290, and defined as follows: “-” = production less than that of the reference polypeptide; and “+” = production at least that of the reference polypeptide, but less than 1.45-fold increased production, as compared to the reference polypeptide.

(562) In these experiments, the five variants in Table 65.2 (i.e., SEQ ID NO: 5348, 5356, 5364, 5372, and 5386) produced rebaudioside A from stevioside and/or rebaudioside M from rebaudioside D with ADP-glucose at greater quantities than SEQ ID NO: 5290. SEQ ID NO: 5372 was selected for further directed evolution.

Example 66

Beta-1,3-ADP-Glycosyltransferase Variants of SEQ ID NO: 5372

(563) In this Example, experiments for evolution and screening of β 1,3-glycosyltransferase (β 1,3GT) polypeptides derived from SEQ ID NO: 5372 for improved glucosylation of steviol glycosides using in situ synthesized ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 5371 (i.e., SEQ ID NO: 5372) was carried out by constructing libraries of variant genes in which mutations associated with improved production identified during the development of the present invention were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a fourteenth round (“Round 14”) of 74 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(564) HTP Assay for Glucose Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(565) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 5371 variants. Pellets were lysed, and the lysate was cleared as described in Example 60, and then diluted 10× into potassium phosphate buffer, pH 6. Assays were conducted with 10 μL lysate in 100 μL reactions and with 10 mM stevioside (>95% purity) or rebaudioside D substrate, 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.15 g/L SUS SFP SEQ ID NO: 1764, and 40 mM sucrose (cane sugar). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 2 h. Reactions were solubilized, quenched, and diluted to ~10 μM steviol glycosides as described in Example 60. Samples were analyzed by RapidFire-MS/MS as described in Example 31, Table 31.1. Glycosyltransferase variant polypeptides that produced rebaudioside A/rebaudioside M from stevioside/rebaudioside D with in situ synthesized ADP-glucose at greater quantities than SEQ ID NO: 5372 were identified. The engineered polypeptides are listed in Table 66.1. Shake-flask scale cultures were grown, lysed, cleared, and lyophilized to powder as described in Example 1 for analysis of variants shown in Table 66.2 relative to SEQ ID NO: 5372.

(566) TABLE-US-00138 TABLE 66.1 β1,3GT Round 14 Variants and RebA and RebM Levels SEQ ID Amino Acid Increased Increased NO: Differences (Relative (nt/aa) to SEQ ID NO: 5372) RebM.sup.a RebA.sup.a 5543/5544 W87M/Q177P/Y249I +++ ++ 5545/5546 I88R/K208E/Y249I/V320I – ++ 5547/5548 W87M/I88R/Q177P/D198E/ +++ + Y249I 5549/5550 W87M/I88R +++ + 5551/5552 N159Q/D198I/K208E/V320I +++ +++ 5553/5554 W87M/D198E/A199K ++ ++ 5555/5556 W87M/D198E +++ ++ 5557/5558 I88R/V320I + +++ 5559/5560 W87M +++ ++ 5561/5562 W87M/I88R/E94K/N159Q/ +++ ++ Q177P/K208E 5563/5564 W87M/K208E +++ ++ 5565/5566 I88R/Q177P/K208E + ++ 5567/5568 I88R/D198I – ++ 5569/5570 W87M/I88R/K208E ++ + 5571/5572 I88R +++ ++ 5573/5574 W87M/I88R/N159Q/D198E +++ + 5575/5576 W87M/K208E/V320I ++ + 5577/5578 Q177P/D198E +++ ++ 5579/5580 Q177P/K208E ++ ++ 5581/5582 N159Q/Q177P/K208E +++ ++ 5583/5584 W87M/N159Q/Q177P +++ ++ 5585/5586 I88R/Q177P/V320I ++ ++ 5587/5588 W87M/N159Q/V320I ++ + 5589/5590 N159Q/Q177P/D198E +++ ++ 5591/5592 W87M/V320I +++ + 5593/5594 I88R/A199K – ++ 5595/5596 I88R/A199K/K208E – ++ 5597/5598 W87M/D198E/K208E/V320I +++ ++ 5599/5600 I88R/K208E – ++ 5601/5602 I88R/Q177P ++ +++ 5603/5604 P272K ND + 5605/5606 D268G ND + 5607/5608 P158D ND + 5609/5610 E2H ND + 5611/5612 E2G ND + 5613/5614 E2P ND + 5615/5616 S374R ND + 5617/5618 S135A ND + 5619/5620 K171T ND + 5621/5622 S214L ND + 5623/5624 R173N ND + 5625/5626 L256P ND + 5627/5628 L437T ND + 5629/5630 L257Q ND + 5631/5632 L437I ND ++ 5633/5634 K222R ND + 5635/5636 D134S ND + 5637/5638 S374K ND ++ 5639/5640 N408D ND + 5641/5642 P158E ND + 5643/5644 E113S ND + 5645/5646 K392D ND + 5647/5648 Y453R ND + 5649/5650 K439A ND + 5651/5652 N195H ND + 5653/5654 K302R ND + 5655/5656 L257A ND + 5657/5658 R412H ND + 5659/5660 E330D ND + 5661/5662 R173P ND + 5663/5664 N399D ND + 5665/5666 A348C ND + 5667/5668 D134G ND + 5669/5670 K289T ND + 5671/5672 S253G ND + 5673/5674 D300G ND + 5675/5676 E73P ND + 5677/5678 F49A ND + 5679/5680 D268H ND ++ 5681/5682 D268A ND + 5683/5684 E445T ND + 5685/5686 E73T ND + 5687/5688 R411Q ND + 5689/5690 L437V ND ++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 5372, and defined as follows: “–” = production less than 0.99-fold; “+” = production at least 0.99-fold, but less than 1.25-fold; “++” = at least 1.25-fold, but less than 1.6-fold; and “+++” = at least 1.6-fold increased production, as compared to the reference polypeptide. ND = not determined.

SFP Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(567) Shake flask powders (SFP) were reconstituted to a concentration of 20 g/L and diluted to 0.006-0.2 g/L SFP in 100 μL total reaction volumes of 50 mM potassium phosphate buffer, pH 6, 4 mM stevioside (95% purity), rebaudioside D, rebaudioside E, or rebaudioside A (97% purity), 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.15 g/L SUS SFP SEQ ID NO: 1764, and 24 mM sucrose (cane sugar). The reactions were incubated at 55 and 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 2 hours. The reactions were solubilized by 20× dilution with water, quenched by 5× dilution in acetonitrile with 0.2% formic acid, cleared by centrifugation, and diluted 5× with water for analysis.

Glycosylated products were detected in by SPE-QQQ as described in Example 31, Table 31.1.

(568) TABLE-US-00139 TABLE 66.2 β 1,3GT Round 14 Shake Flask Powder Variants and RebA, RebI, and RebM Levels Increased Increased SEQ ID Amino Acid Differences Increased Increased RebE to RebD to NO: (nt/aa) (Relative to SEQ ID NO: 5372) RebA.sup.a RebI.sup.a rebM.sup.a rebM.sup.a 5551/5552 N159Q/D198I/K208E/V320I + + + + 5561/5562 W87M/I88R/E94K/N159Q/Q177P/K208E + + + + + 5573/5574 W87M/I88R/N159Q/D198E + + + + + 5577/5578 Q177P/D198E + + + + + 5581/5582 N159Q/Q177P/K208E + + + + + 5589/5590 N159Q/Q177P/D198E + + + + + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 5372, and defined as follows: “-” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 2-fold; “++” = at least 2-fold, but less than 4-fold; and “+++” = at least 4-fold increased production, as compared to the reference polypeptide.

(569) In these experiments, the six variants in Table 66.2 (i.e., SEQ ID NO: 5552, 5562, 6675, 5578, 5582, and 5590) produced more rebaudioside A from stevioside, less rebaudioside I from rebaudioside A, more rebaudioside M from rebaudioside E, and/or rebaudioside M from rebaudioside D with ADP-glucose at greater quantities than SEQ ID NO: 5372. SEQ ID NO: 5562 was the most improved for all four reactions, so it was selected for further directed evolution.

Example 67

Beta-1,3-ADP-Glycosyltransferase Variants of SEQ ID NO: 5562

(570) In this Example, experiments for evolution and screening of β 1,3-glycosyltransferase (β 1,3GT) polypeptides derived from SEQ ID NO: 5562 for improved glucosylation of steviol glycosides using in situ synthesized ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 5561 (i.e., SEQ ID NO: 5562) was carried out by constructing libraries of variant genes in which mutations associated with improved activity identified during the development of the present invention were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a fifteenth round (“Round 15”) of 62 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(571) HTP Assay for Glucose Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(572) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 5562 variants. Pellets were lysed, and lysate was cleared as described in Example 60 and then diluted 5 $\times$ into 50 mM potassium phosphate buffer, pH 6. Assays were conducted with 10 μ L lysate in 100 μ L reactions and with 10 mM stevioside (>95% purity) or rebaudioside D substrate, 0.1-0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.15 g/L SUS SFP SEQ ID NO: 1764, and 24-40 mM sucrose (cane sugar). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 50 or 60 $^{\circ}$ C. in a Thermotron $^{\circ}$ titre-plate shaker with 300 RPM shaking for 2 h. Reactions were solubilized, quenched, and diluted to -10 μ M steviol glycosides as described in Example 60. Samples were analyzed by RapidFire-MS/MS as described in Example 31, Table 31.1. Glycosyltransferase variant polypeptides that produced rebaudioside A/rebaudioside M from stevioside/rebaudioside D with in situ synthesized ADP-glucose at greater quantities than SEQ ID NO: 5562 were identified. The engineered polypeptides from the combinatorial libraries are listed in Table 67.1, and those from the saturation mutagenesis libraries are listed in Table 67.2. Shake-flask scale cultures were grown, lysed, cleared, and lyophilized to powder as described in Example 31 for analysis of variants shown in Table 67.3 relative to SEQ ID NO: 5562.

(573) TABLE-US-00140 TABLE 67.1 β 1,3GT Round 15 Combinatorial Variants and RebA Levels SEQ ID Amino Acid Differences Increased NO: (nt/aa) (Relative to SEQ ID NO: 5562) RebA.sup.a 5691/5692 F64S/G169E/E201P/D347P/K392H + 5693/5694 K171T/E201P/K392H/R417P +++ 5695/5696 G169E/K171T/T172H/F264A/K392H/V435Q + 5697/5698 G169E/T172H/E201P/F264A/D347G ++ 5699/5700 I91T/E201P/F264A/D347G/K392H ++ 5701/5702 G169E/T172H/E201P/D347G/K392H ++ 5703/5704 G169E/K171T/E201P/F264A/K392H/V435Q + 5705/5706 E201P/F264A/D347P/K392H/V435Q ++ 5707/5708 G169E/T172H/E201P/F264A/D347K/K392H/V435R ++ 5709/5710 E201P/F264A/D347K/K392H/R417P/V435R + 5711/5712 G169E/T172H/E201P/R417P/V435R +

5713/5714 F64S/T172H/F264A/D268V/D347P/K392H/R417P + 5715/5716
 E201P/F264A/D347K/R417P/V435R ++ 5717/5718 I91T/K94E/K171T/T172H/E201P/F264A/D347G +
 5719/5720 K171T/T172H/E201P/F264A/K392H/R417P/V435R +++ 5721/5722
 K94E/E201P/F264A/D347G/V435Q + 5723/5724 L107A/S252E/S317R + 5725/5726
 W233K/S252E/V320I +++ 5727/5728 W233K/S252E/S317R/N408R ++ 5729/5730
 K190R/W233K/S252E/V320I/N408R ++ 5731/5732 L107A/K190R/W233K/S252E/S317R/V320I +++
 5733/5734 W233K/S252E/S317R/A427R ++ 5735/5736
 T7P/K190R/W233K/S252E/S317R/V320I/A427R/K428R +++ 5737/5738
 T7P/L107A/W233K/S252E/S253G + 5739/5740 W233K/S252E/V320I/N408R/K428R ++ 5741/5742
 T7P/W233K/S252E/S253G/N408R +++ 5743/5744 W233K/S252E/S253G/N408R/A427L ++ 5745/5746
 T7P/K190R/W233K/S252E/S253G/V320I/A427L + 5747/5748
 T7P/W233K/S252E/S253G/S317R/N408R/A427L +++ 5749/5750
 T7P/L107A/W233K/S252E/S317R/N408R + 5751/5752
 T7P/L107A/K190R/W233K/S252E/S253G/S317R/N408R/A427R ++ 5753/5754
 T7P/W233K/S252E/S317R/V320I/A427L + 5755/5756 E2N/T7P/L107A/W233K/S252E/S253G/V320I
 + 5757/5758 K190R/W233K/S252E/S253G/S317R/V320I/N408R/K428R + 5759/5760
 E2L/T7P/W233K/S252E/S253G/V320I/A427R/K428R + 5761/5762
 W233K/S252E/S253G/S317R/V320I/A427R/K428R +++ 5763/5764 T7P/W233K/S252E/S317R/K428R
 + 5765/5766 E2N/W233K/S252E/S317R/V320I + 5767/5768 L107A/W233K/S252E/S253G ++
 5769/5770 T7P/L107A/W233K/S252E/S317R/V320I + 5771/5772
 T7P/K190R/W233K/S252E/N408R/A427R + 5773/5774 K190R/W233K/S252E ++ 5775/5776
 L107A/K190R/W233K/S252E + 5777/5778 K190R/W233K/S252E/S317R/N408R +++ 5779/5780
 E2N/T7P/L107A/W233K/S252E/S253G/V320I/N408R/K428R + 5781/5782 A348S/S374R/V435R +
 5783/5784 S374R + .sup.aLevels of increased production were determined relative to the reference
 polypeptide of SEQ ID NO: 5562, and defined as follows: “+” = production at least 1.59-fold, but less
 than 2.8-fold; “++” = at least 2.8-fold, but less than 3.1-fold; and “+++” = at least 3.1-fold increased
 production, as compared to the reference polypeptide.

(574) TABLE-US-00141 TABLE 67.2 β 1,3GT Round 15 Saturation Mutagenesis Variants and RebA and
 RebM Levels Amino Acid SEQ ID Differences Increased Increased Increased NO: (Relative to RebA,
 RebA, RebM, (nt/aa) SEQ ID NO: 5562) 50° C..sup.a 60° C..sup.a 60° C..sup.a 5785/5786 L56A - + ++
 5787/5788 S55V - - + 5789/5790 R308Q - ++ +++ 5791/5792 K336Q + - - 5793/5794 G364A + +++
 +++ 5795/5796 E342W + + + 5797/5798 T282S ++ +++ +++ 5799/5800 R308L + +++ ++ 5801/5802
 G364S +++ +++ +++ 5803/5804 E407C + + +++ 5805/5806 L391C + ++ +++ 5807/5808 E407V ++ ++
 +++ 5809/5810 V14I ++ ++ ++ 5811/5812 S255L + + + 5813/5814 M422Q - + + .sup.aLevels of
 increased production were determined relative to the reference polypeptide of SEQ ID NO: 5562, and
 defined as follows: “-” = production less than that of the reference polypeptide; “+” = production at least
 that of the reference polypeptide, but less than 1.25-fold; “++” = at least 1.25-fold, but less than 1.4-fold;
 and “+++” = at least 1.4-fold increased production, as compared to the reference polypeptide.

SFP Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP and then from
 ADP-Glucose to Stevioside and Rebaudioside D

(575) Shake flask powders (SFP) were reconstituted to a concentration of 20 g/L and diluted to 0.006-0.2
 g/L SFP in 100 μ L total reaction volumes of 50 mM potassium phosphate buffer, pH 6, 10 mM stevioside
 (95% purity) or rebaudioside D, 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.15 g/L SUS SFP SEQ
 ID NO: 1764, and 24 mM sucrose (cane sugar). The reactions were incubated at 50 and 60° C. in a
 Thermotron® titre-plate shaker with 300 RPM shaking for 2 hours. The reactions were solubilized by 20 $\times$
 dilution with water, quenched by 5 $\times$ dilution in acetonitrile with 0.2% formic acid, cleared by
 centrifugation, and diluted 5 $\times$ with water for analysis. Glycosylated products were detected in by SPE-
 QQQ as described in Example 31, Table 31.1.

(576) TABLE-US-00142 TABLE 67.3 β 1,3GT Round 15 Shake Flask Powder Variants and RebA and
 RebM Levels Amino Acid SEQ ID Differences Increased Increased Increased NO: (Relative to RebA,
 RebA, RebM, (nt/aa) SEQ ID NO: 5562) 50° C..sup.a 60° C..sup.a 60° C..sup.a 5693/5694
 K171T/E201P/K392H/ + + ++ R417P 5707/5708 G169E/T172H/E201P/ + ++ +++
 F264A/D347K/K392H/ V435R 5713/5714 F64S/T172H/F264A/ + + +++ D268V/D347P/K392H/ R417P

5719/5720 K171T/T172H/E201P/ + ++ ++ F264A/K392H/R417P/ V435R 5731/5732
L107A/K190R/W233K/ - + +++ S252E/S317R/V320I 5735/5736 T7P/K190R/W233K/ - ++ ++
S252E/S317R/V320I/ A427R/K428R 5761/5762 W233K/S252E/S253G/ - + ++ S317R/V320I/A427R/
K428R 5777/5778 K190R/W233K/S252E/ - + +++ S317R/N408R .sup.aLevels of increased production
were determined relative to the reference polypeptide of SEQ ID NO: 5562, at 0.025 g/L shake flask
powder and defined as follows: “-” = production less than that of the reference polypeptide; “+” =
production at least that of the reference polypeptide, but less than 2-fold; “++” = at least 2-fold, but less
than 2.4-fold; and “+++” = at least 2.4-fold increased production, as compared to the reference
polypeptide.

(577) In these experiments, the eight variants in Table 67.3 produced more rebaudioside A from
stevioside and/or more rebaudioside M from rebaudioside D at 50° C. and/or 60° C. than SEQ ID NO:
5562. SEQ ID NO: 5708 was the most improved for converting stevioside to rebaudioside A for both
temperatures and also improved for converting rebaudioside D to rebaudioside M at 60° C., so it was
selected for further directed evolution.

Example 68

Beta-1,3-ADP-Glycosyltransferase Variants of SEQ ID NO: 5708

(578) In this Example, experiments for evolution and screening of β 1,3-glycosyltransferase (β 1,3GT)
polypeptides derived from SEQ ID NO: 5708 for improved glucosylation of steviol glycosides using in
situ synthesized ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 5707
(i.e., SEQ ID NO: 5708) was carried out by constructing libraries of variant genes in which mutations
associated with improved production identified during the development of the present invention were
recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP)
assay described below to provide a sixteenth round (“Round 16”) of 100 engineered GT variant
polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(579) HTP Assay for Glucose Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside
and Rebaudioside D

(580) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO:
5708 variants. Pellets were lysed, and lysate was cleared as described in Example 60, and then diluted
20× into 50 mM potassium phosphate buffer, pH 6. Assays were conducted with 10 μ L lysate in 100 μ L
reactions and with 10 mM stevioside (>95% purity), rebaudioside D, rebaudioside A (>97% purity), or
rebaudioside E substrate, 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.15 g/L SUS SFP SEQ ID
NO: 1764, and 24 mM sucrose (cane sugar). The following reaction conditions were used: 50 mM
potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for
3 h. Reactions were solubilized, quenched, and diluted to -10 μ M steviol glycosides as described in
Example 60. Samples were analyzed by RapidFire-MS/MS as described in Example 31, Table 31.1.
Glycosyltransferase variant polypeptides that produced rebaudioside A/rebaudioside M from
stevioside/rebaudioside D with in situ synthesized ADP-glucose at greater quantities than SEQ ID NO:
5708 were identified. The engineered polypeptides are listed in Table 68.1. Shake-flask scale cultures
were grown, lysed, cleared, and lyophilized to powder as described in Example 31 for analysis of variants
shown in Table 68.2 relative to SEQ ID NO: 5708.

(581) TABLE-US-00143 TABLE 68.1 β 1,3GT Round 16 Combinatorial Variants and RebA Levels SEQ
ID Amino Acid Differences Increased NO: (nt/aa) (Relative to SEQ ID NO: 5708) RebA.sup.a 5815/5816
R173N/K190R/S252E/L257A/K347G/ +++ A427R/L437I 5817/5818 R173N + 5819/5820
W233K/S252E/L437V +++ 5821/5822 R173N/K190R/L257A/A427R/L437I ++ 5823/5824
K347G/R411Q/L437I ++ 5825/5826 W233K/S252E/E285Q/L437V ++ 5827/5828 R173N/S252E +++
5829/5830 E285Q/K347G/L437V + 5831/5832 R173N/S252E/D268H/L437V ++ 5833/5834
L257A/K347G/R411Q/L437V + 5835/5836 S252E/L257A/K347G/L437V ++ 5837/5838
R173N/S252E/K347G/R411Q ++ 5839/5840 W233K/S252E +++ 5841/5842
R173N/W233K/S252E/L257A/A427R + 5843/5844 W233K/S252E/L257A/A427R/L437V + 5845/5846
W233K/S252E/R411Q/L437V +++ 5847/5848 R173N/L257A/S374K/L437V + 5849/5850
K190R/S252E/L257A/R411Q/L437I + 5851/5852 R173N/K190R/L257A/S374K/L437V + 5853/5854
W233K/S374K/L437I + 5855/5856 K347G/A427R/L437V ++ 5857/5858
K190R/W233K/S252E/L257A/K347G/ ++ R411Q/L437I 5859/5860 W233K/S252E/L257A/K347G +++

5861/5862 W233K/S252E/L257A/K347G/R411Q/ +++ L437I 5863/5864 K190R/S252E/L257A/E285Q/A427R ++ 5865/5866 R173N/K190R/W233K/S252E/A427R/ + L437V 5867/5868 S374K/R411Q/L437V + 5869/5870 R173N/W233K/L437I + 5871/5872 R173N/A427R/L437I + 5873/5874 S374K/R411Q/L437I + 5875/5876 K190R/S252E ++ 5877/5878 S252E/L257A ++ 5879/5880 W233K/S374K + 5881/5882 R411Q/L437V + 5883/5884 S374K + 5885/5886 K190R/S374K/A427R/L437V + 5887/5888 L437I ++ 5889/5890 R173N/S374K/L437I ++ 5891/5892 K190R/S252E/L257A/A427R + 5893/5894 W233K/S252E/L257A ++ 5895/5896 W233K/S252E/E285Q + 5897/5898 L163A/L257Q/K302R/G364S ++ 5899/5900 K302R/G364S ++ 5901/5902 E2P/K4T/P158D/L163A/G364S/S449R +++ 5903/5904 E2P/P158D/L163R + 5905/5906 E2P/L163R/G364S/Y453R + 5907/5908 E2P/K4T/L163A/L257Q ++ 5909/5910 G364S/N399D ++ 5911/5912 E2P/K4T/L163A/G364S/Y453R ++ 5913/5914 E2P/K4T/L163A/E330D/S449R ++ 5915/5916 K4T/E113S/P158D/K302R/E330D/ + G364S 5917/5918 E2P/E113S/S449R + 5919/5920 E2P/K4T/K302R + 5921/5922 L163R/K302R/E330D/G364S/S449R + 5923/5924 E330D/G364S + 5925/5926 E2P/K4T/E113S/P158D/L163A/E330D/ +++ S449R 5927/5928 E2P/P158D/L163A/G364S/N399D +++ 5929/5930 E2P/K4T/L163R/L257Q/E330D/N399D/ ++ S449R 5931/5932 E2P/E113S/L163R/S449R + 5933/5934 L163A/G364S ++ 5935/5936 P158D/L163A/G364S ++ 5937/5938 E2P/K4T/P158D/E330D +++ 5939/5940 E2P/E113S + 5941/5942 E2P/K4T/L163R/K222R/K302R/E330D/ + G364S/S449R 5943/5944 E113S/P158D/L163A/G364S/N399D + 5945/5946 K4T/E113S/L163A/G364S/N399D ++ 5947/5948 E2P/K4T/E113S/P158D/L163R/K302R/ + G364S/N399D/S449R 5949/5950 E2P/K4T/L163A/G364S +++ 5951/5952 K4T/P158D/L163A/G364S/N399D ++ 5953/5954 E2P/E113S/L163A + 5955/5956 E2P/K4T/P158D/K222R/L257Q/K302R/ ++ E330D 5957/5958 K4T/E113S/P158D/L163A/E330D/ ++ G364S/N399D 5959/5960 E2P/K4T/E113S/L163R/S449R + 5961/5962 E113S/P158D/L163R/G364S/N399D + 5963/5964 P158D/L163R/K302R/E330D/G364S/ + N399D 5965/5966 K4T/G364S/S449R ++ 5967/5968 E2P/G364S/S449R +++ 5969/5970 E2P/E113S/E330D/N399D + 5971/5972 P158D/L163A/L257Q/E330D/G364S/ + S449R 5973/5974 D134S + 5975/5976 L107A/N195H/R417P/K439P +++ 5977/5978 N195H/P272K/V320I/K439P + 5979/5980 D134S/S135A/N195H/D268A/S317R + 5981/5982 N195H/S317R/V320I + 5983/5984 L107A/N195H/P272K ++ 5985/5986 L107A/S374T/R417P/K439P + 5987/5988 L107A/D134S/S135A/N195H/R412H/ + R417P 5989/5990 L107A/N195H/D268A/P322L/K439P +++ 5991/5992 N408D/R417P + 5993/5994 D268A/R417P + 5995/5996 M87W/W266L ++ 5997/5998 M87W/D198E + 5999/6000 D198E/L292P +++ 6001/6002 M87W/H95L/D198E + 6003/6004 D198I +++ 6005/6006 M87W/P322S +++ 6007/6008 D198I/P244L ++ 6009/6010 M87W + 6011/6012 D198E + 6013/6014 M87W/D198I +++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SED ID NO: 5708, and defined as follows: “+” = production at least 1.05-fold, but less than 1.42-fold; “++” = at least 1.42-fold, but less than 1.6-fold; and “+++” = at least 1.6-fold increased production, as compared to the reference polypeptide.

SFP Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(582) Shake flask powders (SFP) were reconstituted to a concentration of 20 g/L and diluted to 0.006-0.2 g/L SFP in 100 µL total reaction volumes of 50 mM potassium phosphate buffer, pH 6, 10 mM stevioside (>95% purity), rebaudioside D, rebaudioside E, or rebaudioside A (>97% purity), 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.15 g/L SUS SFP SEQ ID NO: 1764, and 24 mM sucrose (cane sugar). The reactions were incubated at 50° C. and 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 3 hours. The reactions were solubilized by 20× dilution with water, quenched by 5× dilution in acetonitrile with 0.2% formic acid, cleared by centrifugation, and diluted 10× with water for analysis. Glycosylated products were detected in by SPE-QQQ as described in Example 31, Table 31.1.

(583) TABLE-US-00144 TABLE 68.2 β1,3GT Round 16 Shake Flask Powder Variants and RebA and RebM Levels Increased Increased Increased Increased SEQ ID Amino Acid Differences RebA, RebA, RebM, RebM from NO: (nt/aa) (Relative to SEQ ID NO: 5708) 50° C..sup.a 60° C..sup.a 60° C..sup.a 60° C..sup.a 5815/5816 R173N/K190R/S252E/L257A/ - +++ - - K347G/A427R/L437I 5839/5840 W233K/S252E - ++ - + 5845/5846 W233K/S252E/R411Q/L437V - + - - 5901/5902 E2P/K4T/P158D/L163A/ + + + + + G364S/S449R 5927/5928 E2P/P158D/L163A/G364S/ + + + + +

N399D 5949/5950 E2P/K4T/L163A/G364S ++ + + ND 5975/5976 L107A/N195H/R417P/K439P ++ ++ + ND 5983/5984 L107A/N195H/P272K ++ + - ND 5989/5990 L107A/N195H/D268A/P322L/ +++ + - ND K439P 5997/5998 M87W/D198E +++ +++ - ND 6009/6010 M87W +++ +++ - ND 6011/6012 D198E ++ + + ND .sup.a Levels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 5708, at 0.025 g/L shake flask powder and defined as follows: “-” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.26-fold; “++” = at least 1.26-fold, but less than 1.5-fold; and “+++” = at least 1.5-fold increased production, as compared to the reference polypeptide. ND = not determined.

(584) In these experiments, the thirteen variants in Table 68.2 produced more rebaudioside A from stevioside and/or more rebaudioside M from rebaudioside D at 50° C. and/or 60° C. than SEQ ID NO: 5708. SEQ ID NO: 5976 had the greatest improvement for conversion of stevioside to rebaudioside A without a loss in rebaudioside D to rebaudioside M activity for both temperatures, so this variant was selected for further directed evolution.

Example 69

Beta-1,3-ADP-Glycosyltransferase Variants of SEQ ID NO: 5976

(585) In this Example, experiments for evolution and screening of β 1,3-glycosyltransferase (β 1,3GT) polypeptides derived from SEQ ID NO: 5976 for improved glucosylation of steviol glycosides using in situ synthesized ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 5975 (i.e., SEQ ID NO: 5976) was carried out by constructing libraries of variant genes in which mutations associated with improved production identified during the development of the present invention were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a seventeenth round (“Round 17”) of 123 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(586) HTP Assay for Glucose Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(587) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 5976 variants. Pellets were lysed, and the lysate was cleared as described in Example 60 and then diluted 20× into buffer. In order to thermally challenge the lysates, they were pre-incubated in a thermocycler for 15 minutes at 65.5° C. Assays were conducted with 10 μ L pre-incubated lysate in 100 μ L reactions and with 10 mM stevioside (>95% purity) or rebaudioside D substrate, 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.1 g/L SUS SFP SEQ ID NO: 1804, and 24 mM sucrose (cane sugar). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. Reactions were solubilized by 20× dilution with water, quenched by 5× dilution in acetonitrile with 0.2% formic acid, and diluted 10× with water for analysis. Samples were analyzed by RapidFire-MS/MS as described in Example 31, Table 31.1.

Glycosyltransferase variant polypeptides that produced rebaudioside A/rebaudioside M from stevioside/rebaudioside D with in situ synthesized ADP-glucose at greater quantities than SEQ ID NO: 5976 were identified. The engineered polypeptides are listed in Table 69.1. Shake-flask scale cultures were grown, lysed, cleared, and lyophilized to powder as described in Example 31 for analysis of variants shown in Table 69.2 relative to SEQ ID NO: 5976.

(588) TABLE-US-00145 TABLE 69.1 β 1,3GT Round 17 Variants and RebA and RebM Levels SEQ ID Amino Acid Differences Increased Increased NO: (nt/aa) (Relative to SEQ ID NO: 5976) RebA.sup.a RebM.sup.a 6015/6016 K4T/D198E/N399D/E407V + + 6017/6018 E2P/V14I/T282S/R308L ++ +++ 6019/6020 T282S/R308L/E342W/G364A/S449R +++ +++ 6021/6022 E2P/V14I/R308L/G364A +++ +++ 6023/6024 E2P/K4T/T282S/E342W/G364A/E407V +++ +++ 6025/6026 K4T/L163A/T282S/R308L/E407V ++ ++ 6027/6028 E2P/T282S/N399D + ++ 6029/6030 P158D/T282S/E407V ++ + 6031/6032 E2P/V14I/P158D/D198E/E407V/S449R +++ +++ 6033/6034 E2P/V14I/R308L ++ ++ 6035/6036 V14I/P158D/L163A/D198E/G364A ++ +++ 6037/6038 E2P/K4T/V14I/T282S/R308L/G364A +++ +++ 6039/6040 D198E/T282S/E407V +++ +++ 6041/6042 E2P/V14I/L163A/R308L/G364A +++ +++ 6043/6044 E2P/R308L/G364A/N399D +++ ++ 6045/6046 E2P/V14I/T282S/N399D ++ ++ 6047/6048 T282S/R308L/S449R + + 6049/6050 K4T/V14I/T282S/R308L/G364A/E407V/S449R +++ +++ 6051/6052

E2P/K4T/V14I/D198E/R308L/E407V +++ ++ 6053/6054 P158D/G364A/N399D/E407V/S449R ++ +
 6055/6056 V14I/T282S/E342W/G364A/L391C ++ ++ 6057/6058 E2P/T282S/R308L/E342W/E407V ++
 +++ 6059/6060 E2P/K4T/P158D/L163A/G364A/N399D/E407V ++ ++ 6061/6062
 L163A/T282S/N399D/E407V ++ + 6063/6064 V14I/P158D/D198E/T282S/E342W/G364A/S449R +++
 ++ 6065/6066 P158D/T282S/E407V/S449R ++ ++ 6067/6068 E2P/K4T/V14I/P158D/T282S/E407V +++
 +++ 6069/6070 V14I/L163A/D198E/T282S/E342W/G364A/S449R +++ ++ 6071/6072
 E2P/L163A/T282S ++ + 6073/6074 E2P/T282S ++ + 6075/6076 V14I/P158D/L163A/T282S/G364A
 +++ + 6077/6078 E2P/K4T/V14I/T282S/R308L/E407V ++ + 6079/6080
 K4T/V14I/L163A/T282S/R308L/E342W/E407V/S449R ++ + 6081/6082
 E2P/L163A/T282S/R308L/G364A +++ ++ 6083/6084 E2P/K4T/L163A/G364A/N399D/S449R ++ +++
 6085/6086 L163A/D198E/T282S/E342W ++ ++ 6087/6088 V14I/P158D/G364A ++ + 6089/6090
 K4T/V14I/D198E/R308L/G364A +++ + 6091/6092 E2P/D198E/T282S/R308L/E342W +++ +++
 6093/6094 K4T/V14I/P158D/T282S/G364A/L391C/E407V +++ ++ 6095/6096
 K4T/V14I/G364A/L391C + ++ 6097/6098 K4T/V14I/L163A/T282S + + 6099/6100
 E2P/K4T/V14I/L163A/T282S/N399D ++ ++ 6101/6102 V14I/E407V ++ ++ 6103/6104
 E2P/K4T/L163A/D198E/T282S/R308L/E342W ++ ++ 6105/6106 L163A/T282S/E407V/S449R +++
 +++ 6107/6108 E2P/P158D/L163A/E407V/S449R + ++ 6109/6110
 E2P/K4T/D198E/G364A/L391C/S449R ++ +++ 6111/6112 V14I/T282S ++ ++ 6113/6114
 V14I/L163A/T282S/R308L + ++ 6115/6116 E2P/R308L ++ + 6117/6118
 E2P/K4T/L163A/T282S/E342W/G364A +++ +++ 6119/6120 D198E/R308L ++ + 6121/6122
 E2P/T282S/R308L/L391C/E407V + + 6123/6124 K4T/V14I/T282S + + 6125/6126
 E2P/K4T/D198E/T282S/S449R +++ +++ 6127/6128 E2P/K4T/V14I/T282S/G364A +++ +++ 6129/6130
 E2P/V14I/T282S ++ + 6131/6132 E2P/T282S/N399D/E407V +++ ++ 6133/6134
 K4T/T282S/G364A/E407V +++ ++ 6135/6136 K4T/V14I/P158D/L163A/D198E/T282S/E407V +++ ++
 6137/6138 E2P/K4T/P158D/D198E/T282S/G364A/E407V +++ +++ 6139/6140
 E2P/V14I/L163A/G364A/E407V +++ +++ 6141/6142 V14I/T282S/N399D/E407V/S449R ++ ++
 6143/6144 K4T/V14I/T282S/E342W/N399D/E407V + + 6145/6146 E2P/V14I/T282S/R308L/G364A
 +++ ++ 6147/6148 E2P/K4T/V14I/P158D/T282S/G364A/S449R +++ ++ 6149/6150
 E2P/D198E/T282S/N399D +++ ++ 6151/6152 P158D/G364A/S449R ++ + 6153/6154
 E2P/K4T/V14I/L163A/D198E ++ + 6155/6156 K4T/T282S/E342W/G364A/E407V +++ + 6157/6158
 L163A/T282S/R308L/G364A/N399D +++ ++ 6159/6160 D198E/T282S/R308L/E342W ++ + 6161/6162
 V14I/T282S/G364A +++ ++ 6163/6164 K4T/G364A ++ + 6165/6166 V14I/T282S/R308L + + 6167/6168
 V14I/T282S/G364A/L391C/E407V/S449R +++ ++ 6169/6170 K4T/P158D/T282S/G364A/N399D ++ +
 6171/6172 L75G + + 6173/6174 P139S + + 6175/6176 L401F/E402L + ++ 6177/6178 P131V ND +
 6179/6180 L138V ND + 6181/6182 P90S ND + 6183/6184 S254Q + + 6185/6186 P139R + + 6187/6188
 M108H ND + 6189/6190 S106Y + + 6191/6192 R12S ND + 6193/6194 P90T ND + 6195/6196 D416S +
 + 6197/6198 P139A + + 6199/6200 A427K ND + 6201/6202 I44V ND + 6203/6204 V365I + +
 6205/6206 S37R + + 6207/6208 K247C + ++ 6209/6210 D258N ND + 6211/6212 E115R + + 6213/6214
 Q71L/R331K + + 6215/6216 L75M + + 6217/6218 D416L + + 6219/6220 D114P + + 6221/6222 D389S
 ND + 6223/6224 S456R ND + 6225/6226 S456* - + 6227/6228 S156C + + 6229/6230 P70K + -
 6231/6232 D389A + + 6233/6234 H248W + +++ 6235/6236 A427R + + 6237/6238 L429W + +
 6239/6240 K247L + + 6241/6242 A433L + +++ 6243/6244 E162A + + 6245/6246 H248L + +++
 6247/6248 K432L + ++ 6249/6250 H248C + ++ 6251/6252 L174P + + 6253/6254 P112N ND +
 6255/6256 D389E + ++ 6257/6258 P90Q - + 6259/6260 W74H + + .sup.aLevels of increased production
 were determined relative to the reference polypeptide of SEQ ID NO: 5976, and defined as follows: “+” =
 production at least that of the reference polypeptide, but less than 2-fold; “++” = at least 2-fold, but less
 than 2.75-fold; and “+++” = at least 2.75-fold increased production, as compared to the reference
 polypeptide.

SFP Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP and then from
 ADP-Glucose to Stevioside and Rebaudioside D

(589) Shake flask powders (SFP) were reconstituted to a concentration of 20 g/L and diluted to 0.006-0.2
 g/L SFP in 100 µL total reaction volumes of 50 mM potassium phosphate buffer, pH 6, 10 mM stevioside
 (>95% purity) or rebaudioside D, 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.1 g/L SUS SFP

SEQ ID NO: 1804, and 24 mM sucrose (cane sugar). The reactions were incubated at 55° C. with no pre-incubation or at 60° C. following 15 minute pre-incubation at 65.5° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 hours. The reactions were solubilized by 20× dilution with water, quenched by 5× dilution in acetonitrile with 0.2% formic acid, cleared by centrifugation, and diluted 10× with water for analysis. Glycosylated products were detected in by SPE-QQQ as described in Example 31, Table 31.1.

(590) TABLE-US-00146 TABLE 69.2 β 1,3GT Round 17 Shake Flask Powder Variants Increased Increased SEQ ID Amino Acid Differences RebA RedA, RebM, NO (Relative to SEQ ID NO: 5976) 55° C.a 60° C..sup.a 60° C..sup.a 6147/6148 E2P/K4T/V14I/P158D/T282S/G364A/S449R + + + 6037/6038 E2P/K4T/V14I/T282S/R308L/G364A + + + + 6137/6138 E2P/K4T/P158D/D198E/T282S/G364A/E407V + + + + 6091/6092 E2P/D198E/T282S/R308L/E342W + + + 6067/6068 E2P/K4T/V14I/P158D/T282S/E407V + + + + 6049/6050 K4T/V14I/T282S/R308L/G364A/E407V/S449R + + + + 6023/6024 E2P/K4T/T282S/E342W/G364A/E407V + + + + 6105/6106 L163A/T282S/E407V/S449R + + + +

.sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 5976, at 0.025 g/L shake flask powder and defined as follows: “+” = production at least 1.3-fold, but less than 2.4-fold; “++” = at least 2.4-fold, but less than 4-fold; and “+++” = at least 4-fold increased production, as compared to the reference polypeptide.

(591) In these experiments, the eight variants in Table 69.2 produced more rebaudioside A from stevioside and/or more rebaudioside M from rebaudioside D at 55° C. and/or 60° C. than SEQ ID NO: 5976. SEQ ID NO: 6138 had the greatest improvement for conversion of stevioside to rebaudioside A under the conditions with pre-incubation followed by the 60° C. assay, and was improved under the other conditions as well, so this variant was selected for further directed evolution.

Example 70

Beta-1,3-ADP-Glycosyltransferase Variants of SEQ ID NO: 6138

(592) In this Example, experiments for evolution and screening of β 1,3-glycosyltransferase (β 1,3GT) polypeptides derived from SEQ ID NO: 6138 for improved glucosylation of steviol glycosides using in situ synthesized ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 6137 (i.e., SEQ ID NO: 6138) was carried out by constructing libraries of variant genes in which mutations associated with improved production identified during the development of the present invention were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide an eighteenth round (“Round 18”) of 100 engineered GT variant polypeptides with glycosyltransferase activity toward ADP-glucose and steviol glycosides.

(593) HTP Assay for Glucose Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(594) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 6137 variants. Pellets were lysed, and lysate was cleared as described in Example 60, and then diluted 20-40× into 50 mM potassium phosphate buffer, pH 6. In order to thermally challenge the lysates, they were pre-incubated in a thermocycler for 15 minutes at 68.6° C. Assays were conducted with 10 μ L pre-incubated lysate in 100 μ L reactions and with 10 mM stevioside (>95% purity) or rebaudioside D substrate, 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.1 g/L SUS SFP SEQ ID NO: 1840, and 24 mM sucrose (cane sugar). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. Reactions were solubilized by 20× dilution with water, quenched by 5× dilution into acetonitrile with 0.2% formic acid, and diluted 10× with water for analysis.

(595) Samples were analyzed by RapidFire-MS/MS as described in Example 31, Table 31.1.

Glycosyltransferase variant polypeptides that produced rebaudioside A/rebaudioside M from stevioside/rebaudioside D with in situ synthesized ADP-glucose at greater quantities than SEQ ID NO: 6138 were identified. The engineered polypeptides are listed in Table 70.1. Shake-flask scale cultures were grown, lysed, cleared, and lyophilized to powder as described in Example 31 for analysis of variants shown in Table 70.2 relative to SEQ ID NO: 6138.

(596) TABLE-US-00147 TABLE 70.1 β 1,3GT Round 18 Variants and RebA and RebM Levels SEQ ID

Amino Acid Differences Increased Production NO: (nt/aa) (Relative to SEQ ID NO: 6138) RebA.sup.a

RebM.sup.a 6261/6262 V14I/E330D ++ +++ 6263/6264 E113S/D158P/L163A/K190R/L257Q/L437I ++
 +++ 6265/6266 D158P/V320I/L437I/S449R ++ ++ 6267/6268
 V14I/D158P/L163A/K190R/E342W/L437I + +++ 6269/6270 V14I/K190R/L257Q/S317R/V320I/P322L
 + ++ 6271/6272 V14I/E113S/D158P/L163A/K190R/L257Q/R308L/E342W ++ +++ 6273/6274
 E113S/D158P/L163A/K190R/R308L/V320I ++ +++ 6275/6276 K190R/S449R +++ +++ 6277/6278
 V14I/E113S/D158P/L163A ++ +++ 6279/6280 V14I/E113S/L163A/K190R/R308L/S317R ++ +++
 6281/6282 V14I/E113S/D158P + ++ 6283/6284 E113S/D158P/K190R/V320I/P322L/S449R ++ +++
 6285/6286 E113S/L257Q/S317R/P322L/L437I + ++ 6287/6288 E113S/D158P/V320I/P322L/L437I +++
 +++ 6289/6290 E113S/D158P/K190R/L257Q/V320I ++ +++ 6291/6292 P322L/E330D ++ +++
 6293/6294 S317R/V320I/P322L/E330D ++ +++ 6295/6296 V14I/S449R ++ +++ 6297/6298
 V14I/E113S/D158P/R308L/S317R/E330D/S449R +++ +++ 6299/6300
 V14I/E113S/D158P/L163A/L437I +++ +++ 6301/6302 D158P/R308L/V320I/L437I +++ +++ 6303/6304
 V14I/D158P/L163A/L257Q/R308L/S449R ++ +++ 6305/6306 D158P/L163A/R308L + +++ 6307/6308
 V14I/L257Q/R308L/V320I/P322L/E330D ++ +++ 6309/6310
 E113S/D158P/L163A/L257Q/R308L/S317R/P322L/L437I/S449R ++ +++ 6311/6312
 V14I/L163A/S317R/V320I + +++ 6313/6314 V14I/D158P/L257Q/R308L/L437I/S449R +++ +++
 6315/6316 L163A/R308L/E330D/L437I/S449R ++ +++ 6317/6318 D158P/L163A ++ +++ 6319/6320
 V14I/D158P/L163A/L257Q + +++ 6321/6322 V14I/D158P/S317R/V320I/E330D/L437I +++ +++
 6323/6324 V14I/E113S/V320I/L437I/S449R +++ +++ 6325/6326 E113S/L257Q/R308L ++ +++
 6327/6328 E113S/D158P/L163A/K190R/R308L/S317R/P322L ++ +++ 6329/6330 V14I/K190R + +++
 6331/6332 D158P/L163A/K190R/L257Q/R308L/E342W/S449R + +++ 6333/6334
 V14I/D158P/L163A/K190R/L257Q/R308L/S317R/E330D/L437I +++ +++ 6335/6336
 V14I/L257Q/R308L/P322L/E330D/L437I +++ +++ 6337/6338 V14I/E113S/L163A/L257Q/L437I +++
 +++ 6339/6340 V14I/S317R + +++ 6341/6342 V14I/D158P +++ +++ 6343/6344
 V14I/E113S/L163A/R308L/S317R/V320I/P322L/E330D/S449R ++ +++ 6345/6346
 V14I/E113S/L163A/L257Q/R308L/E342W/L437I/S449R +++ +++ 6347/6348
 D158P/L163A/P322L/L437I +++ +++ 6349/6350 L29M/D375P + ++ 6351/6352 G84H ++ + 6353/6354
 R88T + + 6355/6356 D375A + + 6357/6358 Q197K + + 6359/6360 I202H + ++ 6361/6362 P83S + +
 6363/6364 I202T + + 6365/6366 G84N ++ + 6367/6368 A85L + + 6369/6370 V19Q + + 6371/6372
 A199M + + 6373/6374 I202Q + ++ 6375/6376 N383V ++ 6377/6378 H81T + + 6379/6380 D375V + +
 6381/6382 A199Y + + 6383/6384 D72S + + 6385/6386 D375P +++ +++ 6387/6388 D72T + + 6389/6390
 P83N + + 6391/6392 R88K + + 6393/6394 V155L ++ ++ 6395/6396 A199H + + 6397/6398 T80P + +
 6399/6400 Q71V + + 6401/6402 P83A + + 6403/6404 E209T + + 6405/6406 A199E + + 6407/6408 P83T
 + + 6409/6410 I202V + + 6411/6412 G84D + + 6413/6414 R88C + + 6415/6416 A199Q +++ ++
 6417/6418 D375T + ++ 6419/6420 R88A + + 6421/6422 P83K + + 6423/6424 R88H + + 6425/6426
 S41A + ND 6427/6428 A366T + ND 6429/6430 A366V ++ ND 6431/6432 E105A + ND 6433/6434
 S273R ++ ND 6435/6436 S273H ++ ND 6437/6438 V263T + ND 6439/6440 P168C + ND 6441/6442
 L243I +++ ND 6443/6444 H95N + ND 6445/6446 A366S + ND 6447/6448 H46S + ND 6449/6450
 S273A + ND 6451/6452 A366L ++ ND 6453/6454 A366C +++ ND 6455/6456 Y249S + ND 6457/6458
 F45L + ND 6459/6460 P168T + ND .sup.aLevels of increased production were determined relative to the
 reference polypeptide of SEQ ID NO: 6138, and defined as follows: “-” = production less than 0.9-fold;
 “+” = production at least 0.9- fold, but less than 1.5-fold; “++” = at least 1.5-fold, but less than 1.9-fold;
 and “+++” = at least 1.9-fold increased production, as compared to the reference polypeptide. ND = not
 determined.

SFP Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP and then from
 ADP-Glucose to Stevioside and Rebaudioside D

(597) Shake flask powders (SFP) were reconstituted to a concentration of 20 g/L and diluted to 0.006-0.2
 g/L SFP in 100 µL total reaction volumes of 50 mM potassium phosphate buffer, pH 6, 10 mM stevioside
 (>95% purity) or rebaudioside D, 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.1 g/L SUS SFP
 SEQ ID NO: 1804, and 24 mM sucrose (cane sugar). The reactions were incubated at 55° C. with no pre-
 incubation or at 60° C. following 15 minute pre-incubation at 68.6° C. in a Thermotron® titre-plate
 shaker with 300 RPM shaking for 4 hours. The reactions were solubilized by 20× dilution in water,

quenched by 5× dilution in acetonitrile with 0.2% formic acid, cleared by centrifugation, and diluted 10× in water for analysis. Glycosylated products were detected in by SPE-QQQ as described in Example 31, Table 31.1.

(598) TABLE-US-00148 TABLE 70.2 β 1,3GT Round 18 Shake Flask Powder Variants and RebA and RebM Levels SEQ ID NO: Amino Acid Differences Increased Increased (nt/aa) (Relative to SEQ ID NO: 6138) RebA, 60° C..sup.a RebM, 60° C..sup.a 6261/6262 V14I/E330D + + 6267/6268 V14I/D158P/L163A/K190R/E342W/L437I + +++ 6287/6288 E113S/D158P/V320I/P322L/L437I +++ +++ 6299/6300 V14I/E113S/D158P/L163A/L437I ++ ++ 6333/6334 V14I/D158P/L163A/K190R/L257Q/R308L/S ++ ++ 317R/E330D/L437I .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 6138, at 0.025 g/L shake flask powder and defined as follows: “–” = production less than 1.08-fold; “+” = production at least 1.08-fold, but less than 1.5-fold; “++” = at least 1.5-fold, but less than 1.7-fold; and “+++” = at least 1.7-fold increased production, as compared to the reference polypeptide.

(599) In these experiments, the five variants in Table 70.2 (SEQ ID NOS: 6262, 6268, 6288, 6300, and 6334) produced more rebaudioside A from stevioside and more rebaudioside M from rebaudioside D at 60° C. with pre-incubation than SEQ ID NO: 6138. SEQ ID NO: 6288 had the greatest improvement for conversion of stevioside to rebaudioside A under the conditions with pre-incubation followed by the 60° C. assay, and was the second most improved for rebaudioside D to M, so this variant was selected for further directed evolution.

Example 71

Beta-1,3-ADP-Glycosyltransferase Variants of SEQ ID NO: 6288

(600) In this Example, experiments for evolution and screening of β 1,3-glycosyltransferase (β 1,3GT) polypeptides derived from SEQ ID NO: 6288 for improved glucosylation of steviol glycosides using in situ synthesized ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 6287 (i.e., SEQ ID NO: 6288) was carried out by constructing libraries of variant genes in which mutations associated with improved production identified during the development of the present invention were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide an nineteenth round (“Round 19”) of 108 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(601) HTP Assay for Glucose Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(602) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 6288 variants. Pellets were lysed, and lysate was cleared as described in Example 60 and then diluted 40× into 50 mM potassium phosphate buffer, pH 6. In order to thermally challenge the lysates, they were pre-incubated either in a thermocycler for 15 minutes at 73.5° C. or at 65° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 16 h. Assays were conducted with 10 μ L pre-incubated lysate in 100 μ L reactions and with 10 mM stevioside (>95% purity) or rebaudioside D substrate, 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.05 g/L SUS SFP SEQ ID NO: 2064, and 24 mM sucrose (cane sugar). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. Reactions were solubilized by 20× dilution in water, quenched by 5× dilution in acetonitrile with 0.2% formic acid, and diluted 10× with water for analysis. Samples were analyzed by RapidFire-MS/MS as described in Example 31, Table 31.1. Glycosyltransferase variant polypeptides that produced rebaudioside A/rebaudioside M from stevioside/rebaudioside D with in situ synthesized ADP-glucose at greater quantities than SEQ ID NO: 6288 were identified. The engineered polypeptides from combinatorial libraries screened with the 15 minute pre-incubation are listed in Table 71.1, and those from saturation mutagenesis libraries screened with the 16 h pre-incubation are listed in Table 71.2. Shake-flask scale cultures were grown, lysed, cleared, and lyophilized to powder as described in Example 1 for analysis of variants shown in Table 71.3 relative to SEQ ID NO: 6288.

(603) TABLE-US-00149 TABLE 71.1 β 1,3GT Round 19 Combinatorial Variants and RebA and RebM Levels SEQ ID NO: Amino Acid Differences Increased Increased (nt/aa) (Relative to SEQ ID NO: 6288) RebA.sup.a RebM.sup.a 6461/6462 S37R/Q71L/K247C/R331K/V365I/D389E/L401F/L429W + +++

6465/6466 P139K/S156C/K247L/D389E/L401F/A433L/S449R + + + + + 6465/6466
 K247L/H248W/5449R + + + + + 6467/6468 P139K/S156C/L174P/D389E/L401F/A427R/A433L/5449R
 ++ + + + + 6469/6470 P139K/K247C/H248W/R331K/L401F/P417Q/K432L/S449R + + + + + 6471/6472
 L174P/D389E/L429W/K432L/A433L/5449R + + + + + 6473/6474 D416S/K432L/A433L + + + + +
 6475/6476 S37R/H248W/D389A/L401F/L429W + + + + + 6477/6478
 Q71L/H248W/V365I/D389A/L401F + + + + + 6479/6480
 P139K/S156C/H248W/D389A/L401F/D416S/A427R/L429W/A433L + + + + + 6481/6482
 Q71L/P139K/D389A/A427R/L429W/K432L/A433L/5449R + + + + 6483/6484
 S37R/Q71L/L125M/L174P/K247C/H248W/A427R + + + + + 6485/6486
 S37R/L174P/L401F/E402L/A433L/5449R + + + + + 6487/6488
 S156C/D389E/L401F/E402L/D416L/K432L/A433L ++ + + + + 6489/6490 S254Q/A427R/A433L + + + +
 6491/6492 S156C/5254Q/R331K/V365I/A427R/K432L/5449R ++ + + + 6493/6494 A427R/K432L/5449R
 +++ + + + + 6495/6496 S37R/P139K/A427R/K432L/A433L/S449R + + + + 6497/6498
 P139K/H248W/E402L/D416L/A427R/A433L/5449R + + + + 6499/6500 D416L/K432L/A433L/5449R +
 + + + + + 6501/6502 K247C/A427R/K432L + + + + + 6503/6504 H248W/L401F/L429W/K432L/A433L +
 + + + + + 6505/6506 H248W/R331K/A427R/L429W/A433L ++ + + + 6507/6508 A427R/K432L +++ + + +
 6509/6510 P139K/5254Q/L401F/D416S/A427R/A433L/5449R/ ++ + + + 6511/6512
 L174P/G329P/K432L/5449R ++ + + + 6513/6514 P139K/L401F/S449R ++ + + + 6515/6516
 P139K/L174P/S254Q ++ + + + 6517/6518 P139K/L174P/V365I/L401F/E402L/A427R/A433L/S449R
 +++ + + + + 6519/6520 D389E/L401F/A427R/K432L +++ + + + + 6521/6522
 Q71L/A427R/K432L/A433L/S449R + + + + + 6523/6524
 S37R/P139K/G329P/V365I/A427R/A433L/S449R ++ + + + + 6525/6526
 Q71L/L174P/G329P/A427R/L429W/K432L/A433L + + + + + 6527/6528
 R331K/V365I/L429W/K432L/A433L/S449R ++ + + + + 6529/6530 D416L/A427R/A433L/S449R + + + + +
 6531/6532 S37R/H248W/R331K/D389E/A427R/A433L + + + + + 6533/6534
 S37R/P139K/L429W/K432L/A433L ++ + + + + 6535/6536 S156C/H248W/L256M + + + + + 6537/6538
 K247C/H248W/L401F ++ + + + + 6539/6540 S37R/R331K/K432L/A433L ++ + + + + 6541/6542
 K247C/R331K/L401F/A427R/K432L/S449R ++ + + + + 6543/6544 D389A/L401F/D416S/K432L/S449R
 ++ + + + 6545/6546 P139K/L174P/H248W/R331K/D389A/L401F/S449R ++ + + + + 6547/6548
 L174P/K247C/H248W/D389A/L401F/K432L/A433L + + + + 6549/6550 S254Q/V365I ++ + + +
 6551/6552 P139K/S156C/V365I + + + + 6553/6554 P139K/S156C/K247C/V365I/L401F/A433L/S449R +
 + + + + + 6555/6556 Q71L/H248W/S449R ++ + + + 6557/6558 Q71L/K432L/A433L ++ + + + 6559/6560
 H248W/D416L ++ + + + 6561/6562 Q71L/S254Q/A433L/S449R ++ + + + 6563/6564 A433L/S449R ++
 + + + + 6565/6566 K247L/L401F ++ + + + 6567/6568 H248W/V365I/D389E/A427R/L429W/K432L/5449R
 ++ + + + 6569/6570 Q71L/L174P/V365I/A427R/K432L + + + + 6571/6572 P139K/S156C/H248W/D389E
 ++ + + + 6573/6574 P139/H248W/S254A/S449R ++ + + + 6575/6576 L174P/V365I ++ + + + 6577/6578
 H248W/S449R ++ + + + 6579/6580 H248W/D416L/S449R/ ++ + + + 6581/6582 K432L + + + + 6583/6584
 P139K/H248W/K432L/A433L ++ + + + + 6585/6586 V365I/L401F/E402L/L429W/K432L/A433L + + + +
 6587/6588 L401F/A427R ++ + + + .sup.aLevels of increased production were determined relative to the
 reference polypeptide of SEQ ID NO: 6288, and defined as follows: “-” = production less than that of the
 reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.9-fold;
 “++” = at least 1.9-fold, but less than 2.5-fold; “+++” = at least 2.5-fold, but less than 7-fold; and “++++”
 = at least 7-fold increased production, relative to reference polypeptide.
 (604) TABLE-US-00150 TABLE 71.2 β 1,3GT Round 19 Saturation Mutagenesis Variants and RebA and
 RebM Levels SEQ ID NO: Amino Acid Differences Increased Increased (nt/aa) (Relative to SEQ ID NO:
 6288) RebA.sup.a RebM.sup.a 6589/6590 M87R + + 6591/6592 M144Q/S449R +++ + + 6593/6594
 I91Q + + 6595/6596 I91N/S449R - + + 6597/6598 I91T/S449R + + 6599/6600 L25M/S449R + + +
 6601/6602 D69M/S449R + + 6603/6604 M87Q ++ + + 6605/6606 M87K ++ + + + 6607/6608 T361C ++ +
 + + + 6609/6610 L25I/S449R + + 6611/6612 M87A/S449R + + + + 6613/6614 M87E ++ + + + 6615/6616
 A153V + + 6617/6618 A153T/S449R +++ + 6619/6620 L25Q/S449R + + + 6621/6622 L25S + + +
 6623/6624 D69A + + 6625/6626 I91V + + + 6627/6628 I91L + + 6629/6630 W233M/S449R ND +++
 6631/6632 S317Y ND +++ 6633/6634 W233V ND + 6635/6636 W233S ND +++ 6637/6638
 A212L/S449R ND + 6639/6640 W233L/S449R ND ++ 6641/6642 V369K ND + 6643/6644 V421I ND +

6645/6646 W233C ND ++ 6647/6648 W233Q/S449R ND +++ 6649/6650 E6P ND - 6651/6652 K347P/S449R ND + 6653/6654 Q159R/S449R ND ++ 6655/6656 Q159K ND +++ 6657/6658 E288P/S449R ND ++ 6659/6660 H172S ND + 6661/6662 T5S/S449R ND + 6663/6664 W233G ND + 6665/6666 Q303C ND + 6667/6668 W233A ND ++ 6669/6670 R10K ND + 6671/6672 W233R ND ++ 6673/6674 H172T/S449R ND + 6675/6676 Q303V ND + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 6288, and defined as follows: “-” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.28-fold; “++” = at least 1.28-fold, but less than 1.5-fold; and “+++” = at least 1.5-fold increased production, relative to the reference polypeptide.

SFP Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(605) Shake flask powders (SFP) were reconstituted to a concentration of 20 g/L and diluted to 0.006-0.2 g/L SFP in 100 µL total reaction volumes of 50 mM potassium phosphate buffer, pH 6, 10 mM stevioside (>95% purity) or rebaudioside D, 0.2 mM ADP (Sigma, >93% purity) co-substrate, 0.05 g/L SUS SFP SEQ ID NO: 2064, and 24 mM sucrose (cane sugar). The reactions were incubated in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 hours at 55° C. with no pre-incubation or at 60° C. following 15 minute pre-incubation at 73.5° C. The reactions were solubilized by 20× dilution in water, quenched by 5× dilution in acetonitrile with 0.2% formic acid, cleared by centrifugation, and diluted 10× with water for analysis. Glycosylated products were detected in by SPE-QQQ as described in Example 31, Table 31.1. A further thermostability characterization was conducted with the clarified shake flask lysates prior to lyophilization, as follows: lysates were diluted 400× in buffer and incubated in a thermocycler at a gradient of 55-70° C. for 16 h. To determine the percent of remaining activity, the pre-incubated lysates were assayed as described above, with either stevioside or rebaudioside D and 4 h incubation at 60° C. The percent activity remaining is expressed as production at the high temperature divided by production at lowest pre-incubated temperature.

(606) TABLE-US-00151 TABLE 71.3 β1,3GT Round 19 Shake Flask Powder Variants and RebA, RebM, and Thermostability Levels SEQ ID Amino Acid Differences Increased Increased Increased NO:

(Relative to RebA, RebA, RebM, % Activity (nt/aa) SEQ ID NO: 6288) 55° C..sup.a 60° C..sup.a 60° C..sup.a Remaining.sup.b

6467/6468 P139K/S156C/L174P/D389E/ + ++ +++ ++

L401F/A427R/A433L/S449R 6493/6494 A427R/K432L/S449R + ++ ++ ++ 6517/6518

P139K/L174P/V365I/L401F/ - + ++ +++ E402L/A427R/A433L/S449R 6519/6520

D389E/L401F/A427R/K432L + ++ ++ +++ 6527/6528 R331K/V365I/L429W/K432L/ - + +++ +

A433L/S449R 6539/6540 S37R/R331K/K432L/A433L + + +++ - 6573/6574

P139K/H248W/S254A/S449R - + +++ + 6583/6584 P139K/H248W/K432L/A433L - + +++ ++

.sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 6288, at 0.025 g/L shake flask powder and defined as follows: “-” = production less than 0.9-fold; “+” = production at least 0.9-fold, but less than 3.2-fold; “++” = at least 3.2-fold, but less than 5.5-fold; and “+++” = at least 5.5-fold increased production, relative to reference polypeptide. .sup.bThe percent levels of activity remaining were determined following 16 h pre-incubation at 70° C. relative to each variant following 16 h pre-incubation at 55° C. and defined as follows: “-” = less than 20% of activity remained following 16 h pre-incubation at 55° C.; “+” = at least 20%, but less than 40% of the activity remained; “++” = at least 40%, but less than 60% of the activity remained; and “+++” = at least 60% of the activity remained.

(607) In these experiments, the eight variants in Table 71.3 produced more rebaudioside A from stevioside and more rebaudioside M from rebaudioside D at 60° C. with pre-incubation than SEQ ID NO: 6288. SEQ ID NO: 6468 had the greatest improvement for conversion of rebaudioside D to rebaudioside M under the conditions with pre-incubation followed by the 60° C. assay and had the greatest percent activity remaining following 16 h thermochallenge, so this variant was selected for further directed evolution.

Example 72

Beta-1,3-ADP-Glycosyltransferase Variants of SEQ ID NO: 6468

(608) In this Example, experiments for evolution and screening of β1,3-glycosyltransferase (β1,3GT) polypeptides derived from SEQ ID NO: 6468 for improved glucosylation of steviol glycosides using in

6747/6748 S37R/S41A/F45L/D72T/V155L/D72T/V155L/K190R/A366C/D375P ++ +++ 6749/6750
D72T/H81T/P83S/G84N/R88T/V155L/K190R ++ ++ 6751/6752 V155L/V263T/A366C/K432L/L433A
++ +++ 6753/6754 S37R/S41A/F45L/D72T/V155L/V263T/D375P ++ +++ 6755/6756
S37R/S41A/F45L/V155L/A366C/K432L/L433A ++ +++ 6757/6758
D72T/H81T/P83S/G84N/R88T/L163A/P168T/V263T/ ++ ++ R331K/D375P 6759/6760
S37R/S41A/F45L/V155L/A366C ++ ++ 6761/6762
S37R/S41A/F45L/D72T/V155L/L163A/P168T/L243I/ + +++
H248W/S273R/R331K/A366V/K432L/L433A 6763/6764
S37R/S41A/F45L/D72T/V155L/K190R/S273R/R331K/ ++ ++ A366V/D375P 6765/6766
S37R/S41A/F45L/L243I/H248W/S273R/R331K + ++ 6767/6768
S37R/S41A/F45L/H81T/P83S/G84N/R88T/L163A/P168T/ ++ ++ V263T/S273R/R331K/A366C/D375P
6769/6770 S37R/S41A/F45L/D72T/V155L/K190R/S273R/A366C ++ +++ 6771/6772
S37R/S41A/F45L/V155L/P168T/H248W/S273R/R331K/D375P + ++ 6773/6774
S37R/S41A/F45L/V155L/L163A/P168T/V263T/R331K/D375P ++ +++ 6775/6776
S37R/S41A/F45L/D72T/V155L/R331K/A366V/D375P/K432L/L433A ++ +++ 6777/6778
S37R/S41A/F45L/V155L/K190R/R331K/A366V/D375P ++ +++ 6779/6780
H81T/P83S/G84N/R88T/V155L/L163A/P168T/S273R/ ++ ++ R331K/D375P 6781/6782
D72T/H81T/P83S/G84N/R88T/L163A/P168T/K190R/ ++ ++ L243I/V263T/R331K/A366C 6783/6784
S37R/S41A/F45L/D72T/V155L/H248W/V263T/S273R/A366C ++ +++ 6785/6786
V155L/P168T/D375P ++ +++ 6787/6788
S41A/F45L/L163A/P168T/L243I/H248W/S273R/A366C/K432L + +++ 6789/6790
S37R/S41A/F45L/V155L/D375P ++ +++ 6791/6792 D72T/L243I/H248W/A366V/K432L/L433A + +++
6793/6794 D72T/L243I/H248W/S273R/A366V/K432L/L433A + +++ 6795/6796
S37R/S41A/F45L/D72T/V155L/V263T/R331K/D375P ++ +++ 6797/6798
D72T/V155L/S273R/R331K/D375P/K432L ++ +++ 6799/6800
D72T/L243I/H248W/V263T/A366V/K432L/L433A ++ +++ 6801/6802
S37R/S41A/F45L/D72T/H81T/P83S/G84N/R88T/V155L/ ++ ++ R331K/A366C/D375P 6803/6804
S37R/S41A/F45L/H81T/P83S/G84N/R88T/V155L/P168T/ ++ ++ R331K/D375P 6805/6806
S37R/S41A/F45L/D72T/H81T/P83S/G84N/R88T/V155L/ ++ ++ R331K/D375P 6807/6808
S37R/S41A/F45L/D72T/V155L/K190R/H248W/V263T/R331K ++ +++ 6809/6810
S37R/S41A/F45L/D72T/V263T/A366C/K432L/L433A ++ +++ 6811/6812
S37R/S41A/F45L/D72T/V155L/S273R/A366C ++ ++ 6813/6814
S37R/S41A/F45L/D72T/H81T/P83S/G84N/R88T/V155L/ ++ ++ V263T/S273R/R331K/D375P/K432L
6815/6816 S37R/S41A/F45L/V263T/R331K/D375P/K432L ++ +++ 6817/6818
S37R/S41A/F45L/D72T/R331K/A366C ++ ++ 6819/6820
H81T/P83S/G84N/R88T/V263T/S273R/R331K/A366C/K432L ++ ++ 6821/6822
H81T/P83S/G84N/R88T/V155L/A366C ++ ++ 6823/6824
S37R/S41A/F45L/D72T/R88T/V155L/P168T/K190R/ ++ ++ R331K/A366C/D375P/K432L 6825/6826
S37R/S41A/F45L/D72T/V155L/L163A/P168T/V263T/ ++ ++ A366C/K432L/L433A 6827/6828
H81T/P83S/G84N/R88T/V155L/V263T/A366V/D375P ++ ++ 6829/6830
S37R/S41A/F45L/D72T/V155L/L163A/P168T/K190R/ ++ ++
L243I/V263T/S273R/R331K/A366C/K432L 6831/6832
S37R/S41A/F45L/D72T/H81T/P83S/G84N/R88T/H248W/D375P ND + 6833/6834
D72T/H81T/P83S/G84N/R88T/V155L/L163A/P168T/ ++ + K190R/A366C 6835/6836
S37R/S41A/F45L/D72T/V155L/K190R/V263T/R331K/A366C ++ ++ 6837/6838
D72T/V155L/K190R/A366C ++ ++ 6839/6840 S37R/S41A/F45L/D72T/K190R/V263T/S273R/R331K
ND ++ 6841/6842 D72T/V155L/K190R/V263T/R331K/A366C ++ ++ 6843/6844
S37R/S41A/F45L/D72T/K190R/A366C ++ ++ 6845/6846
H81T/P83S/G84N/R88T/K190R/V263T/D375P ++ ++ 6847/6848
D72T/H81T/P83S/G84N/R88T/V155L/A366C ++ ++ 6849/6850
D72T/H81T/P83S/G84N/R88T/V155L/S273R/R331K/D375P ++ ++ 6851/6852
S37R/S41A/F45L/R331K/A366V/K432L/L433A ++ + 6853/6854
S37R/S41A/F45L/D72T/R88T/V155L/P168T/K190R/R331K/D375P ++ ++ 6855/6856

S37R/S41A/F45L/D72T/H81T/P83S/G84N/R88T/V155L/ ++ ++ L163A/P168T/K190R/R331K/D375P
 6857/6858 D72T/H81T/P83S/G84N/R88T/V155L/K190R/S273R/ ++ ++ R331K/A366V/K432L
 6859/6860 S41A/F45L/D72T/V155L/N263T/R331K/A366V/D375P/ ++ ++ K432L/L433A 6861/6862
 S37R/S41A/F45L/D72T/H81T/P83S/G84N/R88T/R331K/ ++ ++ A366V/K432L/L433A 6863/6864
 S37R/S41A/F45L/D72T/K190R/R331K/A366C ++ ++ 6865/6866
 S37R/S41A/F45L/H81T/P83S/G84N/R88T/V155L/P168T/ ++ + K190R/L243I/R331K/A366C
 6867/6868 S37R/S41A/F45L/H81T/P83S/G84N/R88T/V155L/V263T/ ++ ++ S273R/R331K/A366C
 6869/6870 S37R/S41A/F45L/D72T/R88T/K190R/V263T/S273R/R331K/A366C ++ ++ 6871/6872
 S37R/S41A/F45L/D72T/V263T/R331K/A366C ++ ++ 6873/6874
 H81T/P83S/G84N/R88T/V155L/K190R/V263T/R331K/A366C ++ ++ 6875/6876
 S37R/S41A/F45L/D72T/V155L/K190R/S273R/A366C/D375P/K432L ++ ++ 6877/6878
 D72T/V155L/L163A/P168T/K190R/A366V/K432L/L433A ++ ++ .sup.aLevels of increased production
 were determined relative to the reference polypeptide of SEQ ID NO: 6468, and defined as follows: “-” =
 production at least that of the reference polypeptide; “+” = at least that of the reference polypeptide, but
 less than 2-fold; “++” = at least 2-fold, but less than 4-fold; “+++” = at least 4-fold, but less than 8-fold;
 and “++++” = at least 8-fold increased production, as compared to the reference polypeptide. ND = not
 determined.

(612) TABLE-US-00153 TABLE 72.2 β 1,3GT Round 20 Saturation Mutagenesis Variants and RebA
 Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 6468)
 RebA.sup.a 6879/6880 L322K +++ 6881/6882 R435S +++ 6883/6884 V9K + 6885/6886 P439G +
 6887/6888 Q265H + 6889/6890 S253D + 6891/6892 S61Q +++ 6893/6894 P2Q + 6895/6896 P2H +
 6897/6898 N408S + 6899/6900 P439A ++ 6901/6902 K428R ++ 6903/6904 D170A + 6905/6906 G96K
 + 6907/6908 D98E + 6909/6910 K171Q + 6911/6912 R296Q + 6913/6914 A118V ++ 6915/6916 L229M
 + 6917/6918 P2K +++ 6919/6920 S253P +++ 6921/6922 K222Q ++ 6923/6924 P439H ++ 6925/6926
 A118T/L120V ++ 6927/6928 T4N + 6929/6930 M183L ++ 6931/6932 Q269N ++ 6933/6934 S253T +++
 6935/6936 P272S + 6937/6938 R435Y ++ 6939/6940 K289N + 6941/6942 S253V + 6943/6944 K222A
 ++ 6945/6946 A438S ++ 6947/6948 R435G ++ 6949/6950 S442T + 6951/6952 R435D ++ 6953/6954
 L322P/V407I + 6955/6956 E226S + 6957/6958 D170V + 6959/6960 P2R + 6961/6962 L322V/V407I
 +++ 6963/6964 R173K ++ 6965/6966 N399S + 6967/6968 N3M ++ 6969/6970 M183I +++ 6971/6972
 K428L + 6973/6974 K302G + 6975/6976 K222R + 6977/6978 R435N ++ 6979/6980 Q269R +++
 6981/6982 R435E +++ 6983/6984 V9S ++ 6985/6986 S442F +++ 6987/6988 Y396T ++ 6989/6990
 K428I +++ 6991/6992 S214R ++ 6993/6994 E405P + 6995/6996 R234S + 6997/6998 P439E +
 6999/7000 N3F + 7001/7002 V390I +++ 7003/7004 P439R + 7005/7006 Y193F ++ 7007/7008 R403V
 ++ 7009/7010 K171L + 7011/7012 P158T + 7013/7014 N408D + 7015/7016 G96C + 7017/7018 K171A
 ++ 7019/7020 Q269M + 7021/7022 D300E + 7023/7024 K222N + 7025/7026 D170G + 7027/7028 P2M
 + 7029/7030 K428Y + 7031/7032 K94R + 7033/7034 L322T ++ 7035/7036 E448Q + 7037/7038 D98T
 ++ 7039/7040 N3L + 7041/7042 R411K ++ 7043/7044 D170H + 7045/7046 Q269L + 7047/7048 K171P
 +++ 7049/7050 E405S ++ 7051/7052 M183P ++ 7053/7054 V395I/P439V +++ 7055/7056 L322S +++
 7057/7058 V9A + 7059/7060 R449L + 7061/7062 Y165L +++ 7063/7064 A438E + 7065/7066 R234T ++
 7067/7068 L229Q + 7069/7070 K428G ++ 7071/7072 N399P +++ 7073/7074 F64M + 7075/7076 E330S
 ++ 7077/7078 N408K + 7079/7080 D434G + 7081/7082 E405T + 7083/7084 N399Q ++ 7085/7086
 R435L ++ 7087/7088 R435V + 7089/7090 P2F ++ 7091/7092 E448K + 7093/7094 F64L + 7095/7096
 K53E/I437T +++ 7097/7098 R411H + 7099/7100 K428V +++ 7101/7102 K428T ++ 7103/7104 R173L +
 7105/7106 D98S + 7107/7108 P439W + 7109/7110 Y396V + 7111/7112 S113P ++ 7113/7114 K428Q
 +++ 7115/7116 V9M + 7117/7118 A438R ++ 7119/7120 D434E +++ 7121/7122 R411T +++ 7123/7124
 R234N ++ 7125/7126 K428S +++ 7127/7128 D300A + 7129/7130 R173S + 7131/7132 K289R +
 7133/7134 L322P +++ 7135/7136 S113G + 7137/7138 R423T + 7139/7140 P439M + 7141/7142 E405A
 + 7143/7144 S214R/K222H + 7145/7146 D72E/D170A ++ 7147/7148 T4S +++ 7149/7150 R449G +
 7151/7152 D134E/P158N +++ 7153/7154 R435K + 7155/7156 L322A ++ 7157/7158 S452T +
 7159/7160 R412K +++ 7161/7162 A118C + 7163/7164 Y444A + 7165/7166 E405D + 7167/7168 L120V
 ++ 7169/7170 S253E +++ 7171/7172 P2V + 7173/7174 D170P ++ 7175/7176 K289G + 7177/7178
 K428E ++ 7179/7180 V390R + 7181/7182 S304P + 7183/7184 S214K ++ 7185/7186 R173I + 7187/7188
 S113D +++ 7189/7190 R435I +++ 7191/7192 D72E/E405S ++ 7193/7194 R435A + 7195/7196 E398S +

7197/7198 I454V ++ 7199/7200 R449S + 7201/7202 K428N +++ 7203/7204 V9G ++ 7205/7206 K289D + 7207/7208 S304K ++ 7209/7210 L322G ++ 7211/7212 A129P + 7213/7214 S253N + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 6468, and defined as follows: “+” = production at least 1.05-fold, but less than 1.14-fold; “++” = at least 1.14-fold, but less than 1.2-fold; and “+++” = at least 1.2-fold increased production as compared to the reference polypeptide.

SFP Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(613) Shake flask powders (SFP) were reconstituted to a concentration of 20 g/L and diluted to 0.006-0.2 g/L SFP in 100 μ L total reaction volumes of 50 mM potassium phosphate buffer, pH 6, 15 mM stevioside (>95% purity) or rebaudioside D, 0.2 mM ADP (Amresco, ultra pure grade) co-substrate, 0.05 g/L SUS SFP SEQ ID NO: 2432, and 37.5 mM sucrose (cane sugar). The reactions were incubated in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 hours at 55° C. with no pre-incubation or at 60° C. following 15 minute pre-incubation at 79° C. The reactions were solubilized by 20 $\times$ dilution with water, quenched by 5 $\times$ dilution in acetonitrile with 0.2% formic acid, cleared by centrifugation, and diluted 15 $\times$ with water for analysis. Glycosylated products were detected in by SPE-QQQ as described in Example 31, Table 31.1. A further thermostability characterization was conducted with the clarified shake flask lysates prior to lyophilization as follows: lysates were diluted 400 $\times$ in buffer and incubated in a thermocycler at a gradient of 60-75° C. for 16 h. To determine the percent activity remaining, the pre-incubated lysates were assayed as described above with either stevioside or rebaudioside D and 4 h incubation at 60° C. The percent of the remaining activity is expressed as production at the high temperature divided by production at lowest pre-incubated temperature.

(614) TABLE-US-00154 TABLE 72.3 β 1,3GT Round 20 Shake Flask Powder Variants and RebA, RebM, and Thermostability Levels Amino Acid Differences Increased Increased Increased SEQ ID (Relative to RebA, RebA, RebM, % Activity NO: (nt/aa) SEQ ID NO: 6468) 55° C..sup.a 60° C..sup.a 60° C..sup.a Remaining.sup.b 6739/6740 F45L/G84N/P168T/K190R/ + +++ ++ ++ A199Q/S254A/S273R/V365I/ A366C 6873/6874 H81T/P83S/G84N/R88T/ + +++ + + V155L/K190R/V263T/R331K/ A366C 6863/6864 S37R/S41A/F45L/D72T/ + ++ ++ + K190R/R331K/A366C 6755/6756 S37R/S41A/F45L/V155L/ + +++ ++ ++ A366C/K432L/L433A 6809/6810 S37R/S41A/F45L/D72T/ + +++ ++ ++ V263T/A366C/K432L/L433A 6743/6744 V155L/P168T/K190R/Q197K/ + ++ ++ + A199Q/A366C 6695/6696 A199Q/V263T/R331K/V365I/ + +++ + ++ A366C 6837/6838 D72T/V155L/K190R/A366C + ++ ++ - 6849/6850 D72T/H81T/P83S/G84N/R88T/ + ++ ++ ++ V155L/S273R/R331K/ D375P 6707/6708 F45L/D72T/G84N/R88T/ + +++ + +++ Q197K/D375P 6681/6682 P83S/R88T/V155L/S273R/ - +++ ++ ++ A366V/D375P 6697/6698 V365I/A366C - + + + 6785/6786 V155L/P168T/D375P - ++ ++ +++ 6725/6726 Q197K/I202H/H248W - + ++ ++ .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 6468, at 0.025 g/L shake flask powder and defined as follows: “-” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.8-fold; “++” = at least 1.8-fold, but less than 2.5-fold; and “+++” = at least 2.5-fold increased production, relative to reference polypeptide. .sup.bThe percent of activity remaining for each variant was determined following 16 h pre-incubation at 70° C., relative to the production of each variant following 16 h pre-incubation at 60° C., and defined as follows: “-” = less than 40% of activity remained following 16 h pre-incubation at 60° C.; “+” = at least 40%, but less than 60% of the activity remained; “++” = at least 60%, but less than 80% of the activity remained; and “+++” = at least 80% of the activity remained.

(615) In these experiments, the fourteen variants in Table 72.3 produced more rebaudioside A from stevioside and more rebaudioside M from rebaudioside D at 60° C. with pre-incubation than SEQ ID NO: 6468, and all but four also outperformed at 55° C. without pre-incubation. SEQ ID NO: 6864 was selected as the best enzyme for the catalysis of glycosyltransfer from ADP-glucose to stevioside and rebaudioside D for the formation of rebaudioside A and rebaudioside M, respectively.

Example 73

Beta-1,3-ADP-Glycosyltransferase Variants of SEQ ID NO: 6864

(616) In this Example, experiments for evolution and screening of β 1,3-glycosyltransferase (β 1,3GT) polypeptides derived from SEQ ID NO: 6864 for improved glucosylation of steviol glycosides using in

situ synthesized ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 6863 (i.e., SEQ ID NO: 6864) was carried out by constructing libraries of variant genes in which mutations associated with improved production identified during the development of the present invention were recombined. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a twenty-first round ("Round 21") of 37 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides. (617) HTP Assay for Glucose Transfer from Sucrose to ADP and then from ADP-Glucose to Rebaudioside A 60

(618) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 6863 variants. Pellets were lysed, and lysate was cleared as described in Example 34 and then diluted 40× into 50 mM potassium phosphate buffer, pH 6. In order to thermally challenge the lysates, they were pre-incubated at 75° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 h. Assays were conducted with 10 µL pre-incubated lysate in 100 µL reactions and with 20 g/L rebaudioside A 60% substrate, 0.1 g/L ADP (Amresco, ultra pure grade) co-substrate, 0.05 g/L SUS SFP SEQ ID NO: 2510, 0.1 g/L β-1,2-glycosyltransferase (β12GT) SEQ ID NO: 4550 and 37.5 mM sucrose (cane sugar). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 h. Reactions were solubilized by 20× dilution with water, quenched by 5× dilution with acetonitrile with 0.2% formic acid, and diluted 15× with water for analysis. Samples were analyzed by RapidFire-MS/MS as described in Example 31, Table 31.1. Glycosyltransferase variant polypeptides that produced rebaudioside A/rebaudioside M from stevioside/rebaudioside D with in situ synthesized ADP-glucose at greater quantities than SEQ ID NO: 6864 were identified. The engineered polypeptides are listed in Table 73.1. Shake-flask scale cultures were grown, lysed, cleared, and lyophilized to powder as described in Example 31 for analysis of variants shown in Table 73.2 relative to SEQ ID NO: 6864.

(619) TABLE-US-00155 TABLE 73.1 β1,3GT Round 21 Combinatorial Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 6864) RebM.sup.a
7361/7362 R109W/M144Q/A153V/V155L/Q159K/L433A + 7363/7364 M87K/M144Q +++ 7365/7366 M87K/M144Q/Q159K/T361C + 7367/7368 M87K/L433A + 7371/7372 M144Q +++ 7373/7374 M87K/M144Q/Q159K/T361C/L433A + 7375/7376 R88T/M144Q/Q159K +++ 7377/7378 R88T/M144Q ++ 7379/7380 T5S/D69A/I91Q/A212L/Q303V ++ 7381/7382 V263T/E288P/Q303V/S317Y ++ 7383/7384 T5S/E6P/D69A/E288P/Q303V ++ 7385/7386 T5S/L25Q/E288P + 7387/7388 T5S/E6P/I91Q/E288P/S317Y +++ 7389/7390 T5S/I91Q/S317Y/V421I + 7391/7392 L25Q/I91Q/V263T/E288P/Q303V + 7393/7394 E288P + 7395/7396 T5S/L25Q/I91Q/V263T + 7397/7398 T5S/I91Q/Q303V ++ 7399/7400 T5S/E288P ++ 7401/7402 E6P/A212L/E288P/Q303V + 7403/7404 L25Q/I91T/Q303V/S317Y/V369K + 7405/7406 L25Q/V263T/S317Y +++ 7407/7408 E6P/I91Q/A212L/E288P/Q303V/V369K/V421I + 7409/7410 I91Q/E288P/S317Y/V369K/V421I ++ 7411/7412 L25Q/I91Q/A212L/E288P + 7413/7414 A212L/E288P + 7415/7416 I91Q/Q303V +++ 7417/7418 I91Q/E288P/Q303V/S317Y/V369K ++ 7419/7420 I91Q/S317Y ++ 7421/7422 T5S/I91Q/E288P/Q303V +++ 7423/7424 T5S/E6P/L25I/S317Y ++ 7425/7426 I91Q/V263T/S317Y/V369K ++ 7427/7428 I91Q/S317Y/V369K + 7429/7430 T5S/D69A/I91Q/A212L/E288P + 7431/7432 T5S/L25Q/I91Q/A212L/Q303V/S317Y ++ 7433/7434 E6P/E288P + 7435/7436 L25Q/I91Q/S317Y/V369K + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 6864, and defined as follows: "+" = at least 1.12-fold, but less than 1.23-fold; "++" = at least 1.23-fold, but less than 1.33-fold; and "+++" = at least 1.33-fold increased production, as compared to the reference polypeptide.

SFP Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(620) Shake flask powders (SFP) were reconstituted to a concentration of 20 g/L and diluted to 0.003-0.1 g/L SFP in 100 µL total reaction volumes of 50 mM potassium phosphate buffer, pH 6, 15 mM stevioside (>95% purity) or rebaudioside D, 0.2 mM ADP (Amresco, ultra pure grade) co-substrate, 0.05 g/L SUS SFP SEQ ID NO: 2510, and 37.5 mM sucrose (cane sugar). The reactions were incubated in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 hours at 60° C. The reactions were solubilized by 20× dilution with water, quenched by 5× dilution with acetonitrile with 0.2% formic acid,

cleared by centrifugation, and diluted 15× with water for analysis. Glycosylated products were detected in by SPE-QQQ as described in Example 31, Table 31.1. A further thermostability characterization was conducted with the clarified shake flask lysates prior to lyophilization as follows: lysates were diluted 400× in 50 mM potassium phosphate buffer, pH 6, and incubated in a thermocycler at a gradient of 60-75° C. for 16 h. To determine the percent of activity remaining, the pre-incubated lysates were assayed as described above with either stevioside or rebaudioside D and 4 h incubation at 60° C. The percent of activity remaining is expressed as production at the high temperature divided by production at the lowest pre-incubated temperature.

(621) TABLE-US-00156 TABLE 73.2 β 1,3GT Round 21 Shake Flask Powder Variants and RebA, RebM, and Thermostability Levels Increased Increased SEQ ID NO: Amino Acid Differences RebM, RebA, % Activity (nt/aa) (Relative to SEQ ID NO: 6864) 60° C..^a 60° C..^a Remaining..^b 7369/7370 M87K/M144Q + + + 7375/7376 R88T/M144Q/Q159K + + - 7387/7388 T5S/E6P/I91Q/E288P/S317Y + + + 7405/7406 L25Q/V263T/S317Y + - + 7423/7424 T5S/E6P/L25I/S317Y - - + 7431/7432 T5S/L25Q/I91Q/A212L/Q303V/S317Y - - + .^aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 6864, at 0.0125 g/L shake flask powder and defined as follows: “-” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.15-fold; “++” = at least 1.15-fold increased production, relative to reference polypeptide. .^bThe percent of activity remaining for each variant was determined following 24 h pre-incubation at 73.5° C., relative to the production from each variant following 24 h pre-incubation at 60° C. and is defined as follows: “-” = less than 20% of activity remained following 16 h pre-incubation at 60° C.; “+” = at least 20%, but less than 40% activity remained; and “++” = at least 40% activity remained.

(622) In these experiments, five variants in Table 73.2 produced more rebaudioside A from stevioside and more rebaudioside M from rebaudioside D at 60° C. than SEQ ID NO: 6864. SEQ ID NO: 7388 was selected as the best enzyme for the catalysis of glycosyltransfer from ADP-glucose to stevioside and rebaudioside D for the formation of rebaudioside A and rebaudioside M, respectively.

Example 74

Beta-1,3-ADP-Glycosyltransferase Variants of SEQ ID NO: 7388

(623) In this Example, experiments for evolution and screening of β 1,3-glycosyltransferase (β 1,3GT) polypeptides derived from SEQ ID NO: 7388 for improved glucosylation of steviol glycosides using in situ synthesized ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 7387 (i.e., SEQ ID NO: 7388) was carried out by constructing libraries of variant genes in which mutations associated with improved production identified during the development of the present invention were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a twenty-second round (“Round 22”) of 88 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(624) HTP Assay for Glucose Transfer from Sucrose to ADP and then from ADP-Glucose to Rebaudioside A 60

(625) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 7387 variants. Pellets were lysed, and lysate was cleared as described in Example 60, and then diluted 20× into 50 mM potassium phosphate buffer, pH 6. In order to thermally challenge the lysates, they were pre-incubated at 75° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 h. Assays were conducted with 10 μ L pre-incubated lysate in 100 μ L reactions and with 20 g/L rebaudioside A 60% substrate, 0.05 g/L ADP (Amresco, ultra pure grade) co-substrate, 0.05 g/L SUS SFP SEQ ID NO: 2510, 0.08 g/L β -1,2-glycosyltransferase (β 12GT) SEQ ID NO: 7324, and 40 g/L sucrose (cane sugar). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 16 h. Reactions were solubilized by 20× dilution with water, quenched by 5× dilution with acetonitrile with 0.2% formic acid, and diluted 20× with water for analysis. Samples were analyzed by RapidFire-MS/MS as described in Example 31, Table 31.1. Glycosyltransferase variant polypeptides that produced rebaudioside M from rebaudioside A 60 with in situ synthesized ADP-glucose at greater quantities than SEQ ID NO: 7388 were identified. The engineered polypeptides are listed in Table 74.1 and Table 74.2. Shake-flask scale cultures were grown,

lysed, cleared, and lyophilized to powder as described in Example 31 for analysis of variants shown in Table 74.3 relative to SEQ ID NO: 7388.

(626) TABLE-US-00157 TABLE 74.1 β 1,3GT Round 22 Combinatorial Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 7388) RebM.sup.a
7945/7946 M87K/M144Q + 7947/7948 K53E/W233Q/Q269R/V390I/V395I/Y396T ++ 7949/7950 V9G/M144Q/K331R + 7951/7952 M87K/M144Q/Y396T ++ 7953/7954 K53E/M144Q/W233Q/Q269R/K331R/K428I ++ 7955/7956 Q269R ++ 7957/7958 R37S/W233S + 7959/7960 M183L/W233Q/R234N/K331R/K428I + 7961/7962 V9G/R37S/S113D/Y396T ++ 7963/7964 V9G/M87K + 7965/7966 R37S/S113D/M144Q/Y396T + 7967/7968 Q269R/K428I/I437L ++ 7969/7970 S113D/W233Q/R234N + 7971/7972 W233Q/Y396T/N399Q + 7973/7974 K53E/R234N ++ 7975/7976 M144Q/R234N/Q269R + 7977/7978 D134E/K222A/V263T ++ 7979/7980 V263T ++ 7981/7982 D69A/D134E ++ 7983/7984 D69A/V263T/D434E/A438S/P439H + 7985/7986 K171P/V263T ++ 7987/7988 L322S + 7989/7990 D69A/V263T ++ 7991/7992 H81T +++ 7993/7994 K222A/V263T/R435I/S442F + 7995/7996 H81T/K222A/V263T/L322S/R435E/S442F ++ 7997/7998 H81T/D134E +++ 7999/8000 H81T/L433A/R435I/A438R/S442F ++ 8001/8002 D69A +++ 8003/8004 P2K/N3M/L433A/R435E/S442F + 8005/8006 D69A/H81T + 8007/8008 H81T/K222A +++ 8009/8010 K222A +++ 8011/8012 P2K/H81T + 8013/8014 D69A/P439H +++ 8015/8016 S61Q/L243I/D300A/R308L/V407I/R411T + 8017/8018 C156S/Y165L/H248W/D300A/Q303V/R308L ++ 8019/8020 L120V/C156S/Q159K/H248W/D300A/R308L +++ 8021/8022 L120V/Q159K/Q197K/V365I/R411T + 8023/8024 L120V/S253T/D300A/Q303V/R308L/V407I + 8025/8026 S61Q/C156S/Q159K/L163A/Y165L/L243I/H248W/S253T/D300A/ ++ Q303V/R308L 8027/8028 R308L + 8029/8030 A85V/C156S/Q159K/L243I/H248W/S253T/R308L/E405P/V407I/ +++ R411T 8031/8032 S61Q/L120V/C156S/L163A/Y165L/Q197K/Q303V/R308L + 8033/8034 S61Q/C156S/Q197K/S253E +++ 8035/8036 Q197K/D300A/R308L/R411T ++ 8037/8038 Q159K/L163A/Y165L/Q197K/S214R/L243I/D300A/Q303V/ + R308L/V407I 8039/8040 C156S/S214R/R308L/R411T +++ 8041/8042 S61Q/S214R + 8043/8044 L120V/Q159K/Q197K/R308L/V407I/R411T ++ 8045/8046 S61Q/C156S/L163A/L243I/H248W/D300A/Q303V/R308L/V407I +++ 8047/8048 S61Q/C156S/L163A/D300A/V365I/R411T + 8049/8050 L120V/C156S/Q159K/E169D/Q197K/S214R/Q303V/R308L/V365I/ ++ E405P/V407I 8051/8052 L120V/R308L/V407I/R411T + 8053/8054 C156S/H248W/S253T/R308L +++ 8055/8056 L120V/C156S/H248W/Q303V/R308L/R411T ++ 8057/8058 Q197K/S253T/R308L/V407I + 8059/8060 S61Q/L120V/D300A/Q303V/V407I +++ 8061/8062 L120V/Q159K/Y165L/Q197K ++ 8063/8064 S61Q/V365I/E405P + 8065/8066 L120V/Q197K/S253T/D300A/R308L + 8067/8068 S61Q/S214R/D300A/R308L + 8069/8070 L163A/Q197K/S253E/D300A/Q303V/R308L/V365I + 8071/8072 L120V/Q159K + 8073/8074 S61Q/L163A/D300A/Q303V/R308L/E405P/V407I/R411T +++ 8075/8076 S61Q/Y165L/H248W/S253T/V407I/R411T + 8077/8078 S61Q/D300A/Q303V/R308L +++ 8079/8080 C156S/Q197K/H248W/D300A/R411T ++ 8081/8082 D300A/Q303V/R308L + 8083/8084 D300A/R308L/E405P/R411T + 8085/8086 Q197K/D300A/Q303V/R308L/V365I + 8087/8088 S61Q/L120V/Q159K/D300A/R308L/V407I +++ 8089/8090 S61Q/D300A/Q303V/E405P + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 7388, and defined as follows: “+” = at least 1.19-fold, but less than 1.26-fold; “++” = at least 1.26-fold, but less than 1.37-fold; and “+++” = at least 1.37-fold increased production, as compared to the reference polypeptide.

(627) TABLE-US-00158 TABLE 74.2 β 1,3GT Round 22 Saturation Mutagenesis Variants and RebM Levels SEQ ID NO: Amino Acid Differences (nt/aa) (Relative to SEQ ID NO: 7388) Increased RebM.sup.a 8339/8340 R76S + 8341/8342 A107L ++ 8343/8344 C338V ++ 8345/8346 T72P +++ 8347/8348 S61D +++ 8349/8350 L56D + 8351/8352 A107V ++ 8353/8354 S61E ++ 8355/8356 R88M + 8357/8358 R88L ++ 8359/8360 C156S + 8361/8362 A41E + 8363/8364 K139N + 8365/8366 V407T + 8367/8368 M87E + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 7388, and defined as follows: “+” = at least 1.04-fold, but less than 1.14-fold; “++” = at least 1.14-fold, but less than 1.18-fold; and “+++” = at least 1.18-fold increased production, as compared to the reference polypeptide.

SFP Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP and then from

ADP-Glucose to Stevioside and Rebaudioside D

(628) Shake flask powders (SFP) were reconstituted to a concentration of 20 g/L and diluted to 0.005-0.15 g/L SFP in 100 μ L total reaction volumes of 50 mM potassium phosphate buffer, pH 6, 15 mM stevioside (>95% purity) or rebaudioside D, 0.1 g/L ADP (Amresco, ultra pure grade) co-substrate, 0.05 g/L SUS SFP SEQ ID NO: 2510, and 37.5 mM sucrose (cane sugar). The reactions were incubated in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 hours at 60° C. The reactions were solubilized by 20 $\times$ dilution with water, quenched by 5 $\times$ dilution with acetonitrile with 0.2% formic acid, cleared by centrifugation, and diluted 15 $\times$ with water for analysis. Glycosylated products were detected in by SPE-QQQ as described in Example 31, Table 31.1. A further thermostability characterization was conducted with the clarified shake flask lysates prior to lyophilization as follows: lysates were diluted 20 $\times$ in phosphate buffer and incubated in a thermocycler at a gradient of 60-75° C. for 24 h. To determine the percent of activity remaining, the pre-incubated lysates were assayed with 20 g/L rebaudioside A 60, 0.05 g/L ADP, 40 g/L sucrose, 0.05 g/L SUS SFP SEQ ID NO: 2510, and 0.08 g/L β 1,2GT SFP SEQ ID NO: 7324, in 50 mM potassium phosphate buffer at pH 6, with 16 h incubation at 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking. Reactions were solubilized, quenched, and diluted as described for the high throughput assay. The percent activity remaining was expressed as production at the high temperature divided by production at the lowest pre-incubated temperature.

(629) TABLE-US-00159 TABLE 74.3 β 1,3GT Round 22 Shake Flask Powder Variants and RebA, RebM from RebD, and RebM from RebA60, and Thermostability Levels Increased SEQ ID Amino Acid Differences Increased Increased RebM (from NO: (Relative to RebM, RebA, A60), % Activity (nt/aa) SEQ ID NO: 7388) 60° C..sup.a 60° C..sup.a 60° C..sup.a Remaining.sup.b 7947/7948

K53E/W233Q/Q269R/ + + + + - V390I/V395I/Y396T 7949/7950 V9G/M144Q/K331R - + + + -
7951/7952 M87K/M144Q/Y396T + + + + + 7967/7968 Q269R/K428I/I437L + - + - 7977/7978
D134E/K222A/V263T + + + + + 8007/8008 H81T/K222A + + + + + 8029/8030 A85V/C156S/Q159K/
+ + + - + + + + L243I/H248W/S253T/R308L/ E405P/V407I/R411T 8033/8034 S61Q/C156S/Q197K/ + + +
+ + + + S253E 8039/8040 C156S/S214R/R308L/ + + + + + - R411T 8059/8060

S61Q/L120V/D300A/ + + + + Q303V/V407I 8087/8088 S61Q/L120V/Q159K/ + + + + + + +
D300A/R308L/V407I .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 7388, at 0.019 g/L shake flask powder and defined as follows: “-” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.21-fold; “++” = at least 1.21-fold, but less than 1.45-fold; and “+++”, at least 1.45-fold increased production, relative to the reference polypeptide. .sup.bThe percent of activity remaining for each variant was determined following 24 h pre-incubation at 73.5° C., relative to the production from each variant following 24 h pre-incubation at 60° C. and is defined as follows: “-” = less than 33% of activity remained following 24 h pre-incubation at 60° C.; “+” = at least 33%, but less than 50% of the activity remained; and “++” = at least 50% activity remained.

(630) In these experiments, all eleven variants in Table 74.3 produced more rebaudioside M from rebaudioside A 60 than SEQ ID NO: 7388, and eight variants also produced more rebaudioside A from stevioside and rebaudioside M from rebaudioside D. SEQ ID NOS: 8034 and 8088 performed the best overall. SEQ ID NO: 8088 was selected as the best enzyme for the catalysis of glycosyltransfer from ADP-glucose to stevioside and rebaudioside D for the formation of rebaudioside A and rebaudioside M, respectively, because of its superior performance in the one-pot assay.

Example 75

Beta-1,3-ADP-Glycosyltransferase Variants of SEQ ID NO: 8088

(631) In this Example, experiments for evolution and screening of β 1,3-glycosyltransferase (β 1,3GT) polypeptides derived from SEQ ID NO: 8088 for improved glucosylation of steviol glycosides using in situ synthesized ADP-glucose are described. Directed evolution of the GT encoded by SEQ ID NO: 8087 (i.e., SEQ ID NO: 8088) was carried out by constructing libraries of variant genes in which mutations associated with improved production identified during the development of the present invention were recombined and in which certain structural features were subjected to saturation mutagenesis. These libraries were then plated, grown, and screened using the high-throughput (HTP) assay described below to provide a twenty-third round (“Round 23”) of 80 engineered GT variant polypeptides with glucosyltransferase activity toward ADP-glucose and steviol glycosides.

(632) HTP Assay for Glucose Transfer from Sucrose to ADP and then from ADP-Glucose to Rebaudioside A 60

(633) Assays were performed on 96-well plates of cleared *E. coli* culture lysates expressing SEQ ID NO: 8087 variants. Pellets were lysed, and lysate was cleared as described in Example 34, and then diluted 10× into 50 mM potassium phosphate buffer, pH 6. In order to thermally challenge the lysates, they were pre-incubated at 75° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 1 h. Assays were conducted with 10 µL pre-incubated lysate in 100 µL reactions and with 20 g/L rebaudioside A 60% substrate, 0.025 g/L ADP (Amresco, ultra pure grade) co-substrate, 0.05 g/L SUS SFP SEQ ID NO: 8420, 0.08 g/L β-1,2-glycosyltransferase (β12GT) SEQ ID NO: 7784, and 40 g/L sucrose (cane sugar). The following reaction conditions were used: 50 mM potassium phosphate buffer, pH 6, 60° C. in a Thermotron® titre-plate shaker with 300 RPM shaking for 16 h. Reactions were solubilized by 20× dilution with water, quenched by 5× dilution with acetonitrile with 0.2% formic acid, and diluted 20× with water for analysis. Samples were analyzed by RapidFire-MS/MS as described in Example 31, Table 31.1. Glycosyltransferase variant polypeptides that produced rebaudioside M from rebaudioside A 60 with in situ synthesized ADP-glucose at greater quantities than SEQ ID NO: 8088 were identified. The engineered polypeptides are listed in Table 75.1 and Table 75.2. Shake-flask scale cultures were grown, lysed, cleared, and lyophilized to powder as described in Example 1 for analysis of variants shown in Table 75.3 relative to SEQ ID NO: 8088.

(634) TABLE-US-00160 TABLE 75.1 β1,3GT Round 23 Combinatorial Variants and RebM Levels SEQ ID NO: Amino Acid Differences Increased (nt/aa) (Relative to SEQ ID NO: 8088) RebM.sup.a

8481/8482 M87K/Q91L/V120L/W233Q + 8483/8484 M87K/Q91L/K94C/V263T/E389L + 8485/8486 M87A/Q91L/K94C/W233Q/E259T + 8487/8488 M87K/Q91L/K94C + 8489/8490 Q91L/K94C/W233Q/E259T + 8491/8492 M87A/Q91L/V120L/W233Q + 8493/8494 M87K/Q91L/W233Q + 8495/8496 Q91L/V120L ++ 8497/8498 W233Q/E259T/E389L +++ 8499/8500 Q91L/W233Q ++ 8501/8502 V263T/E389L + 8503/8504 Q91L/W233Q/E389L +++ 8505/8506 K94C/W233Q/R411T +++ 8507/8508 M87A/R435E ++ 8509/8510 Q91L/W233Q/E259T/E389L ++ 8511/8512 L163A/W233Q + 8513/8514 M87K/Q91L/K94C/V120L + 8515/8516 M87A/K94C ++ 8517/8518 M87A +++ 8519/8520 M87A/Q91L/L163A ++ 8521/8522 W233Q/A438S + 8523/8524 E259T + 8525/8526 M87A/Q91L/K94C/W233Q ++ 8527/8528 W233Q/E259T/V263T ++ 8529/8530 Q91L/V120L/W233Q + 8531/8532 M87A/K428I/D431M/R435E ++ 8533/8534 V263T +++ 8535/8536 L322S + 8537/8538 M87A/Q91L/V120L/W233Q/R411T/D431M/R435E/I437L ++ 8539/8540 M87K/Q91L/L163A/W233Q/V263T +++ 8541/8542 Q91L/K94C/V120L/W233Q/E389L/D431M/A438S + 8543/8544 M87A/Q91L/E259T/V263T/E389L/K428I/D431M/R435I/I437L +++ 8545/8546 M144Q/E389L + 8547/8548 M87A/W233Q/L322S/E389L/R411T + 8549/8550 M87K/K94C/M144Q/V263T/K428I/R435E +++ 8551/8552 Q91L/K94C ++ 8553/8554 M87K/Q91L + 8555/8556 E389L ++ 8557/8558 M87K +++ 8559/8560 Q91L/K94C/V120L/W233Q ++ 8561/8562 W233Q +++ 8563/8564 K428I/D431M/R435E ++ 8565/8566 M87A/Q91L/L322S + 8567/8568 M87K/Q91L/V263T/E389L ++ 8569/8570 Q91L ++ 8571/8572 M87A/W233Q/E389L +++ 8573/8574 M87K/L163A/W233Q ++ 8575/8576 M87K/Q91L/L163A/E389L ++ 8577/8578 M87A/W233Q +++ 8579/8580 M87K/Q91L/K94C/W233Q + 8581/8582 M87K/Q91L/E389L +++ 8583/8584 M87A/Q91L ++ 8585/8586 M87A/Q91L/W233Q +++ 8587/8588 M87K/Q91L/L163A/W233Q ++ 8589/8590 M87A/E259T + 8591/8592 M87A/K94C/V263T + 8593/8594 M87K/Q91L/M144Q/E259T + 8595/8596 M87A/Q91L/W233Q/E389L + 8597/8598 M87A/V263T + 8599/8600 M87A/W233Q/E259T + 8601/8602 T72P/R76S/A107L/C156S + 8603/8604 K53E/H81T/H195Q/Q197K + 8605/8606 D69A/T72P/R76S/A107L + 8607/8608 R37S/T72P/H195Q/K331R + 8609/8610 H195Q/Q197K + 8611/8612 T72P/R76S/A107L/H195Q/Q197K + 8613/8614 R37S/T72P/R76S/H81T + 8615/8616 T72P/Q269R + 8617/8618 Q197K + 8619/8620 R37S/T72P/R76S/A107L/C156S/K331R + 8621/8622 H81T/A107L/H195Q + .sup.aLevels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 8088, and defined as follows: “+” = at least 1.3-fold, but less than 1.55-fold; “++” = at least 1.55-fold, but less than 1.8-fold; and “+++” = at least 1.8-fold increased production, as compared to the reference polypeptide.

(635) TABLE-US-00161 TABLE 75.2 β1,3GT Round 23 Saturation Mutagenesis Variants and RebM

Levels SEQ ID NO: Amino Acid Differences (nt/aa) (Relative to SEQ ID NO: 8088) Increased RebM.sup.a 9223/9224 P324D + 9225/9226 V413L + 9227/9228 V451Q ++ 9229/9230 P324G + 9231/9232 S252P + 9233/9234 S111T + 9235/9236 S255T + 9237/9238 L328T + 9239/9240 S55G + .sup.a Levels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 8088, and defined as follows: “+” = at least 1.05-fold, but less than 1.2-fold; and “++” = at least 1.2-fold increased production, as compared to the reference polypeptide.

SFP Characterization Assay and Analysis for Glucosyl Transfer from Sucrose to ADP and then from ADP-Glucose to Stevioside and Rebaudioside D

(636) Shake flask powders (SFP) were reconstituted to a concentration of 20 g/L and diluted to 0.005-0.15 g/L SFP in 100 μ L total reaction volumes of 50 mM potassium phosphate buffer, pH 6, 15 mM stevioside (>95% purity) or rebaudioside D, 0.1 g/L ADP (Amresco, ultra pure grade) co-substrate, 0.05 g/L SUS SFP SEQ ID NO: 8420, and 37.5 mM sucrose (cane sugar). The reactions were incubated in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 hours at 60° C. The reactions were solubilized by 20 $\times$ dilution with water, quenched by 5 $\times$ dilution with acetonitrile with 0.2% formic acid, cleared by centrifugation, and diluted 15 $\times$ with water for analysis. Glycosylated products were detected in by SPE-QQQ as described in Example 31, Table 31.1. A one-pot reaction was conducted with 0.01-0.3 g/L SFP in 100 μ L total reaction volumes of 50 mM potassium phosphate buffer, pH 6, 20 g/L RebA60, 0.025 g/L ADP (Amresco, ultra pure grade) co-substrate, 0.05 g/L SUS SFP SEQ ID NO: 8420, 0.12 g/L β 1,2GT SFP SEQ ID NO: 7784, and 40 g/L sucrose (cane sugar). The reactions were incubated in a Thermotron® titre-plate shaker with 300 RPM shaking for 16 hours at 60° C. The reactions were solubilized by 20 $\times$ dilution with water, quenched by 5 $\times$ dilution with acetonitrile with 0.2% formic acid, cleared by centrifugation, and diluted 20 $\times$ with water for analysis.

(637) TABLE-US-00162 TABLE 75.3 β 1,3GT Round 23 Shake Flask Powder Variants and RebA, RebM from RebD, and RebM from RebA60, and Thermostability Levels SEQ ID Amino Acid Differences Increased Increased Increased RebM NO: (Relative to RebM, RebA, (from A60), (nt/aa) SEQ ID NO: 8088) 60° C..sup.a 60° C..sup.a 60° C..sup.a 8491/8492 M87A/Q91L/V120L/W233Q ++ + ++ 8503/8504 Q91L/W233Q/E389L + + ++ 8571/8572 M87A/W233Q/E389L + ++ + 8597/8598 M87A/V263T ++ + ++ 8611/8612 T72P/R76S/A107L/H195Q/Q197K ++ - +++ 8621/8622 H81T/A107L/H195Q + - ++ .sup.a Levels of increased production were determined relative to the reference polypeptide of SEQ ID NO: 8088, at 0.019 g/L shake flask powder and defined as follows: “-” = production less than that of the reference polypeptide; “+” = production at least that of the reference polypeptide, but less than 1.1-fold; “++” = at least 1.1-fold, but less than 1.25-fold; and “+++” , at least 1.25-fold increased production, relative to reference polypeptide.

(638) In these experiments, all eleven variants in Table 75.3 produced more rebaudioside M from rebaudioside A 60 than SEQ ID NO: 8088, and eight variants also produced more rebaudioside A from stevioside and rebaudioside M from rebaudioside D. SEQ ID NO: 8598 was selected as the best enzyme for the catalysis of glycosyltransfer from ADP-glucose to stevioside and rebaudioside D for the formation of rebaudioside A and rebaudioside M, respectively, because of its superior performance in the one-pot assay.

Example 76

Thermal Tolerance of Engineered Glycosyltransferases and Recycling Enzyme

(639) Three assays were performed to determine the thermal tolerance of an engineered β -1,3-glycosyltransferase (β 1,3GT, SEQ ID NO: 6864), β -1,2-glycosyltransferase (β 1,2GT, SEQ ID NO: 4550), and sucrose synthase recycling enzyme (SUS, SEQ ID NO: 2510) using shake flask powders (SFP).

(640) First, a multi-day stability assay at 60° C. was performed by diluting each enzyme SFP in 50 mM potassium phosphate pH 6 (β 1,3GT, 0.5 g/L; β 1,2GT, 0.25 g/L; SUS, 0.1 g/L) and incubating for variable times from 0-48 h prior to assaying. The amount of activity remaining was determined by assaying under the conditions described in this Example. For β 1,3GT, 10 μ L of the 0-48 h pre-incubated shake flask powder was used in 100 μ L total reaction volumes of 50 mM potassium phosphate buffer, pH 6, 15 mM stevioside (>95% purity) or rebaudioside D, 0.2 mM ADP (Amresco, ultra pure grade) co-substrate, 0.05 g/L SUS SFP SEQ ID NO: 2432, and 37.5 mM sucrose (cane sugar). The reactions were incubated in a Thermotron® titre-plate shaker with 300 RPM shaking for 4 hours at 60° C., then solubilized by 20 $\times$ dilution with water, quenched by 5 $\times$ dilution into acetonitrile with 0.2% formic acid, cleared by

centrifugation, and diluted 15× with water for analysis by SPE-QQQ as described in Example 31, Table 31.1. At 28 h, 60% of the un-preincubated activity for rebaudioside D to rebaudioside M remained, and at 48 h, 39% activity remained; for stevioside to rebaudioside A, 34% remained after 28 h and 19% remained after 48. Thus, the β 1,3GT SEQ ID NO: 6864 has a half-life of 20-40 h at 60° C. For β 1,2GT, the assay was performed similarly with 20 mM rebaudioside A (>97% purity) and 50 mM sucrose. At 24 h, 50-74% of the un-preincubated activity remained, and at 46 h, 33% of the activity remained. Thus, the β 1,2GT SEQ ID NO: 4550 has a half-life of 29-58 h at 60° C. For SUS, the assay was performed similarly with 15 mM rebaudioside A (>97% purity), 37.5 mM sucrose, 9 mM fructose, and 0.5 g/L β 1,2GT SEQ ID NO: 4550 as the coupled enzyme. At 24 h, 89% of the un-pre-incubated activity remained, and at 48 h, 86% of the activity remained. Thus, the SUS SEQ ID NO: 2510 has a half-life of >100 h at 60° C. All three of these half-lives at 60° C. represent a large difference from wild-type enzymes, which do not have significant stability above ambient temperature.

(641) Second, identical stocks of each of the three enzymes were preincubated for 24 h across a temperature gradient of 60-69.1° C. (SUS), 59.9-79.9° C. (β 1,2GT), or 59.7-75.1° C. in a thermocycler, and enzymes were assayed as described above to determine the activity remaining relative to the lowest pre-incubation temperature (~60° C. for all three). For β 1,3GT, 20% of the activity was retained following pre-incubation at 73.5° C. relative to preincubation at 59.7° C. For β 1,2GT, 17% of the activity was retained following pre-incubation at 73.6° C. For SUS, 85% of the activity was retained following pre-incubation at 69.1° C. Stability at 24 h up to temperatures >70° C. allows for a wide range of fermentation downstream processing temperatures and for a wide range of steviol glycoside conversion temperatures. By heating the steviol glycoside conversion reaction, the risk of microbial contamination is reduced and steviol glycoside substrate and product solubilities are enhanced, increasing the rate of conversion. Additionally, the intrinsic reaction rate is slightly increased by increasing temperature.

(642) Third, β 1,3GT, 0.1 g/L; β 1,2GT, 0.025 g/L; and SUS, 0.01 g/L were assayed in single substrate conversion assays as described above without pre-incubation with incubation at 55-65° C. to determine whether the enzymes would be robust in this temperature range. For all three enzymes, there was <36% increase or decrease in the conversion measured after 4 h at 65° C. relative to 55° C.

Example 77

Assay of Engineered Glycosyltransferases and Recycling Enzyme with Additional NDPs and NDP-Glucoses

(643) Multiple nucleoside diphosphates (NDPs) and nucleoside diphosphate-glucoses (NDP-glucoses) can be used with the engineered glycosyltransferases and recycling system.

(644) To determine the promiscuity of the engineered β -1,3-glycosyltransferase (β 1,3GT, SEQ ID NO: 6864) and β -1,2-glycosyltransferase (β 1,2GT, SEQ ID NO: 4550) for alternate NDP-glucose donors, three commercially available NDP-glucoses were tested: ADP-glucose (Amresco, ultra high grade), GDP-glucose (Sigma, >97% purity), and TDP-glucose (Carbosynth, >95% purity) with enzyme shake flask powders (SFP). For β 1,3GT, the reactions were carried out with 0.025 g/L SFP in 1 mM stevioside (>95% purity), 1 mM NDP-glucose, 50 mM potassium phosphate buffer, pH 6. At 1, 2, and 3 h, the reactions were solubilized by 4× dilution in water, quenched by 5× dilution in acetonitrile with 0.2% formic acid, and diluted 5× in water for analysis. For β 1,2GT, the reactions were carried out with 0.0025 g/L SFP in 1 mM rebaudioside A (>97% purity), but were otherwise identical to the method described above. The results are summarized in Table 68.1 as the percent (%) conversion of steviol glycoside substrate in the first hour.

(645) TABLE-US-00163 TABLE 77.1 NDP-Glucose Utilization by Engineered Glycosyltransferases
SEQ ID NO: (nt/aa) ADP-Glucose.sup.a GDP-Glucose.sup.a TDP-Glucose.sup.a 4550 +++++ +++ ++
6864 + + - .sup.a Levels of NDP-glucose utilization were determined as mmol steviol glycoside converted per gram shake flask powder and defined as follows: “-” = activity less than 10 mmol/g; “+” = activity between 10 and 30 mmol/g; “++” = activity between 30 and 50 mmol/g; “+++” = activity between 50 and 70 mmol/g; and “++++” = at least 70 mmol/g activity.

(646) To determine the promiscuity of the engineered β -1,2-glycosyltransferase (β 1,2GT, SEQ ID NO: 4550) and sucrose synthase recycling enzyme (SUS, SEQ ID NO: 2510) for alternate NDP co-factors, four commercially available NDPs were tested: ADP (Sigma, >95%), CDP (Sigma, >95%), GDP (Sigma, >96%), IDP (Sigma, >96%). The reactions were carried out with 0.001 g/L SUS SFP and 0.1 g/L β 1,2GT

SFP in 10 mM rebaudioside A (>97% purity), 0.2 mM NDP, 50 mM potassium phosphate buffer, pH 6. At 1, 2, and 3 h, the reactions were solubilized by 40× dilution in water, quenched by 5× dilution in acetonitrile with 0.2% formic acid, and diluted 5× in water for analysis. A second sucrose synthase (SEQ ID NO: 72) was also tested, using 10 μL of purified protein glycerol stock in the 100 μL reaction. The results are summarized in Table 77.2 as NDP turnovers (mmol rebaudioside D/mmol cofactor) in the first hour. As these data indicate the use of ADP/ADP-glucose provides useful reaction conditions. In addition, these compounds are beneficial from an economic perspective, as compared to other options (e.g., UDP/UDP-glucose).

(647) TABLE-US-00164 TABLE 77.2 NDP Utilization by Engineered Glycosyltransferase and Recycling Enzyme SEQ ID NO: (nt/aa) ADP.sup.a CDP.sup.a GDP.sup.a IDP.sup.a 2510 +++ - - - 72 ++ - + + .sup.aLevels of NDP utilization were determined as mmol steviol glycoside converted per mmol cofactor in 1 h and defined as follows: “-” = activity less than 1; “+” = between 1 and 2; “++” = between 2 and 3; and “+++” = greater than 3 cofactor turnovers.

Example 78

Process to Convert Rebaudioside A 60% to Rebaudioside M

(648) Reb A 60 (~1:2 mixture of stevioside and rebaudioside A [reb A] by mass) has unexpectedly high solubility in water and sucrose solutions. The individual aqueous solubility of stevioside and Reb A is reported to be in the range of 3-5 g/L. Surprisingly, 100-200 g/L solutions of Reb A 60 were prepared either in water or in 200 g/L of sucrose and no precipitate developed in said solutions upon standing at room temperature for one week. Remarkably, during the course of the reactions, both the Reb D intermediate and the Reb M product remained soluble (as evident by the homogenous reaction mixture) at levels (ca. 30-50 g/L) far exceeding their reported solubility limits (~0.3-0.5 and 3-5 g/L, respectively). The efficiency of the process was greatly enhanced by the unpredictably high solubility of the Reb A 60 starting material and the Reb D intermediate.

(649) A process to convert Reb A 60 to rebaudioside M was developed with an engineered β-1,3-glycosyltransferase (β1,3GT, SEQ ID NO: 6138), β-1,2-glycosyltransferase (β1,2GT, SEQ ID NO: 3696), and sucrose synthase (SUS, SEQ ID NO: 1846). A recycling stock solution was prepared, consisting of 0.2 g/L SUS, 0.1 g/L ADP, and 200 g/L sucrose in 50 mM pH 6, potassium phosphate buffer. β1,2GT was dissolved in this solution to 1.6 g/L, and a separate stock of β1,3GT was dissolved in the recycling stock solution to 2.0 g/L. Then, 100 mg of Reb A 60 was placed in a one dram vial and 0.5 mL each of the 1.6 g/L β1,2GT stock and 2.0 g/L β1,3GT stock were added. The resulting homogenous solution was stirred at 55° C. Precipitate gradually developed and at 24 h the reaction mixture was a thick white slurry. HPLC analysis showed the presence of 90-94% Reb M (~120-130 g/L Reb M).

(650) In another experiment, the above reaction was carried out in the presence of 0.5 g/L of ADP and 10 mM of EDTA. Under these conditions, the β1,2GT and β1,3GT loadings were both decreased to 0.4 g/L, while still reaching >90% conversion to Reb M. When the lyophilized enzyme powders were produced at 10-15 L scale fermentation and processed downstream with 60° C. heat treatment and ultrafiltration, 95% of the stevioside and rebaudioside A in Reb A 60 were converted to rebaudioside M under certain ratios of β1,2GT stock to β1,3GT stock.

(651) A process to convert Reb A 60 to rebaudioside M was developed with another set of engineered β-1,3-glycosyltransferase (β1,3GT, SEQ ID NO: 6864), β-1,2-glycosyltransferase (β1,2GT, SEQ ID NO: 4550), and sucrose synthase (SUS, SEQ ID NO: 2510) enzymes. A recycling stock solution was prepared, consisting of 0.2 g/L of SUS, 0.5 g/L of ADP and 200 g/L of sucrose in 50 mM pH 6, potassium phosphate buffer with 10 mM of EDTA. β1,2GT was dissolved in this solution to 0.6 g/L, and a separate stock of β1,3GT was dissolved in the recycling stock solution to 1.2 g/L. Then, 100 mg of Reb A 60 was placed in a one dram vial and 0.5 mL each of the 0.6 g/L β1,2GT stock and 1.2 g/L β1,3GT stock were added. The resulting homogenous solution was stirred at 60° C. Precipitate gradually developed and at 24 h the reaction mixture was a thick white slurry. HPLC analysis showed the presence of 90-92% Reb M (~120-130 g/L Reb M). This reaction was scaled to 2 grams Reb A 60 in 20 mL with 0.6 g/L β1,2GT, 0.2 g/L SUS, and 1.2 g/L β1,3GT with 0.5 g/L ADP and 10 mM EDTA at 60° C., and it was determined that 87.1% RebM was obtained by area under curve using JECFA method. For purification, the reaction mix was centrifuged at 40° C., the supernatant was decanted, and the pellet was resuspended with 1 volume deionized water, centrifuged at 20° C., and the supernatant was decanted. This wash was repeated for four

total washes and the pellet was lyophilized, yielding 94.3% RebM as determined by area under curve using JECFA method.

(652) All publications, patents, patent applications and other documents cited in this application are hereby incorporated by reference in their entireties for all purposes to the same extent as if each individual publication, patent, patent application or other document were individually indicated to be incorporated by reference for all purposes.

(653) While various specific embodiments have been illustrated and described, it will be appreciated that various changes can be made without departing from the spirit and scope of the invention.

Claims

1. An engineered glycosyltransferase comprising a polypeptide sequence that has at least 85% sequence identity to SEQ ID NO: 4256 and having glycosyltransferase activity.
2. The engineered glycosyltransferase of claim 1, wherein said polypeptide sequence of said engineered glycosyltransferase comprises at least a substitution or substitution set at position 7, 8/375, 9, 10, 12, 50/137/189/375, 53, 62/153/155/156/159, 62/153/155/156/159/427, 62/153/155/199/406, 65, 68, 99, 106, 110, 115, 116-131, 132, 136, 137/164/375, 153, 153/155/156, 153/155/156/159/199/238/406, 153/155/237/238/239/406, 153/155/238/239, 153/155/427, 153/156/159, 153/156/199/237/427, 153/156/427, 153/159/237/238/352, 153/176/181/427, 155/156, 155/156/176/181/199, 155/156/176/238/427, 155/181/199/238/406, 156/199, 156/199/427, 164/375/433, 170, 178, 190, 192, 194, 200, 220, 223/375, 238, 242, 245, 257, 272, 280, 302, 304, 335, 375, 385, 395, 399, 402, 408, 412, 416, 423, 427, 445, 447, or 449, wherein said positions are relative SEQ ID NO: 4256.
3. The engineered glycosyltransferase of claim 1, wherein said engineered glycosyltransferase is an NDP-glycosyltransferase selected from ADP-glucose-dependent glycosyltransferases (AGTs), CDP-glucose-dependent glycosyltransferases (CGTs), GDP-glucose-dependent glycosyltransferase (GGTs), TDP-glucose-dependent glycosyltransferases (TGTs), and IDP-glucose-dependent glycosyltransferase (IGTs).
4. A composition comprising the engineered glycosyltransferase of claim 1.
5. A method for glycosylation of a substrate comprising: (a) providing at least one substrate, the engineered glycosyltransferase of claim 1, and NDP-glucose; and (b) contacting said substrate with said glycosyltransferase in the presence of NDP-glucose under conditions such that said substrate is glycosylated to produce at least one glycosylated product.
6. The method of claim 5, wherein said glycosylated product comprises rebaudioside M, wherein said substrate comprises rebaudioside I.
7. The method of claim 5, wherein said glycosylated product comprises rebaudioside M, wherein said substrate comprises at least one stevioside substrate.
8. The method of claim 5, wherein said glycosylated product comprises rebaudioside D, wherein said substrate comprises at least one stevioside substrate.
9. The method of claim 5, wherein said NDP-glucose is selected from ADP-glucose, CDP-glucose, TDP-glucose, GDP-glucose, and IDT-glucose.
10. The method of claim 5, further comprising at least one engineered sucrose synthase and at least one additional glycosyltransferase.
11. The engineered glycosyltransferase of claim 1, wherein said polypeptide sequence of said engineered glycosyltransferase comprises at least one substitution or substitution set 8V/375L, 50V/137R/189G/375L, 62A/153A/155M/156W/159V, 62A/153A/155M/156W/159V/427T, 62A/153A/155M/199S/406Q, 137R/164E/375L, 153A, 153A/155M/156W, 153A/155M/156W/159V/199S/238G/406Q, 153A/155M/237L/238T/239F/406Q, 153A/155M/238T/239F, 153A/155M/427T, 153A/156W/159V, 153A/156W/199S/237L/427T, 153A/156W/427T, 153A/159V/237L/238G/352A, 153A/176Q/181L/427S, 155M/156W, 155M/156W/176Q/181L/199S, 155M/156W/176Q/238T/427S, 155M/181L/199S/238G/406Q, 156W/199S, 156W/199S/427T, 164E/375L/433D, 223A/375L, 375L, 427S, 7H, 9G, 10D, 12S, 53N, 65F, 68I, 99R, 106K, 106R, 1065, 110G, 115A, 115T, 116I, 131C, 1315, 132G, 132R, 132T, 136D, 136G, 136R, 1365, 170E, 178L, 190T, 192A, 192L, 192P, 194F, 200S, 220M, 220Q, 220S, 238I, 238L, 242L, 245P, 257H, 272H, 280W, 302P, 302S, 304I, 335K, 335R, 385A, 385C, 385P, 385S, 395Q, 399K, 402R,

- 408E, 412R, 416R, 423L, 445H, 445K, 445R, 447L, 449K, 449L, 449Q, or 449R, wherein the positions are relative to the reference sequence of SEQ ID NO: 4256.
12. The engineered glycosyltransferase of claim 1, wherein said polypeptide sequence of said engineered glycosyltransferase has at least 86% sequence identity to SEQ ID NO: 4256.
13. The engineered glycosyltransferase of claim 1, wherein said polypeptide sequence of said engineered glycosyltransferase has at least 90% sequence identity to SEQ ID NO: 4256.
14. The engineered glycosyltransferase of claim 1, wherein said polypeptide sequence of said engineered glycosyltransferase has at least 95% sequence identity to SEQ ID NO: 4256.
15. The engineered glycosyltransferase of claim 1, wherein said polypeptide sequence of said engineered glycosyltransferase has at least 99% sequence identity to SEQ ID NO: 4256.
16. The engineered glycosyltransferase of claim 1, wherein said polypeptide sequence of said engineered glycosyltransferase comprises 65D/N/S/C/F/G/P/Q/R/S, 116I/V, 194F/P, 200S/A/P/R/T/V/E, 257H, 385A/C/L/P/R/S, 408P/A/L/T/E, 423A/R/L, or 375L, or combinations thereof, wherein the positions are relative to the reference sequence of SEQ ID NO: 4256.
17. The engineered glycosyltransferase of claim 12, wherein said polypeptide sequence of said engineered glycosyltransferase comprises 65D/N/S/C/F/G/P/Q/R/S.
18. The engineered glycosyltransferase of claim 12, wherein said polypeptide sequence of said engineered glycosyltransferase comprises 65N.
19. The engineered glycosyltransferase of claim 12, wherein said polypeptide sequence of said engineered glycosyltransferase comprises 116I/V.
20. The engineered glycosyltransferase of claim 12, wherein said polypeptide sequence of said engineered glycosyltransferase comprises 194F/P.
21. The engineered glycosyltransferase of claim 12, wherein said polypeptide sequence of said engineered glycosyltransferase comprises 200S/A/P/R/T/V/E.
22. The engineered glycosyltransferase of claim 12, wherein said polypeptide sequence of said engineered glycosyltransferase comprises 200A.
23. The engineered glycosyltransferase of claim 12, wherein said polypeptide sequence of said engineered glycosyltransferase comprises 257H.
24. The engineered glycosyltransferase of claim 12, wherein said polypeptide sequence of said engineered glycosyltransferase comprises 385A/C/L/P/R/S.
25. The engineered glycosyltransferase of claim 12, wherein said polypeptide sequence of said engineered glycosyltransferase comprises 385P.
26. The engineered glycosyltransferase of claim 12, wherein said polypeptide sequence of said engineered glycosyltransferase comprises 408P/A/L/T/E.
27. The engineered glycosyltransferase of claim 12, wherein said polypeptide sequence of said engineered glycosyltransferase comprises 423A/R/L.
28. The engineered glycosyltransferase of claim 12, wherein said polypeptide sequence of said engineered glycosyltransferase comprises 375L.
29. The engineered glycosyltransferase of claim 12, wherein said polypeptide sequence of said engineered glycosyltransferase comprises 65N, 116I/V, 194F/P, 200A, 257H, 375L, 385P, 408P or 423R, or combinations thereof.
30. The method of claim 5, wherein said glycosylated product comprises rebaudioside M, wherein said substrate comprises rebaudioside A.
31. The method of claim 7, further comprising at least one engineered sucrose synthase and at least one engineered beta-1,3-glycosyltransferase.
32. The method of claim 8, further comprising at least one engineered sucrose synthase and at least one engineered beta-1,3-glycosyltransferase.
33. The method of claim 30, further comprising at least one engineered sucrose synthase and at least one engineered beta-1,3-glycosyltransferase.
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